CAI Services for Developers

Lab Guide

Version: CAIS-R42-DEV-202404

Informatica University

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CAI Services for Developers

Version: CAIS-R42-DEV-202404

April 2024

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Document Conventions

This guide uses the following formatting conventions:

If you see…	It means…	Example
>	Indicates a sub menu to navigate to.	Click Repository > Connect. In this example, you should click the Repository menu or button and choose Connect.
boldfaced text	Indicates text you need to type or enter.	Click the Rename button and name the new source definition **S_EMPLOYEE**.
UPPERCASE	Database tables and column names are shown in all UPPERCASE.	T_ITEM_SUMMARY
italicized text	Indicates a variable you must replace with specific information.	Connect to the Repository using the assigned *login_id*.
Note:	The following paragraph provides additional facts.	**Note:** You can select multiple objects to import by using the Ctrl key.
Tip:	The following paragraph provides suggested uses or a Velocity best practice.	**Tip:** The m_ prefix for a mapping name is…

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Module 3: Create a Basic Process
Lab 3-1: Create a Basic Process to Display User Input

Overview:
A Process Designer is a suite of tools for building cloud-based processes.

In this lab, you will learn to use the process designer to create a basic process that accepts text as an input and returns the same text as an output.

Objectives:
- Create a basic Process
- Invoke the Process

Duration:
15 Minutes

Tasks:

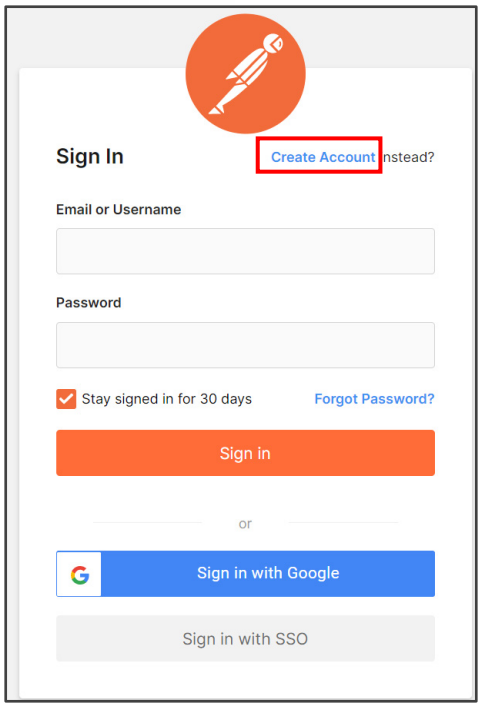
Note: The screen in your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

Save the CAI Lab Prep Files_Learners Folder

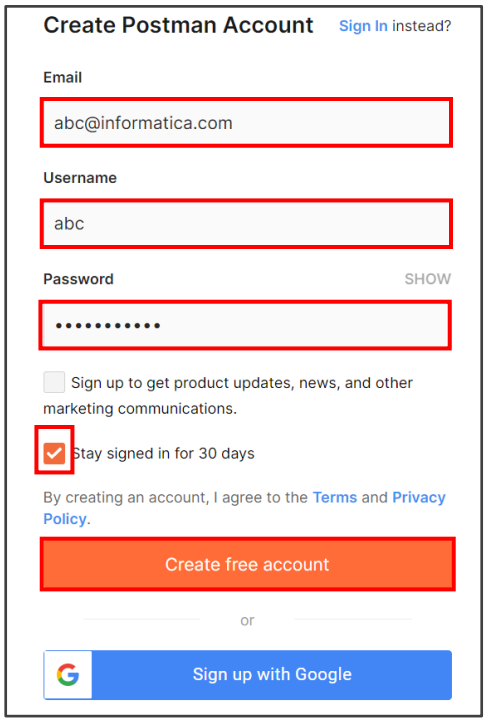
At the beginning of the class, you will receive a **CAI Lab Prep Files_Learners.zip** file from your instructor. Make sure you have unzipped the **CAI Lab Prep Files_Learners.zip** file and saved it on your desktop.

Create Postman Account:

1. Open a web browser and enter the following link to create an account on Postman: https://identity.getpostman.com/
2. Click **Create Account**.

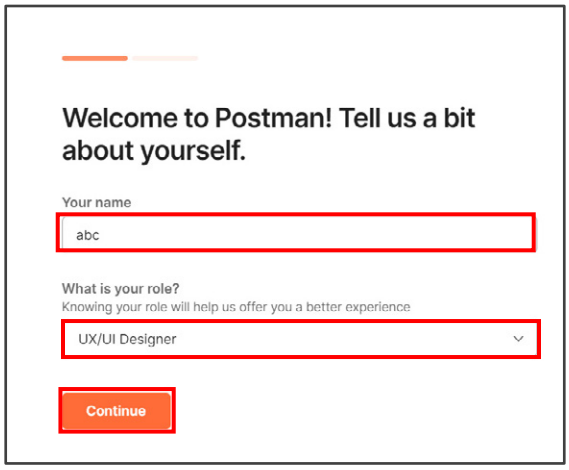


3. In the Email field, enter a valid email address.
4. In the Username field, enter a username.
5. In the Password field, enter your password.
6. Make sure that the **Stay signed in for 30 days** checkbox is selected.
7. Click **Create free account**.

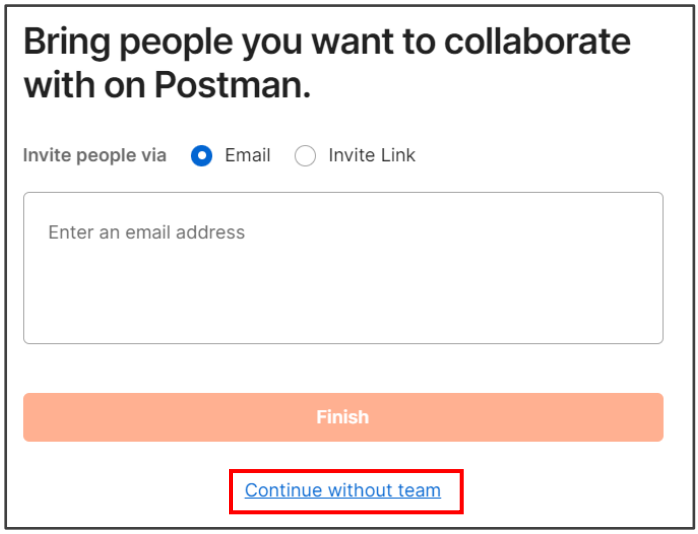


Note: You can also click the Sign up with Google tab and enter your Google credentials to sign up on Postman.

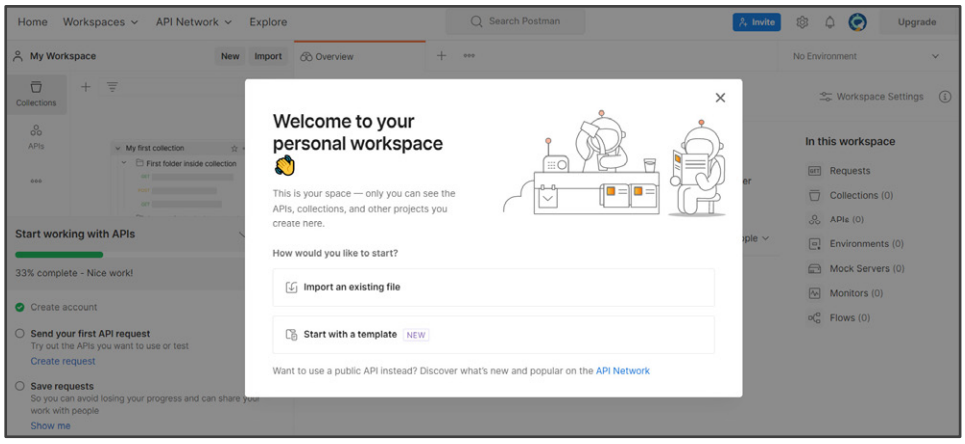
8. Enter the required details and click **Continue**.



9. Click the **Continue without team** link.



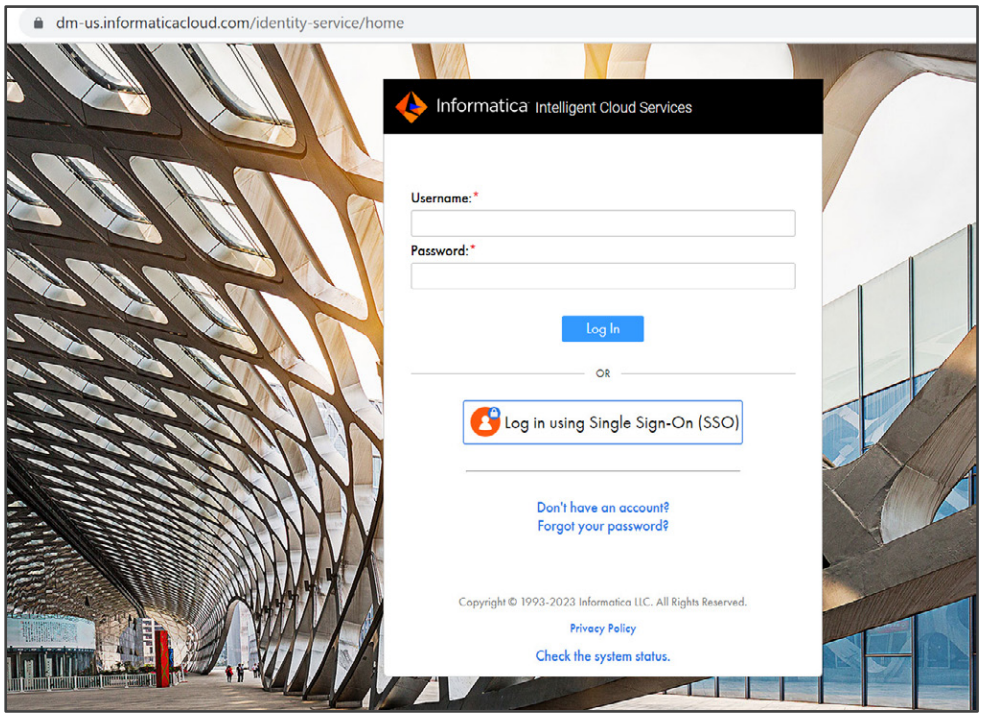
You have successfully created your account on Postman.



10. Close the popup window.

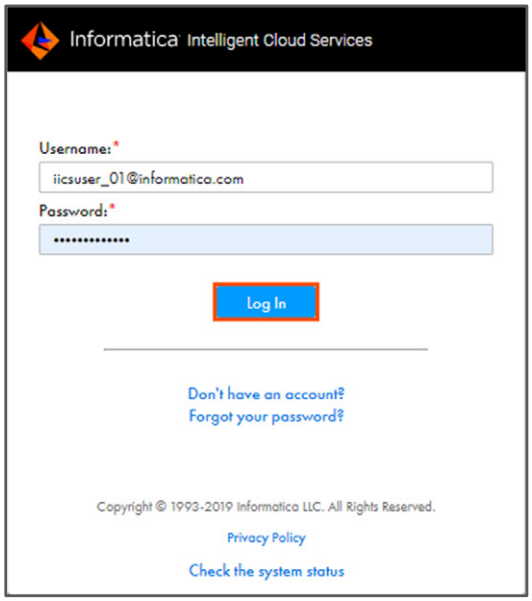
Create a Process

11. Open a web browser and enter the **IICS** link as given below:
    https://dm-us.informaticacloud.com/

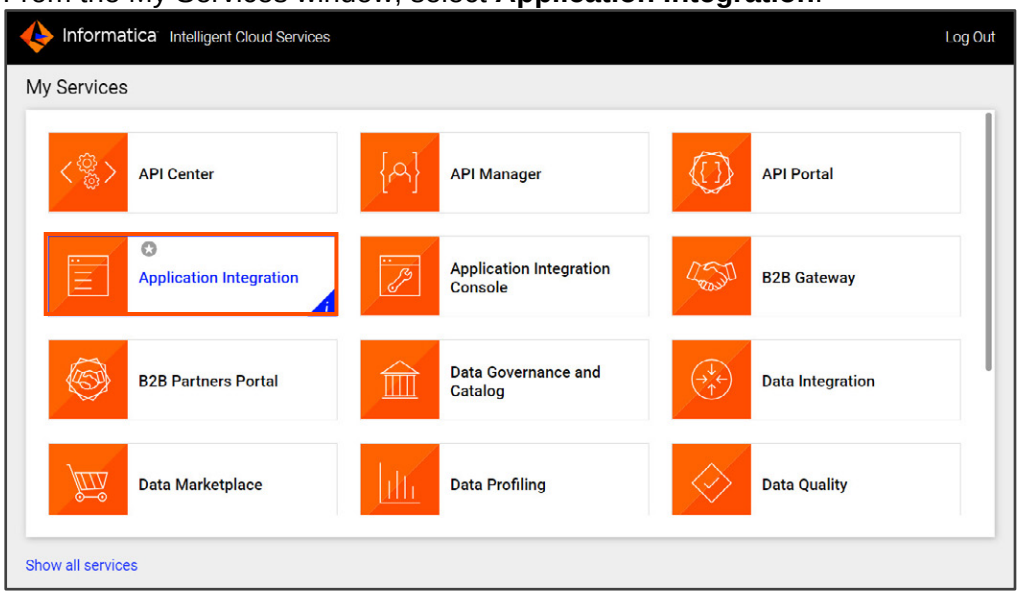


Note: Bookmark the IICS link as **Informatica Cloud**. You can use this bookmark for subsequent labs.

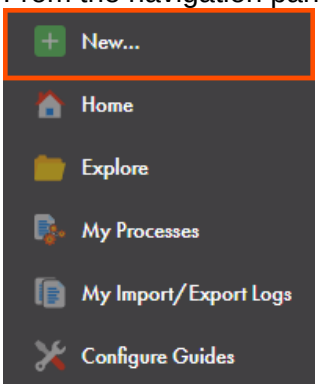
12. Enter the IICS **Username** and **Password** provided by the instructors, and click **log in**.



13. From the My Services window, select **Application Integration**.

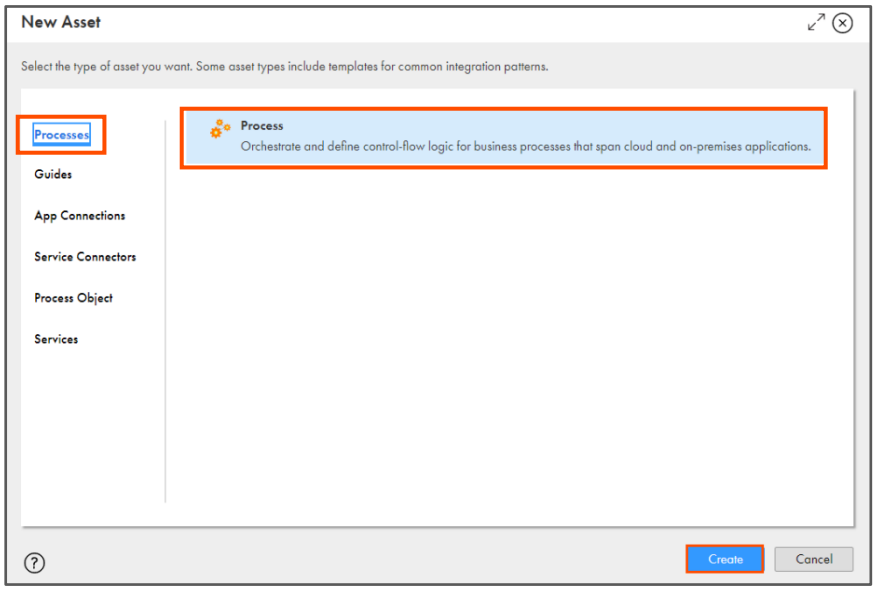


14. From the navigation pane, select **New**.



Recommended: When you create a new asset, open your project folder, and click **New**. This sets the asset location to your project folder by default.

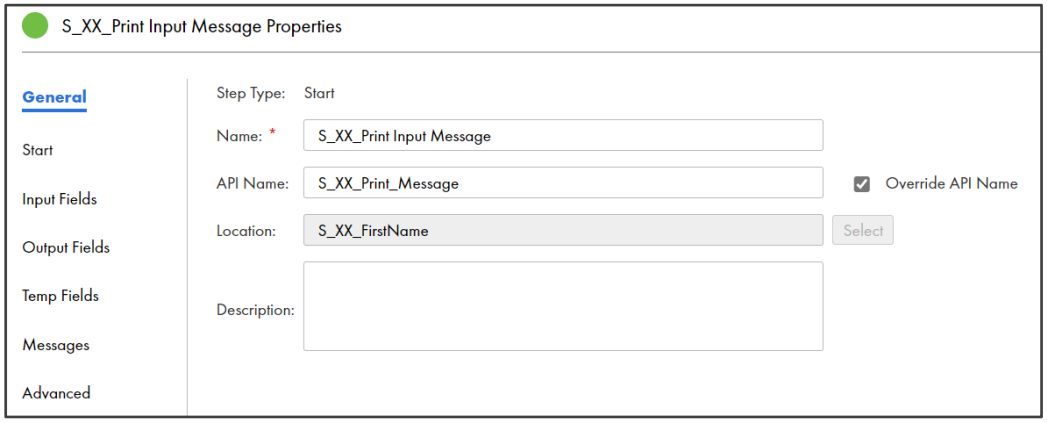
15. Retain **Processes** > **Process** and click **Create**.



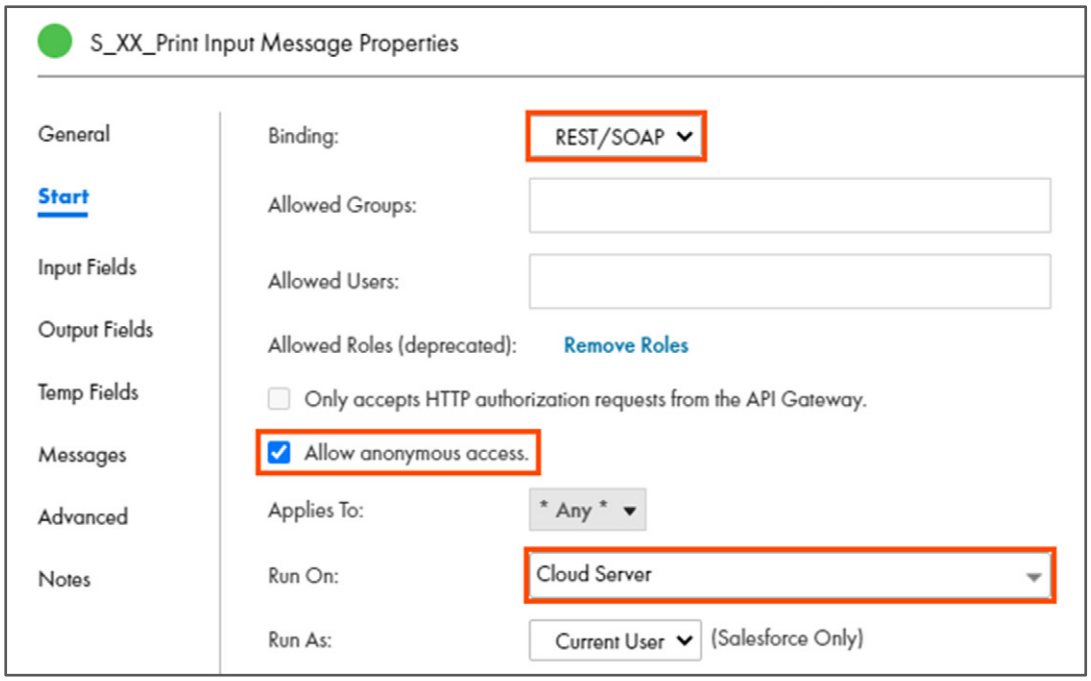
16. In the General properties tab, in the Name field, enter **S_XX_Print Input Message**.
    Note: Here, XX is your unique student number.

Provide a customized API name for the process:

17. To provide a customized API Name, select the **Override API Name** option and specify the name. For example, provide **S_XX_Print_Message**.
    Note: The only allowed characters for the API Name are alphanumeric, underscore, and hyphen.
18. In the Location field, select your project folder **S_XX_FirstName**.



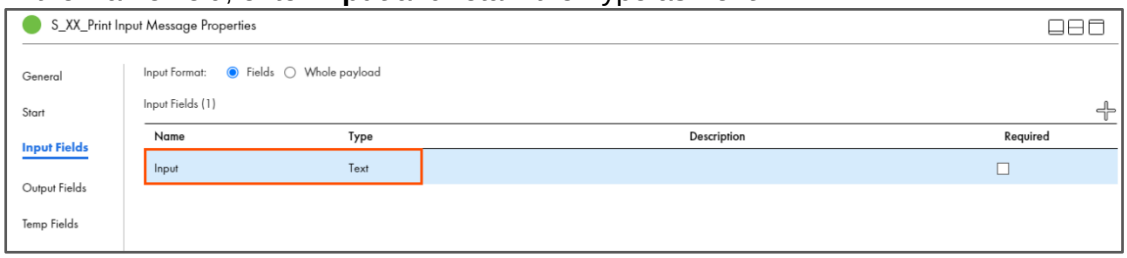
    Note: This ensures that the process is stored in your project folder.
19. In the Start tab, retain the Binding option as **REST/SOAP**.
20. Select the **Allow anonymous access** option and retain the Run On option as **Cloud Server**.



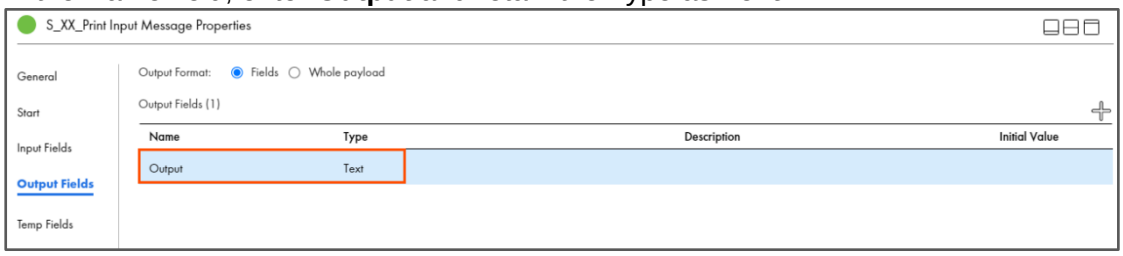
Note: The Allow anonymous access option enables all users to access the process.

Now, you need to create an input field of type text and assign a value (the user input) to it using the Assignment step. You will also create an output field that takes the input value from the Assignment step and displays it as an output. The output field must have the same type as the input field.

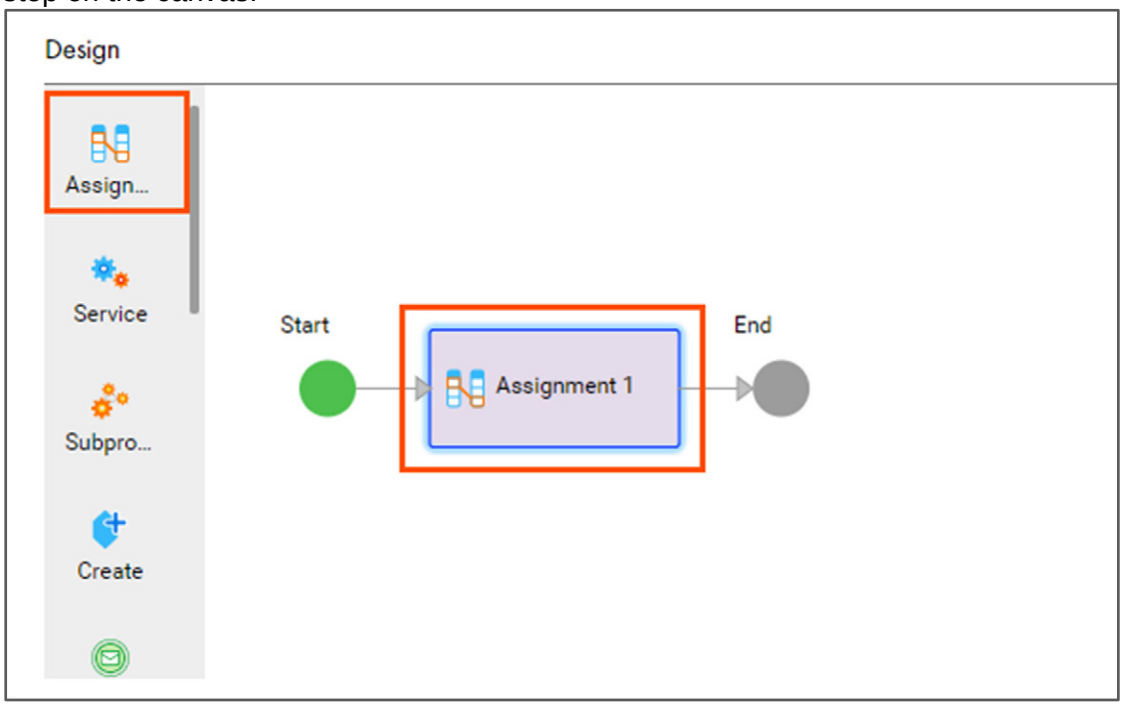
21. In the Input Fields tab, to add a new field, click .
22. In the Name field, enter **Input** and retain the Type as **Text**.



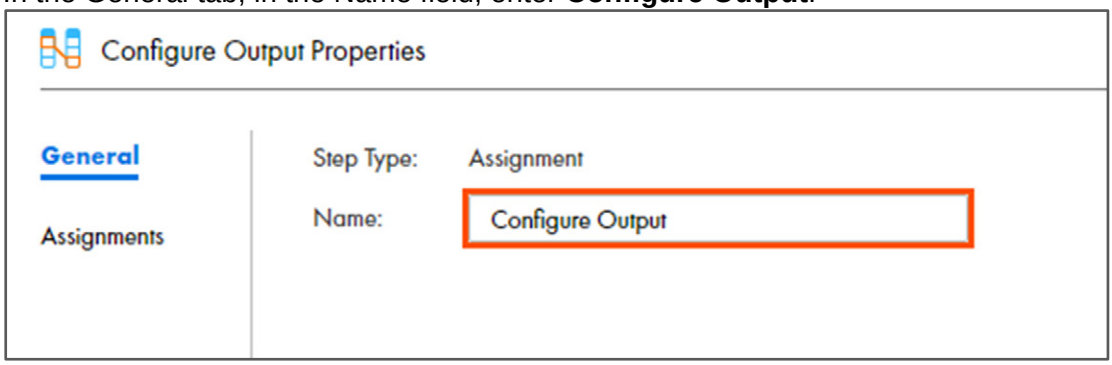
23. In the Output Fields tab, to add a new output field, click .
24. In the Name field, enter **Output** and retain the Type as **Text**.



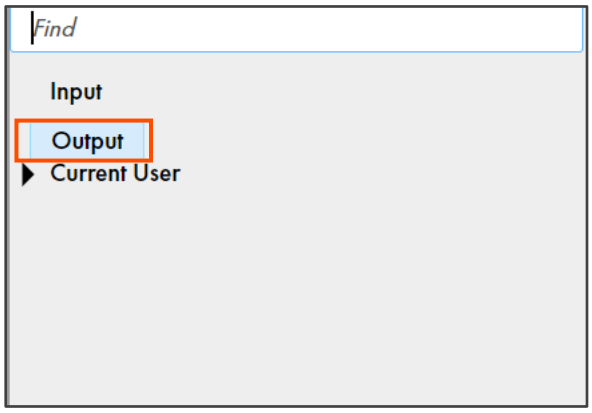
25. From the palette, drag and drop an **Assignment** step between the Start and the End step on the canvas.



26. Select the **Assignment** step.
27. In the General tab, in the Name field, enter **Configure Output**.

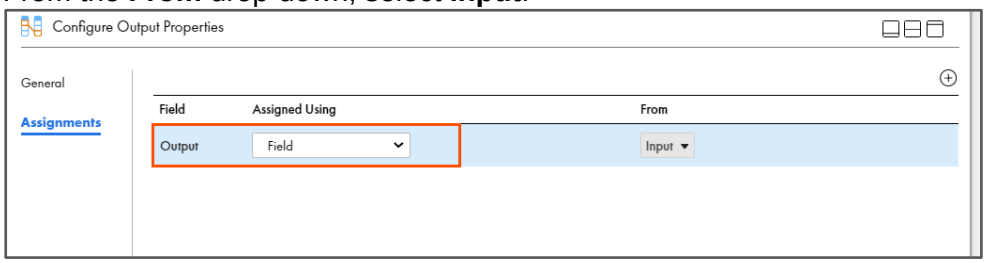


28. In the Assignments tab, click the plus **(+)** icon, and select **Output**.

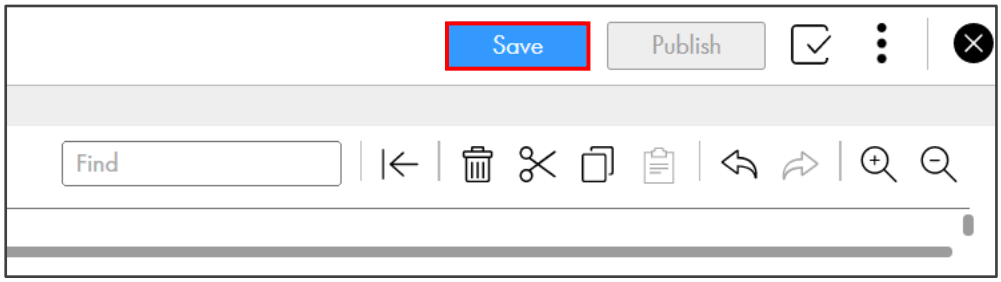


Note: This field takes values from the input field and displays them as an output.

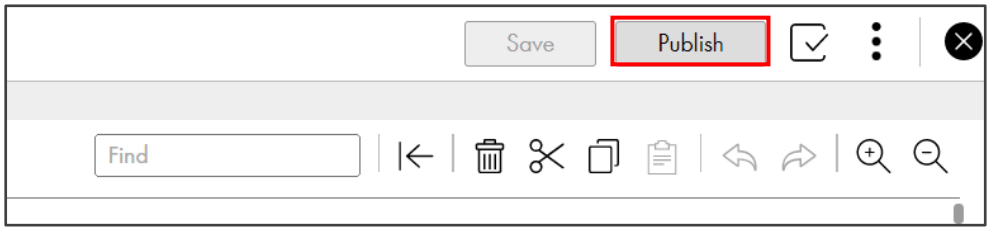
29. From the **Assigned Using** drop-down, select **Field**.
30. From the **From** drop-down, select **Input**.

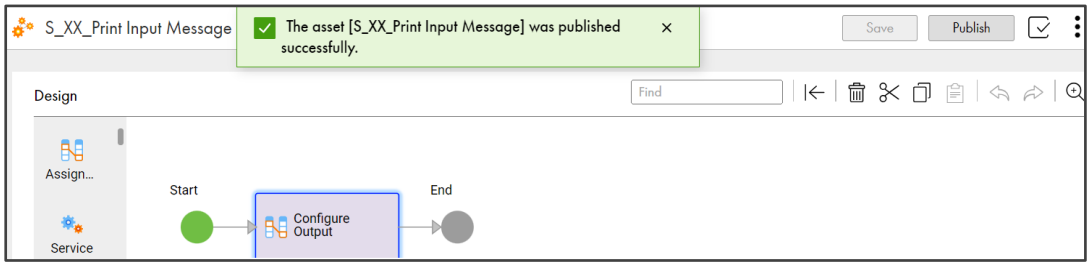


31. Click **Save**.



32. Click **Publish** to publish the process.

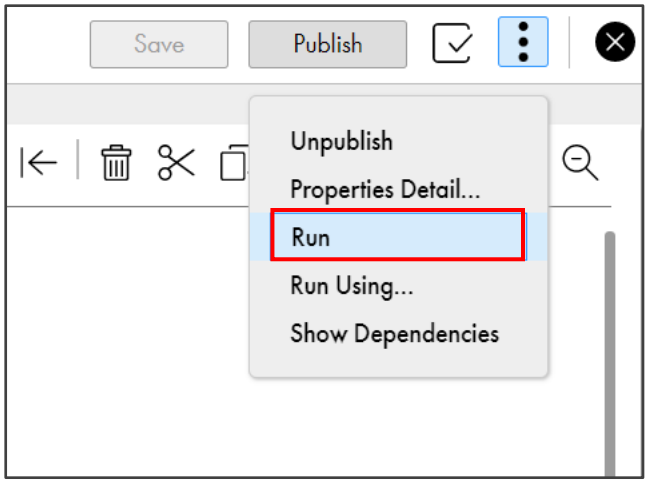




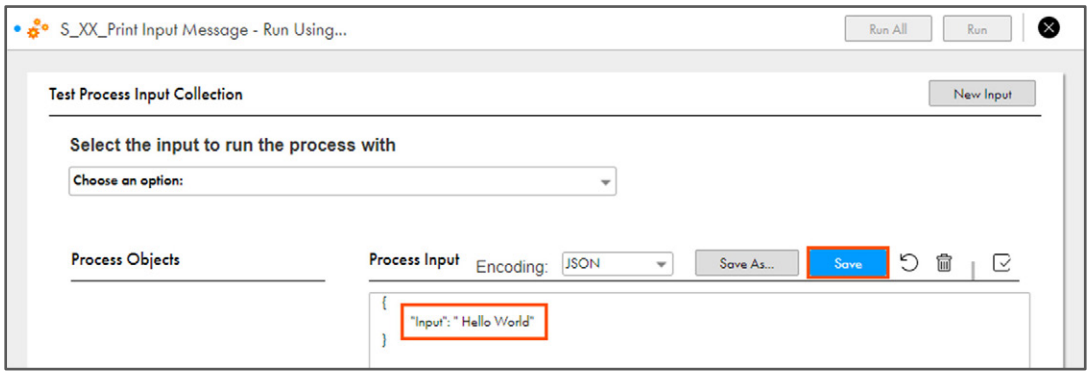
The process is published successfully.

Invoke the Process from the Interface:

33. To run the process, click ⋮, and select **Run**.

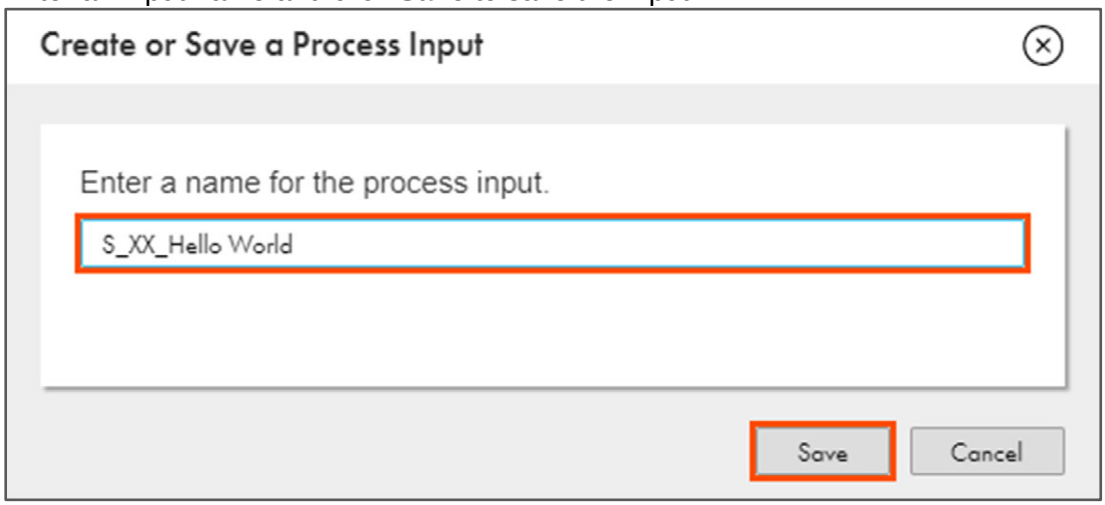


34. In the Process Input section, enter the input as **Hello World**, and click **Save**.

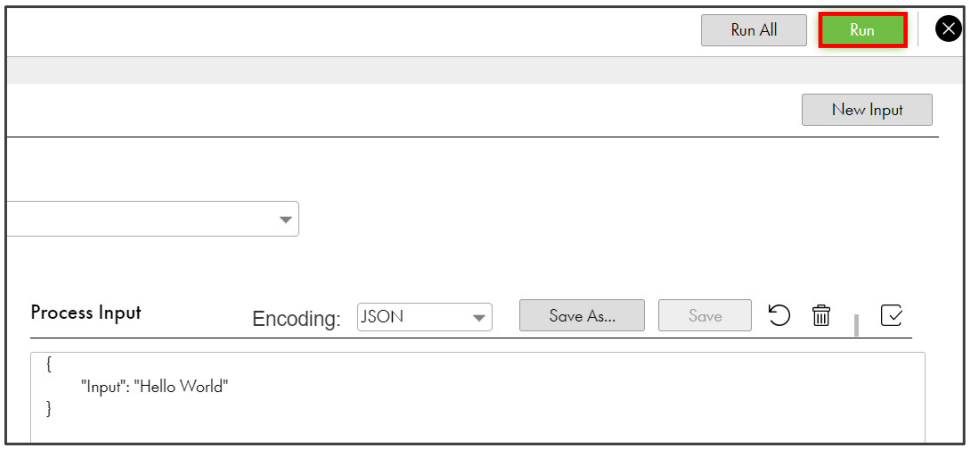


Note: This input will be available as the output.

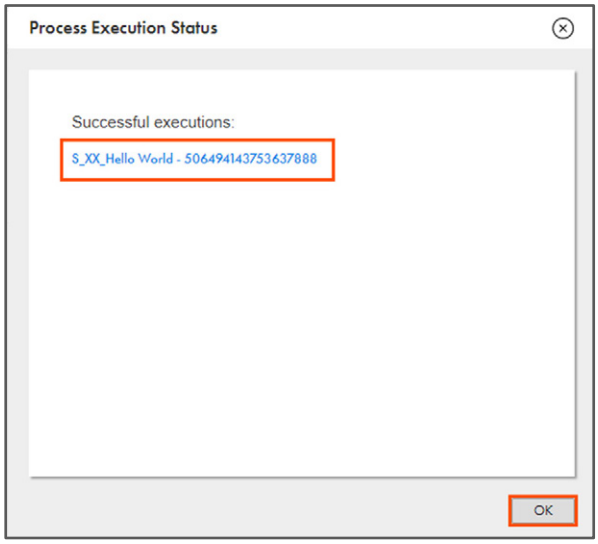
35. Enter an input name and click **Save** to save the input.



36. After the input is added and validated successfully, click **Run**.

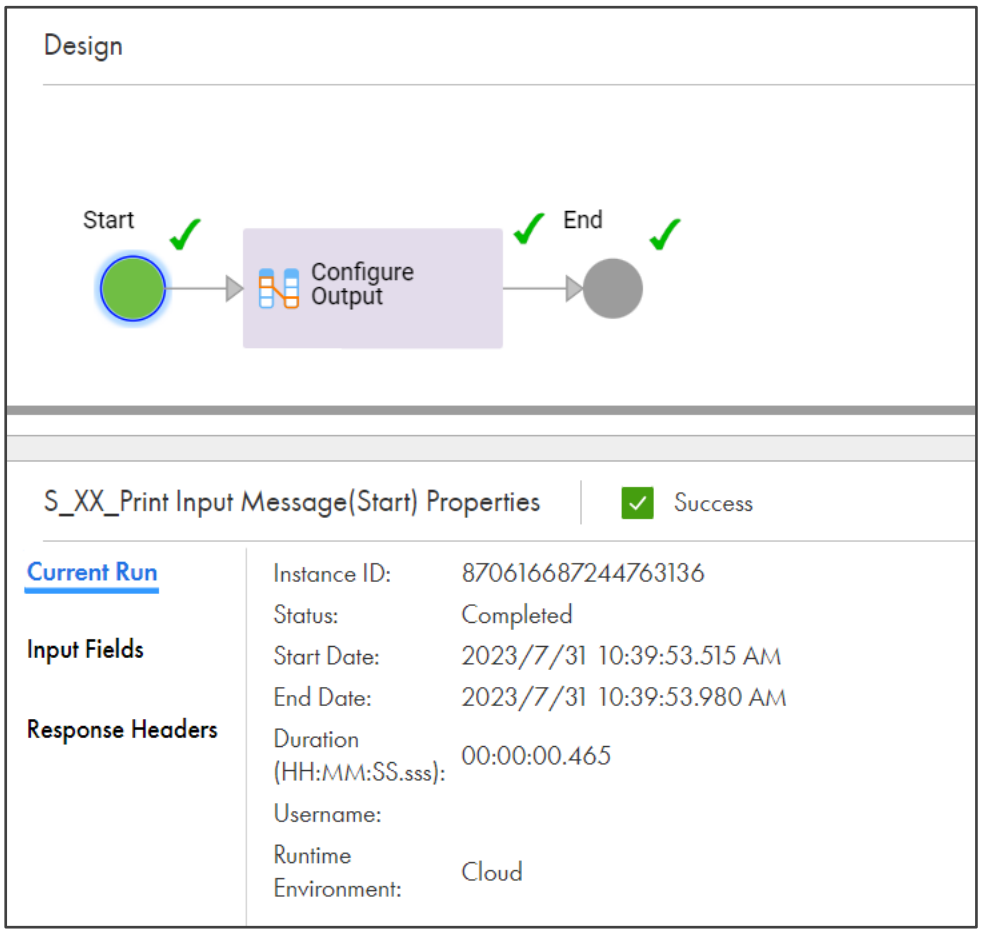


37. To check the Process Execution Status, click the **Process Execution link**, and then click **OK**.



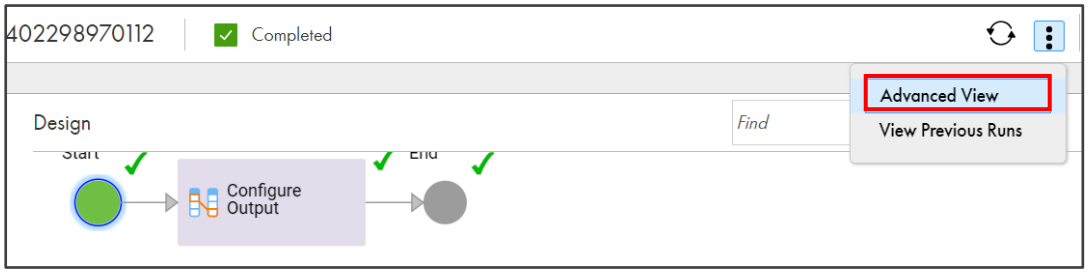
Note: If you do not wish to check the status, click **OK**.

The execution details are as shown below:

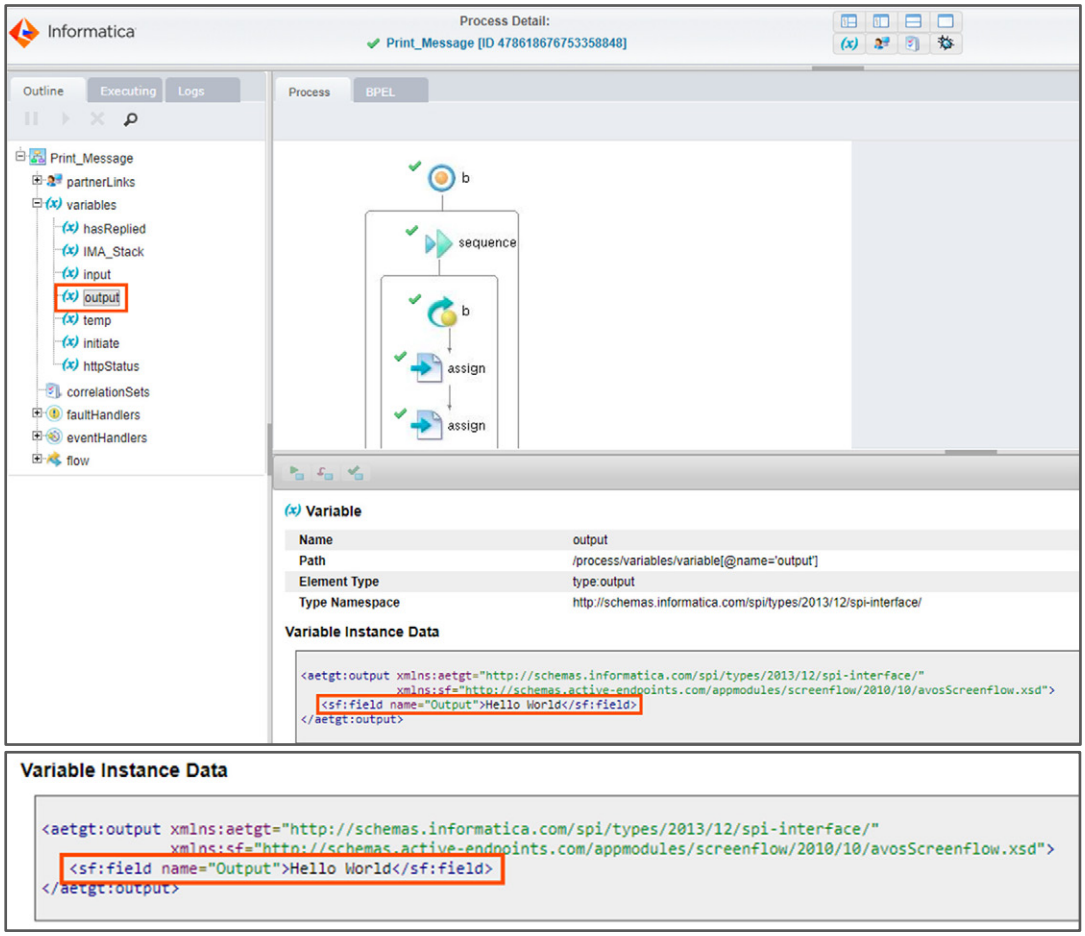


Note: If you are not able to see this screen, click **Refresh**.

38. To check the output, from the **Actions** menu, select **Advanced View**.



39. Expand the **variables** section and select **output**.
    In the Variable Instance Data section, you can check the output.

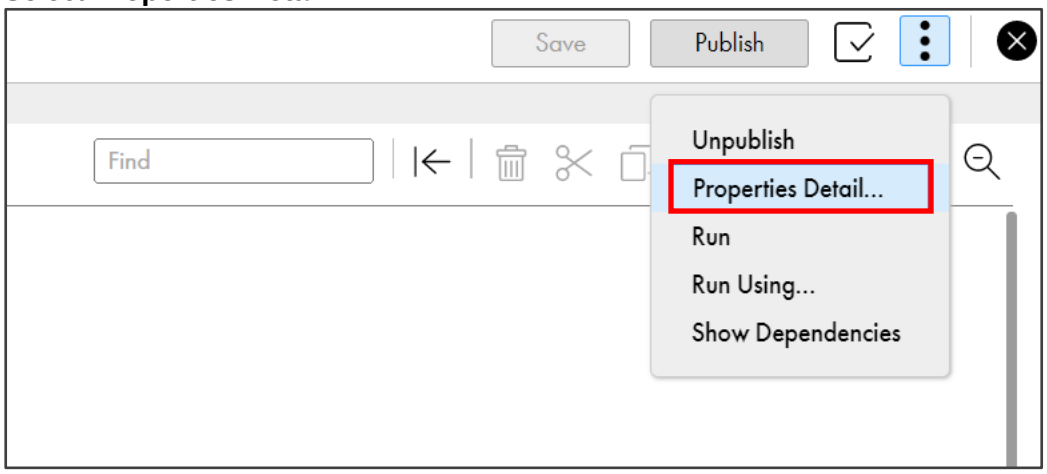


40. Close all the instances of the process and the Advanced View tab.

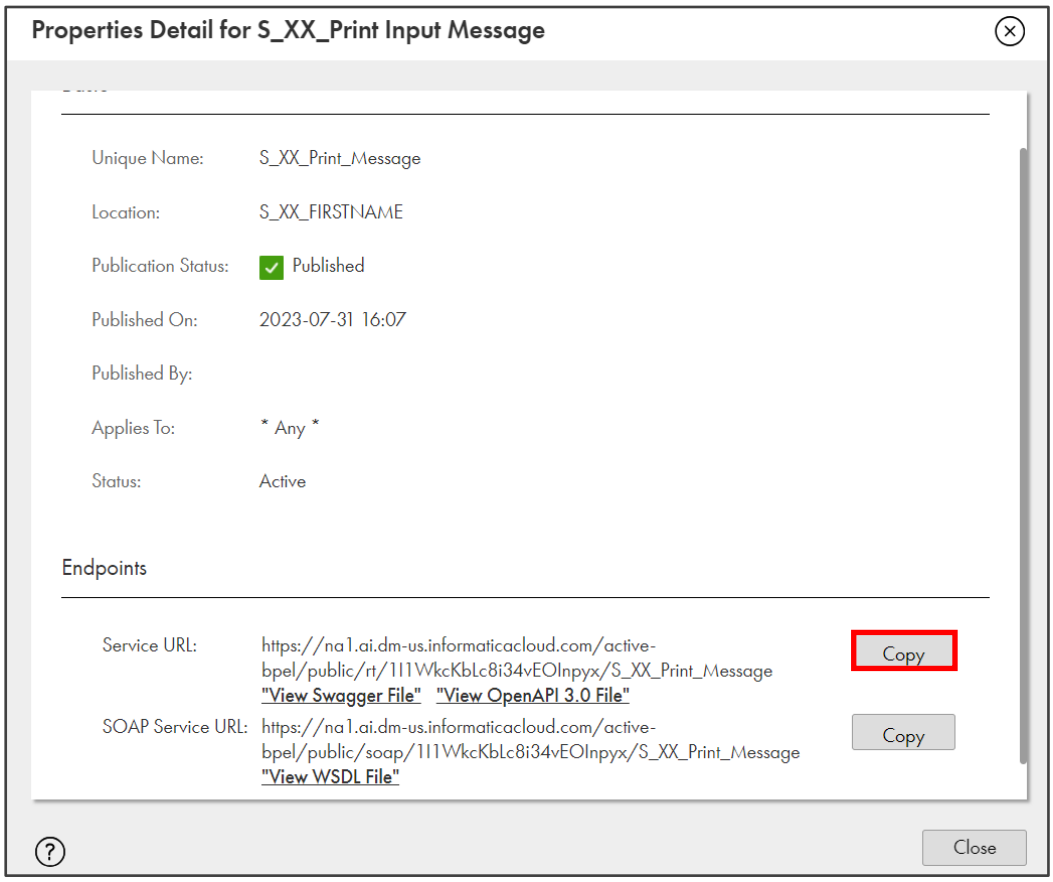
    You have successfully created a basic process and invoked it from the interface.

Invoke the Process Using Service URL:

41. In CAI, navigate back to the **S_XX_Print Input Message** process, and click .
42. Select **Properties Detail**.



43. From the Properties Detail page, to copy the Service URL, click **Copy**.



44. Open the Notepad and enter the copied Service URL.
45. In a web browser, enter the following URL and press Enter:

    <Service URL>?Input=Hello World

    OR

    From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **01_LabGuide_CAI_CreateABasicProcess_3-1**. Copy the command mentioned under **Step A** and paste it into the web browser.

    Note: Replace <Service URL> with the URL that you have copied in Notepad. **DO NOT CLOSE THE NOTEPAD.**

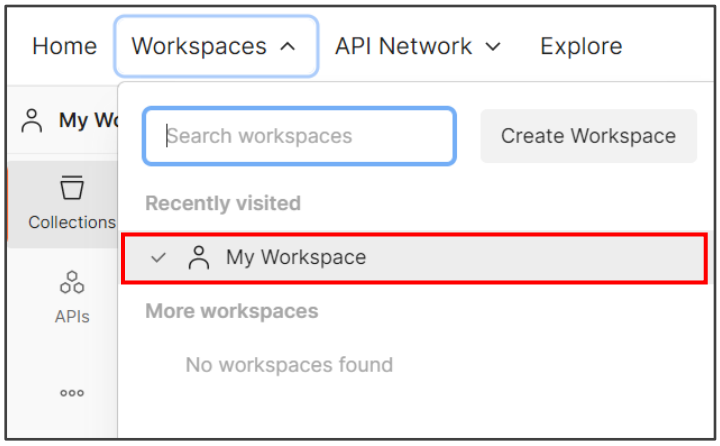
    For Example: https://na1.ai.dm-us.informaticacloud.com/active-bpel/public/rt/hnLOSorFiEqfZgh7aWGiIF/Print_Message?Input=Hello%20World
46. Observe that you get the output as shown below:

```
{"Output":"Hello World"}
```

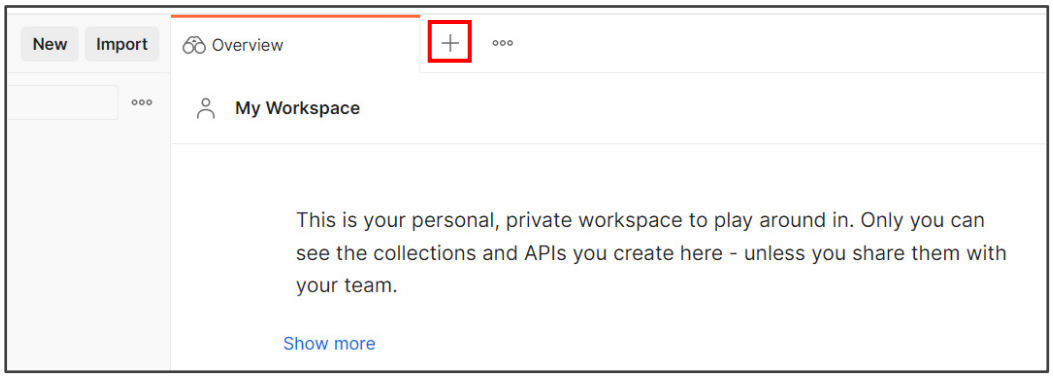
47. Close the browser.

Invoke the Process Using Postman

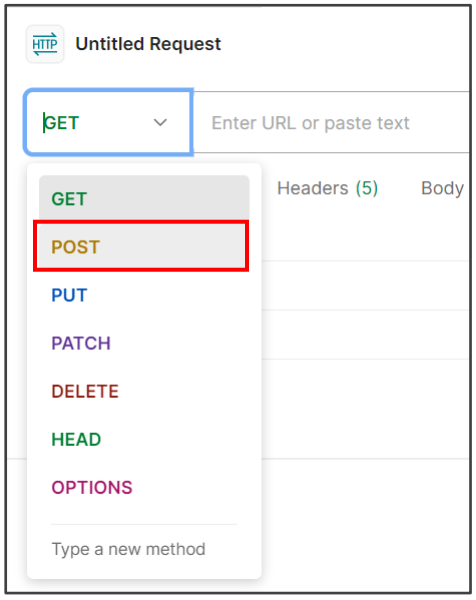
48. Open a web browser and enter the following link to open Postman: https://identity.getpostman.com/
49. Log in to Postman using your credentials.
50. From the Workspace drop-down, select **My Workspace**.



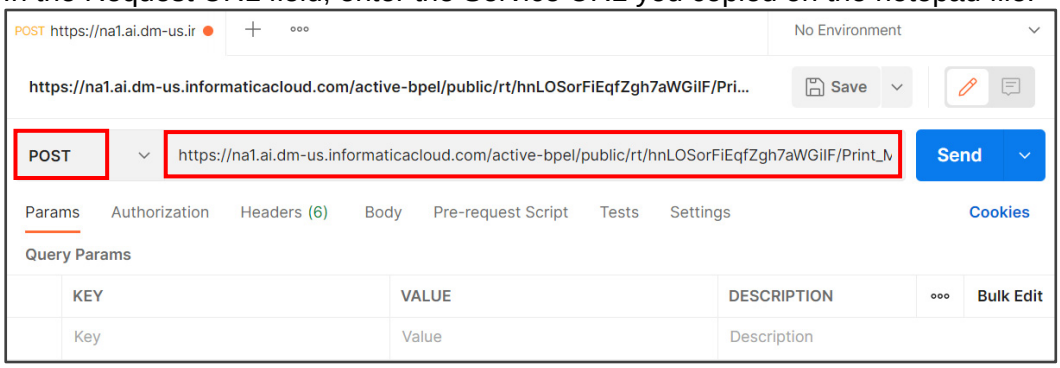
51. Click the **+** icon.



52. In the Workspace section, from the drop-down, select **POST**.



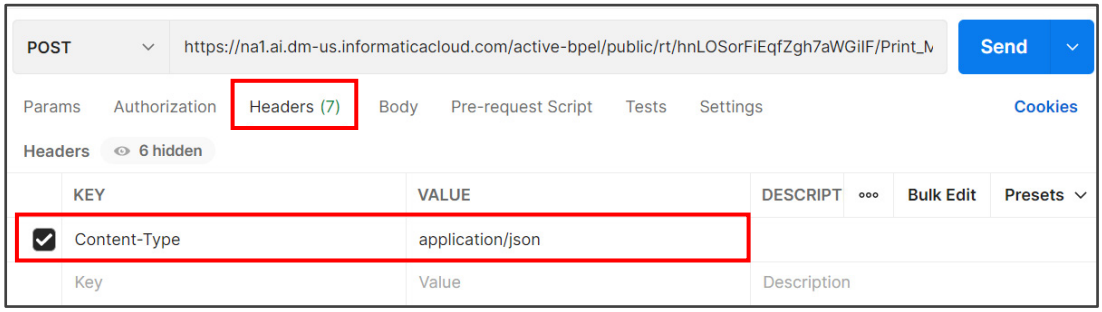
53. In the Request URL field, enter the Service URL you copied on the notepad file.



54. Select the **Authorization** tab.
55. From the TYPE drop-down, select **No Auth**.

56. Select the **Headers** tab.
57. Enter Key and Value as shown in the table below:

Key	Value
Content-Type	application/json

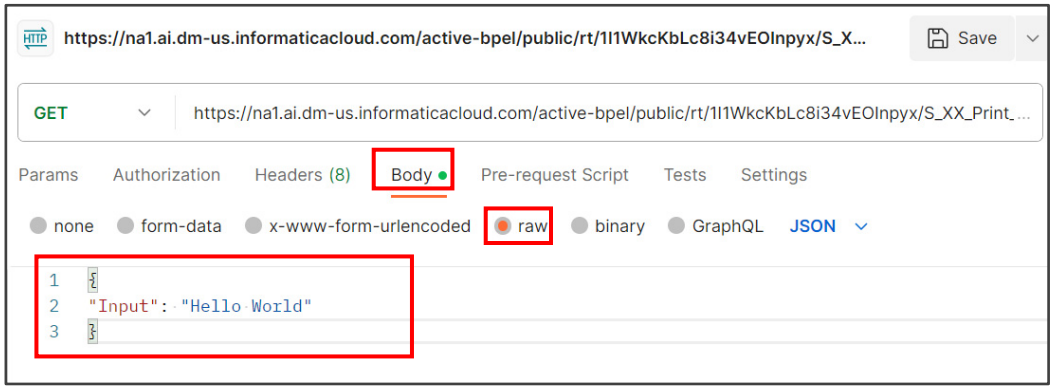


58. Select the **Body** tab and select **raw**.
59. Enter the following text body:

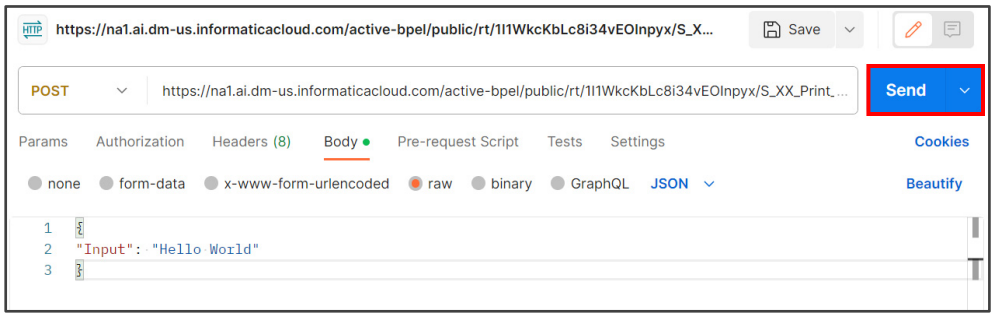
    {"Input": "Hello World"}

    OR

    From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **01_LabGuide_CAI_CreateABasicProcess_3-1**. Copy the command mentioned under **Step B** and paste it into the body field.



60. To view the response, click **Send**.



61. Verify that the status of the response is **200 OK**.
62. Examine the details in the response.

This concludes the lab.

Module 4: Process Objects
Lab 4-1: Create Process Objects

Overview:
Process objects combine sets of structured data into groups called objects. These objects can be used in processes. The named process objects help you identify the inputs and outputs of a Process.

This is **part 1** of the set of three labs. The primary objective of this exercise is to obtain the conversion rates of specific currencies such as United States Dollars (USD), Australian Dollars (AUD), Great Britain Pounds (GBP), and Canadian Dollars (CAD).

In this lab, you will create four Process Objects to collect input values and input types to request a conversion and get a response output. You will use these Process Objects in the ExchangeRate process which will be explained in the subsequent modules.

Note: This lab is continued in **Lab 7-2: Create a JDBC Connection** of Module 7 and **Lab 10-1: Invoke a Synchronous Web Service Call** of Module 10.

Objectives:
- Create a Process Object – BaseRatePO
- Create a Process Object – ConversionRatePO
- Create a Process Object – ConversionRequestPO
- Create a Process Object – ConversionResponsePO

Duration:
15 Minutes

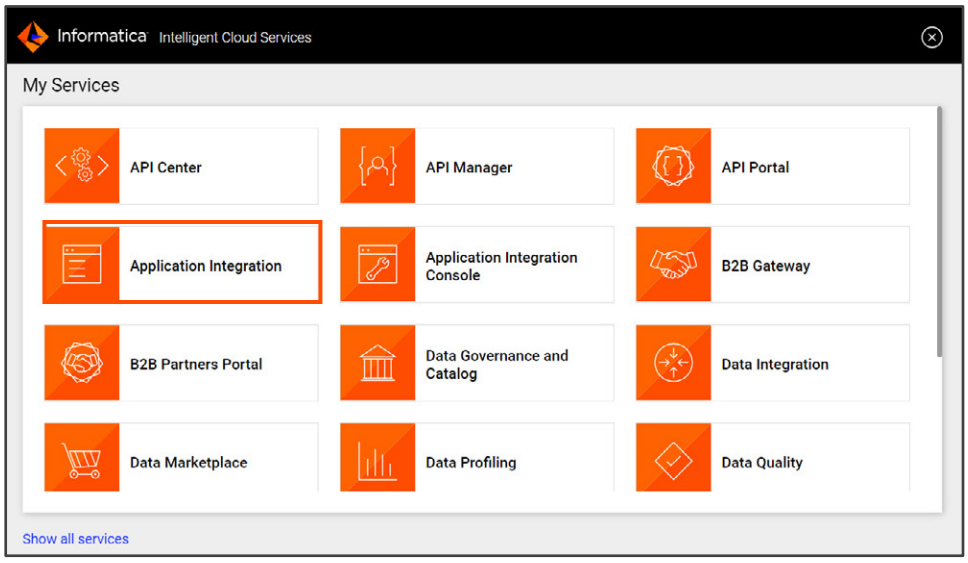
Tasks:

Note: The screen in your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

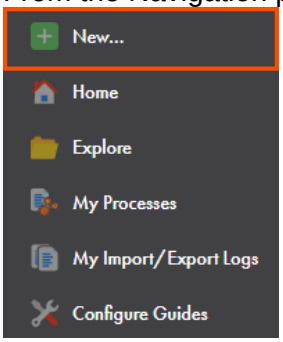
Create a Process Object – BaseRatePO:

This process object includes the type and value of the currency. It will be used as a reference later in the ConversionRequestPO and ConversionResponsePO process objects.

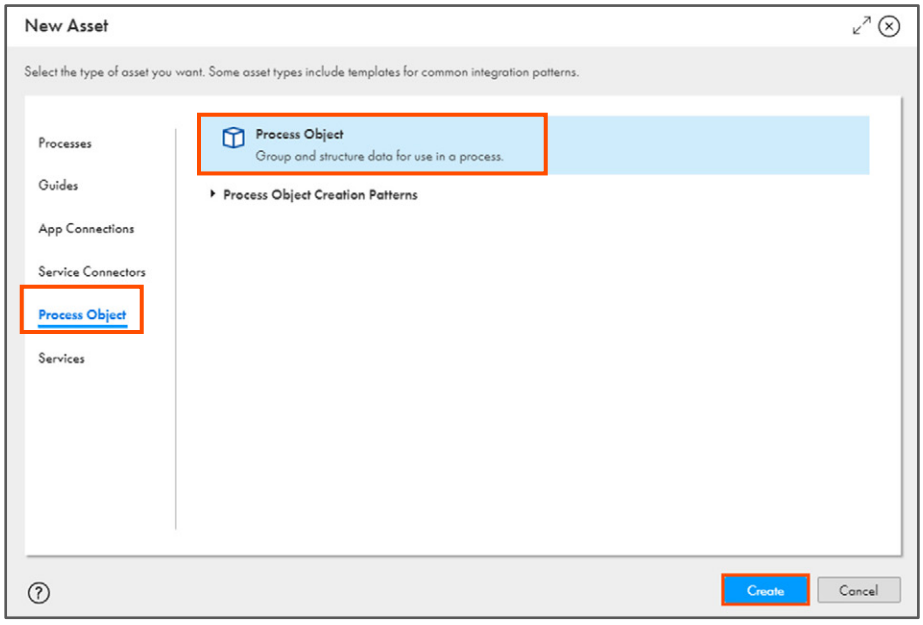
1. Log into IICS and access the **Application Integration** service.



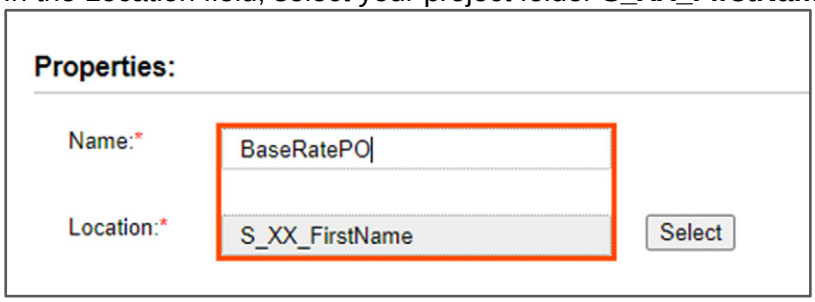
2. From the Navigation pane, click **New**.



3. In the New Asset window, from the Process Object tab, select **Process Object**, and click **Create**.

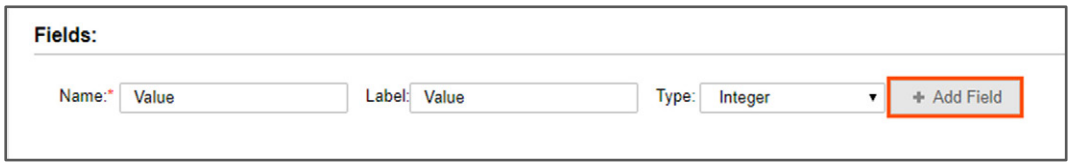


4. In the Process Object Properties page, in the Name field, enter **BaseRatePO**.
5. In the Location field, select your project folder **S_XX_FirstName**.

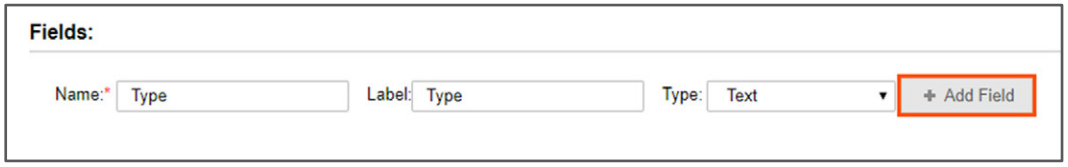


For BaseRatePO Process Object two fields are required – **Value** and **Type**.

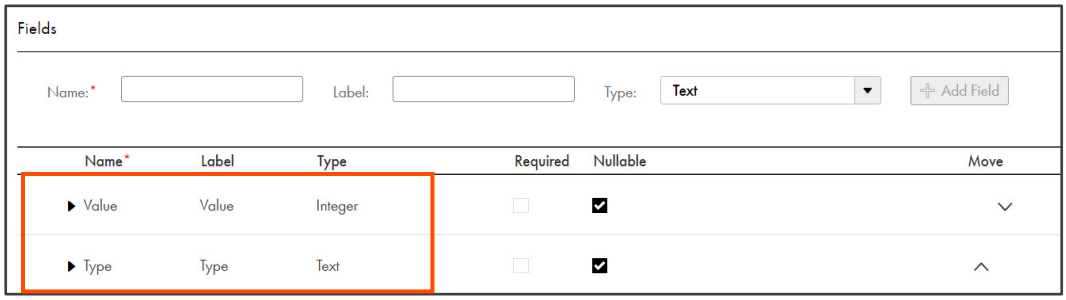
6. For the first field, enter Name and Label as **Value**.
7. From the Type drop-down, select **Integer**.
8. Click **Add Field**.



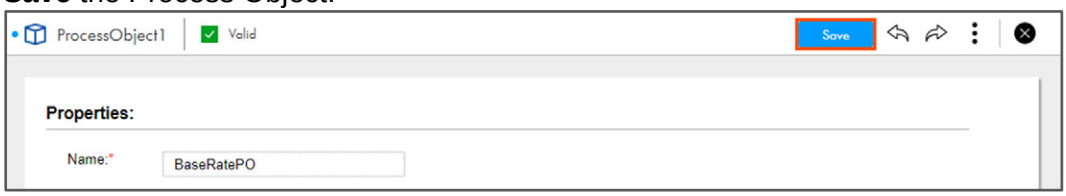
9. For the second field, enter Name and Label as **Type**.
10. From the Type drop-down, select **Text**.
11. Click **Add Field**.



Fields should look like as shown in the image below:



12. **Save** the Process Object.

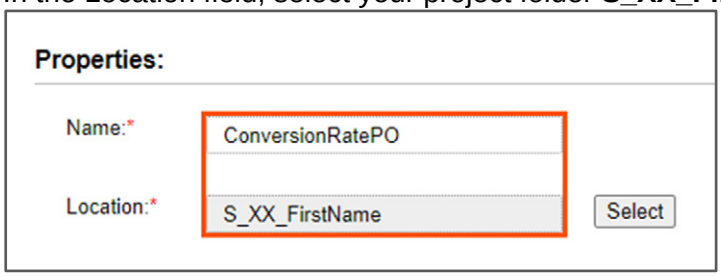


You have successfully designed a Process Object. Similarly, you will design the other three Process Objects.

Create a Process Object – ConversionRatePO:

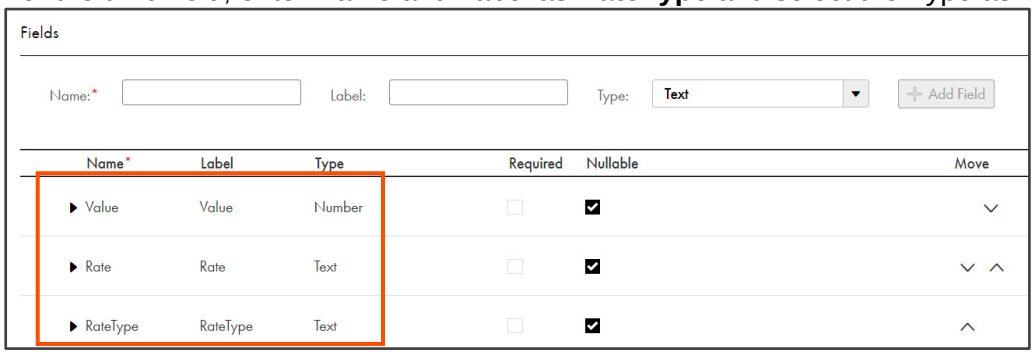
This process object is used to display the conversion rates as compared to the base rates. It includes the fields – Value, Rate, and RateType. This process object will also be used as a reference later in the ConversionRequestPO and ConversionResponsePO process objects.

13. Create another process object and in the Name field, enter **ConversionRatePO**.
14. In the Location field, select your project folder **S_XX_FirstName**.



For this Process Object, three fields are required – **Value**, **Rate**, and **RateType**.

15. Add the first field with the Name and Label as **Value** and the Type as **Number**.
16. For the second field, enter Name and Label as **Rate** and select the Type as **Text**.
17. For the third field, enter Name and Label as **RateType** and select the Type as **Text**.



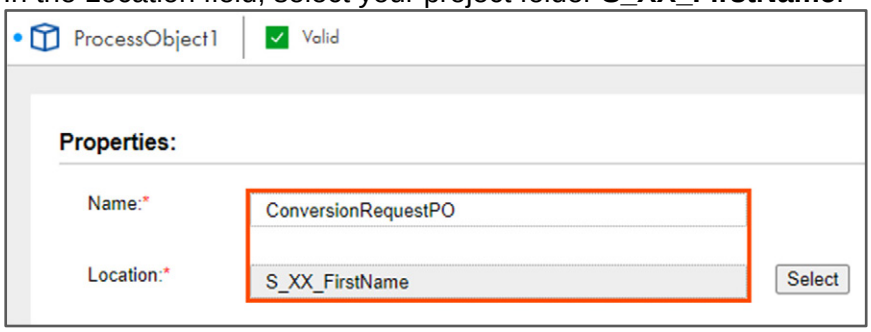
18. **Save** the Process Object.

Create a Process Object – ConversionRequestPO:

This process object uses BaseRatePO and ConversionRatePO to obtain the base rate and conversion rates of the requested currency. This will be used as an input into the process.

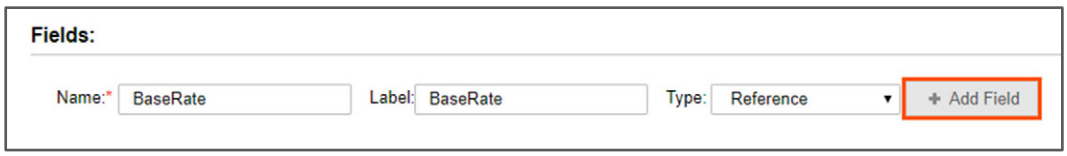
19. Create another process object and in the Name field, enter **ConversionRequestPO**.

20. In the Location field, select your project folder **S_XX_FirstName**.



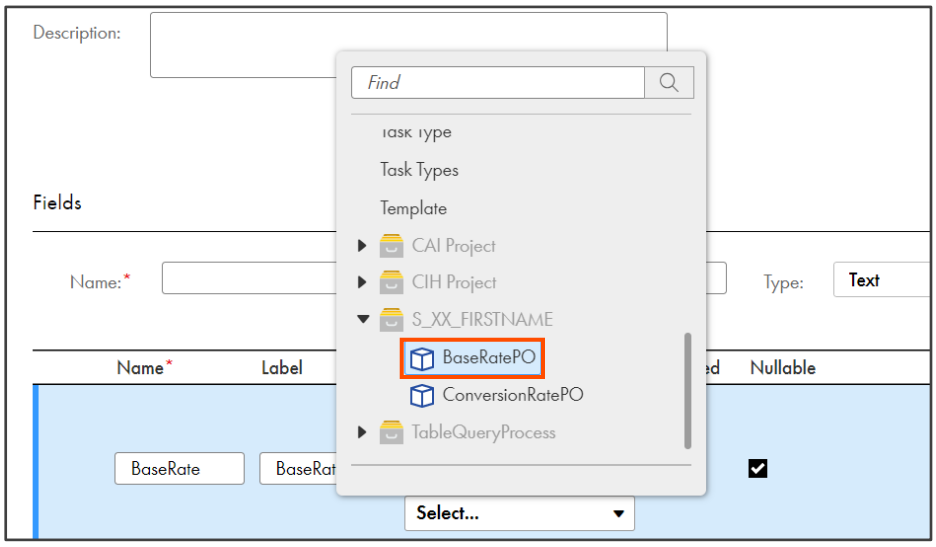
You will now add two object fields that refer to the two previously created Process Objects – BaseRatePO and ConversionRatePO.

21. Add the first field with the Name and Label as **BaseRate** and the Type as **Reference**.
22. Click **Add Field**.

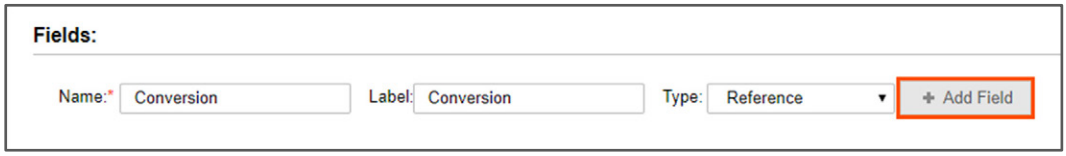


A new field is created and in the Type field, a new option **Reference** is available.

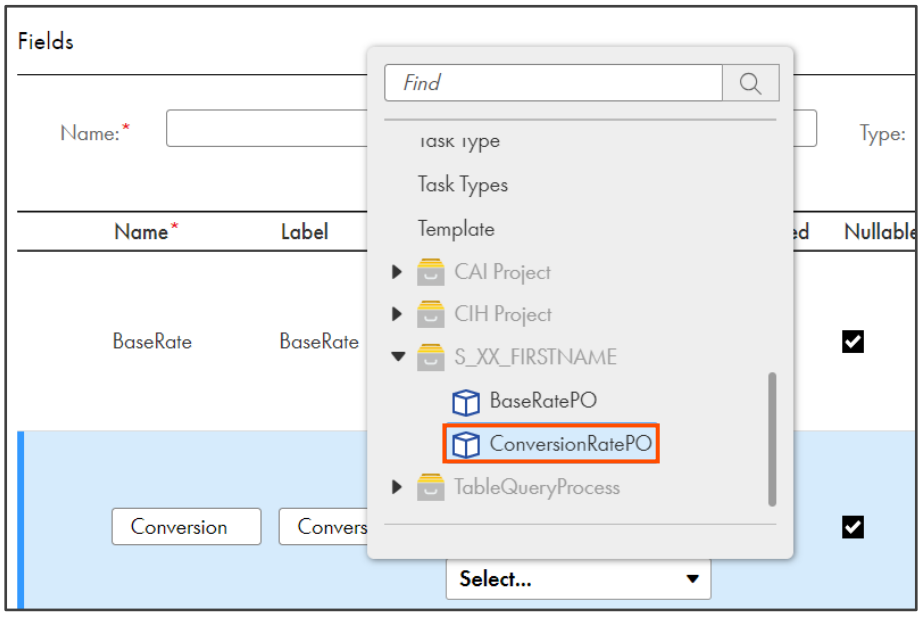
23. Under the Type field, from the drop-down, expand your project folder **S_XX_FirstName**, and select **BaseRatePO**.



24. Add the second field with the Name and Label as **Conversion** and the Type as **Reference**.



25. Under the Type field, from the drop-down, expand your project folder **S_XX_FirstName**, and select **ConversionRatePO**.



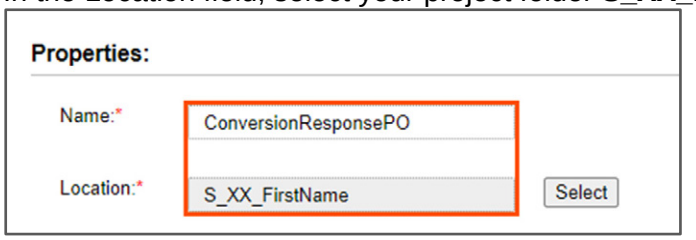
Note: ConversionRatePO might take time to appear in the drop-down. If you can't see ConversionRatePO in the drop-down, log out from IICS and log in again. If you still can't see it in the drop-down, clear your browse cache, and try it again.

26. **Save** the Process Object.

Create a Process Object – ConversionResponsePO:

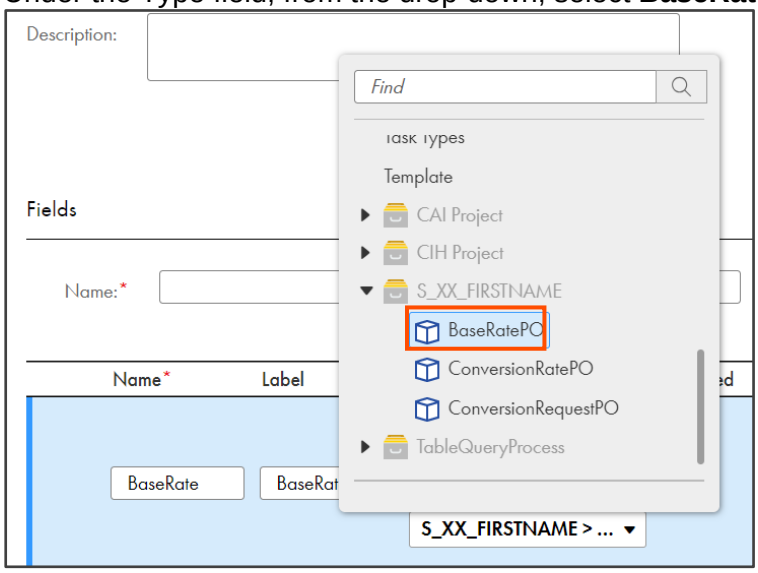
This process object is used to display the conversion rates of the previous request.

27. Create another process object and in the Name field, enter **ConversionResponsePO**.
28. In the Location field, select your project folder **S_XX_FirstName**.

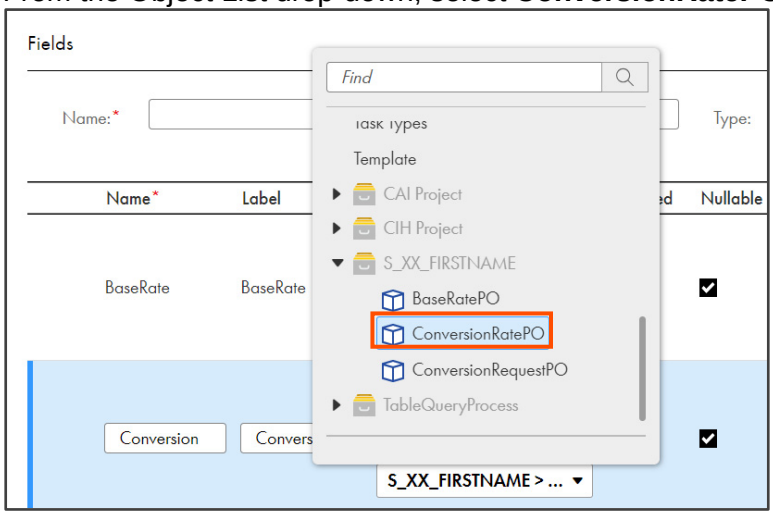


29. Add the first field with the Name and Label as **BaseRate** and the Type as **Reference**.

30. Under the Type field, from the drop-down, select **BaseRatePO**.



31. Add the second field with the Name and Label as **Conversion** and the Type as **Object List**.
32. From the Object List drop-down, select **ConversionRatePO**.



33. **Save** the Process Object.

This concludes the lab.

Module 5: Service Connectors
Lab 5-1: Create a Service Connector

Overview:
A service connector allows you to connect to third-party services such as the Thomas Bayer service.

This lab is **part 1** of a set of three labs. The overall goal of the exercise is to obtain customer and order details such as customer name and order total. The process includes – creating a service connector, creating a connection using the service connector, and then creating a process to obtain the desired output.

In this lab, you will create a Service Connector from a publicly available REST Web Service.

Note: This lab is continued in **Lab 7-1: Create an App Connection using Service Connector** and **Lab 7-3: Create a Customer and Order Details Process** of Module 7.

Objectives:
- Create a new Service Connector
- Define Service Connector input parameters
- Define a Service Connector Action and Binding
- Test the Service Connector

Duration:
25 Minutes

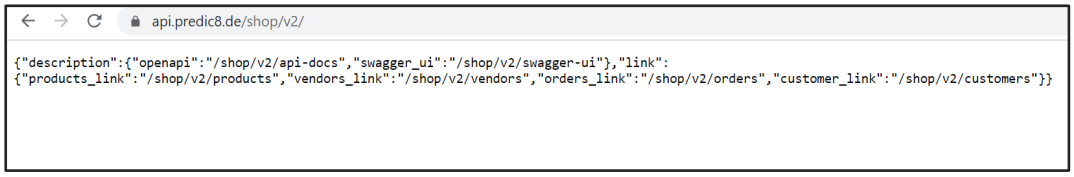
Tasks

Examine the web service:

1. To examine the list of top-level resources available in the web service, enter the following URL in the browser:
   https://api.predic8.de/shop/v2/

   OR

   From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **03_LabGuide_CAI_Create a Service Connector_5-1**. Copy the command mentioned under **Step A** and paste it into the browser.



Examine the Customer resource:

2. To examine the Customer resource, enter the following URL in a new tab on the browser:

   https://api.predic8.de/shop/v2/customers/

   OR

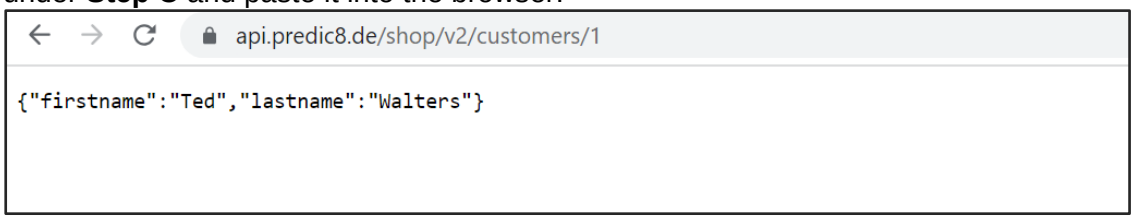
   From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **03_LabGuide_CAI_Create a Service Connector_5-1.** Copy the command mentioned under **Step B** and paste it into the browser.

3. To examine the details of a customer, enter the following URL in a new tab on the browser:

   https://api.predic8.de/shop/v2/customers/1

   OR

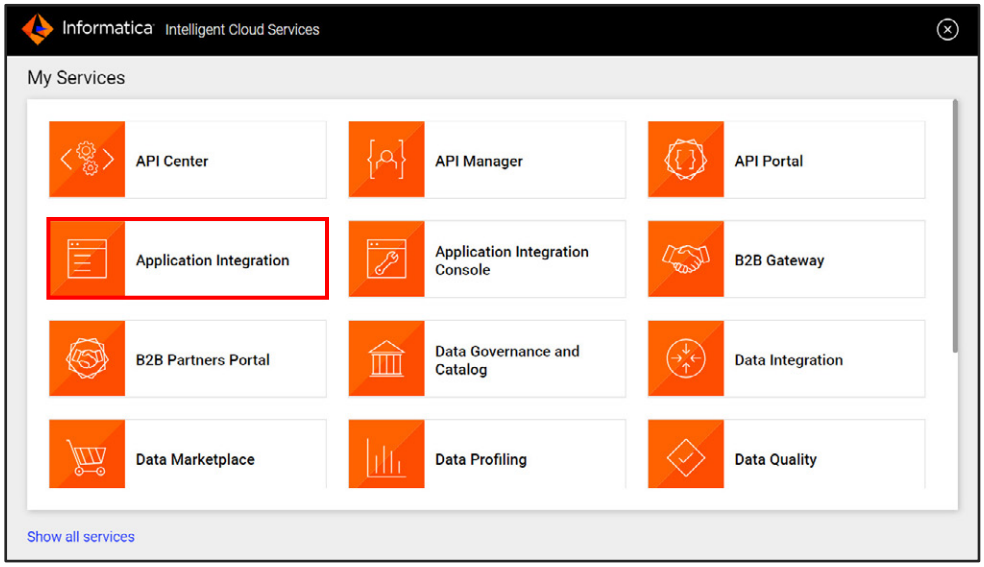
   From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **03_LabGuide_CAI_Create a Service Connector_5-1.** Copy the command mentioned under **Step C** and paste it into the browser.



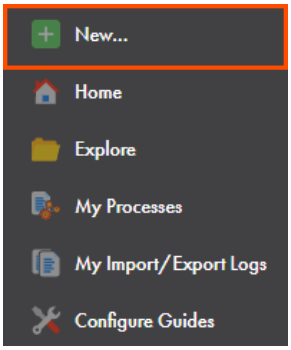
   Note: The updated URL passes the Order id. A response returns the information about the customer with **Order id=1**. The API responds with two fields: firstname and lastname.

Create a new Service Connector:

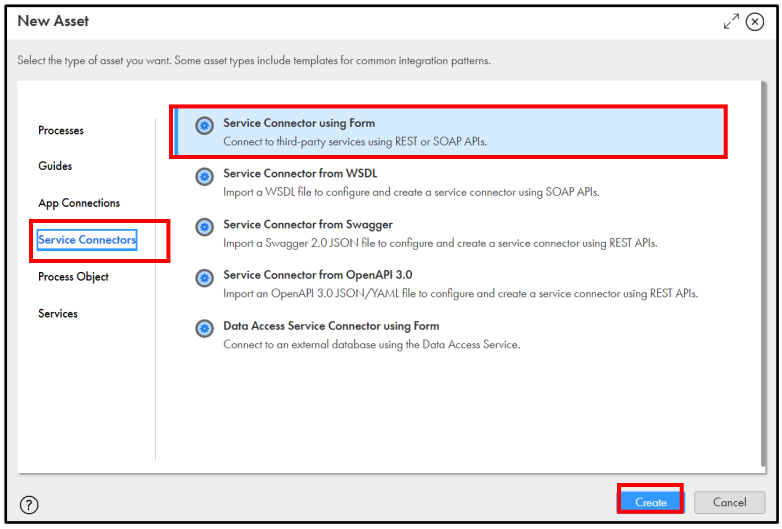
4. Log into IICS and access the **Application Integration** service.



5. To create a new Service Connector, from the navigation pane, click **New**.

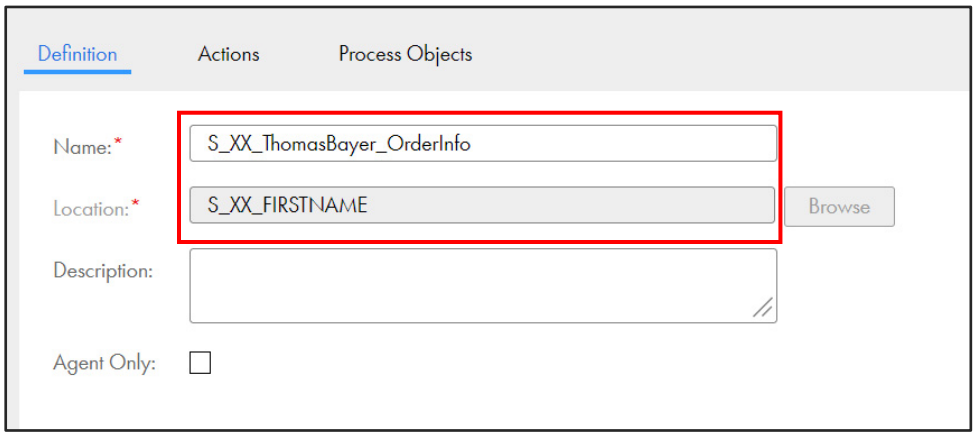


6. From the Service Connectors tab, select **Service Connector using Form**.
7. Click **Create**.

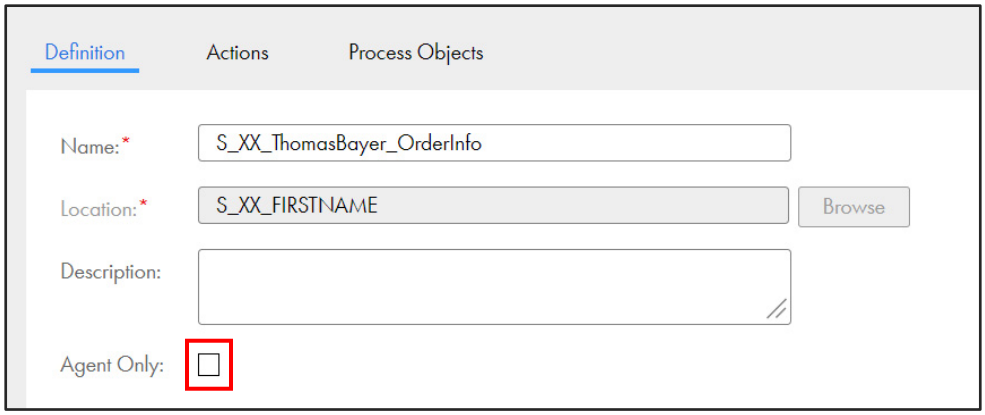


Note: A new Service Connector page appears.

8. In the service connector Name field, enter **S_XX_ThomasBayer_OrderInfo**.
   Note: Here, XX refers to your initials.
9. In the Location field, select your project folder **S_XX_FirstName**.



   Note: Here, XX is the unique student number.
10. Verify that the **Agent Only** checkbox is cleared.



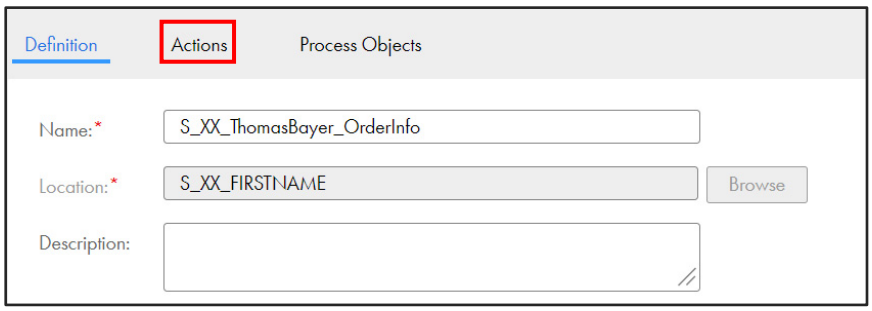
   Note: If you select Agent Only, then you can run it only on the secure agent machine and not on the cloud server.

Create Actions:

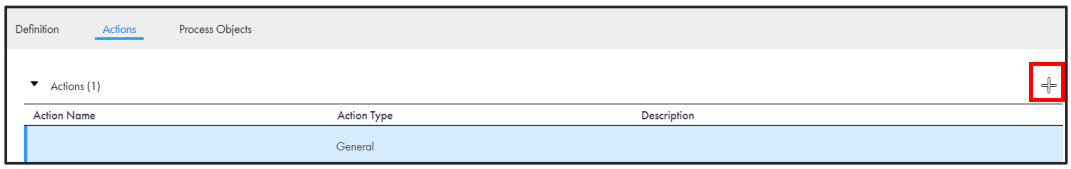
In this section, you will create three actions – OrderDetails, CustomerDetails, and ItemDetails. These actions will be used in the later labs to extract the corresponding information from the web service.

Create the OrderDetails Action:

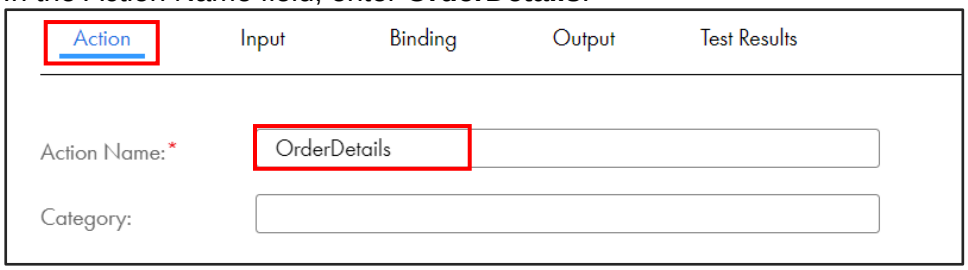
11. Click the **Actions** tab.



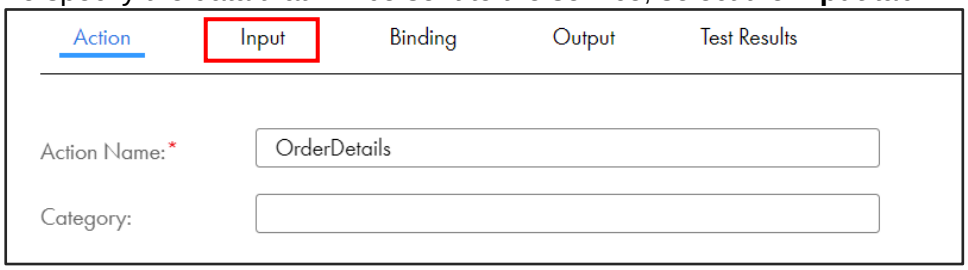
12. To add an Action, click .



13. In the Action Name field, enter **OrderDetails**.

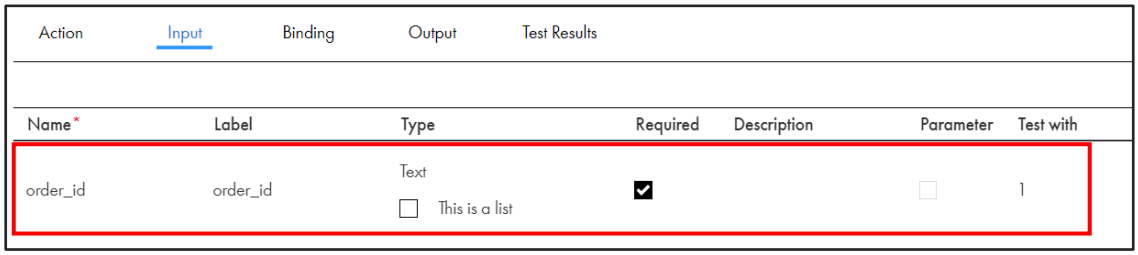


14. To specify the data that will be sent to the service, select the **Input** tab.



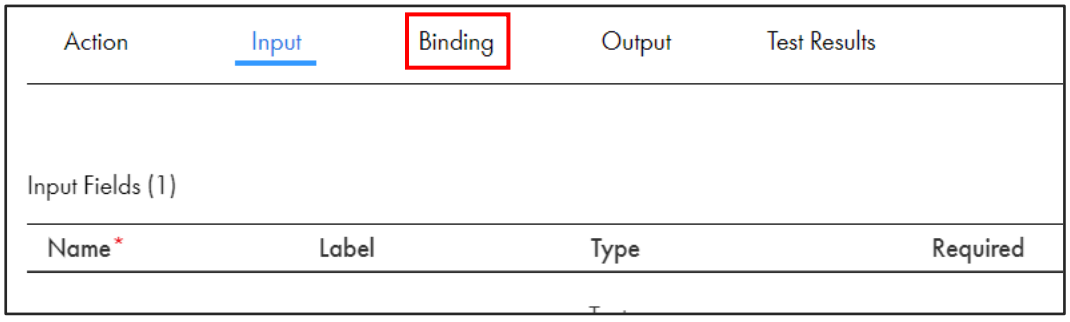
15. Click to add an input field.
16. Create the input field as shown in the table below:

Name	Label	Type	Required	Parameter	Test With
order_id	order_id	Text	Yes	No	1



Note: '1' is just a sample number that will be used to test the service connector.

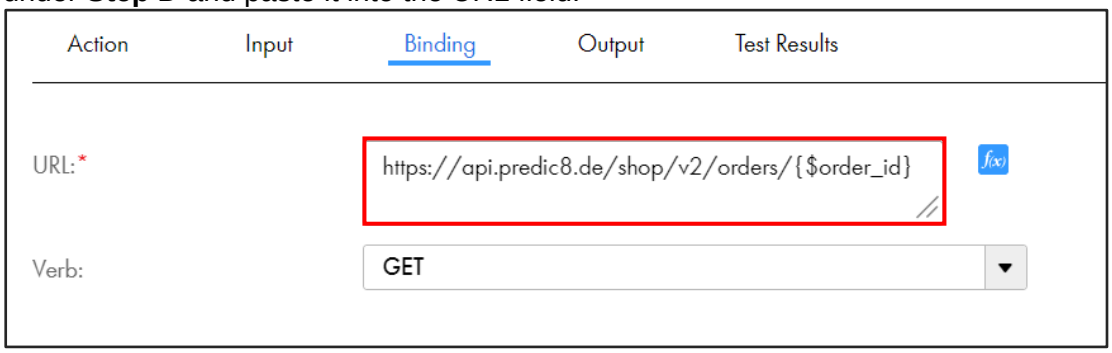
17. To specify the mode in which the data will be sent, select the **Binding** tab.



18. In the URL field, enter the following URL:
    https://api.predic8.de/shop/v2/orders/{$order_id}

    OR

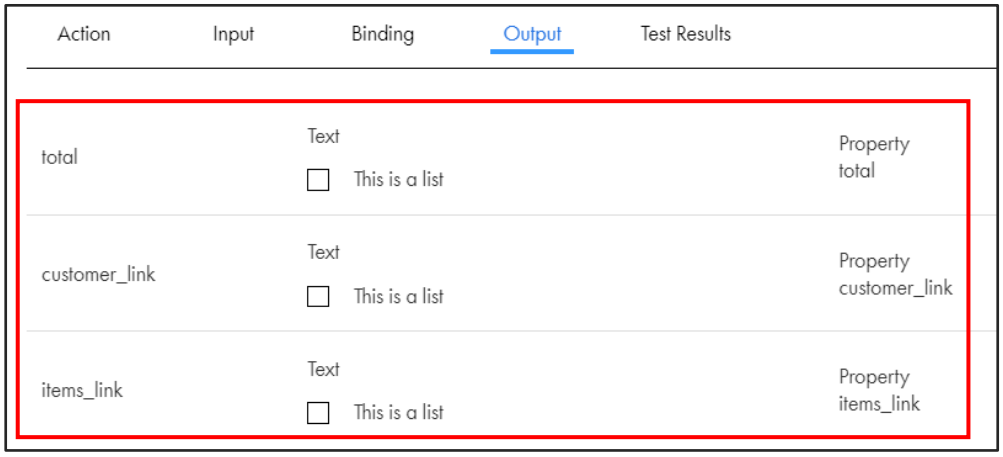
    From the **CAI Lab Prep Files_Learners**, open the notepad file named **03_LabGuide_CAI_Create a Service Connector_5-1.** Copy the command mentioned under **Step D** and paste it into the URL field.



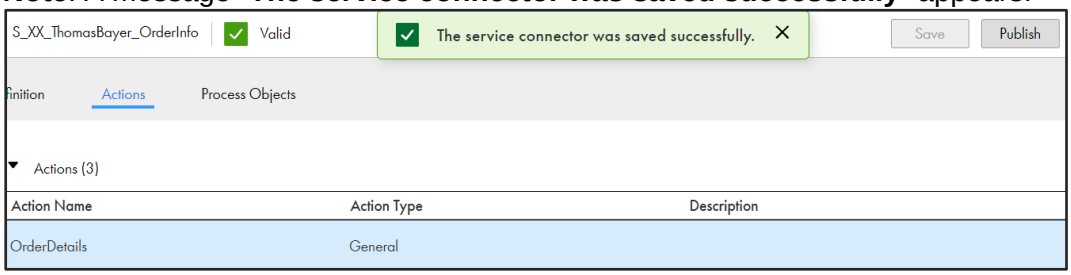
19. To specify the output fields, select the **Output** tab.

20. To create the output fields, click .
21. Create the Output Fields as shown in the table below:

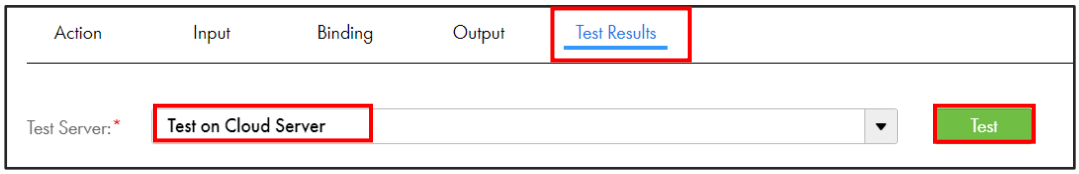
Name	Type	Get From
total	Text	Property-> total
customer_link	Text	Property-> customer_link
Items_link	Text	Property-> items_link



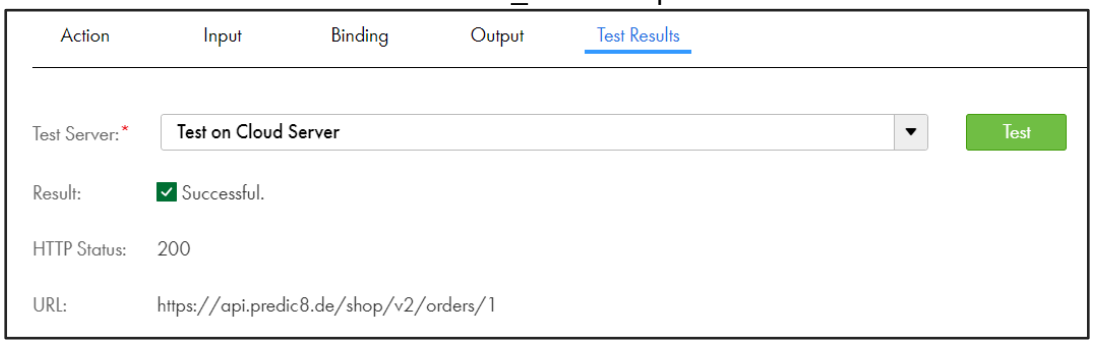
22. To save the connector, click **Save**.
    Note: A message "**The service connector was saved successfully**" appears.



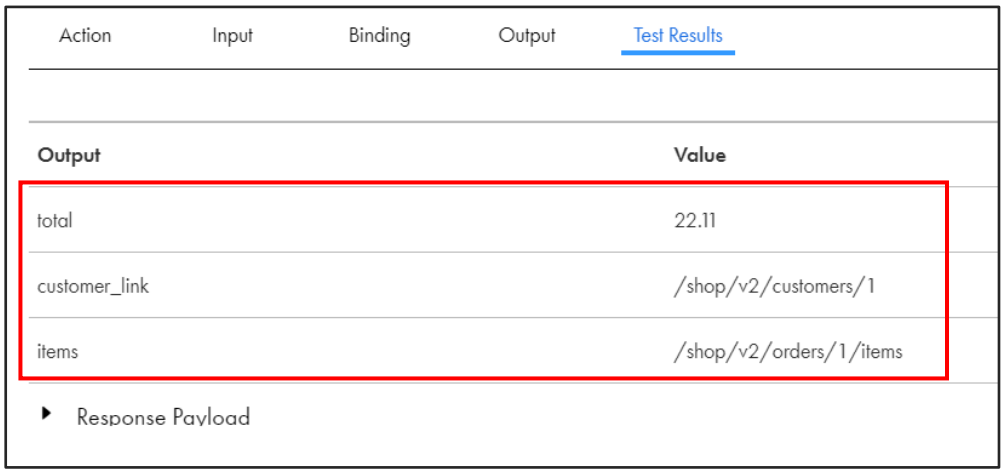
23. To test the service connector, click the **Test Results** tab.
24. From the Test Server drop-down, retain **Test on Cloud Server**.
25. Click **Test**.



26. In the Test Results tab, verify that the result of the test is **Successful**.
    Note: The URL mentions that the order_id field requested Information for id=1.

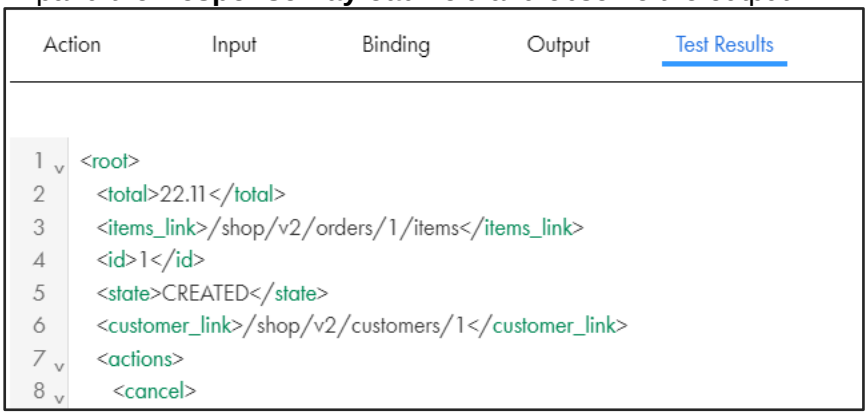


27. From the Output Fields section, observe the output.



Note: The output value in the image is subject to change based on the live data available in the API.

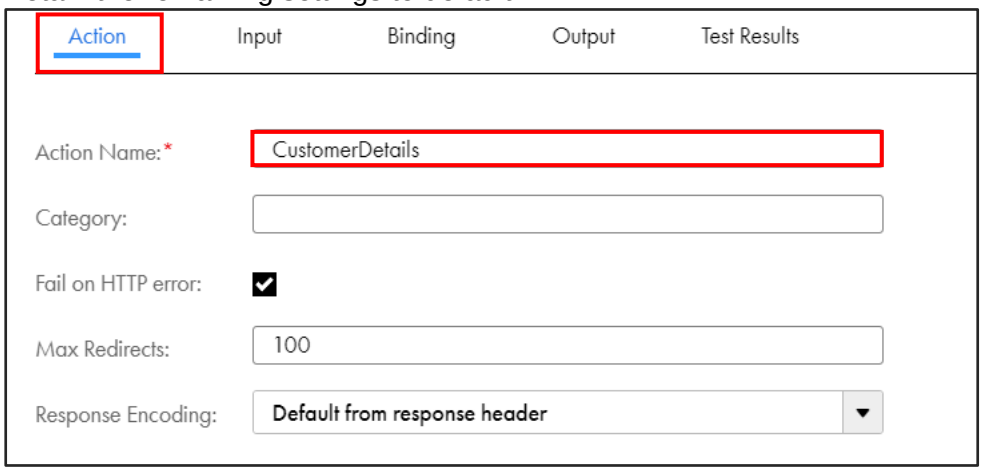
28. Expand the **Response Payload** field and observe the output.



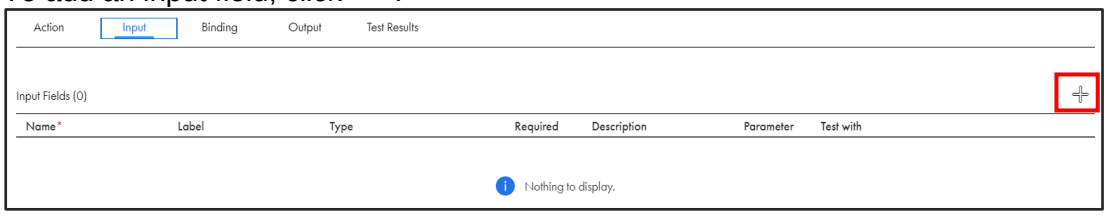
Note: Response payload displays the response received from the service. Remember, you have entered the Test With value as “1”. The service has responded with the total, items_link, and customer_link of order with id “1”.

Create the CustomerDetails Action:

29. Select the **Actions** tab.
30. To add an action, click .
31. In the Action tab of the service connector, enter Action Name as **CustomerDetails**.
32. Retain the remaining settings to default.

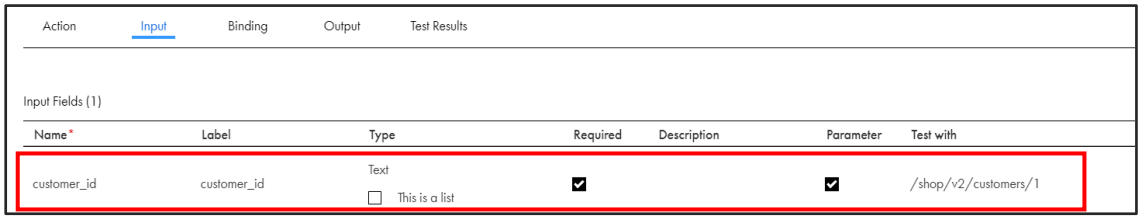


33. To specify the data that will be sent to the service, select the **Input** tab.
34. To add an input field, click .



35. Create the input Fields as shown in the table below:

Name	Label	Type	Required	Parameter	Test With
customer_id	customer_id	Text	Yes	Yes	/shop/v2/customers/1

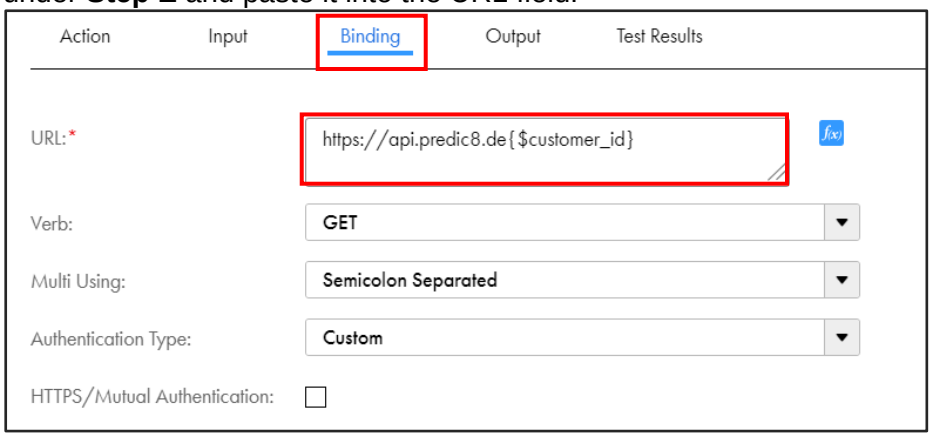


36. To specify the mode in which the data will be sent, select the **Binding** tab.
37. In the URL field, enter the following URL:

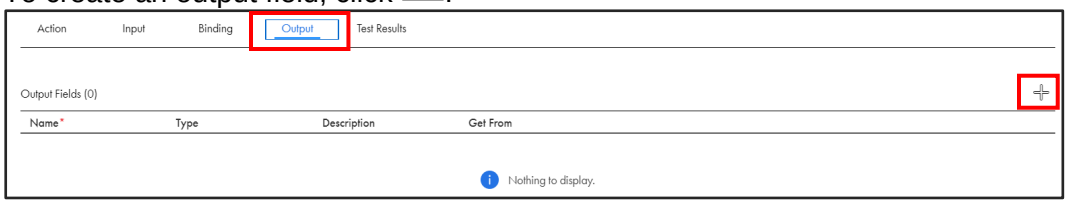
    https://api.predic8.de{$customer_id}

    OR

    From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **03_LabGuide_CAI_Create a Service Connector_5-1.** Copy the command mentioned under **Step E** and paste it into the URL field.

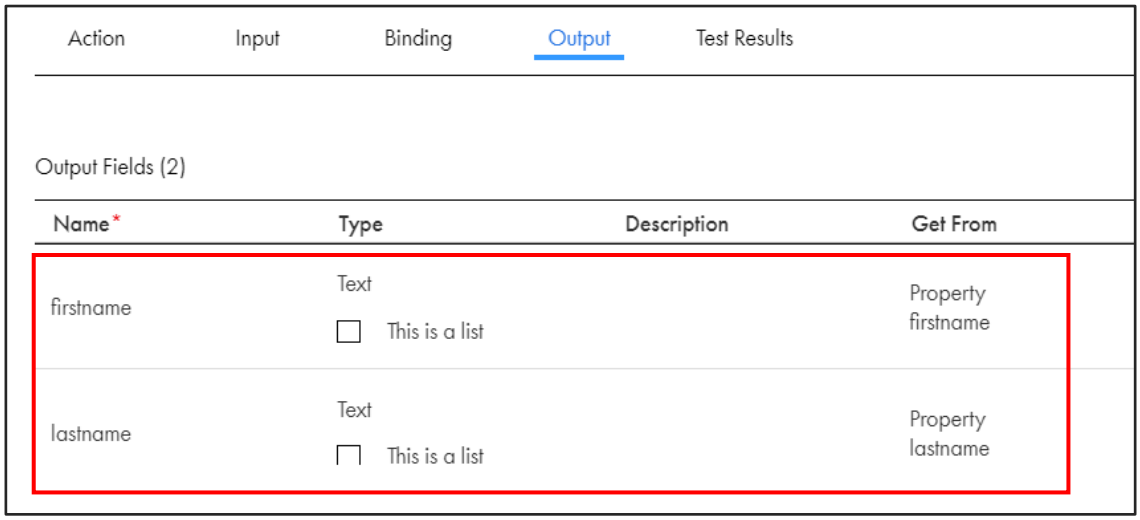


38. To define the output format of the response, select the **Output** tab.
39. To create an output field, click .

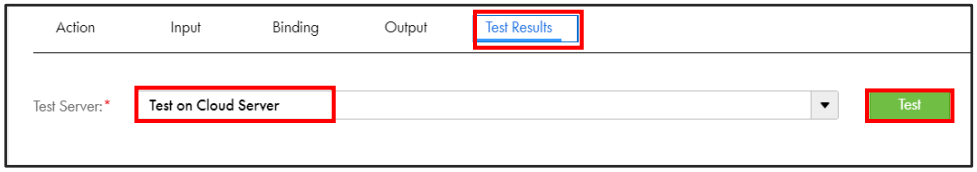


40. Create the Output Fields, as shown in the table below:

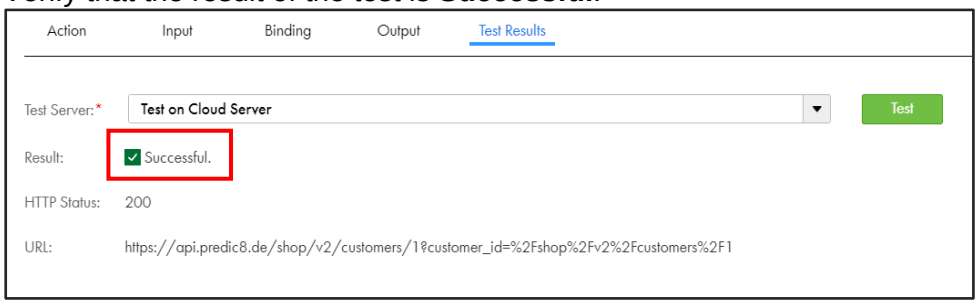
Name	Type	Get From
firstname	Text	Property-> firstname
lastname	Text	Property-> lastname



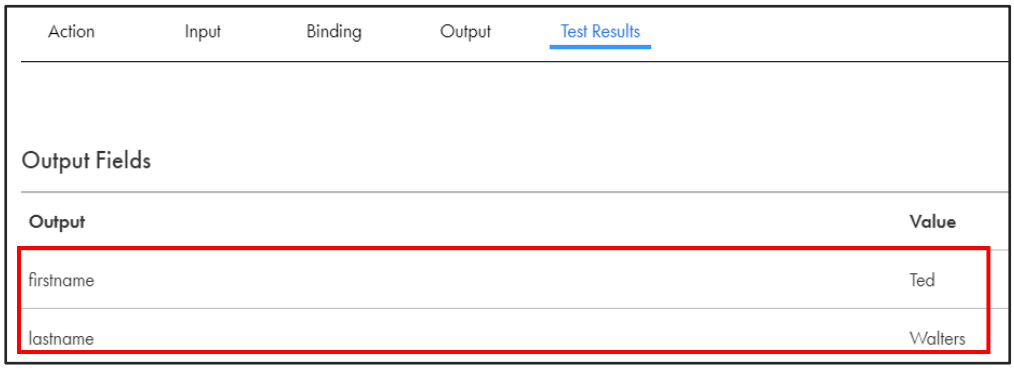
41. To test the service connector, select the **Test Results** tab.
42. From the Test Server field, retain **Test on Cloud Server**.
43. Click **Test**.



44. Verify that the result of the test is **Successful**.

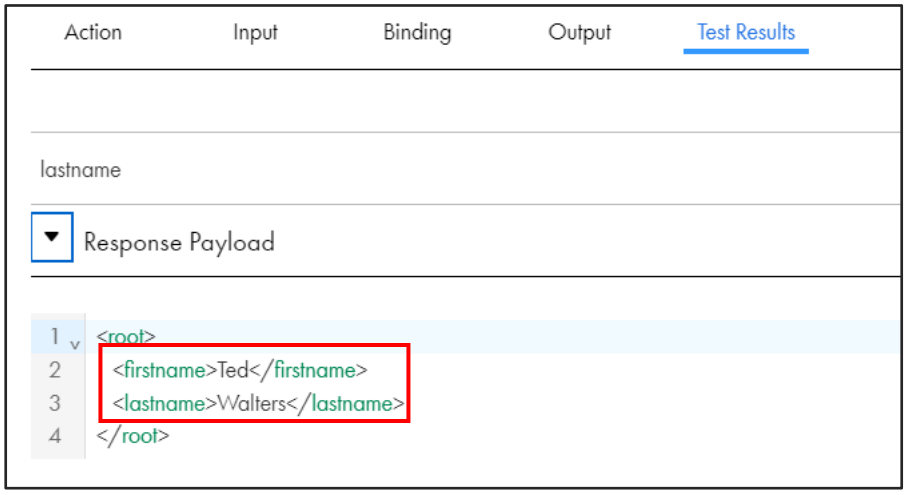


45. Observe the **Output Fields** section. It lists the output fields with the output values defined in the Service Connector action.



Note: The output value in the image is subject to change based on the live data available in the API.

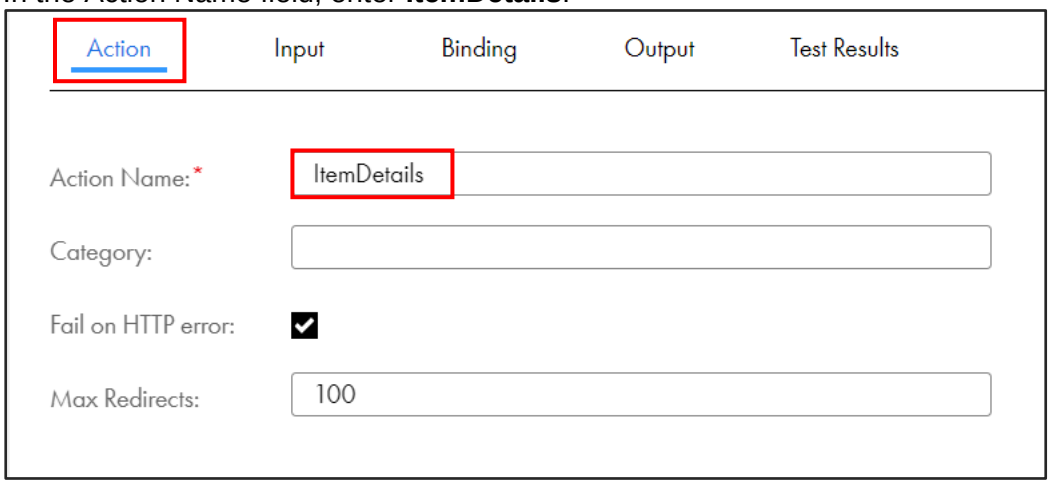
46. Expand the **Response Payload** field and observe the output.



Create the ItemDetails Action:

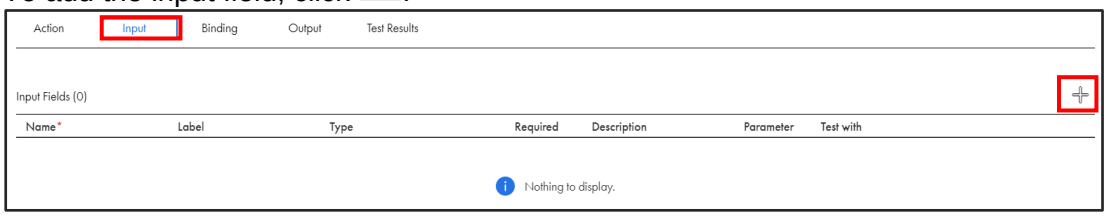
47. Select the **Actions** tab.
48. To add the action, click .
49. In the Action tab of the service connector, specify the Action Name as **ItemDetails**.

50. In the Action Name field, enter **ItemDetails**.



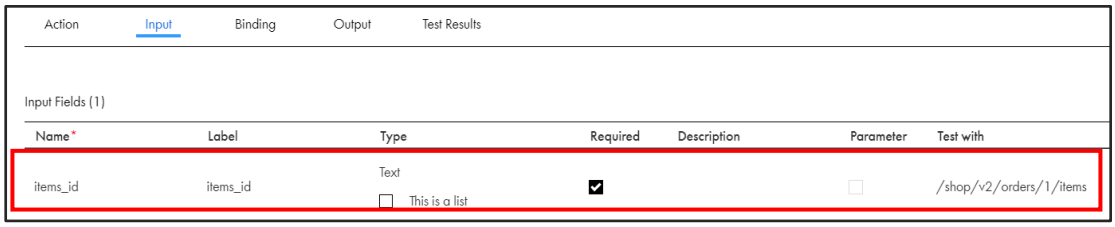
51. To specify the data that will be sent to the service, select the **Input** tab.

52. To add the input field, click .



53. Create the Input Fields as shown in the table below:

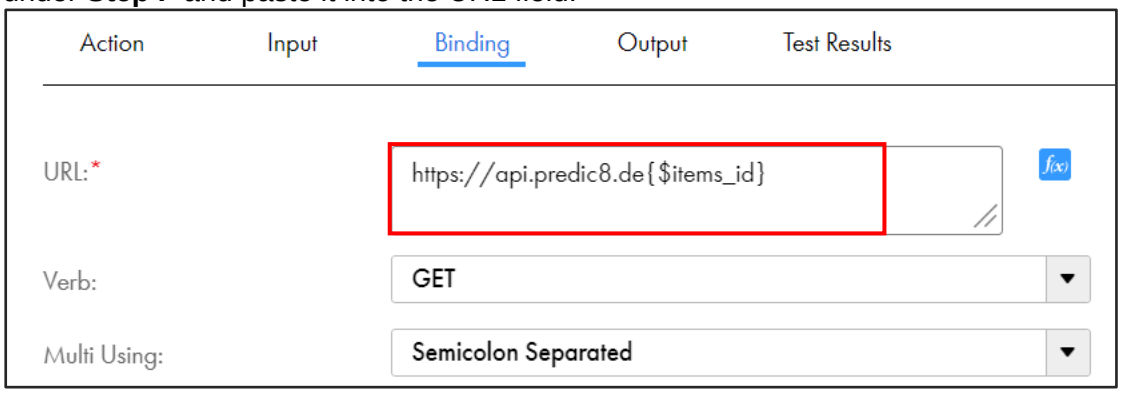
Name	Label	Type	Required	Parameter	Test With
items_id	items_id	Text	Yes	No	/shop/v2/orders/1/items



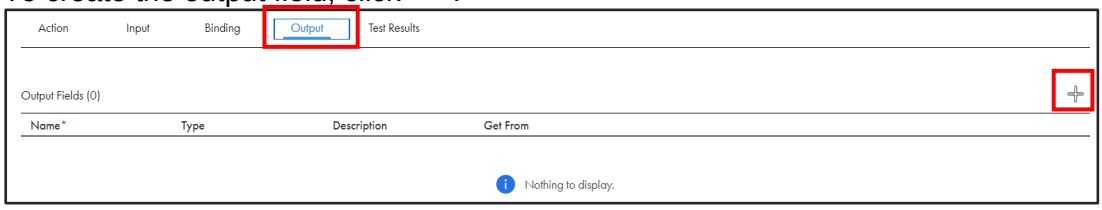
54. To specify the mode in which the data will be sent, select the **Binding** tab.
55. In the URL field, enter the following URL:
    https://api.predic8.de{$items_id}

    OR

    From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **03_LabGuide_CAI_Create a Service Connector_5-1.** Copy the command mentioned under **Step F** and paste it into the URL field.

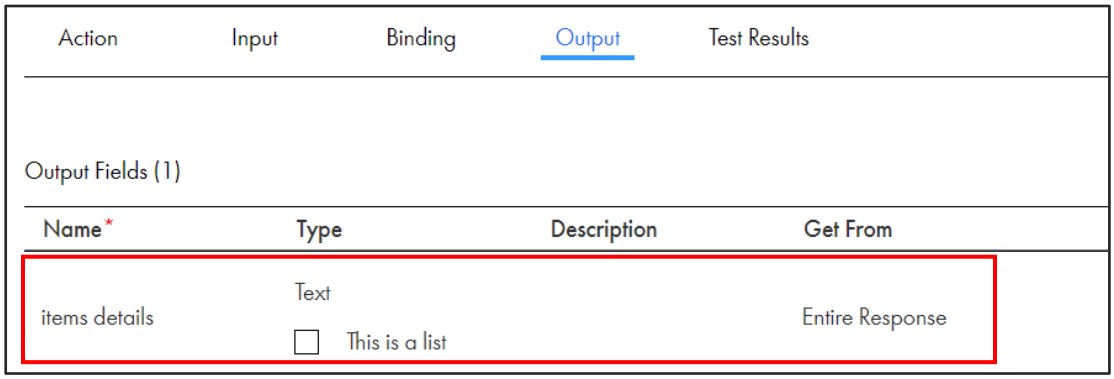


56. To define the output format of the response, select the **Output** tab.
57. To create the output field, click .

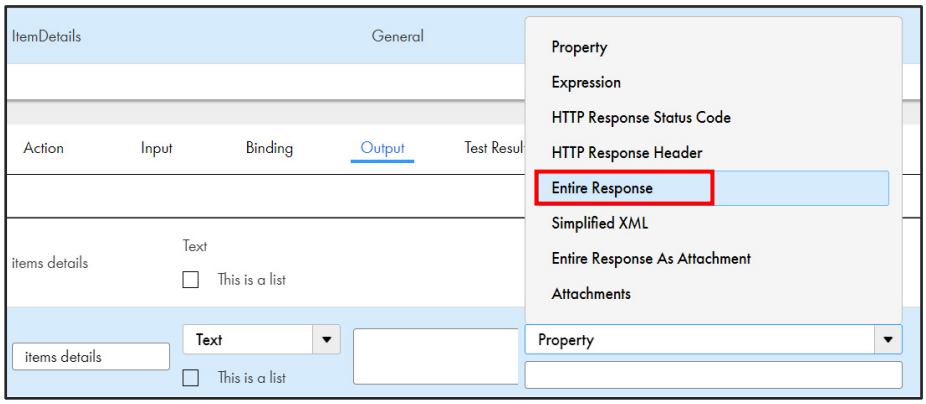


58. Create the Output Field, as shown in the table below:

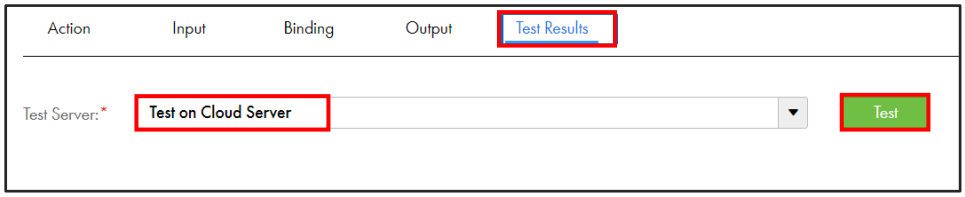
Name	Type	Get From
items details	Text	Entire Response



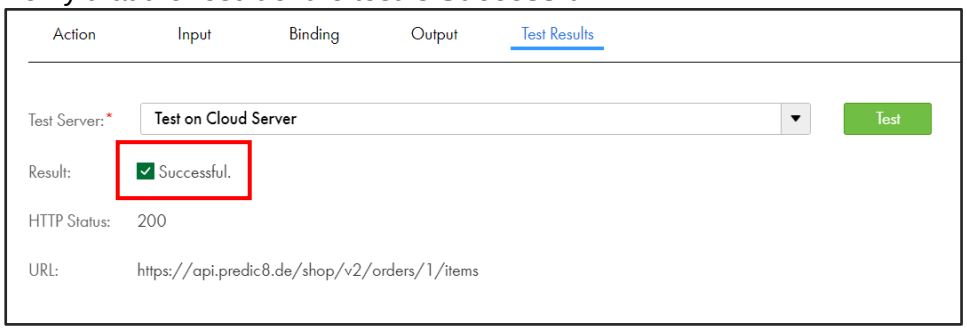
Note: You can select the “Entire Response” option from the **Get From** drop-down.



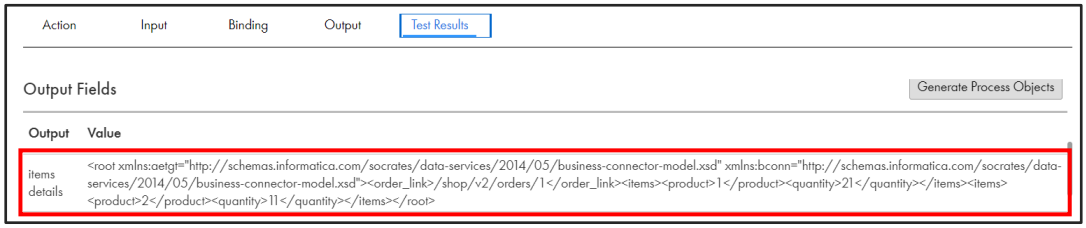
59. To test the service connector, select the **Test Results** tab.
60. From the Test Server field, retain **Test on Cloud Server**.
61. Click **Test**.



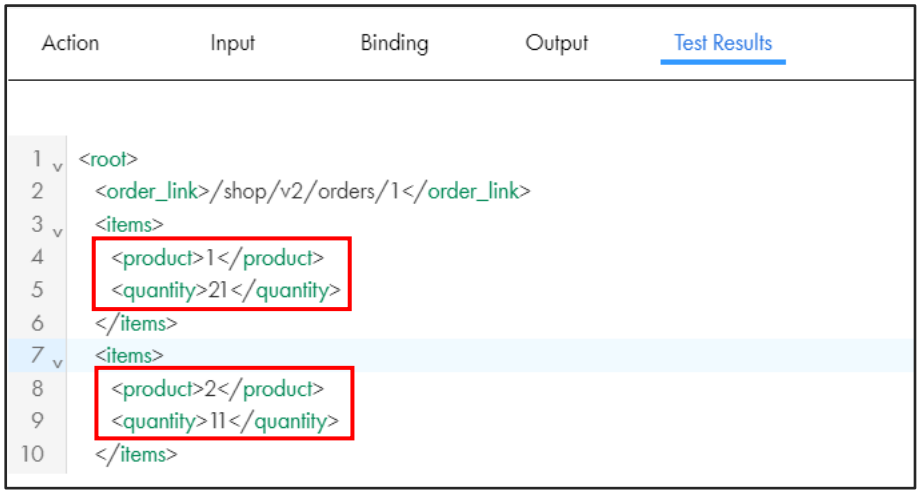
62. Verify that the result of the test is **Successful**.



63. Observe the **Output Fields** section. You can see the entire string of ordered items for order_id “1”.
Note: The output value in the image is subject to change based on the live data available in the API.

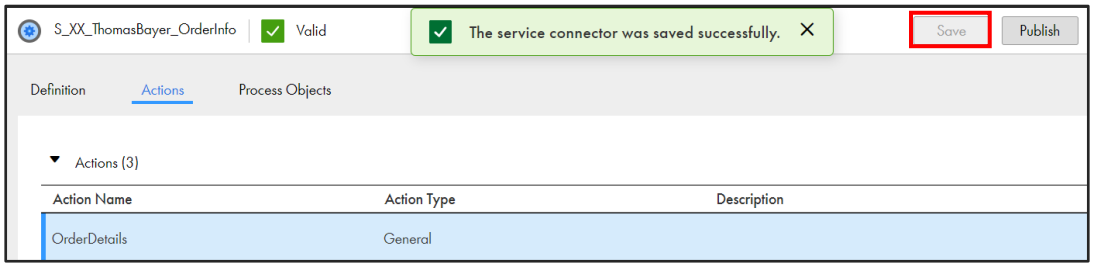


64. Expand the **Response Payload** field and observe the output.

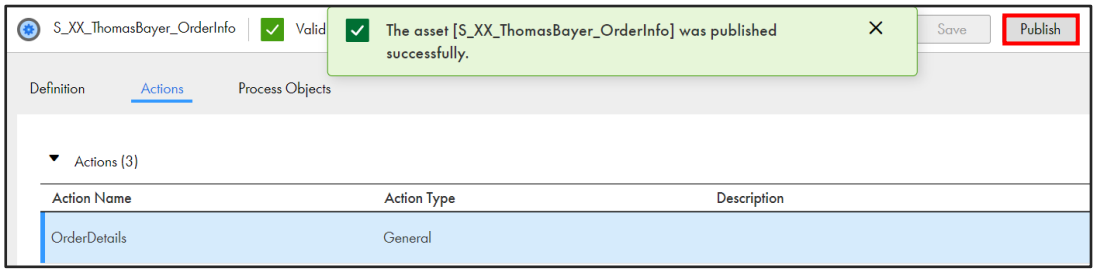


Note: Response payload displays the product numbers along with their quantities.

65. Click **Save** to save the connection.



66. Click **Publish** to publish the connection.



Note: A message **The asset was published successfully** appears.

This concludes the lab.

Module 5: Service Connectors
Lab 5-2: Create a Service Connector from WSDL

Overview:
In this lab, you will create a Service Connector from a WSDL file.

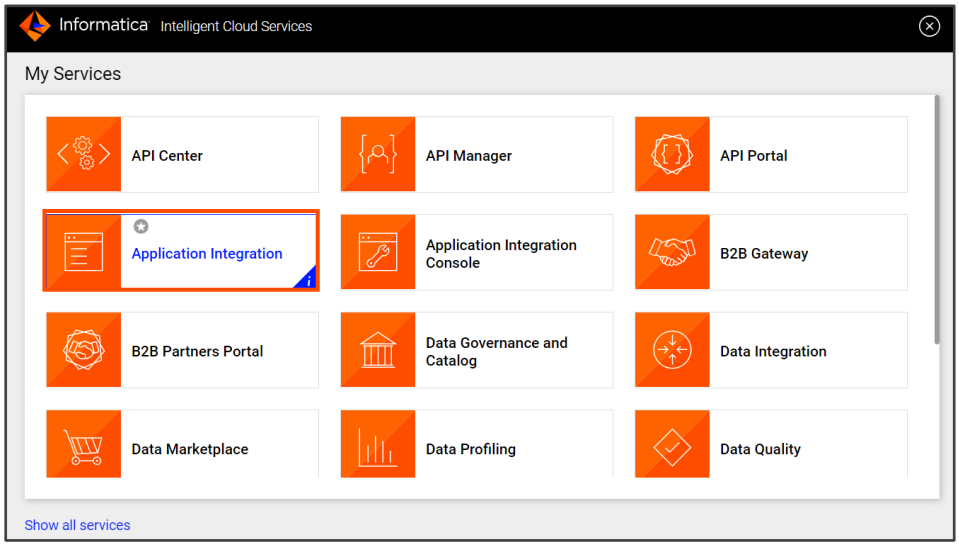
Objectives:
- Import a WSDL File
- Modify an existing Action
- Test the Service Connector

Duration:
25 Minutes

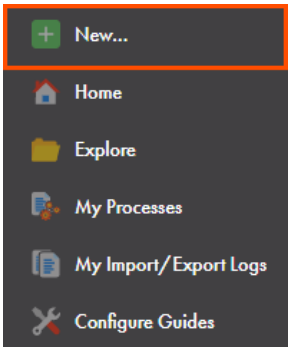
Tasks

Import a WSDL File:

1. Log into IICS and access the **Application Integration** service.

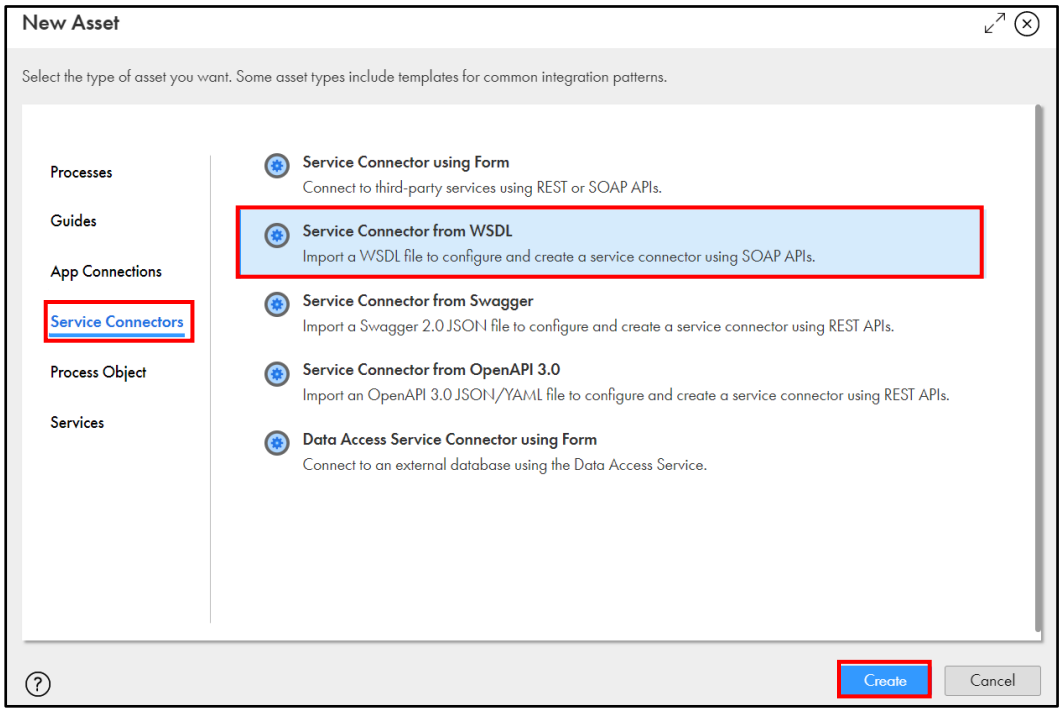


2. To create a new Service Connector, from the navigation pane, click **New**.



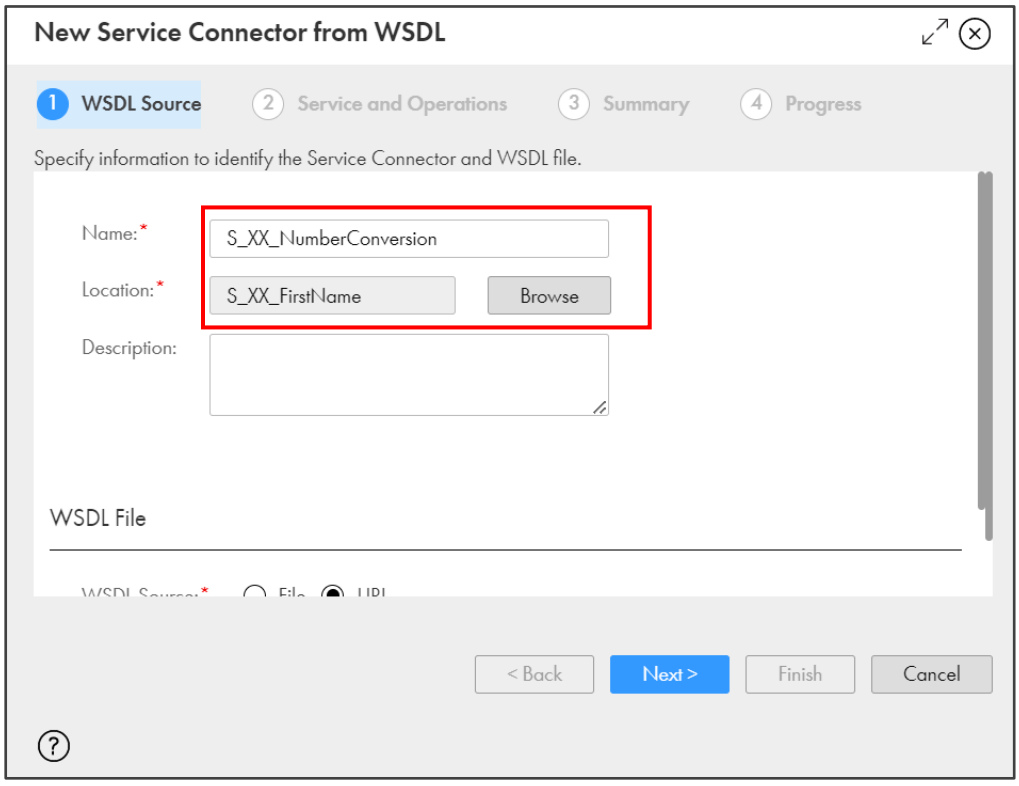
3. From the Service Connectors tab, select **Service Connector from WSDL**.

4. Click **Create**.



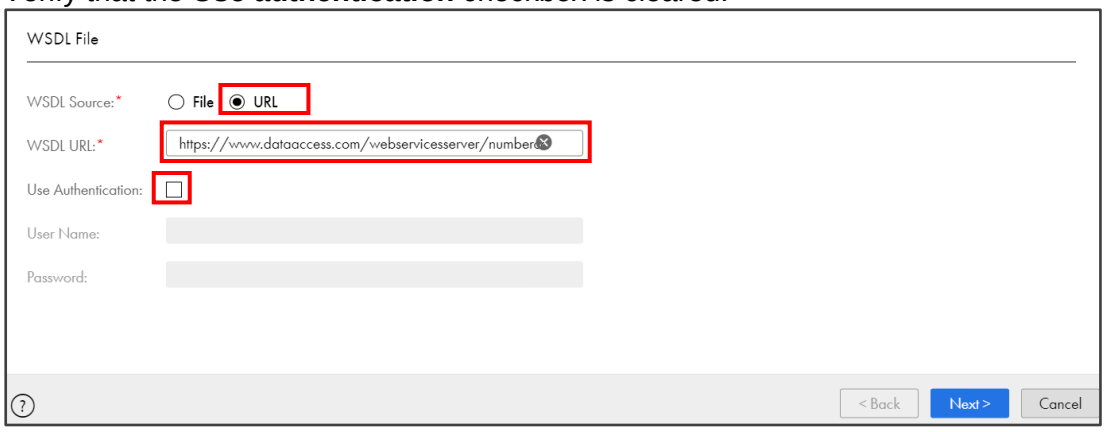
Note: A new Service Connector page appears.

5. In the service connector Name field, enter **S_XX_NumberConversion**.
   Note: Here, XX is the unique student number.
6. In the Location field, select your project folder **S_XX_FirstName**.

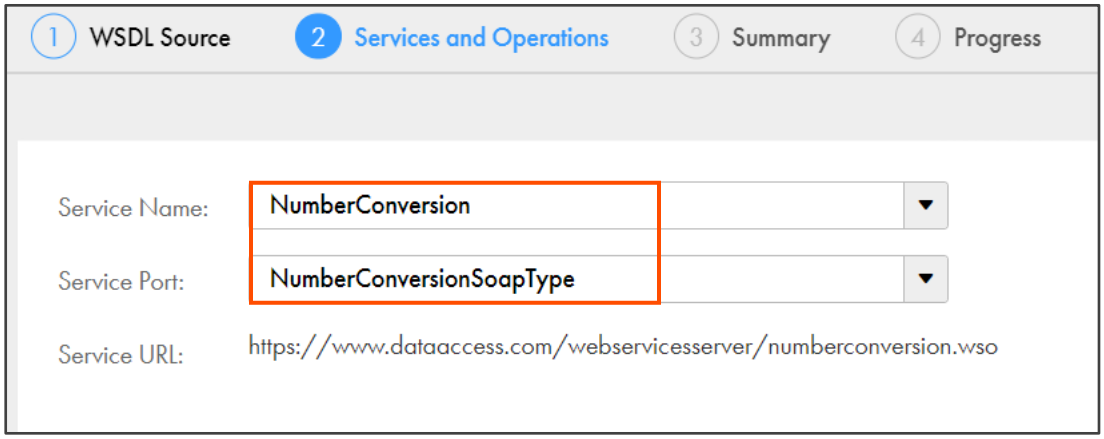


Note: Here, XX is the unique student number.

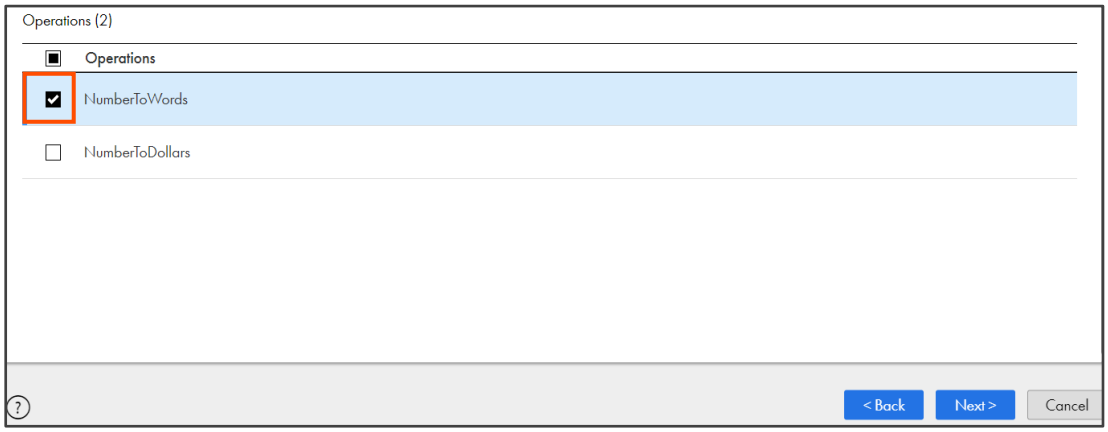
7. From WSDL Source, select **URL**.
8. In the WSDL URL field, enter the following URL:
   https://www.dataaccess.com/webservicesserver/numberconversion.wso?WSDL
9. Verify that the **Use authentication** checkbox is cleared.



10. Click **Next**.
11. Retain the values in **Service Name** and **Service Port** fields.



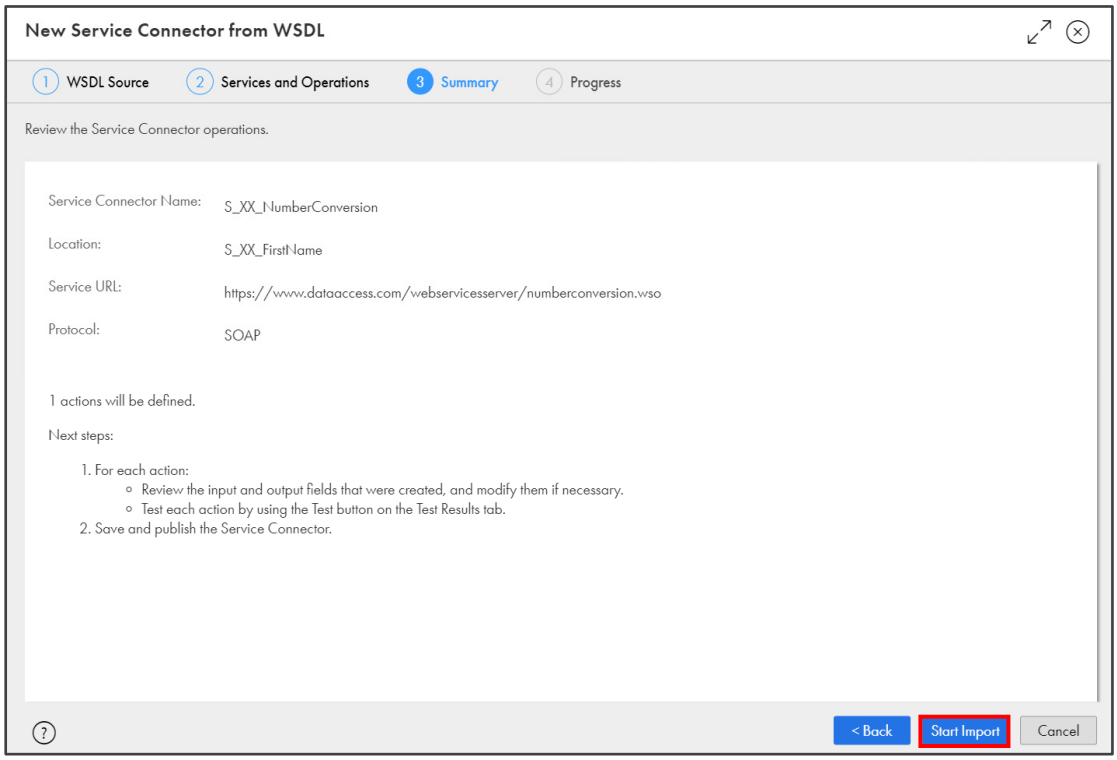
12. In the Operations section, make sure that the **NumbertoWords** option is checked.



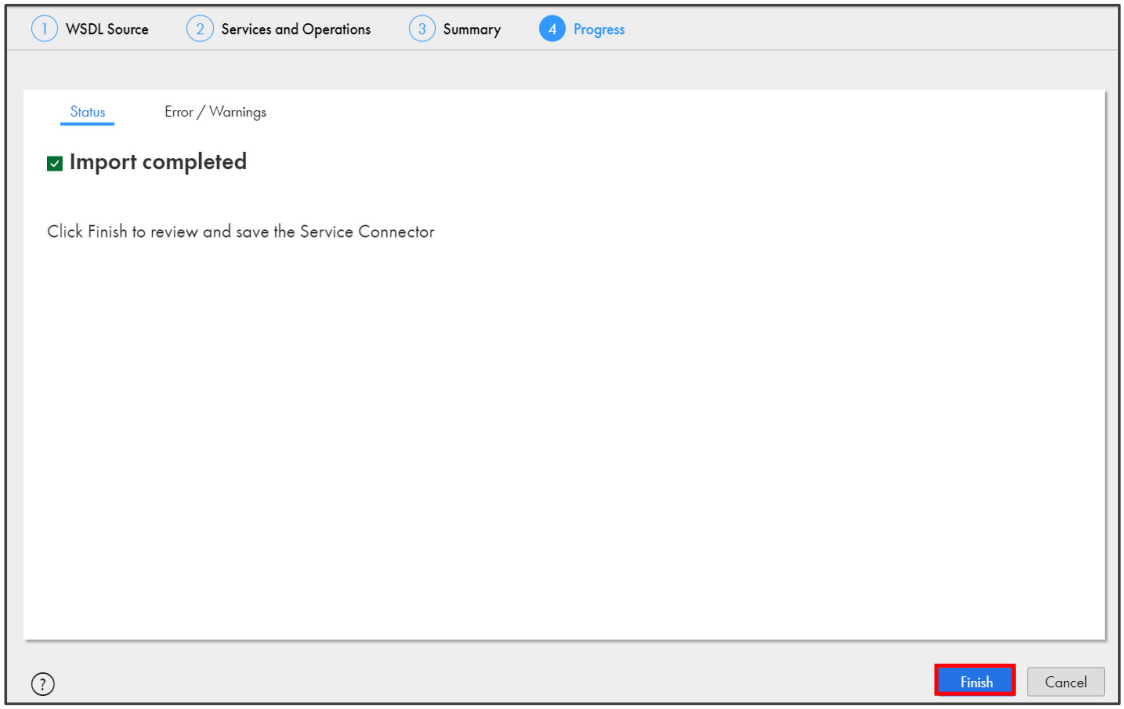
Note: Ensure that the NumberToDollars option is unchecked.

13. Click **Next**.

14. Click **Start Import**.

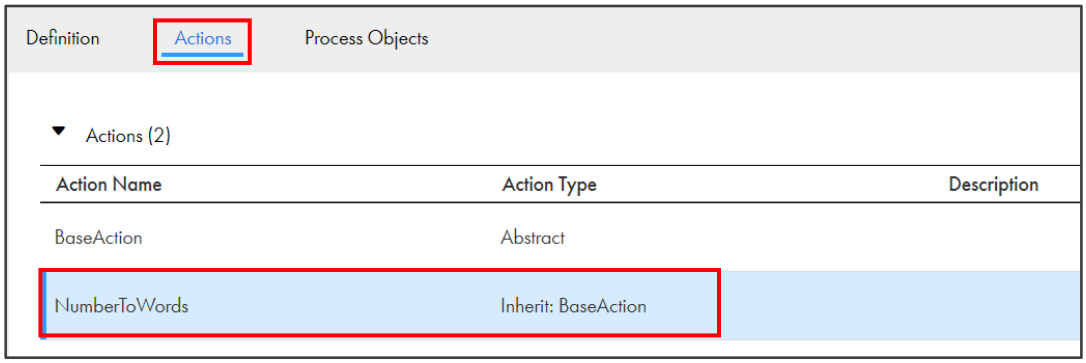


15. Click **Finish**.

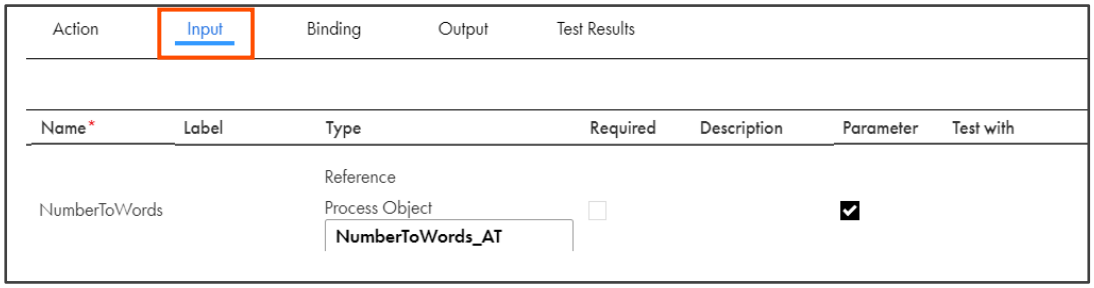


Modify the NumberToWords Action:

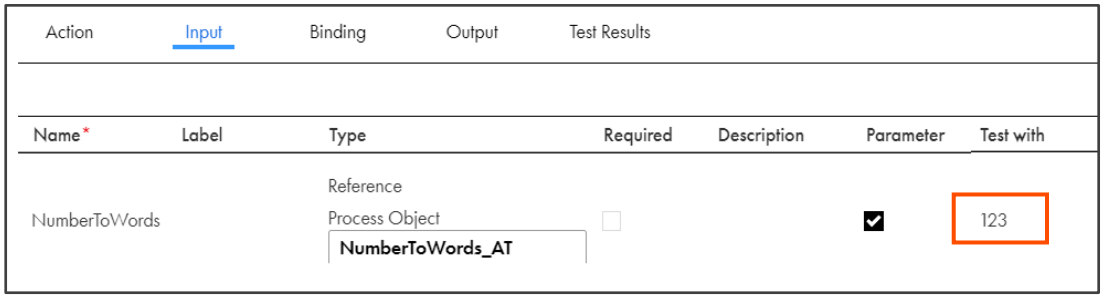
16. Click the **Actions** tab.
17. Select the **NumberToWords** Action.



18. Select the **Input** tab.

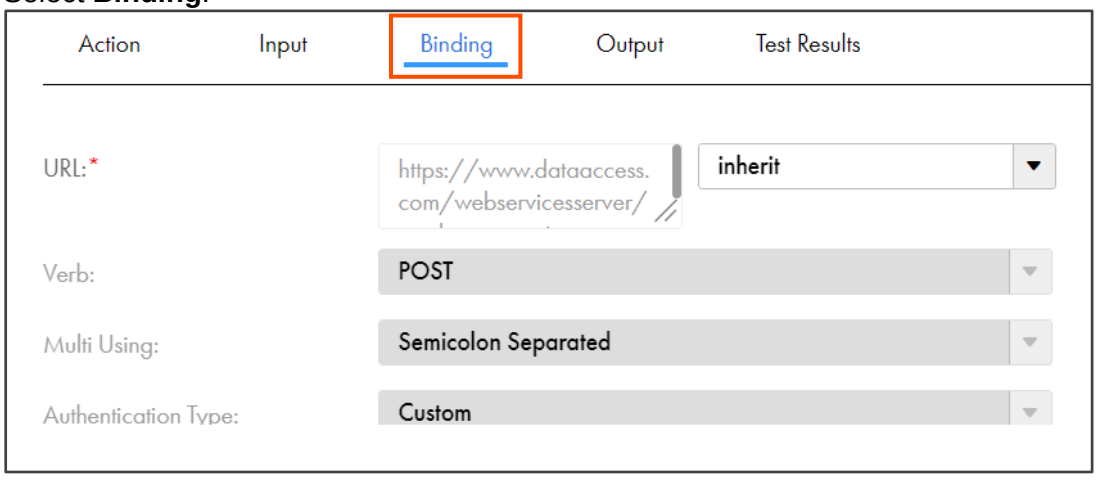


19. Click the **NumberToWords** row and in the Test With field, enter **123**.

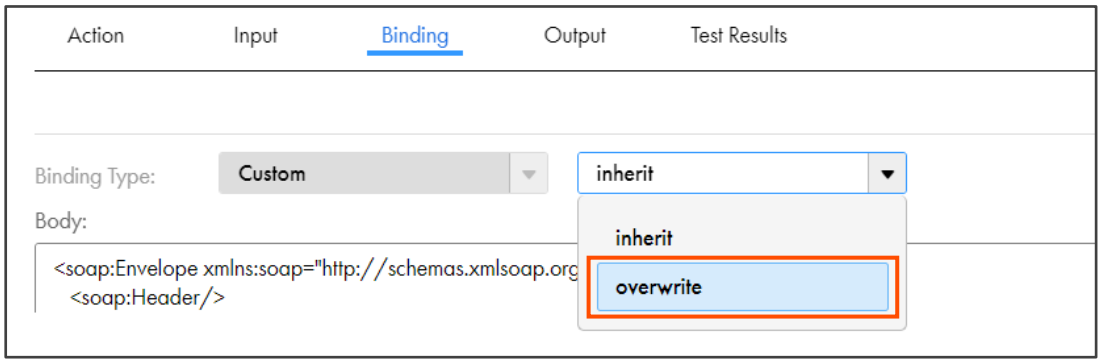


Note: You can replace this value with any number which you want to convert to words.

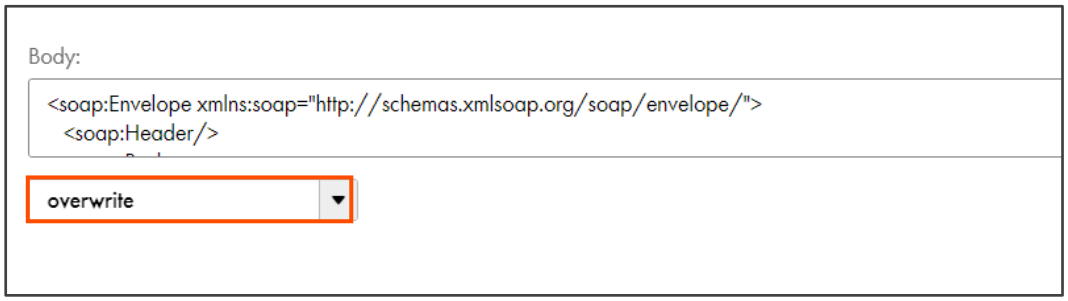
20. Select **Binding**.



21. Scroll down and from the Binding Type drop-down, select **overwrite**.



22. Scroll down and make sure that the **overwrite** option is selected in the drop-down.



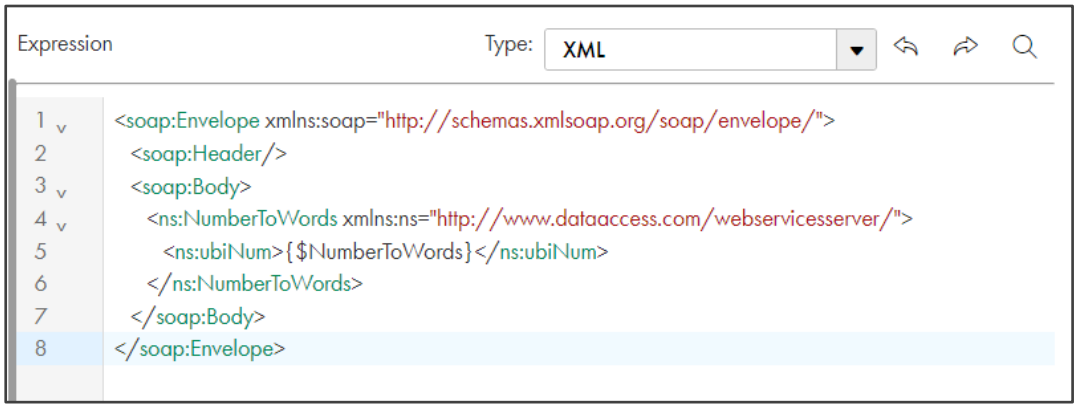
23. Click $f(x)$.

24. In the Expression section, replace the text with the following:

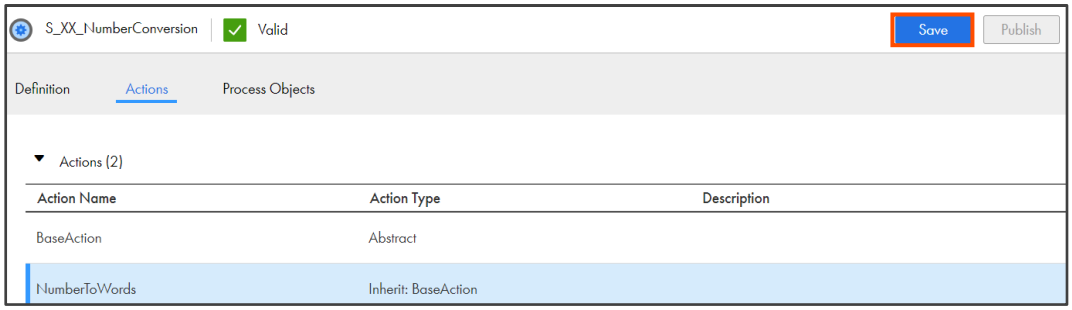
```
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Header/>
  <soap:Body>
    <ns:NumberToWords
xmlns:ns="http://www.dataaccess.com/webservicesserver/">
      <ns:ubiNum>{$NumberToWords}</ns:ubiNum>
    </ns:NumberToWords>
  </soap:Body>
</soap:Envelope>
```

OR

From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **04_LabGuide_CAI_Create a Service Connector from WSDL_5-2**. Copy the command mentioned in the file and paste it into the body field.

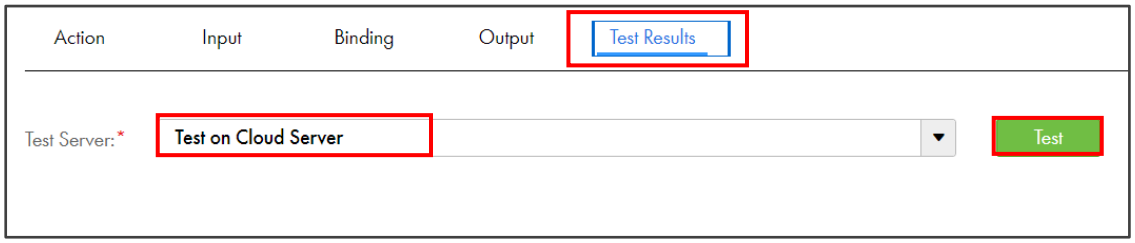


25. Click **OK**.
26. Click **Save**.

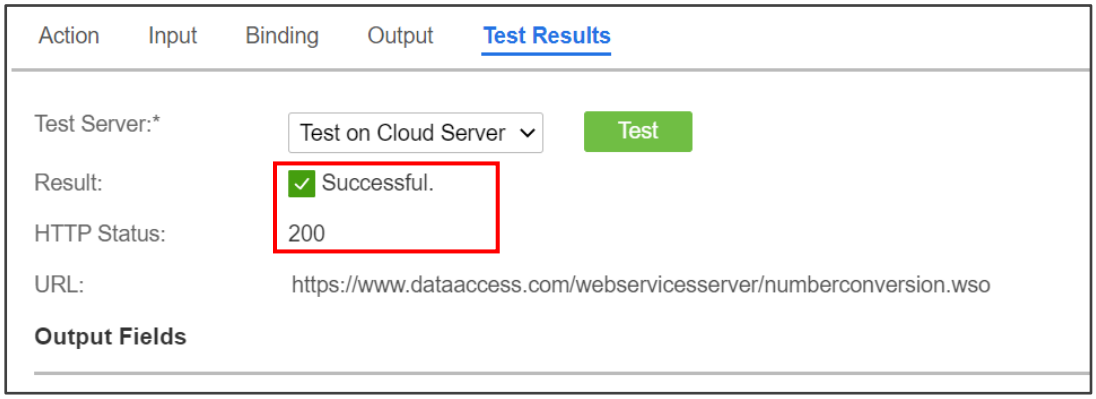


Test Action

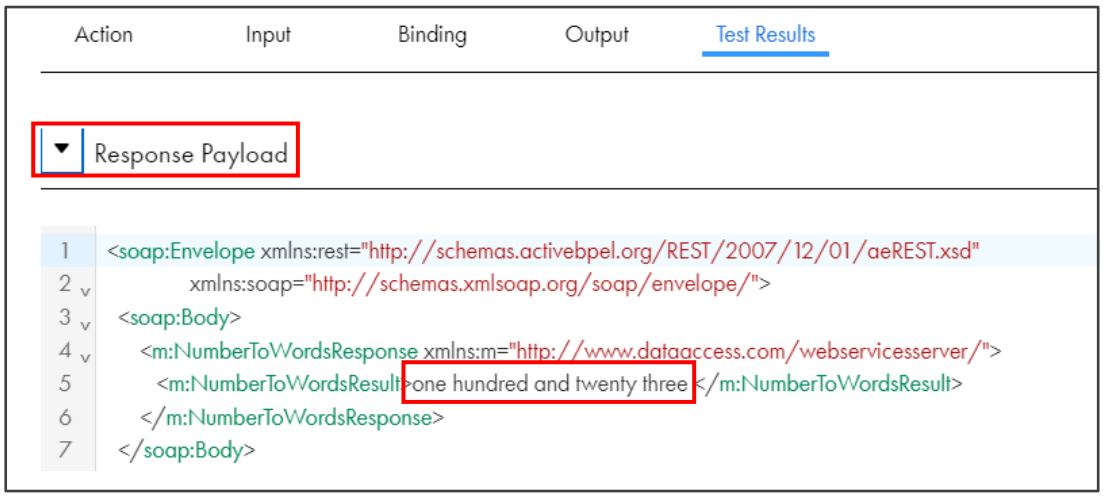
27. Select **Test Results**.
28. From the Test Server drop-down, retain **Test on Cloud Server** and click **Test**.



29. In the Test Results tab, verify that the result of the test is **Successful**.
    Note: Note that HTTPS Status is 200.

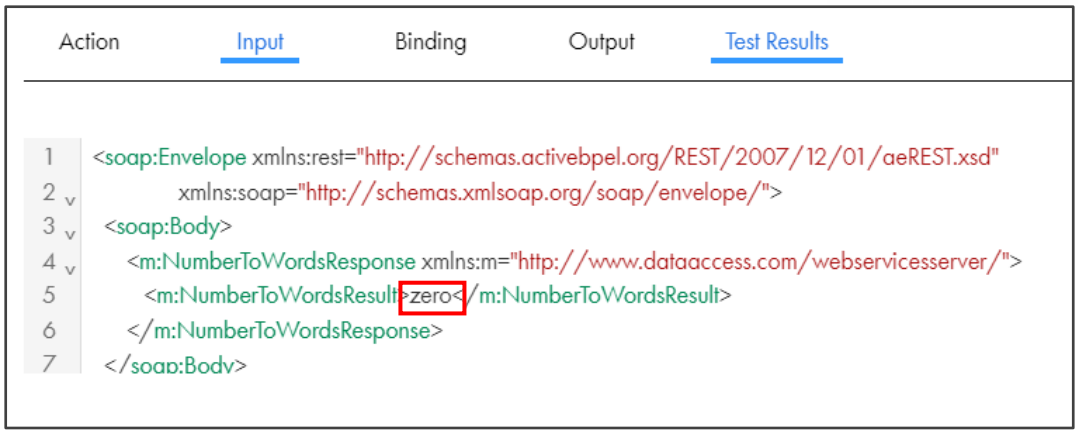


30. From the Output Fields section, scroll down and expand **Response Payload**.
31. Observe the output.

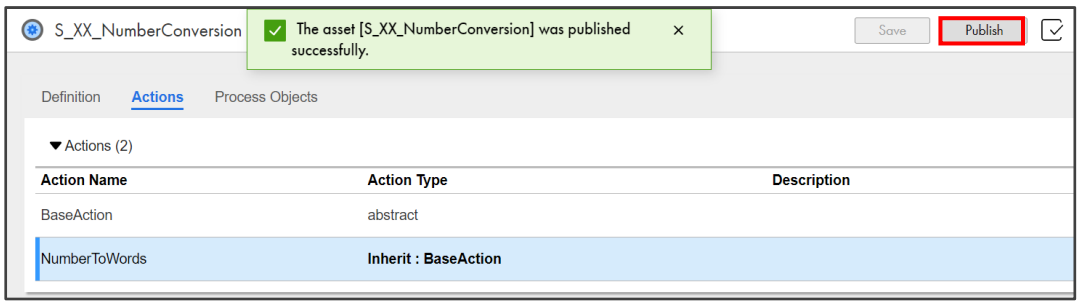


Note: Response Payload displays the response received from the service. Remember, you have entered the Test With value as “123”. The service has converted the number “123” into words “one hundred and twenty three”.

If Response Payload displays ‘zero’, check the Input tab and make sure that the **Test With** option is not empty. If the **Test With** option is empty, enter “123” and test it again.



32. Click **Publish**.

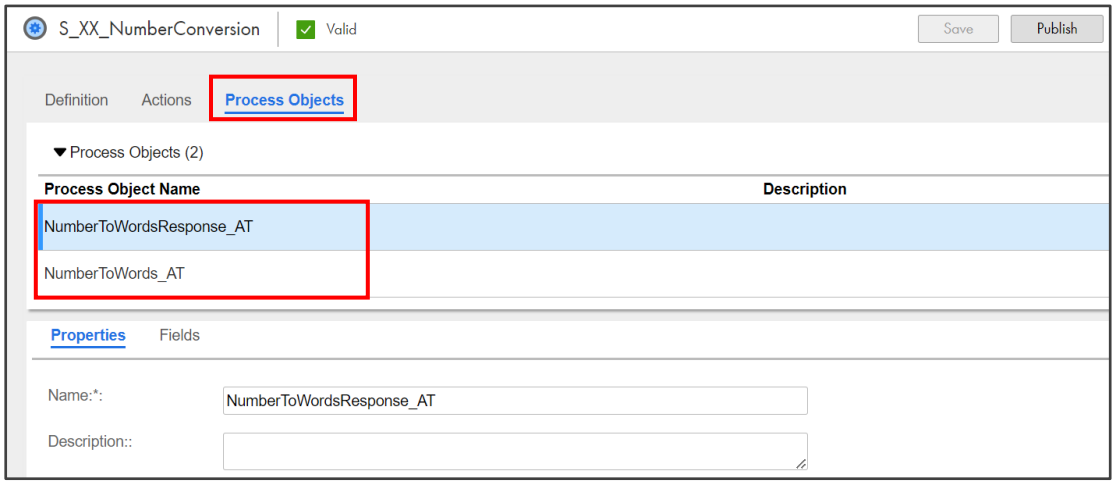


A message "**The asset [asset_name] was published successfully"** appears.

Service Connector Process Objects

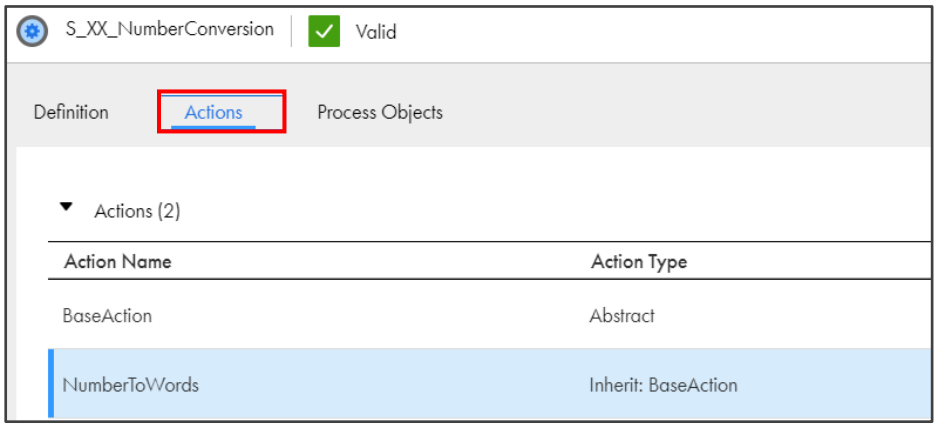
A service connector process object groups data. The WSDL URL you imported to generate a service connector generates two process objects.

33. To view service connector process objects, click the **Process Objects** tab.

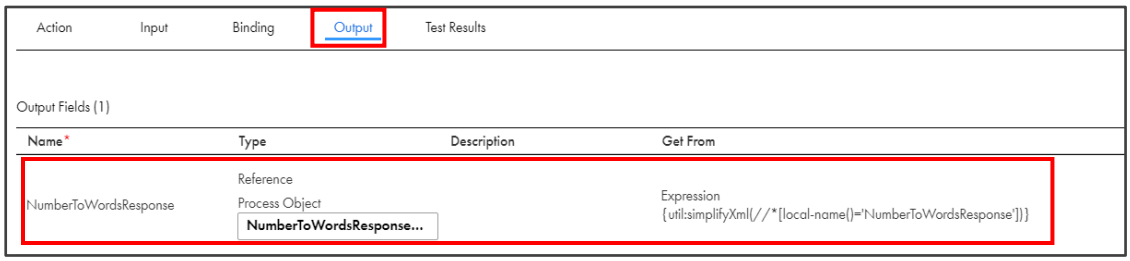


You can observe two Process Objects: **NumberToWordsResponse_AT** and **NumberToWords_AT**.

34. Click the **Actions** tab.

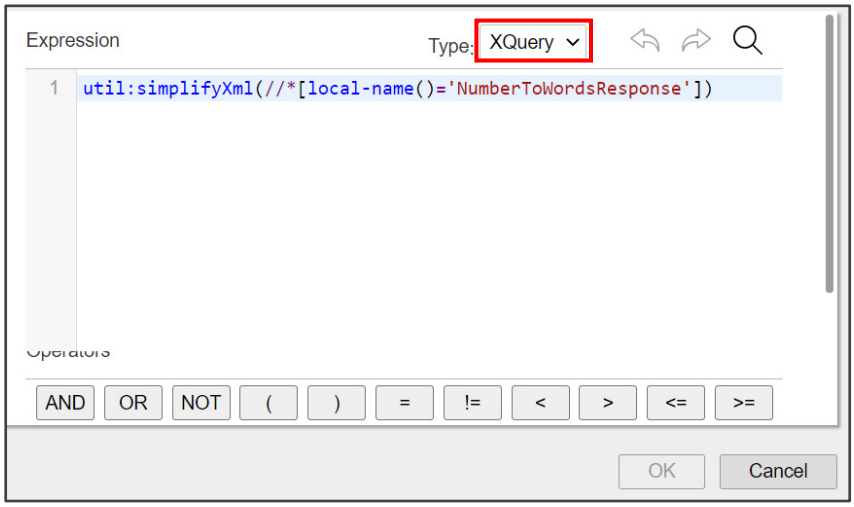


35. To check the Expression, click the **Output** tab.
36. Click the **NumberToWordsResponse** row.



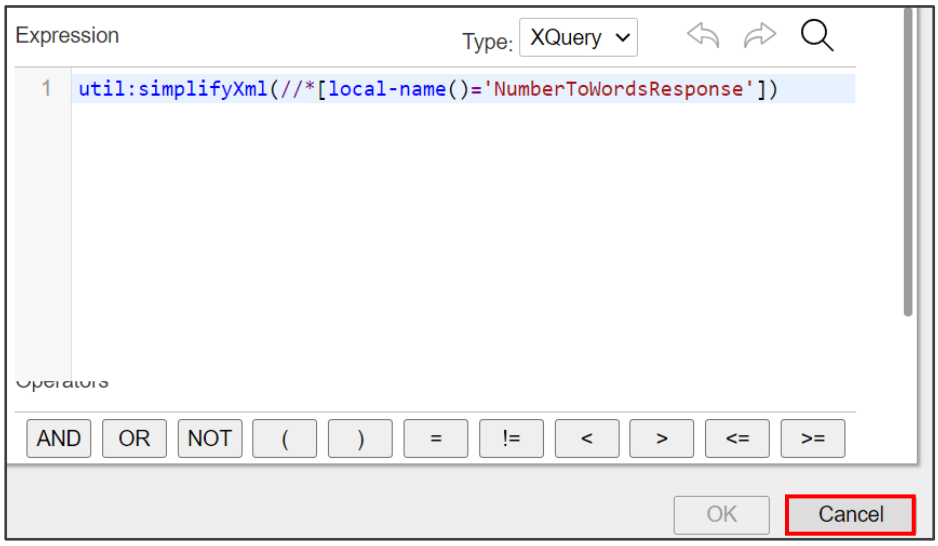
37. Click .

Observe the Expression Type is **XQuery**.



Note: You will learn about XQuery in **Module 9: Using XQuery in CAI**.

38. Click **Cancel**.



39. Close the Service Connector.

This concludes the lab.

Module 5: Service Connectors
Lab 5-3: Create and Test a Swagger File

Overview:

Informatica Intelligent Cloud Services supports Swagger specification version 2.0. When you generate a Swagger file in Informatica Intelligent Cloud Services, you send an API call to the service using a sample request. If you do not have permission to send an API call to the service, you can generate a Swagger file using a sample request and a sample response without submitting an API call.

You cannot modify a Swagger file once it is created. If you want to make changes to a Swagger file, create a new Swagger file.

Note: The Swagger file generation functionality is available for the convenience of the REST V2 customers. Informatica does not warranty the compatibility of the Swagger file for all customer scenarios.

Objectives:

- Sign up to Access OpenWeatherAPI
- Create Swagger files
- Test the Swagger file

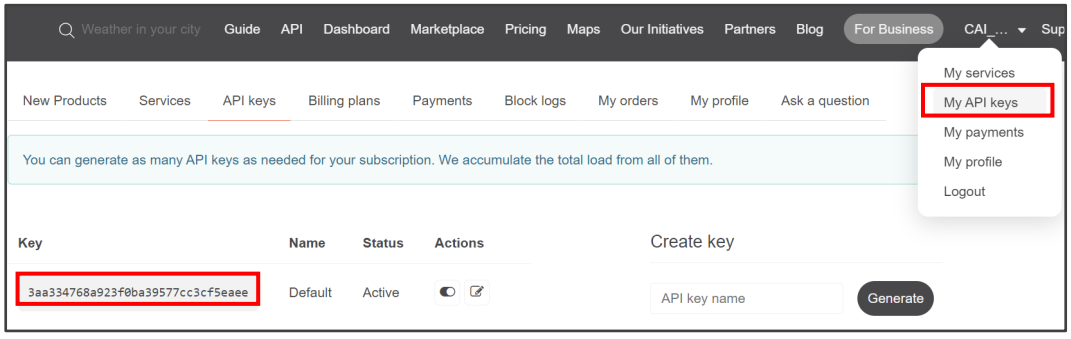
Duration:

15 Minutes

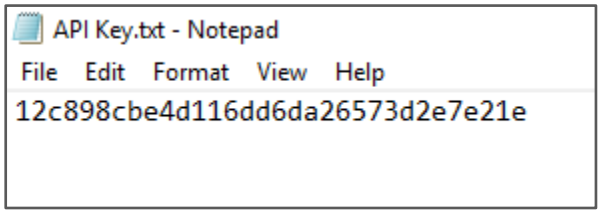
Tasks

Sign up to Access OpenWeatherAPI

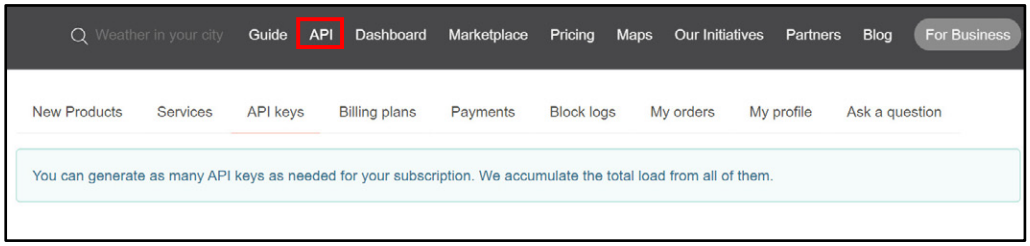
1. Open a web browser and enter the following URL in a new tab:
   https://home.openweathermap.org/users/sign_up
   Note: You can bookmark this link for future use.
2. Create a new account in OpenWeather using the link provided.
   Note: After you create an account, you can access your unique API key.
3. To access your unique API Key, click on your user ID, and select **MY API Keys**.
4. Copy the API key from the Key field.



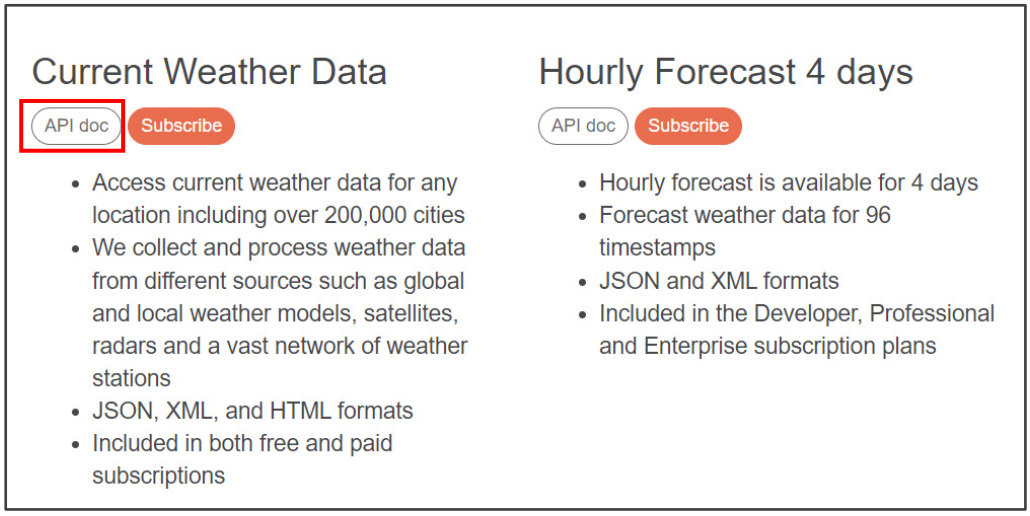
5. Paste the API key in a notepad and save the notepad file on your system.



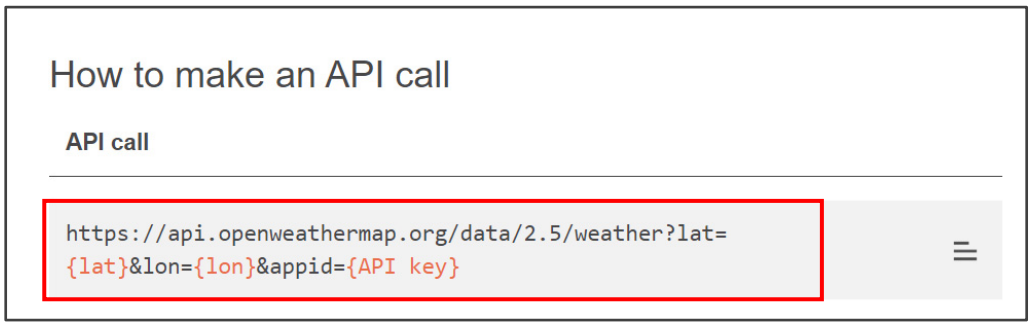
6. Click **API**.



7. Scroll down and locate **Current Weather Data**. Click **API doc**.



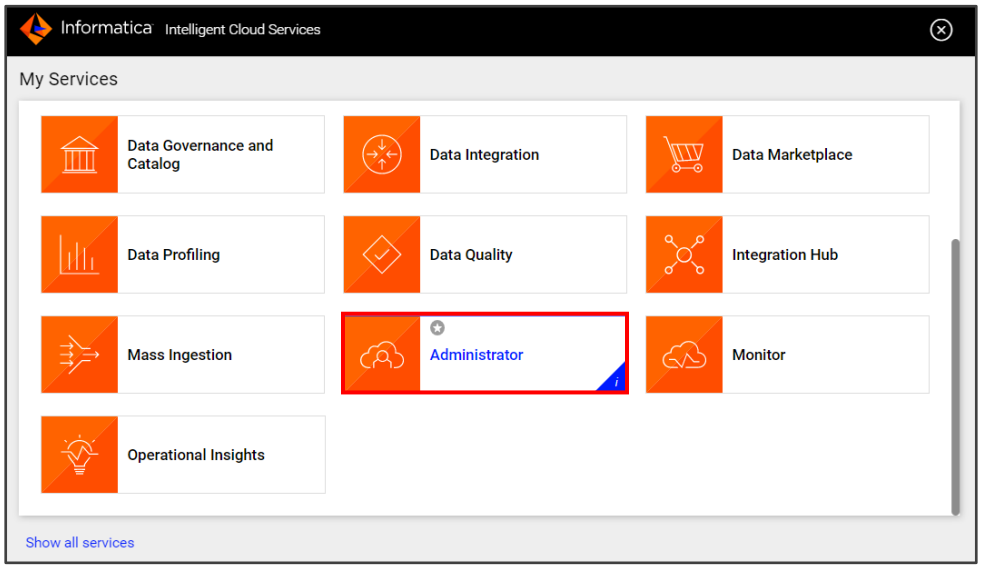
8. From the API Call section copy the URL and save it in a notepad file.



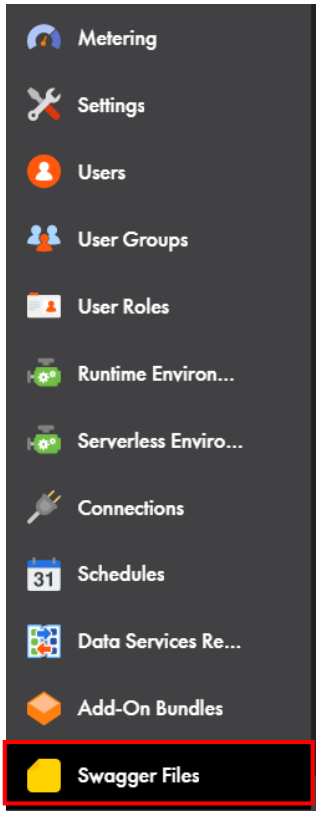
Note: We will be using this URL while generating the Swagger file.

Create the CurrentForecastWeather Swagger File

9. Log into IICS and access the **Administrator** service.

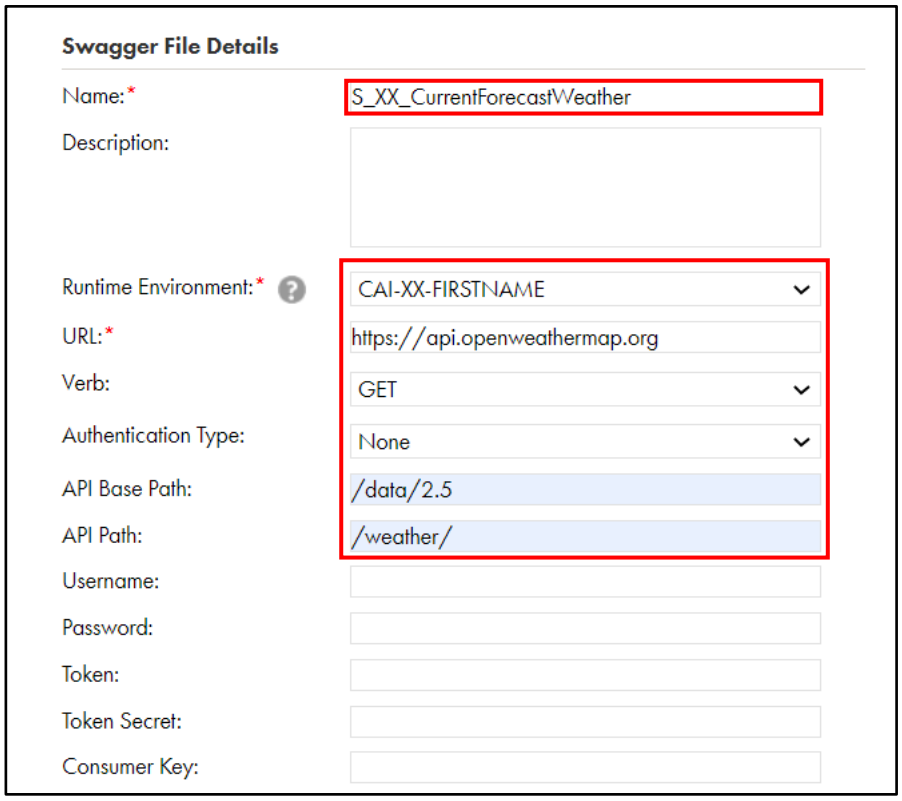


10. To create a Swagger file, from the navigation pane, click **Swagger Files**.



11. Click **New**.
12. In the Name field, enter **S_XX_CurrentForecastWeather**.
    Note: Here, XX is the unique student number.
13. In the Runtime Environment field, select your secure agent.

14. In the URL field, enter the following URL: https://api.openweathermap.org
15. In the Verb field, select **Get**.
16. In the Authentication Type field, retain **None**.
17. In the API Base Path field, enter **/data/2.5**.
18. In the API Path field, enter **/weather/**.



19. In the Accept field, select **application/json**.
20. In the Query Params field, enter the following parameter:

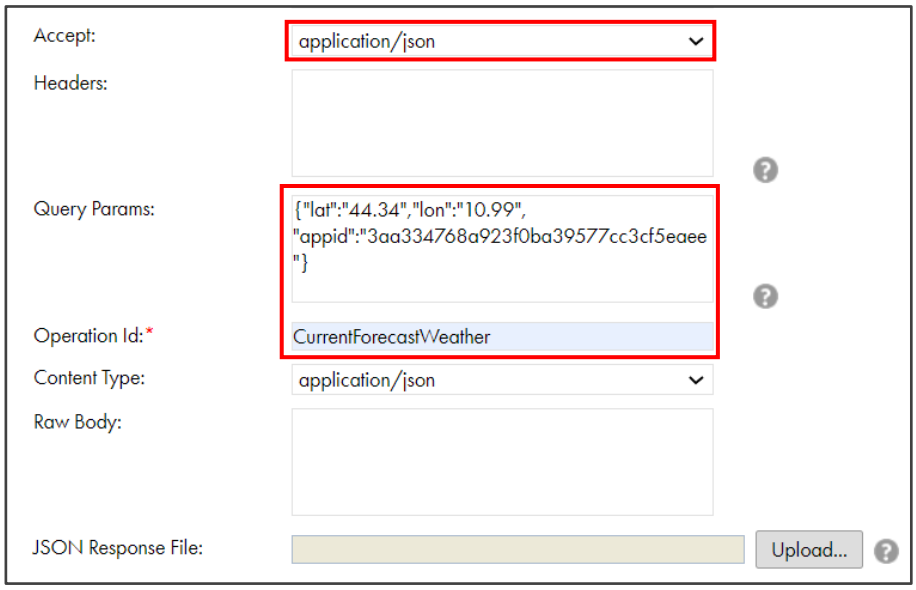
    {"lat":"<latitude>","lon":"<longitude>", "appid":"<API Key>"}

    Note: Replace latitude and longitude with the digits of your choice and API Key with the key you copied to the notepad.

    For example:

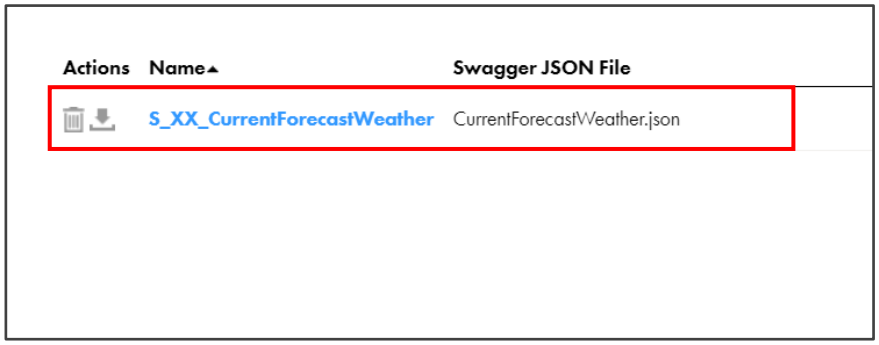
    {"lat":"44.34","lon":"10.99", "appid":"3aa334768a923f0ba39577cc3cf5eaee"}

21. In the Operation Id field, enter **CurrentForecastWeather**.



22. Click **Save**.

    Note: A message **Swagger saved successfully** appears. Click the **Swagger Files** tab to check the file.

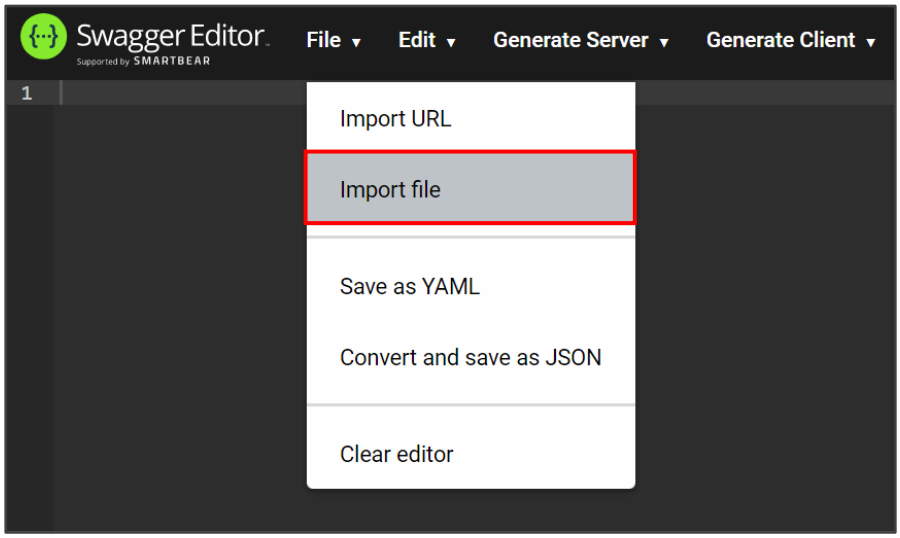


23. Download S_XX_CurrentForecastWeather and save it on your machine.

Test the CurrentForecastWeather Swagger File

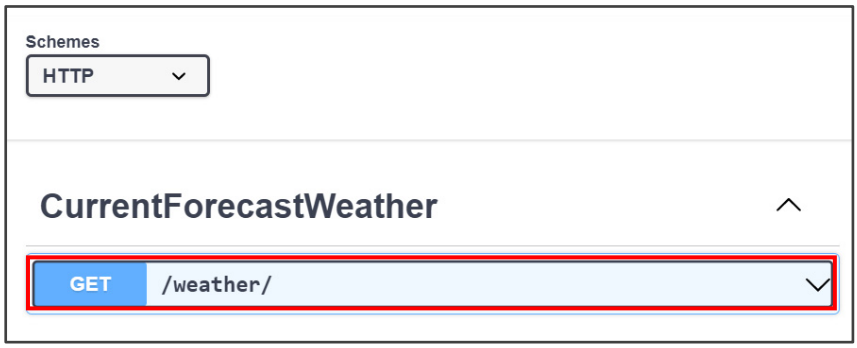
In the last step, you have created the S_XX_CurrentForecastWeather Swagger file. Now, you will test the correctness of the S_XX_CurrentForecastWeather Swagger file.

24. Open your web browser and enter the following URL: https://editor.swagger.io/
25. From the main menu, select **File** > **Import file**.

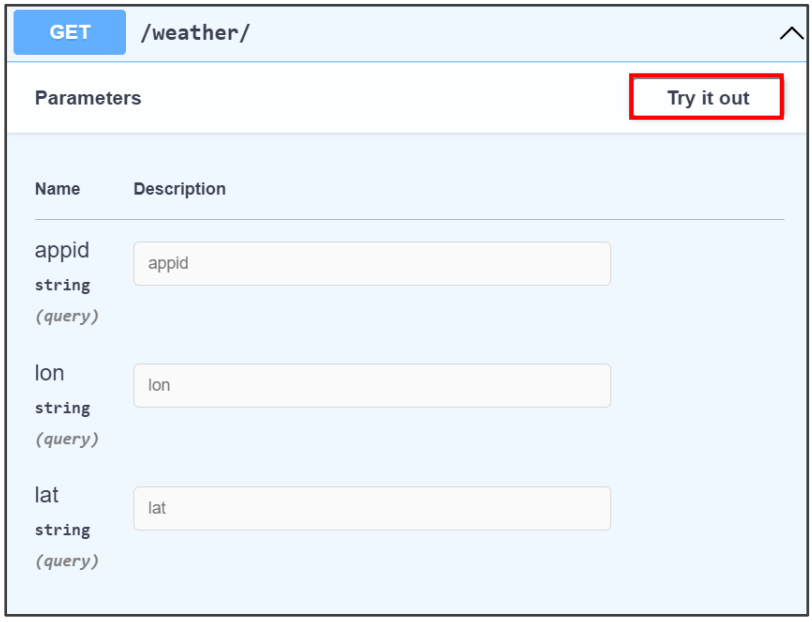


26. Select the **CurrentForecastWeather.json** file.
27. You can see a list of errors. Click **Hide** to ignore the errors.

28. Click the **Get/weather field** to expand it.

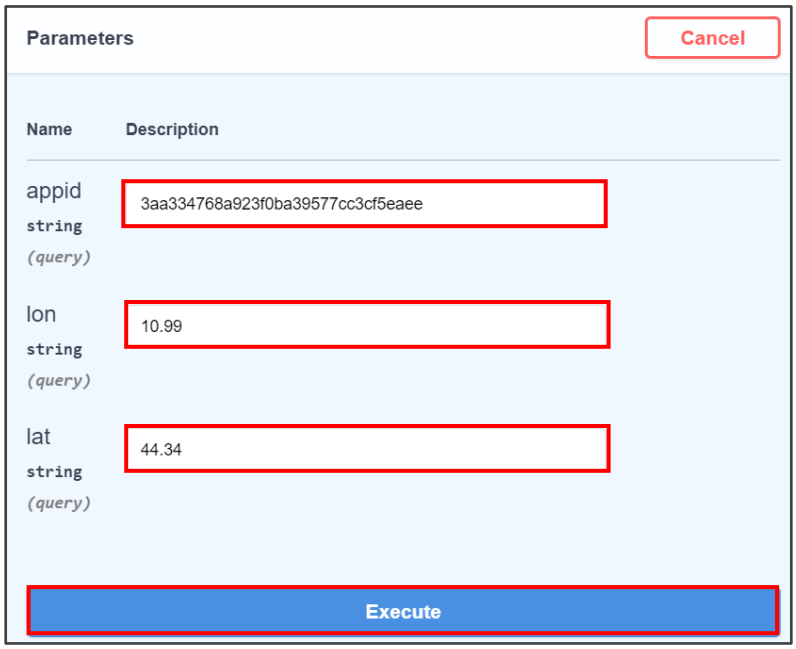


29. Click **Try it out**.



30. In the appid field, enter the API Key that you have copied to the notepad.
31. In the lon filed, enter **10.99**.
32. In the lat field, enter **44.34**.

33. Click **Execute**.



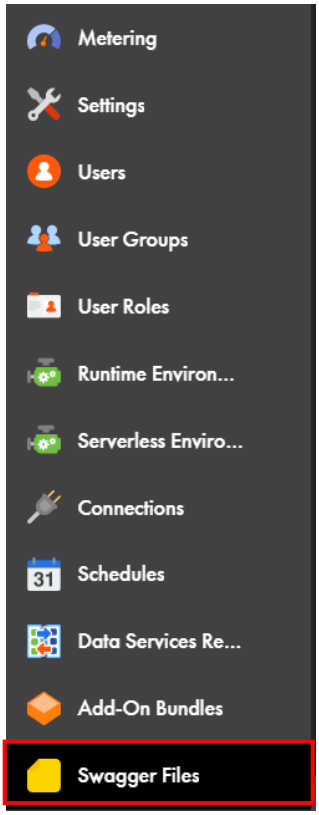
34. In the Response body section, observes the latitude and longitude values. You can see the same values which you have passed in the previous step.

Skill Application:

Create a Swagger File

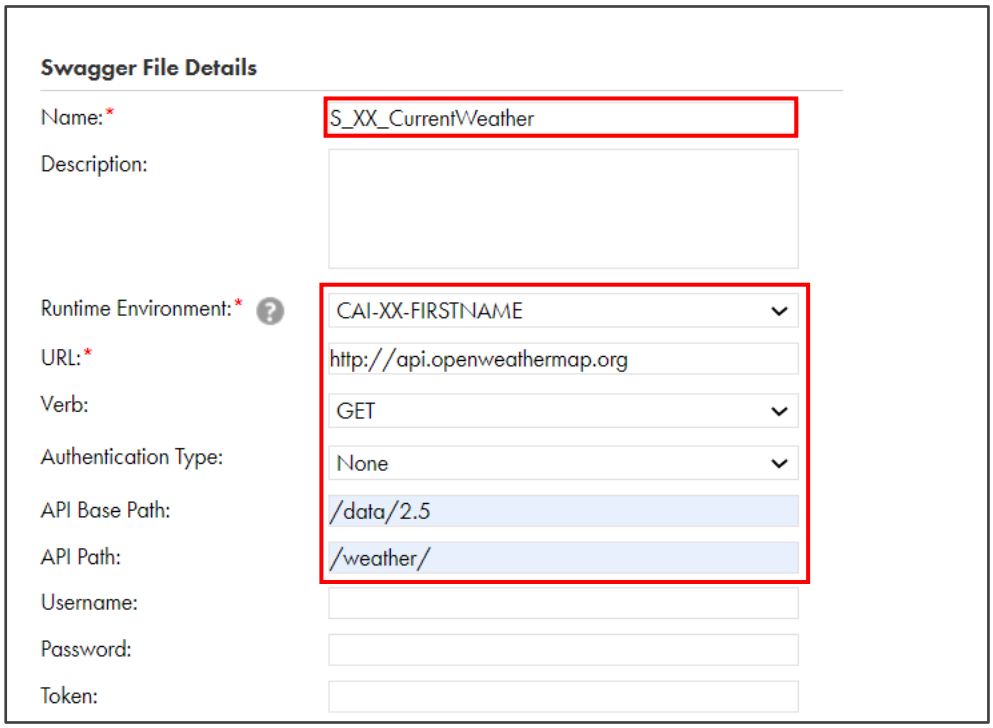
In this section of lab, create a new Swagger file and test it using the https://editor.swagger.io/ URL. Use the following parameters to create and test the Swagger file:

1. Navigate back to IICS and make sure that the Administrator service is opened.
2. To create a Swagger file, from the navigation pane, click **Swagger Files**.



3. Click **New**.
4. In the Name field, enter **S_XX_CurrentWeather**.
   Note: Here, XX is the unique student number.
5. In the Runtime Environment field, select your secure agent.
6. In the URL field, enter the following URL: https://api.openweathermap.org
7. In the Verb field, select **Get**.
8. In the Authentication Type field, retain **None**.
9. In the API Base Path field, enter **/data/2.5**.

10. In the API Path field, enter **/weather/**.

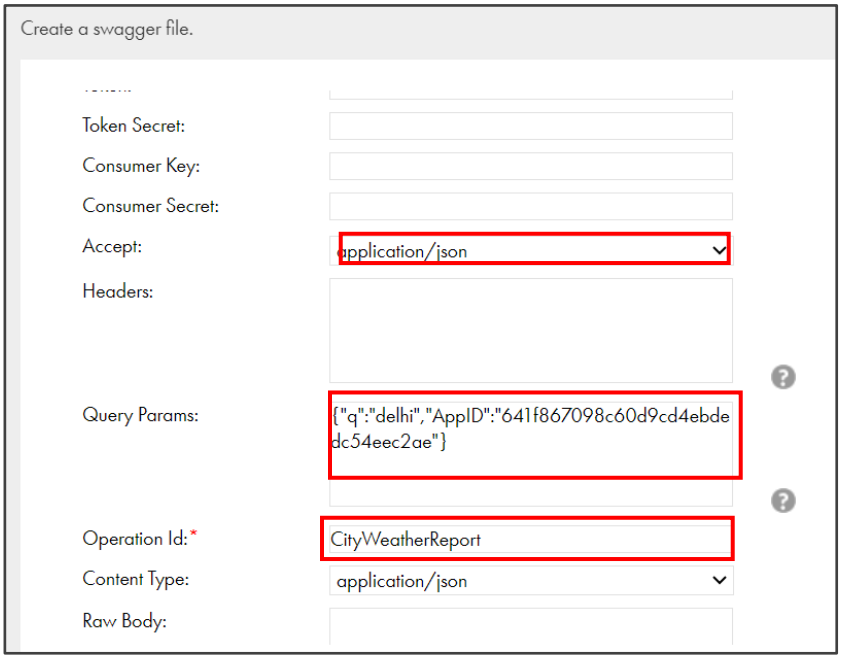


11. In the Accept field, select **application/json**.
12. In the Query Params field, enter the following parameter:

    {“q”:”<City Name>”,“AppID”:“<API Key>”}

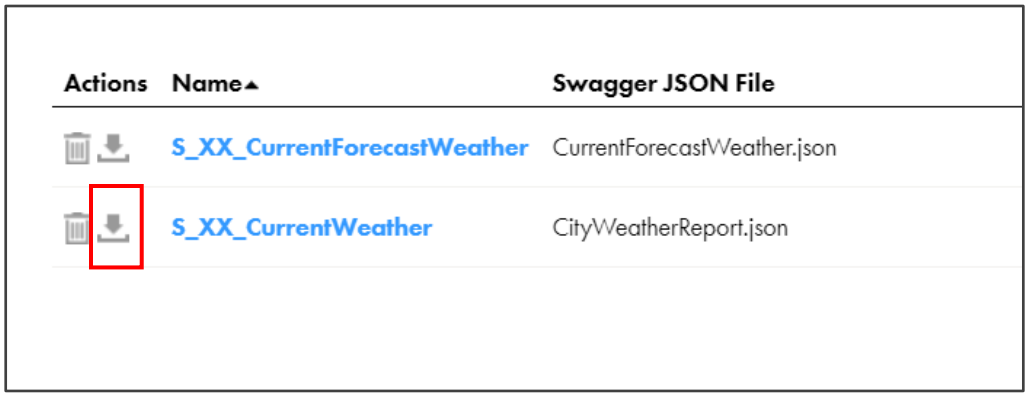
    Note: Replace City Name with the city of your choice and API Key with the key you copied on the notepad. For example:

    {"q":"delhi","AppID":"3aa334768a923f0ba39577cc3cf5eaee"}
13. In the Operation Id field, enter **CityWeatherReport**.



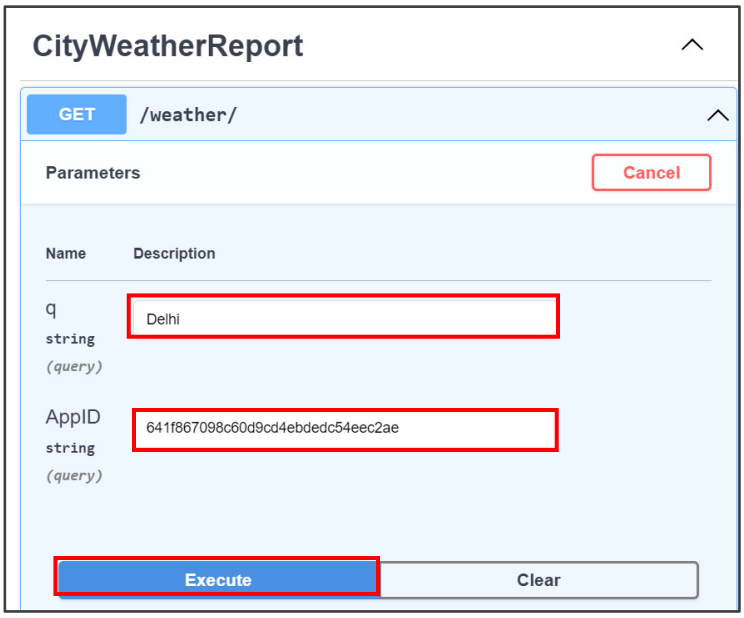
14. Save and download the Swagger file.

Note: A message **Swagger saved successfully** appears. Click the **Swagger Files** tab to check the file.



15. Test the Swagger file using the https://editor.swagger.io/ URL.
16. In the q field, enter **Delhi**.

    Note: Instead of Delhi, you can enter a different city in the q field.
17. In the appid field, enter the API Key that you have copied to the notepad.
18. Click **Execute**.



19. In the Response body filed, you will get the current weather report of the city (Delhi).

Response body


```
  "weather": [
    {
      "id": 711,
      "main": "Smoke",
      "description": "smoke",
      "icon": "50d"
    }
  ],
  "base": "stations",
  "main": {
    "temp": 299.2,
    "feels_like": 299.2,
    "temp_min": 299.2,
    "temp_max": 299.2,
    "pressure": 1014,
    "humidity": 44
  },
  "visibility": 1800,
  "wind": {
    "speed": 2.06,
    "deg": 280
  },
  "clouds": {
```


Response body


```
    "humidity": 44
  },
  "visibility": 1800,
  "wind": {
    "speed": 2.06,
    "deg": 280
  },
  "clouds": {
    "all": 1
  },
  "dt": 1676536782,
  "sys": {
    "type": 1,
    "id": 9165,
    "country": "IN",
    "sunrise": 1676510961,
    "sunset": 1676551284
  },
  "timezone": 19800,
  "id": 1273294,
  "name": "Delhi",
  "cod": 200
}
```

This concludes the lab.

Module 5: Service Connectors
Lab 5-4: Create a Service Connector from a Swagger File

Overview:
You can import a Swagger JSON file to easily create a service connector with multiple operations. Use the New Service Connector from the Swagger wizard to import a Swagger JSON file and create a service connector. After you specify the Swagger JSON file, Application Integration parses the file and loads the operations.

Objectives:
- Import a Swagger JSON File
- Modify an existing Action
- Test the Service Connector

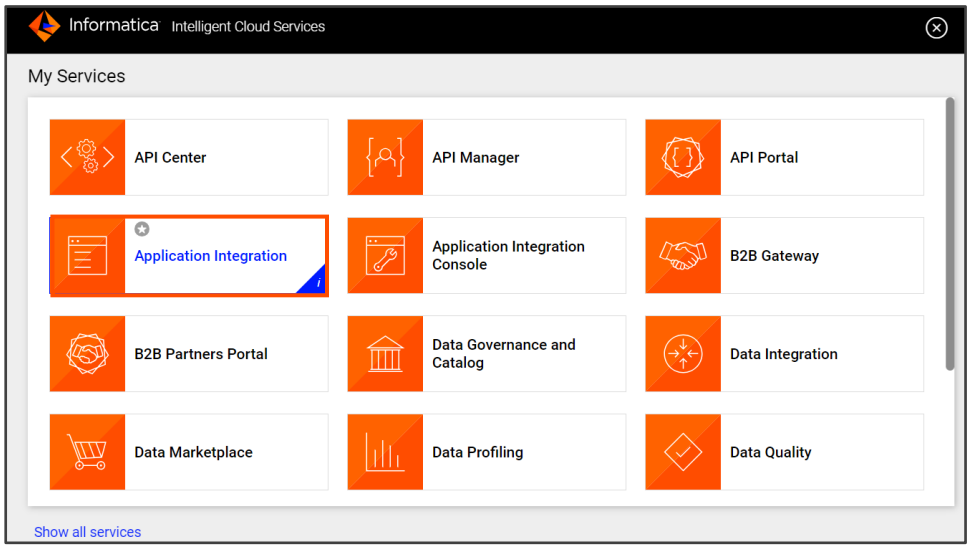
Duration:
15 Minutes

Tasks

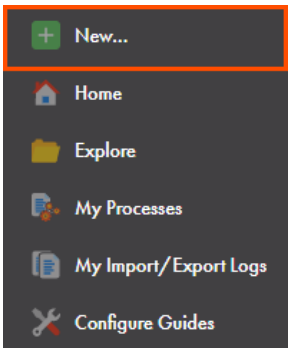
Note: The screen in your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

Import a Swagger File:

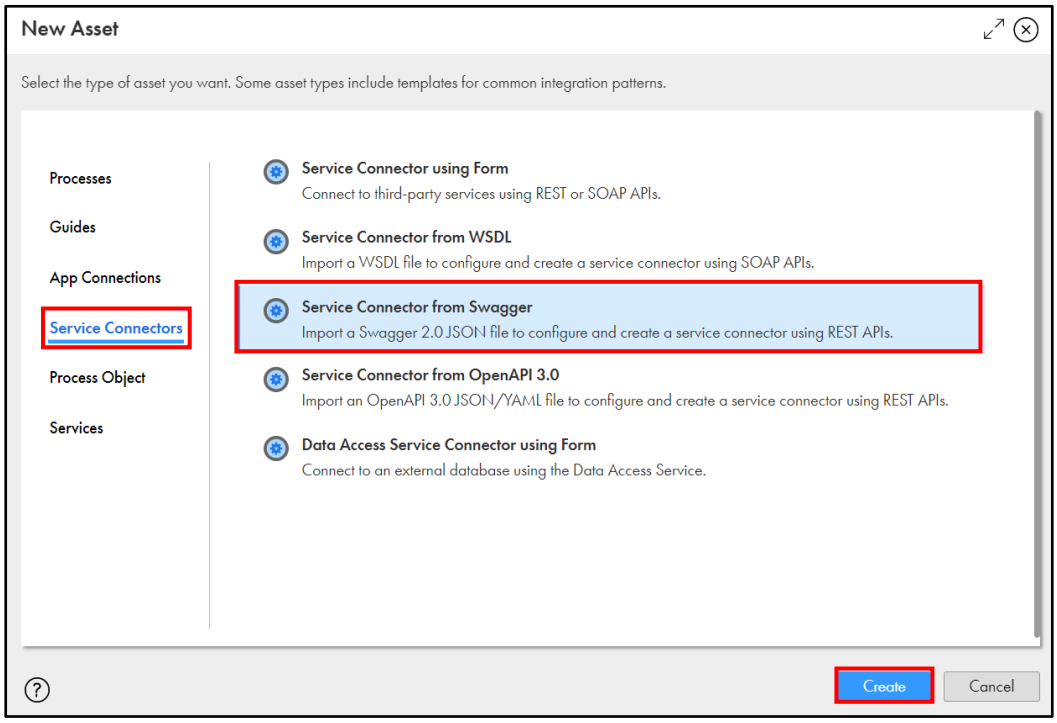
1. Log into IICS and access the **Application Integration** service.



2. To create a new Service Connector, from the navigation pane, click **New**.



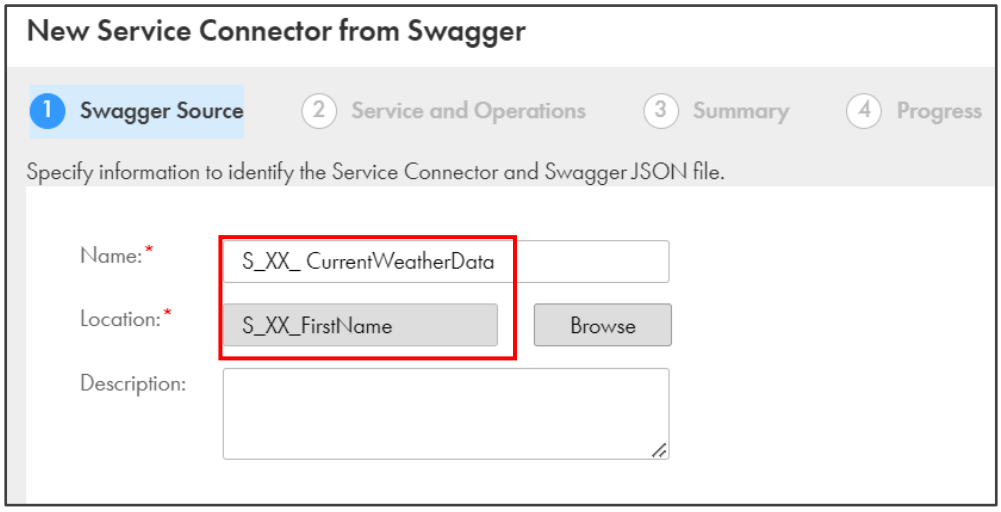
3. From the Service Connectors tab, select **Service Connector from Swagger**.
4. Click **Create**.



Note: A new Service Connector page appears.

5. In the service connector Name field, enter **S_XX_ CurrentWeatherData**.
Note: Here, XX is the unique student number.

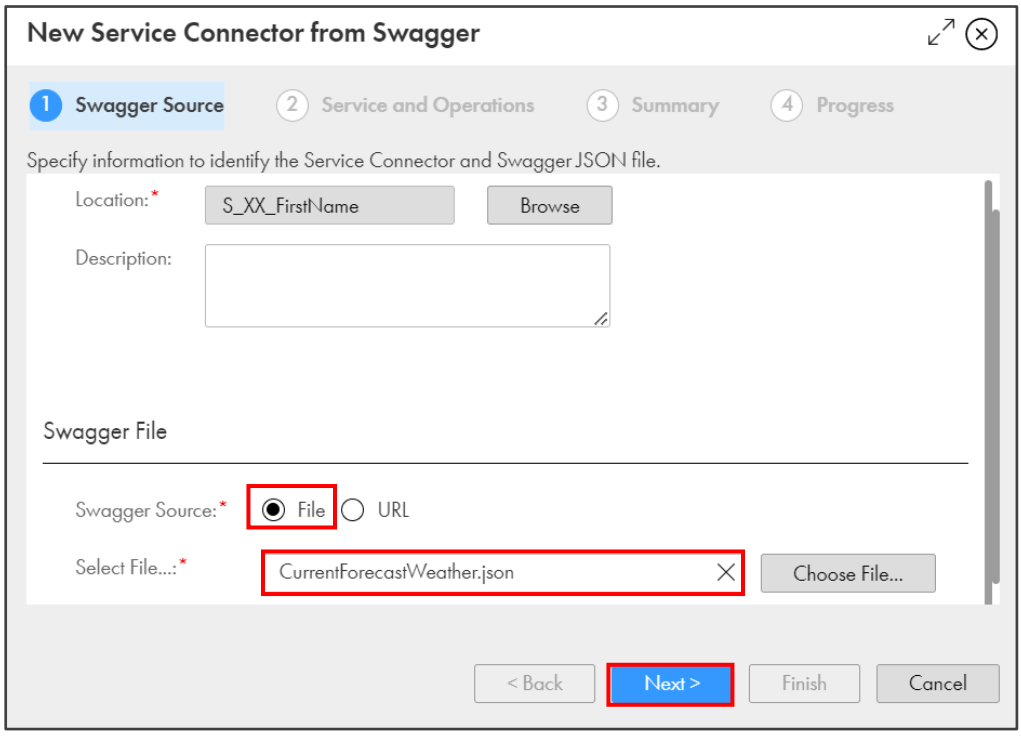
6. In the Location field, select your project folder **S_XX_FirstName**.



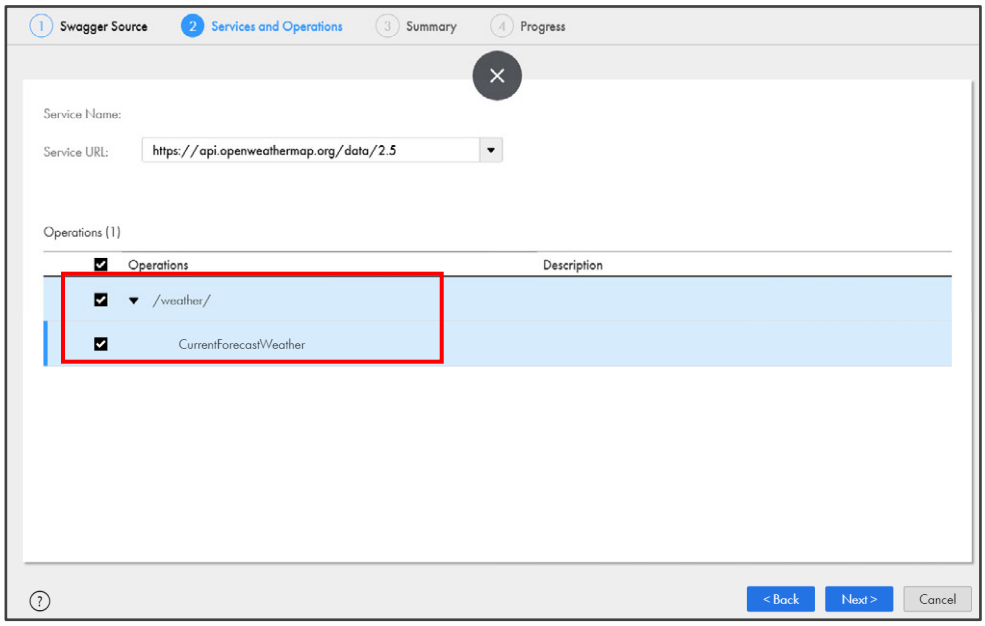
Note: Here, XX is the unique student number.

7. From Swagger Source, select **File**.
8. Select the CurrentForecastWeather.json file.

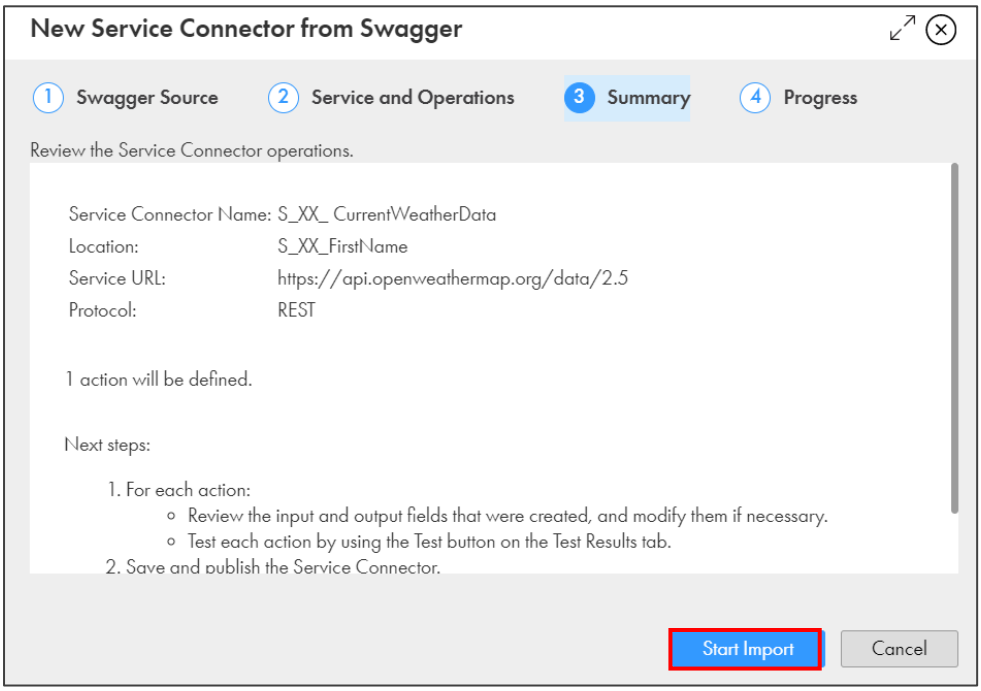
   Note: you have created the CurrentForecastWeather.json file in the previous lab.
9. Click **Next**.



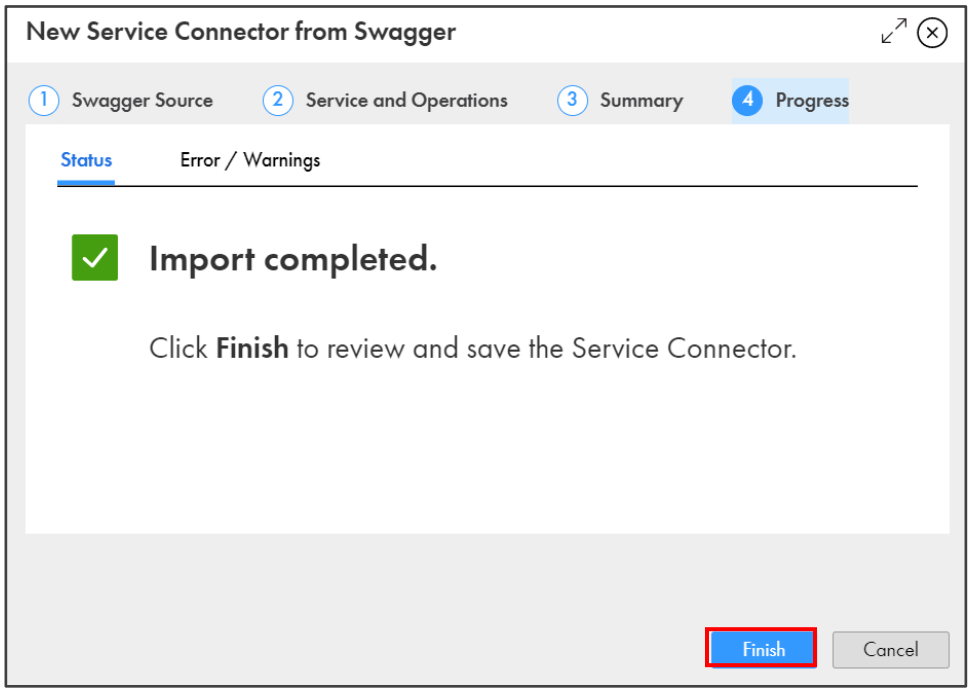
10. Expand **weather** and ensure that the **CurrentForecastWeather** checkbox is selected.



11. Click **Next**.
12. Click **Start Import**.



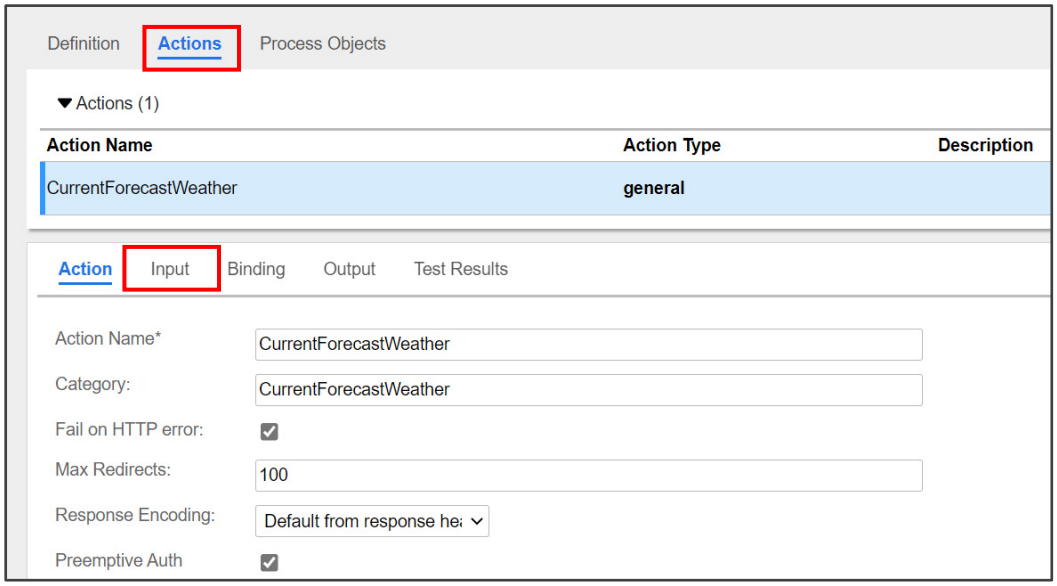
13. Click **Finish**.



14. Click **Save**.
    Note: A message **The Service Connector was saved successfully** appears.

Add Parameters to the Service Connector:

15. Click **Actions** and select the Input tab.
16. From the Action Name, select **CurrentForecastWeather**.
    Note: There is only one action in the Action Name section.



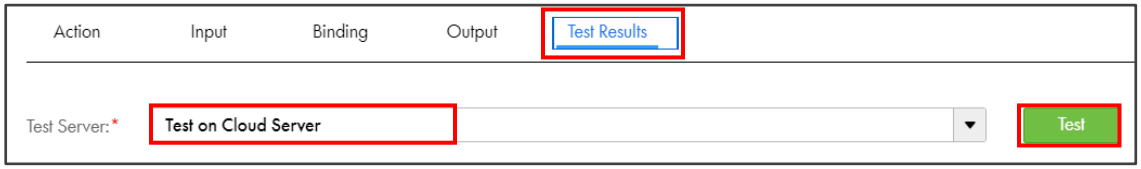
17. In the appid **Test with** field, enter the API Key which you have saved in the notepad file.
18. In the lon **Test with** field, enter **10.99**.

19. In the lat **Test with** field, enter **44.34**.

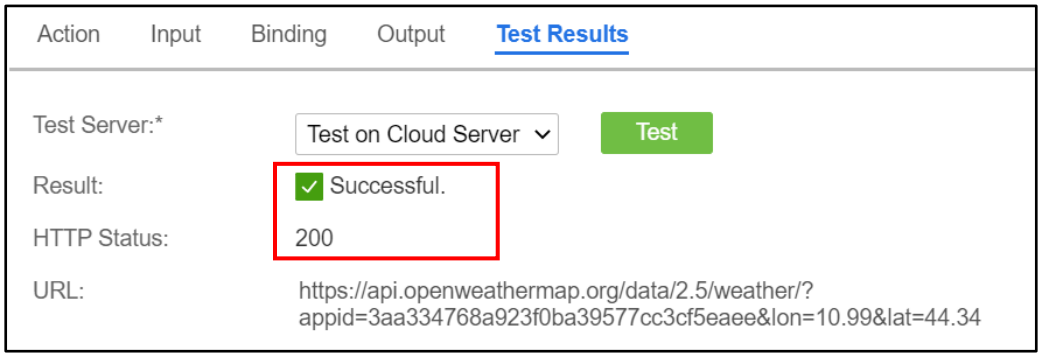
20. Click **Save**.

Test Action:

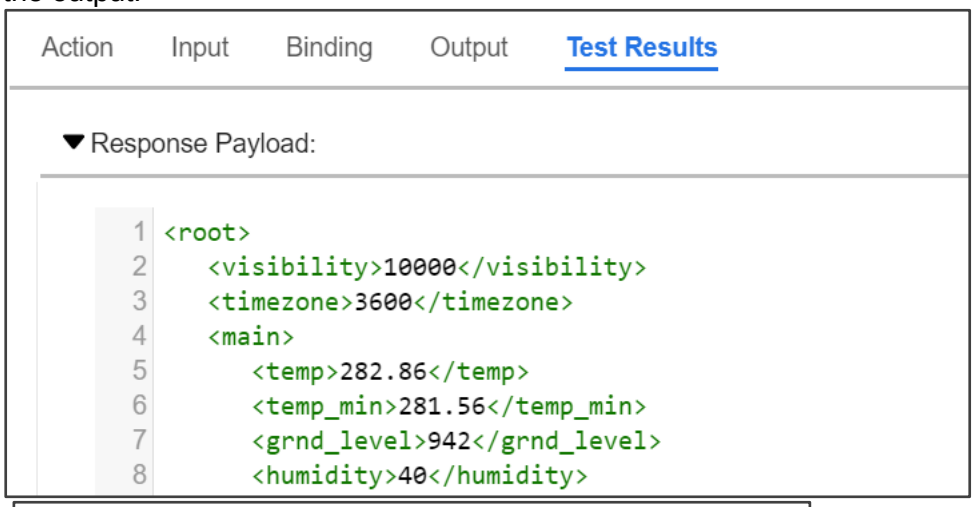
21. Select **Test Results**.
22. From the Test Server drop-down, retain **Test on Cloud Server** and click **Test**.

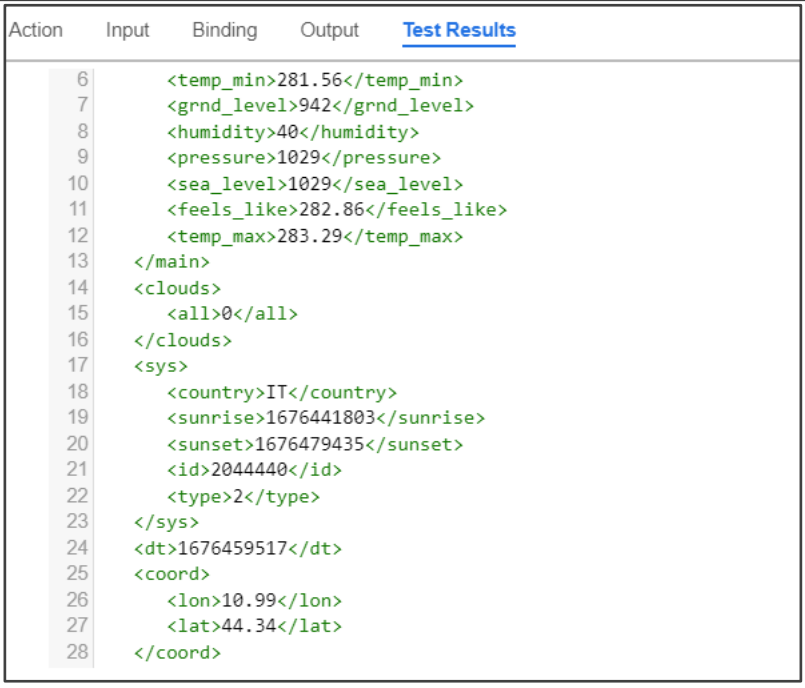


23. In the Test Results tab, verify that the result of the test is **Successful**.
Note: Note that HTTPS Status is 20

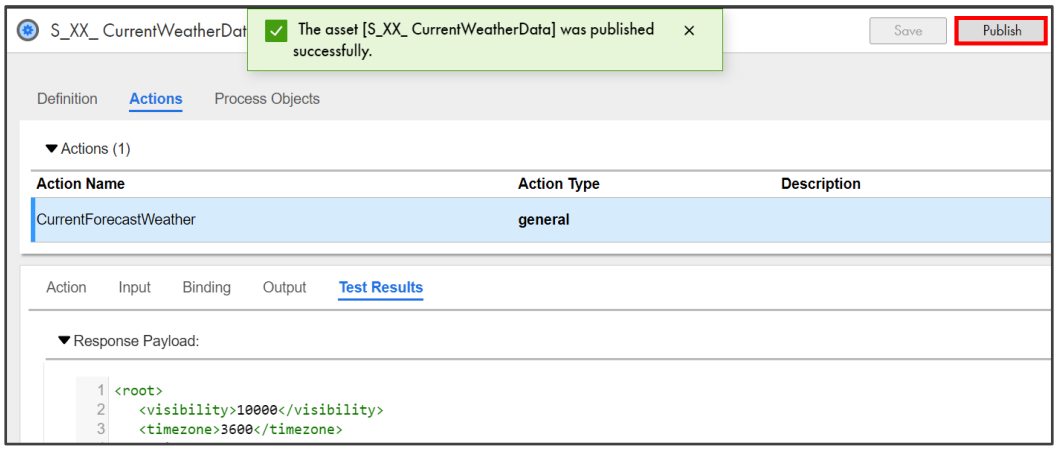


24. From the Output Fields section, scroll down to the **Response Payload** field and observe the output.





25. Click **Publish**.



Note: A message **The asset [S_XX_ CurrentWeatherData] was published successfully** appears.

This concludes the lab.

Module 5: Service Connectors
Lab 5-5: Create a Service Connector from OpenAPI 3.0

Overview:

You can import an OpenAPI 3.0 JSON or YAML file to easily create a service connector with multiple operations. Use the New Service Connector from OpenAPI 3.0 wizard to import an OpenAPI 3.0 file and create a service connector. After you specify the OpenAPI 3.0 file, Application Integration parses the file and loads the operations.

In this lab, you will create a Service Connector from OpenAPI 3.0. To do so, you will use the OpenAPI 3.0 URL that you have created in **Lab 3-1: Create a Basic Process to Display User Input**.

Objectives:

- Obtain the OpenAPI 3.0 URL
- Create a Service Connector from OpenAPI 3.0
- Add Parameter to an existing Action
- Test the Service Connector

Duration:

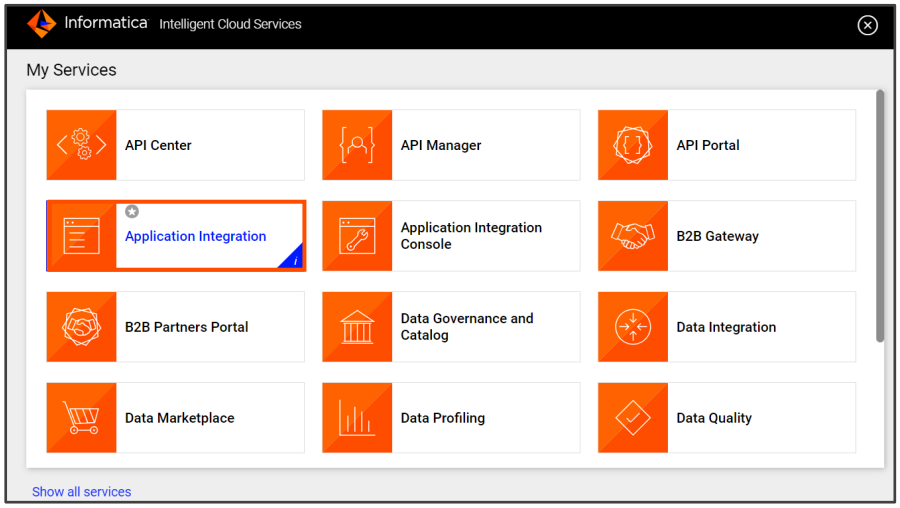
10 Minutes

Tasks

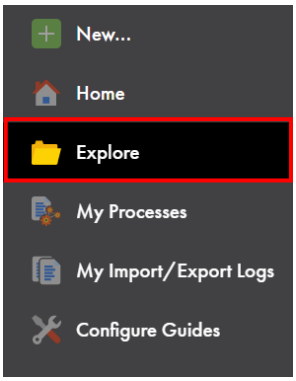
Note: The screen in your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

Obtain the OpenAPI 3.0 URL:

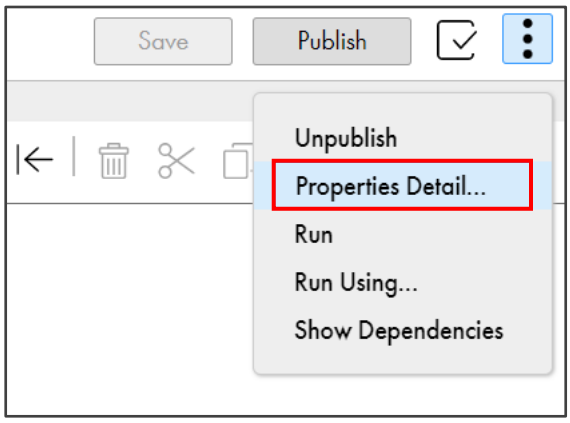
1. Log into IICS and access the **Application Integration** service.



2. Click **Explore**.



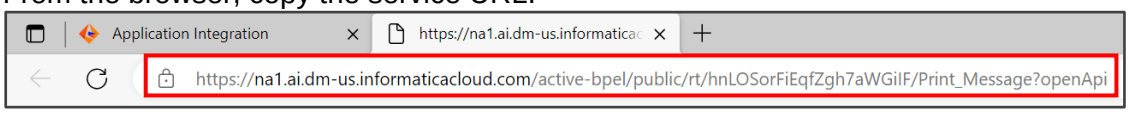
3. Navigate to your project folder and open the **S_XX_Print Input Message** process.
4. From the Actions menu, select **Properties Detail..**.



5. From Service URL, click **View OpenAPI 3.0 File**.

Note: The URL will open in a new browser.

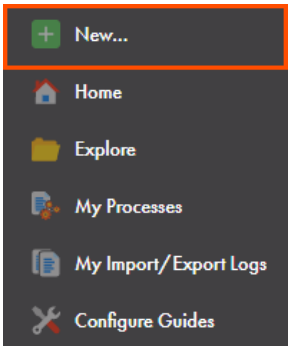
6. From the browser, copy the service URL.



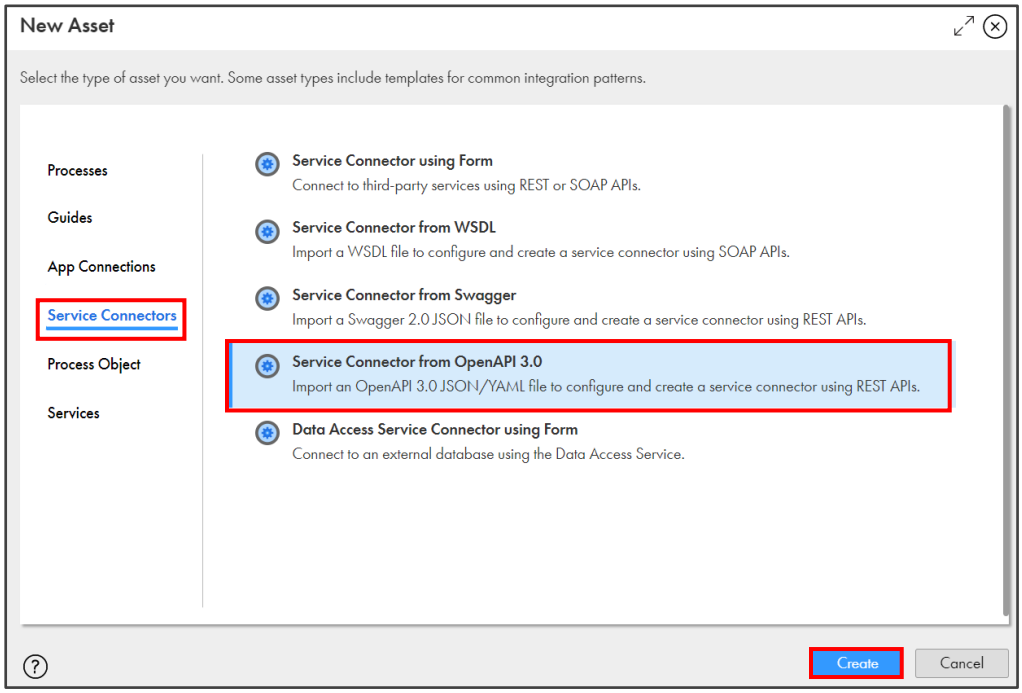
7. Open a notepad file and paste the URL.

Create a Service Connector from OpenAPI 3.0

8. Go back to the IICS Application Integration service page and from the navigation pane, click **New**.



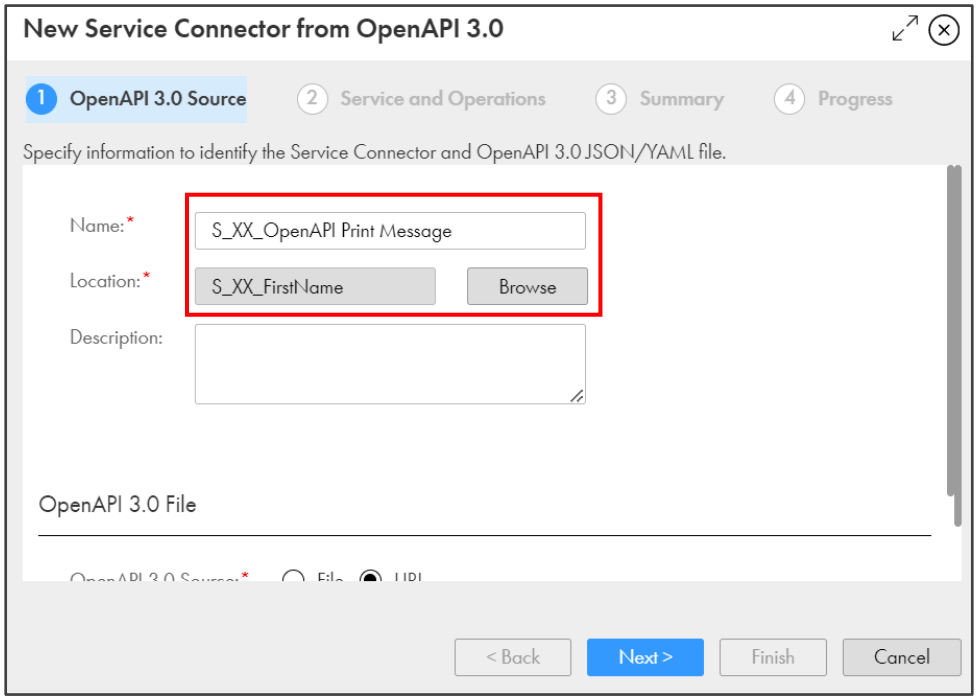
9. From the Service Connectors tab, select **Service Connector from OpenAPI 3.0**.
10. Click **Create**.



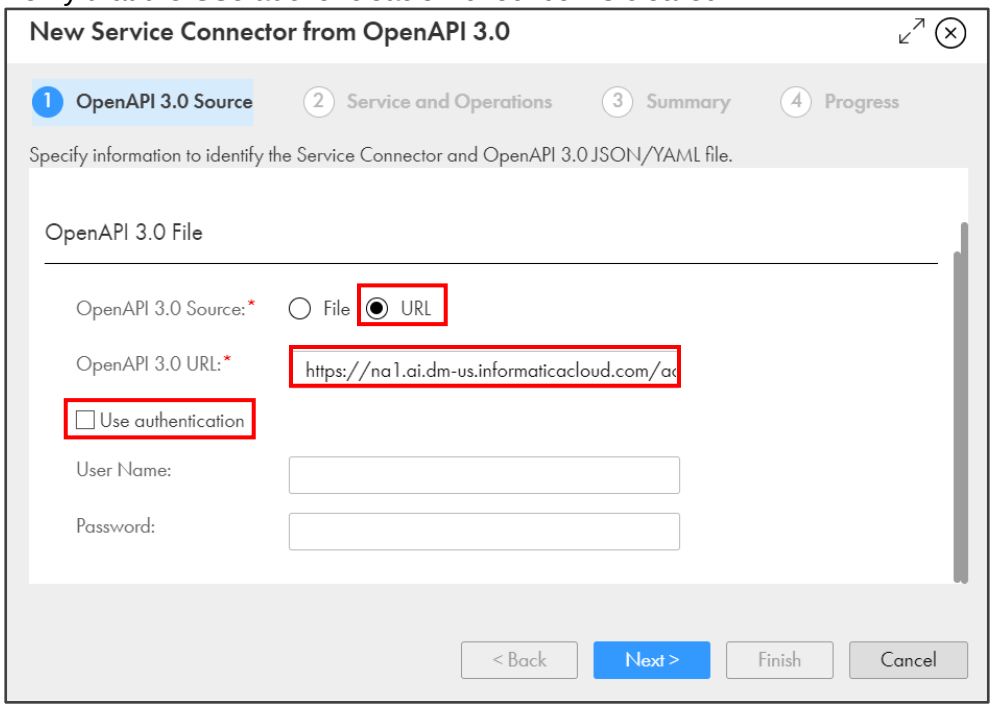
Note: A new Service Connector page appears.

11. In the service connector Name field, enter **S_XX_OpenAPI Print Message**.
Note: Here, XX is the unique student number.

12. In the Location field, select your project folder **S_XX_FirstName**.

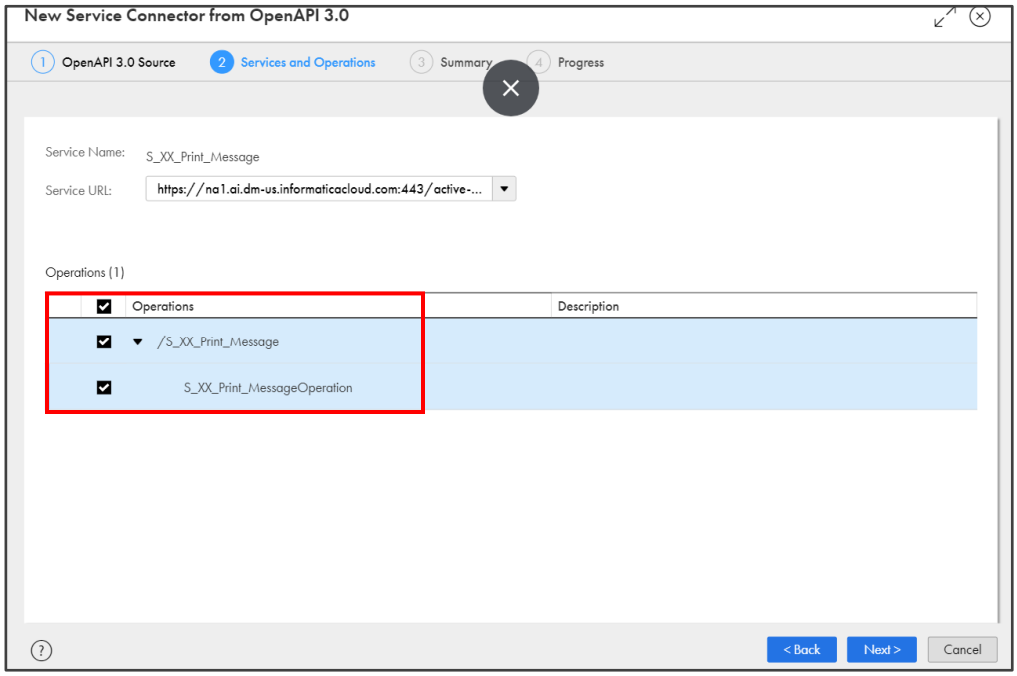


13. From OpenAPI 3.0 Source, select **URL**.
14. Copy the URL from the notepad and paste it in the OpenAPI 3.0 URL field.
15. Verify that the **Use authentication** checkbox is cleared.

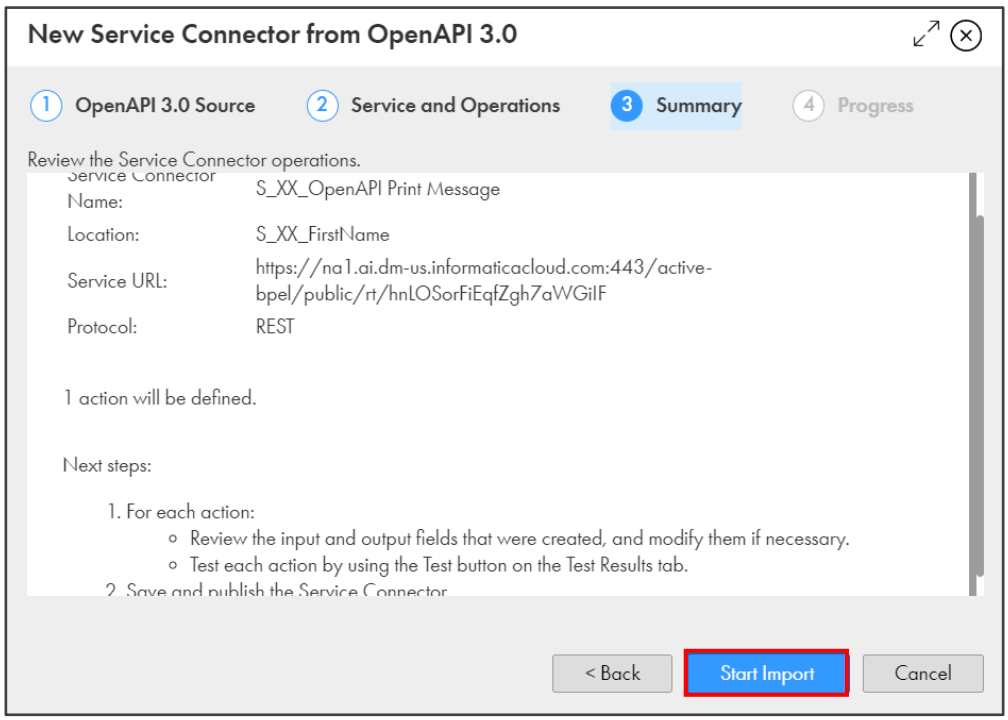


16. Click **Next**.

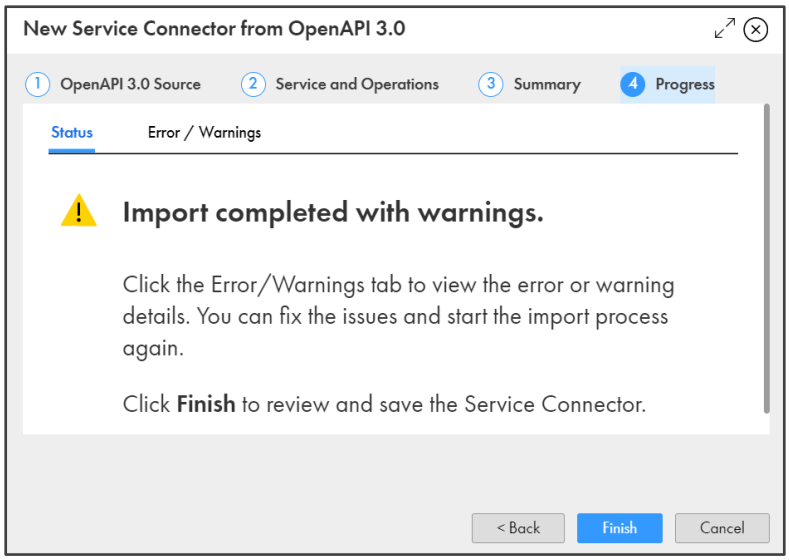
17. Expand **Print_Message** and verify that **Print_Message** and **Print_MessageOperation** checkboxes are selected.



18. Click **Next**.
19. Click **Start Import**.



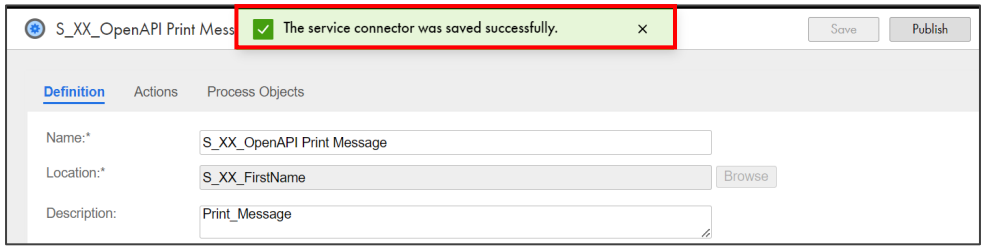
Note: You can see a message "**Import Completed with warnings**". You can ignore this message as it will not stop you to move further.



Note: If an error occurs during import, the wizard displays the error under Error/Warnings on the Progress tab. You can take corrective actions and import the OpenAPI 3.0 file again.

20. Click **Finish**.
21. Click **Save**.

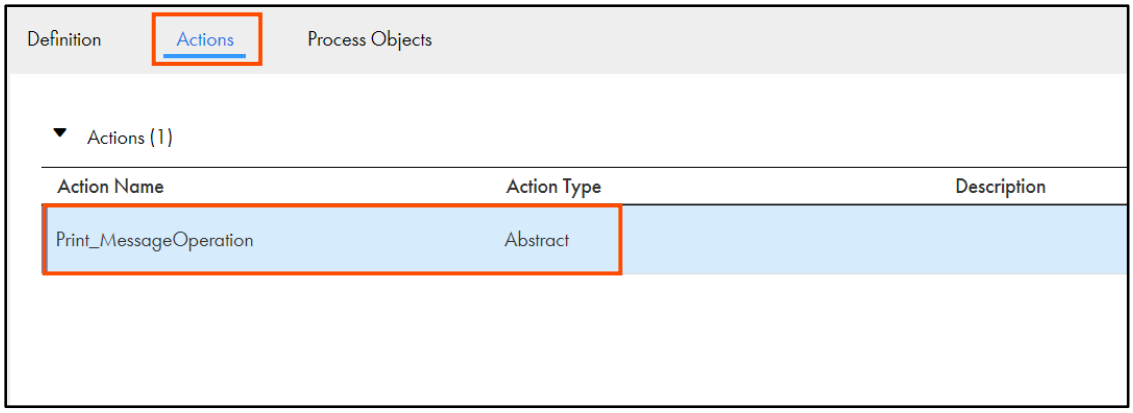
    Note: A message **The service connector was saved successfully** appears.



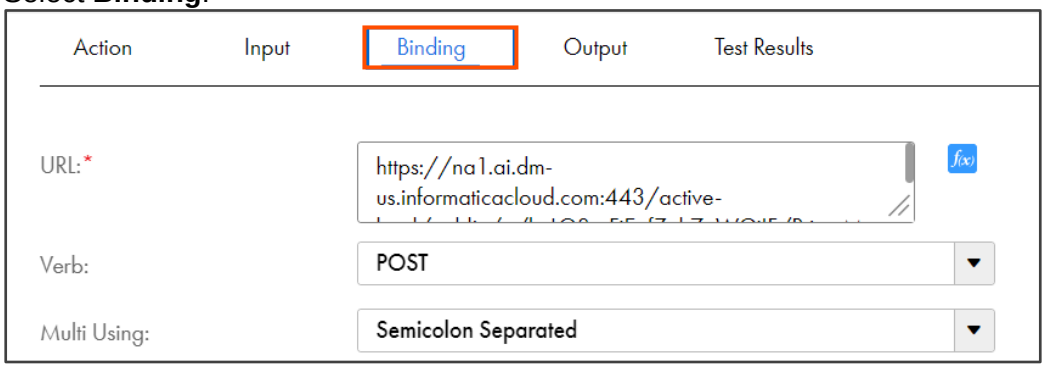
Add Parameters to the Action:

1. Click the **Actions** tab.
2. Make sure that the **Print_MessageOperation** Action is selected.

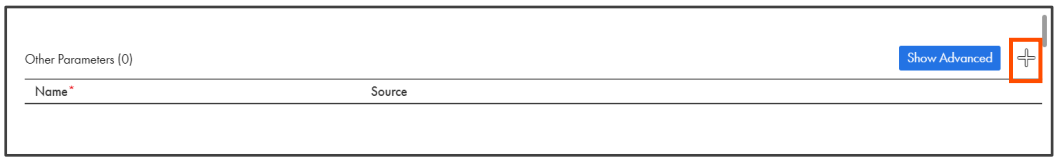
   Note: You can see only one action in the Action Name section.



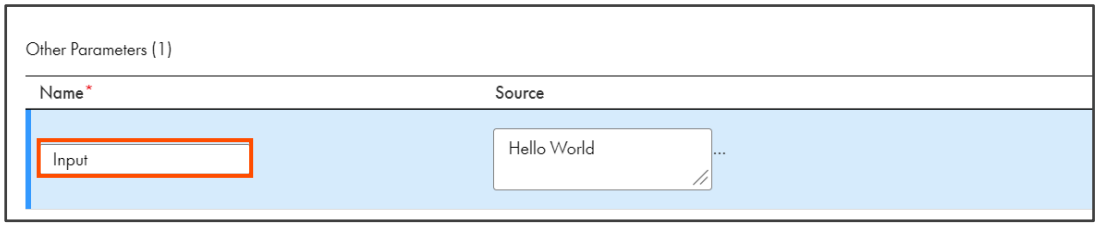
3. Select **Binding**.



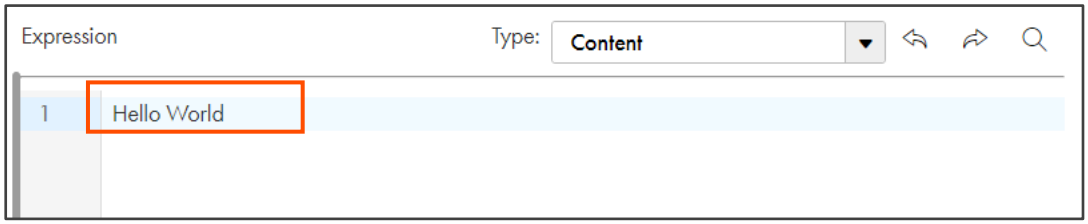
4. Scroll down and from the Other Parameters section, click the '**+**' icon to add a parameter.



5. In the Name field, enter **Input**.



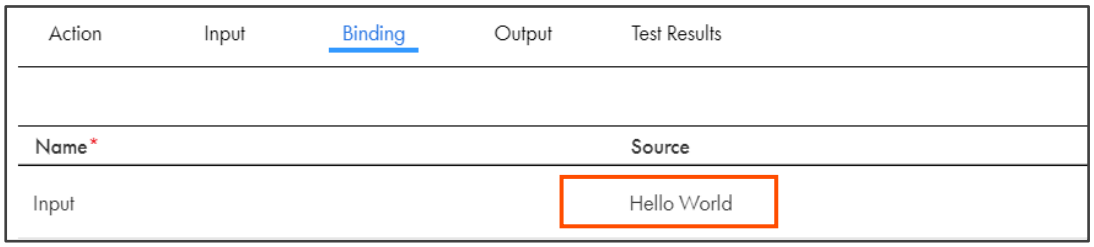
6. In the Source field, click .
7. In the Expression field, enter **Hello World**.



Note: In the Source field, make sure to copy and paste the text that you want to enter. Sometimes, it does not accept the entire text that you want to type.

8. Click **OK**.

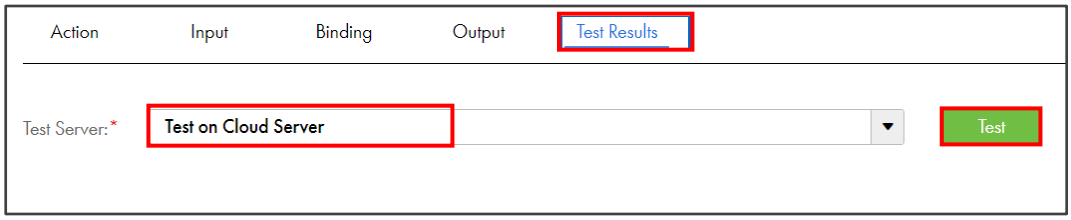
Note: In the source field, make sure that the entire text has been entered.



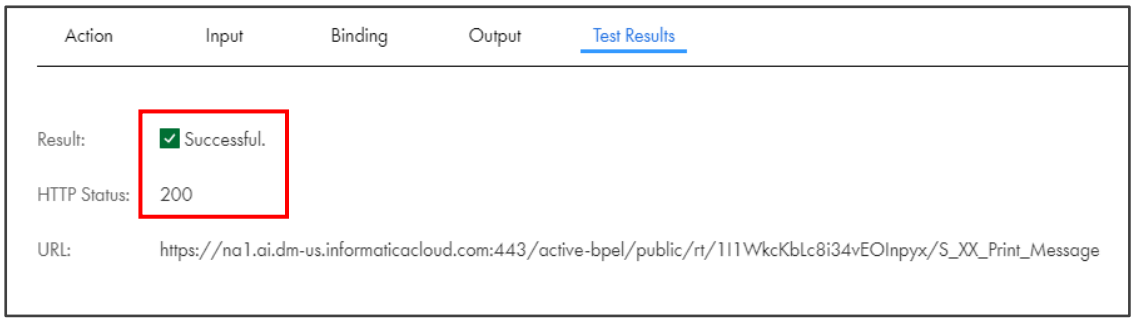
9. Click **Save**.

Test Action:

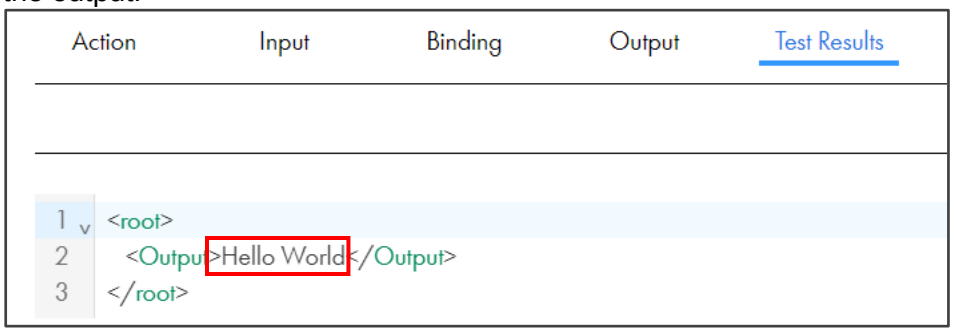
1. Select **Test Results**.
2. From the Test Server drop-down, retain **Test on Cloud Server** and click **Test**.



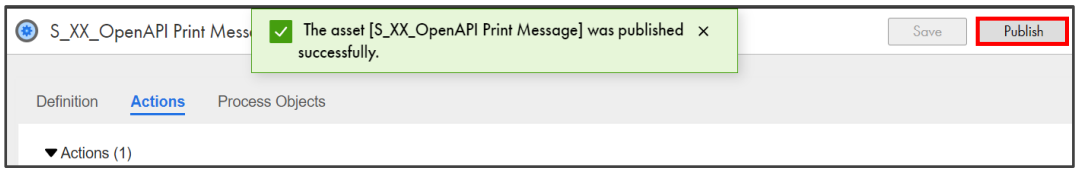
3. In the Test Results tab, verify that the result of the test is **Successful**. Observe that the HTTPS Status is **200**.



4. From the Output Fields section, scroll down to the **Response Payload** field and observe the output.



5. Click **Publish**.



A message "**The asset [S_XX_OpenAPI Print Message] was published successfully**" appears.

This concludes the lab.

Module 6: Data Access Service Connectors
Lab 6-1: Create a Data Access Service Connector Using Form

Overview:
You can create a Data Access Service Connector on a Secure Agent to access data directly from a database. The databases include IBM DB2, Microsoft SQL Server, MySQL, Oracle, and

You can create a Data Access Service Connector by defining the database properties and specifying variables to interact directly with the database. Using this service, you can create an invoke activity that executes SQL CRUD operations, such as select, insert, update, or delete on a specified data source, and receives responses based on the columns in the database table, expression, result set, and attachments from the data source.

In this lab, you will create a Data Access Service Connector to fetch the employee records from the database.

Objectives:
- Create a Data Access Service Connector
- Define properties and actions for the Data Access Service Connector
- Test the Data Access Service Connector

Duration:
15 Minutes

Note: The screen on your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

Tasks:

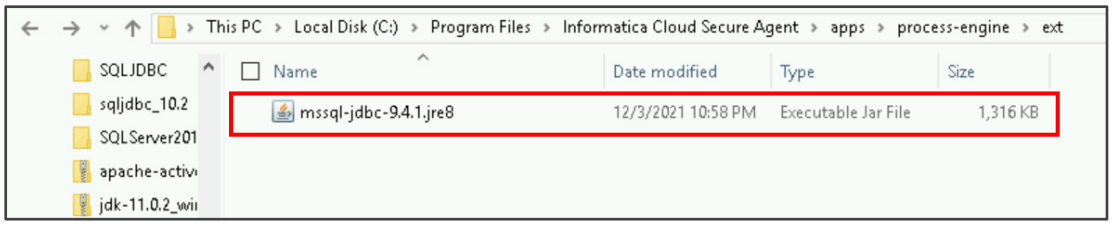
IMPORTANT: In this lab, some steps need to be performed by the Instructor on the Server Agent (Please check **Steps 1 to 3**).

STEPS FOR THE INSTRUCTOR

Before creating a Data Access Service Connector, you must copy the required JDBC JAR files in the Secure Agent Installation directory and restart the Secure Agent.

Copy the JDBC JAR File

1. On the desktop, from the **CAI Lab Prep Files** folder, copy the **mssql-jdbc-9.4.1.jre8** file.
2. Paste the jar file in the following location:
   C:\Program Files\Informatica Cloud Secure Agent\apps\process-engine\ext



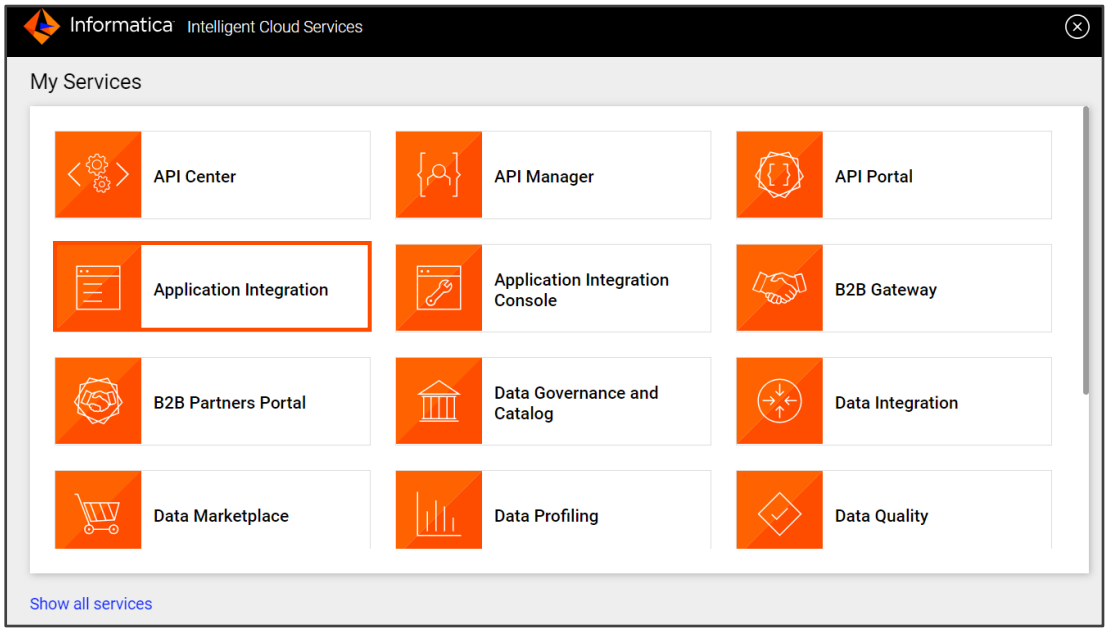
Note: C:\Program Files\Informatica Cloud Secure Agent\apps\process-engine\ext is the Secure Agent Installation directory.

3. After you paste the jar file in the Secure Agent Installation directory, you must restart the Secure Agent and wait for it to come up and running.

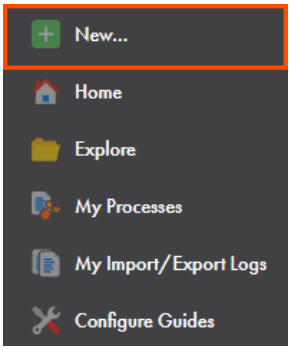
STEPS FOR THE LEARNERS

Create a Data Access Service Connector:

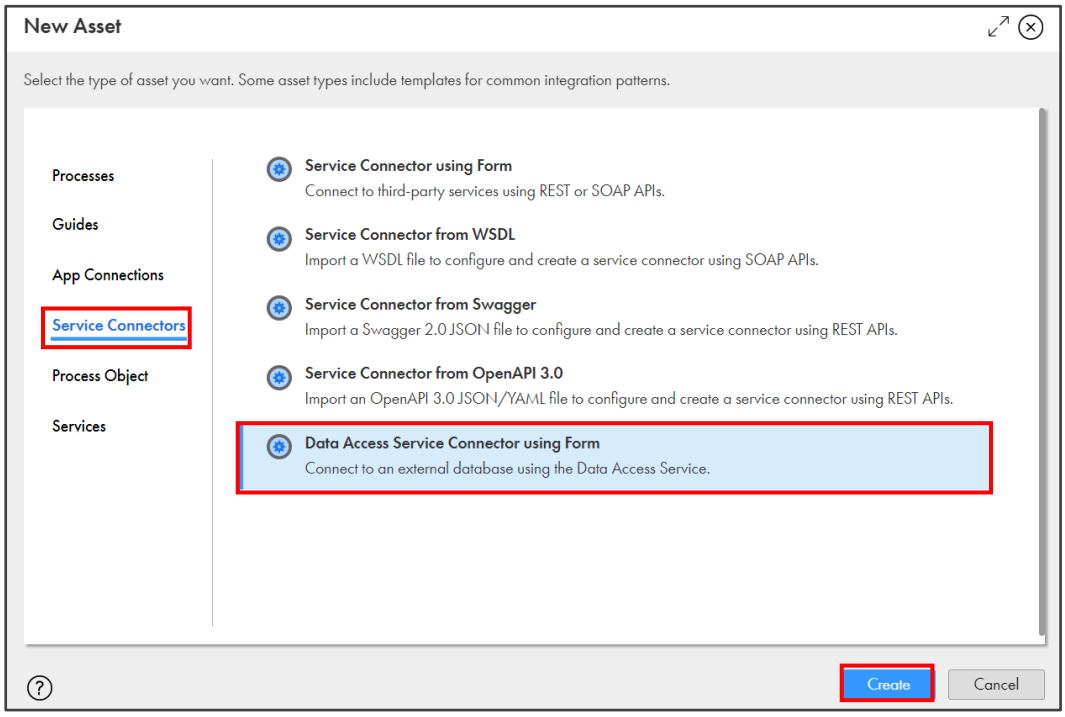
4. Log into IICS and access the **Application Integration** service.



5. To create a new Service Connector, from the navigation pane, click **New**.



6. From the Service Connectors tab, select **Data Access Service Connector using Form**.
7. Click **Create**.



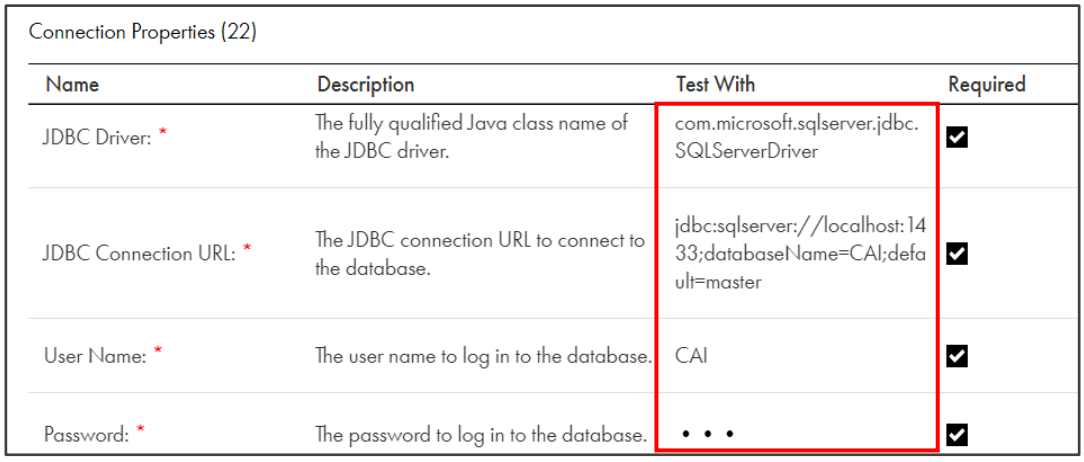
Note: A new Service Connector page appears.

Define Properties

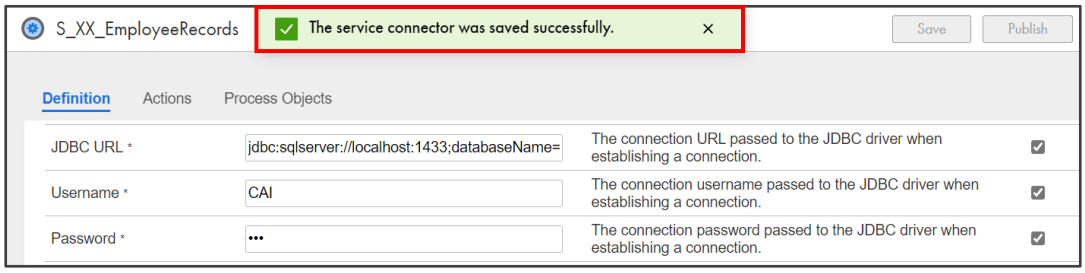
8. In the service connector Name field, enter **S_XX_EmployeeRecords**.
   Note: Here, XX is the unique student number.
9. In the Location field, select your project folder **S_XX_FirstName**.

Note: Here, XX is the unique student number.

10. In the **Test With** field of the JDBC Driver row, enter the following: **com.microsoft.sqlserver.jdbc.SQLServerDriver**
11. In the **Test With** field of the JDBC URL row, enter the following URL: **jdbc:sqlserver://localhost:1433;databaseName=CAI;default=master**
12. In the **Test With** field of the Username and Password rows, enter **CAI**.

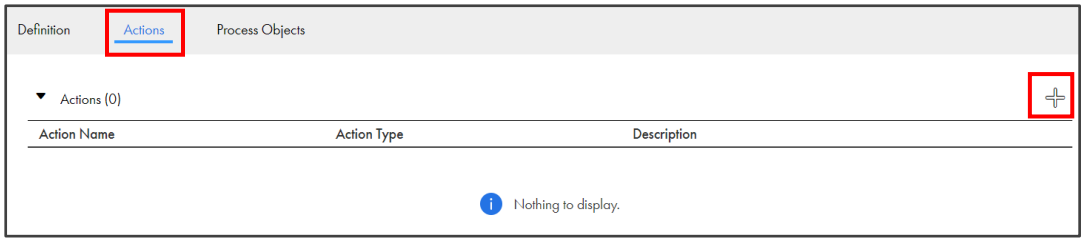


13. Click **Save**.
    Note: A message **The service connector was saved successfully** appears.

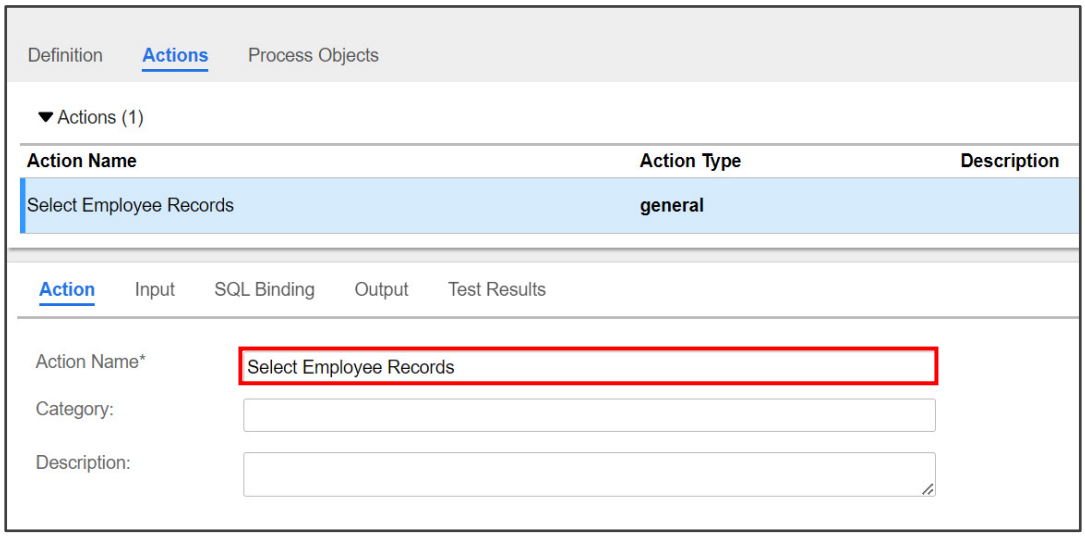


Define Actions

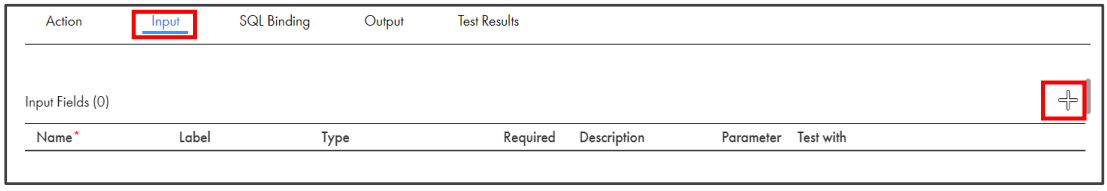
1. Click the **Actions** tab.
2. Click to add a new action.



3. In the Action Name field, enter **Select Employee Records**.

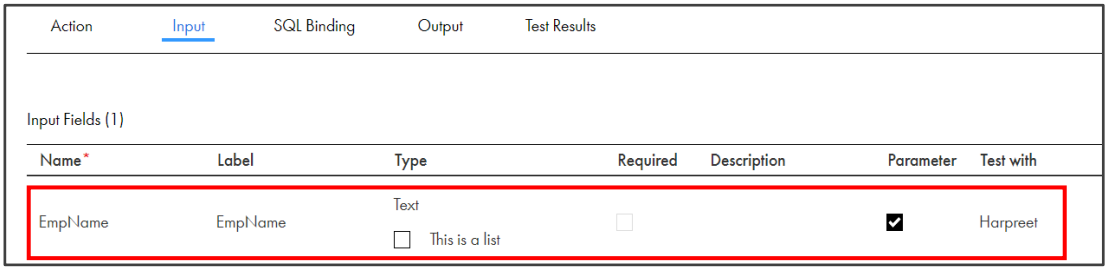


4. Click the **Input** tab and click the '**+**' icon to add the input fields.

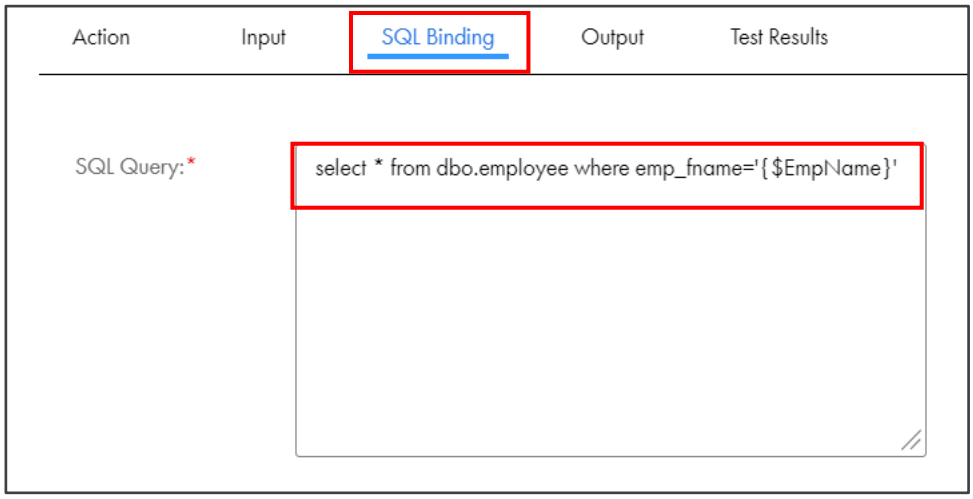


5. In the **Input Fields** section, enter the following values:

Property	Value
Name	EmpName
Label	EmpName
Type	Text
Test with	Harpreet

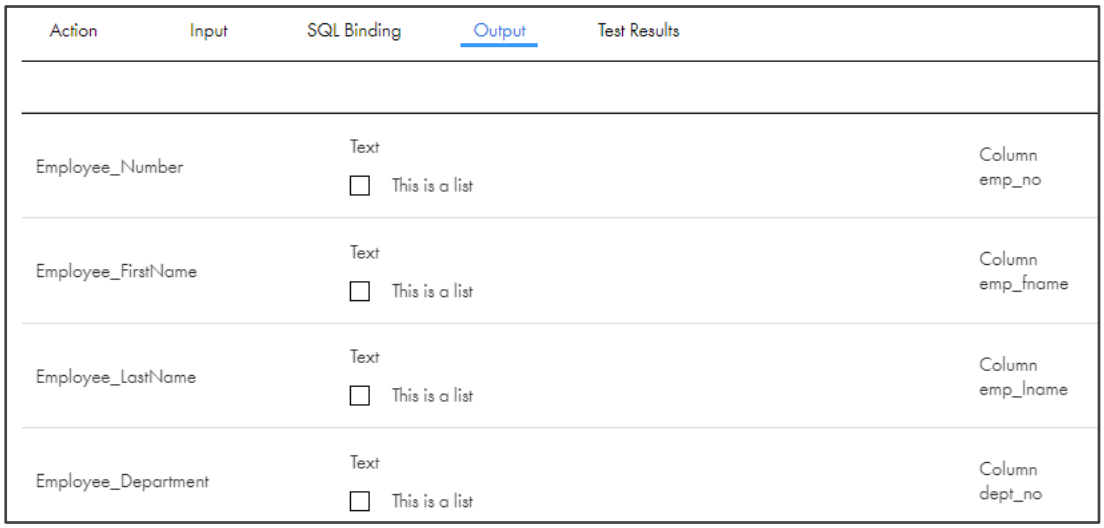


6. Click the **SQL Binding** tab and in the SQL field, enter the following query:
   select * from dbo.employee where emp_fname='{$EmpName}'
   OR
   From the CAI Lab Prep Files_Learners folder, open the notepad file named **08_LabGuide_CAI_Create Data Access Service Connector Using Form_6-1**. Copy the command mentioned in the notepad and paste it in the SQL field.



7. Click the **Output** tab and click the '**+**' icon to add the output fields.
8. In the **Output Fields** section, enter the following values:

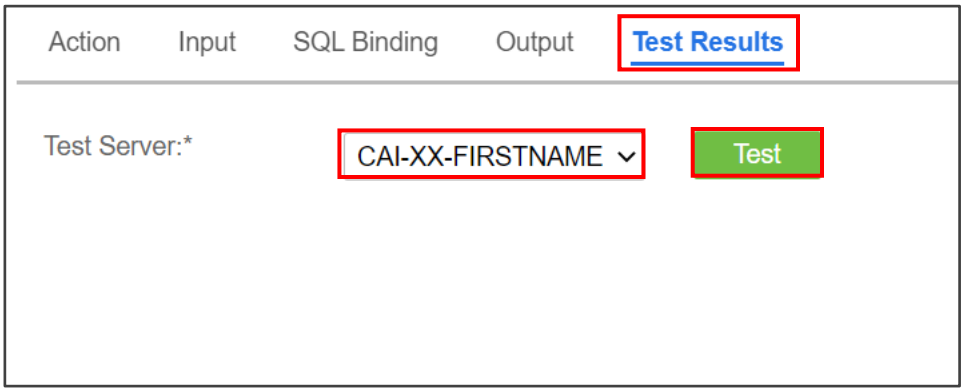
Name	Type	Get From	Value in Get From
Employee_Number	Text	Column	emp_no
Employee_FirstName	Text	Column	emp_fname
Employee_LastName	Text	Column	emp_lname
Employee_Department	Text	Column	dept_no



9. Click **Save**.

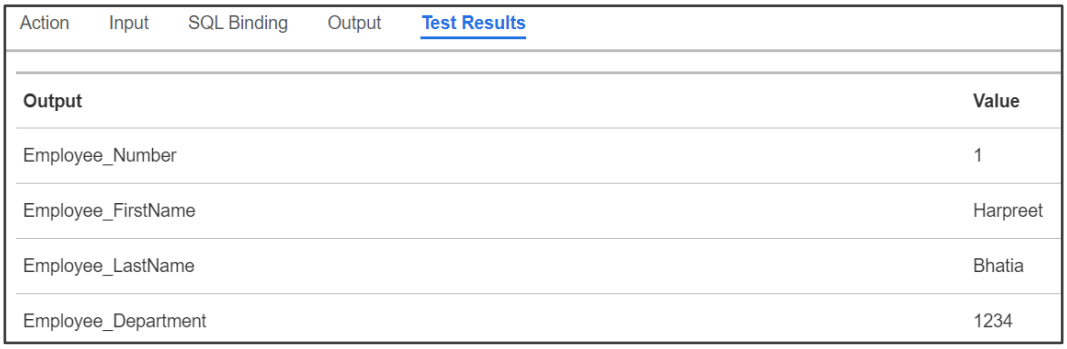
Test Action:

10. Select **Test Results**.
11. From the Test Server drop-down, select your Secure Agent group and click **Test**.

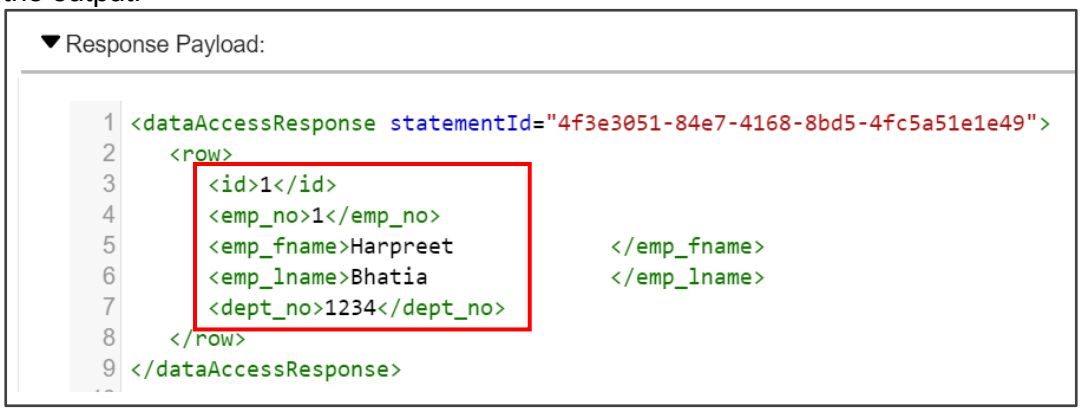


12. In the Output Fields section, you should see the following values:

Output	Value
Employee_Number	1
Employee_FirstName	Harpreet
Employee_LastName	Bhatia
Employee_Department	1234



13. From the Output Fields section, scroll down to the **Response Payload** field and observe the output.



14. Click **Publish**.

Note: A message **The asset was published successfully** appears.

This concludes the lab.

Module 6: Data Access Service Connectors
Lab 6-2: Create a Currency Exchange Data Access Service Connector

Overview:
You can create a Data Access Service Connector on a Secure Agent to access data directly from a database. The databases include IBM DB2, Microsoft SQL Server, MySQL, Oracle, and

You can create a Data Access Service Connector by defining the database properties and specifying variables to interact directly with the database. Using this service, you can create an invoke activity that executes SQL CRUD operations, such as select, insert, update, or delete on a specified data source, and receives responses based on the columns in the database table, expression, result set, and attachments from the data source.

In this lab, you will create a Data Access Service Connector for currency exchange and perform various actions on the database table.

Objectives:
- Create a Data Access Service Connector
- Define properties and actions for the Data Access Service Connector
- Test the Data Access Service Connector

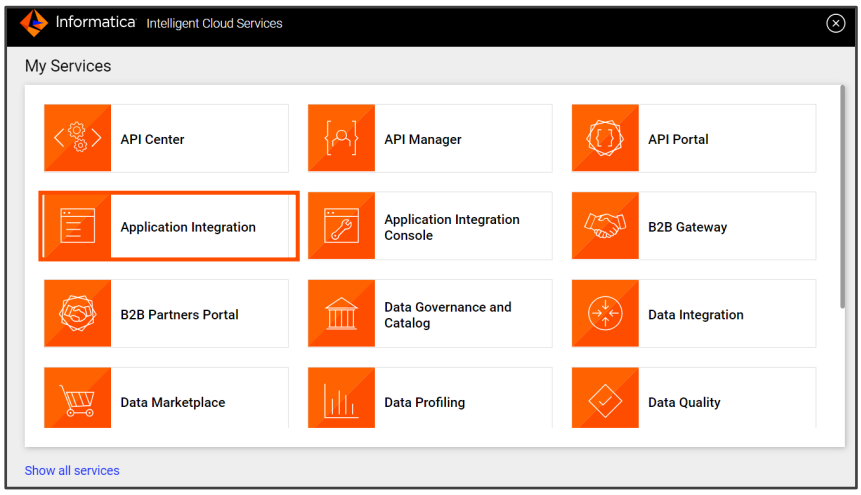
Duration:
30 Minutes

Note: The screen on your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

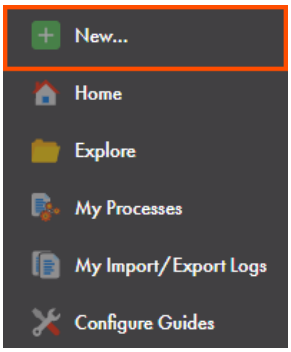
Tasks:

Create a Data Access Service Connector:

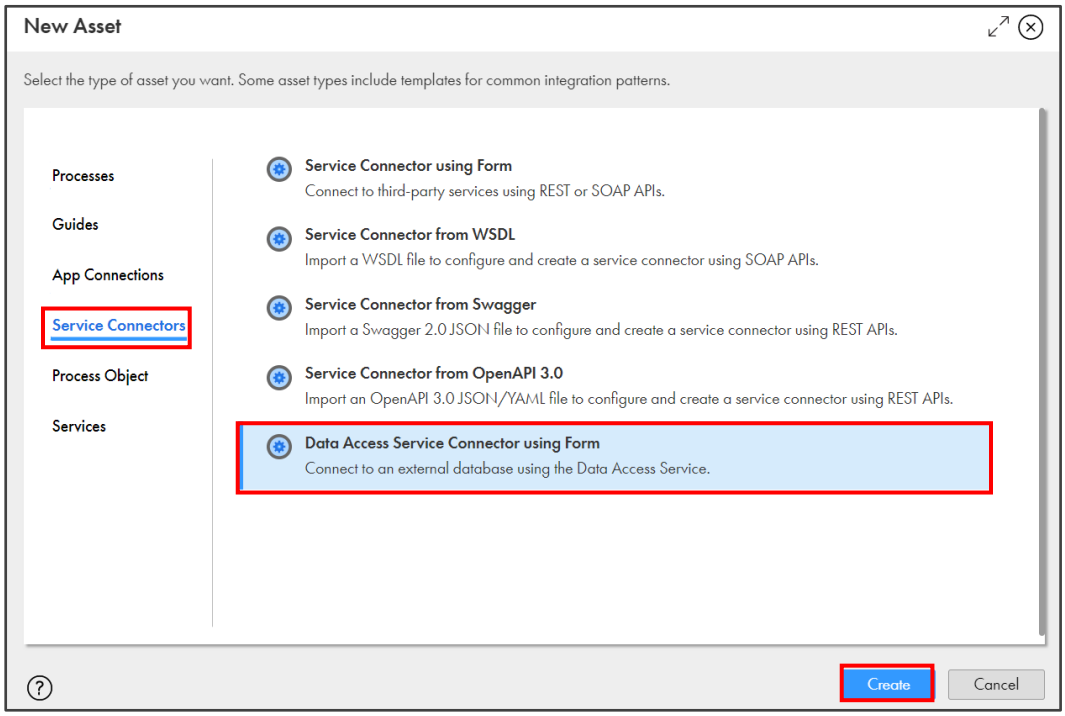
1. Log into IICS and access the **Application Integration** service.



2. To create a new Service Connector, from the navigation pane, click **New**.



3. From the Service Connectors tab, select **Data Access Service Connector using Form**.
4. Click **Create**.

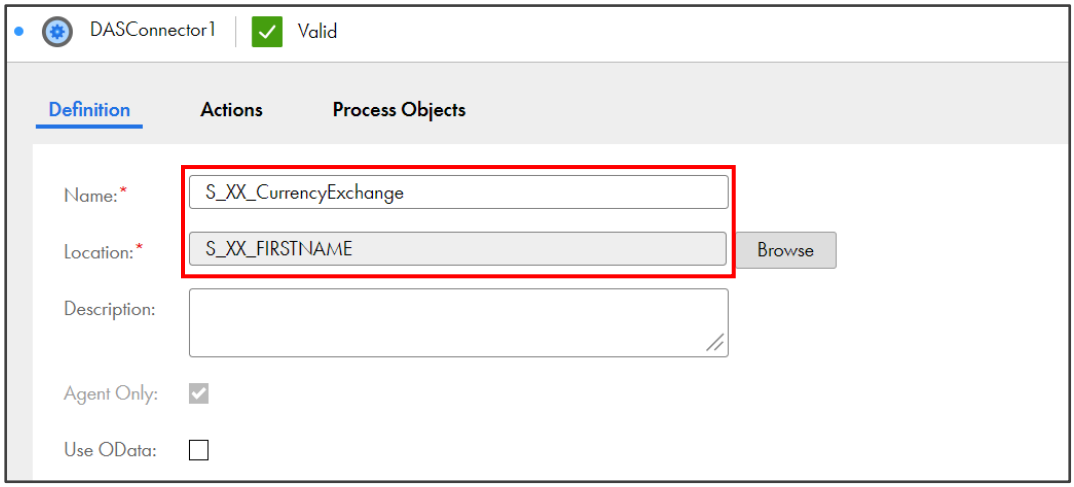


Note: A new Service Connector page appears.

Define Properties

5. In the service connector Name field, enter **S_XX_CurrencyExchange**.
   Note: Here, XX is the unique student number.

6. In the Location field, select your project folder **S_XX_FirstName**.



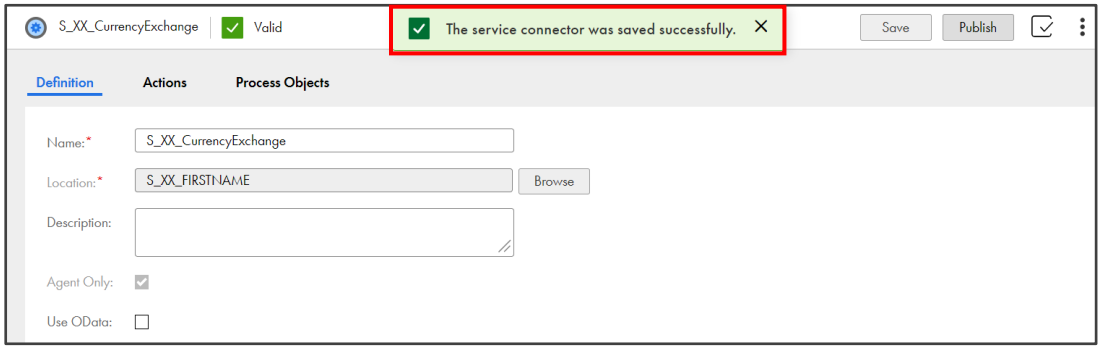
Note: Here, XX is the unique student number.

7. Under Connection Properties, in the **Test With** field of the JDBC Driver row, enter the following: **com.microsoft.sqlserver.jdbc.SQLServerDriver**
8. In the **Test With** field of the JDBC Connection URL row, enter the following URL: **jdbc:sqlserver://localhost:1433;databaseName=CAI;default=master**
9. In the **Test With** field of the Username and Password rows, enter **CAI**.

Connection Properties (22)

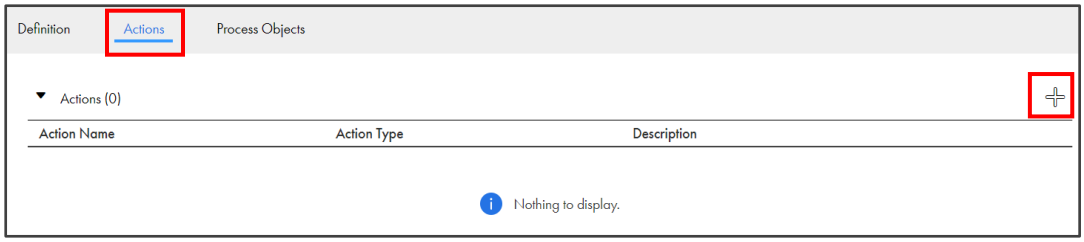
Name	Description	Test With	Required
JDBC Driver: *	The fully qualified Java class name of the JDBC driver.	com.microsoft.sqlserver.jdbc.SQLServerDriver	☑
JDBC Connection URL: *	The JDBC connection URL to connect to the database.	jdbc:sqlserver://localhost:1433;databaseName=CAI;default=master	☑
User Name: *	The user name to log in to the database.	CAI	☑
Password: *	The password to log in to the database.	• • •	☑

10. Click **Save**.
    Note: A message **The service connector was saved successfully** appears.

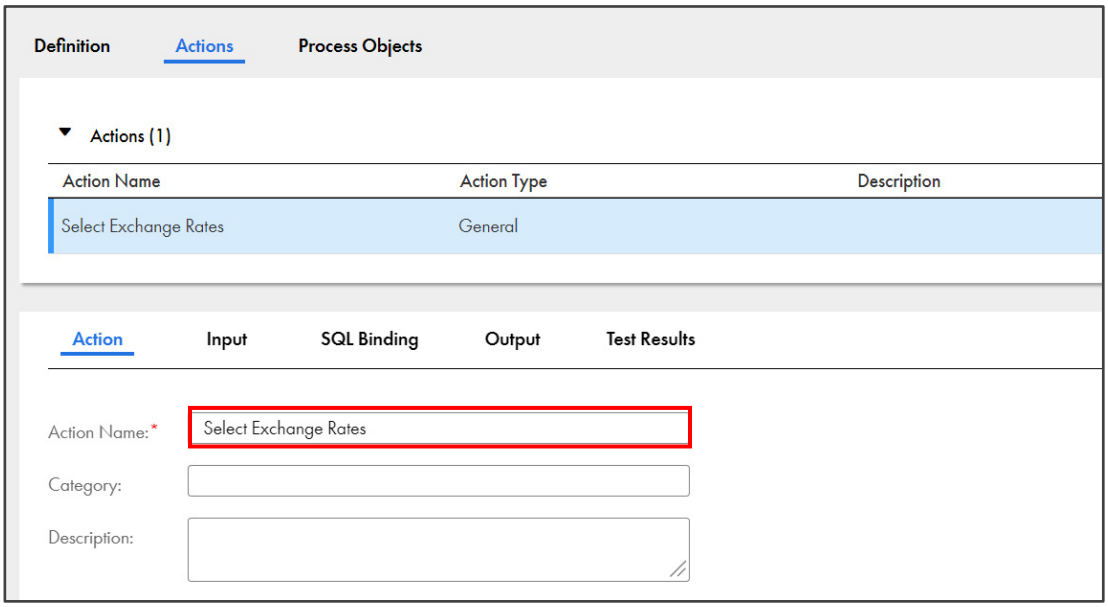


Define an Action – Select Exchange Rates

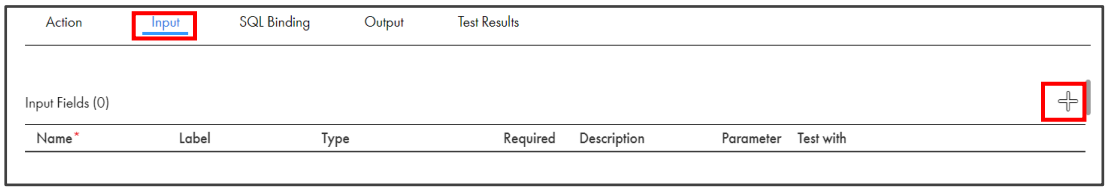
11. Click the **Actions** tab.
12. Click to add a new action.



13. In the Action Name field, enter **Select Exchange Rates**.

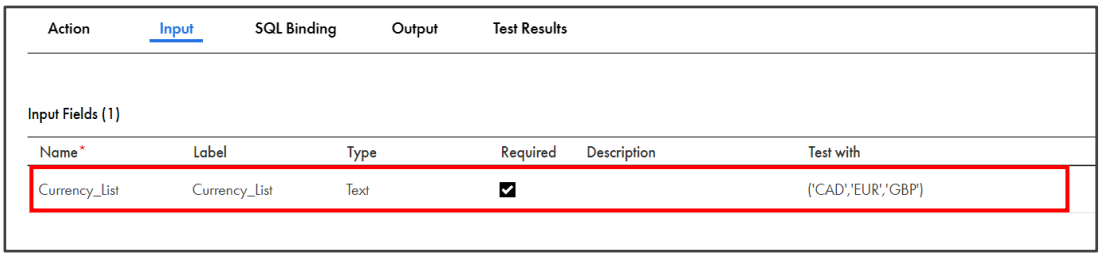


14. Click the **Input** tab and click the '**+**' icon to add the input fields.



15. In the **Input Fields** section, enter the following values:

Property	Value
Name	Currency_List
Label	Currency_List
Type	Text
Required	Checked
Test with	('CAD','EUR','GBP')

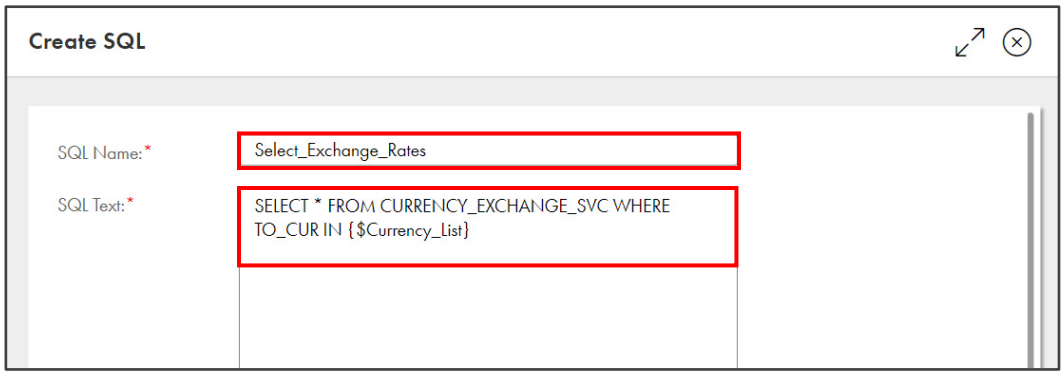


16. Click the **SQL Binding** tab and click .
17. In the SQL Name field, enter **Select_Exchange_Rates**.
18. In the SQL Text field, enter the following query:

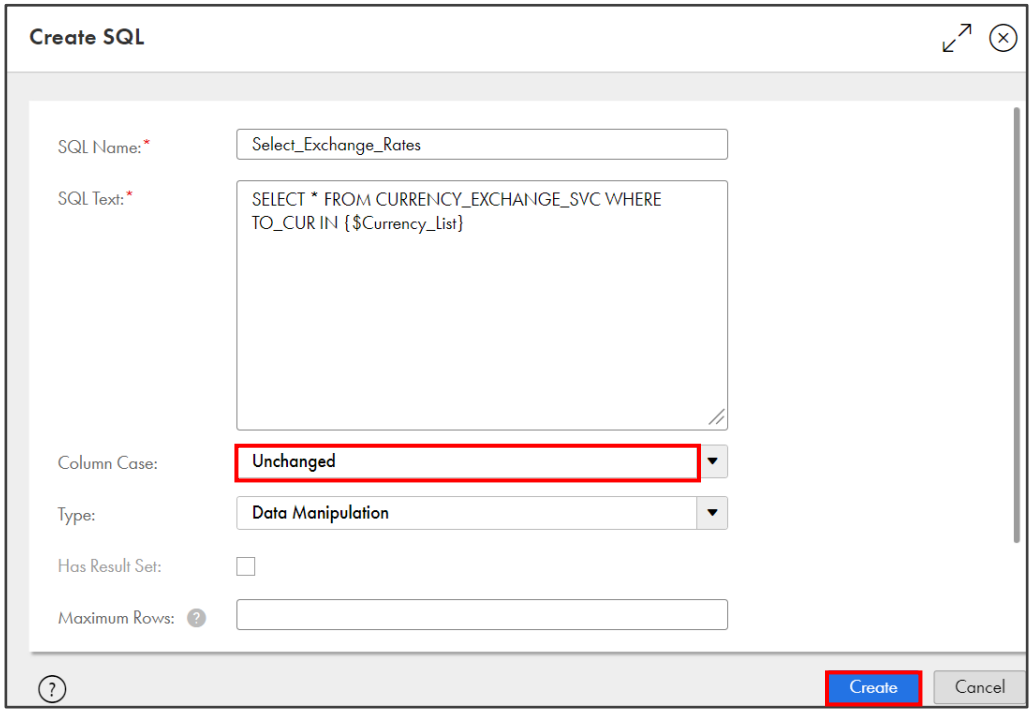
    SELECT * FROM CURRENCY_EXCHANGE_SVC WHERE TO_CUR IN {$Currency_List}

    OR

    From the CAI Lab Prep Files_Learners folder, open the notepad file named **09_LabGuide_CAI_Create a Currency Exchange Data Access Service Connector_6-2**. Copy the command mentioned under the **STEP A** and paste it in the SQL field.

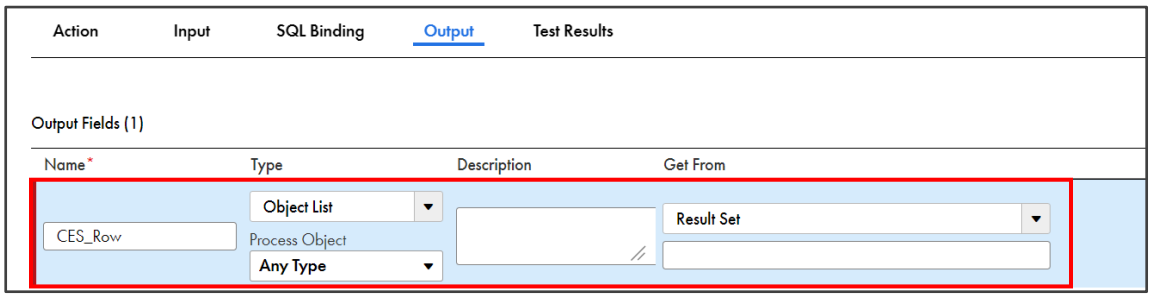


19. From the Column Case drop-down, select **Unchanged**.
20. Click **Create**.



21. Click **Save**.
22. Click the **Output** tab and click the '**+**' icon to add the output fields.
23. In the **Output Fields** section, enter the following values:

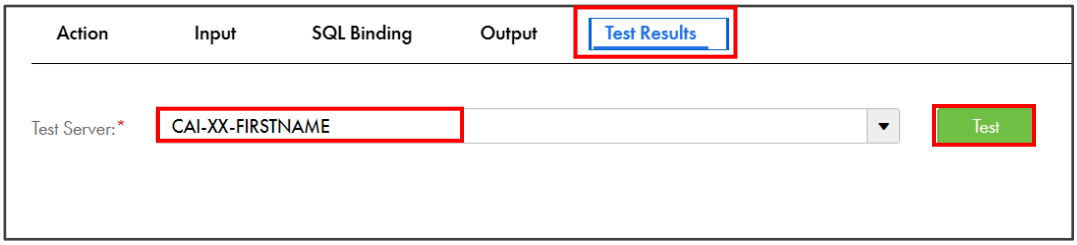
Name	Type	Process Object	Get From
CES_Row	Object List	Any Type	Result Set



24. Click **Save**.

Test Action:

25. Select **Test Results**.
26. From the Test Server drop-down, select your Secure Agent group and click **Test**.

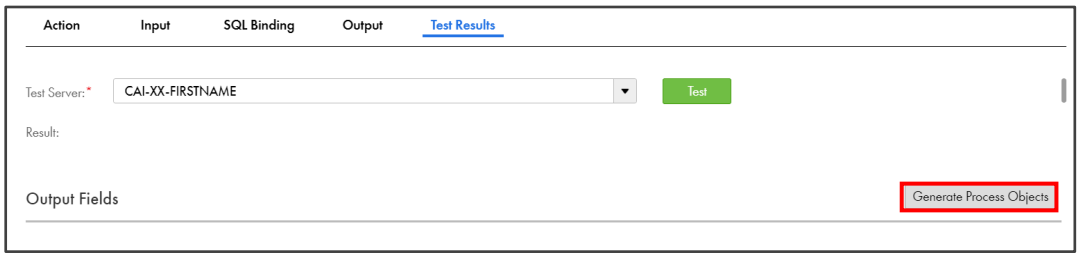


27. From the Output Fields section, scroll down to the **Response Payload** field and observe the output. You should see the exchange rate of three currencies that you specified.

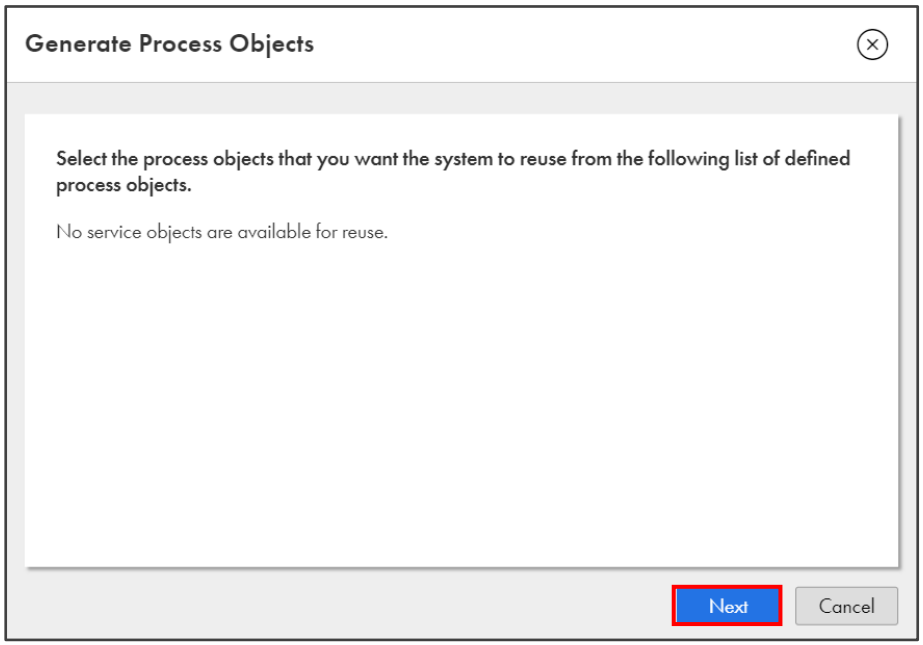
Generate Process Objects:

Now, we will generate a process object to capture the metadata from the CURRENCY_EXCHANGE_SVC table.

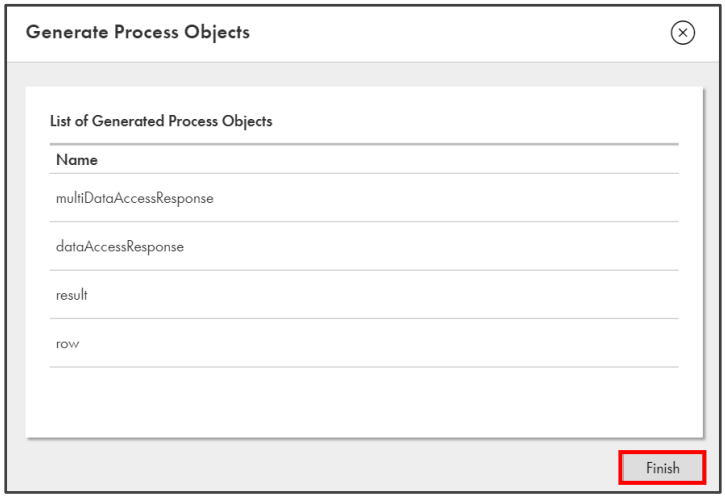
28. Click **Generate Process Objects**.



29. Click **Next**.

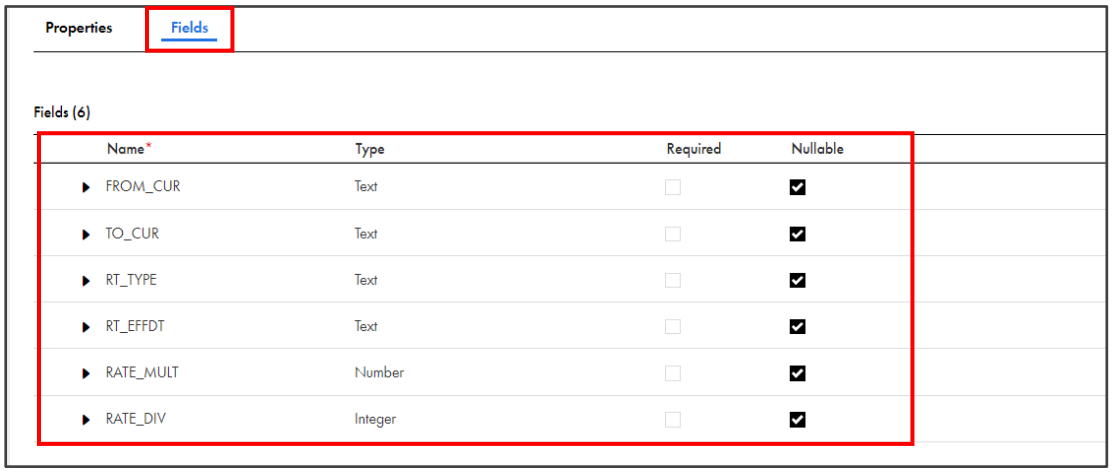


30. Click **Finish**.

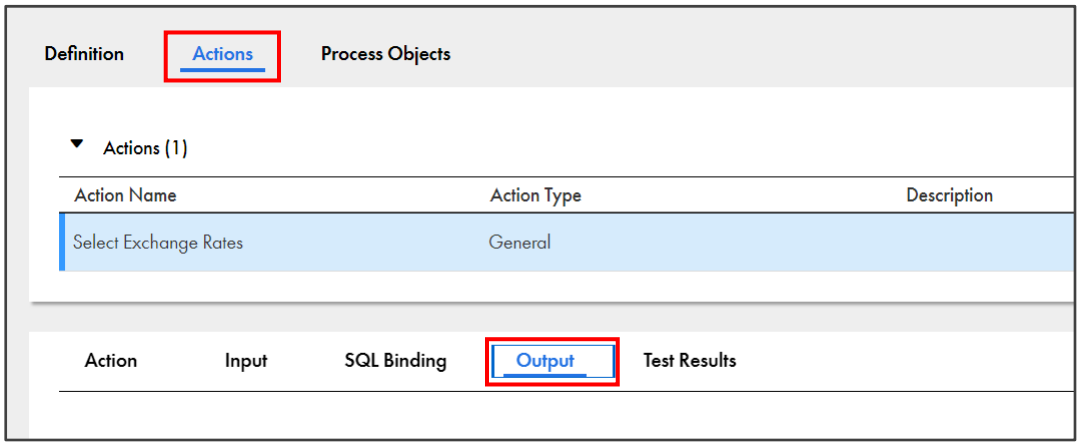


31. Click **Process Objects** and select **row**.

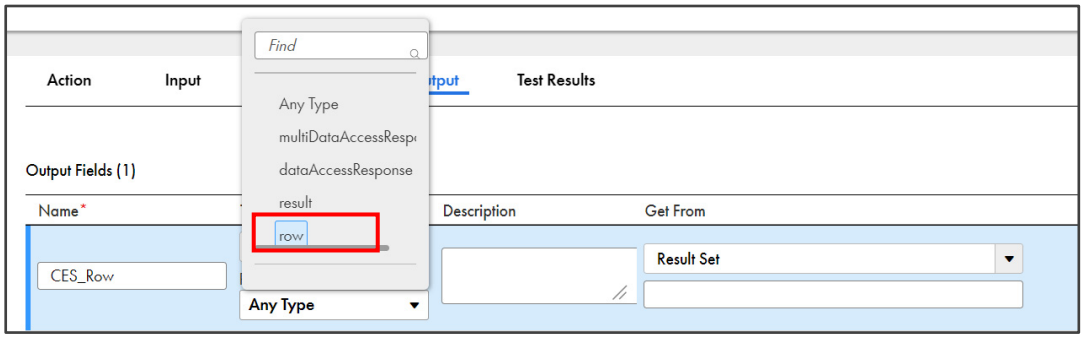
32. Click **Fields**. In the Fields tab, you can see the schema from the table.



33. Go back to the **Actions** tab and click **Output**.

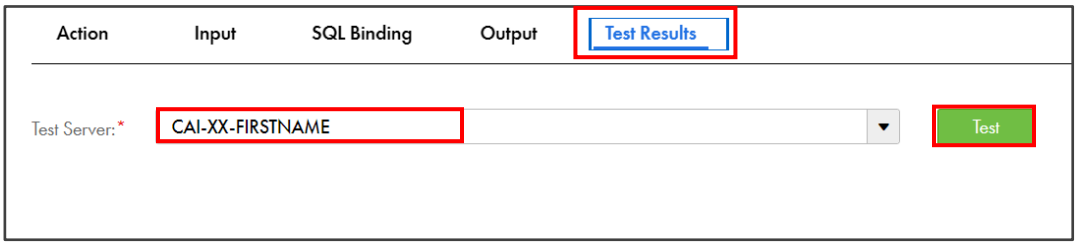


34. From the Process Object drop-down, select **row**.

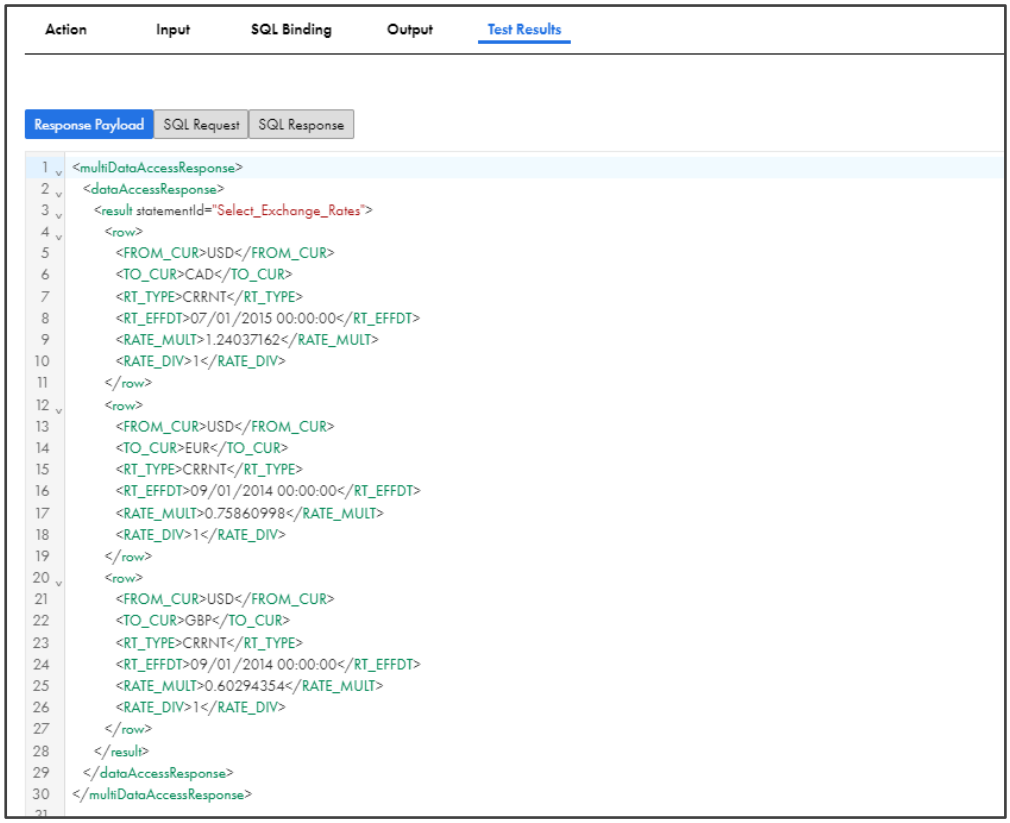


35. Click **Test Results**.
36. From the Test Server drop-down, ensure that your Secure Agent group is selected.

37. Click **Test**.



38. In the Output Fields section, make sure that nothing has broken. You should be able to see the exchange rate of three currencies that you specified:

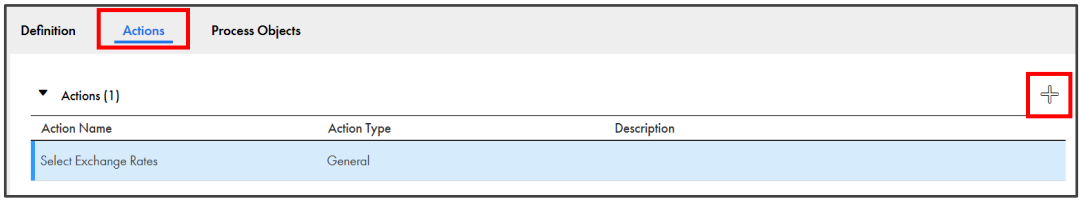


39. Click **Save**.

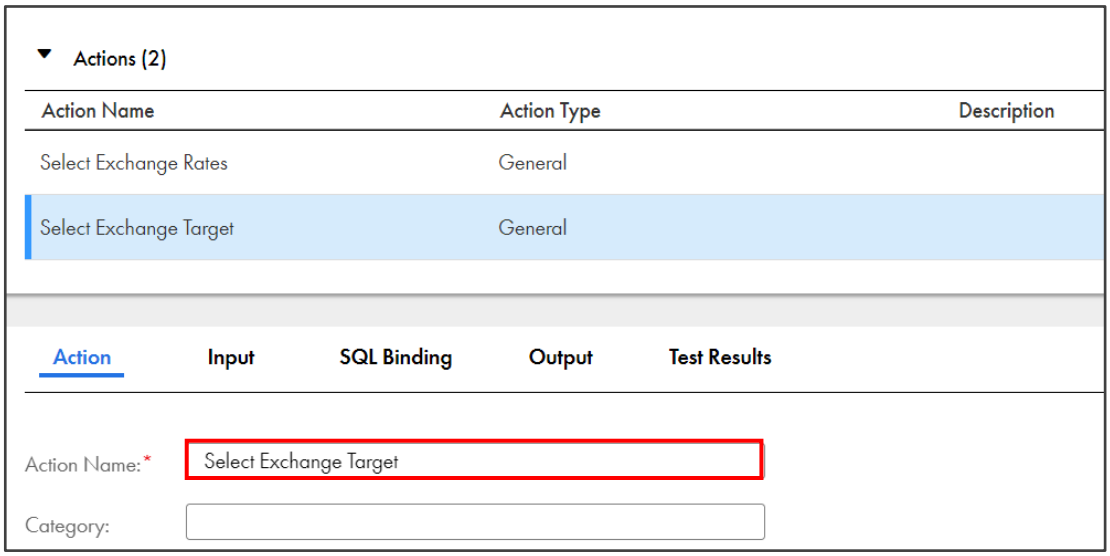
Define a New Action – Select Exchange Target:

This Action will enable you to view the data in the CURRENCY_EXCHANGE_TARGET table without having access to the database.

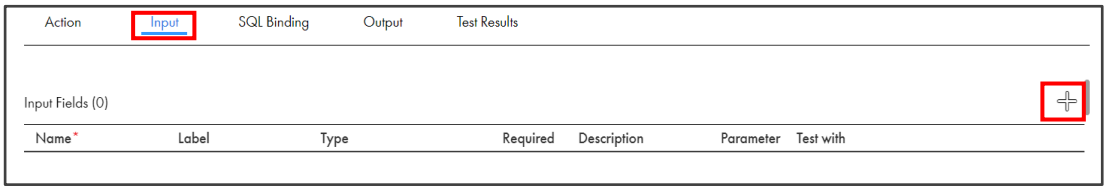
40. Click the **Actions** tab.
41. Click to add a new action.



42. In the Action Name field, enter **Select Exchange Target**.

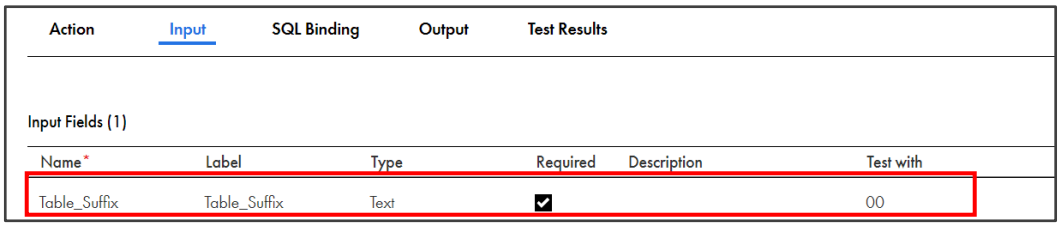


43. Click the **Input** tab and click the '**+**' icon to add the input fields.

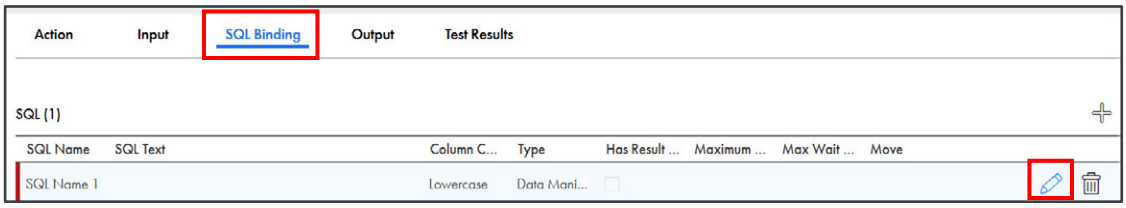


44. In the **Input Fields** section, enter the following values:

Property	Value
Name	Table_Suffix
Label	Table_Suffix
Type	Text
Required	Checked
Test with	Your student number



45. Click the **SQL Binding** tab and click the pencil icon to edit.

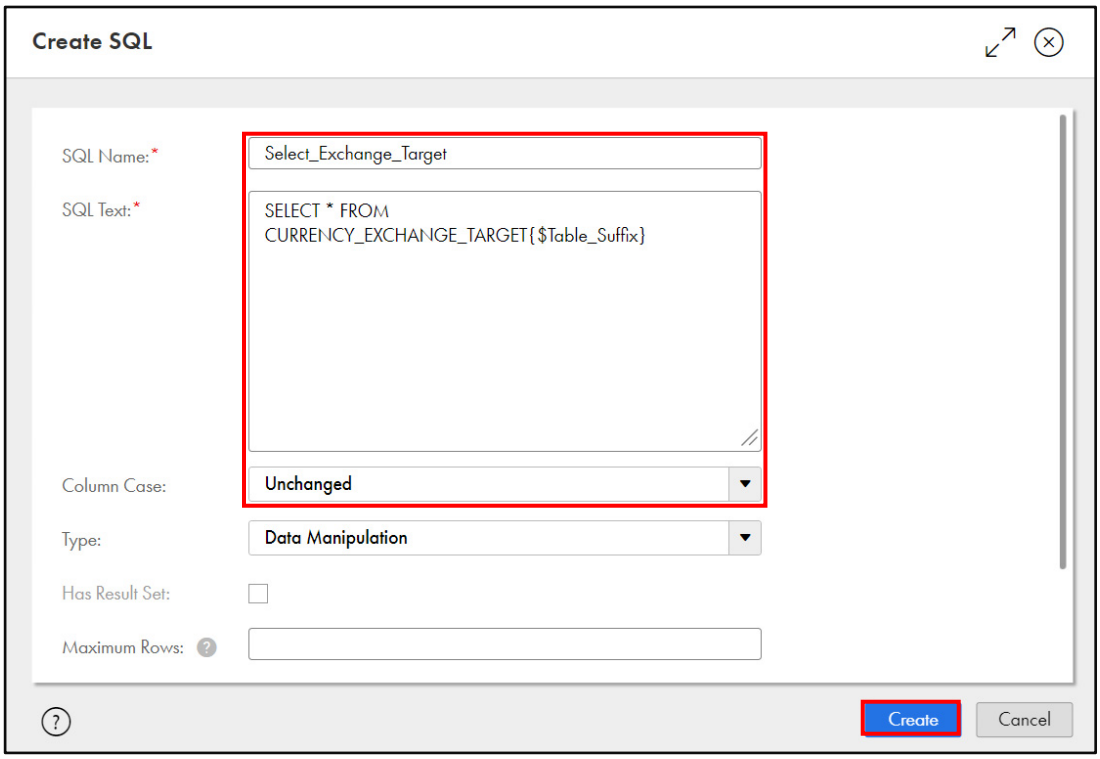


46. In the SQL Name field, enter **Select_Exchange_Target**.
47. In the SQL Text field, enter the following query:

    SELECT * FROM CURRENCY_EXCHANGE_TARGET{$Table_Suffix}

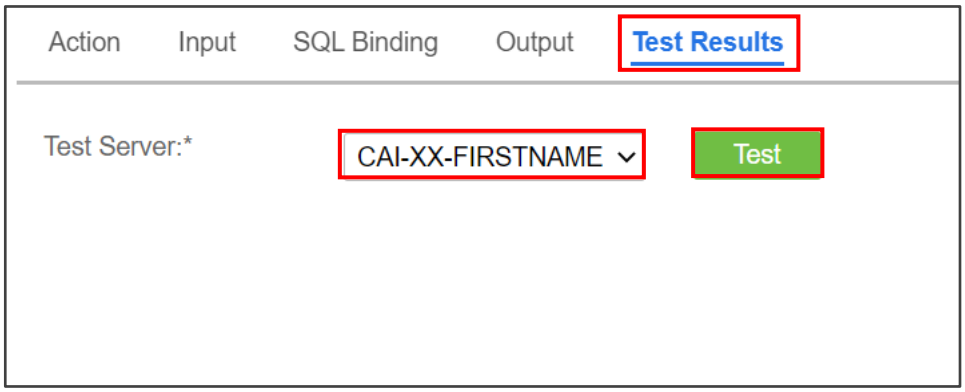
    OR

    From the CAI Lab Prep Files_Learners folder, open the notepad file named **09_LabGuide_CAI_Create a Currency Exchange Data Access Service Connector_6-2**. Copy the command mentioned under the **STEP B** and paste it in the SQL field.
48. From the Column Case drop-down, select **Unchanged**.
49. Click **Create**.

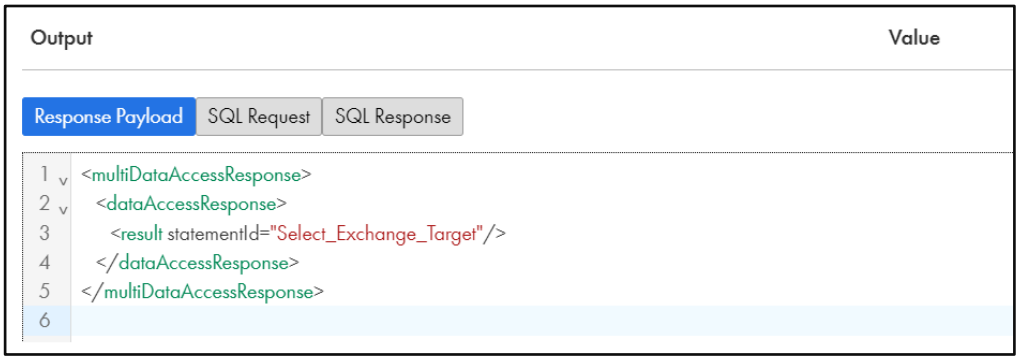


50. Click **Save**.
51. Select **Test Results**.

52. From the Test Server drop-down, select your Secure Agent group and click **Test**.



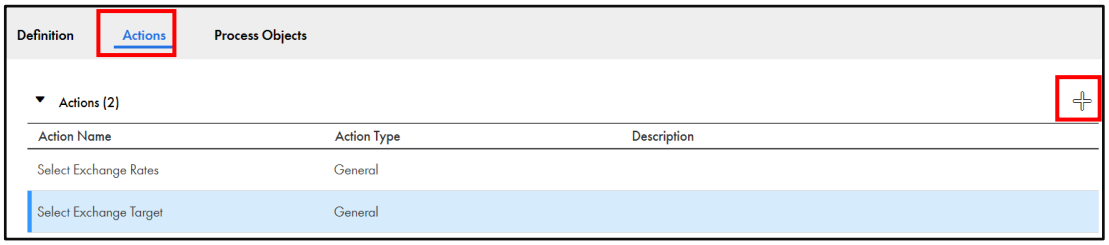
Note: At the moment, there is no data in the CURRENCY_EXCHANGE_TARGET table. So, you will not see any data in the Response Payload section:



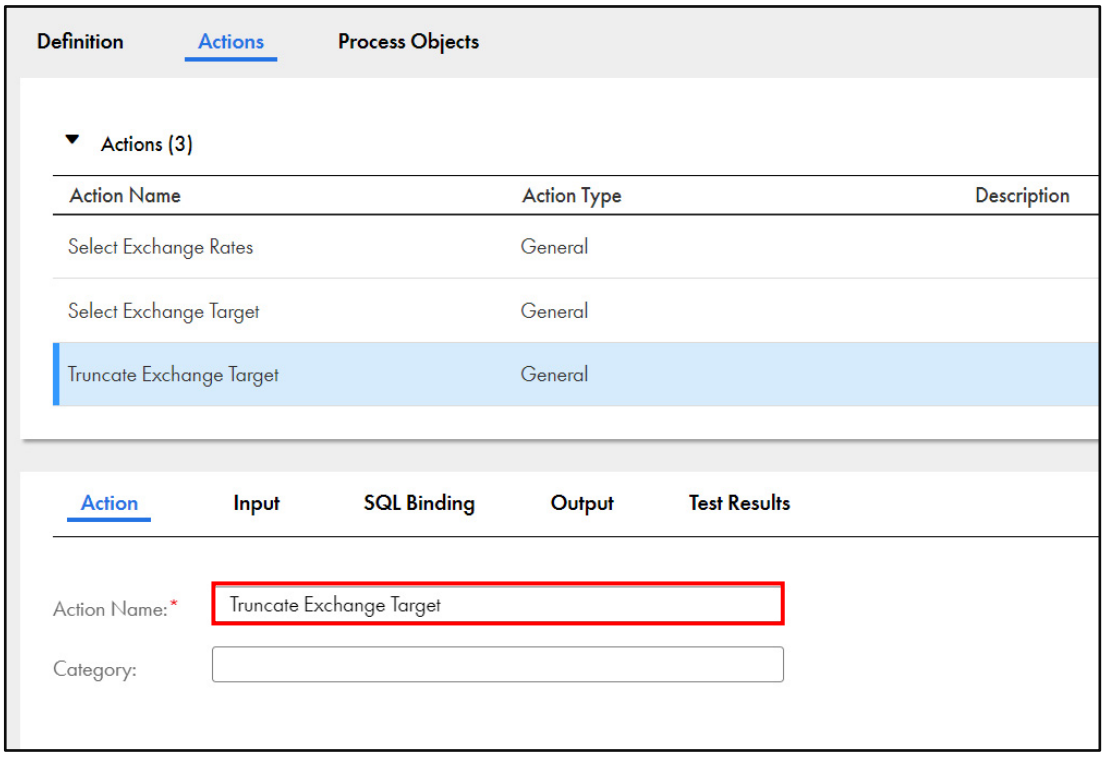
Define a New Action – Truncate Exchange Target:

In this section, we will create a new action Truncate Exchange Target which will truncate the CURRENCY_EXCHANGE_TARGET table.

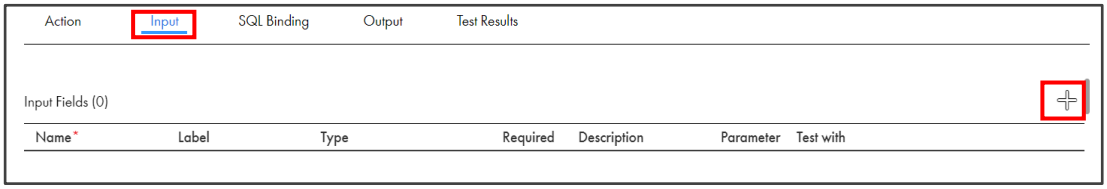
53. Click the **Actions** tab.
54. Click to add a new action.



55. In the Action Name field, enter **Truncate Exchange Target**.

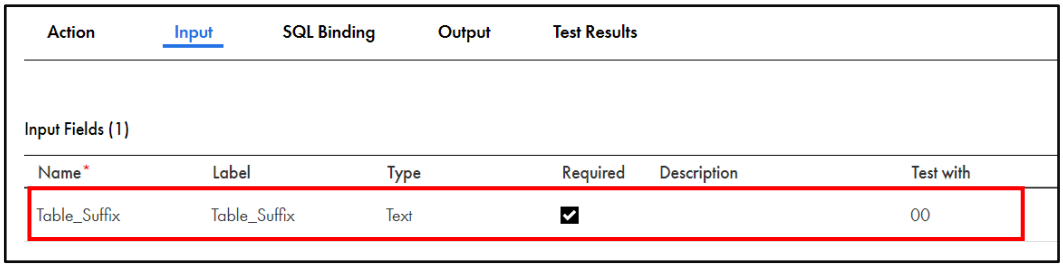


56. Click the **Input** tab and click the '**+**' icon to add the input fields.



57. In the **Input Fields** section, enter the following values:

Property	Value
Name	Table_Suffix
Label	Table_Suffix
Type	Text
Required	checked
Test with	Your student number

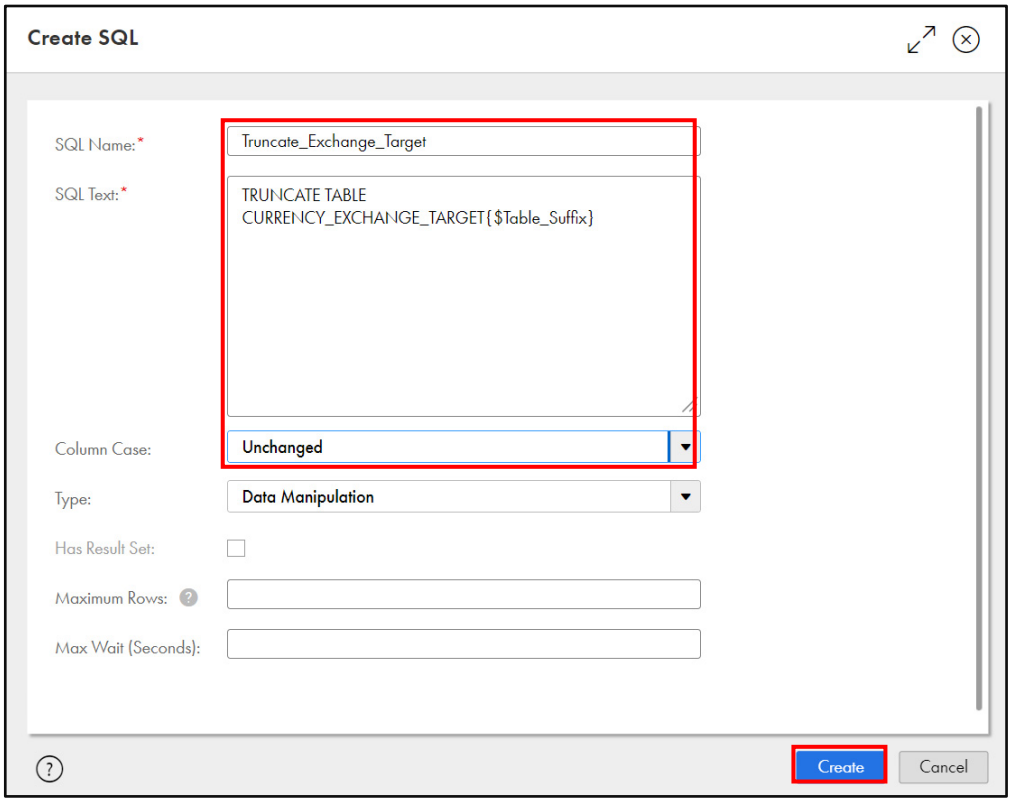


58. Click the **SQL Binding** tab and click the pencil icon to edit the SQL fields.
59. In the SQL Name field, enter **Truncate_Exchange_Target**.
60. In the SQL Text field, enter the following query:

    TRUNCATE TABLE CURRENCY_EXCHANGE_TARGET{$Table_Suffix}

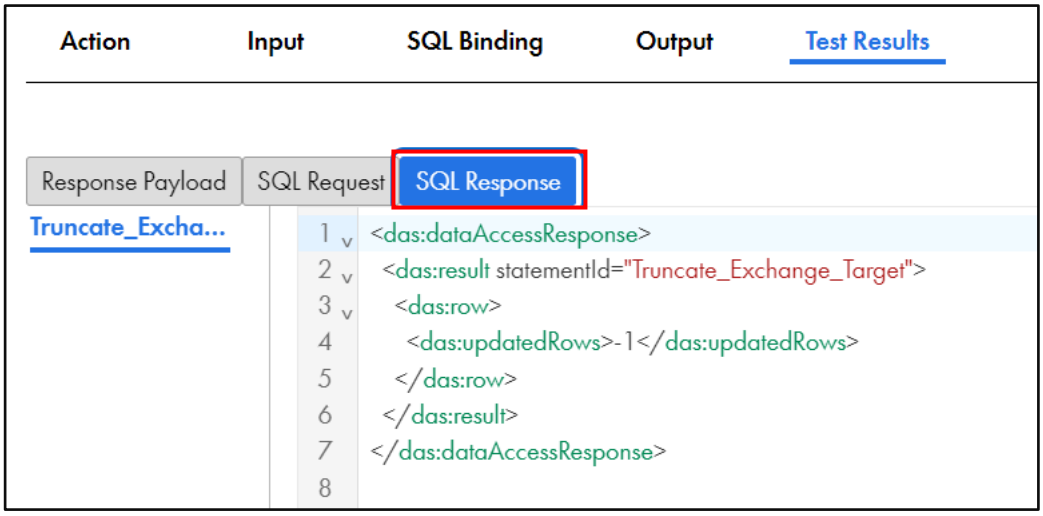
    OR

    From the CAI Lab Prep Files_Learners folder, open the notepad file named **09_LabGuide_CAI_Create a Currency Exchange Data Access Service Connector_6-2**. Copy the command mentioned under the **STEP C** and paste it in the SQL field.
61. From the Column Case drop-down, select **Unchanged**.
62. Click **Create**.



63. Click **Save**.
64. Select **Test Results**.
65. From the Test Server drop-down, ensure that your Secure Agent group is selected and click **Test**.

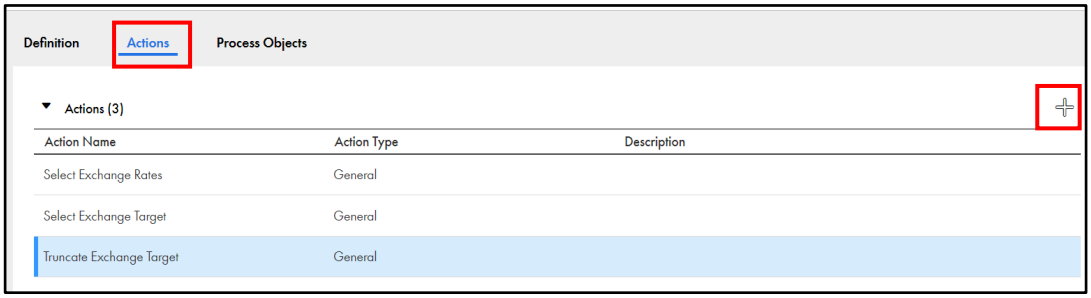
66. In the Output Fields section, click **SQL Response**. Ensure that there is no error in the SQL Response section.



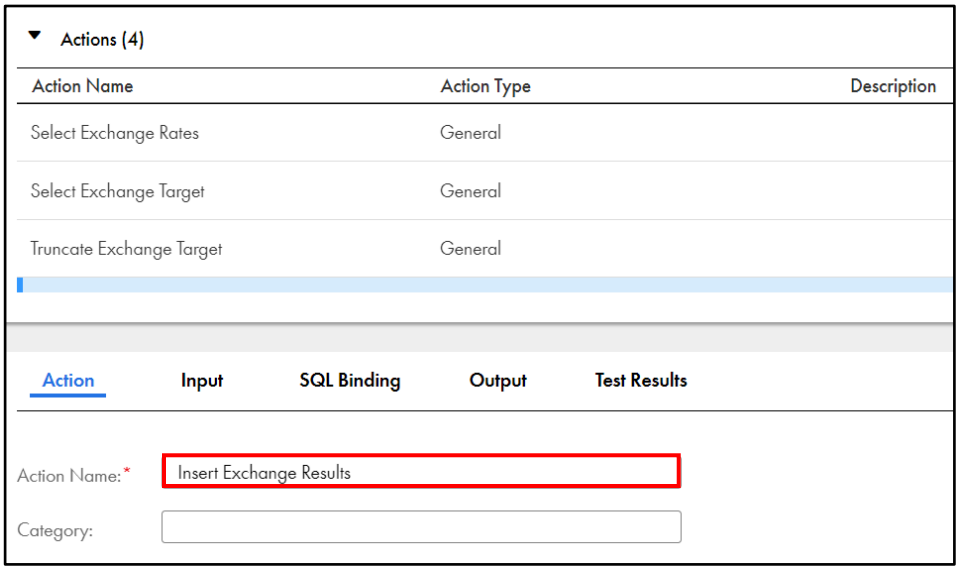
Define a New Action – Insert Exchange Results:

In this section, we will create a new action Insert Exchange Results which will enable you to insert data the CURRENCY_EXCHANGE_TARGET table.

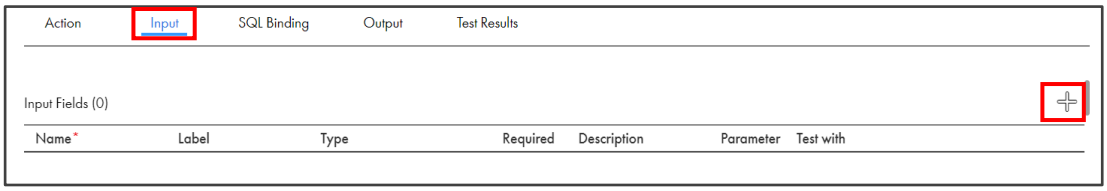
67. Click the **Actions** tab.
68. Click to add a new action.



69. In the Action Name field, enter **Insert Exchange Results**.

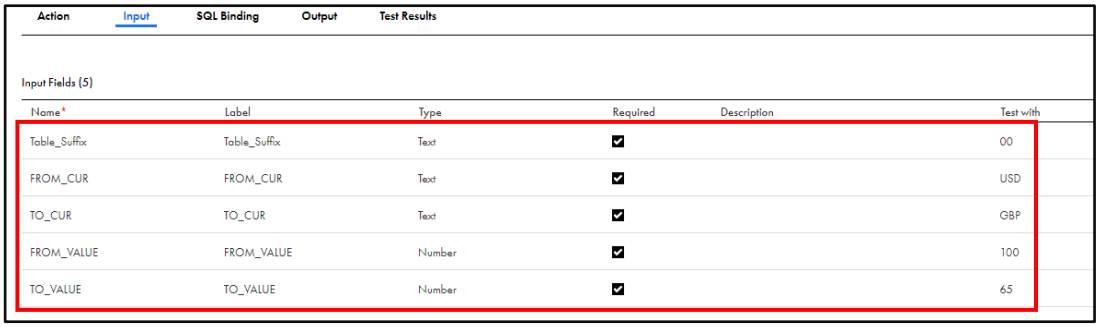


70. Click the **Input** tab and click the '**+**' icon to add the input fields.



71. In the **Input Fields** section, enter the following values:

Name	Label	Type	Required	Test with
Table_Suffix	Table_Suffix	Text	checked	Your student number
FROM_CUR	FROM_CUR	Text	checked	USD
TO_CUR	TO_CUR	Text	checked	GBP
FROM_VALUE	FROM_VALUE	Number	checked	100
TO_VALUE	TO_VALUE	Number	checked	65



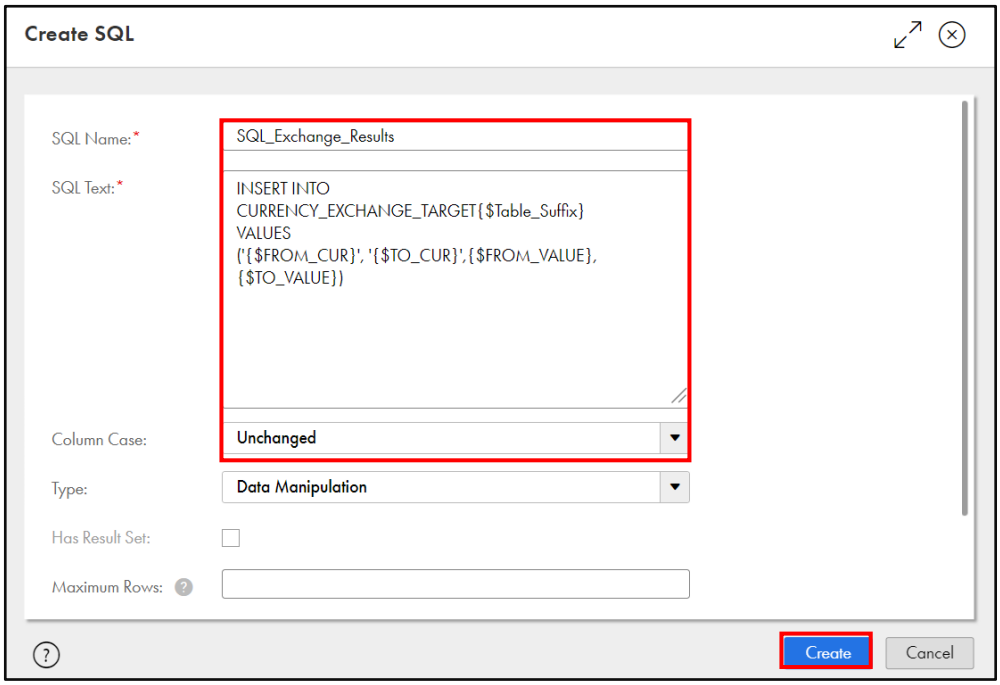
Note: Each time click the '+' icon to enter a new row in the Input Fields section.

72. Click the **SQL Binding** tab and in the SQL Name field, enter **SQL_Exchange_Results**.
73. In the SQL Text field, enter the following query:

    INSERT INTO CURRENCY_EXCHANGE_TARGET{$Table_Suffix} VALUES ('{$FROM_CUR}', '{$TO_CUR}',{$FROM_VALUE},{$TO_VALUE})

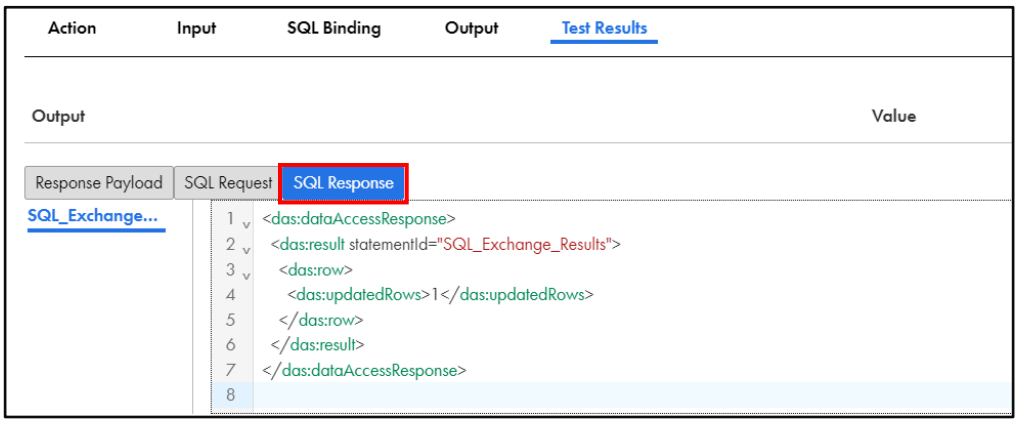
    OR

    From the CAI Lab Prep Files_Learners folder, open the notepad file named **09_LabGuide_CAI_Create a Currency Exchange Data Access Service Connector_6-2**. Copy the command mentioned under the **STEP D** and paste it in the SQL field.
74. From the Column Case drop-down, select **Unchanged**.
75. Click **Create**.



76. Click **Save**.
77. Select **Test Results**.
78. From the Test Server drop-down, ensure that your Secure Agent group is selected and click **Test**.

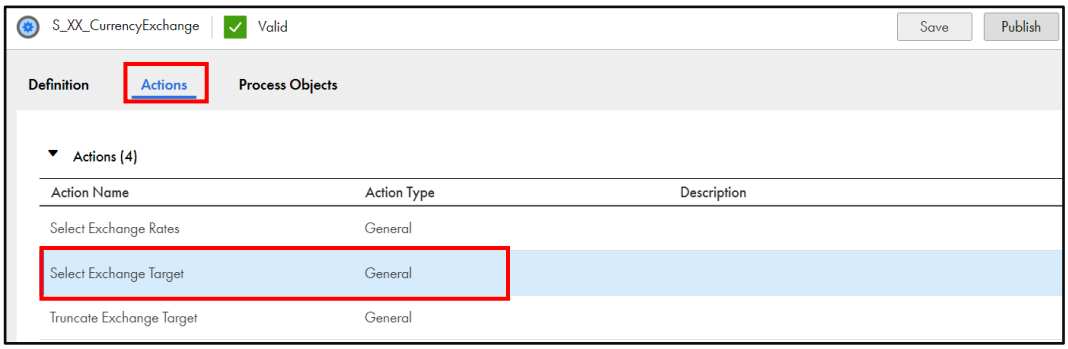
79. In the Output Fields section, click **SQL Response**. Make sure that there is no error in the SQL Response section.



Test the Result

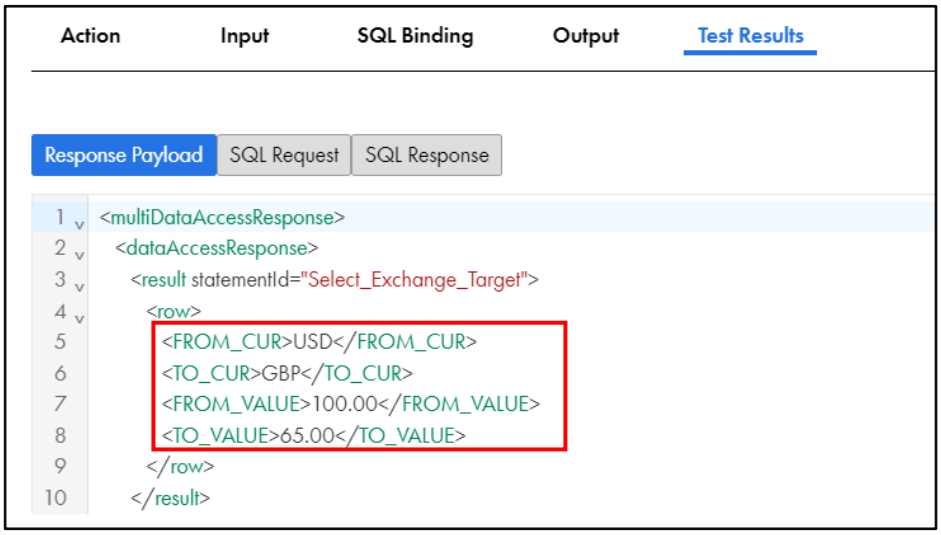
In this section, we will test if the data is successfully inserted into the CURRENCY_EXCHANGE_TARGET table.

80. From the Actions section, select the **Select Exchange Target** action.



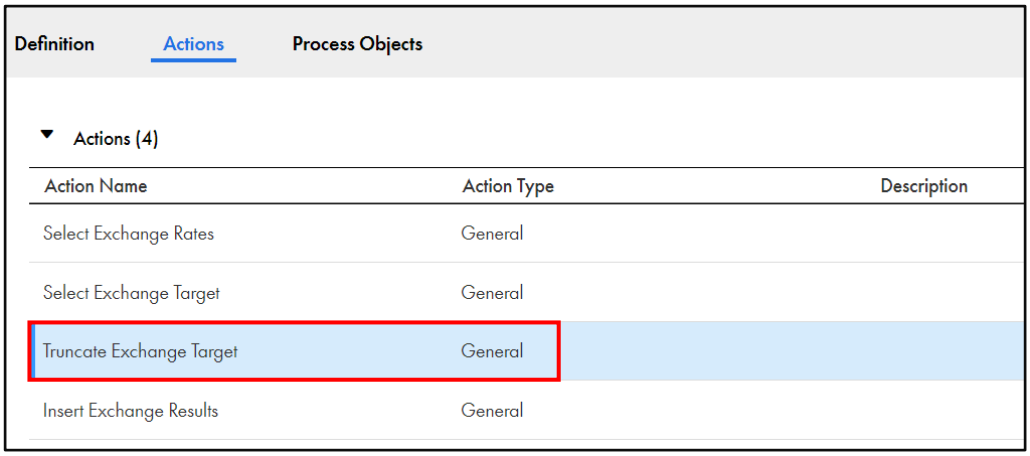
81. Select **Test Results**.
82. From the Test Server drop-down, make sure that your Secure Agent group selected. Click **Test**.

83. In the Output Fields section, you can see the rows are inserted into the CURRENCY_EXCHANGE_TARGET table.



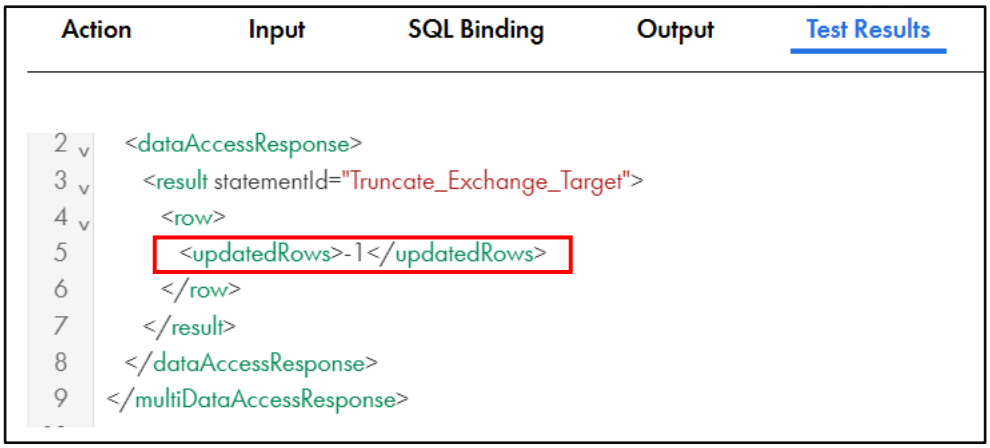
Now, you will clear the inserted rows from the CURRENCY_EXCHANGE_TARGET table and test the results.

84. From the Actions section, select the **Truncate Exchange Target** action.

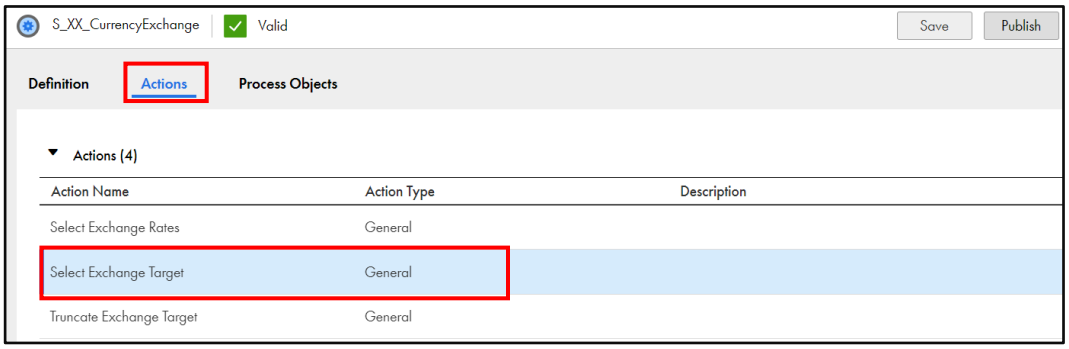


85. Select **Test Results**.
86. From the Test Server drop-down, make sure that your Secure Agent group is selected. Click **Test**.

Note: In the Output Fields section, you will notice that one row is updated.

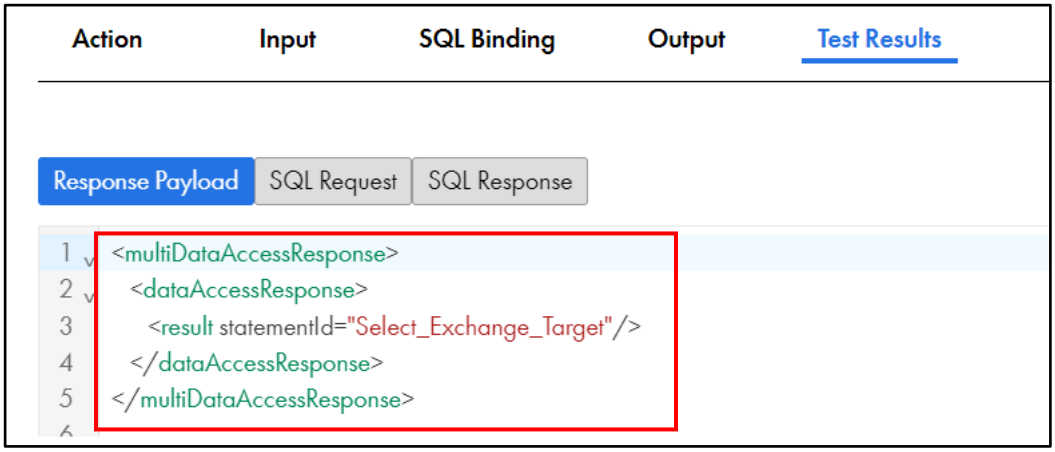


87. Now, go back to **Select Exchange Target** action to check if the data is cleared or not.

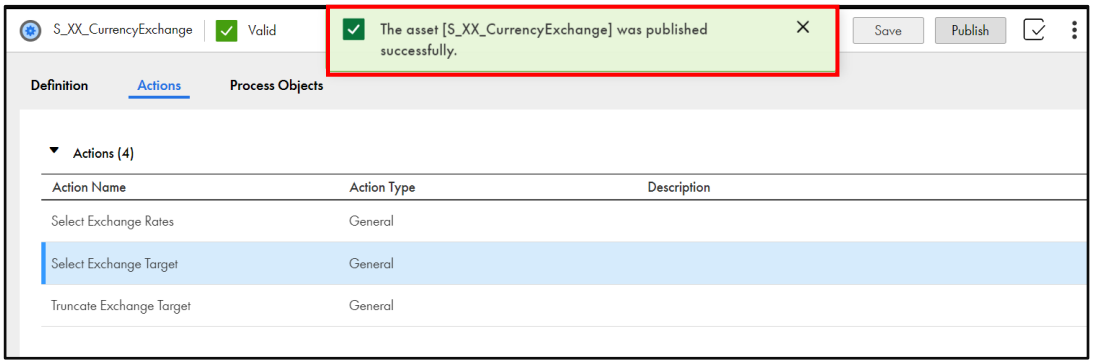


88. Select **Test Results**.
89. From the Test Server drop-down, make sure that your Secure Agent group selected. Click **Test**.

90. In the Output Fields section, you can see the data has been deleted from the CURRENCY_EXCHANGE_TARGET table.



91. Click **Save**.
92. Click **Publish** to publish the Data Access Service Connector.



Note: A message **The asset was published successfully** appears.

This concludes the lab.

Module 6: Data Access Service Connectors
Lab 6-3: Using Currency Exchange Data Access Service Connector in a Process

Overview:

This is **part 2** of the previous lab. In Lab 6-2: Create a Currency Exchange Data Access Service Connector, you created a Currency Exchange Data Access Service Connector. In this lab, you will use the Currency Exchange Data Access Service Connector in a Process. To do this, first you will create an App Connection. Then, you will create a process using the Data Access Service Connector.

Objectives:

- Create an App Connection
- Create a Process using the Data Access Service Connector
- Publish and execute the Process

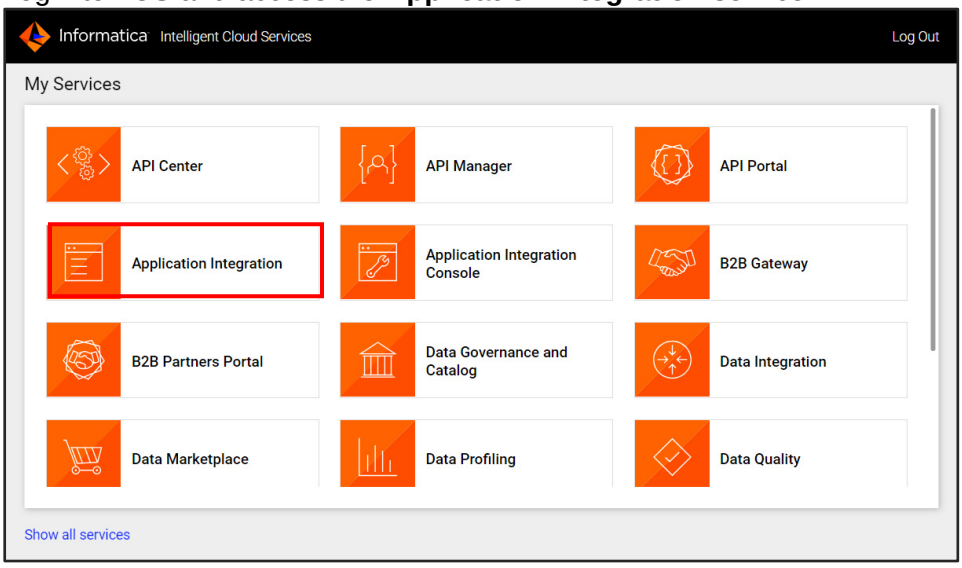
Duration:

45 Minutes

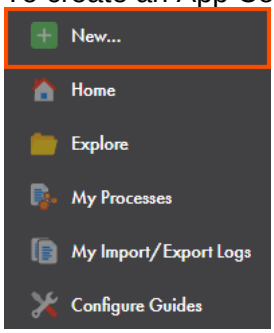
Tasks

Create a connection:

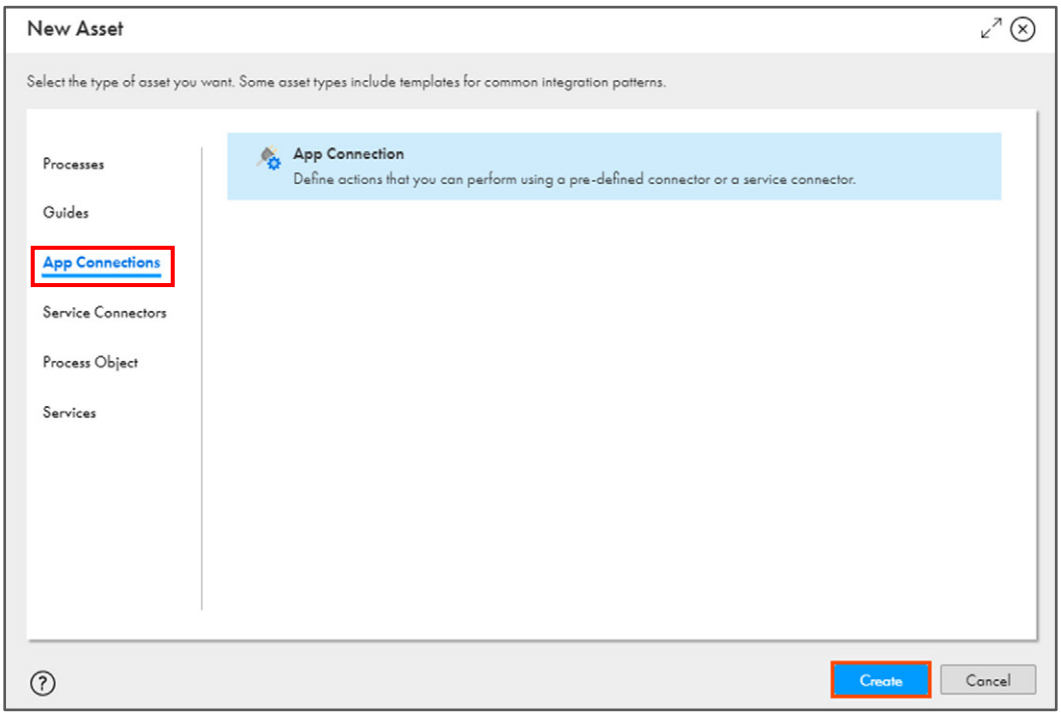
1. Log into IICS and access the **Application Integration** service.



2. To create an App Connection, from the navigation pane, select **New**.



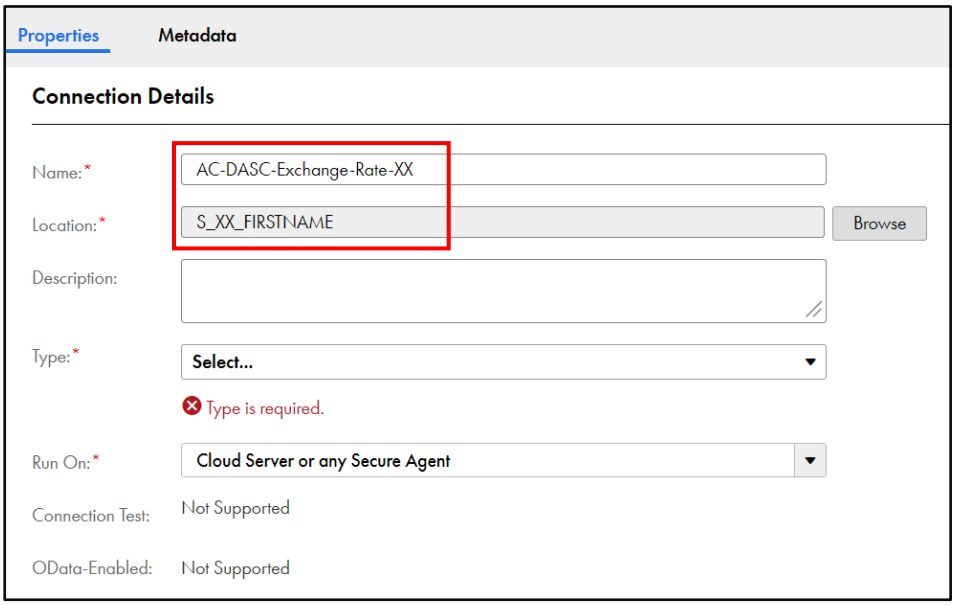
3. In the New Asset window, from the App Connections tab, select **App Connection**, and click **Create**.



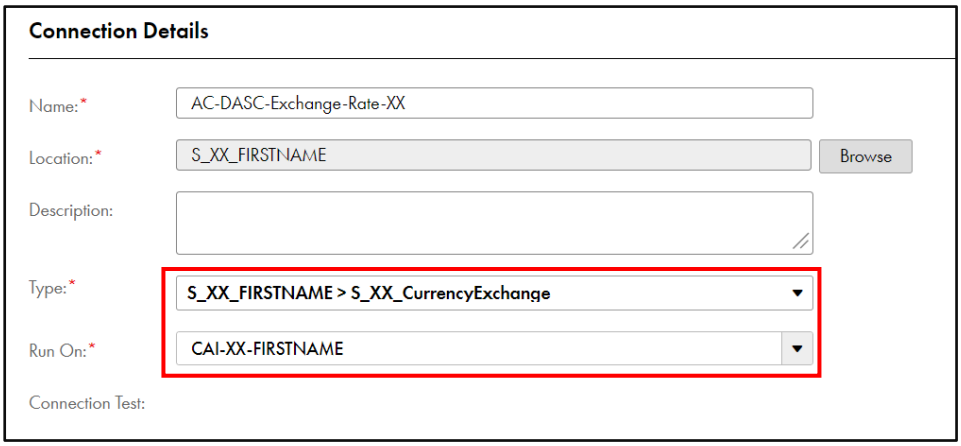
4. In the Connection Details page, enter the connection name as **AC-DASC-Exchange-Rate-XX**.

   Note: Replace XX with your student number.

5. In the Location field, select your project folder **S_XX_FirstName**.



6. From the Type drop-down, locate your project folder **S_XX_FirstName** and select the **S_XX_CurrencyExchange** data access service connector.
7. From the Run On drop-down, select your Secure Agent group.

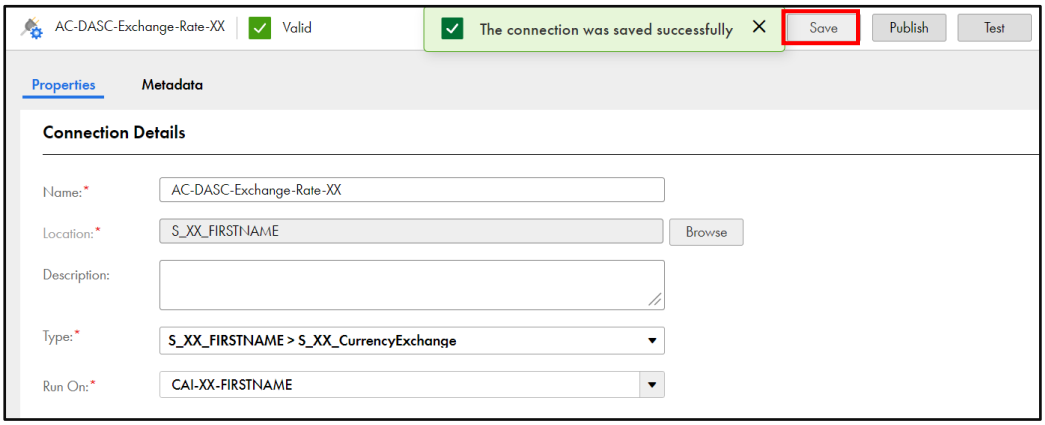


Now, you will be specifying the JDBC Connection Properties. You might be thinking that you have already specified the Connection Properties while creating the Data Access Service Connector. The properties that you entered while creating the Data Access Service Connector was only for testing purpose. The Connection Properties which you will enter in this step will be used by the process.

8. In the JDBC Driver field, enter the following:
   com.microsoft.sqlserver.jdbc.SQLServerDriver
9. In the JDBC Connection URL field, enter the following URL:
   jdbc:sqlserver://localhost:1433;databaseName=CAI;default=master

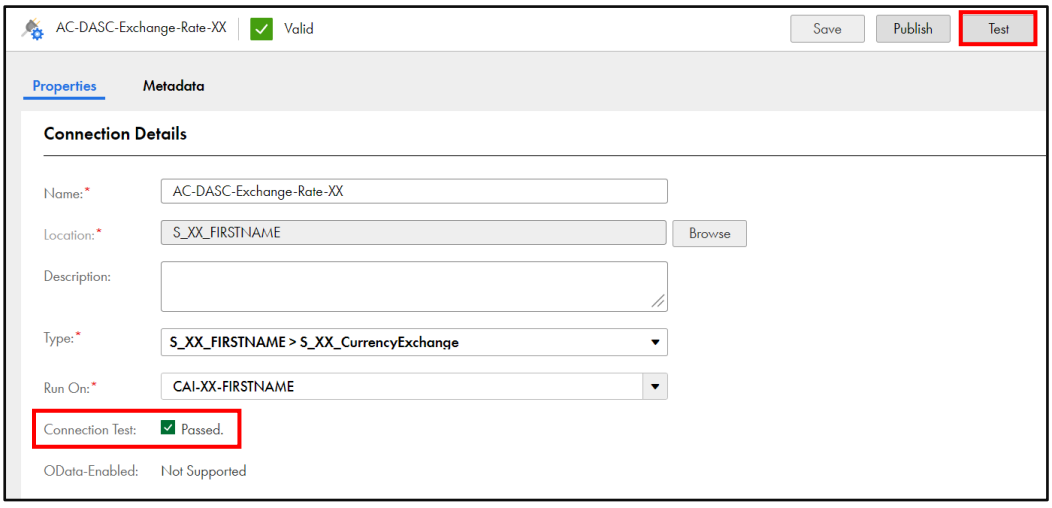
10. In the Username and Password fields, enter **CAI**.

11. Click **Save**.

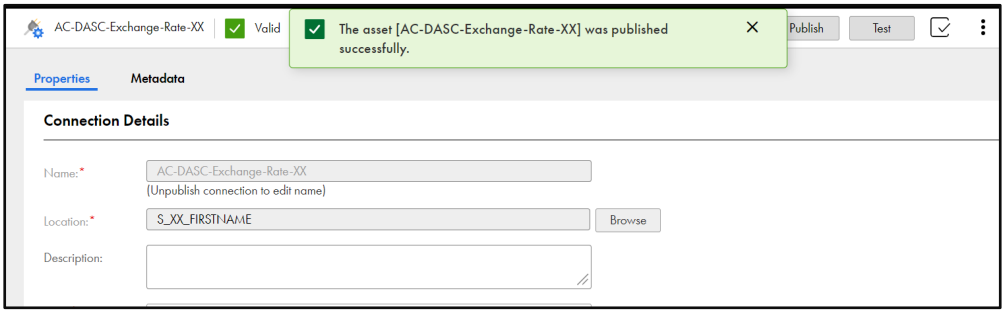


Note: A message **The service connector was saved successfully** appears.

12. Click **Test**. The Connection Test filed will changed as **Passed**.



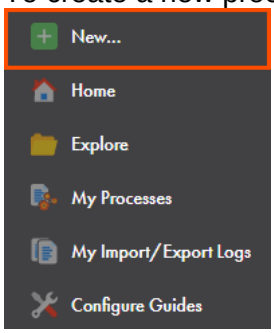
13. Click **Publish** to publish the connection.



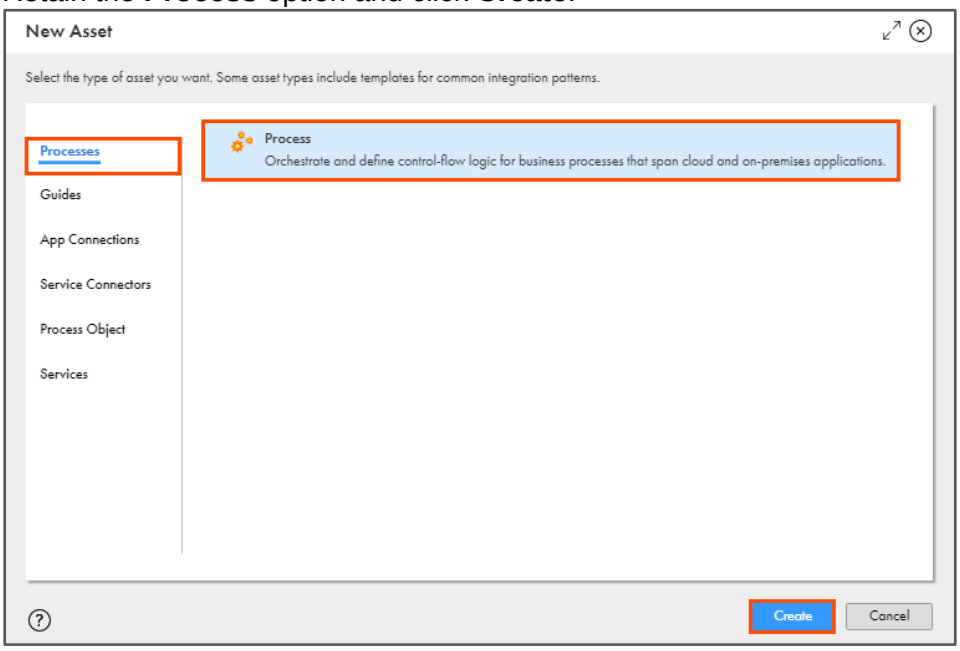
Note: A message **The asset was published successfully** appears.

Create a Process

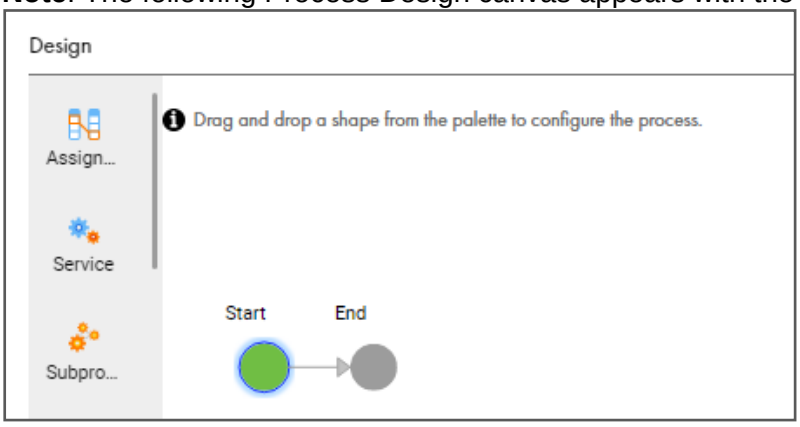
14. To create a new process, from the navigation pane, select **New**.



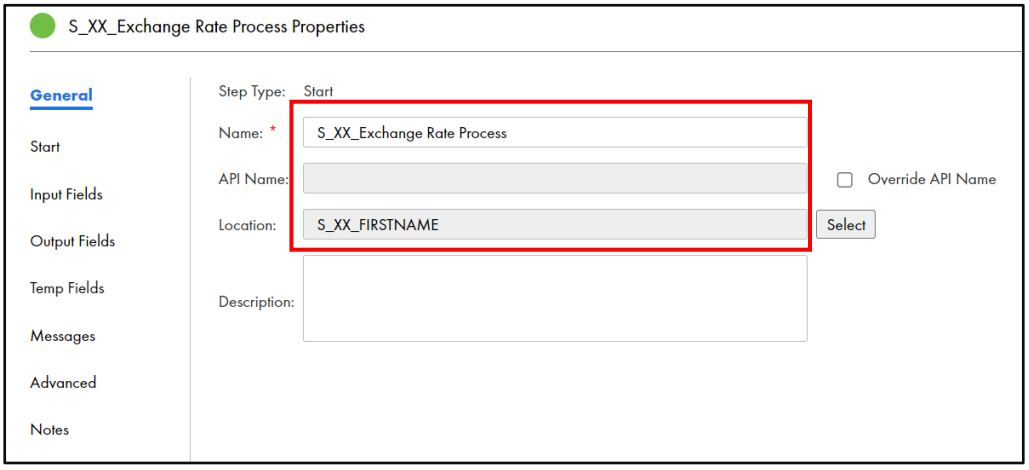
15. Retain the **Process** option and click **Create**.



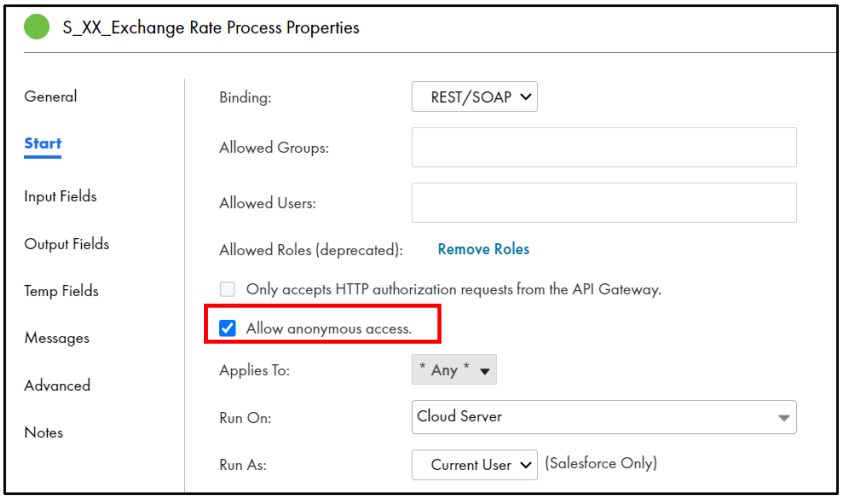
Note: The following Process Design canvas appears with the Start and End steps.



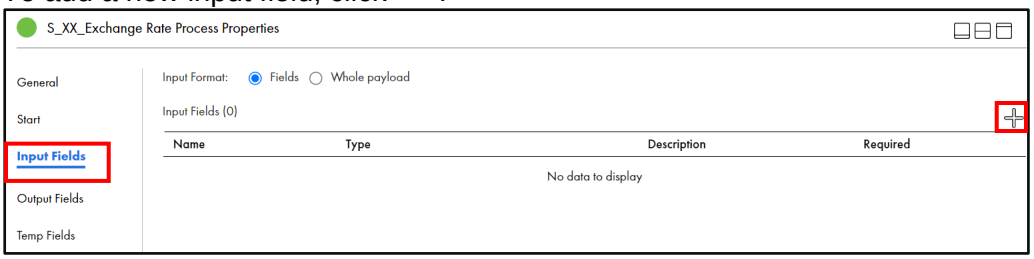
16. From the process canvas, select the **Start** step.
17. In the General tab, enter the name as **S_XX_Exchange Rate Process**.
    Note: Replace XX with your student number.
18. Skip the API Name option.
19. In the Location field, select your project folder **S_XX_FirstName**.



20. From the properties pane, select **Start**.
21. Select the **Allow anonymous access** option.



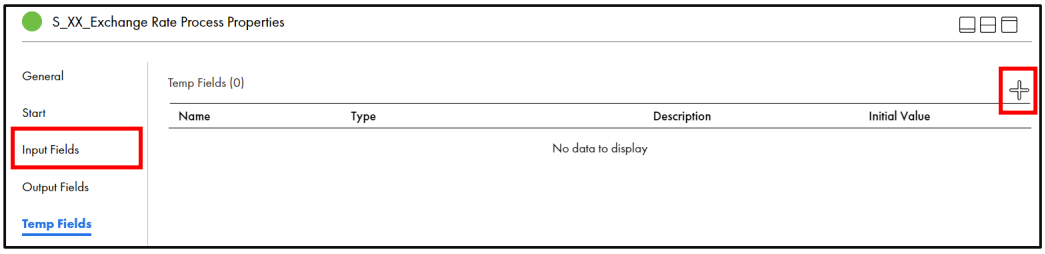
22. From the properties pane, select **Input Fields**.
23. To add a new input field, click .



24. Create the input field, as shown in the table below:

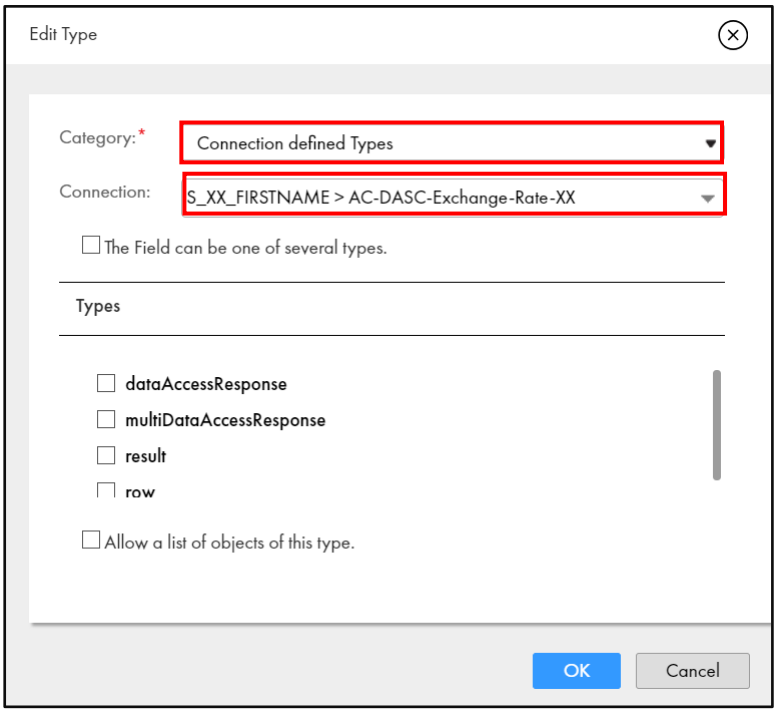
Name	Type
ExchangeRequest	More types>Custom Types> S_XX_FIRSTNAME> ConversionRequestPO

25. From the properties pane, select **Temp Fields**.
26. To add new temp field, click .

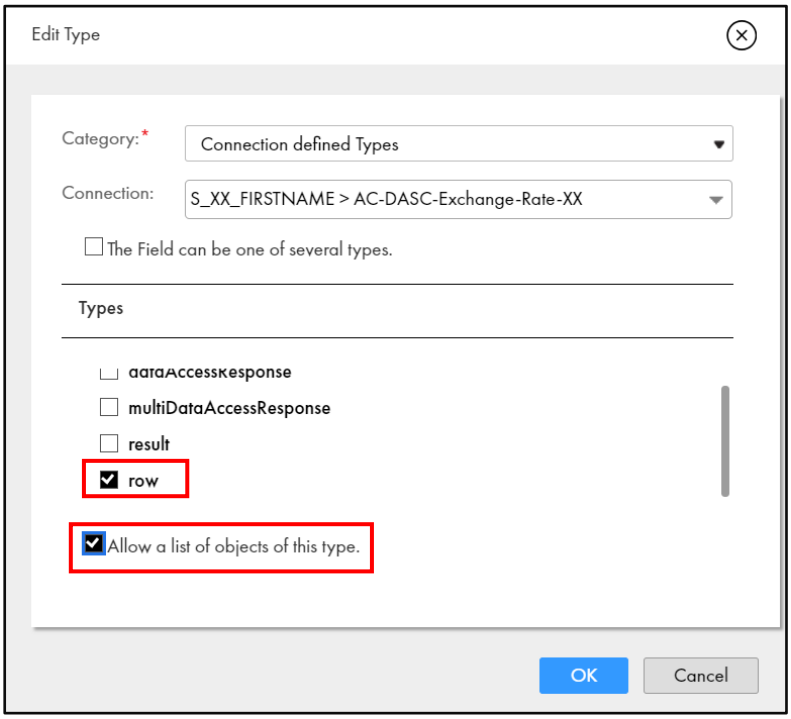


27. In the Name field, enter **SQL_Currency_List**. This field will be used in the WHERE In clause of XQuery.
28. Retain Type as Text.

29. Similarly, add another temp field named as **ListofCurrencyExchangeRates**.
30. From the Type drop-down, select **More types..**.
31. From the Category drop-down, select **Connection defined Types**.
32. From the Connection drop-down, select **S_XX_FIRSTNAME** > **AC-DASC-Exchange-Rate-XX**.

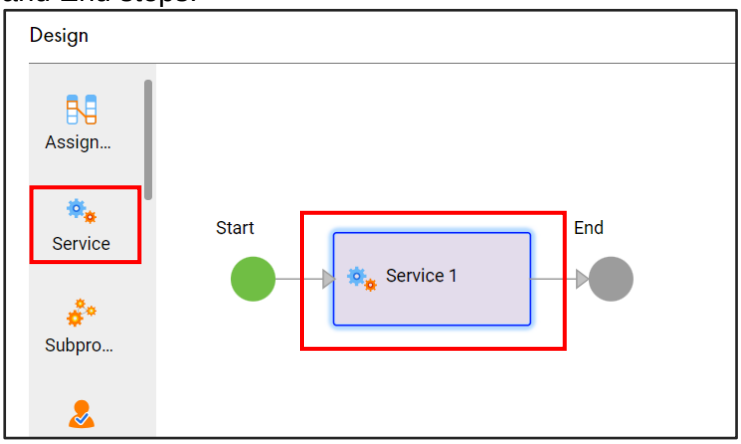


33. Select the **row** and **Allow a list of objects of this type** checkboxes.

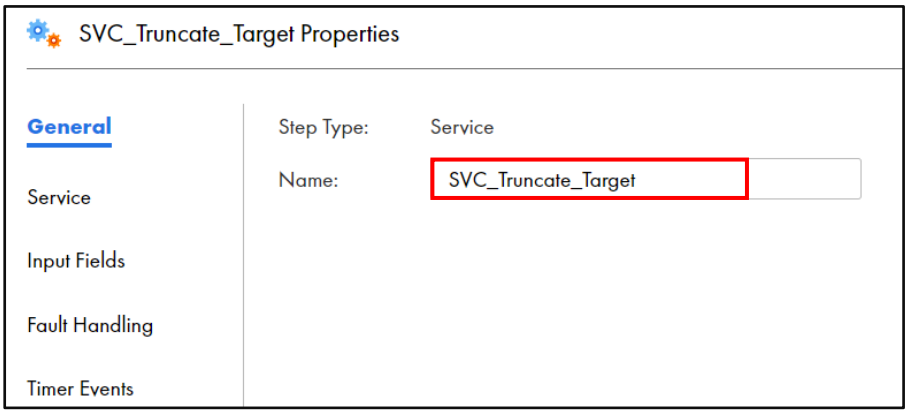


34. Click **OK**.

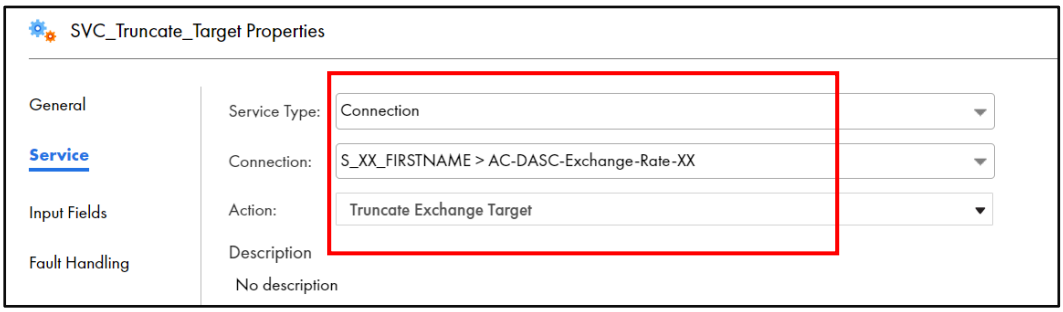
35. From the Design palette, drag and drop a **Service** step on the link between the Start and End steps.



36. To configure the service, from the process canvas, select the **Service** step.
37. In the General section of the Service properties, enter the Name as **SVC_Truncate_Target**.



38. From the properties pane, select **Service**.
39. From the Service Type drop-down, select **Connection**.
40. From the Connection drop-down, expand your project folder, and select **AC-DASC-Exchange-Rate-XX.**
41. From the Action drop-down, select **Truncate Exchange Target**.
Note: This action is obtained from the service connector.

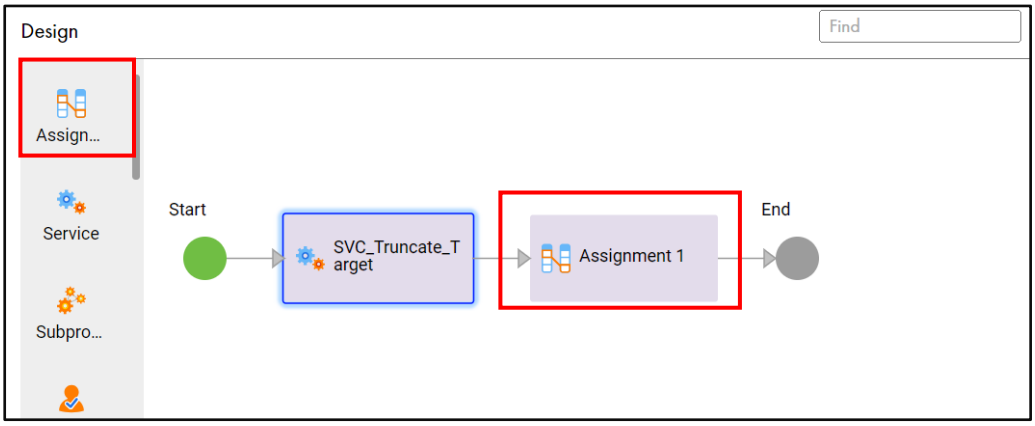


42. From the properties pane, select **Input Fields**.
43. In the Value field, enter your student number.

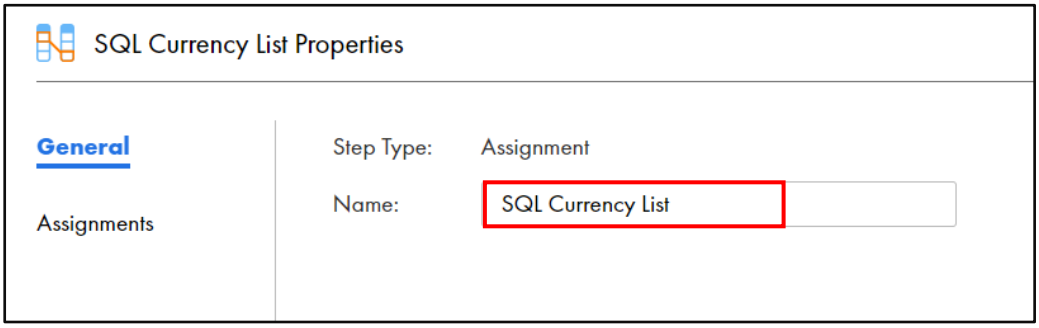
44. Click **Save**.

Now that the information is extracted, it needs to be displayed as an output. For this, you will use an Assignment step.

45. From the Design palette, drag and drop an **Assignment** step on the link between the SVC_Truncate_Target and End steps.

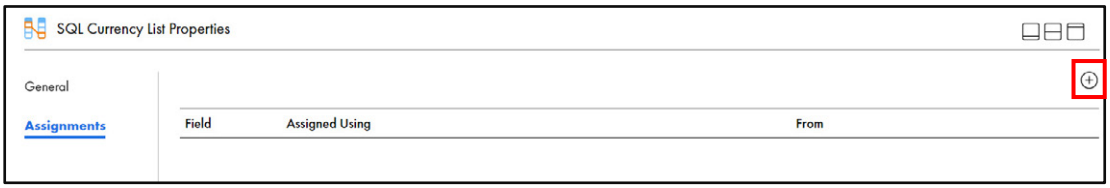


46. To configure the step, from the process canvas, select the **Assignment** step.
47. In the General section of Assignment properties, enter the Name as **SQL Currency List**.

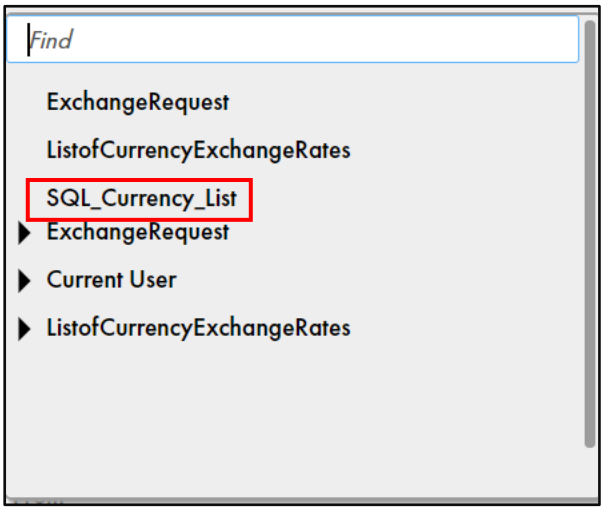


48. From the properties pane, select **Assignments**.

49. Click the **+** icon.

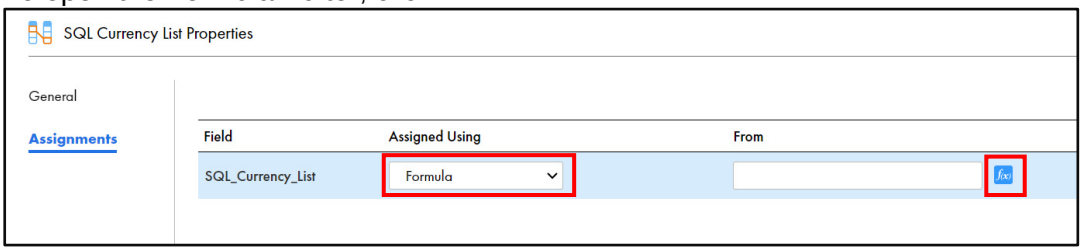


50. From the drop-down, select **SQL_Currency_List**.



51. From the Assigned Using drop-down, select **Formula**.

52. To open the Formula Editor, click .



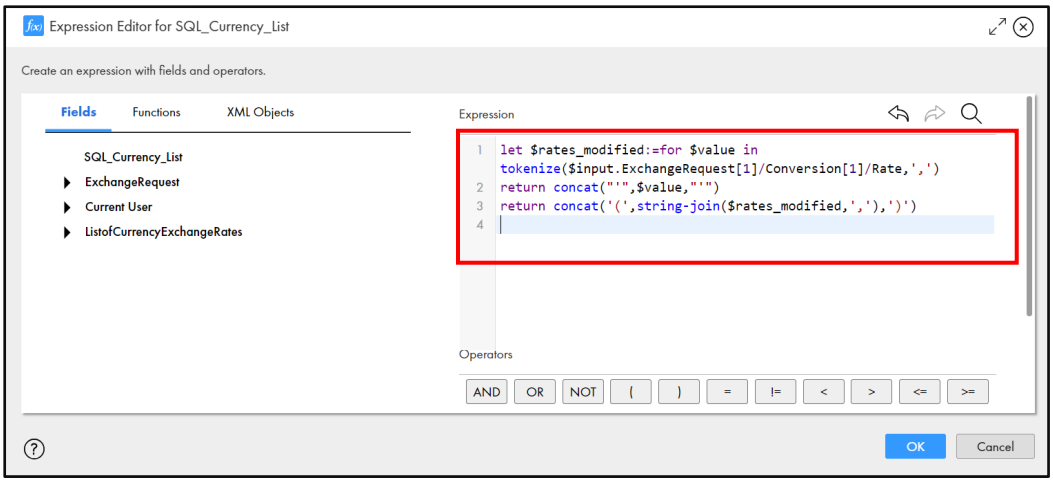
Note: A new Expression Editor for the SQLCondition window appears.

53. In the Expression field, enter the following expression:

let $rates_modified:=for $value in
tokenize($input.ExchangeRequest[1]/Conversion[1]/Rate,',')
return concat("'",$value,"'")
return concat('(',string-join($rates_modified,','),')')

OR

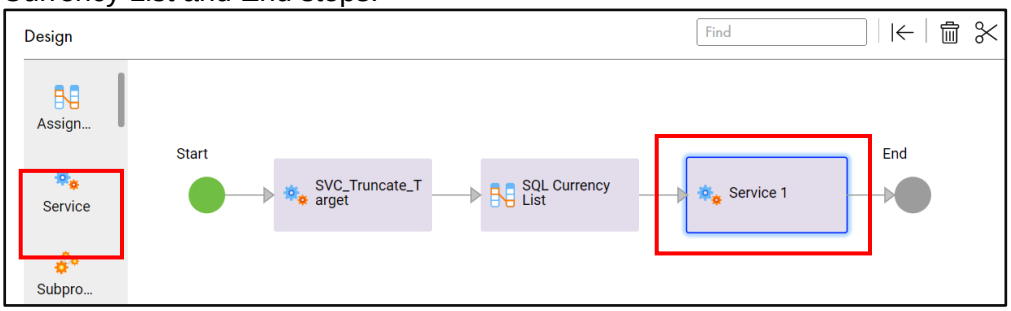
From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **10_Using Currency Exchange Data Access Service Connector in a Process_6-3**. Copy the command mentioned under **Step A** and paste it into the Expression field.



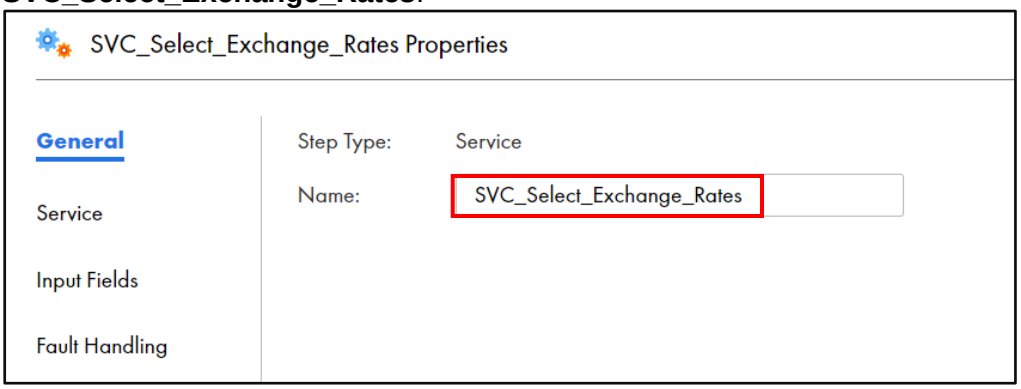
54. Click **OK**.
55. **Save** the process.

Next, you need to add one more service step that will retrieve the currency exchange rates.

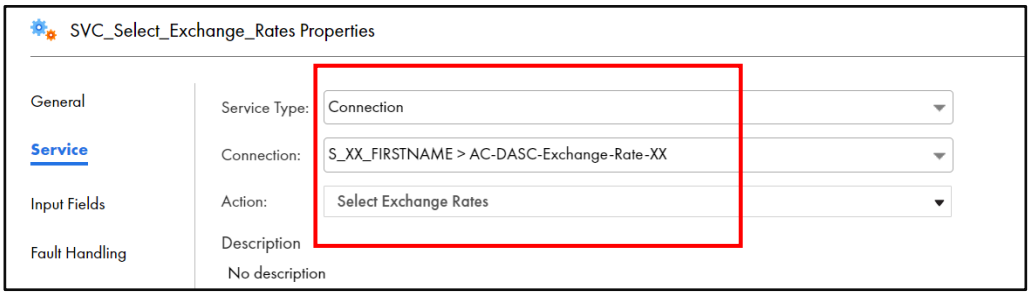
56. From the Design palette, drag and drop a **Service** step on the link between the SQL Currency List and End steps.



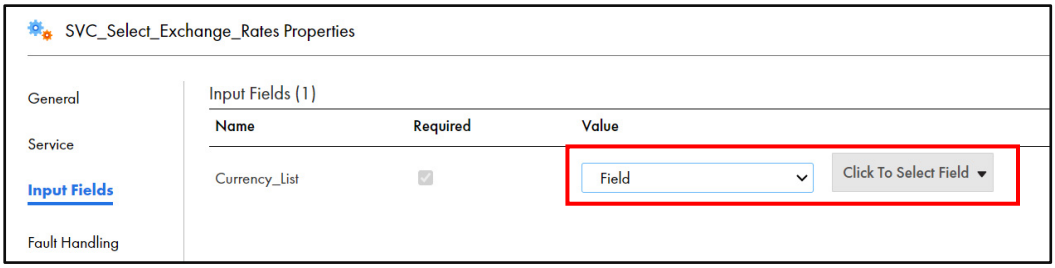
57. To configure the service, from the process canvas, select the **Service** step.
58. In the General section of the Service properties, enter the Name as **SVC_Select_Exchange_Rates**.



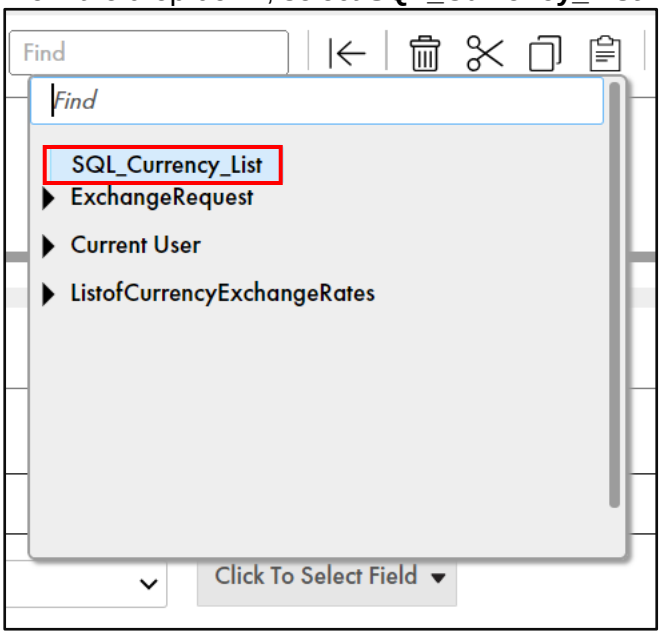
59. From the properties pane, select **Service**.
60. From the Service Type drop-down, select **Connection**.
61. From the Connection drop-down, expand your project folder, and select **AC-DASC-Exchange-Rate-XX.**
62. From the Action drop-down, select **Select Exchange Rates**.



63. From the properties pane, select **Input Fields**.
64. From the Value drop-down, select **Field**.
65. Click on **Click To Select Field**.

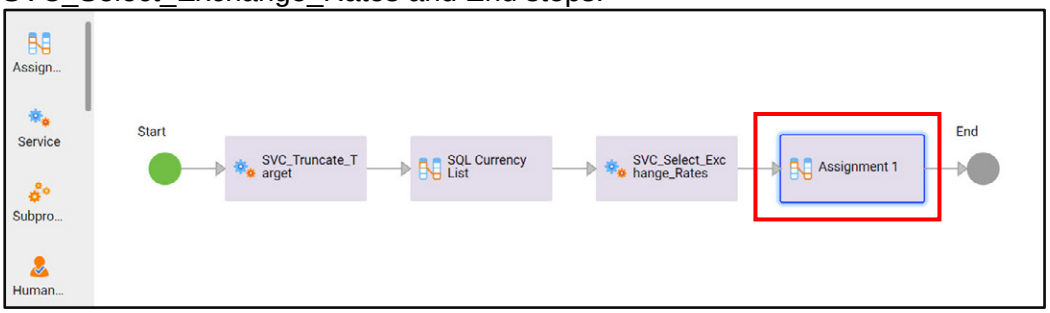


66. From the drop-down, select **SQL_Currency_List**.

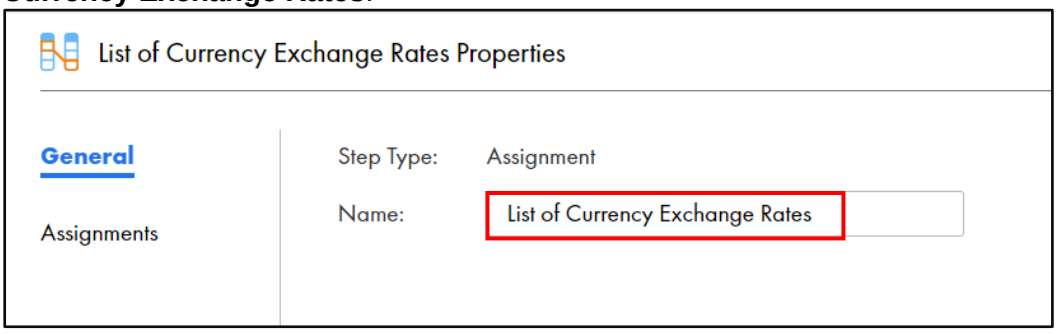


Now, you will use create an Assignment step to display the list of currency exchange rates.

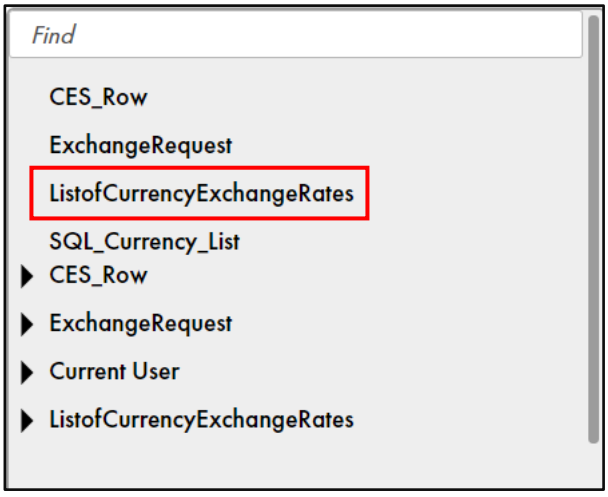
67. From the Design palette, drag and drop an **Assignment** step on the link between the SVC_Select_Exchange_Rates and End steps.



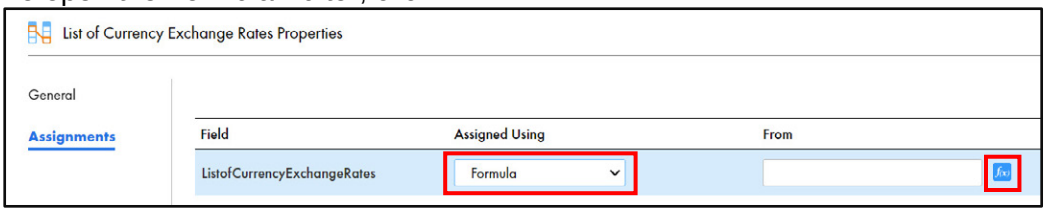
68. To configure the step, from the process canvas, select the **Assignment** step.
69. In the General section of Assignment properties, enter the Name as **List of Currency Exchange Rates**.



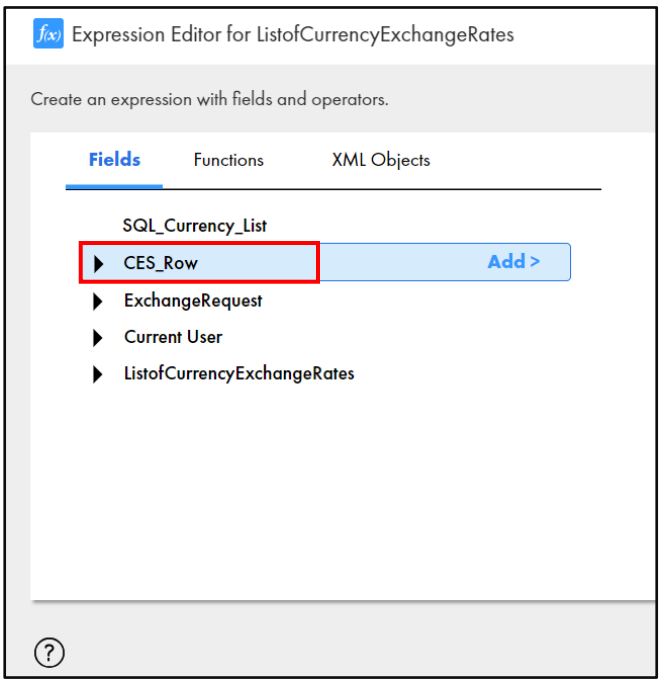
70. From the properties pane, select **Assignments**.
71. Click the **+** icon and from the drop-down, select **ListofCurrencyExchangeRates**.



72. From the Assigned Using drop-down, select **Formula**.
73. To open the Formula Editor, click .



74. Select **CES_row**.



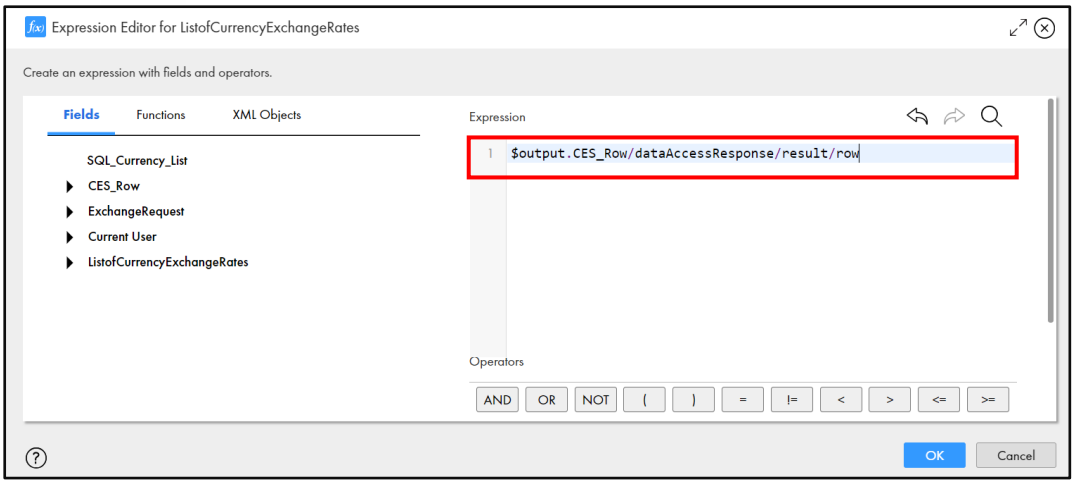
Note: If you can't see the CES_row, click Cancel to close the formula window. Save and close the process. Click the Explore tab and reopen the process.

75. In the Expression field, enter the following expression (in continuation):
    /dataAccessResponse/result/row

    (You must see the complete expression as -
    $output.CES_Row/dataAccessResponse/result/row)

    OR

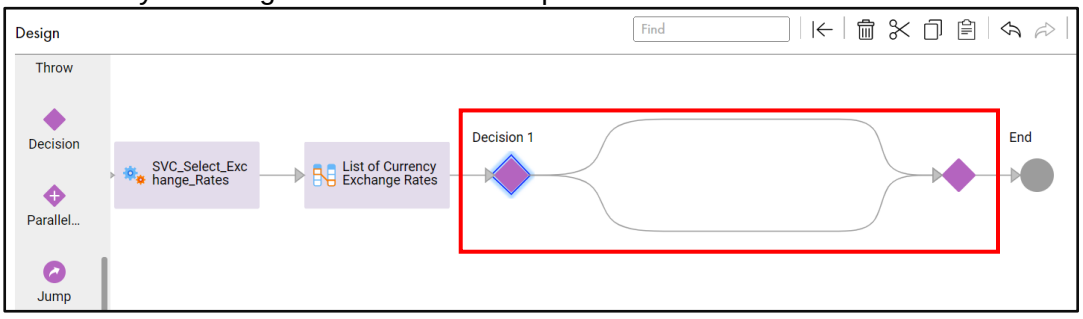
    From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **10_Using Currency Exchange Data Access Service Connector in a Process_6-3**. Copy the command mentioned under **Step B** and paste it into the Expression field.



76. Click **OK**.
77. **Save** the process.

Next, you will add a Decision Step.

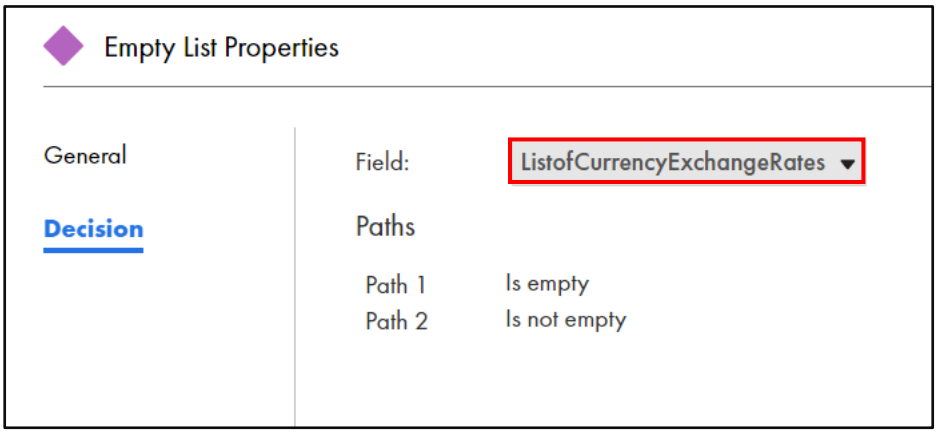
78. From the Design palette, drag and drop a **Decision** step on the link between the List of Currency Exchange Rates and End steps.



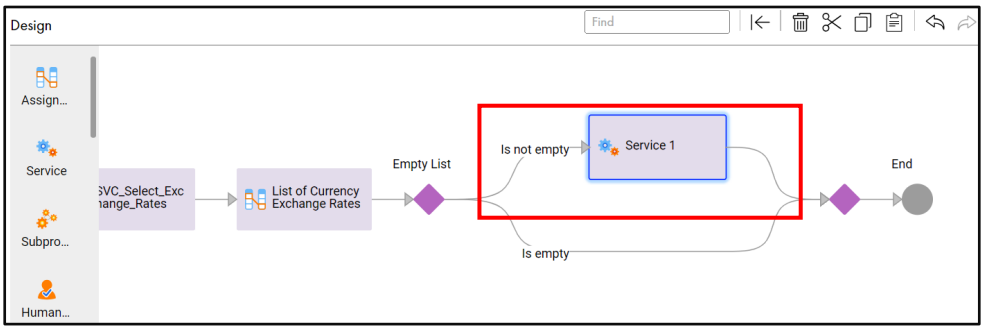
79. To configure the step, from the process canvas, select the **Decesion** step.
80. In the General section of Decesion properties, enter the Name as **Empty List**.

81. From the properties pane, select **Decision**.

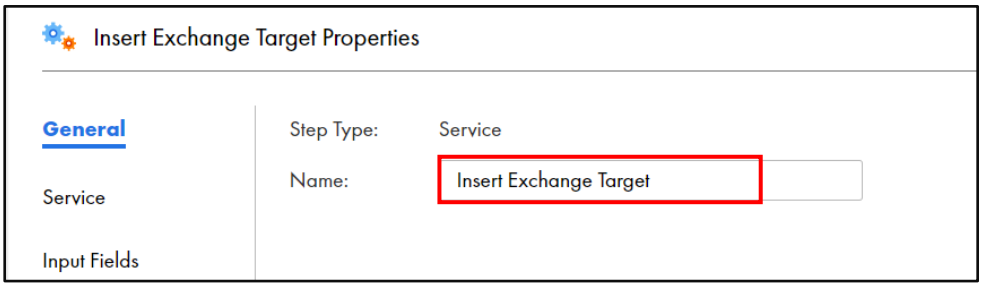
82. From the field drop-down, select **ListofCurrencyExchangeRates**.



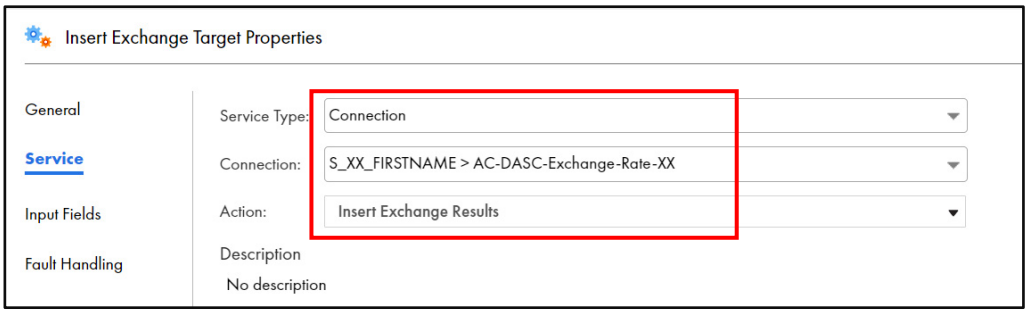
83. From the Design palette, drag and drop a **Service** step on the Is not empty branch of the Decision step.



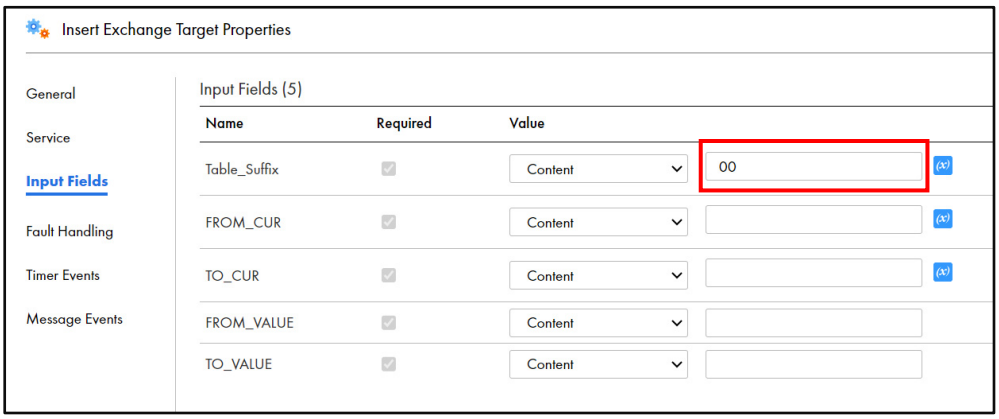
84. In the General section of Service properties, enter the Name as **Insert Exchange Target**.



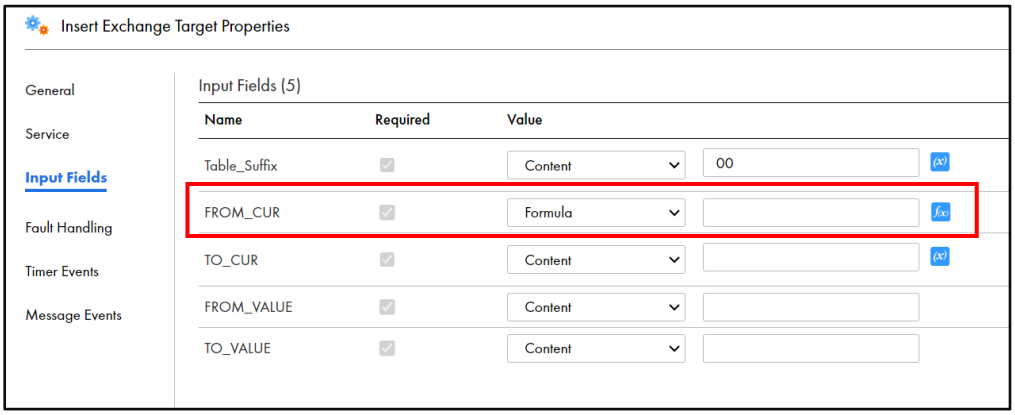
85. From the properties pane, select **Service**.
86. From the Service Type drop-down, select **Connection**.
87. From the Connection drop-down, expand your project folder, and select **AC-DASC-Exchange-Rate-XX.**
88. From the Action drop-down, select **Insert Exchange Results**.



89. From the properties pane, select **Input Fields**.
90. In the Table_Suffix field, enter your student number.

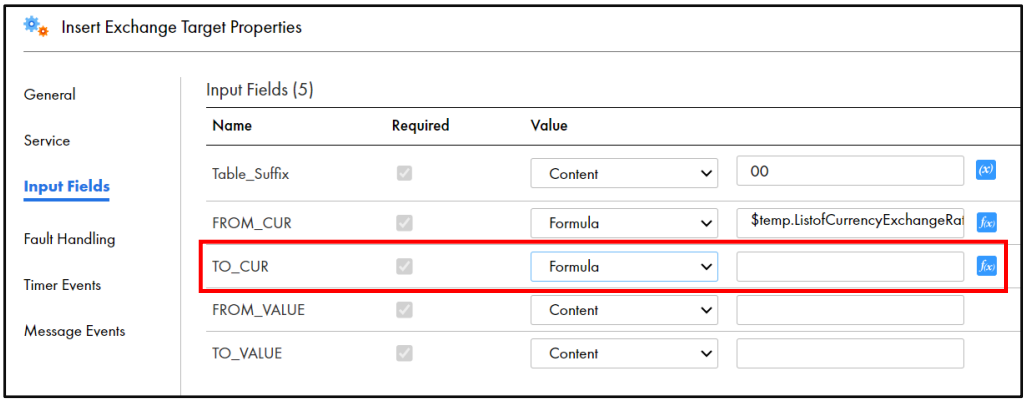


91. In the **FROM_CUR** field, from the Value drop-down, select **Formula**.
92. To open the Formula Editor, click .



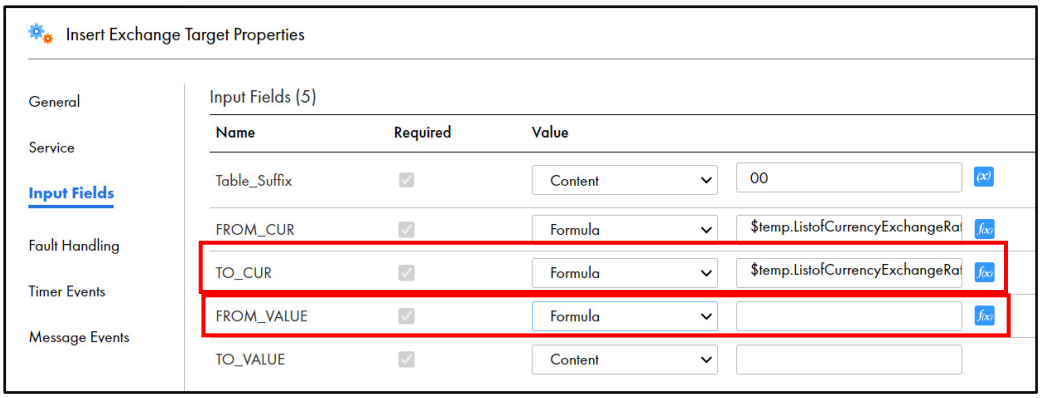
93. Expand **ListofCurrencyExchangeRates**.
94. Select **FROM_CUR** and click **Add>**. Click **OK**.

95. In the TO_CUR field, from the Value drop-down, select **Formula**.
96. To open the Formula Editor, click .



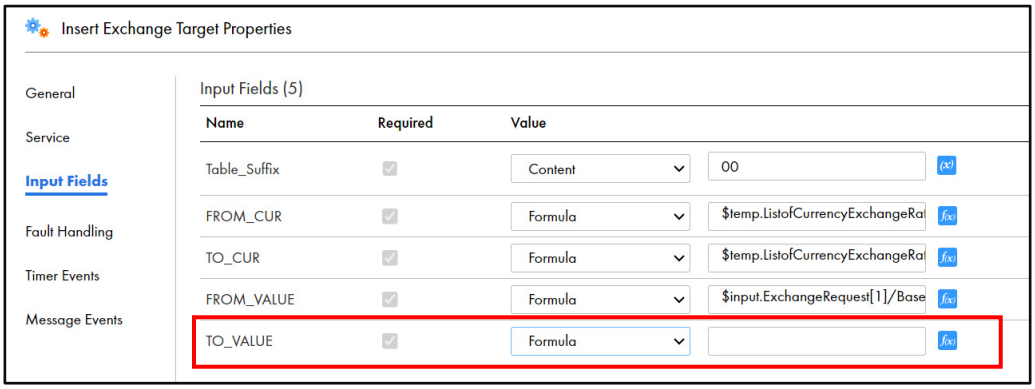
97. Expand **ListofCurrencyExchangeRates**.
98. Select **TO_CUR** and click **Add>**. Click **OK**.

99. In the FROM_VALUE field, from the Value drop-down, select **Formula**.
100. To open the Formula Editor, click .

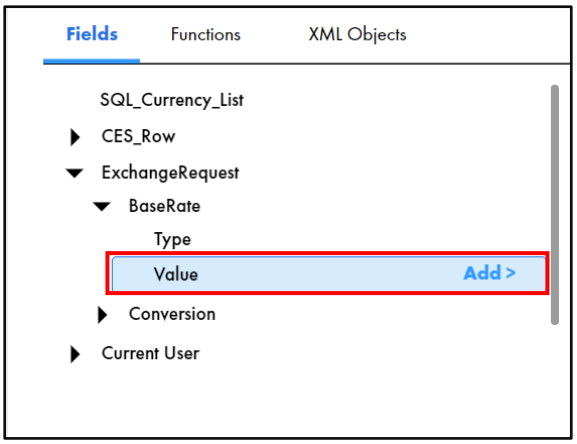


101. Select **ExchangeRequest** > **BaseRate** > **Value**.
102. Click **Add>**. Click **OK**.

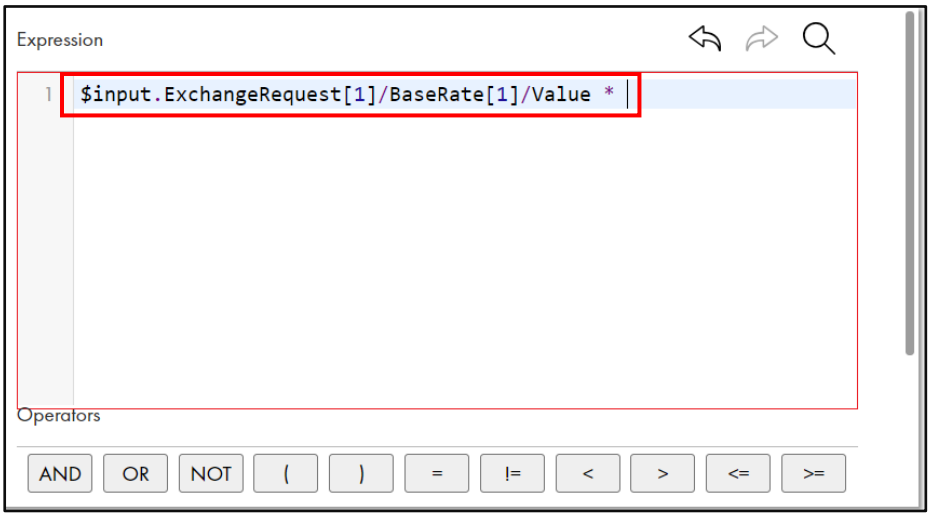
103. In the TO_VALUE field, from the Value drop-down, select **Formula**.
104. To open the Formula Editor, click .



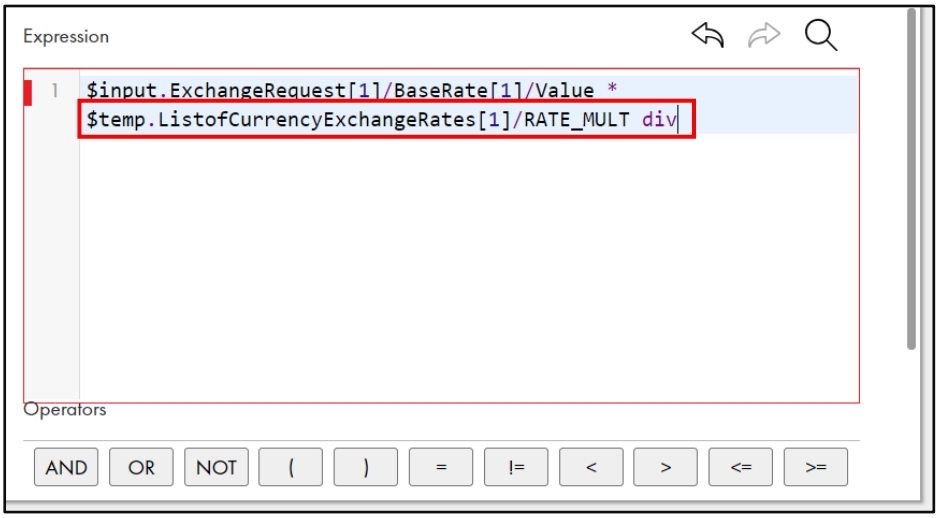
105. Select **ExchangeRequest** > **BaseRate** > **Value**.
106. Click **Add>**.



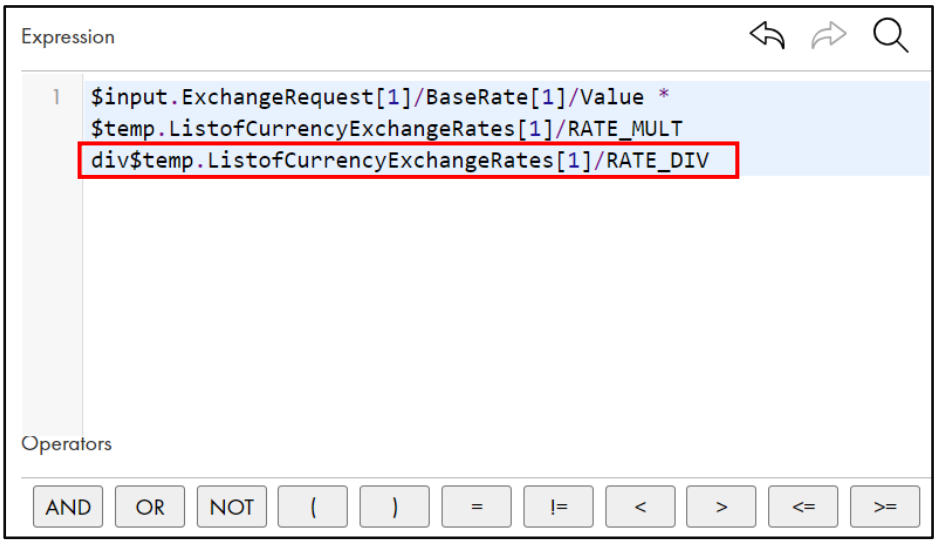
107. In the expression window, enter ***** in the end of the expression.



108. Expand **ListofCurrencyExchangeRates**, select **RATE_MULT**, and click **Add>**.
109. In the expression window, enter **div** in the end of the expression.



110. Expand **ListofCurrencyExchangeRates**, select **RATE_DIV**, and click **Add>**.



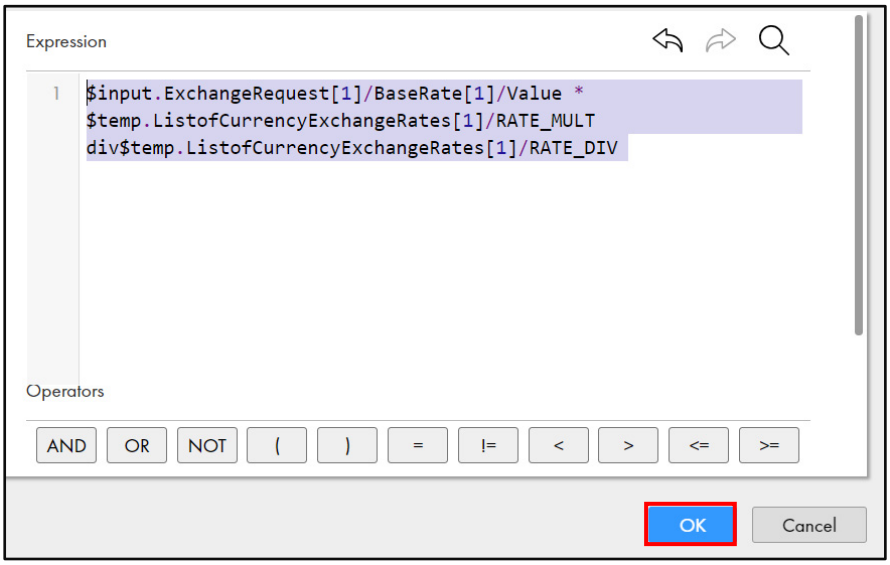
Note: In the end, your expression should look like this:

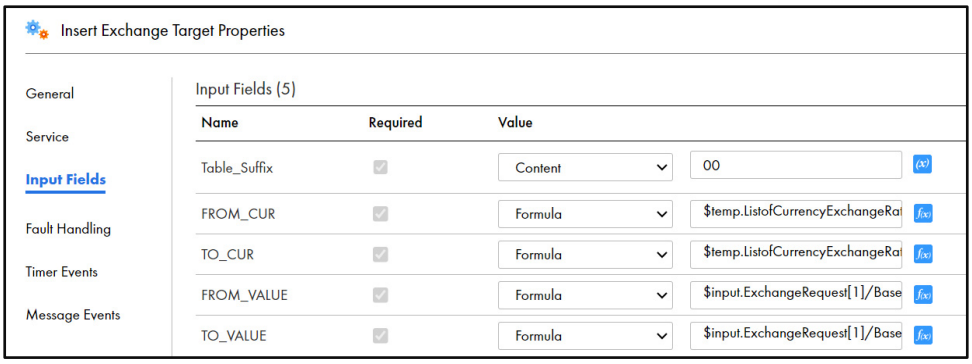
**$input.ExchangeRequest[1]/BaseRate[1]/Value *
$temp.ListofCurrencyExchangeRates[1]/RATE_MULT
div$temp.ListofCurrencyExchangeRates[1]/RATE_DIV**

OR

From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **10_Using Currency Exchange Data Access Service Connector in a Process_6-3**. Copy the command mentioned under **Step C** and paste it into the Expression field.

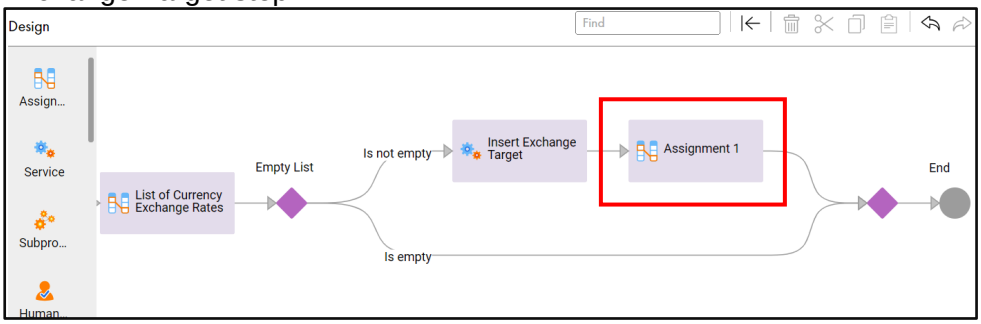
111. Click **OK**.



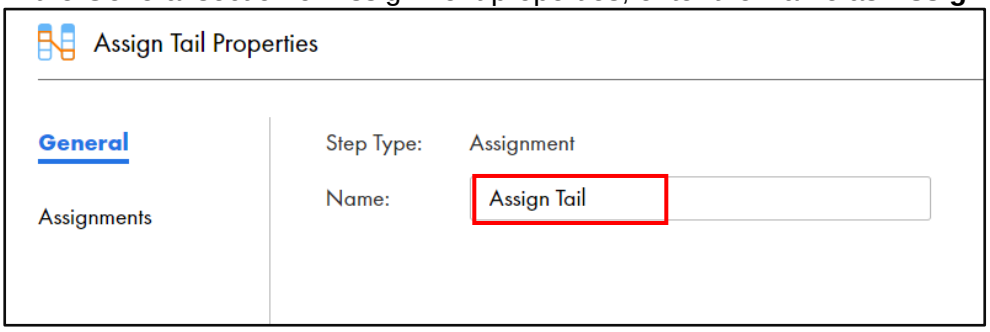


Next, you will add an Assignment Step to apply the Tail function to the list of exchange rates.

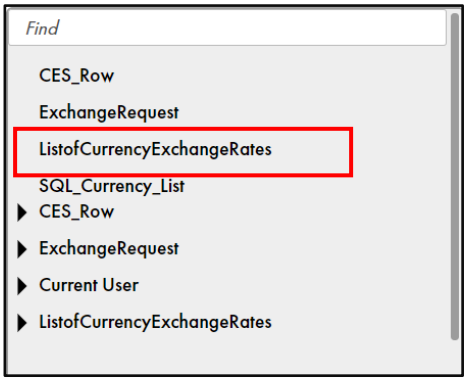
112. From the Design palette, drag and drop an **Assignment** step next to the Insert Exchange Target step.



113. To configure the step, from the process canvas, select the **Assignment** step.
114. In the General section of Assignment properties, enter the Name as **Assign Tail**.



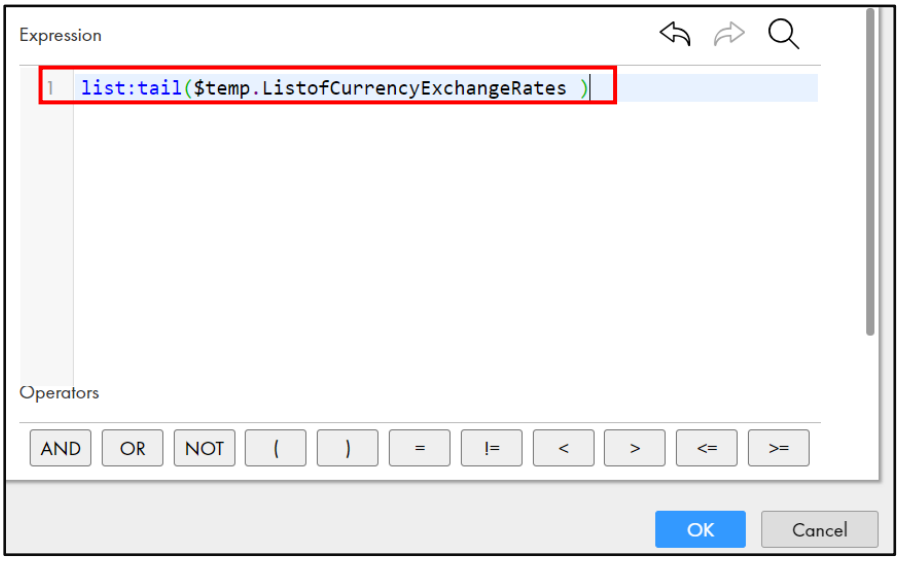
115. From the properties pane, select **Assignments**.
116. Click the **+** icon and from the drop-down, select **ListofCurrencyExchangeRates**.



117. From the Assigned Using drop-down, select **Formula**.
118. To open the Formula Editor, click .
119. In the Expression window, enter the following:
     list:tail($temp.ListofCurrencyExchangeRates)

     OR

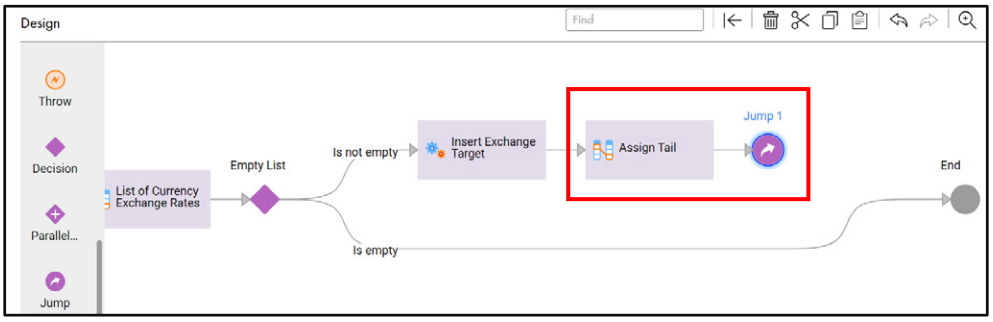
     From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **10_Using Currency Exchange Data Access Service Connector in a Process_6-3**. Copy the command mentioned under **Step D** and paste it into the Expression field.



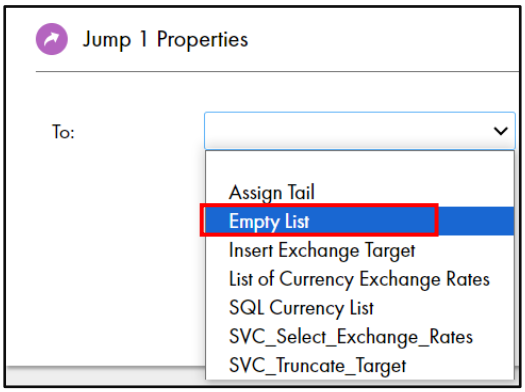
120. Click **OK**.

Next, you will add a Jump step. It will take you back to the Decision step when the list is empty.

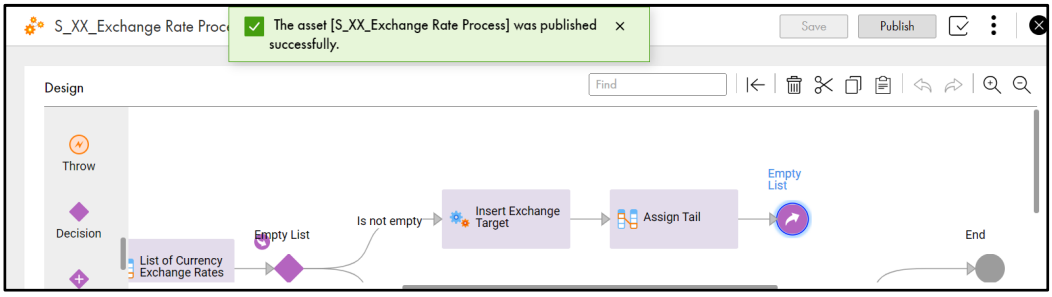
121. From the Design palette, drag and drop a **Jump** step next to the Assign Tail step.



122. From the Jump properties, select **Empty List** from the To drop-down.

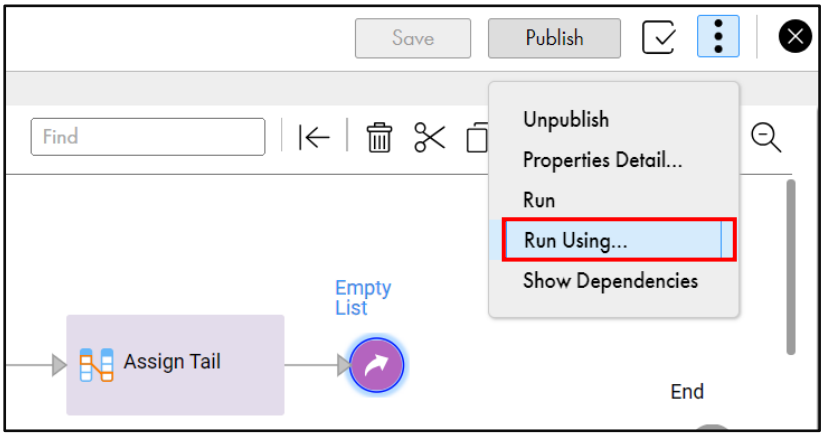


123. Save and publish the process.



Execute The Process

124. click ⋮ and select **Run Using…**.

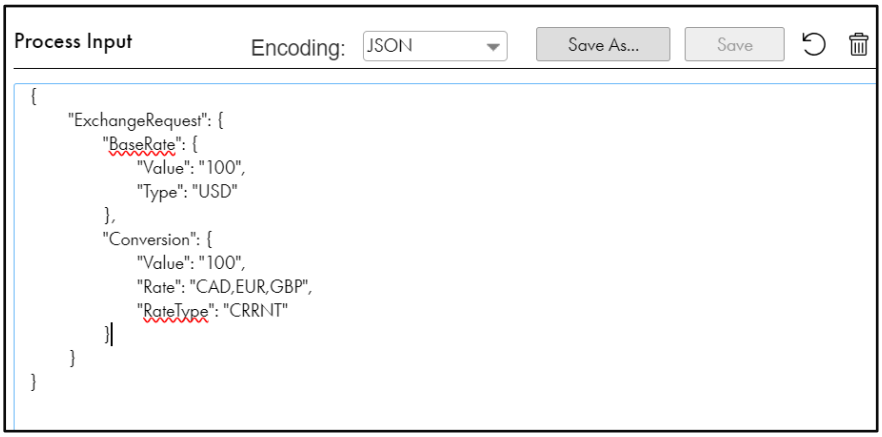


125. In the ExchangeRequest section, in the following inputs:

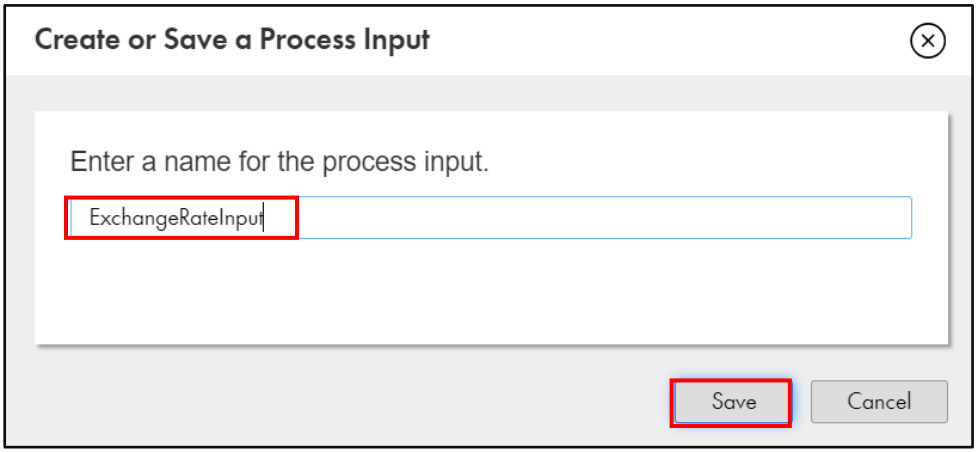
Value	100
Type	USD

126. In the Conversion section, enter the following inputs:

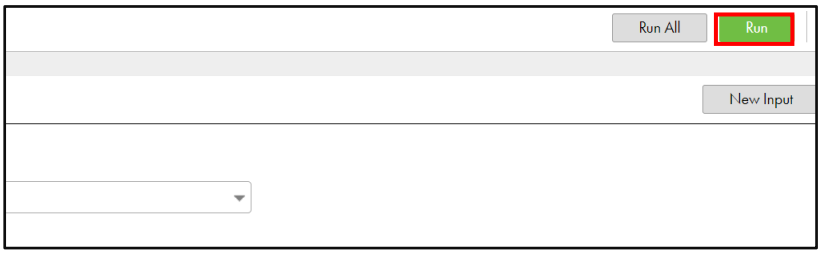
Value	100
Rate	CAD,EUR,GBP
RateType	CRRNT



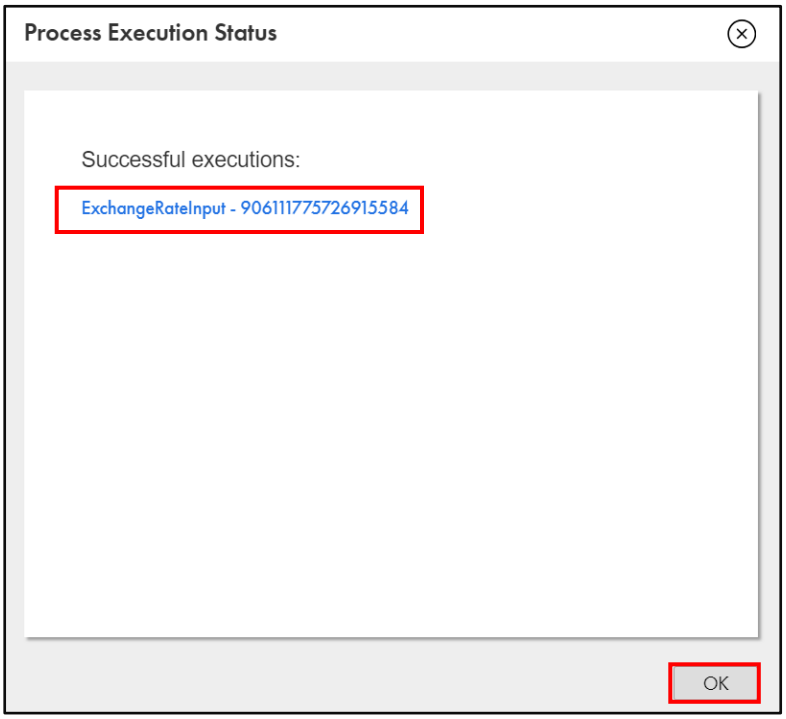
127. Click **Save As**.
128. Save the input as **ExchangeRateInput** and click **Save**.



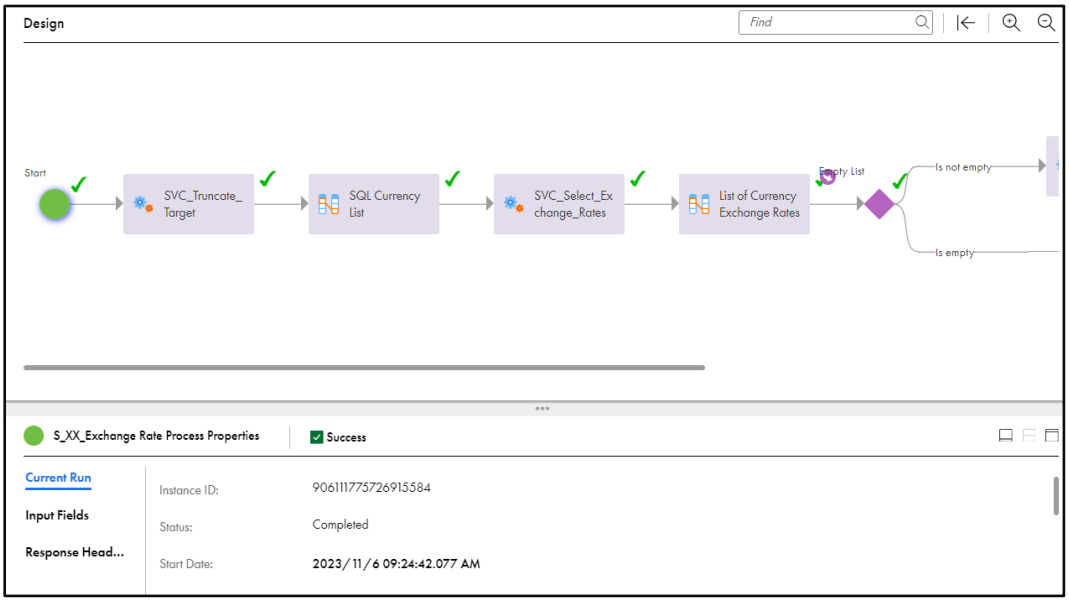
129. Click **Run**.



130. Select the Process Execution link and click **OK**.

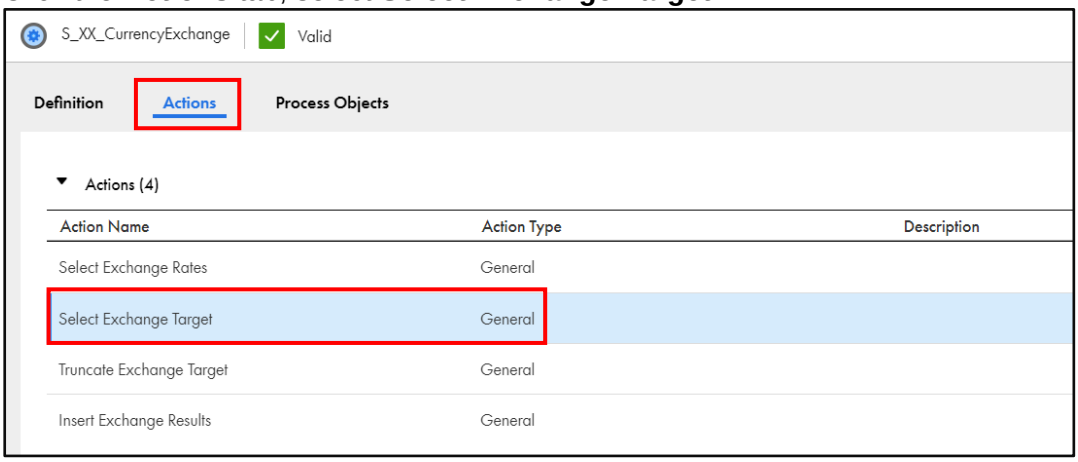


You can see the successful execution of the process.

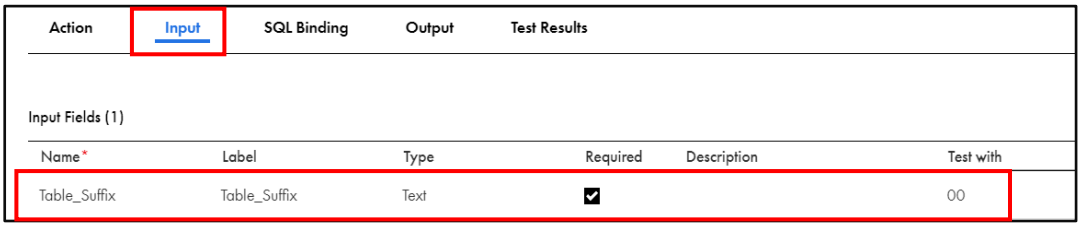


Test the Results

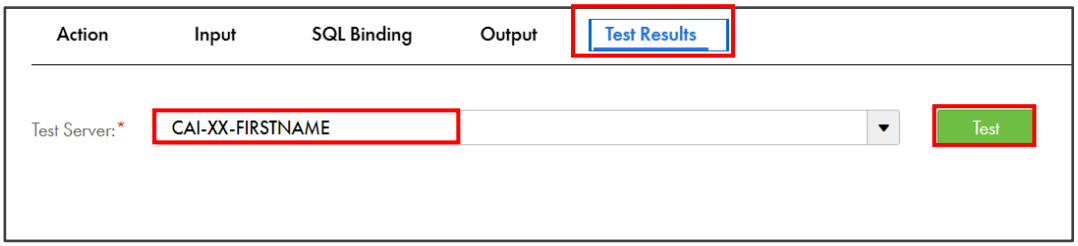
131. To test the results, from your work folder open the **S_XX_CurrencyExchange** data access service connector.
132. Click the **Actions** tab, select **Select Exchange Target**.



133. Click **Input** and ensure that your student number is entered in the **Test with** field.



134. Go to the **Test Results** tab.
135. From the Test Server drop-down, select your Secure Agent group and click **Test**.



136. From the Output Fields section, scroll down to the **Response Payload** field and observe the output. You should see the exchange rate of three currencies that you specified.

This concludes the lab.

Module 7: App Connections
Lab 7-1: Create an App Connection Using Service Connector

Overview:
An App Connection defines a link to a data source, a database, or a Service Connector. In CAI, an App Connection can use any published out-of-the-box connector or Service Connector.

This is **part 2** of the set of three labs. In this lab, you will create an app connection that utilizes the service connector created in the exercise **Lab 5-1: Create a Service Connector**.

Objectives:
- Create an App Connection using Service Connector
- Publish the App Connection

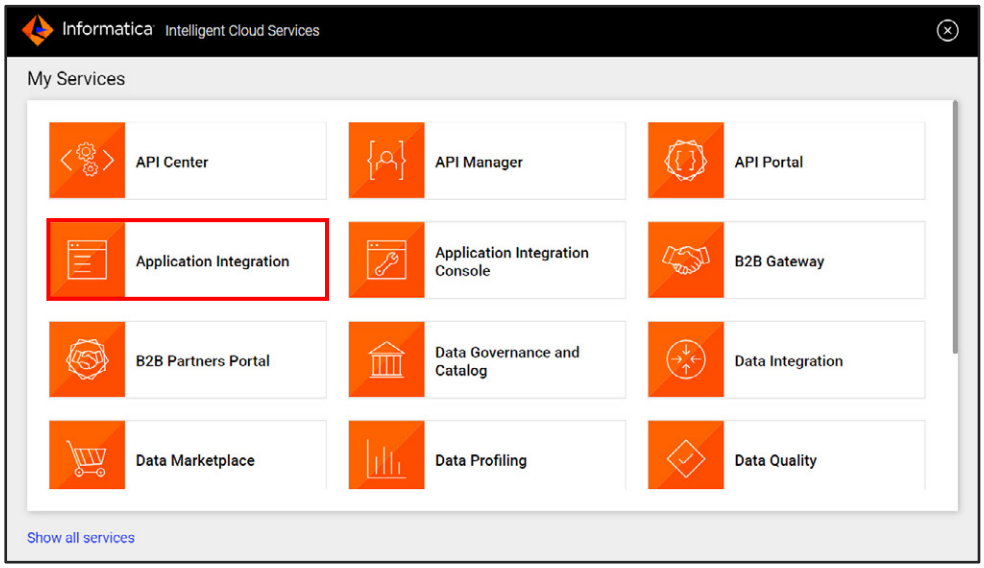
Duration:
10 Minutes

Tasks

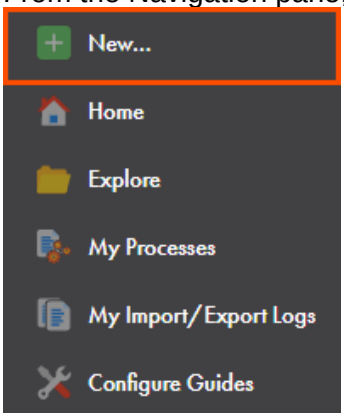
Note: The screen on your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

Create a connection:

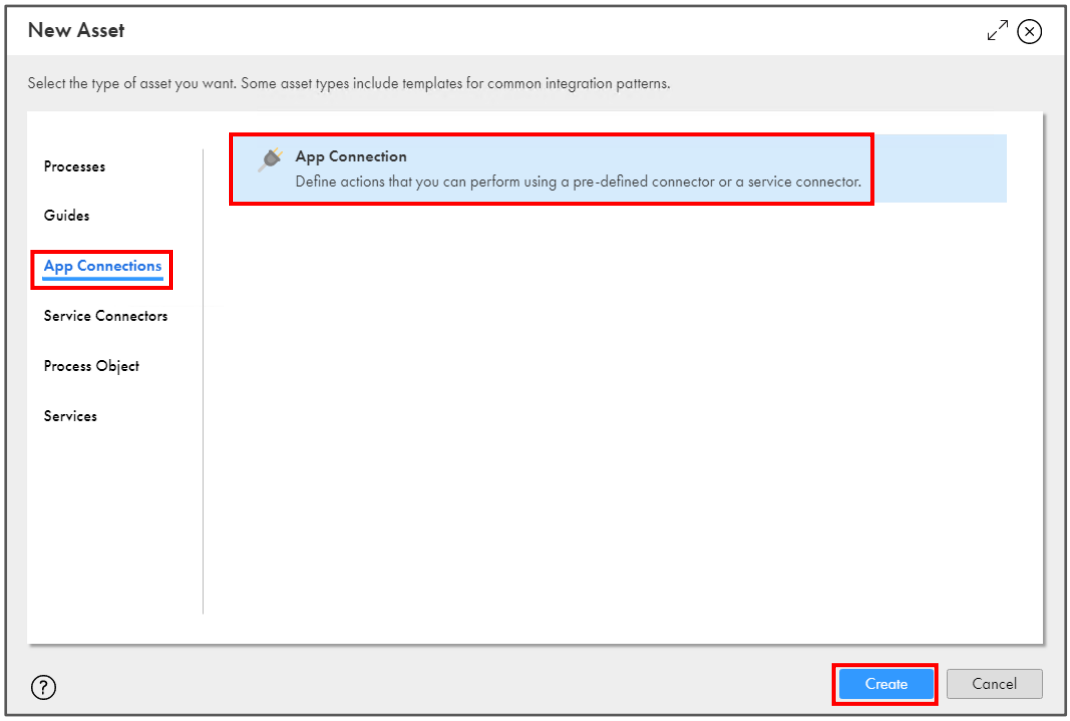
1. Log into IICS and access the **Application Integration** service.



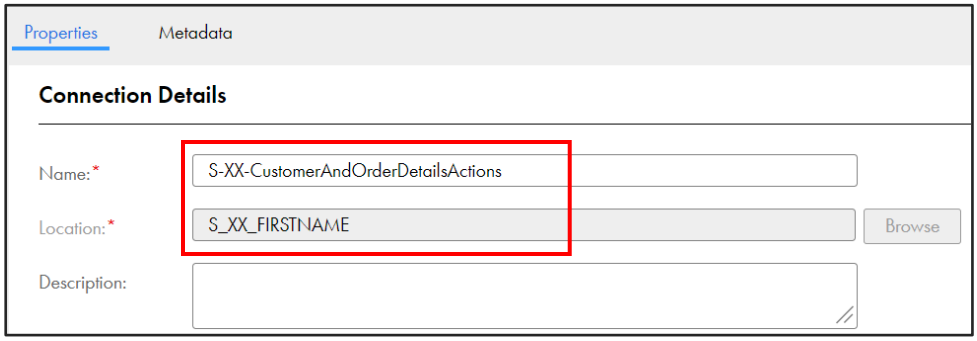
2. From the Navigation pane, click **New**.



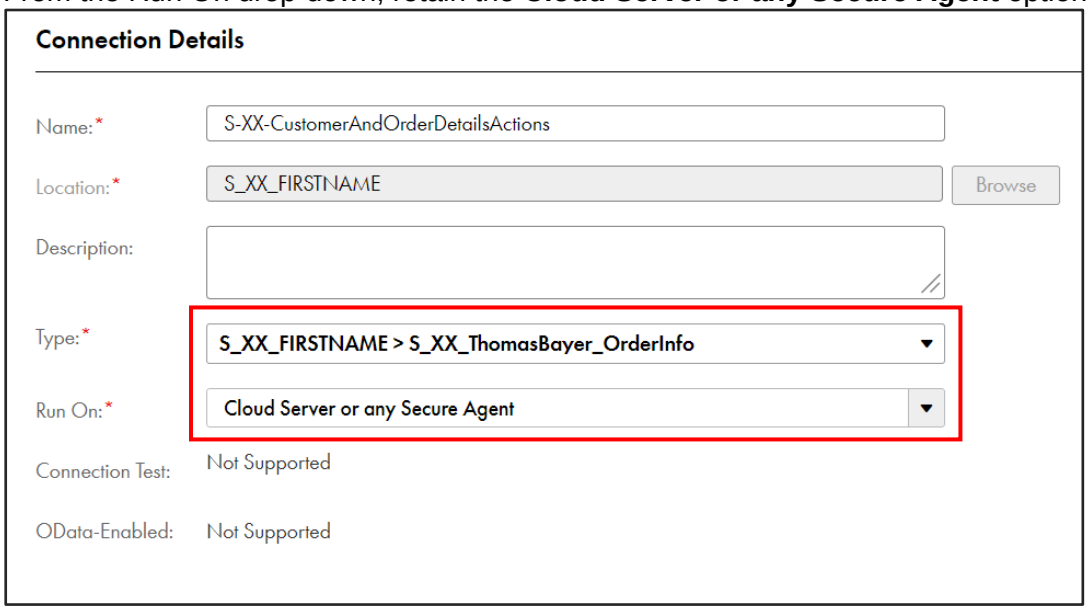
3. In the New Asset window, from the App Connections tab, select **App Connection**, and click **Create**.



4. In the Connection Details page, enter the connection name as **S-XX-CustomerAndOrderDetailsActions**.
5. In the Location field, select your project folder **S_XX_FirstName**.

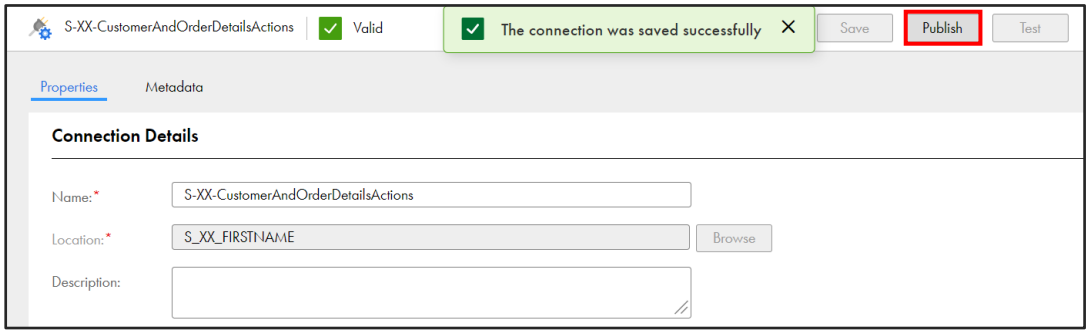


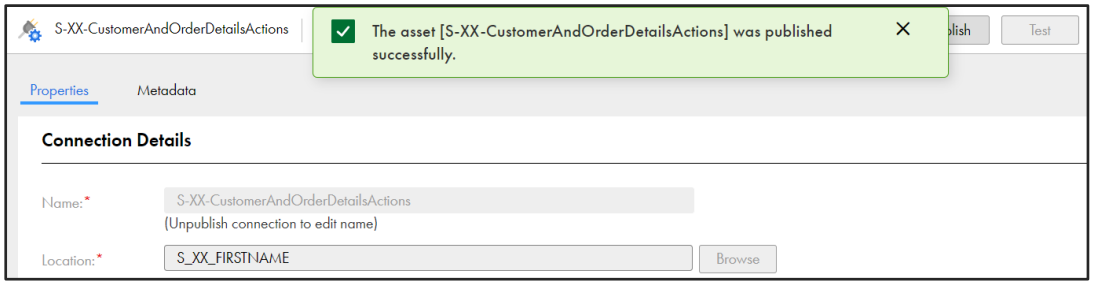
6. From the Type drop-down, expand your project folder, **S_XX_FirstName**, and select **S_XX_ThomasBayer_OrderInfo**.
7. From the Run On drop-down, retain the **Cloud Server or any Secure Agent** option.



8. Click **Save**.

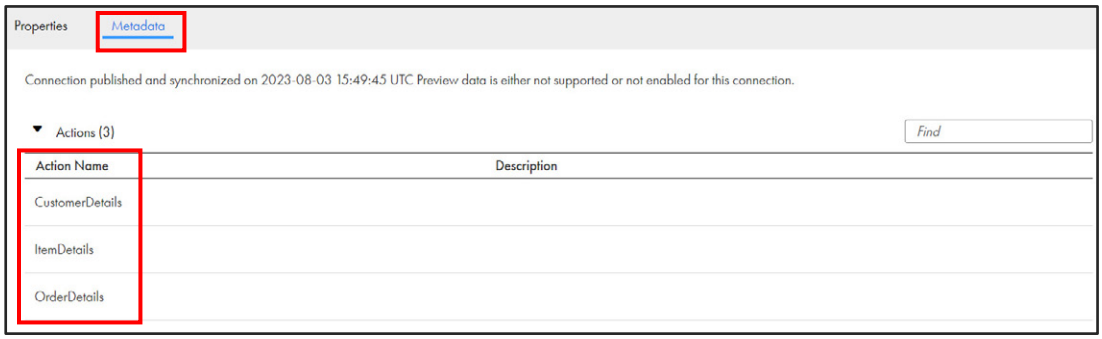
9. Click **Publish**.





You get a notification that "**The asset S-XX-CustomerAndOrderDetailsActions was published successfully.**"

10. In the Metadata tab, observe that the Actions from the Service Connector are added to the Actions section.



Note: If you can't see the Actions in the metadata tab, refresh the page. If you still can't see the Actions, then unpublish the connection and publish it once again.

This concludes the lab.

Module 7: App Connections

Lab 7-2: Create a JDBC Connection

Overview:
Informatica CAI allows you to connect to the databases with the help of the JDBC connector. The databases include IBM DB2, Microsoft SQL Server, MySQL, Oracle, and PostgreSQL.

In this lab, you will create a connection using the built-in JDBC Connector and use that connection in the ExchangeRate process to fetch data from the database table and perform conversion based on the user input values.

Note: This lab is **part 2** of a set of three labs. Earlier, you created process objects (Lab 4-1: Create Process Objects). This lab is continued in **Lab 10-1: Invoke a Synchronous Web Service Call** of Module 10.

Objectives:
- Create a JDBC connection
- Publish the connection

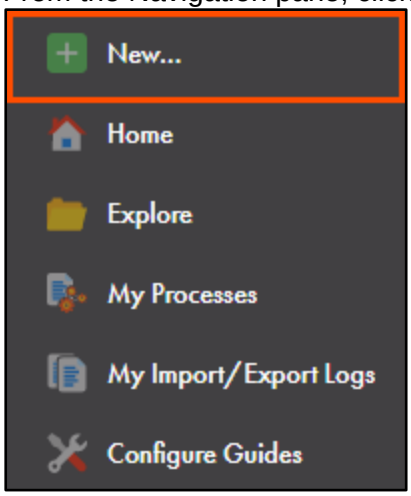
Duration:
15 Minutes

Tasks:

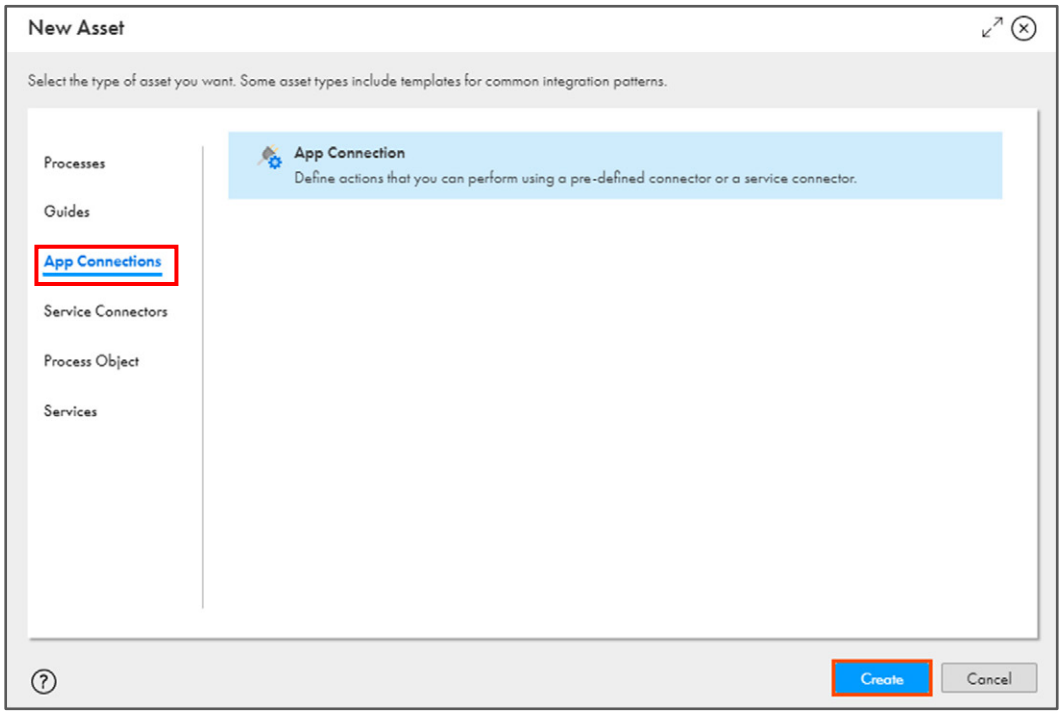
Note: The screen on your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

Create a connection:

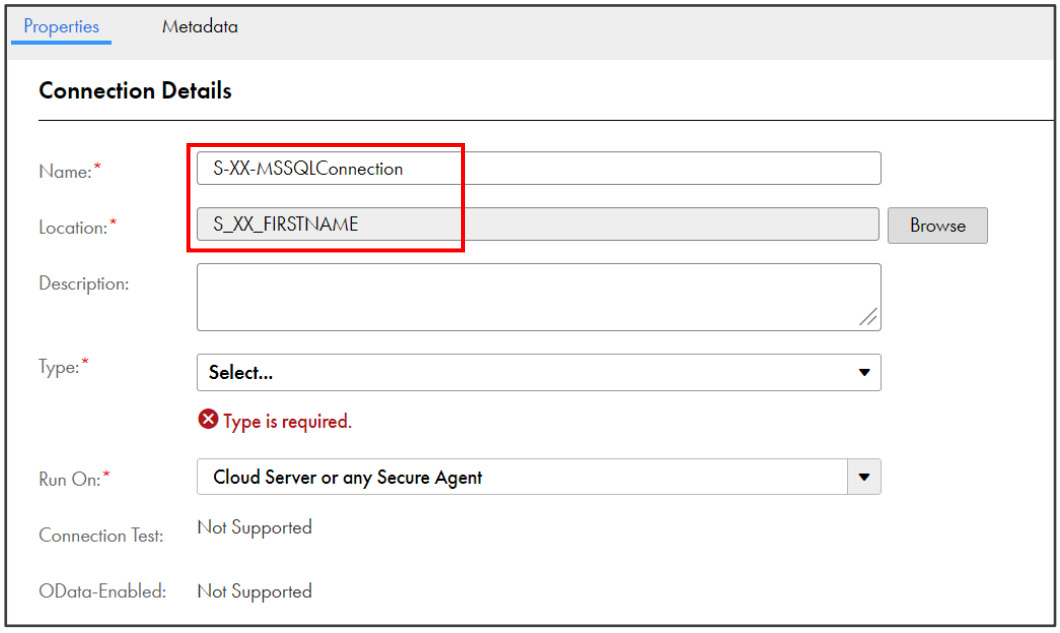
1. Log in to IICS and access the **Application Integration** service.
2. From the Navigation pane, click **New**.



3. In the New Asset window, from the App Connections tab, select **App Connection**, and click **Create**.

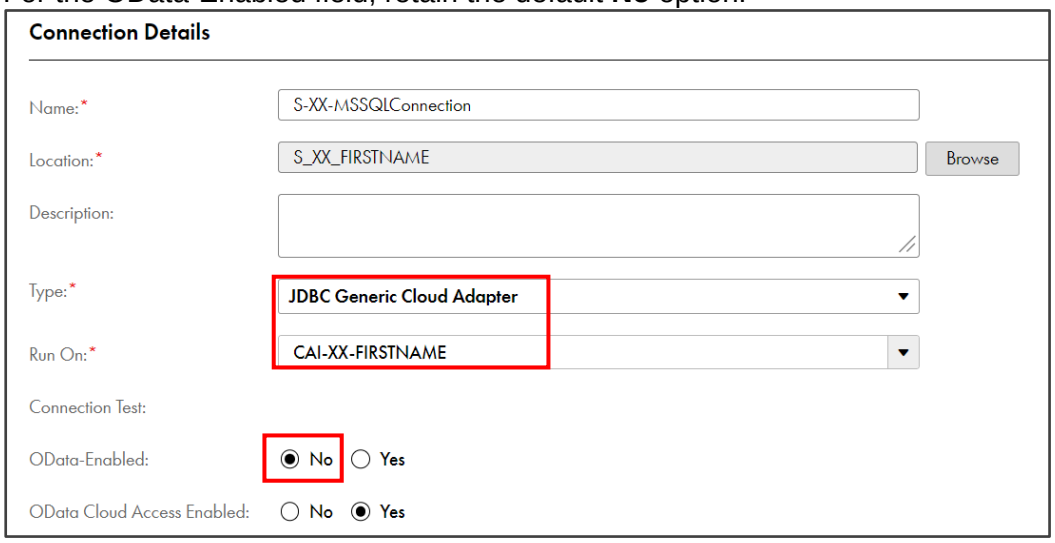


4. In the Connection Details page, enter the connection name as **S-XX-MSSQLConnection**.
5. In the Location field, select your project folder **S_XX_FirstName**.



6. From the Type drop-down, select **JDBC Generic Cloud Adapter**.
7. From the Run On drop-down, select your Secure Agent named in the format **CAI-XX-FirstName**.

8. For the OData-Enabled field, retain the default **No** option.



9. To configure JDBC connection properties, in the JDBC Connection URL field, enter **jdbc:sqlserver://localhost:1433;databaseName=CAI;default=master**

   Note: Hover over and click on the field to enter the value.

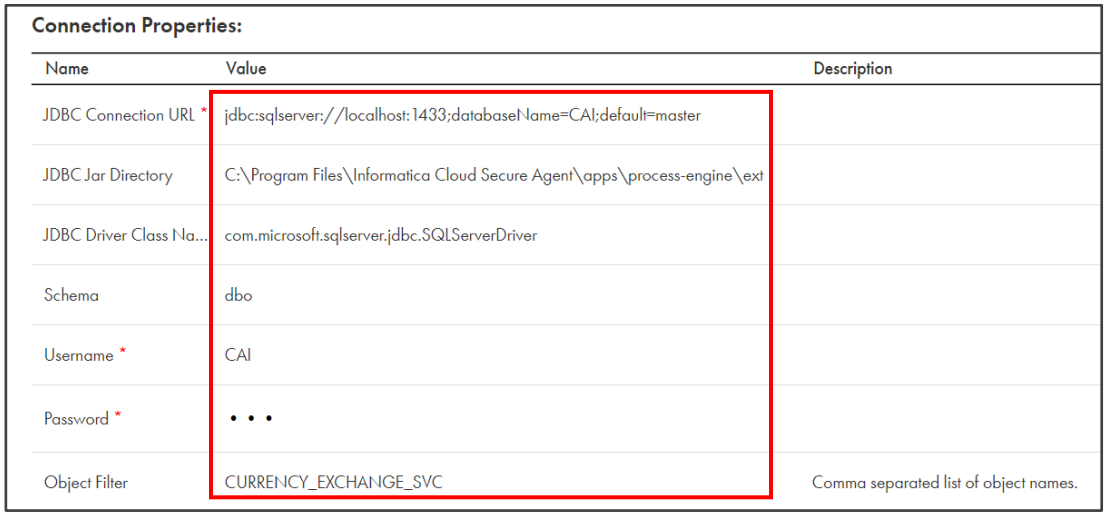
10. In the JDBC Jar Directory field, enter **C:\Program Files\Informatica Cloud Secure Agent\apps\process-engine\ext**

11. In the JDBC Driver Class Name field, enter **com.microsoft.sqlserver.jdbc.SQLServerDriver**

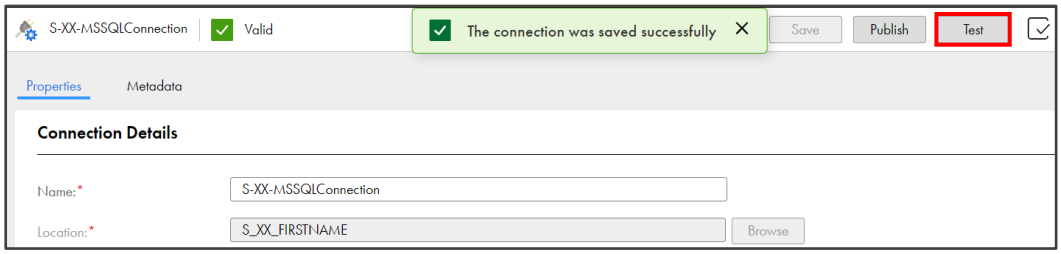
12. In the Schema field, enter **dbo**.

13. Enter **CAI** in the Username and Password fields.

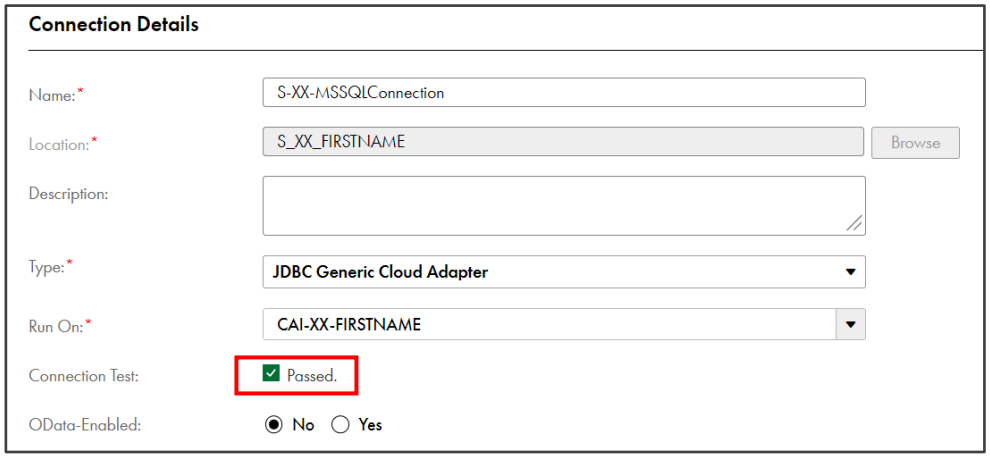
14. In the Object Filter field, enter **CURRENCY_EXCHANGE_SVC**



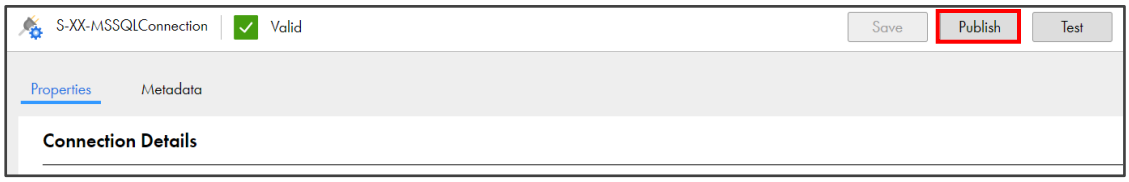
15. **Save** the connection.
16. **Test** the connection.



The Connection Test field mentions **Passed**.

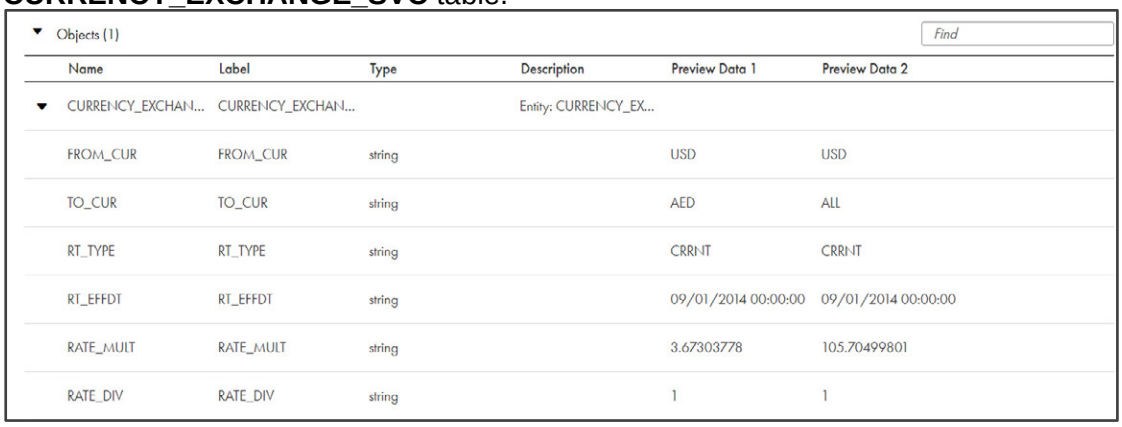


17. **Publish** the connection.



You get a notification that '**The asset [S-XX-MSSQLConnection] was published successfully**'.

18. In the Metadata tab, observe that the Objects section displays the data preview for the **CURRENCY_EXCHANGE_SVC** table.



Objects (1) — Find

Name	Label	Type	Description	Preview Data 1	Preview Data 2
CURRENCY_EXCHAN...	CURRENCY_EXCHAN...		Entity: CURRENCY_EX...		
FROM_CUR	FROM_CUR	string		USD	USD
TO_CUR	TO_CUR	string		AED	ALL
RT_TYPE	RT_TYPE	string		CRRNT	CRRNT
RT_EFFDT	RT_EFFDT	string		09/01/2014 00:00:00	09/01/2014 00:00:00
RATE_MULT	RATE_MULT	string		3.67303778	105.70499801
RATE_DIV	RATE_DIV	string		1	1

Note: Refresh the page if you can't see the data preview.

You have successfully established a connection between Informatica CAI and the database.

This concludes the lab.

Module 7: App Connections
Lab 7-3: Create a Customer and Order Details Process

Overview:

This is **part 3** of the set of three labs. In Lab 5-1: Create a Service Connector, you created a service connector from a publicly available REST Web Service. You also created an app connection (Lab 7-1: Create an App Connection using Service Connector) using the same service connector.

In this lab, you will create a process to obtain the customer and order details through the Order id, using the third-party OData service.

Objectives:

- Create a new Process
- Add and configure service steps
- Publish and execute the Process

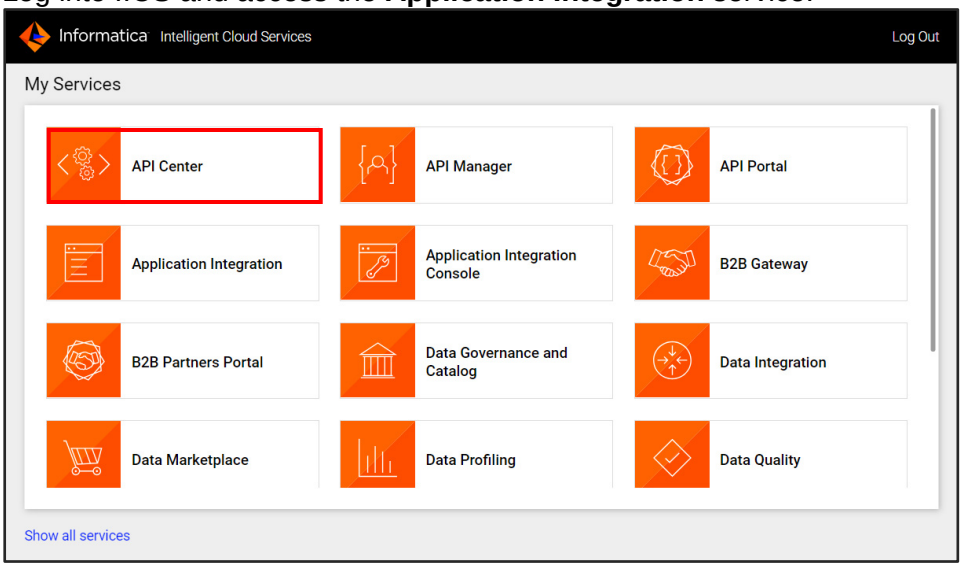
Duration:

30 Minutes

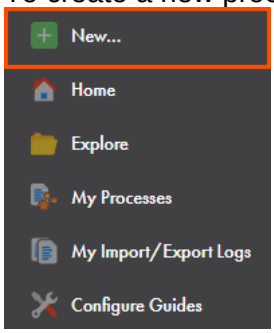
Tasks

Create a new process:

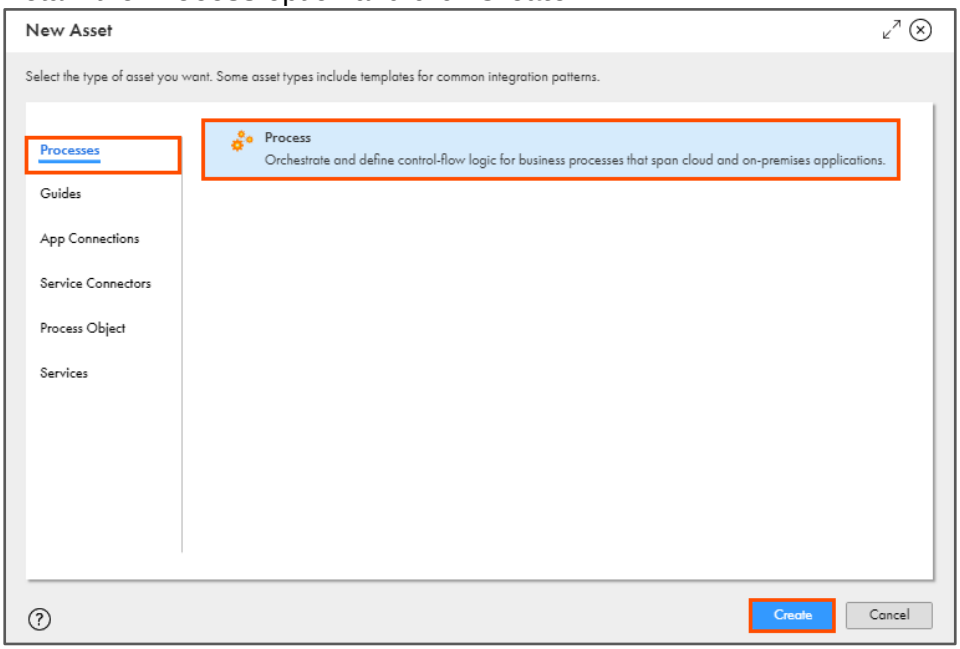
1. Log into IICS and access the **Application Integration** service.



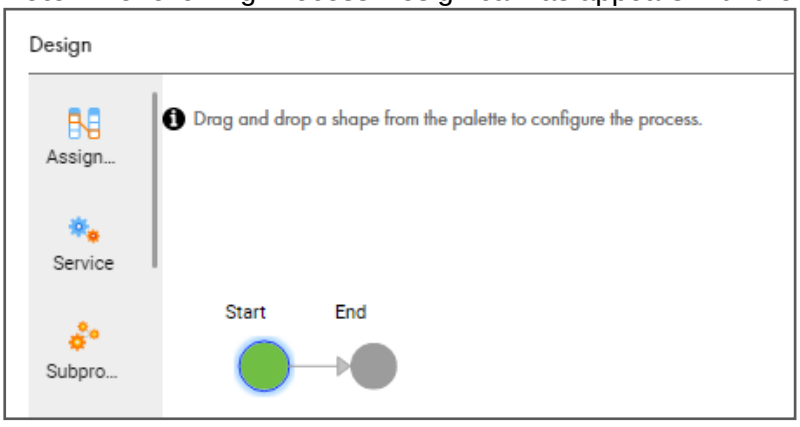
2. To create a new process, from the navigation pane, select **New**.



3. Retain the **Process** option and click **Create**.

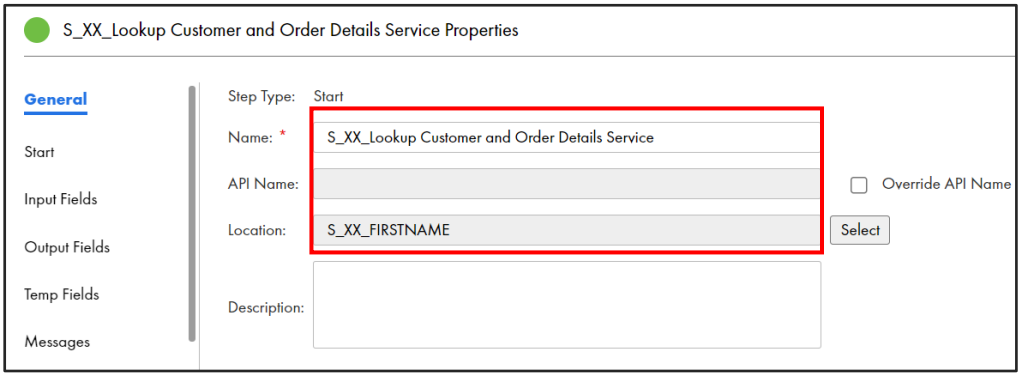


Note: The following Process Design canvas appears with the Start and End steps.

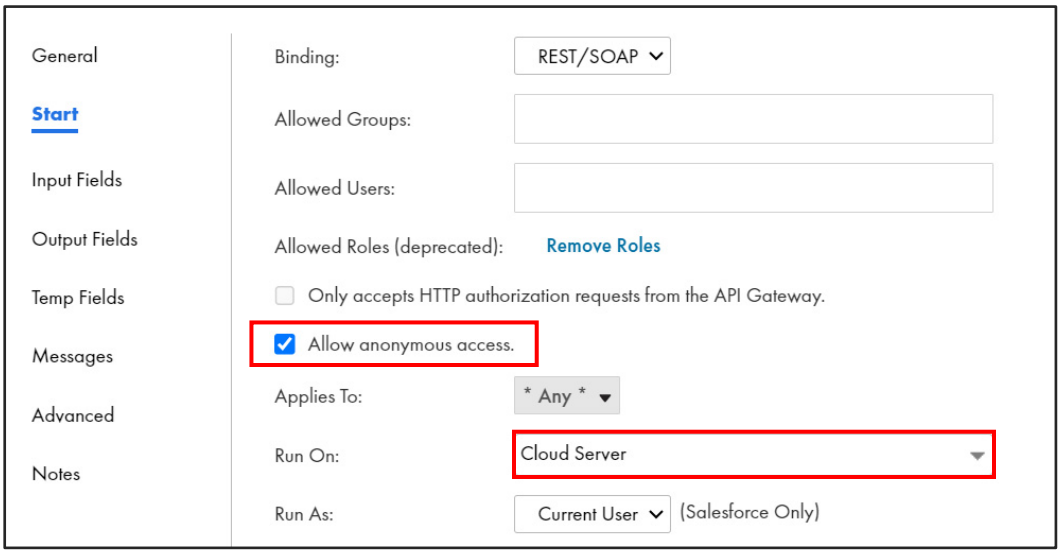


4. From the process canvas, select the **Start** step.
5. In the General tab, enter the name as **S_XX_Lookup Customer and Order Details Service**.

   Note: Replace XX with your student number.
6. Skip the API Name option.
7. In the Location field, select your project folder **S_XX_FirstName**.

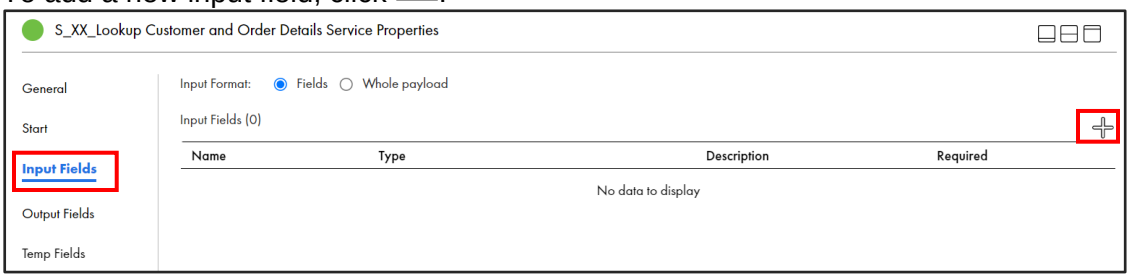


8. From the properties pane, select **Start**.
9. Select the **Allow anonymous access** option.
10. From the Run On drop-down, retain **Cloud Server**.
    Note: You can select either Cloud Server or your secure agent. In this case, you are selecting Cloud Server as the previous service connector was tested and published on cloud.



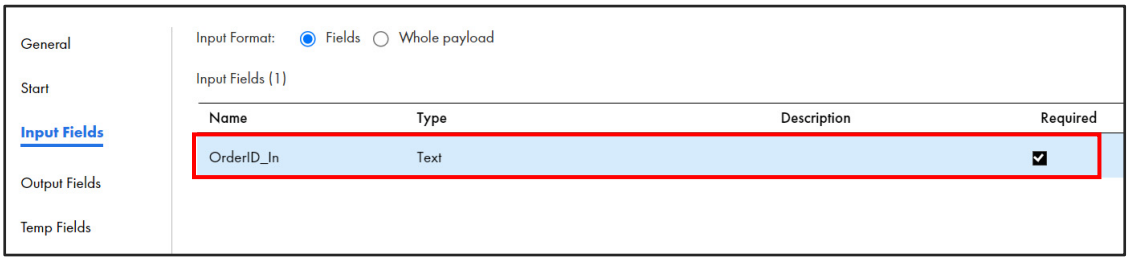
As this process involves accepting an Order id from the user and displaying the relevant details for that order, you need to create an input field that will accept the Order id from the user.

11. From the properties pane, select **Input Fields**.
12. To add a new input field, click .



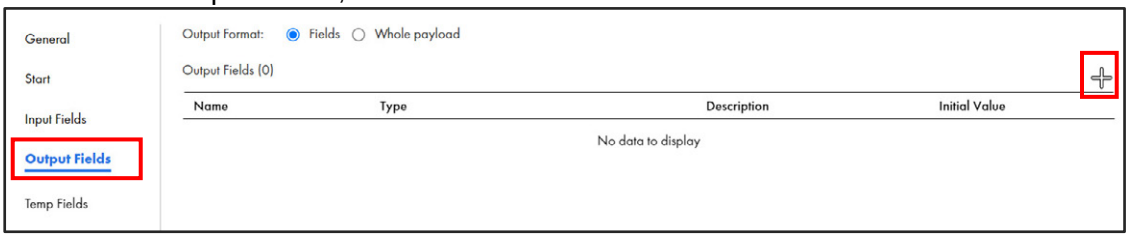
13. Create the input field, as shown in the table below:

Name	Type	Required
OrderID_In	Text	Yes



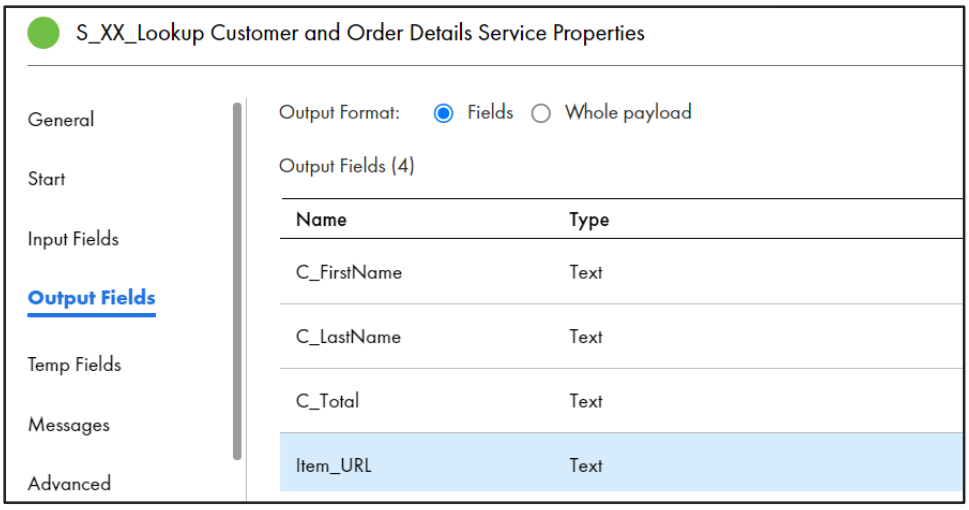
Next, you will create the output fields that will contain the details of the specified order.

14. From the properties pane, select **Output Fields**.
15. To add new Output Fields, click .



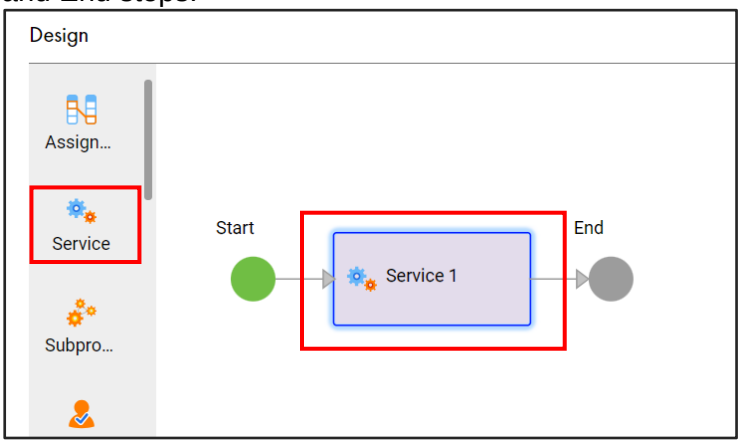
16. Create the Output Fields, as shown in the table below:

Name	Type
C_FirstName	Text
C_LastName	Text
C_Total	Text
Item_URL	Text

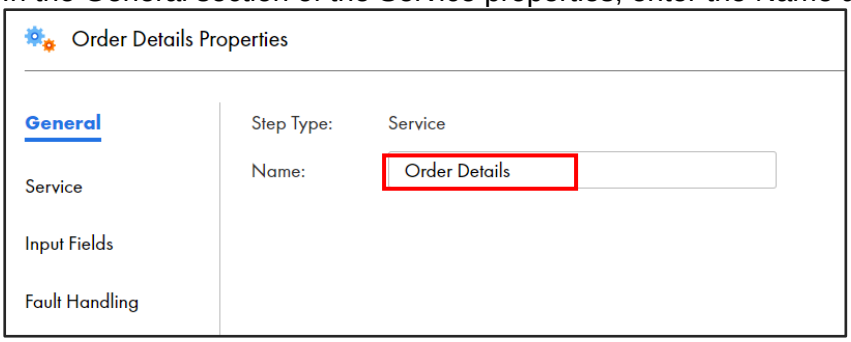


Next, you will add a service step that connects to the third-party service using the connection created in Lab 7-1: Create an App Connection using Service Connector.

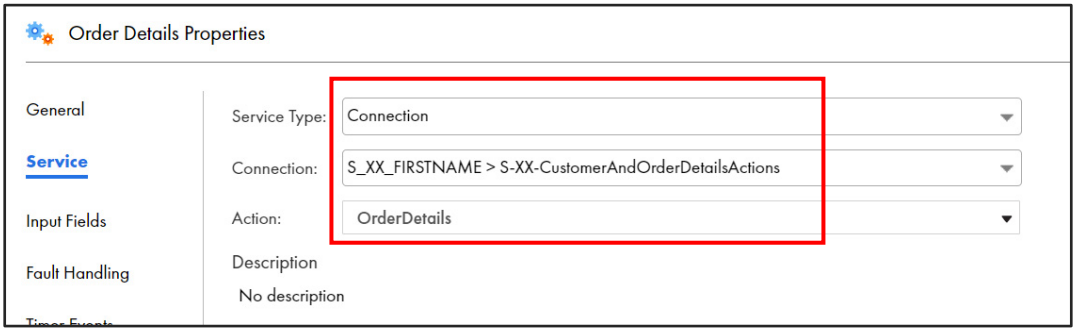
17. From the Design palette, drag and drop a **Service** step on the link between the Start and End steps.



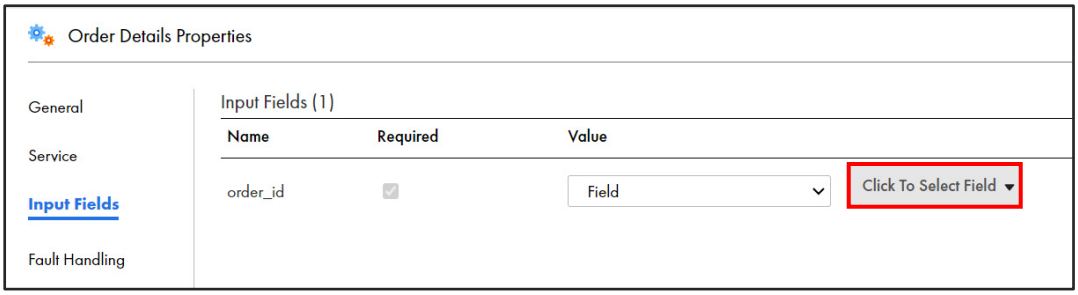
18. To configure the service, from the process canvas, select the **Service** step.
19. In the General section of the Service properties, enter the Name as **Order Details**.



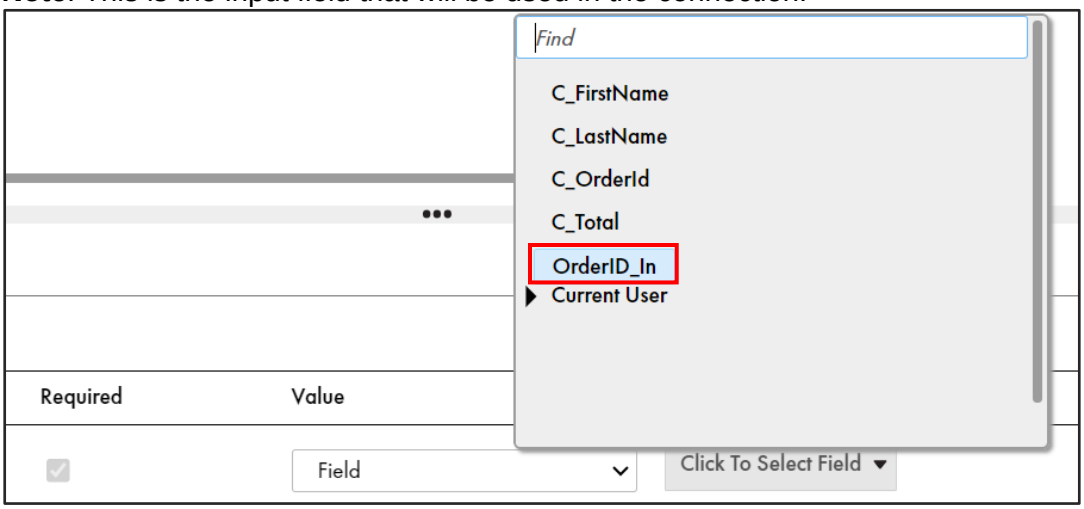
20. From the properties pane, select **Service**.
21. From the Service Type drop-down, select **Connection**.
22. From the Connection drop-down, expand your project folder, and select **S-XX-CustomerAndOrderDetailsActions.**
23. From the Action drop-down, select **OrderDetails**.
    Note: This action is obtained from the service connector.



24. From the properties pane, select **Input Fields**.
    Note: If you do not find the order_id field already present, click the **+** icon, and select **order_id**.
25. From the Value drop-down, select **Field**.
26. Click **Click To Select Field**.



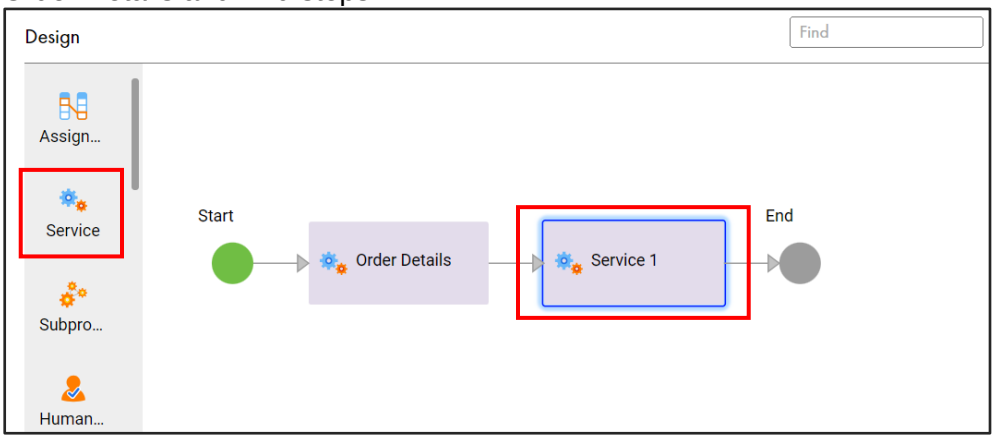
27. From the list, select **OrderID_In**.
    Note: This is the input field that will be used in the connection.



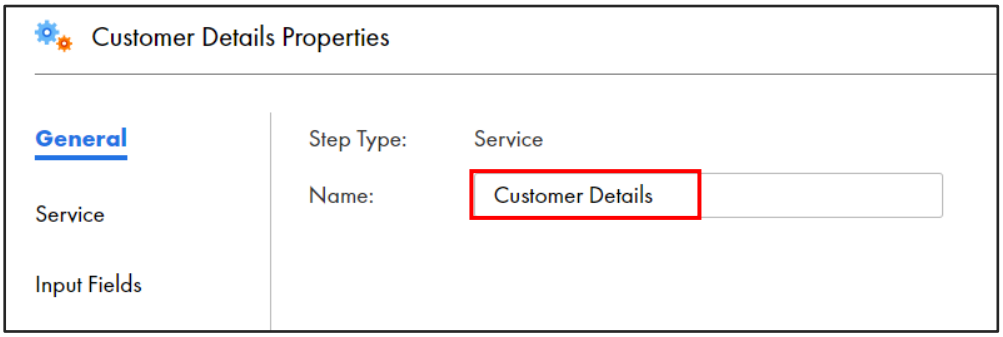
28. Click **Save**.

Next, you need to add a service step that will retrieve the customer details of the specified order.

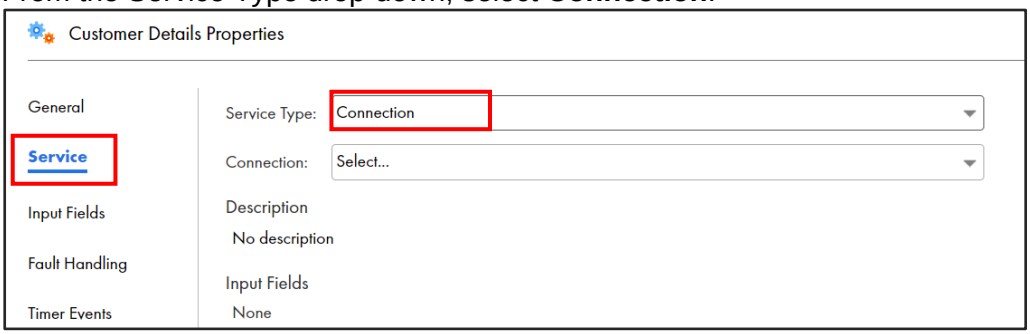
29. From the Design palette, drag and drop a **Service** step on the link between the Order Details and End steps.



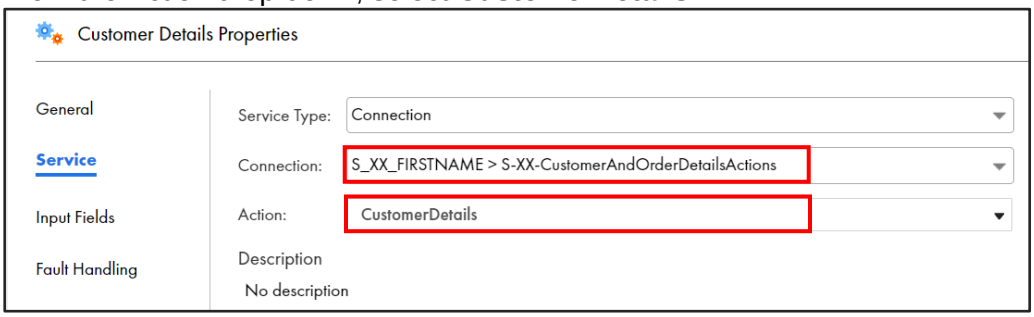
30. To configure the service, from the process canvas, select the **Service** step.
31. In the General section of the Service properties, enter the Name as **Customer Details**.



32. From the properties pane, select **Service**.
33. From the Service Type drop-down, select **Connection**.

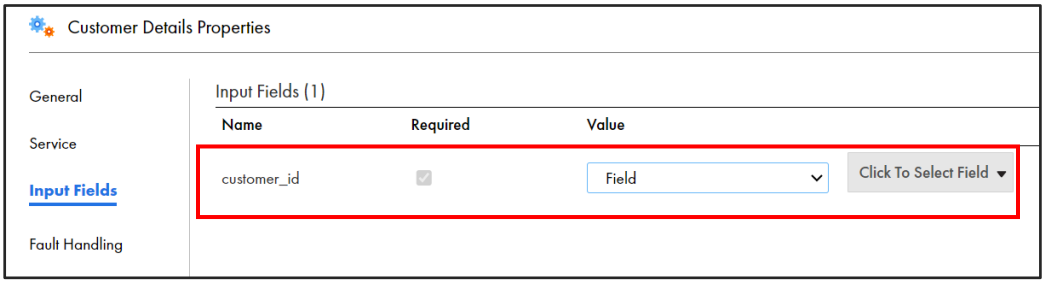


34. From the Connection drop-down, expand **your project folder**, and select **S-XX-CustomerAndOrderDetailsActions**.
35. From the Action drop-down, select **CustomerDetails**.

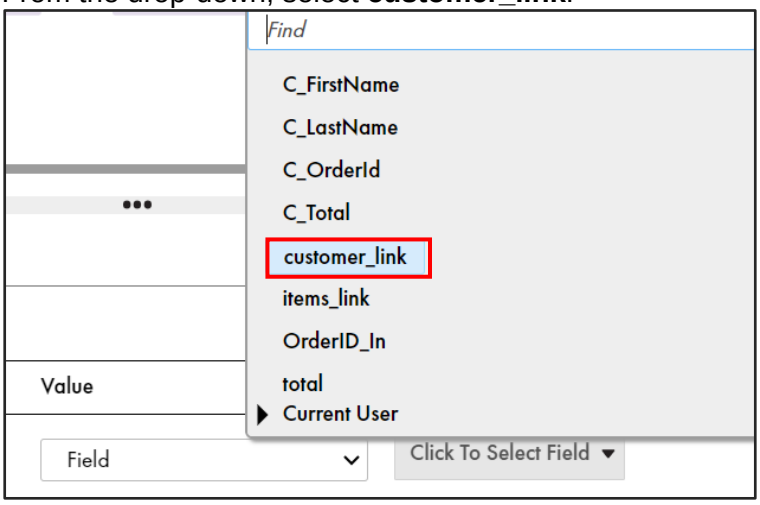


Note: This action is obtained from the service connector.

36. From the properties pane, select **Input Fields**.
Note: If you do not find the customer_id field already present, click the **+** icon, and select **customer_id**.
37. From the Value drop-down, select **Field**.
38. Click on **Click To Select Field**.

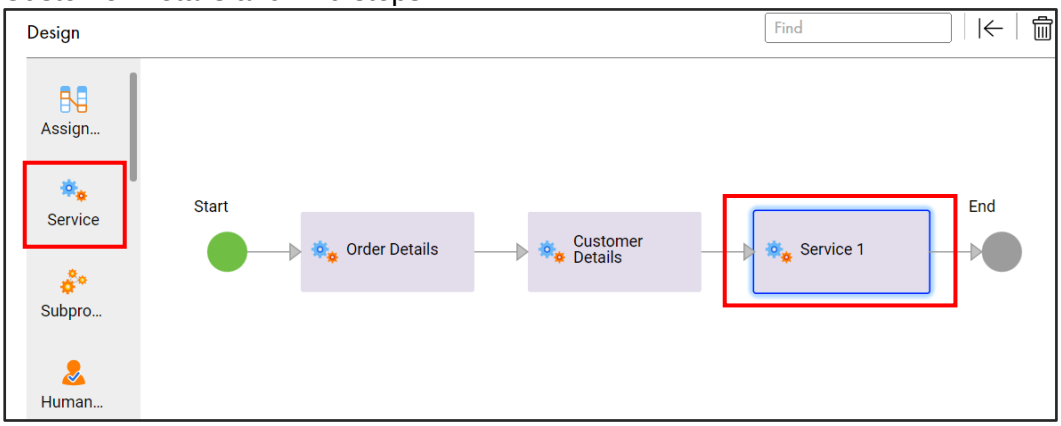


39. From the drop-down, select **customer_link**.

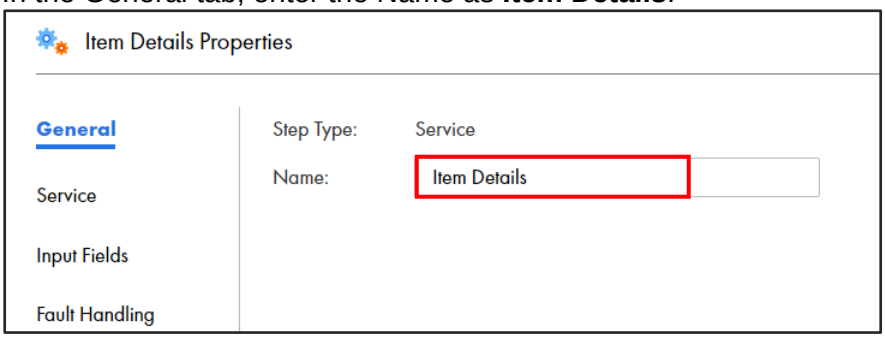


Now, you will add a service step to extract the item details of the specified order.

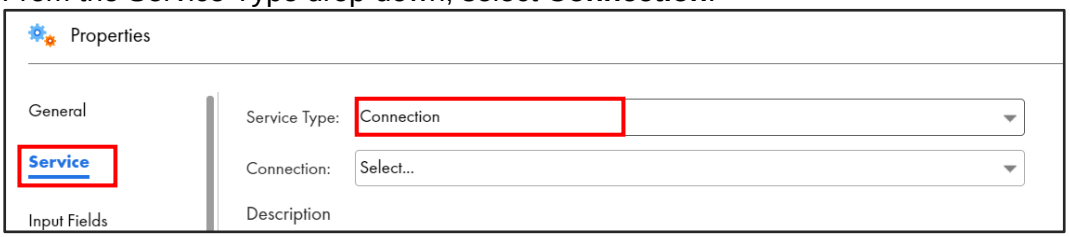
40. From the Design palette, drag and drop a **Service** step on the link between Customer Details and End steps.



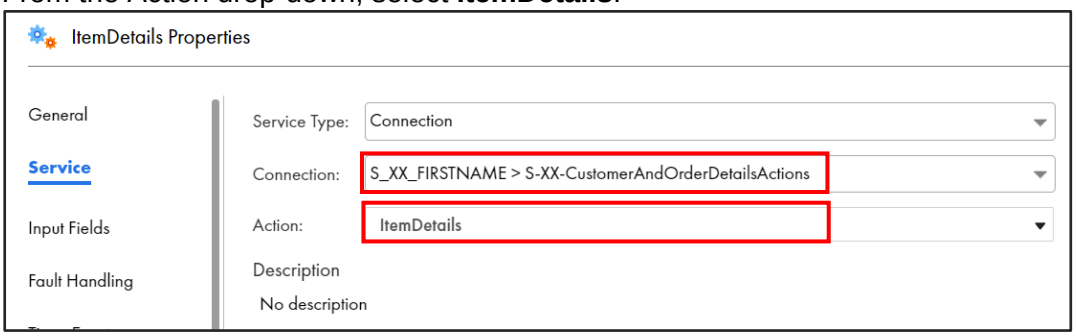
41. To configure the service, from the process canvas, select the **Service** step.
42. In the General tab, enter the Name as **Item Details**.



43. From the properties pane, select **Service**.
44. From the Service Type drop-down, select **Connection**.



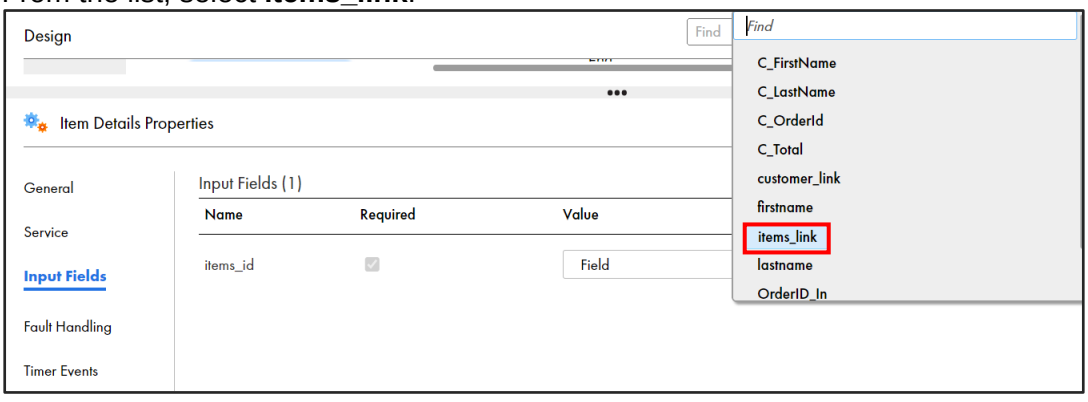
45. From the Connection drop-down, expand your project folder, and select **S-XX-CustomerAndOrderDetailsActions**.
46. From the Action drop-down, select **ItemDetails**.



Note: This action is obtained from the service connector.

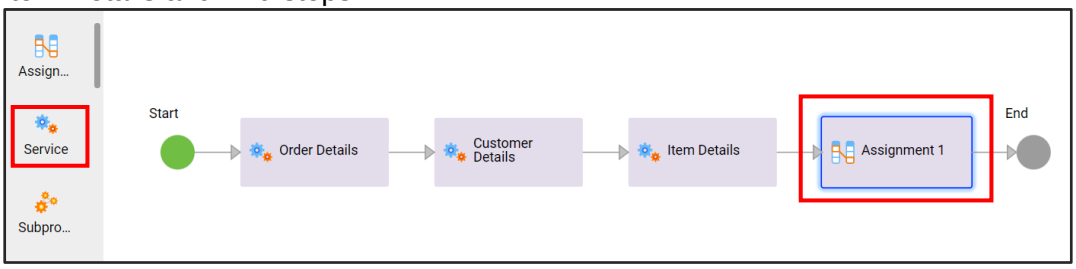
47. From the properties pane, select **Input Fields**.
Note: If you do not find the items_id field already present, click the **+** icon, and select **items_id**.
48. From the Value drop-down, select **Field**.
49. Click on **Click To Select Field**.

50. From the list, select **items_link**.

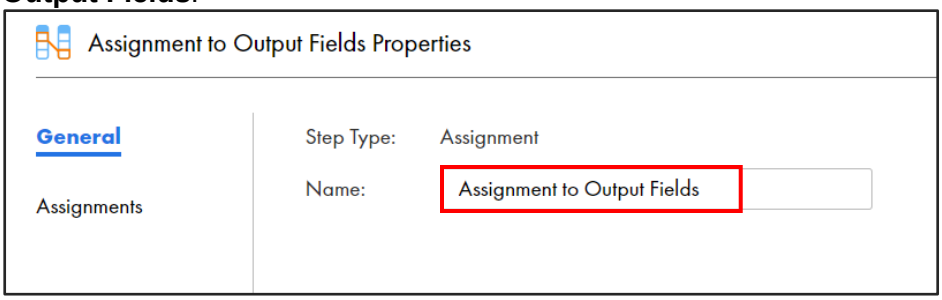


Now that the information is extracted, it needs to be displayed as an output. For this, you will use an Assignment step.

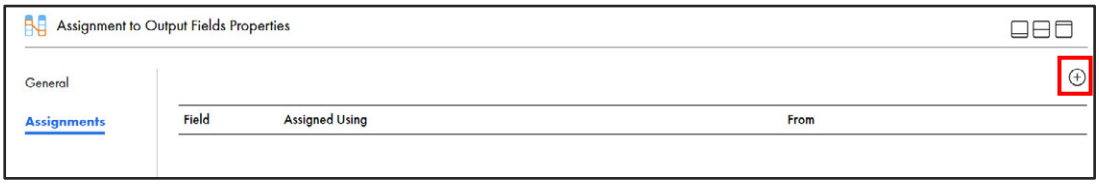
51. From the Design palette, drag and drop an **Assignment** step on the link between the Item Details and End steps.



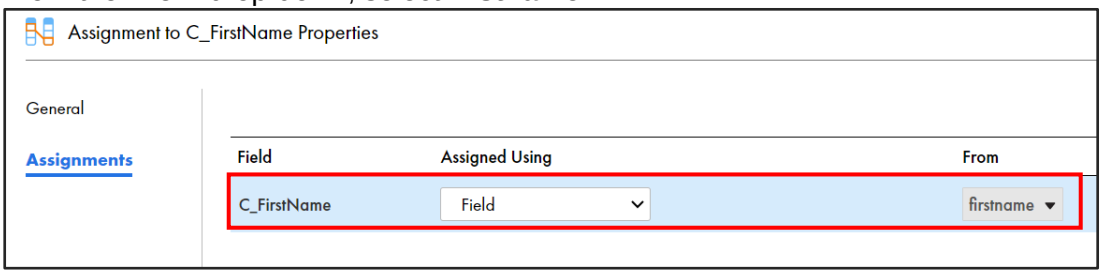
52. To configure the step, from the process canvas, select the **Assignment** step.
53. In the General section of Assignment properties, enter the Name as **Assignment to Output Fields**.



54. From the properties pane, select **Assignments**.
55. Click the **+** icon.

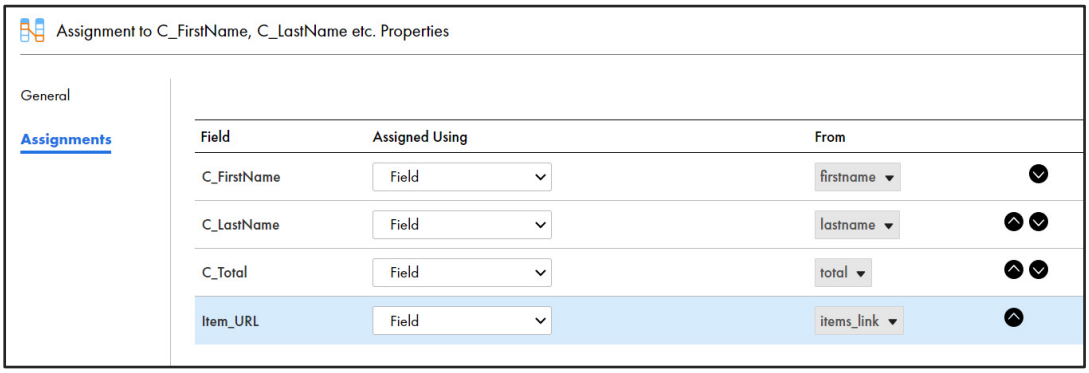


56. From the drop-down, select **C_FirstName**.
57. From the Assigned Using drop-down, select **Field**.
58. From the **From** drop-down, select **firstname**.

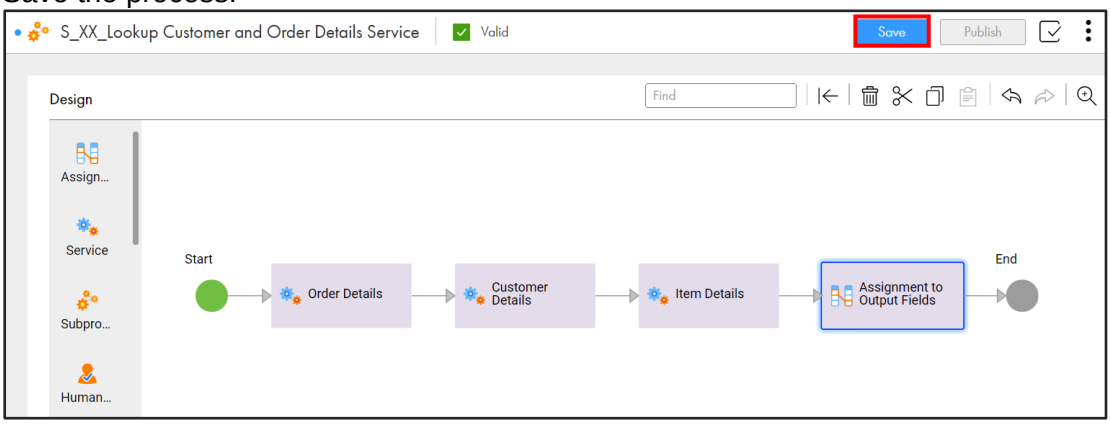


59. Similarly, add the additional fields listed in the table below:

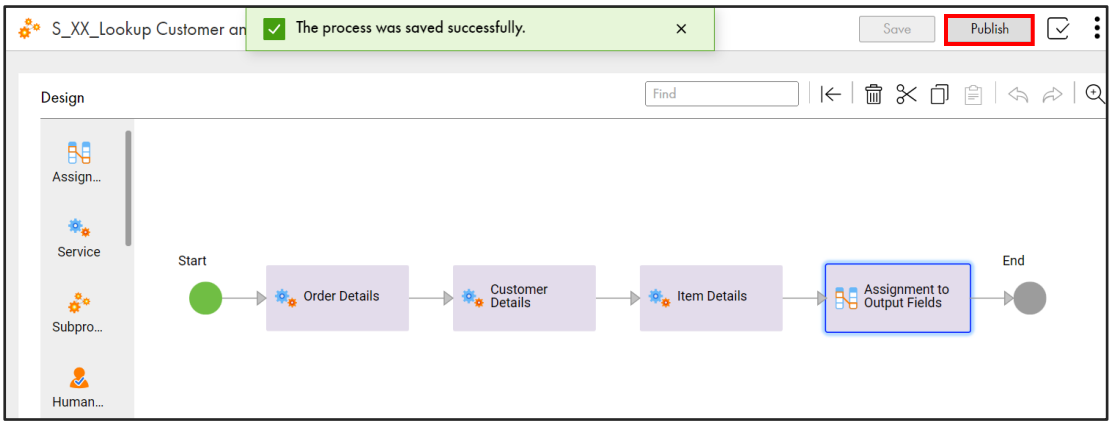
Add Field	Assigned Using	From
C_LastName	Field	lastname
C_Total	Field	total
Item_URL	Field	Items_link



60. Save the process.

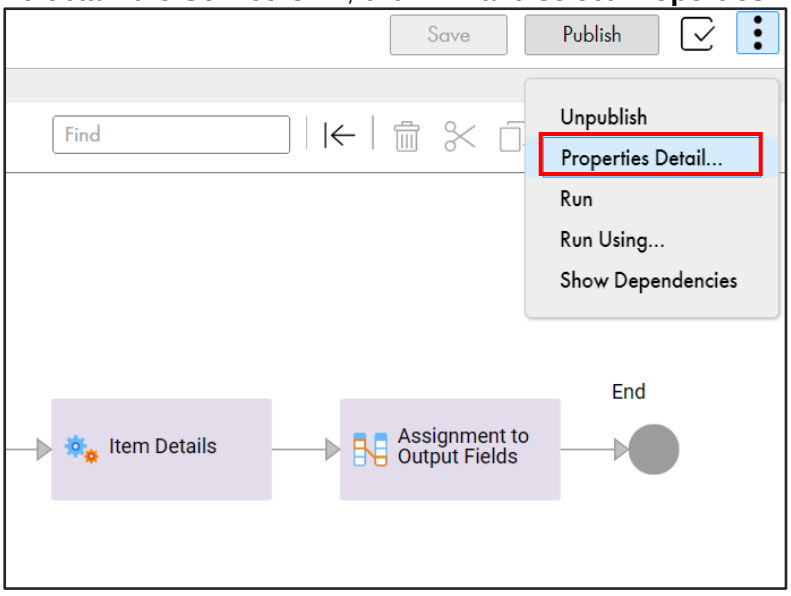


61. Publish the process.

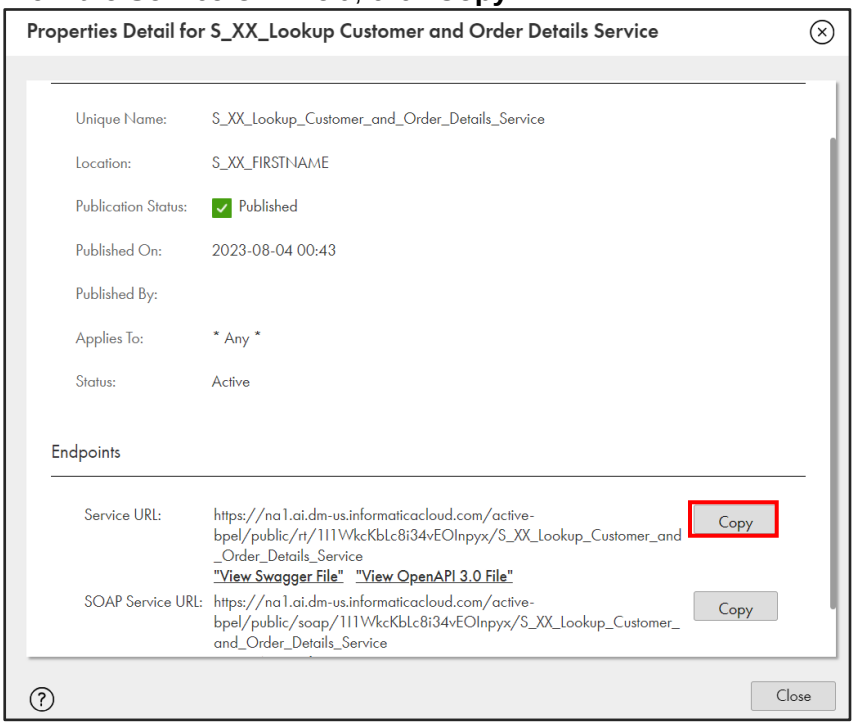


Execute the Process:

62. To obtain the Service URL, click and select **Properties Detail**.

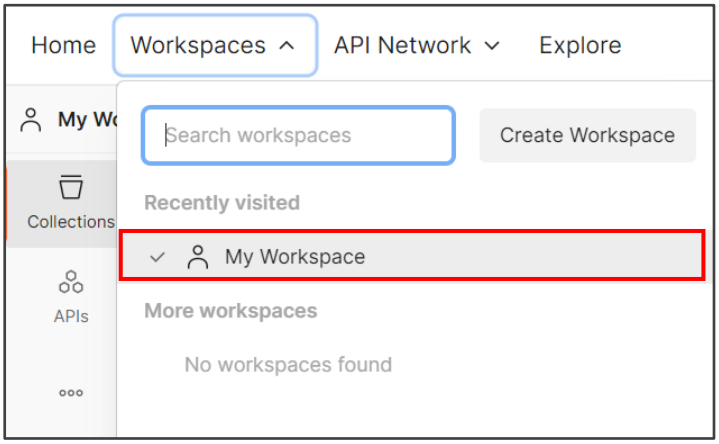


63. From the Service URL field, click **Copy**.

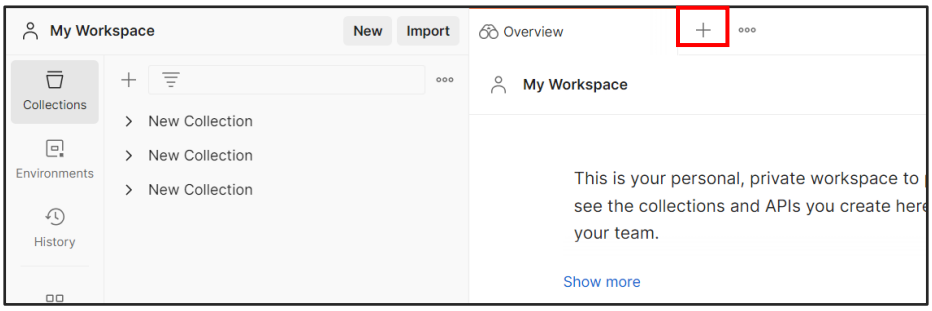


Now, run the process in Postman.

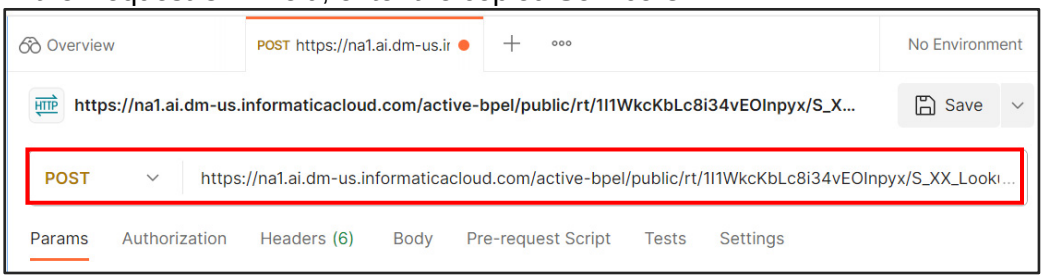
64. Open a web browser and enter the following link to open Postman: https://identity.getpostman.com/
65. Log in to Postman using your credentials.
66. From the Workplace drop-down, select **My Workspace**.



67. Click the '**+**' icon.

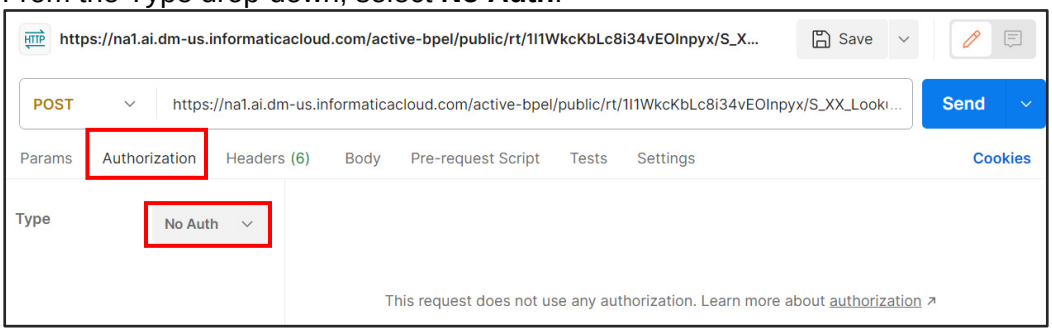


68. From the drop-down, select **POST**.
69. In the Request URL field, enter the copied Service URL.



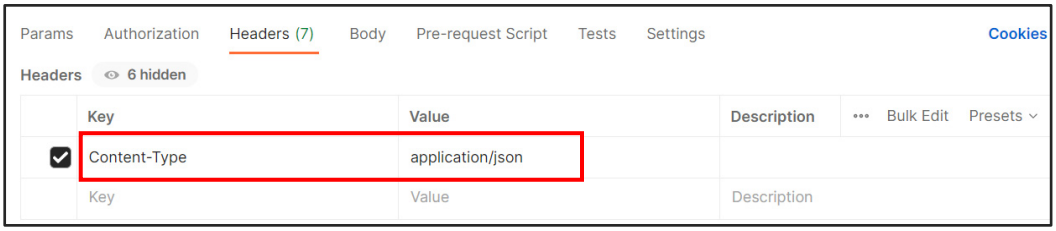
70. Select the **Authorization** tab.

71. From the Type drop-down, select **No Auth**.



72. Select the **Headers** tab.
73. Enter Key and Value as shown in the table below:

Key	Value
Content-Type	application/json



74. Select the **Body** tab and select **raw**.
75. Enter the following text body:

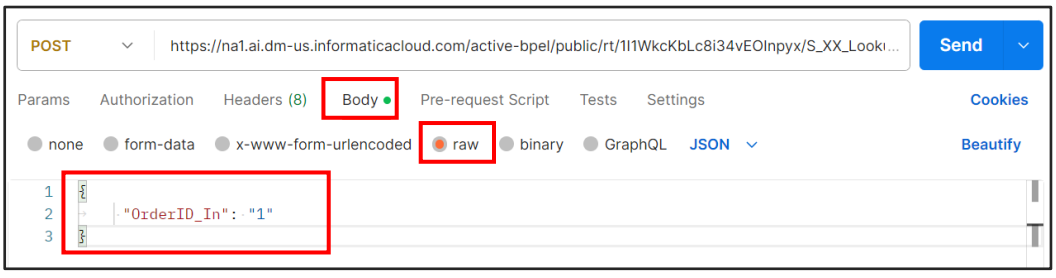
{

"OrderID_In": "1"

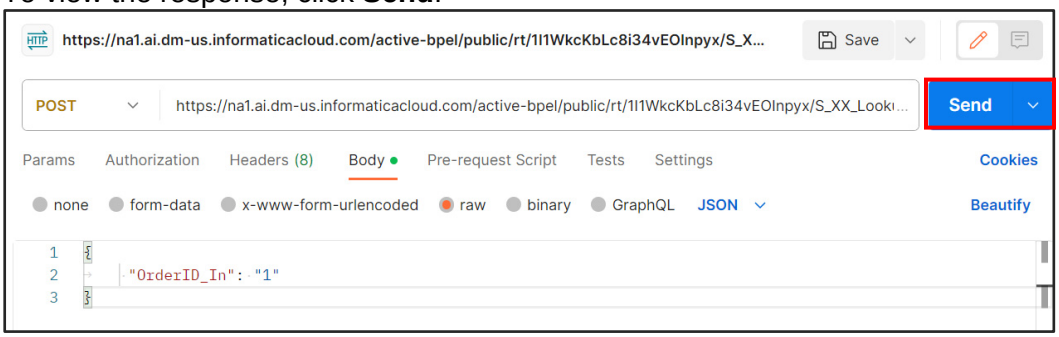
}

OR

From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **13_LabGuide_CAI_CustomerAndOrdersDetails_7-3**. Copy the command mentioned under **Step A** and paste it into the body field.



76. To view the response, click **Send**.



77. Verify that the status of the response is **200 OK**.
78. Examine the details in the response section.
    Note: The C_Total value may differ based on the live data available in the API.

    Note: Observe that you get the customer and order details for order id '1'.

This concludes the lab.

Module 7: App Connections
Lab 7-4: Create a File Connection

Overview:
The File Connector provides connectivity between Application Integration and file systems to monitor file systems for new files, move files, read and write content from files, and select options to handle processed files. You can use the File Connector to write log files that are needed as part of a larger integration or to monitor a file system for new .csv files, read the delimited content from the files, or use the generated XML in a set of process objects.

In this lab, you will create a File connection using the built-in File Connector.

Objectives:
- Create a File connection
- Publish the connection

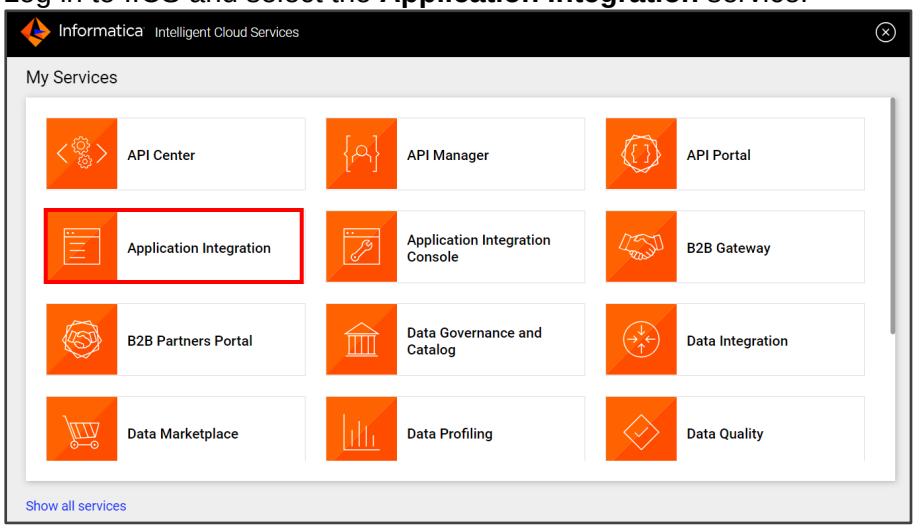
Duration:
15 Minutes

Tasks:

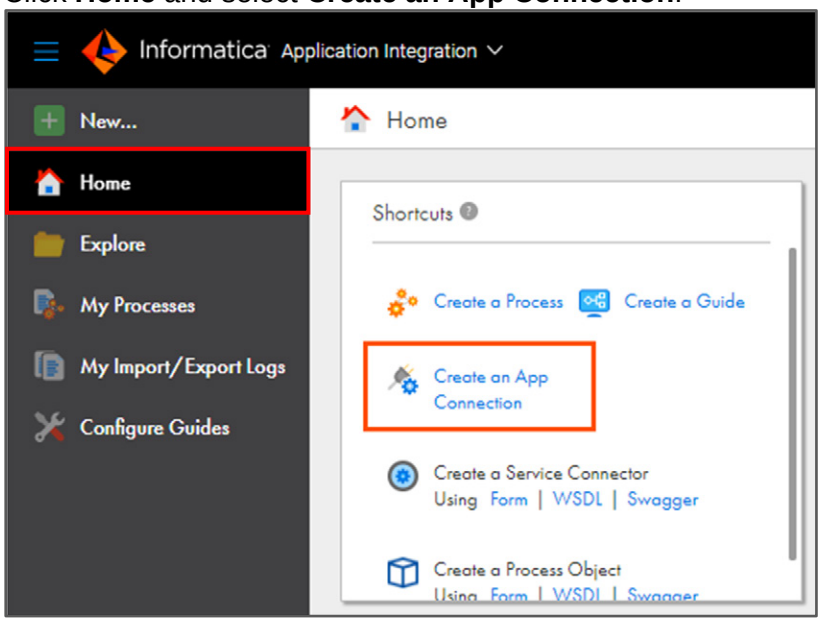
Note: The screen in your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

Create a connection:

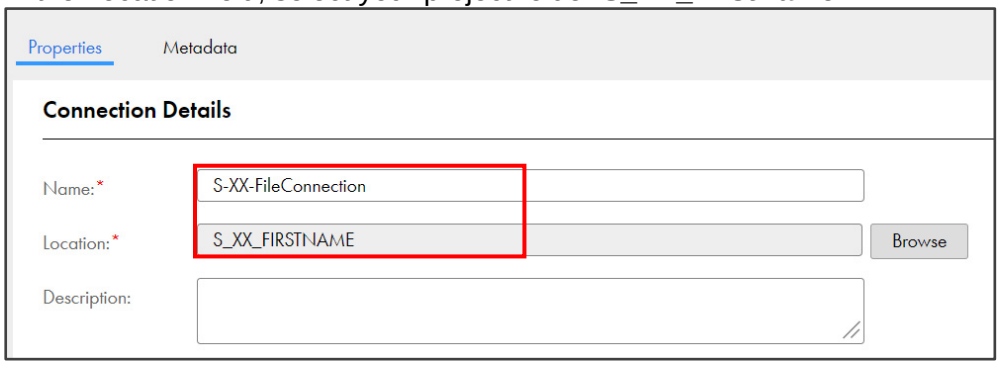
1. Log in to IICS and select the **Application Integration** service.



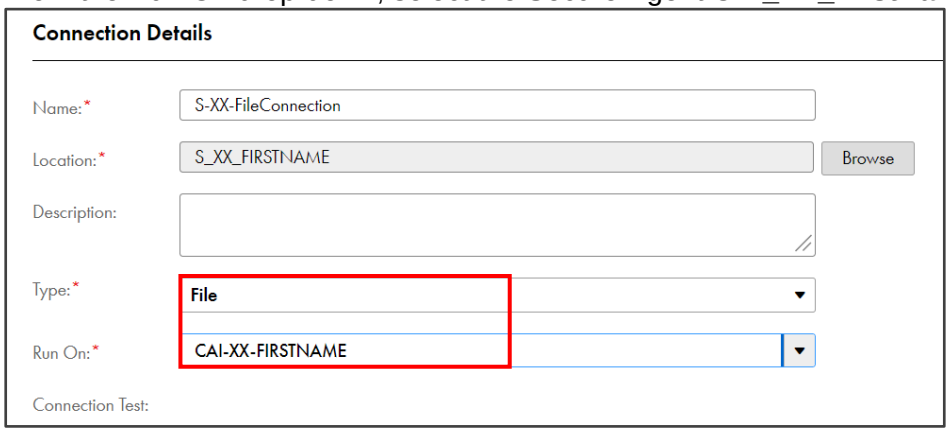
2. Click **Home** and select **Create an App Connection**.



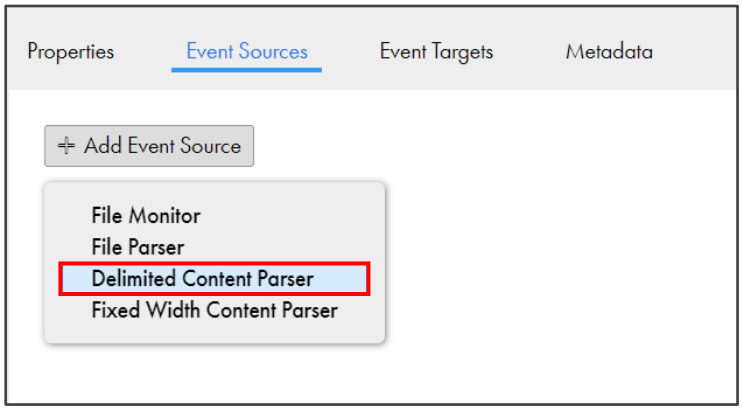
3. In the Connection Details page, enter the connection name as **S-XX-FileConnection**.
4. In the Location field, select your project folder **S_XX_FirstName**.



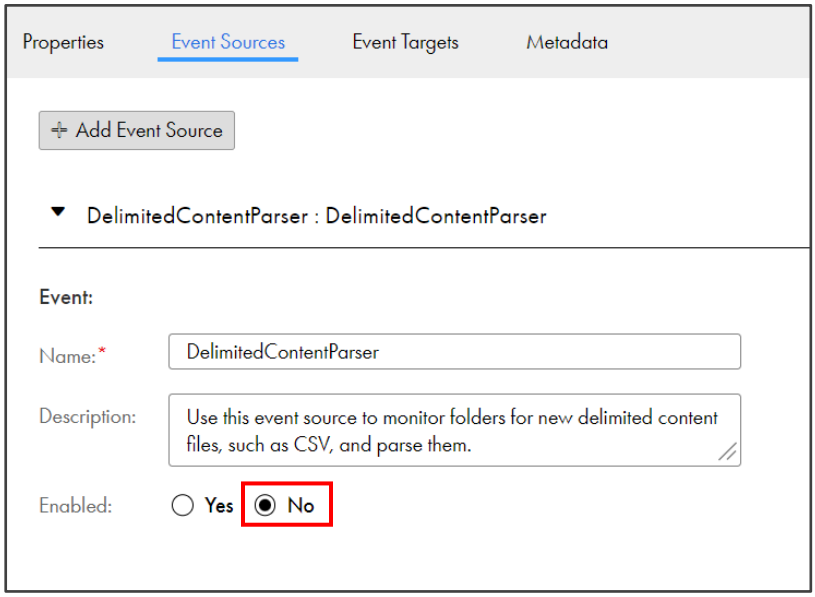
5. From the Type drop-down, select **File**.
6. From the Run On drop-down, select the Secure Agent **CAI_XX_FirstName**.



7. To configure the Event Sources, click the **Event Sources** tab.
8. To add different Event Sources, click "**Add Event Source"** and select "**Delimited Content Parser**."

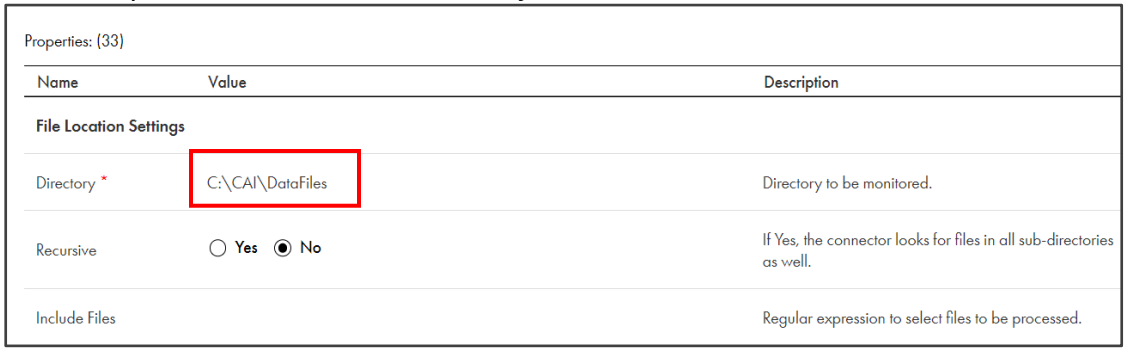


9. Retain the values for the **Name** and **Description** fields.
10. For the "Enabled" option, select **No**.

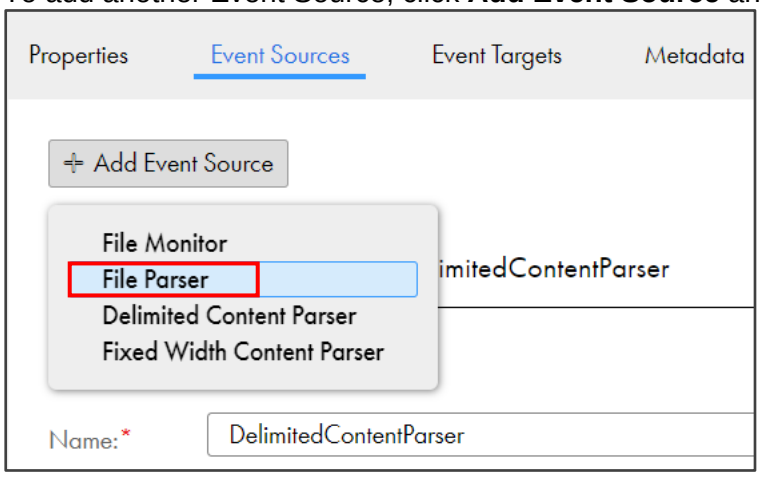


Note: When you select No, the event source is disabled until you are ready to use it. If you select Yes, the event source is available immediately after it is published.

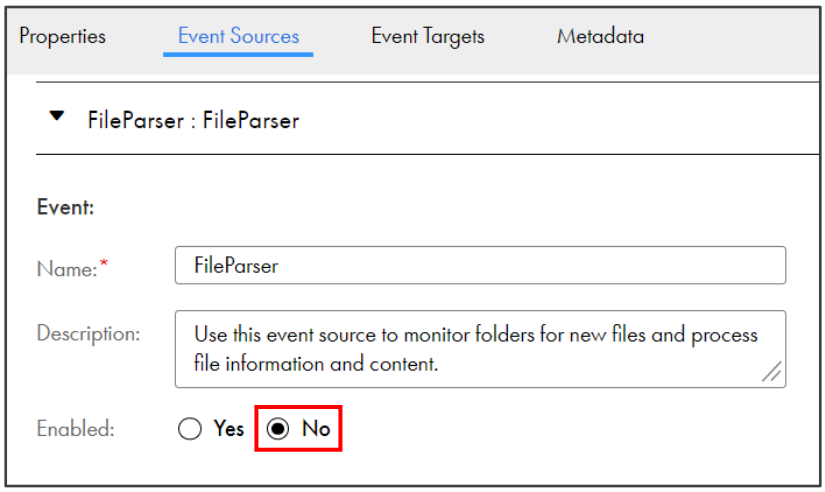
11. In the Properties section, in the Directory field, enter **C:\CAI\DataFiles**.



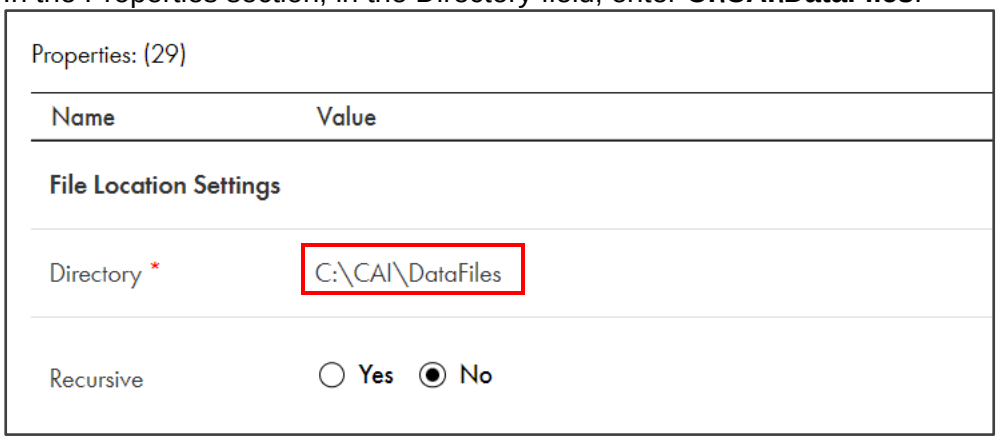
12. To add another Event Source, click **Add Event Source** and select **File Parser**.



13. Retain the values for the **Name** and **Description** fields.
14. For the Enabled option, select **No**.



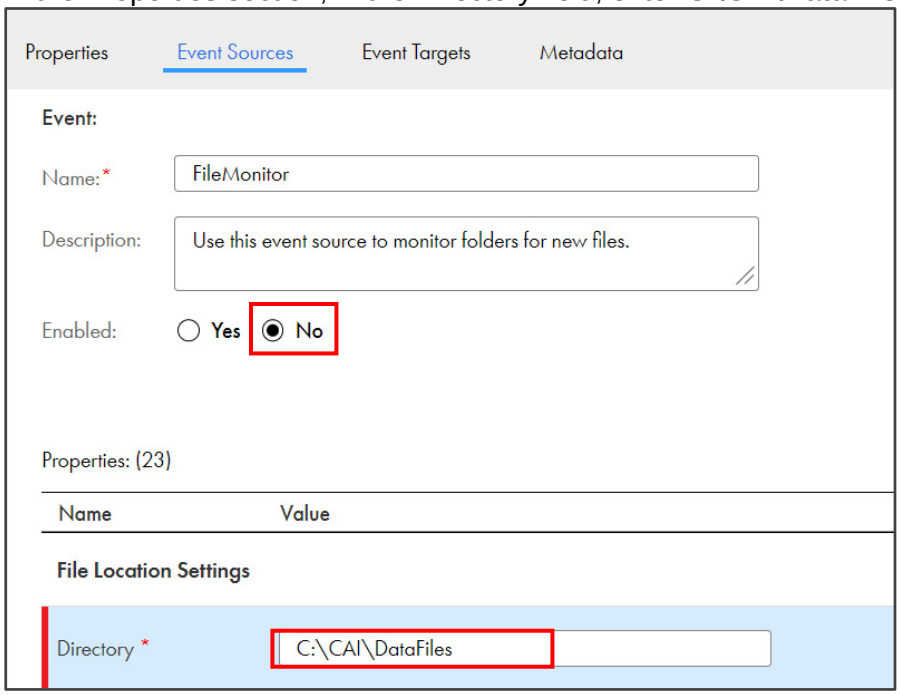
15. In the Properties section, in the Directory field, enter **C:\CAI\DataFiles**.



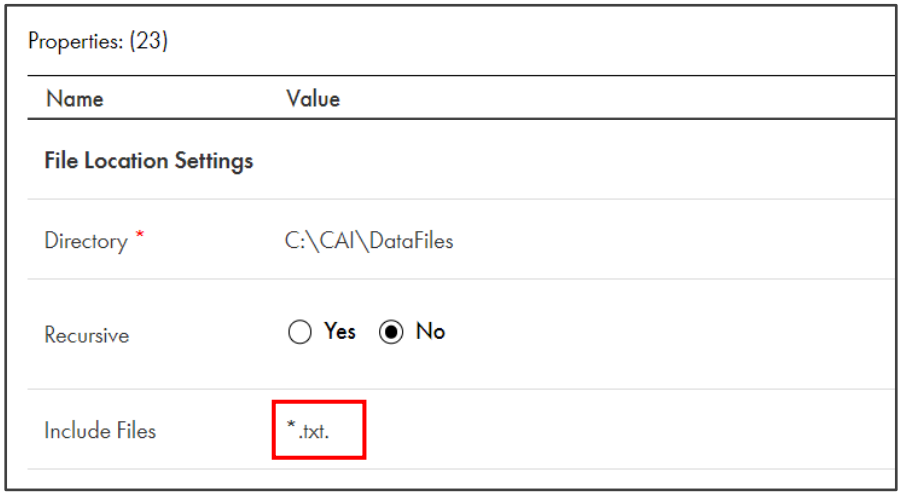
Note: Click the Directory field row to enter the value in the field.

16. Add another Event Source and select **File Monitor**.
17. Retain the values for the **Name** and **Description** fields.
18. For the “Enabled” option, select **No**.

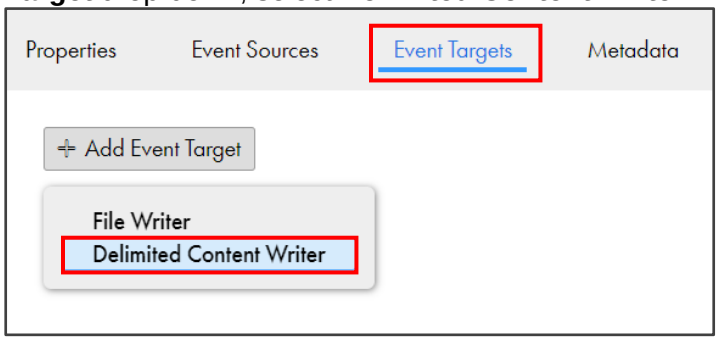
19. In the Properties section, in the Directory field, enter **C:\CAI\DataFiles**.



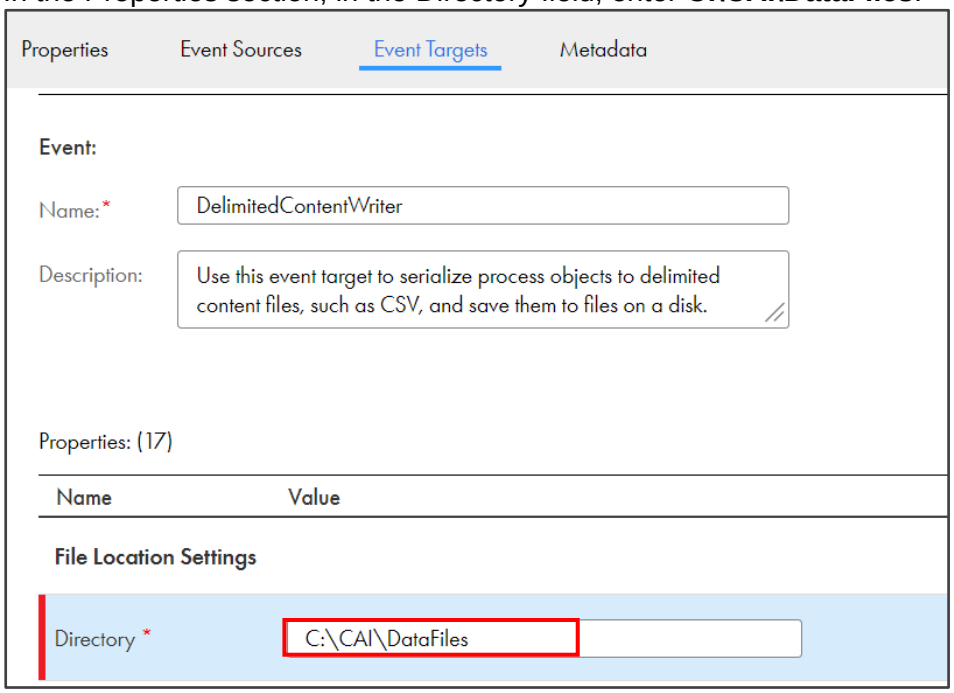
20. In the Include Files field, add an extension ***.txt**.



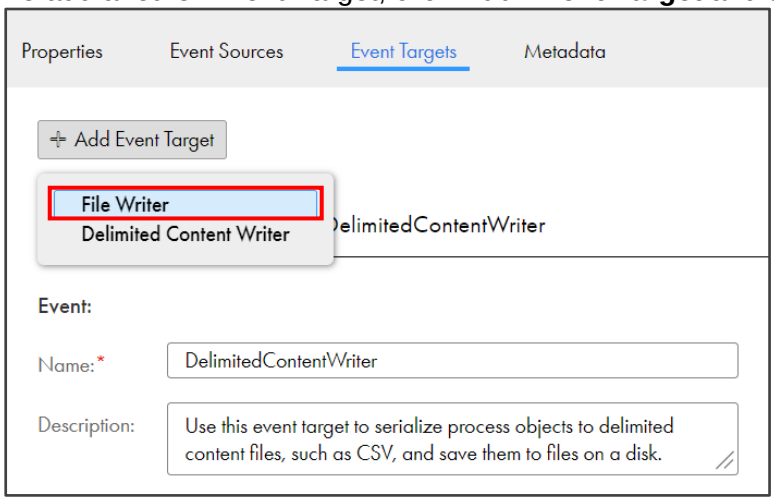
21. To configure the Event Targets, click the **Event Targets** tab, and from the **Add Event Target** drop-down, select **Delimited Content Writer**.



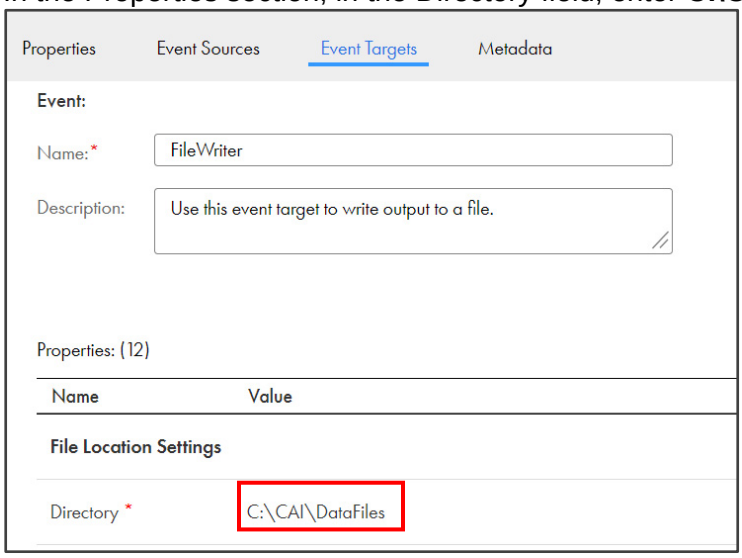
22. Retain the values for the **Name** and **Description** fields.
23. In the Properties section, in the Directory field, enter **C:\CAI\DataFiles**.



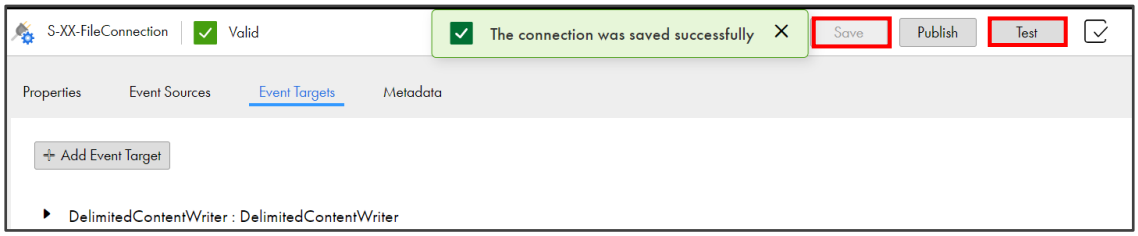
24. To add another Event Target, click **Add Event Target** and select **File Writer**.



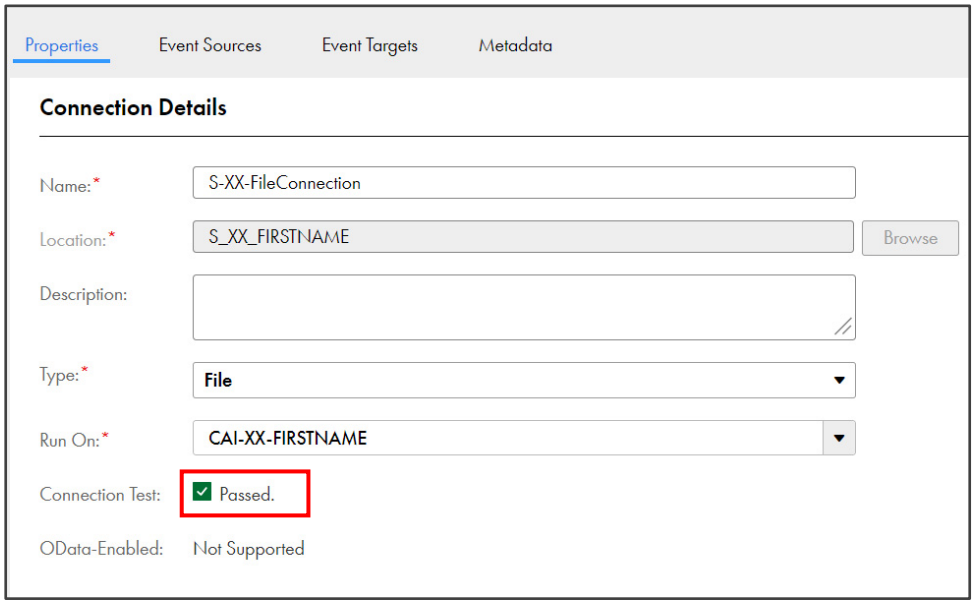
25. Retain the values for the **Name** and **Description** fields.
26. In the Properties section, in the Directory field, enter **C:\CAI\DataFiles**.



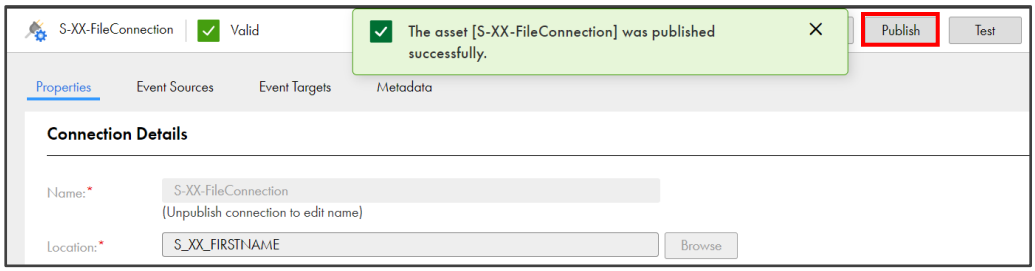
27. **Save** and **Test** the connection.



28. In the Properties tab, the Connection Test field mentions "**Passed"**.

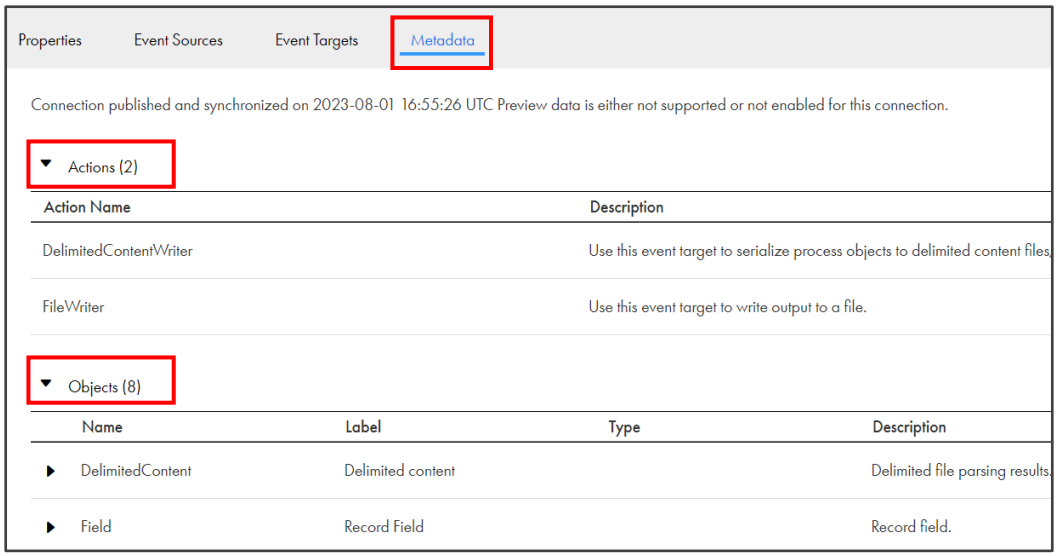


29. **Publish** the connection.



You will get a notification that '**The asset [S-XX-FileConnection] was published successfully.**'.

30. In the **Metadata** tab, observe that the Actions and Objects are added.



This concludes the lab.

Module 7: App Connections

Lab 7-5: Create a Salesforce Connection

Overview:
Salesforce is a cloud-based Customer Relationship Management (CRM) solution for sales teams to manage contacts and sales activities. You can use Salesforce to store and manage contacts and data of the sales activities of your organization.

In this lab, you will install the Managed Package and create a Salesforce connection.

Objectives:

- Install Managed Package
- Create a Salesforce connection

Duration:
10 Minutes

Important: Before you start this lab, ensure that you have a functional Salesforce Developer account.

If you do not have a Salesforce Developer account, create one by using the following URL: https://developer.salesforce.com/signup
On the sign-up page, provide your details and enter the validation code that is sent to your email.

Troubleshooting Tip: If at any time the browser hangs and the desired output is not obtained (assets do not get published, windows do not open, and so on), try closing the browser window and opening a fresh one.

Tasks:

Note: Make sure you are using your sub-org, which was provided to you by the instructor at the beginning of this class.

The screen on your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

Install Salesforce Managed Package:

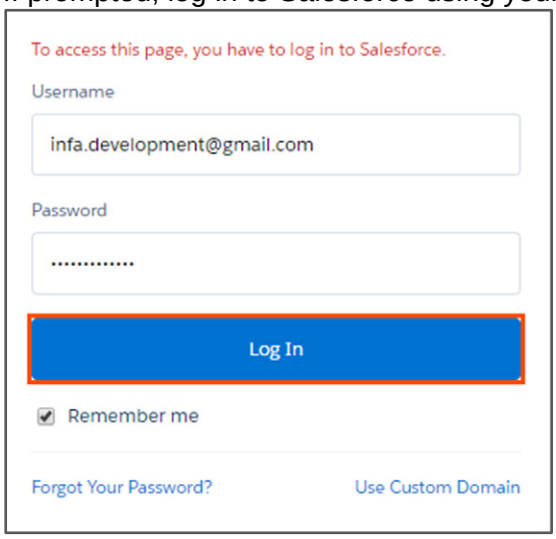
1. To install the Managed Package, open a web browser, and enter the link mentioned below:
   https://login.salesforce.com/packaging/installPackage.apexp?p0=04t1J000000gOtF

   OR,

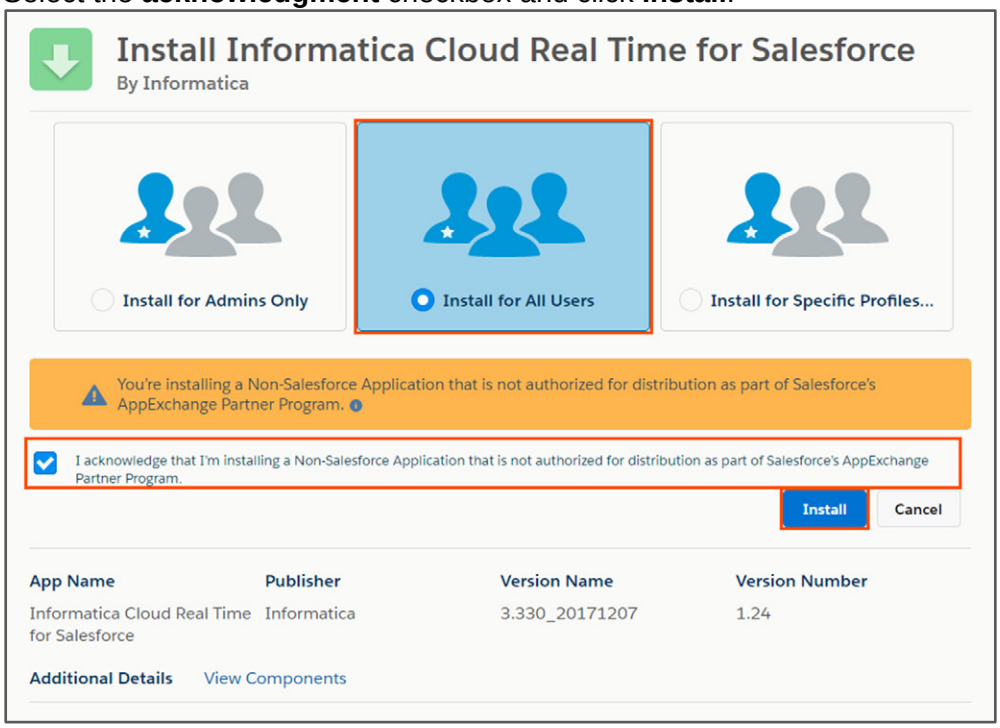
   From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **18_LabGuide_CAI_Create a Process with File Listener Using Flat File Connection_7-8**. Copy the command mentioned under **Step A** and paste it into the body field.

Note: You will learn more about Managed Packages in Module 6 of this course.

2. If prompted, log in to Salesforce using your Salesforce credentials.



3. On the installation page, select the **Install for All Users** option.
4. Select the **acknowledgment** checkbox and click **Install**.

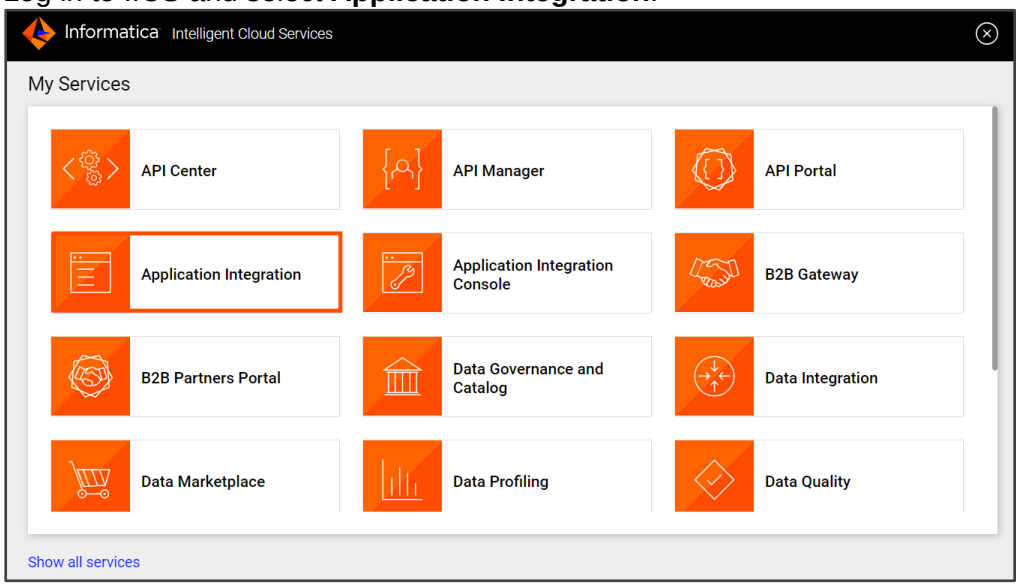


Note: It takes 3-5 minutes to install the package. When the installation is complete, you will receive a confirmation email to your registered Salesforce account.

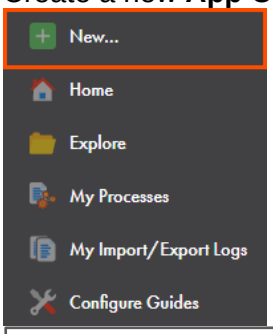
5. Click **Done**.

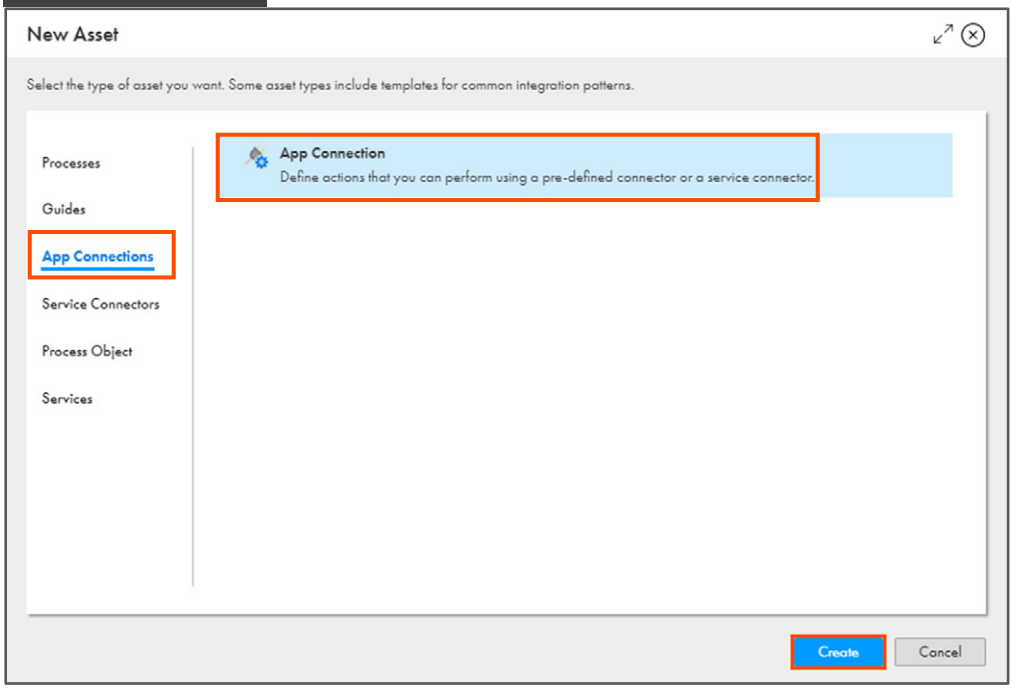
Create a Salesforce connection:

6. Log in to IICS and select **Application Integration**.

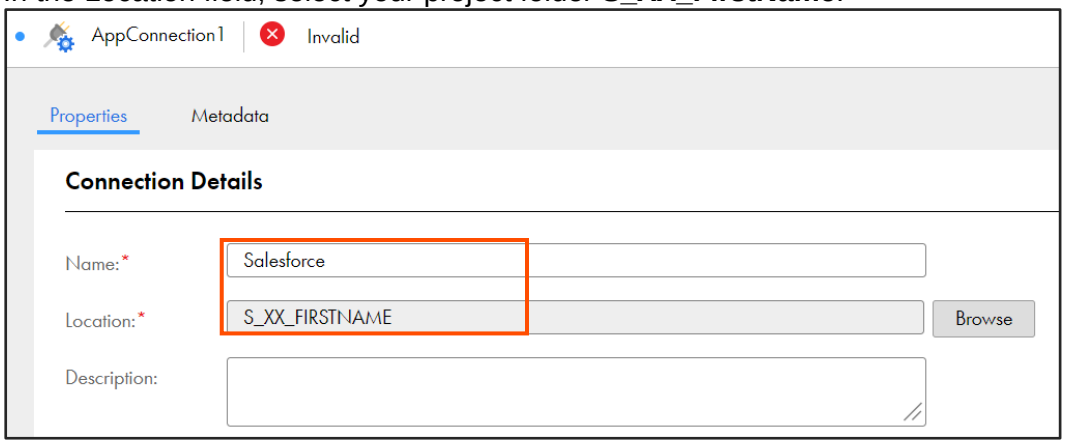


7. Create a new **App Connection**.

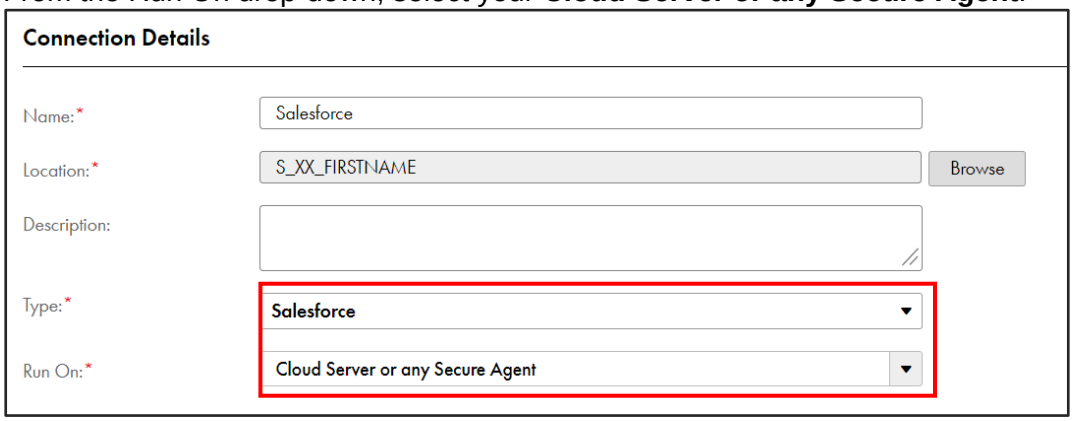




8. In the Connection Details page, enter the connection name as **Salesforce**.
   Note: To use this connection to create Guides, it is mandatory to name the connection as **Salesforce**.
9. In the Location field, select your project folder **S_XX_FirstName**.

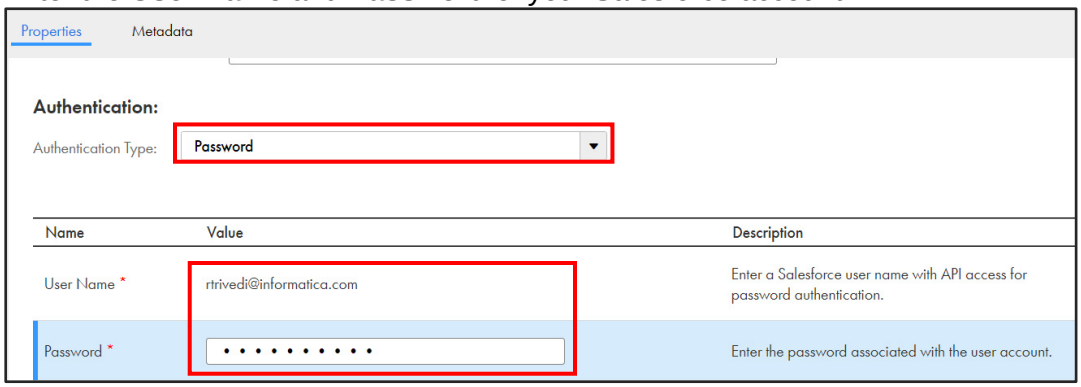


10. From the Type drop-down, select **Salesforce**.
11. From the Run On drop-down, select your **Cloud Server or any Secure Agent**.

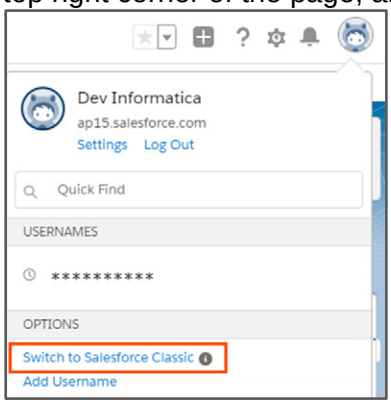


Note: Publishing could be faster when we publish on the cloud server instead of any secure agent. That is why, we selected the run-time environment as **Cloud Server or any Secure Agent** and not Secure Agent.

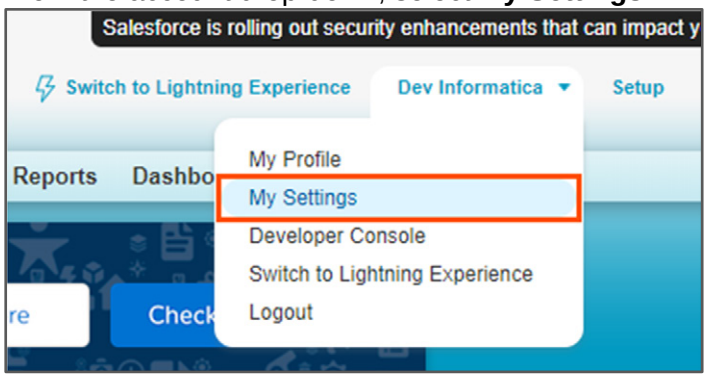
12. Retain the Authentication Type as **Password**.
13. Enter the **User Name** and **Password** of your Salesforce account.



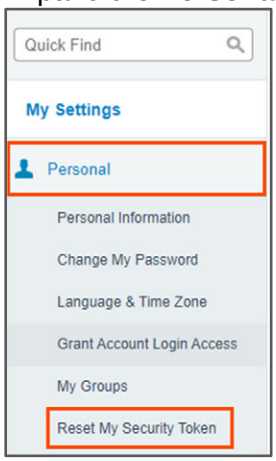
14. To obtain the Security Token, navigate back to your Salesforce account.
Note: If you do not have the Salesforce Classic view, then you need to switch to the Classic view. To switch from Lightning to Classic view, click the **My Profile** icon on the top right corner of the page, and select **Switch to Salesforce Classic**.



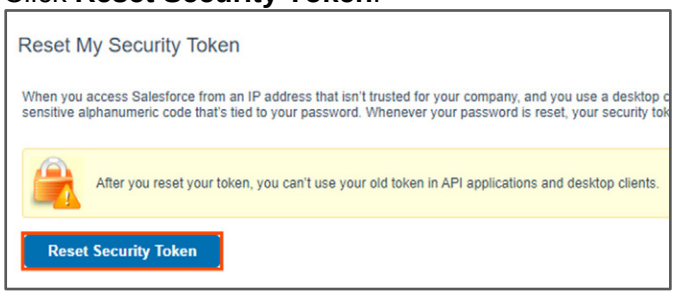
15. From the account drop-down, select **My Settings**.



16. Expand the **Personal** option and select **Reset My Security Token**.

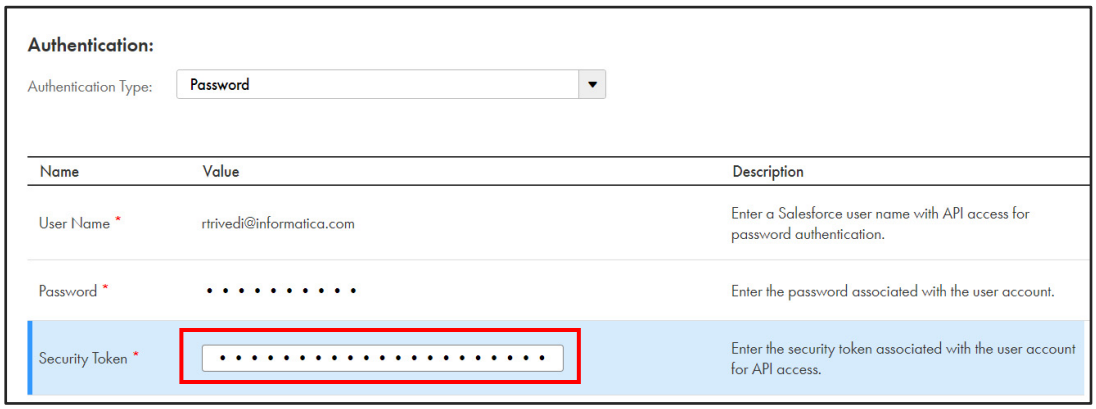


17. Click **Reset Security Token**.

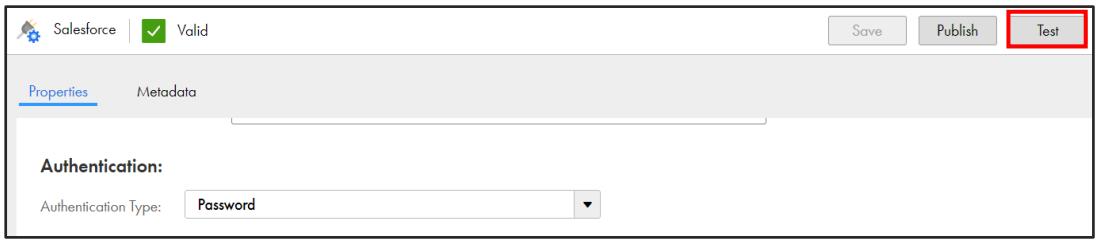


Note: When you reset the security token, it is mailed to your registered e-mail account. This security token validates the Salesforce connection properties and configures the connection.

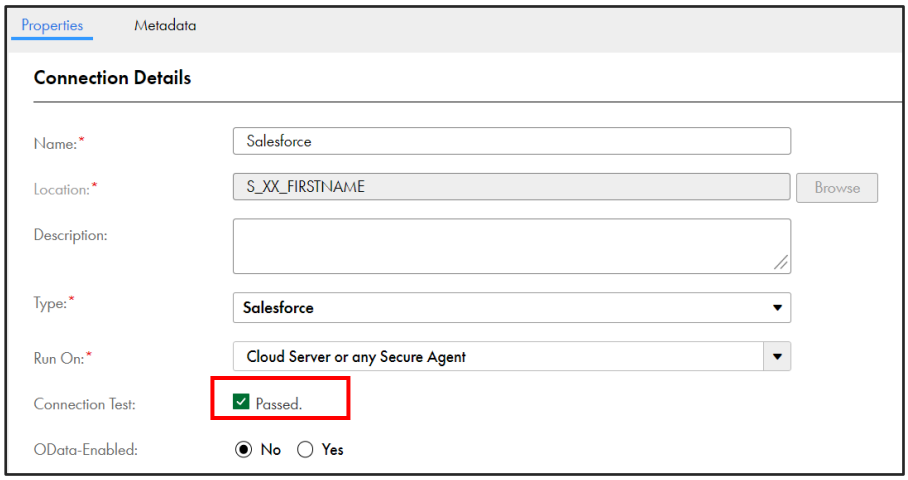
18. Copy the Security Token from the e-mail and paste it into the Security Token field in IICS.



19. **Save** the connection.
20. **Test** the connection.



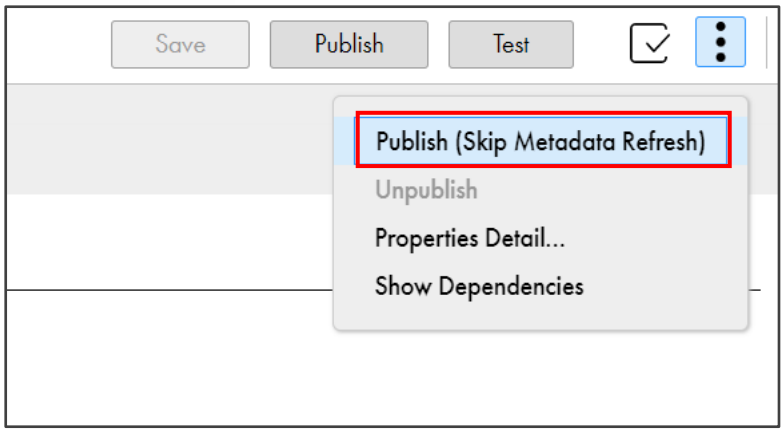
The connection test mentions **Passed**.



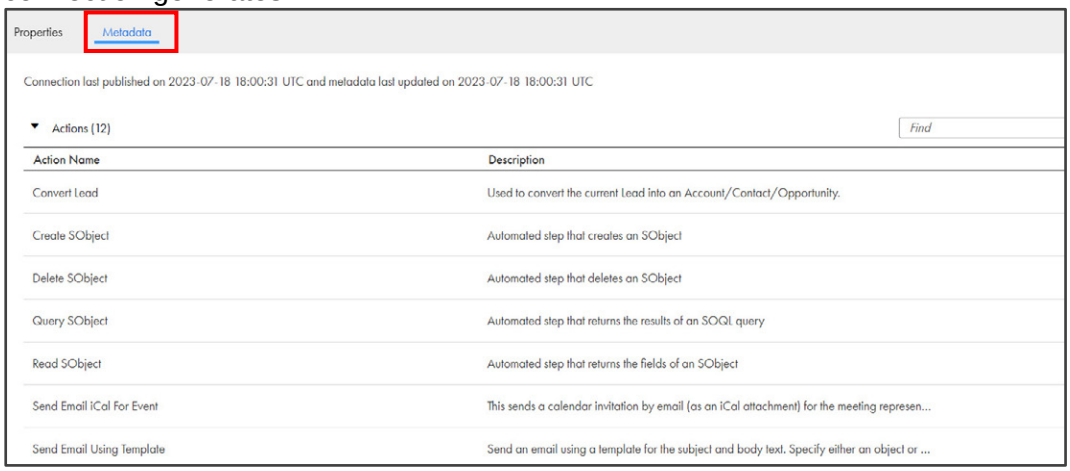
21. **Publish** the connection.

Note: Sometimes, publishing takes a very long time. Wait for 5-6 minutes for the connection to be published.

If you get the "**Publishing timed out"** error, click **OK,** and then use the "Publish" **(Skip Metadata Refresh)** option to publish.



22. After you publish the connection, the **Metadata** tab lists all the process objects that the connection generates.



23. **Close** the Salesforce tab.

This concludes the lab.

Module 7: App Connections
Lab 7-6: Create a Kafka Connection

Overview:
Kafka is a distributed publish-subscribe messaging system that is fast, scalable, and durable. It stores messages in the form of topics. Producers write data to a topic and consumers read data from the topic. Kafka topics are partitioned and replicated across multiple nodes thereby allowing distributed processing. Kafka uses ZooKeeper to manage the Kafka cluster.

In this lab, you will create a Kafka connection using the built-in Kafka connector and use it in a process to read and write data.

Objectives:
- Start Apache ZooKeeper and Kafka server
- Create a topic
- Create a Kafka connection
- Publish the connection
- Create and publish a process to write data to a topic

Duration:
35 Minutes

Tasks:

IMPORTANT: In this lab, some steps need to be performed by the Instructor on the Server Agent (Please check **Steps 1 to 7**). After Step 86, there is a demo that should be performed by the instructor.

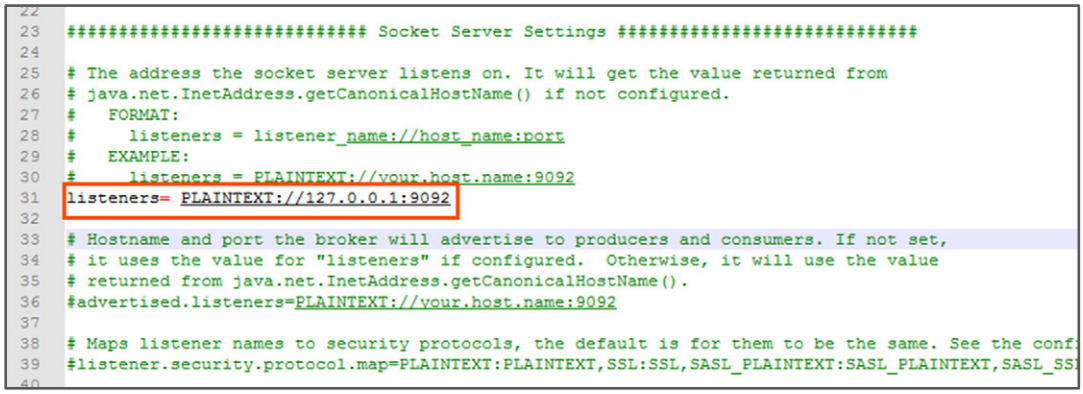
Note: The screen on your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

STEPS FOR THE INSTRUCTOR

Notes:

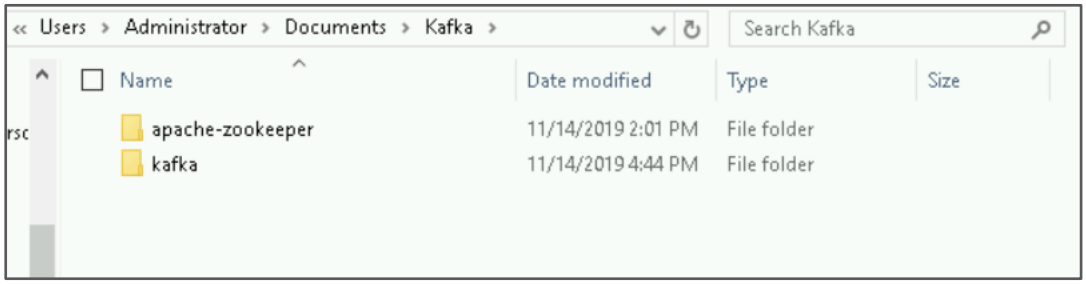
- For this lab, Kafka Server and Apache ZooKeeper are already installed in your system.
- In the **C:\Users\Administrator\Documents\Kafka\kafka\config** folder, open the **server.properties** file. Scroll down and ensure that the listeners field value is as mentioned below:

listeners= PLAINTEXT://127.0.0.1:9092



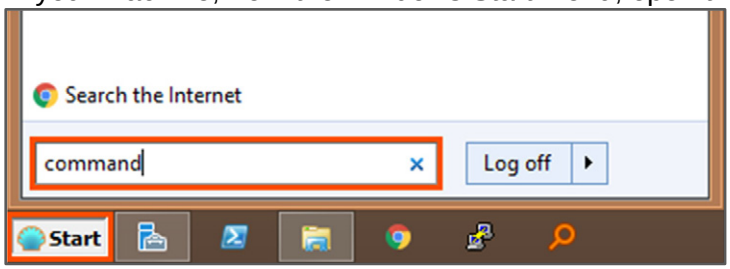
(The above-mentioned port and IP address are the default values where the Kafka Server will be installed. They will change in the scenarios where you do not have the default port as free. In such cases, it will take the next available free port.)

- Navigate to the Kafka installation directory **C:\Users\Administrator\Documents\Kafka**, and make sure you DO NOT see a folder named **kafkaData**. In case you see the kafkaData folder, delete it.

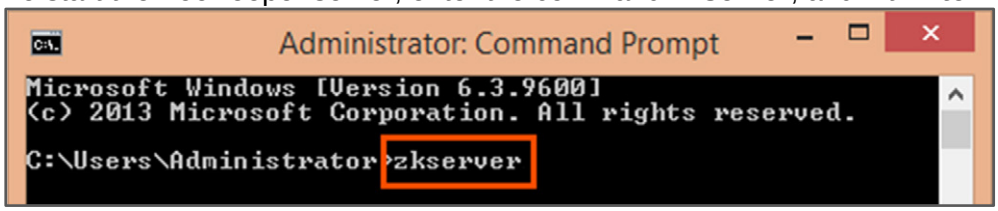


Start Apache ZooKeeper and Kafka server:

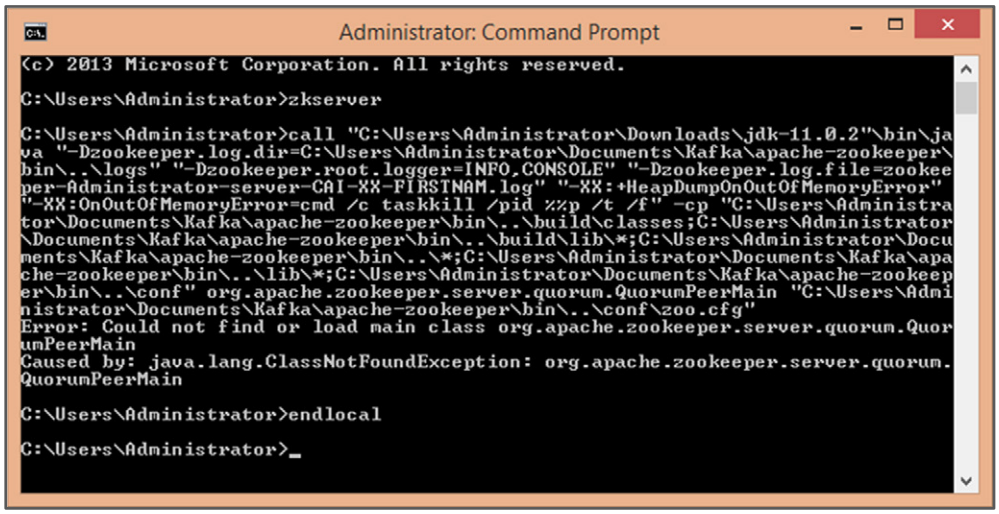
1. In your machine, from the Windows Start menu, open the **Command Prompt**.



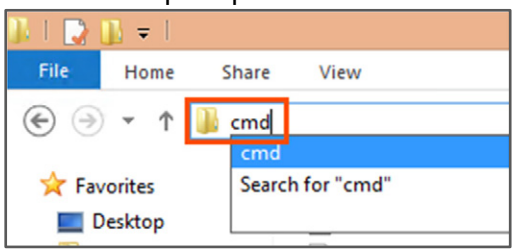
2. To start the ZooKeeper server, enter the command **zkserver**, and hit **Enter**.



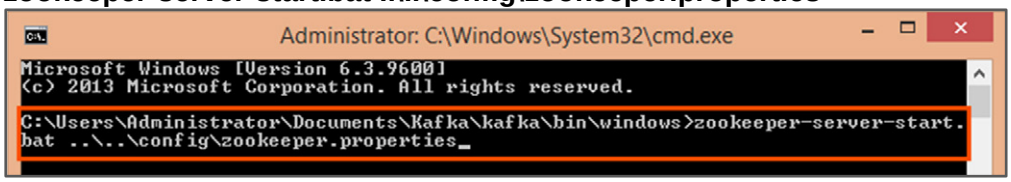
Result:



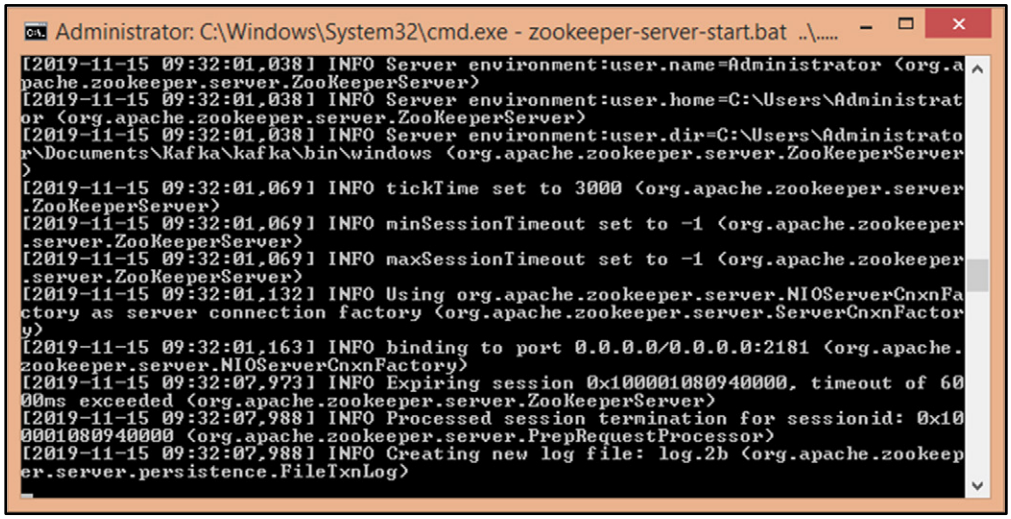
3. From the **C:\Users\Administrator\Documents\Kafka\kafka\bin\windows** folder, open a command prompt.



4. To execute the ZooKeeper server internally, enter the following command: **zookeeper-server-start.bat ..\..\config\zookeeper.properties**

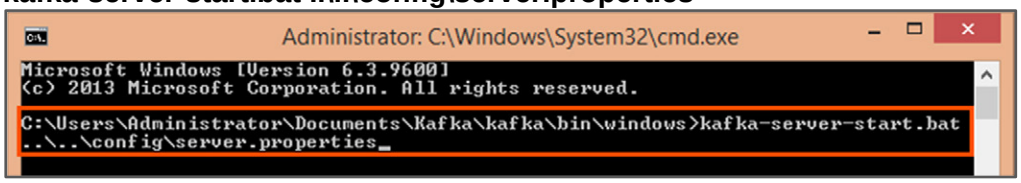


Result:

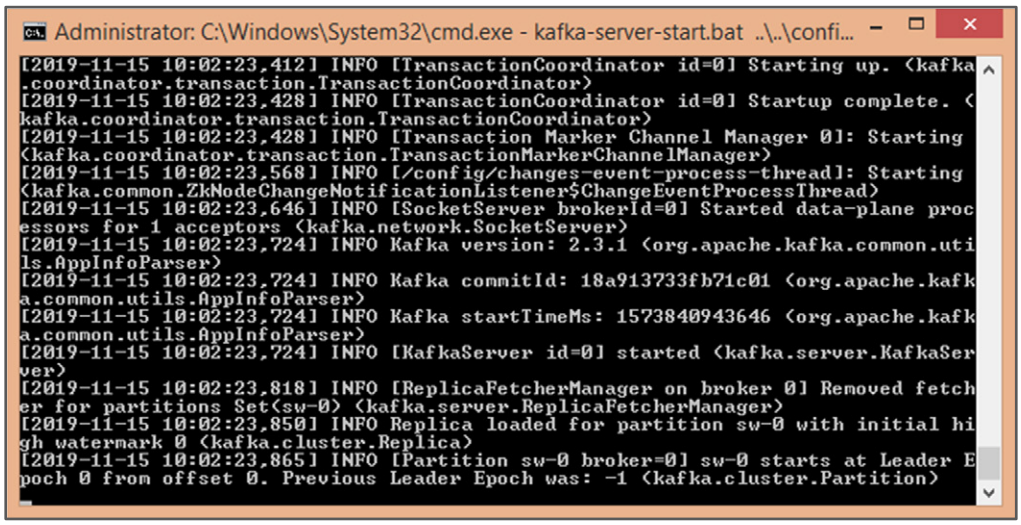


Important: Do not close any of the command prompts till the end of this lab.

5. In the same **C:\Users\Administrator\Documents\Kafka\kafka\bin\windows** folder, open another command prompt, and enter the following command to start the Kafka server:
kafka-server-start.bat ..\..\config\server.properties



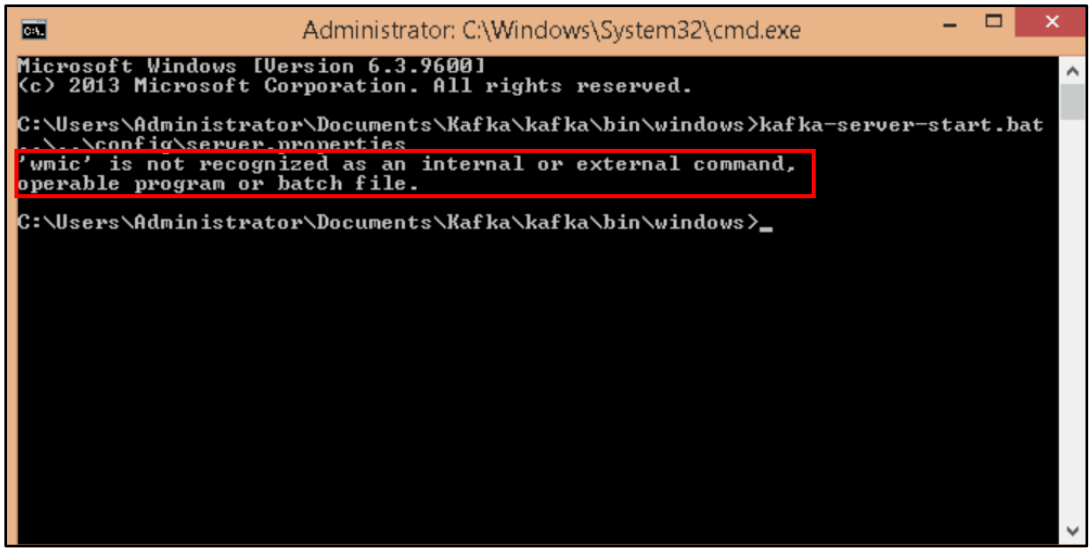
6. **Result**:



Important: Do not close any of the command prompts till the end of this lab.

Note: Sometimes, you can notice the following error while executing the above command:

ERROR: "wmic is not recognized as an internal or external command, operable program or batch file."



If you get the above error, then:

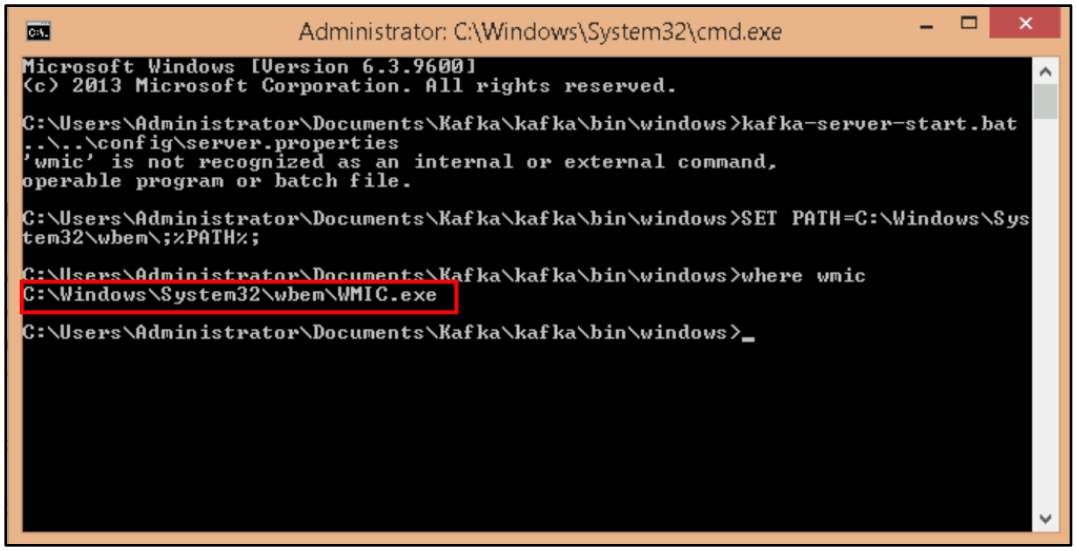
1. Run the following command in the same command window for adding the folder containing wmic to the PATH environment variable:

   SET PATH=C:\Windows\System32\wbem\;%PATH%;

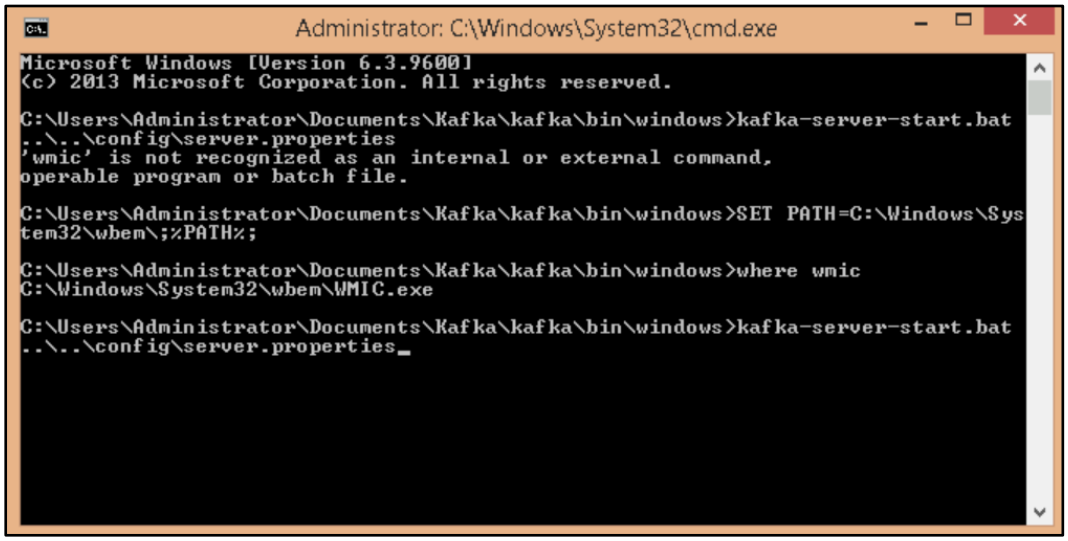
2. After setting the path variable using the command above, run the following command to confirm whether *wmic* is picked properly:
   where wmic

   The above command should return the following as output:

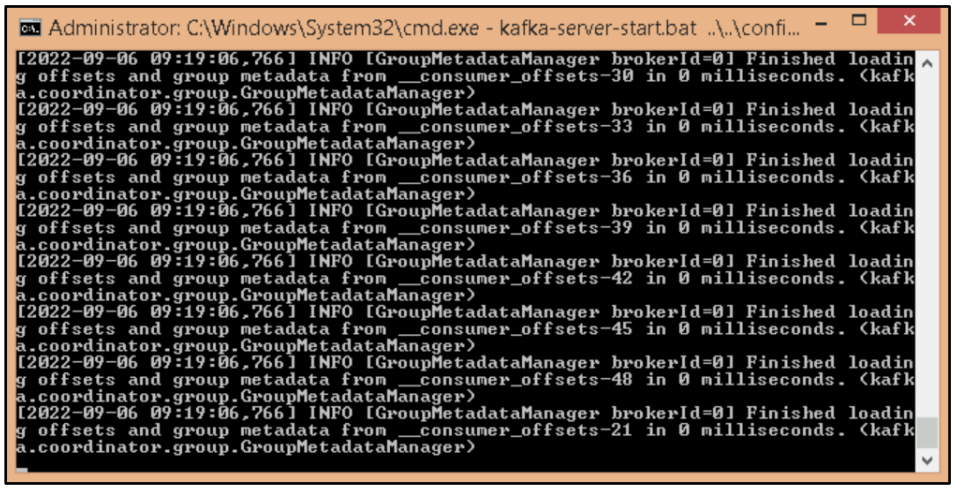
   C:\Windows\System32\wbem\WMIC.exe



3. **Do Not close the command prompt**. In the same command prompt, enter the **kafka-server-start.bat ..\..\config\server.properties** command and press **Enter**.



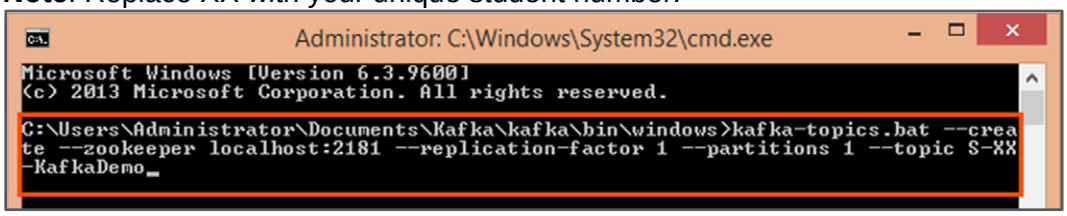
Result:



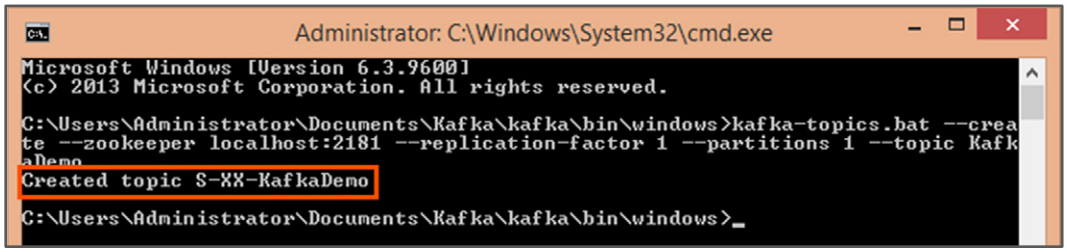
Create a topic:

You need to create a topic for each learner. For example, if there are 10 learners in the class, you need to create 10 topics. Use the following step to create a topic:

7. In the same **C:\Users\Administrator\Documents\Kafka\kafka\bin\windows** folder, open another command prompt, and enter the following command:
   kafka-topics.bat --create --zookeeper localhost:2181 --replication-factor 1 --partitions 1 --topic S-XX-KafkaDemo
   Note: Replace XX with your unique student number.



Result:



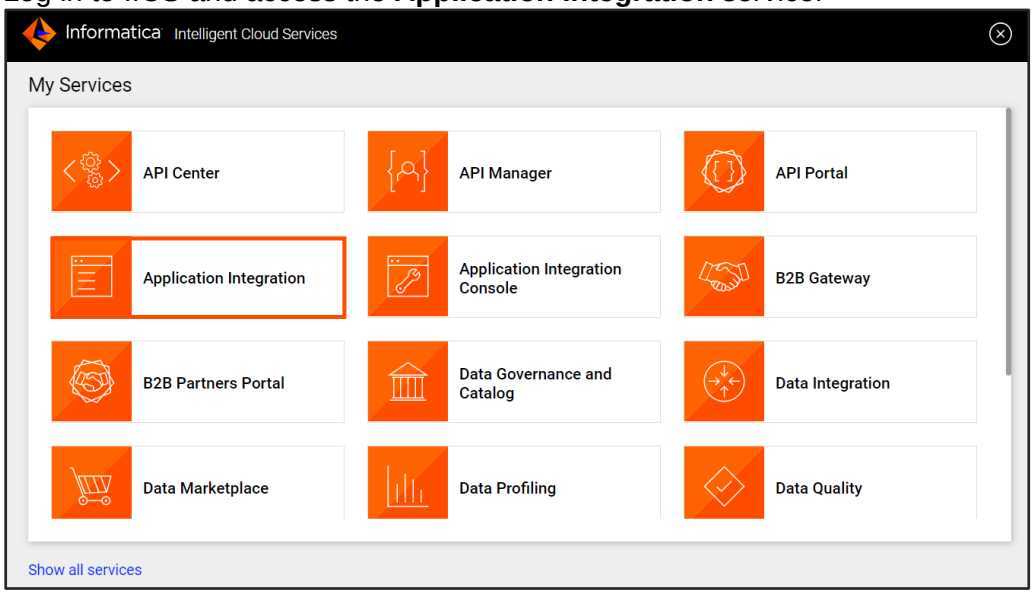
Important: Do not close any of the command prompts till the end of this lab.

NOTE: DO NOT CLOSE THE KAFKA CONNECTION. We will use this connection while executing Lab 13-3: Start and Stop an Event Source in a Listener-based Connection.

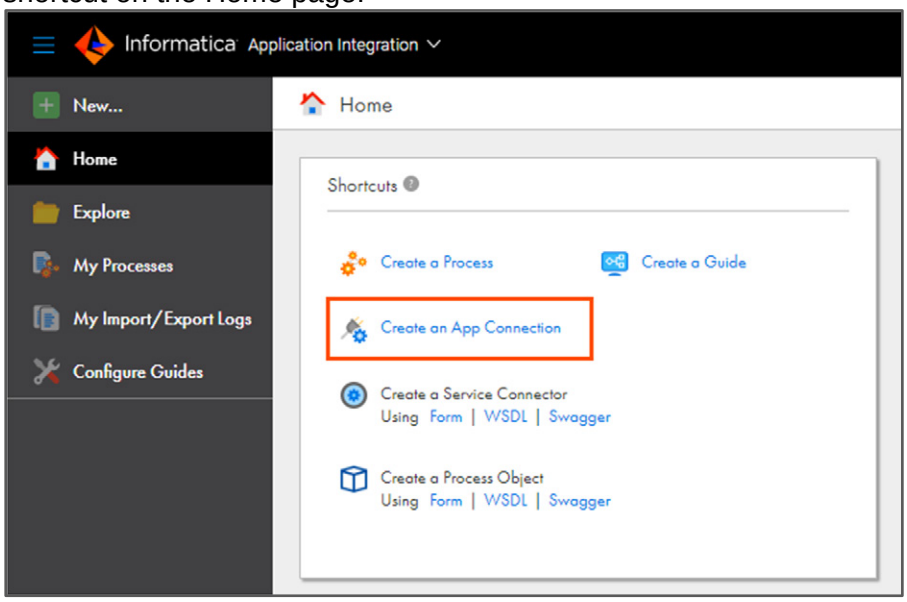
STEPS FOR THE LEARNERS

Create a Kafka connection:

8. Log in to IICS and access the **Application Integration** service.



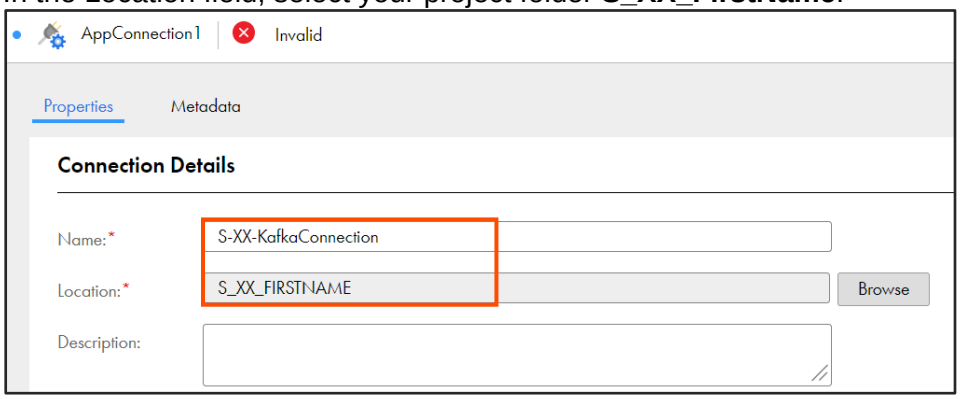
9. Create a new **App Connection**. You can also create a new App Connection using the shortcut on the Home page.



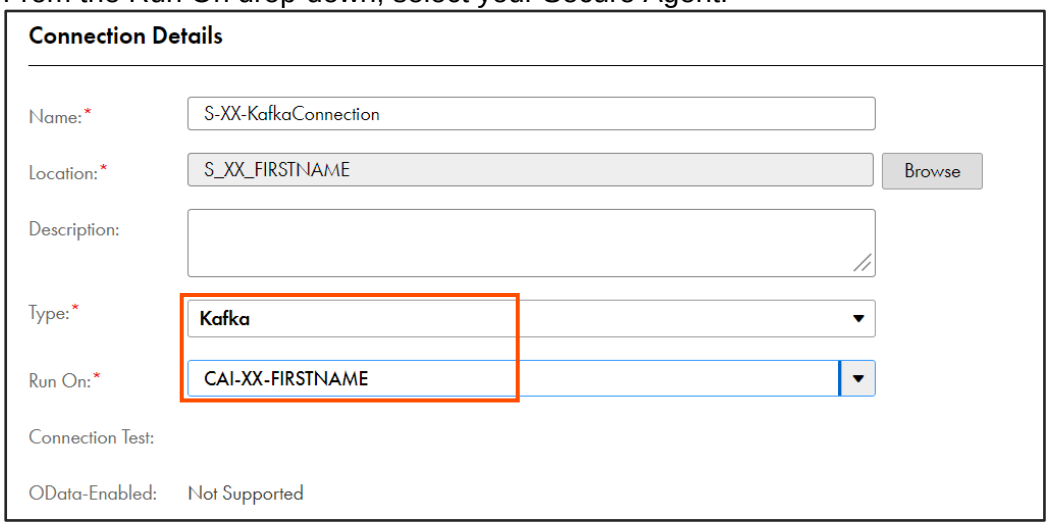
10. In the Name field, enter **S-XX-KafkaConnection**.

    Note: Here, XX is the unique student number.

11. In the Location field, select your project folder **S_XX_FirstName**.



12. From the Type drop-down, select **Kafka**.
13. From the Run On drop-down, select your Secure Agent.

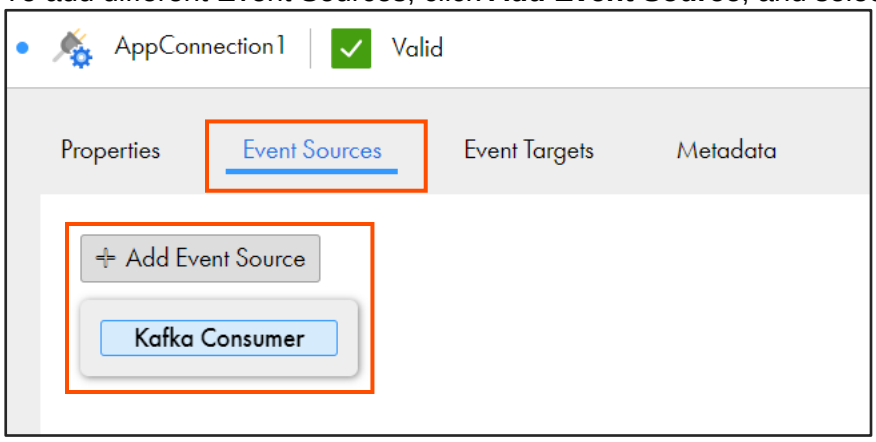


Note: Observe that there are two additional tabs–Event Sources and Event Targets along with Properties and Metadata tabs.

14. In the Connection Settings section, enter the Kafka Host as **localhost**, and Kafka Port as **9092**.

Note: This is the same port number that you checked in the notes section at the beginning of this lab.

15. To configure the Event Sources, click the **Event Sources** tab.
16. To add different Event Sources, click **Add Event Source**, and select **Kafka Consumer**.



17. For the **Name** and **Description** fields, retain the default values.
18. For the Enabled option, retain **Yes**.

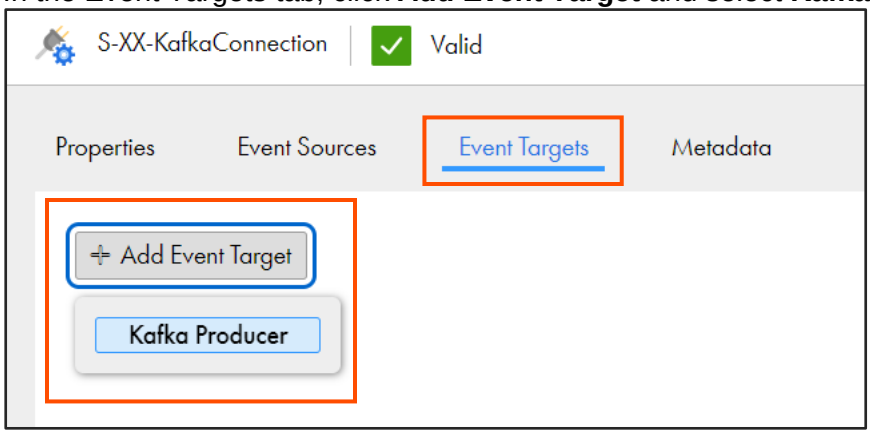
Note: When you select No, the event source is disabled until you are ready to use it. If you select Yes, the event source is available immediately after it is published.

19. In the Connection Settings section, in the Topic field, enter **S-XX-KafkaDemo**.

Note: Here, XX is the unique student number.

20. In the Group Id field, enter a Group Id. For this lab, you can enter your student number or date of birth. For example, you can enter the value 1.

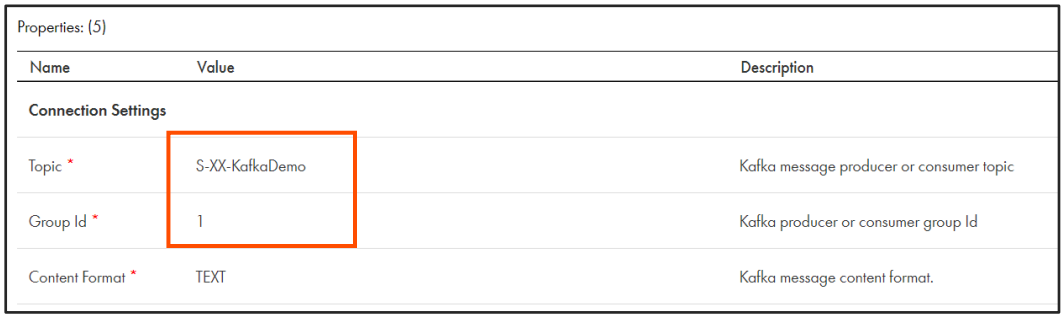
21. **Save** the connection.
22. In the Event Targets tab, click **Add Event Target** and select **Kafka Producer**.



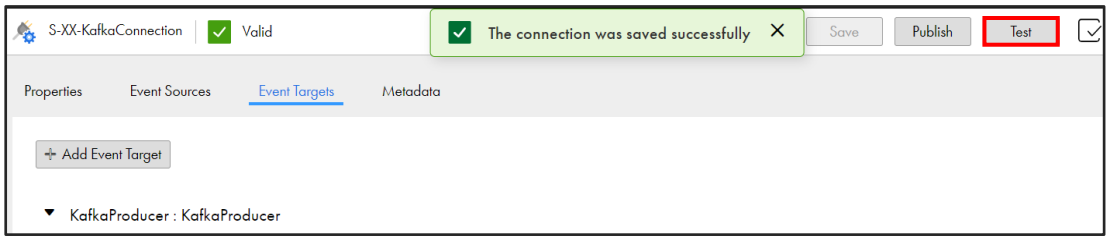
23. For the **Name** and **Description** fields, retain the default values.
24. In the Connection Settings section, in the Topic field, enter **S-XX-KafkaDemo**.

    Note: Here, XX is the unique student number.

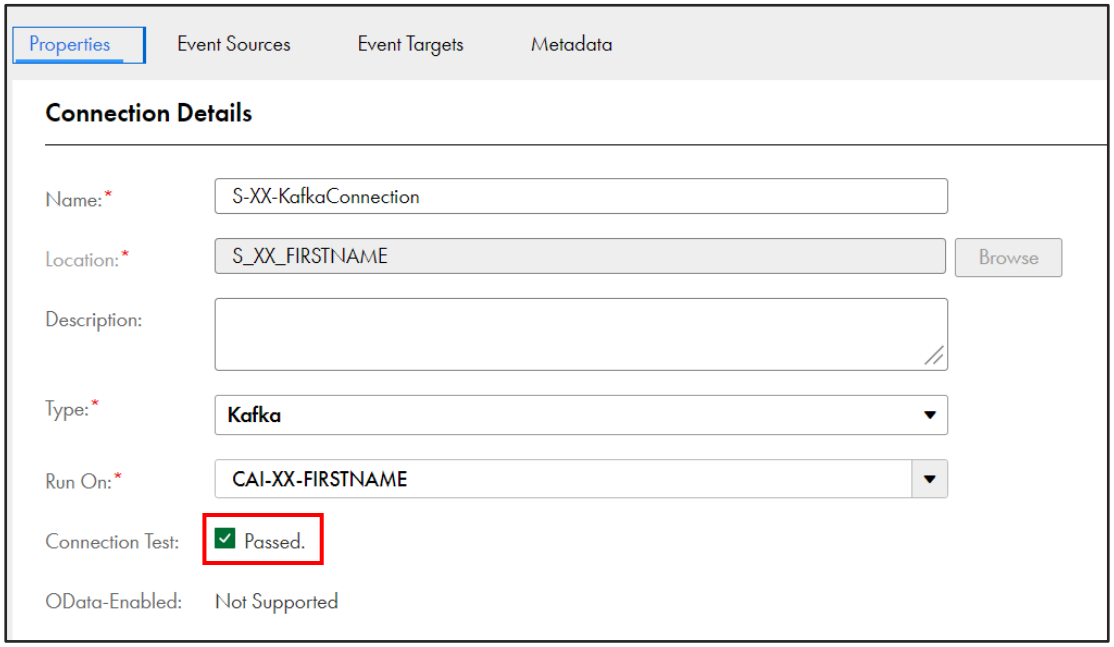
25. In the Group Id field, enter a Group Id. For this lab, you can enter your student number or date of birth. For example, you can enter the value 1.



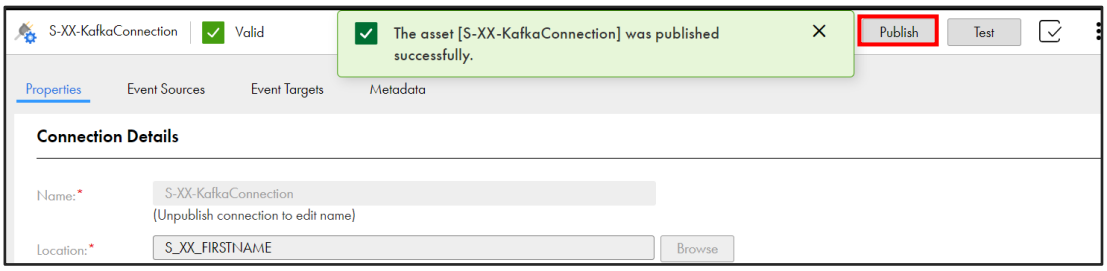
26. **Save** and **Test** the connection.



Note: In the Properties tab, observe that the Connection Test field mentions **Passed**.



27. **Publish** the connection.

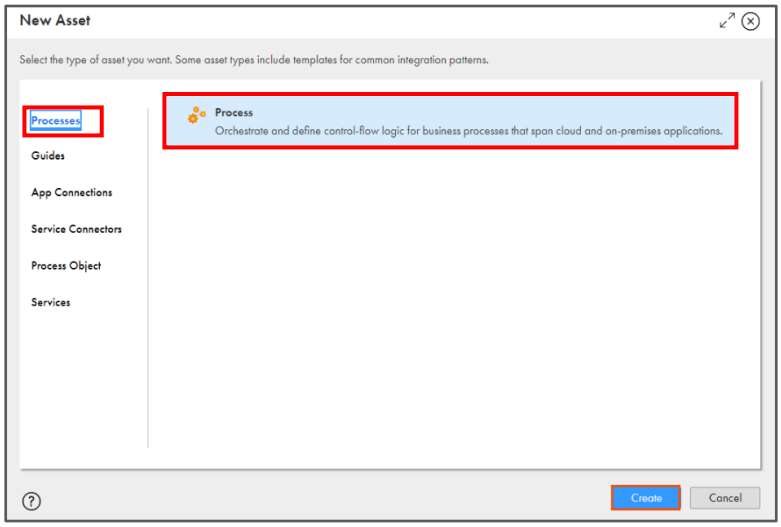


You get a notification – '**The asset [S-XX-KafkaConnection] was published successfully**'.

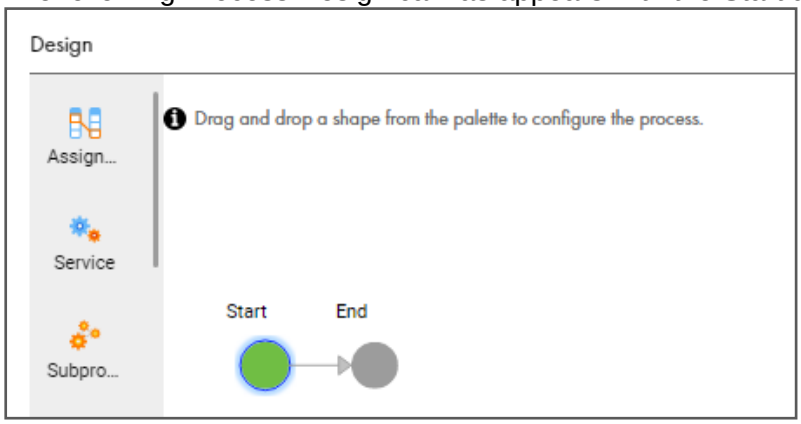
Create a Write process:

You will now create a process to write a message to the topic.

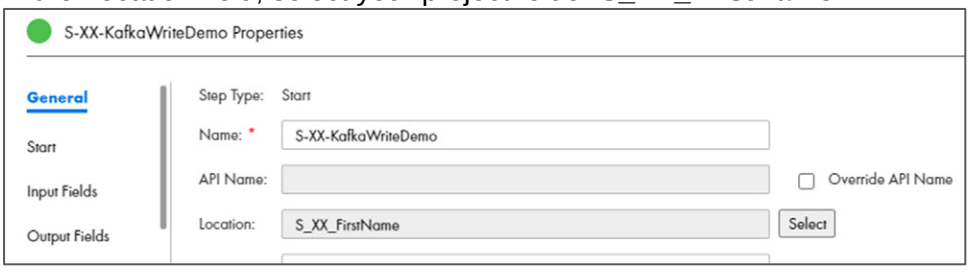
28. Create a new **Process**.



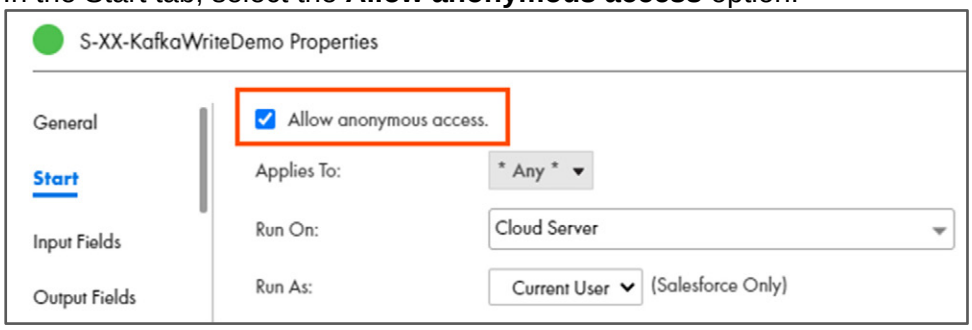
The following Process Design canvas appears with the Start and End steps.



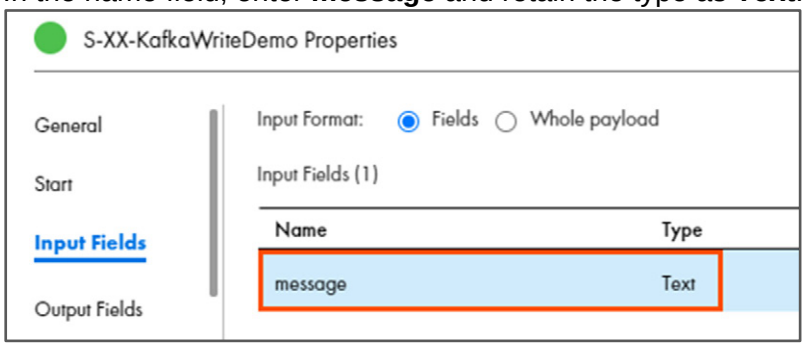
29. From the process canvas, select the **Start** step.
30. In the General tab, enter the name as **S-XX-KafkaWriteDemo**.
31. Skip the API Name option.
32. In the Location field, select your project folder **S_XX_FirstName**.



33. In the Start tab, select the **Allow anonymous access** option.

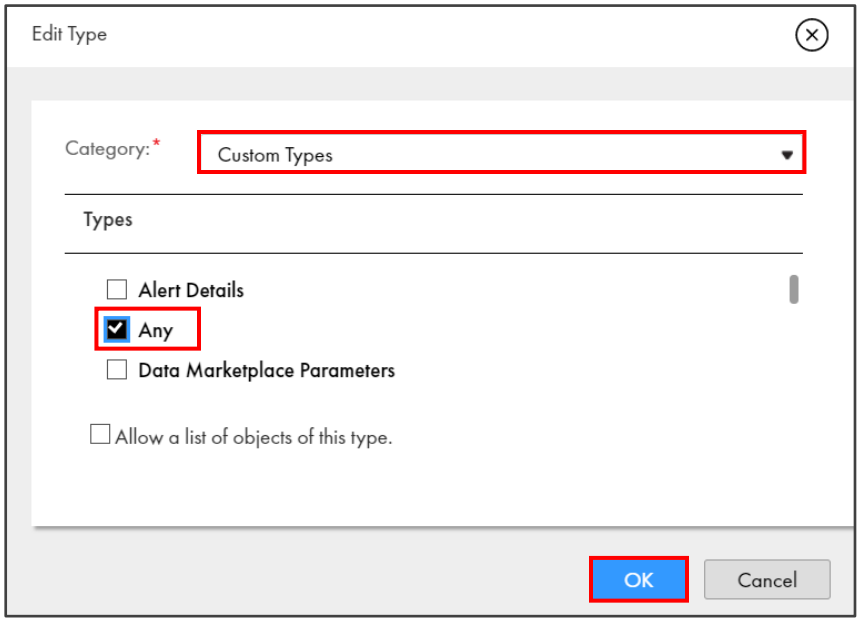


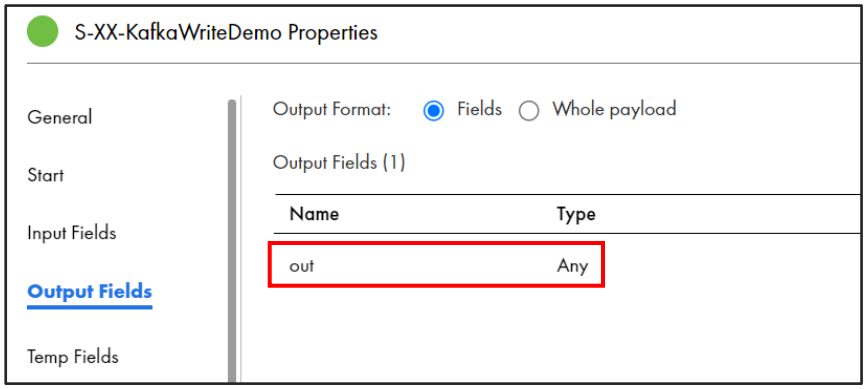
34. In the Input Fields tab, to add a new input field, click .
35. In the name field, enter **message** and retain the type as **Text**.



36. In the Output Fields tab, to add a new output field, click .
37. In the name field, enter **out**.
38. From the Type drop-down, select **More types** and from the Category drop-down, select **Custom Types**.

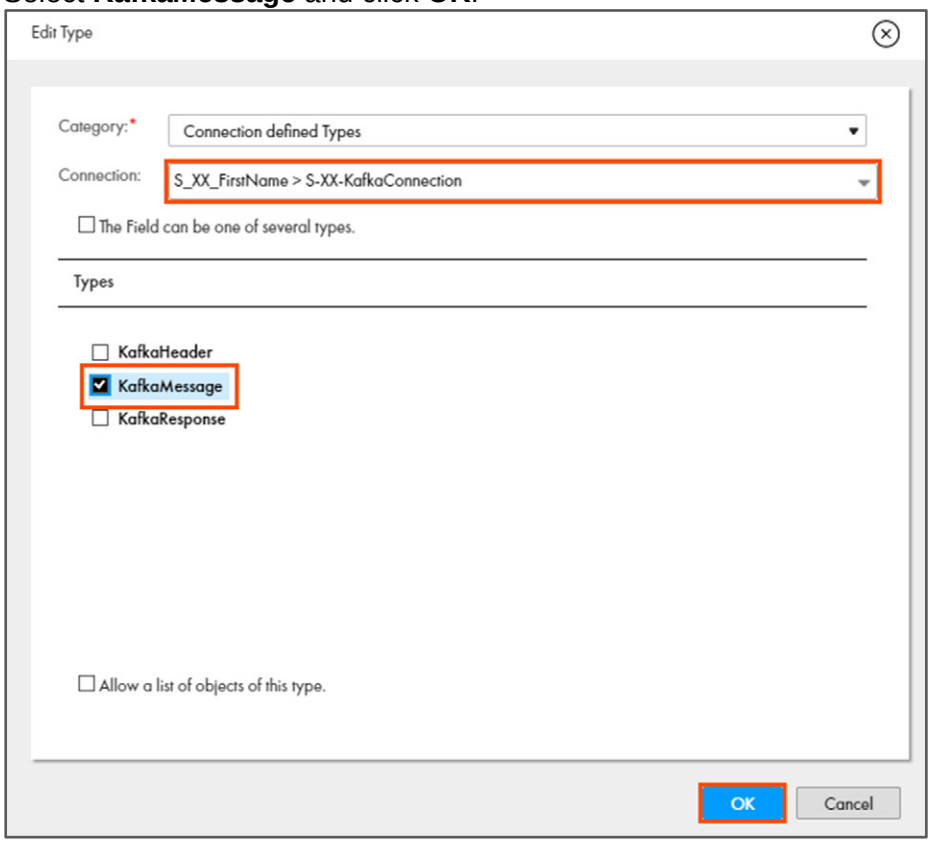
39. Select **Any** and click **OK**.



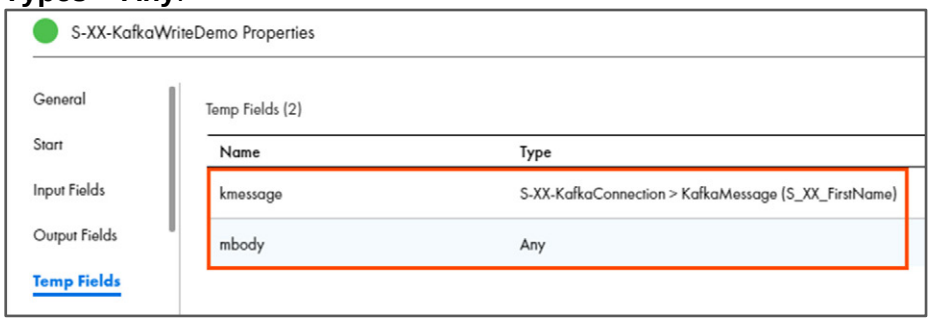


40. In the Temp Fields tab, to add a new temp field, click .
41. In the Name field, enter **kmessage**.
42. From the Type field, select **More types** and from the Category field, select **Connection defined Types**.
43. From the Connection drop-down, navigate to your project folder, and select **S-XX-KafkaConnection**.

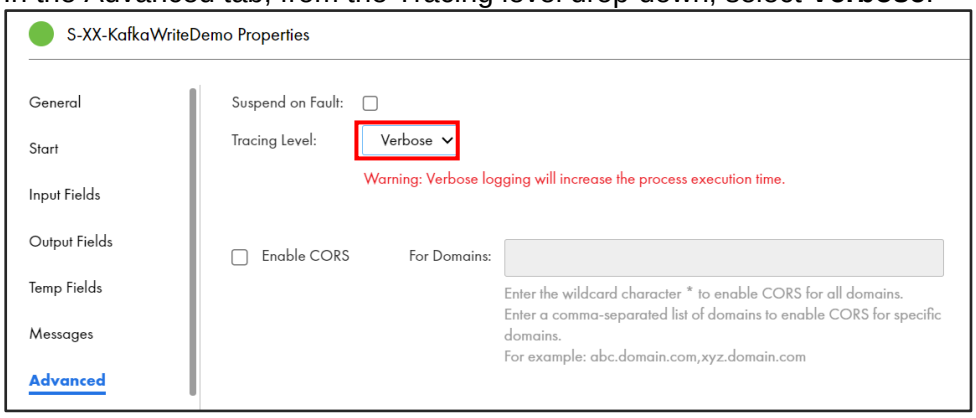
44. Select **KafkaMessage** and click **OK**.



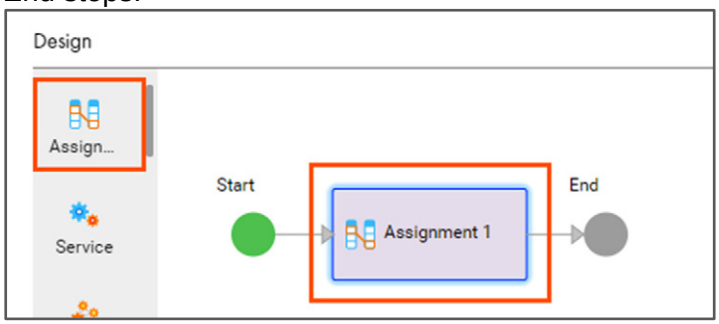
45. Similarly, add another temp field named **mbody**, and Type as **More Types** > **Custom Types** > **Any**.



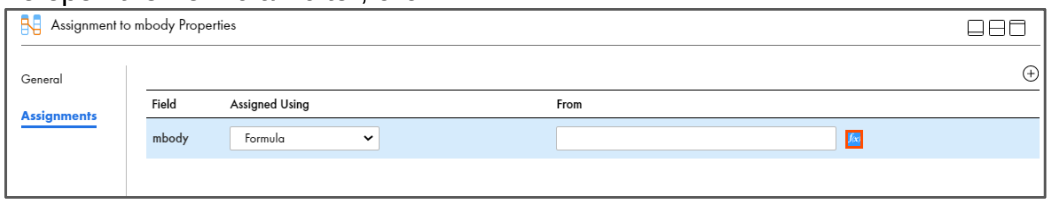
46. In the Advanced tab, from the Tracing level drop-down, select **Verbose**.



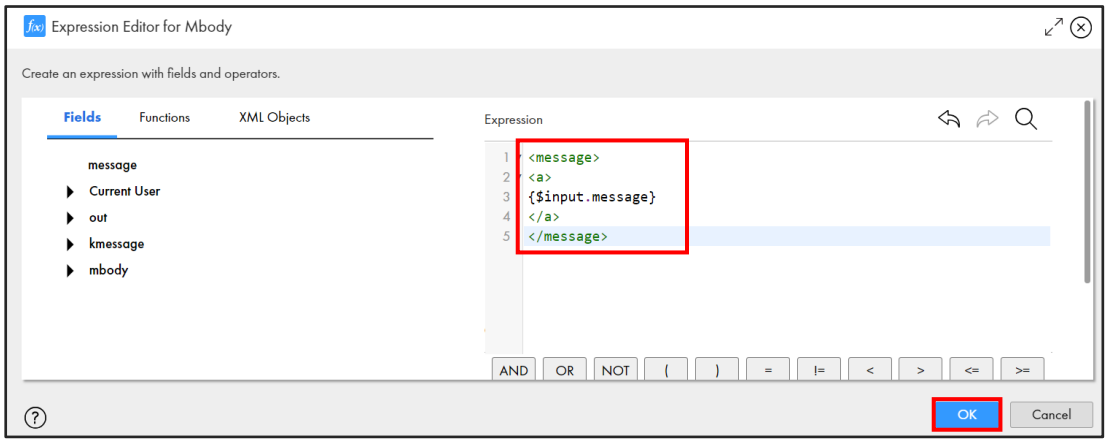
47. On the process canvas, drag and drop an **Assignment** step between the Start and the End steps.



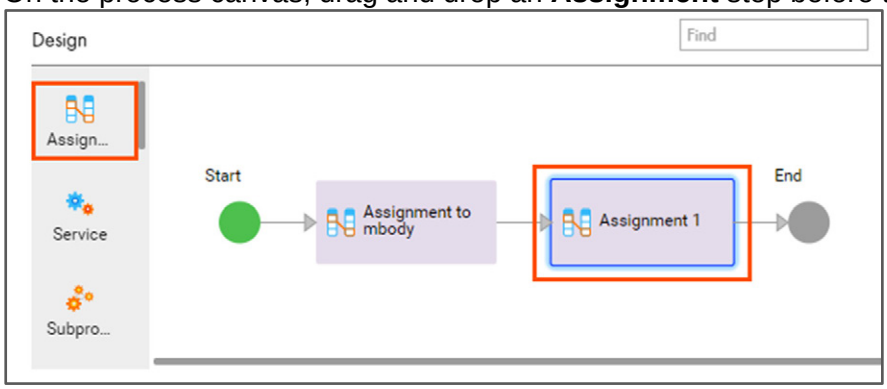
48. Retain the default name in the General tab.
49. In the Assignments tab, click plus (**+**) icon.
50. From the Add Field drop-down, select **mbody** and from the Assigned Using drop-down, select **Formula**.
51. To open the Formula Editor, click .



52. In the Expression Editor window, enter the formula mentioned below:
    <message>
    <a>
    {$input.message}
    ****
    </message>
53. Click **OK**.



54. On the process canvas, drag and drop an **Assignment** step before the End step.

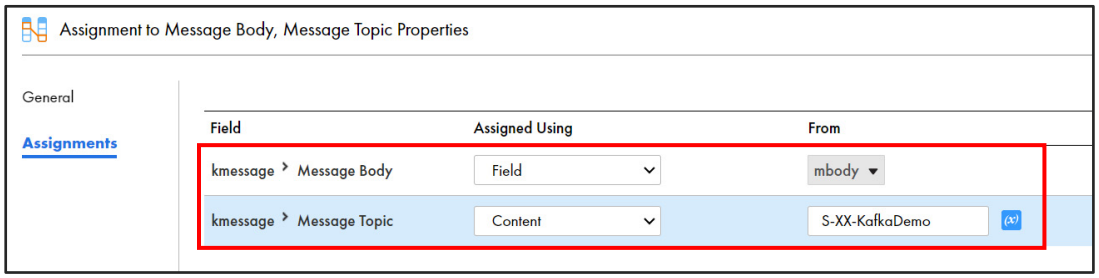


55. In the Assignments tab, click the **plus** icon.
56. From the kmessage drop-down, select **Message Body**.

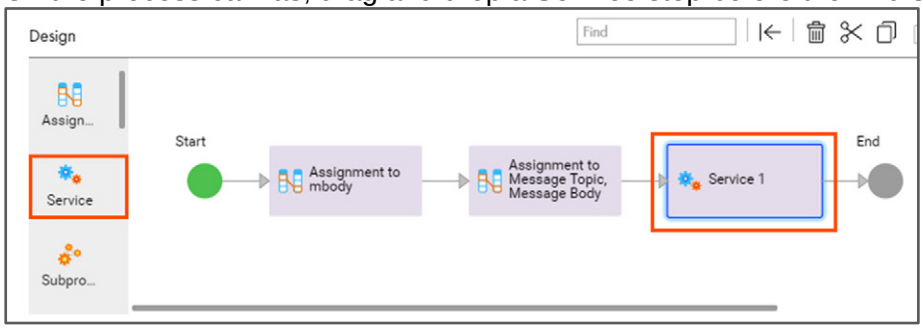
57. Similarly, add another field **kmessage** > **Message Topic**.

58. Assign values to the fields as mentioned in the table:

Field	Assigned Using	From
Message Body	Field	mbody
Message Topic	Content	S-XX-KafkaDemo

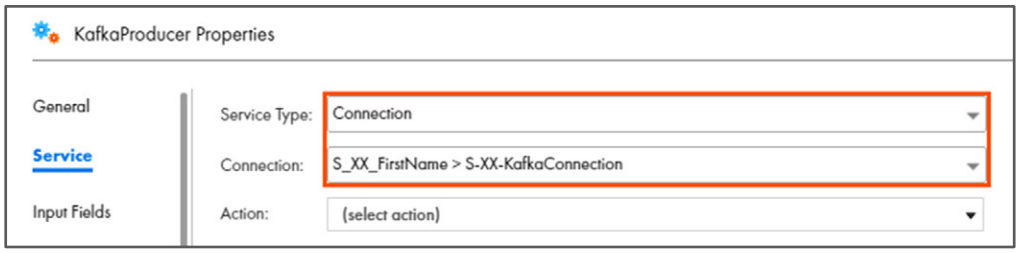


59. On the process canvas, drag and drop a **Service** step before the End step.

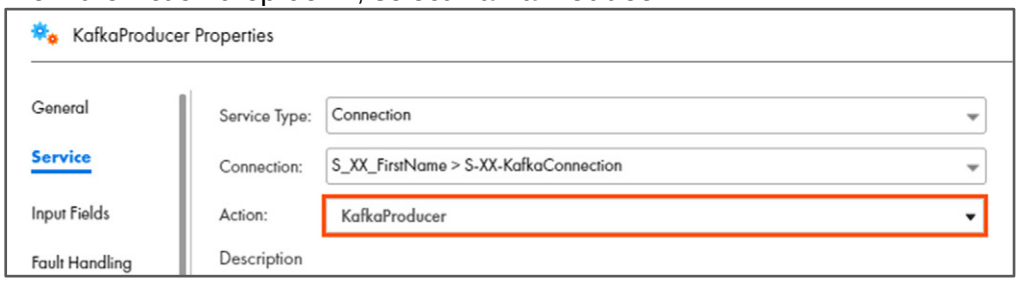


60. In the Name field, enter **KafkaProducer**.

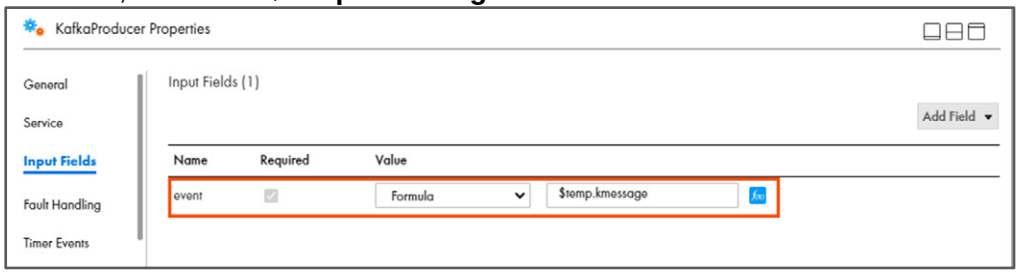
61. In the Service tab, from the Service Type drop-down, select **Connection**.
62. From the Connection drop-down, from your project folder, select **S-XX-KafkaConnection**.



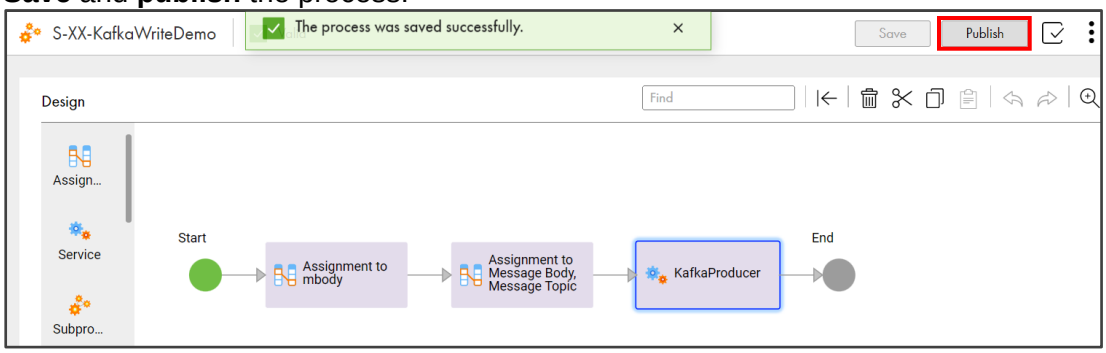
63. From the Action drop-down, select **KafkaProducer**.



64. In the Input Fields tab, for the event input field, from the Value drop-down, select **Formula**, and enter **$temp.kmessage**.



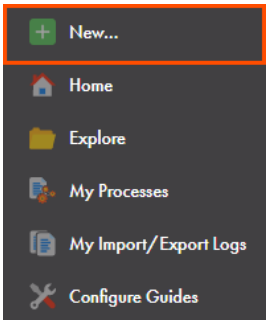
65. **Save** and **publish** the process.

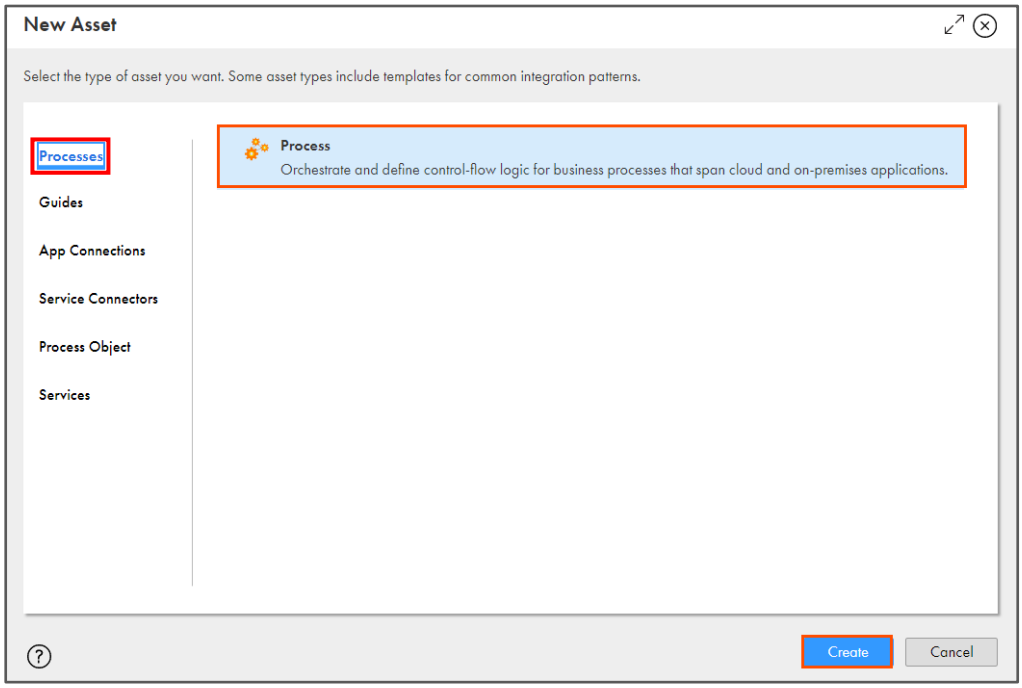


Create a Read process:

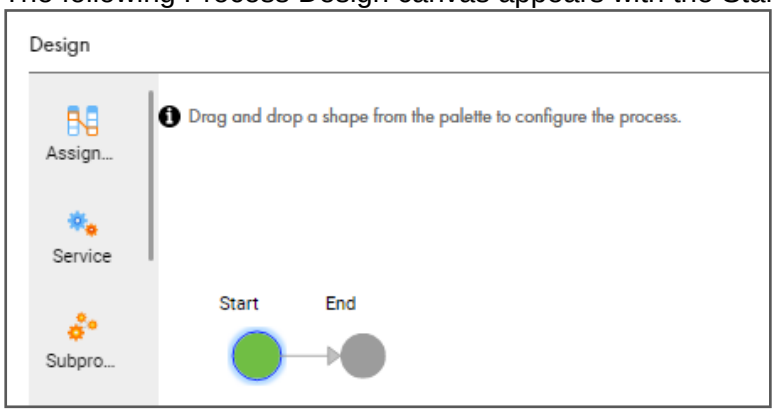
Now, you will read the data from the topic. This is needed in order to read a message from the Kafka topic.

66. Create a new **Process**.

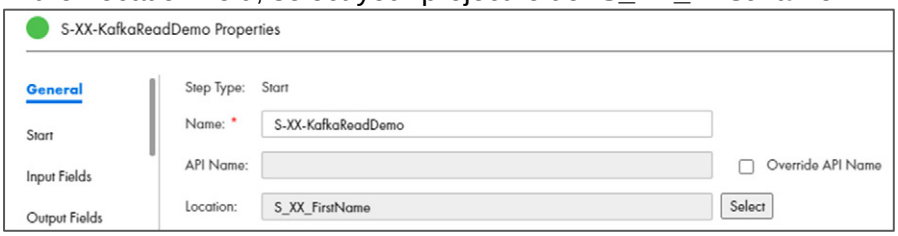




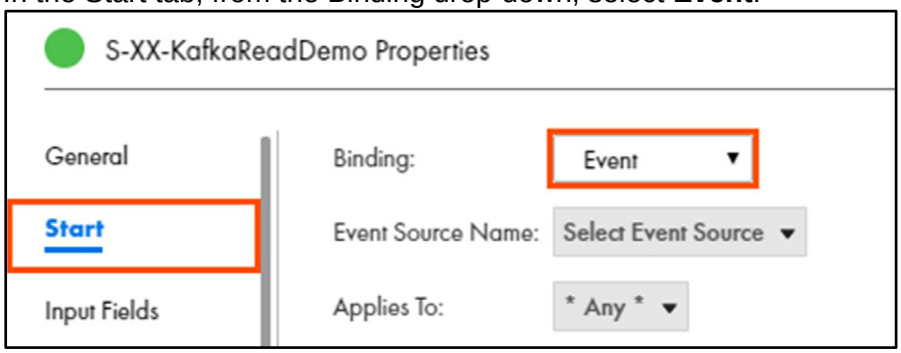
The following Process Design canvas appears with the Start and End steps.



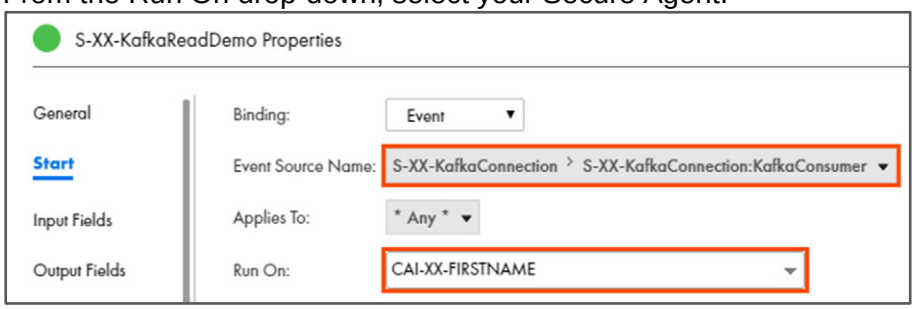
67. From the process canvas, select the **Start** step.
68. In the General tab, enter the name as **S-XX-KafkaReadDemo**.
69. Skip the API Name option.
70. In the Location field, select your project folder **S_XX_FirstName**.



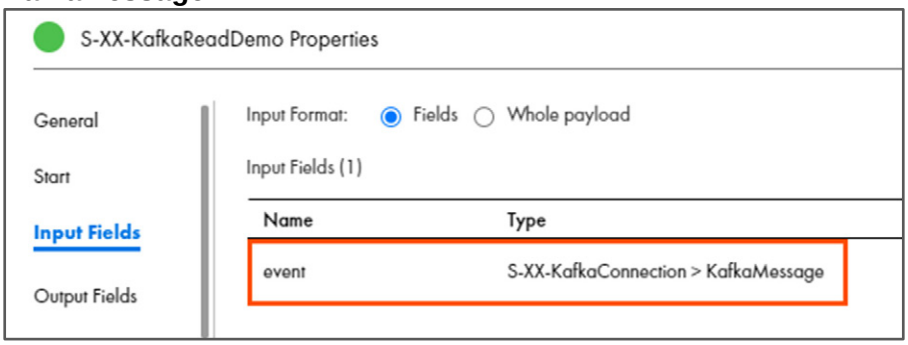
71. In the Start tab, from the Binding drop-down, select **Event**.



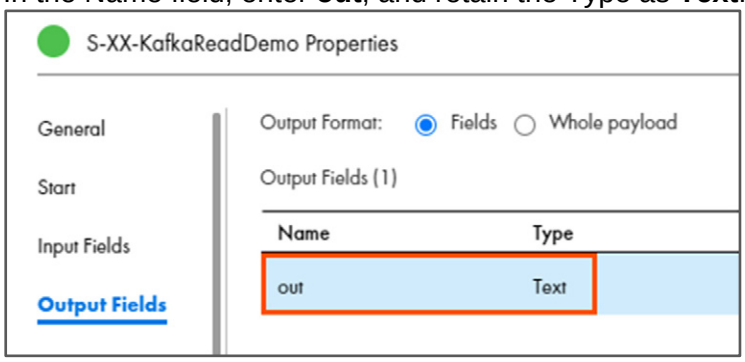
72. From the Event Source Name field, select **S-XX-KafkaConnection** > **S-XX-KafkaConnection: KafkaConsumer**.
73. From the Run On drop-down, select your Secure Agent.



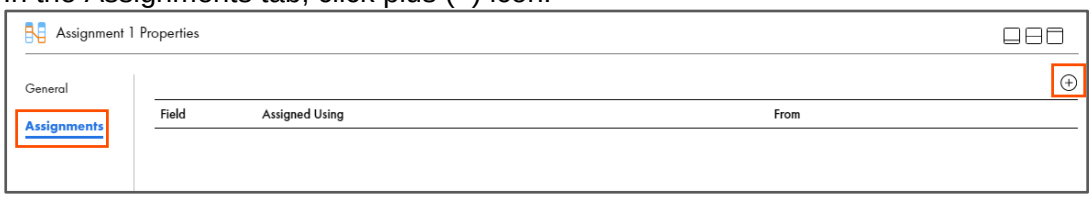
74. Ensure that the **Input Fields** tab contains the **event** field.
    Note: If the event field does not exist, perform the following steps:
    i. In the Input Fields tab, click **Add Field**.
    ii. In the Name field, enter **event**, and select **S-XX-KafkaConnection** > **KafkaMessage**.



75. In the **Output fields** tab, click the plus (**+**) icon to add a field.
76. In the Name field, enter **out**, and retain the Type as **Text**.

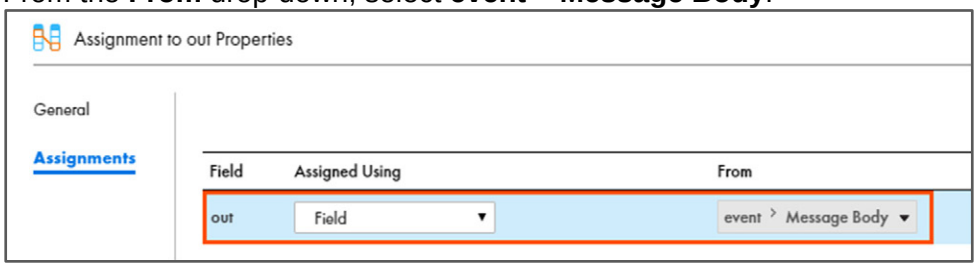


77. On the process canvas, drag and drop an **Assignment** step between the Start and the End steps.
78. In the Assignments tab, click plus (**+**) icon.



79. Select **out**.
80. From the Assigned Using drop-down, select **Field**.

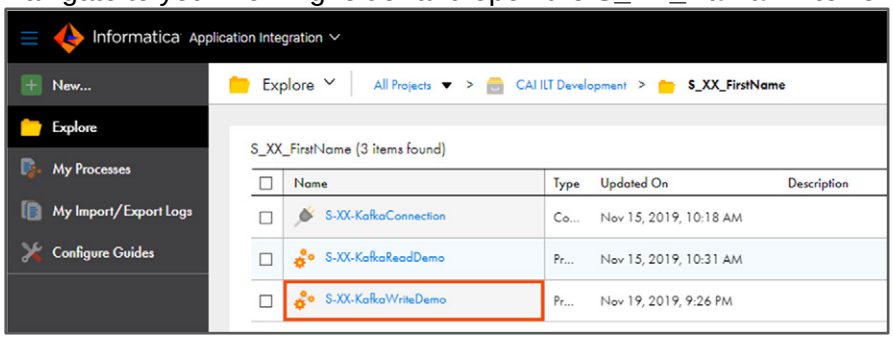
81. From the **From** drop-down, select **event** > **Message Body**.



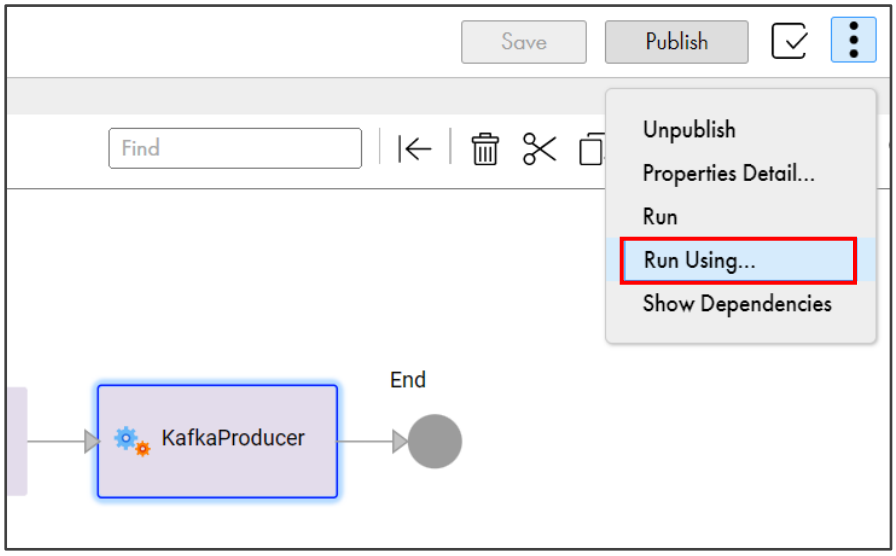
82. **Save** and **publish** the process.
Note: As this process is event-driven, it keeps on pulling the Kafka topic, and as soon as there is a message in the topic it automatically triggers this process.

Execute the process to read and write data to the topic:

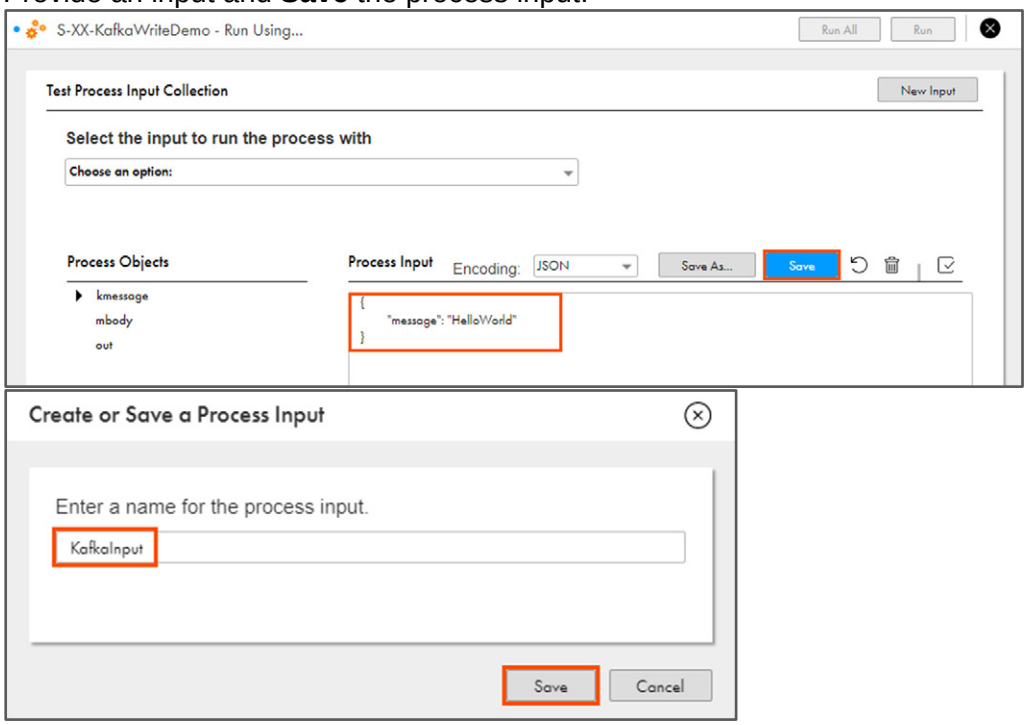
83. Navigate to your working folder and open the **S_XX_KafkaWriteDemo** process.



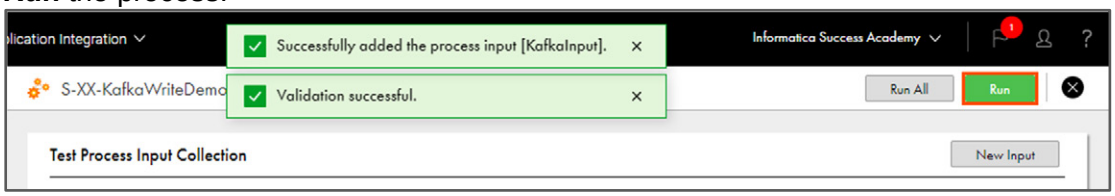
84. From the **Actions** menu, select **Run Using…**.



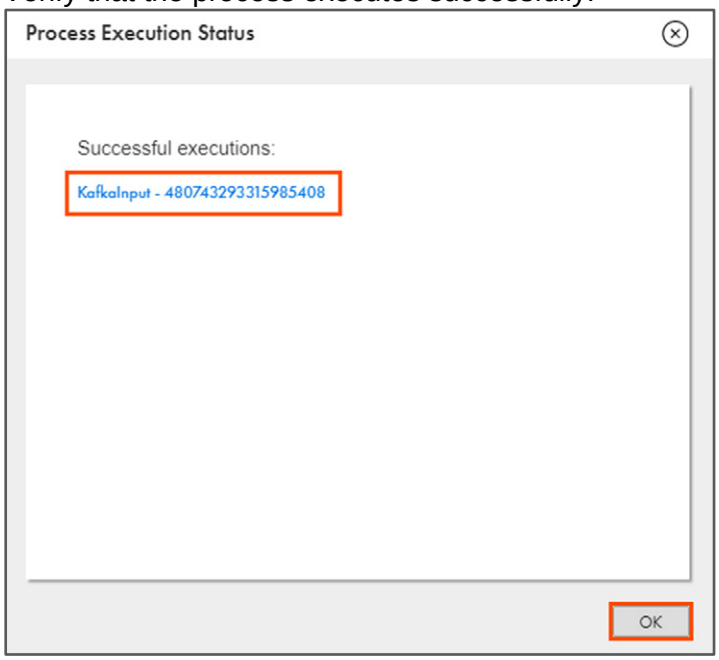
85. Provide an input and **Save** the process input.



86. **Run** the process.



87. Verify that the process executes successfully.



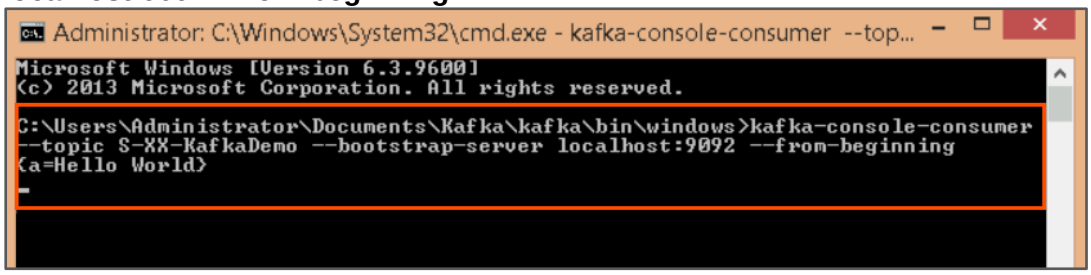
Note: As soon as you write the KafkaWriteDemo process, it will insert a message to the

topic. The KafkaReadDemo process that you have created will be automatically triggered as well since you have a message within the topic. Instructor, please give a demo to validate the write process:

DEMO

Note: This demo should be presented by the instructor.

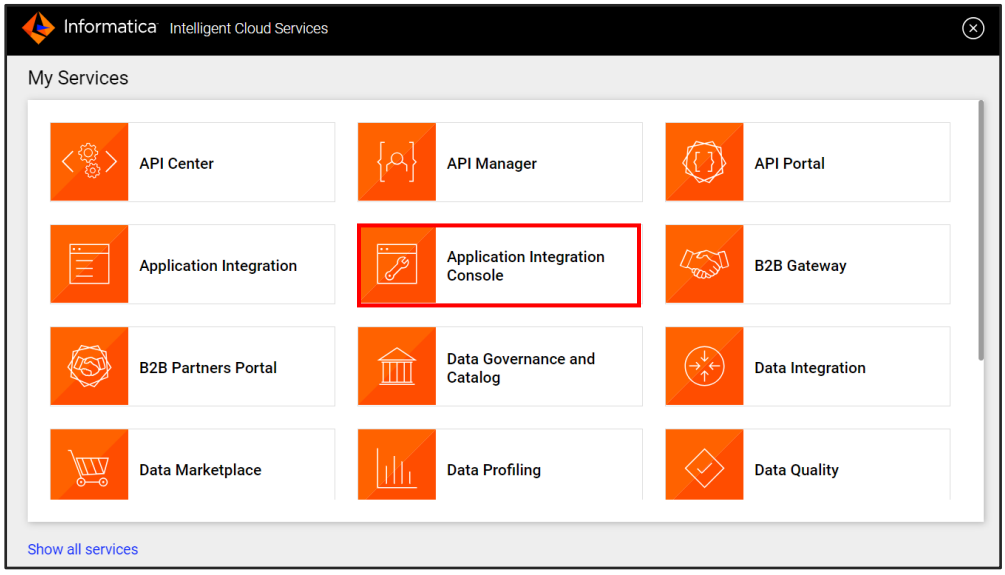
- To validate the write process, from the **C:\Users\Administrator\Documents\Kafka\kafka\bin\windows** folder, open a command prompt.
- Execute the following command: **kafka-console-consumer --topic S-XX-KafkaDemo --bootstrap-server localhost:9092 --from-beginning**



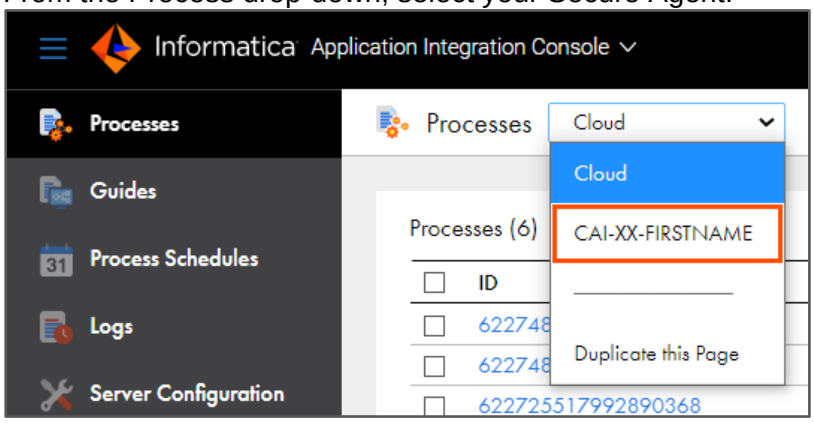
Note: Observe that the message that you entered in the input field is obtained in the command prompt. You can change your input message, run the process again, and check the values in this command prompt. You can close all the command prompts later.

As the read process is automatically invoked by the read process, you can verify that the read process ran in the Application Integration Console service.

88. From the Processes drop-down, navigate to **Application Integration Console**.



89. From the Process drop-down, select your Secure Agent.



90. Verify that the Read process is in Completed state.

This concludes the lab.

Module 7: App Connections

Lab 7-7: Create Flat File Connection

Overview:
In this lab, you will create a connection using the Flat File. This connection can be invoked from a process.

Objectives:

- Create a Flat File connection

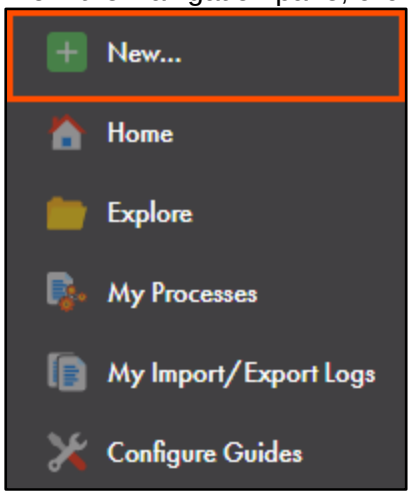
Duration:
15 Minutes

Tasks:

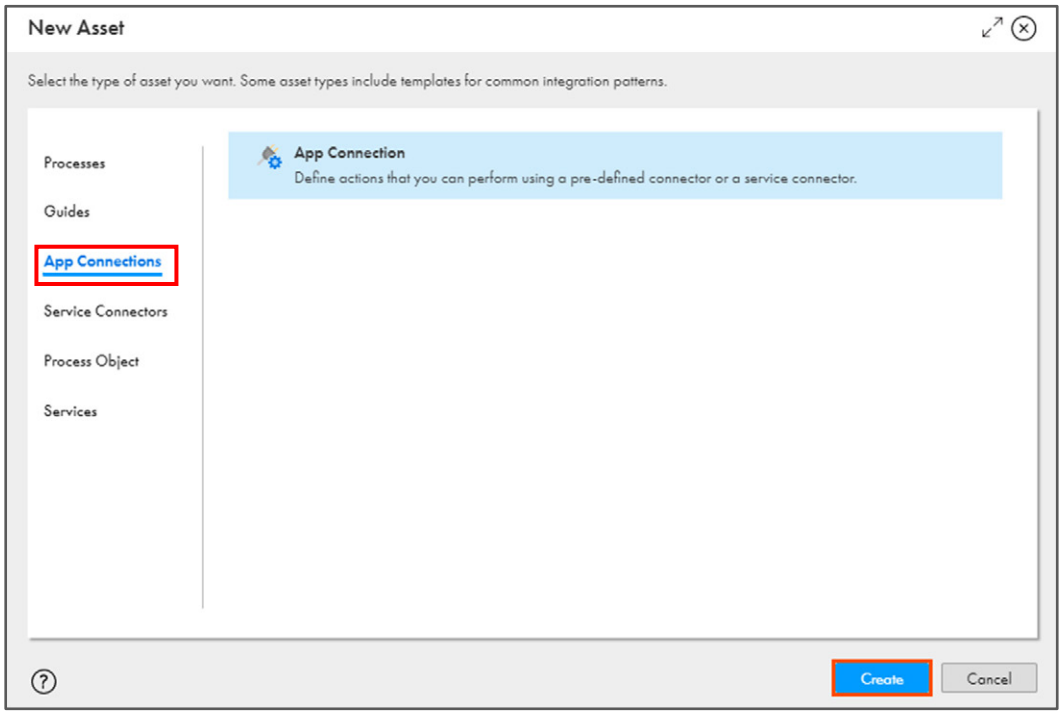
Note: The screen on your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

Create a connection:

1. Log in to IICS and access the **Application Integration** service.
2. From the Navigation pane, click **New**.



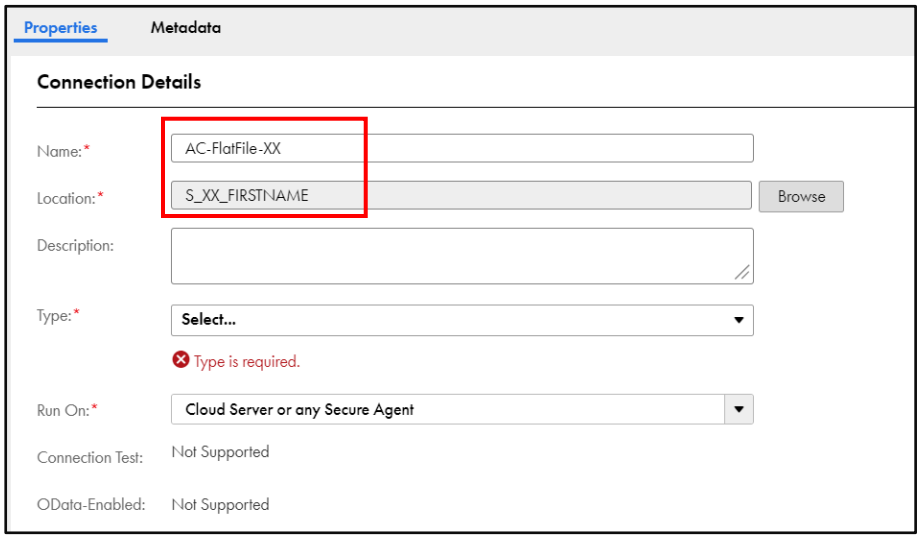
3. In the New Asset window, from the App Connections tab, select **App Connection**, and click **Create**.



4. In the Connection Details page, enter the connection name as **AC-FlatFile-XX**.

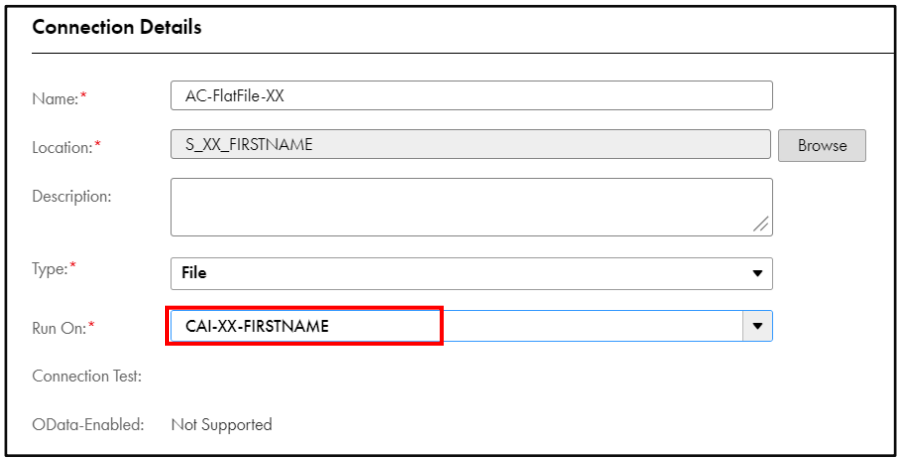
   Note: Replace XX with your student number.

5. In the Location field, select your project folder **S_XX_FirstName**.

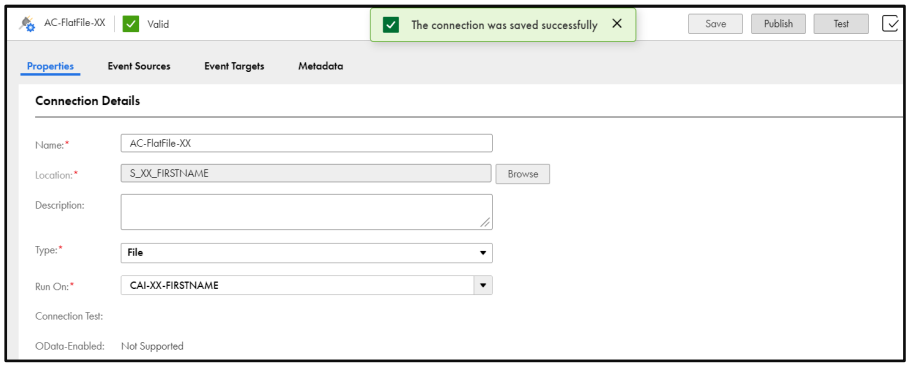


6. From the Type drop-down, select **File**.

7. From the Run On drop-down, select your Secure Agent group.

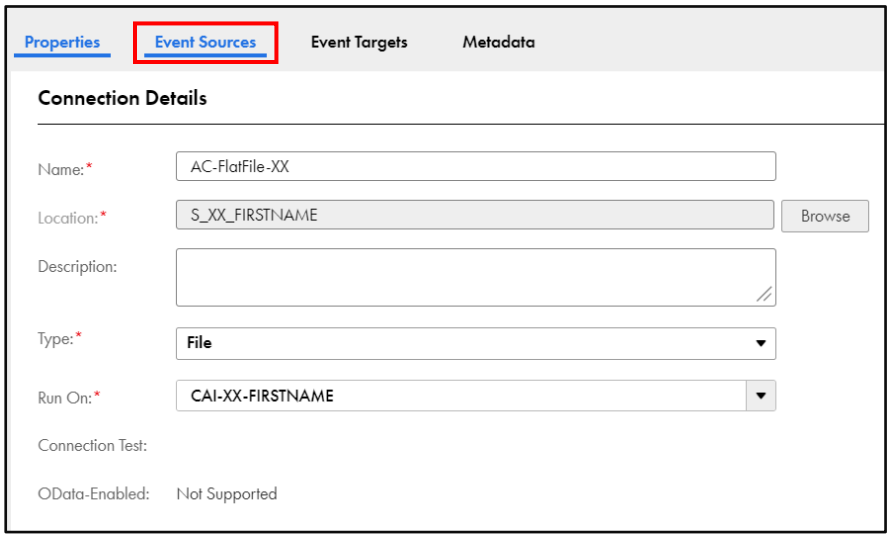


8. Click **Save** to save the connection.

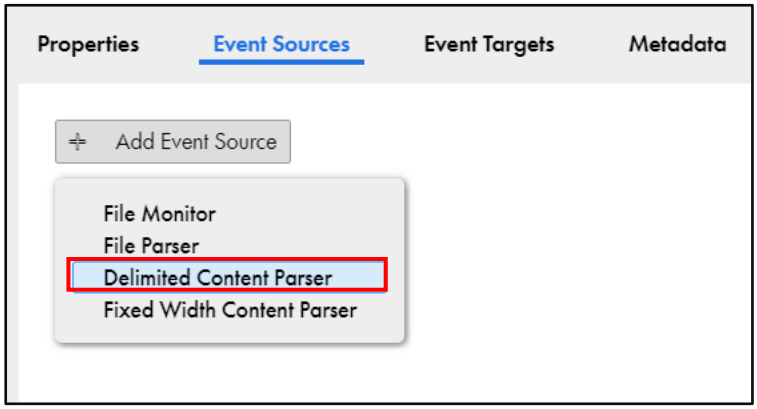


You get a notification that '**The connection was saved successfully**'.

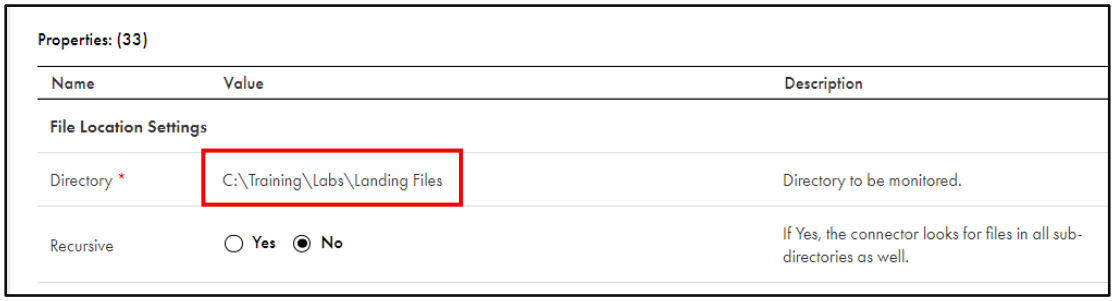
9. Click Event Source.



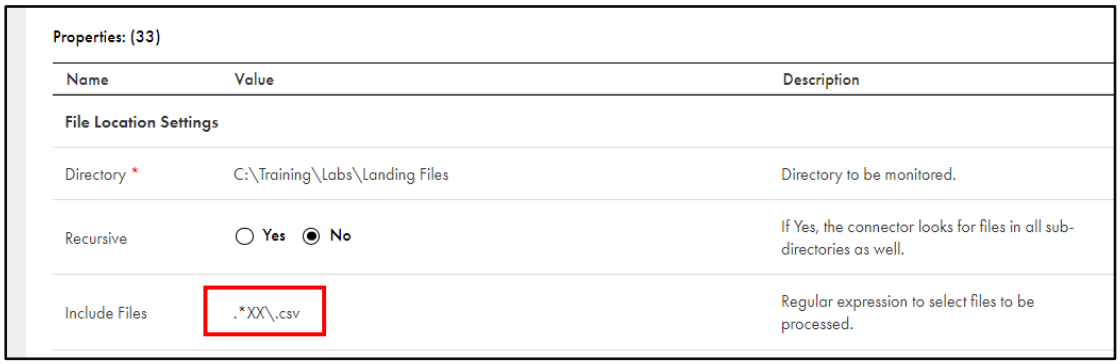
10. Click the **Add Event Source** tab and select **Delimited Content Parser**.



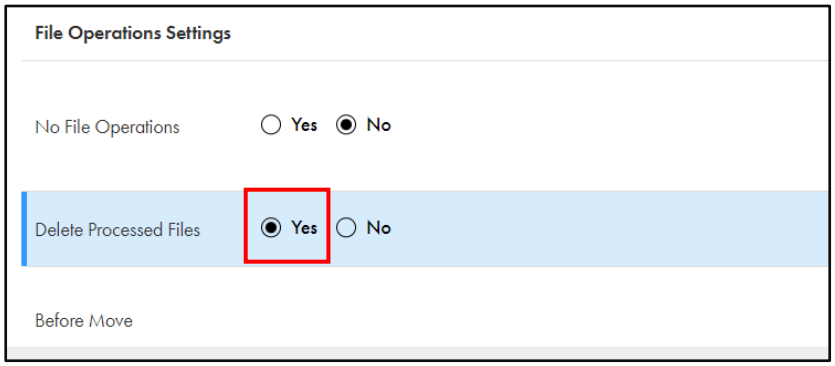
11. Scroll down to the Properties section and in the Directory, enter **C:\Training\Labs\Landing Files**.



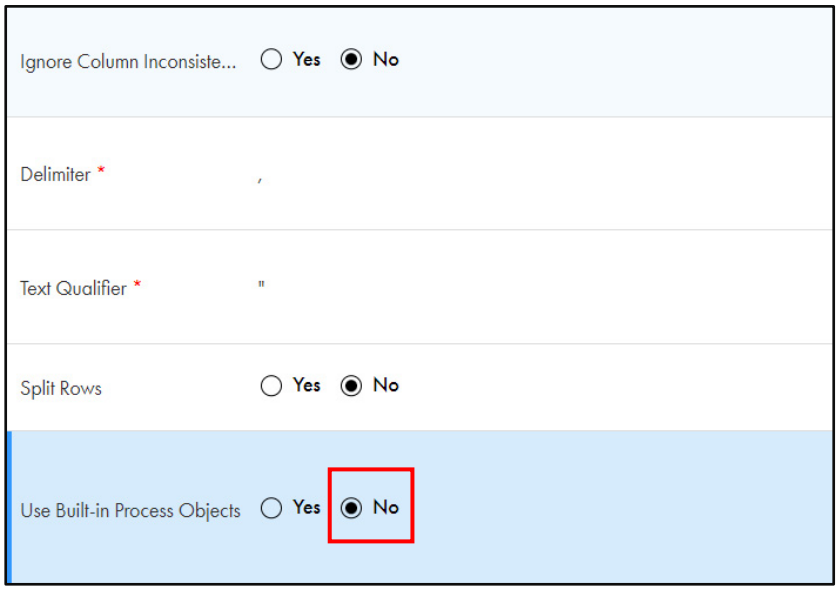
12. In the Include Files field, enter **.*XX\.csv**. Replace XX with your student number.



13. Scroll down to the File Operations Settings section and for the Delete Processed Files option, select **Yes**.

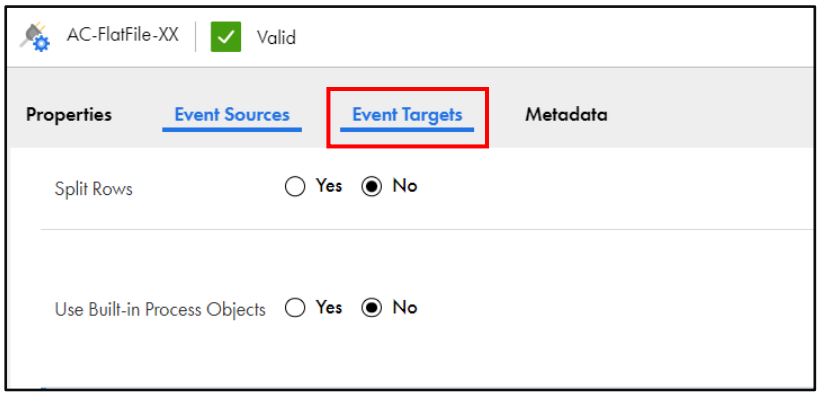


14. Scroll down to the File Parsing Settings section and for the Use Built-in Process Objects, select **No**.

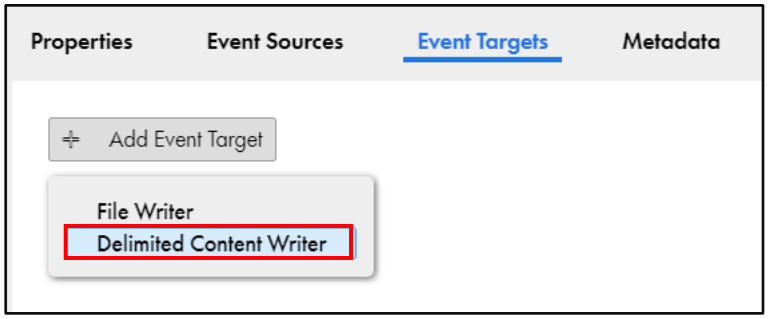


15. In the Custom Object field, enter: **Id, First Name, Last Name, Country, City, Phone, Email, Nickname, Password**.

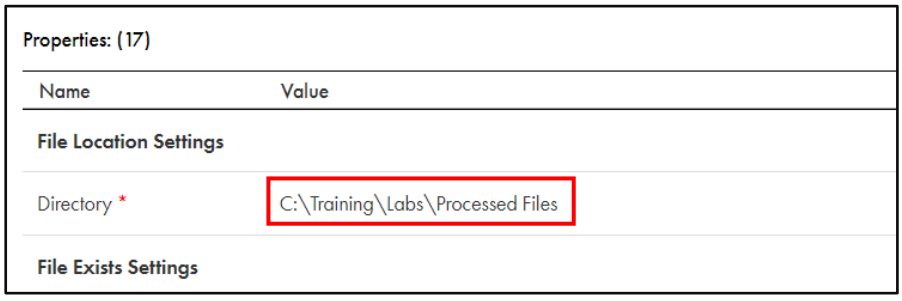
16. Save the connection.
17. Click the **Event Target** tab.



18. Click Add Event Target and select Delimited Content Writer.



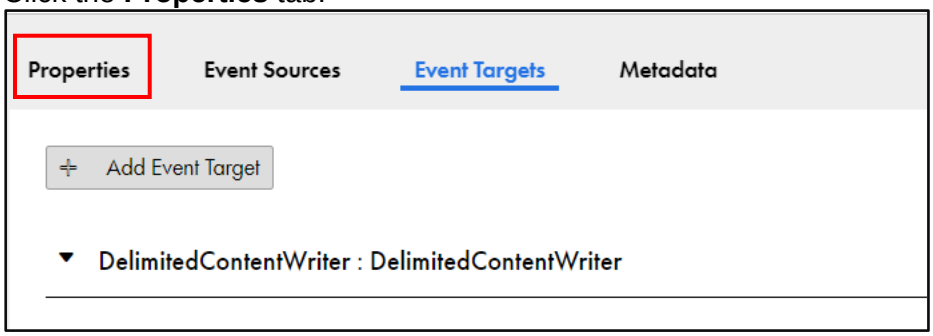
19. In the Directory field, enter **C:\Training\Labs\Processed Files**.



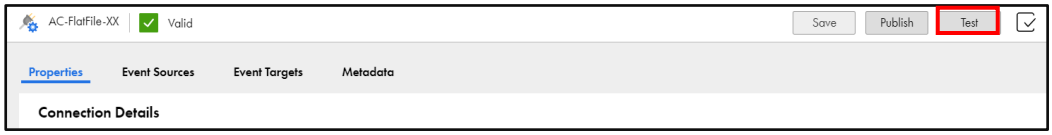
20. Save the connection.

Test Connection

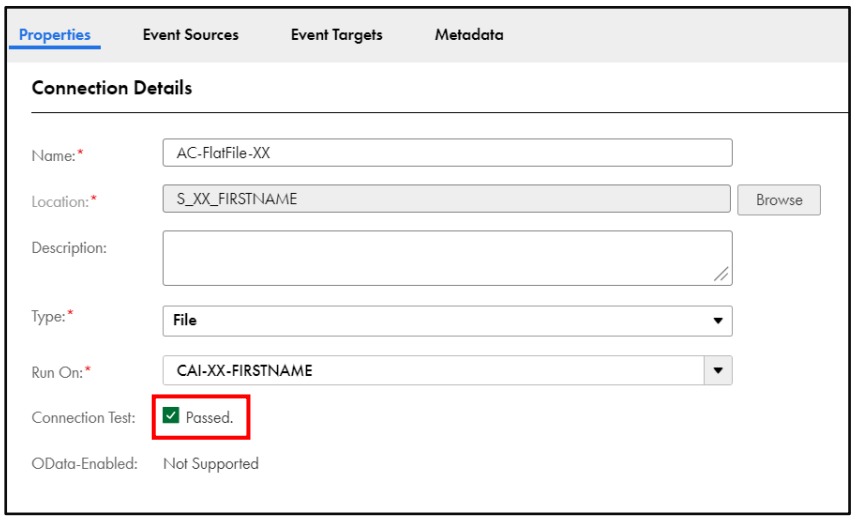
21. Click the **Properties** tab.



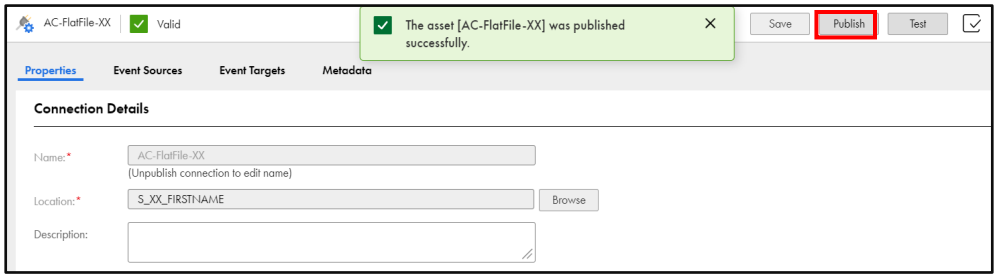
22. Click Test.



Your Connection Test must be Passed.



23. Click Publish to publish the connection.



Note: You will get a notification that '**The asset was published successfully'**.

This concludes the lab.

Module 7: App Connections

Lab 7-8: Create a Process with File Listener Using a Flat File Connection

Overview:

In Lab 7-7, that is, in the previous demonstration, we created a flat file application connection. We will use that connection within a process and gets triggered upon a file arrival. The process will automatically read a file as soon as it arrives in the target folder. The application connection functions as a file observer.

Objectives:

- Create a Process
- Configure a File Listener
- Run the mapping and view results

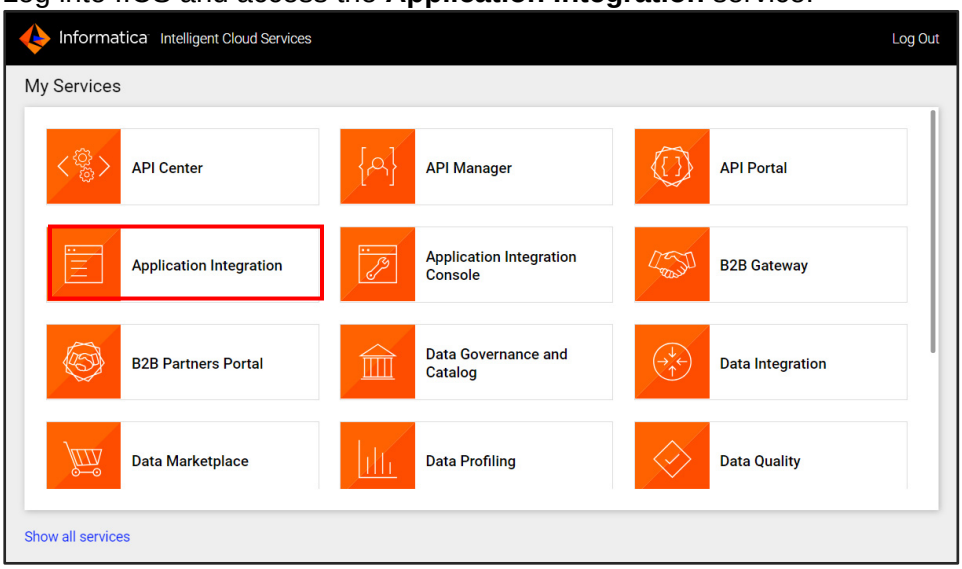
Duration:

45 Minutes

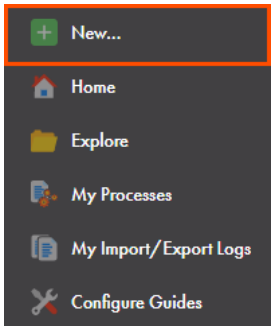
Tasks

Create a connection:

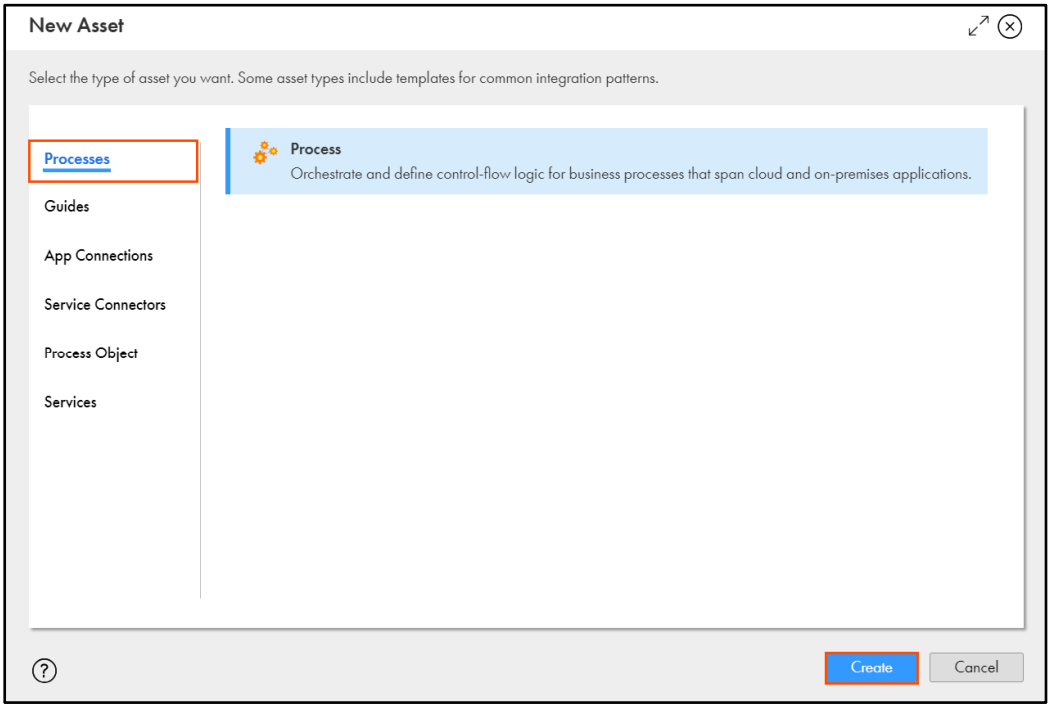
1. Log into IICS and access the **Application Integration** service.



2. To create a New Process, from the navigation pane, select **New**.

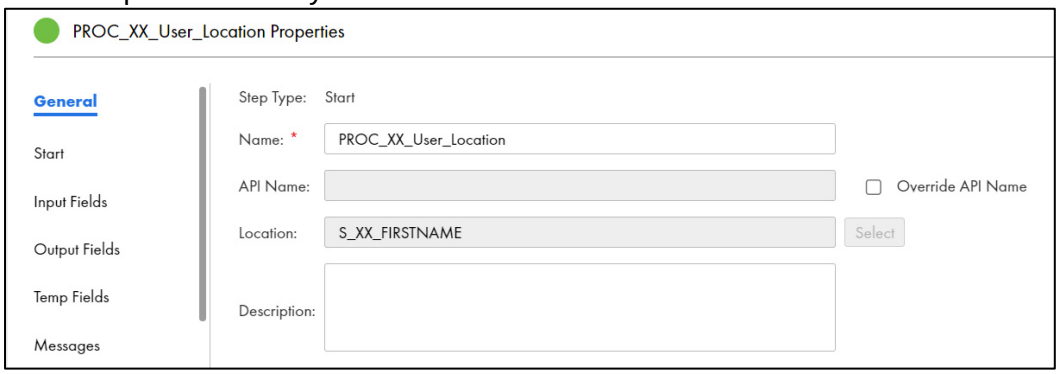


3. In the New Asset window, from the App Connections tab, select **Processes**, and click **Create**.

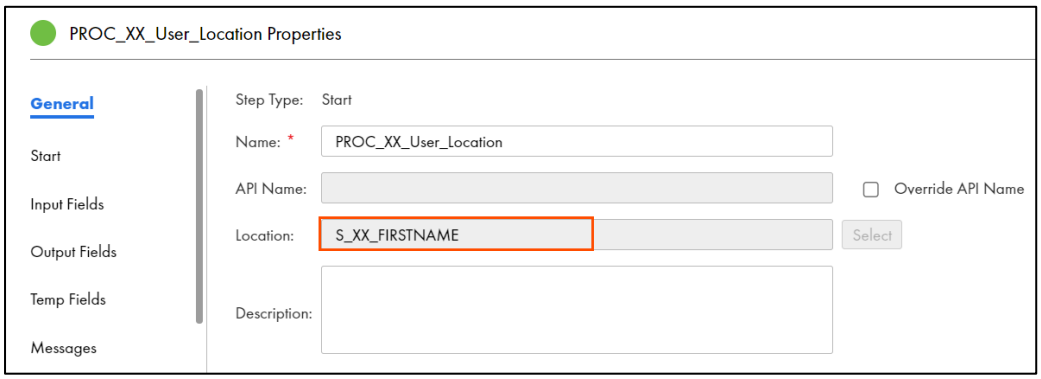


4. In the Process Properties page, in the General section, enter the connection name as **PROC_XX_User_Location**

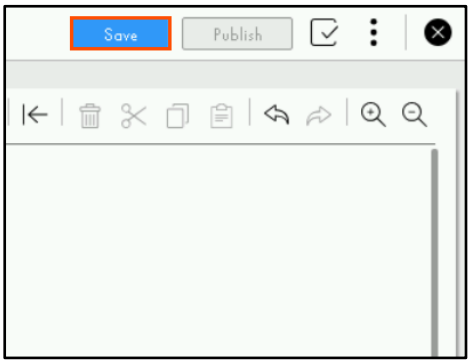
Note: Replace XX with your student number.



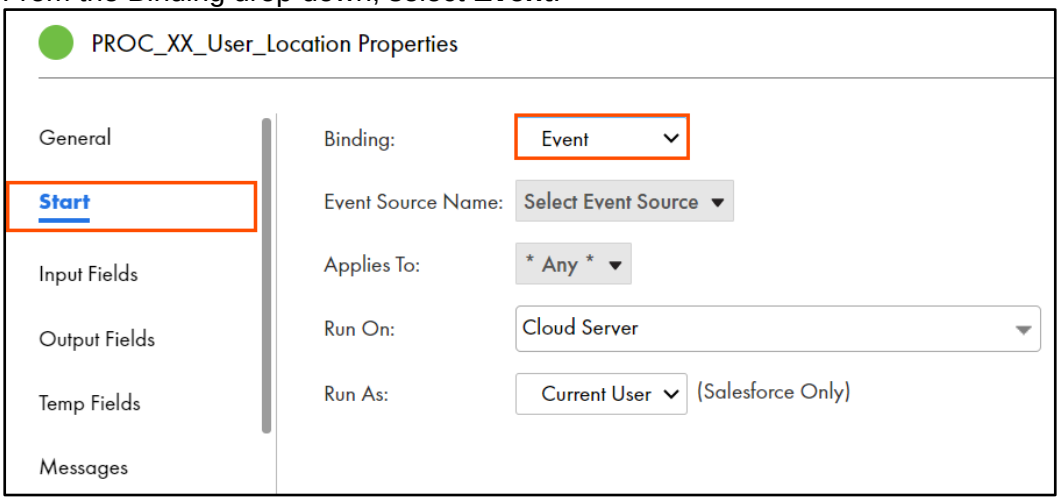
5. In the Location field, select your project folder **S_XX_FirstName**.



6. Click **Save** to save the File.



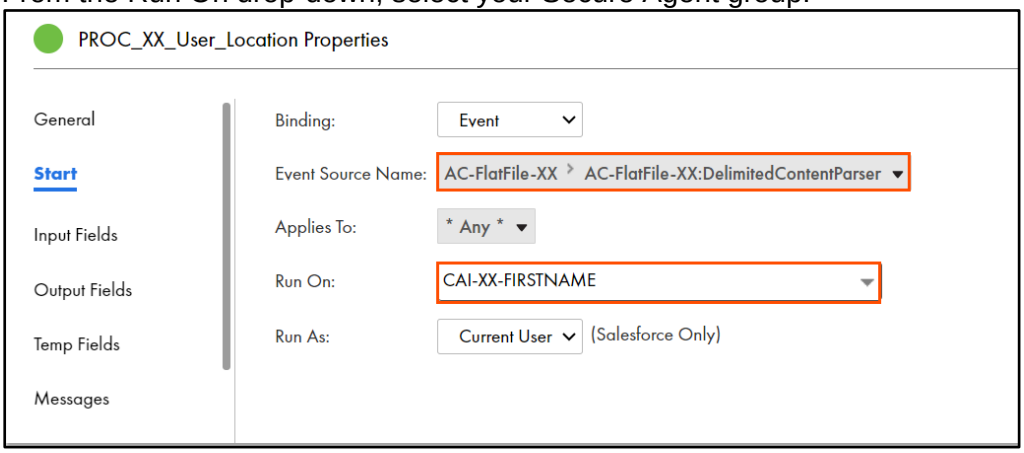
7. From the Properties pane, select **Start**.
8. From the Binding drop-down, select **Event**.



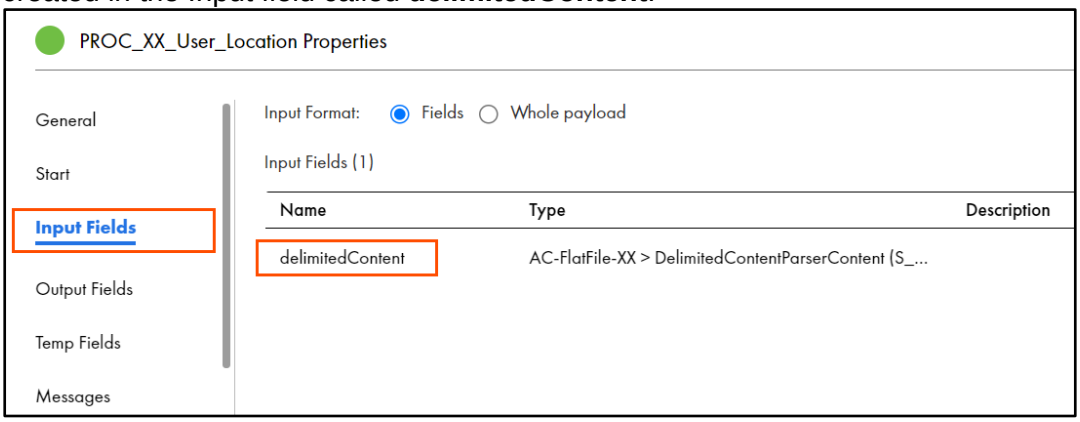
9. To specify the application connection, from the Event Source Name, select AC-Flat-XX and under that, select **DelimiterContentParser**.

   Note: You must select your Application Connection from the drop down.

10. From the Run On drop-down, select your Secure Agent group.

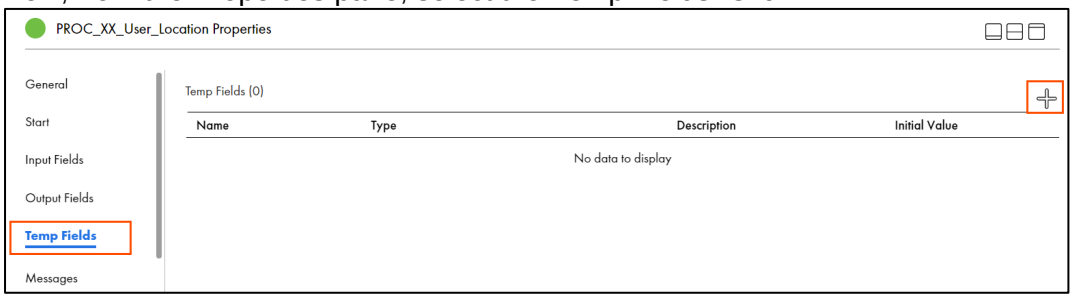


11. **Save** the Process.
12. From the Properties pane, select the Input Fields and notice that it automatically created in the Input field called **delimitedContent**.



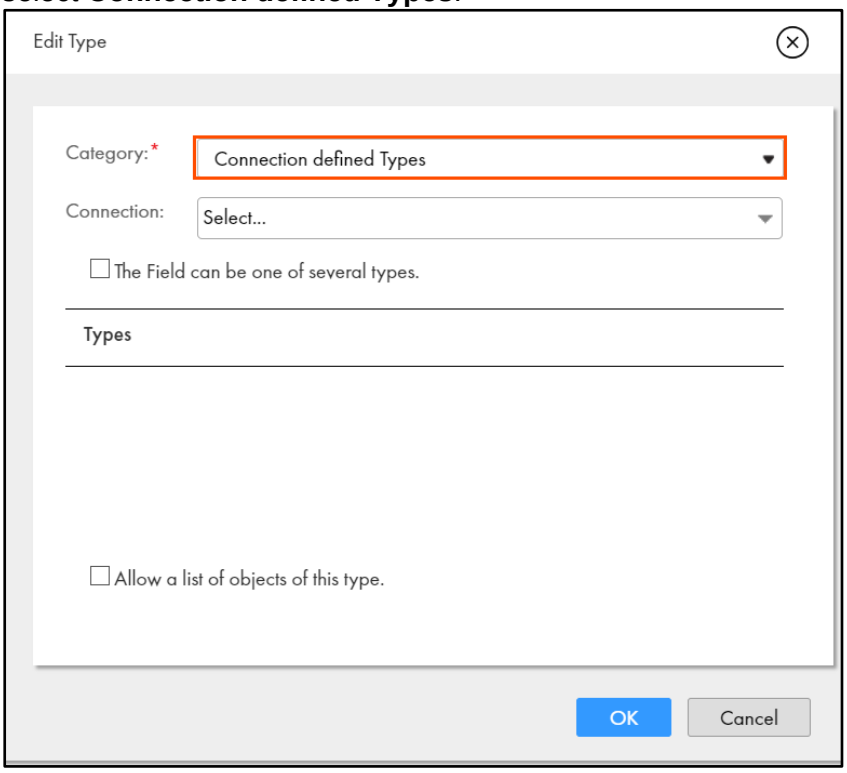
Note: This is where the data will be directed.

13. Now, from the Properties pane, select the Temp Fields. Click .

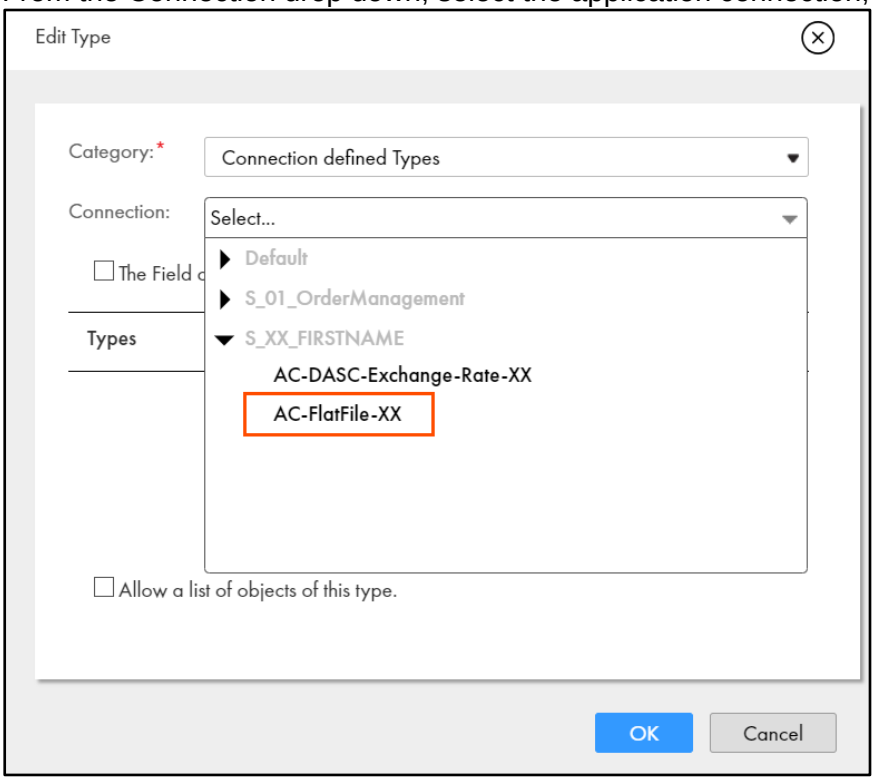


14. Double click on the Name field and enter the name as **Input Records.**

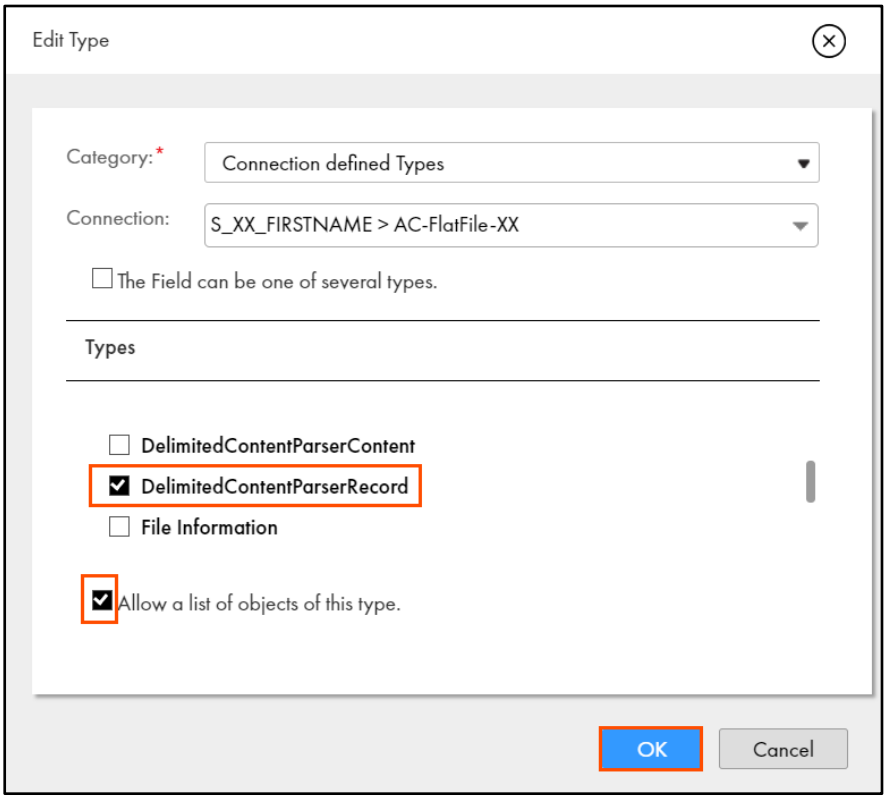
15. From the Type drop-down, select **More types**, and from the Category drop down, select **Connection defined Types**.



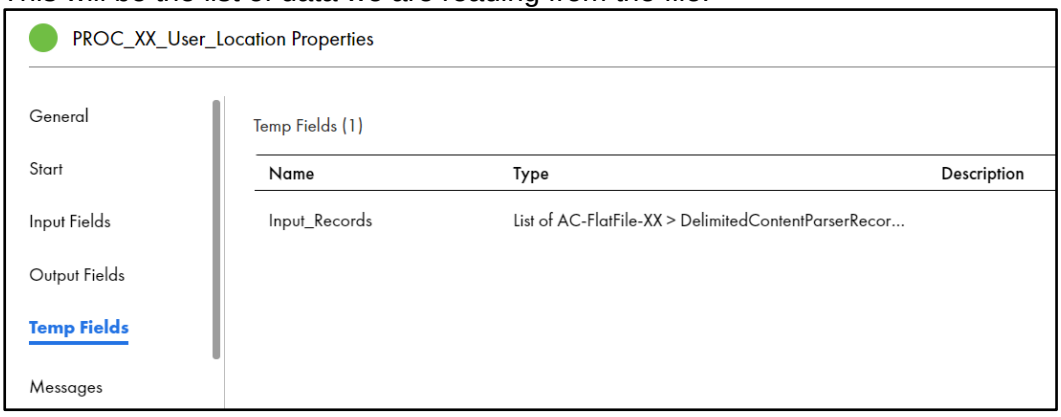
16. From the Connection drop down, select the application connection, **AC-FlatFile-XX**.



17. In the Type, check **DelimitedContentParserRecord**. Also, check the **Allow a list of objects of this type**.
18. Click **OK**.

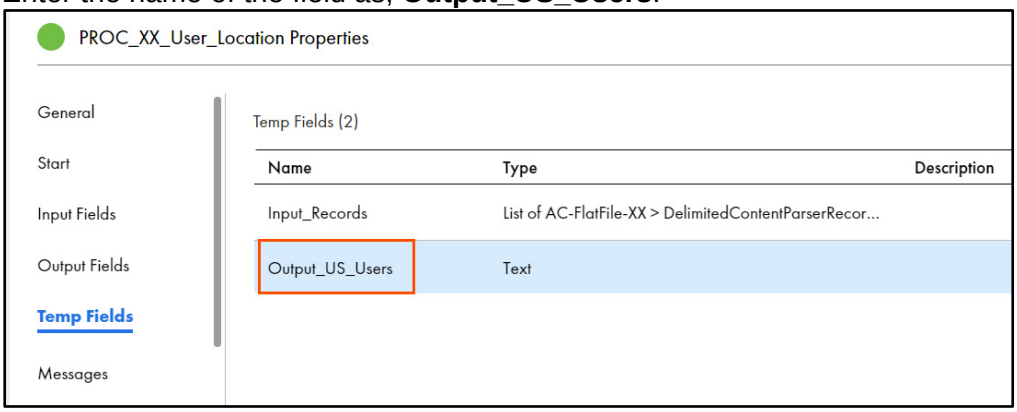


19. This will be the list of data we are reading from the file.

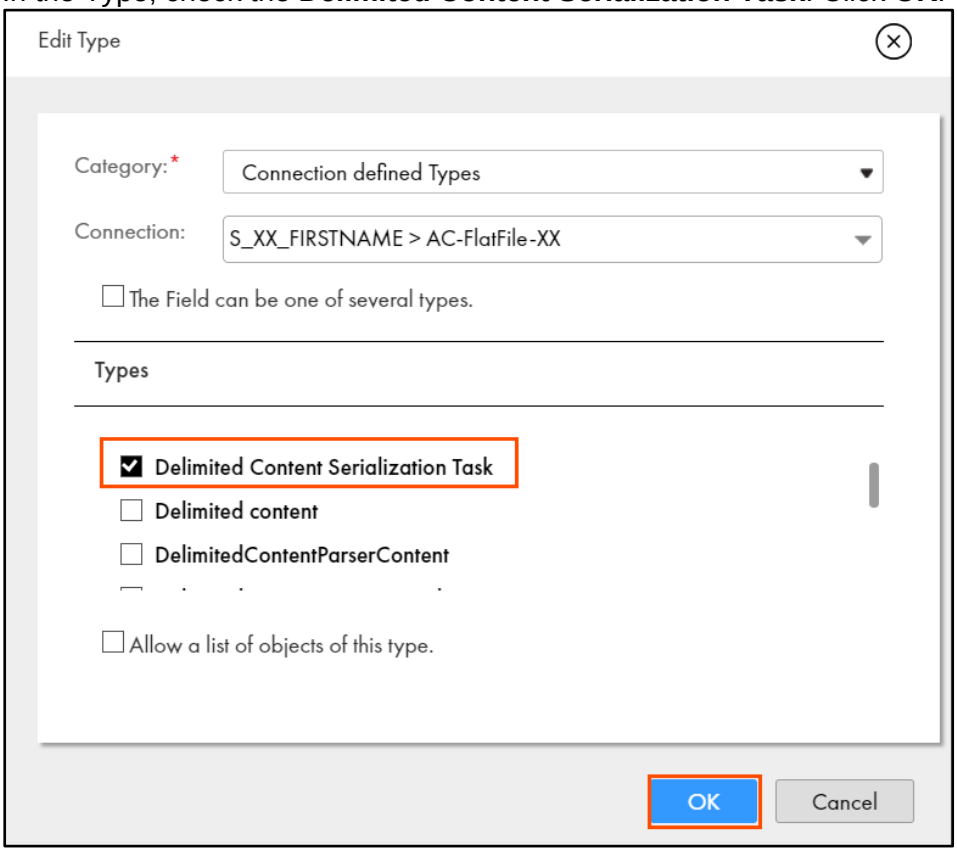


20. To add another field, click .

21. Enter the name of the field as, **Output_US_Users**.

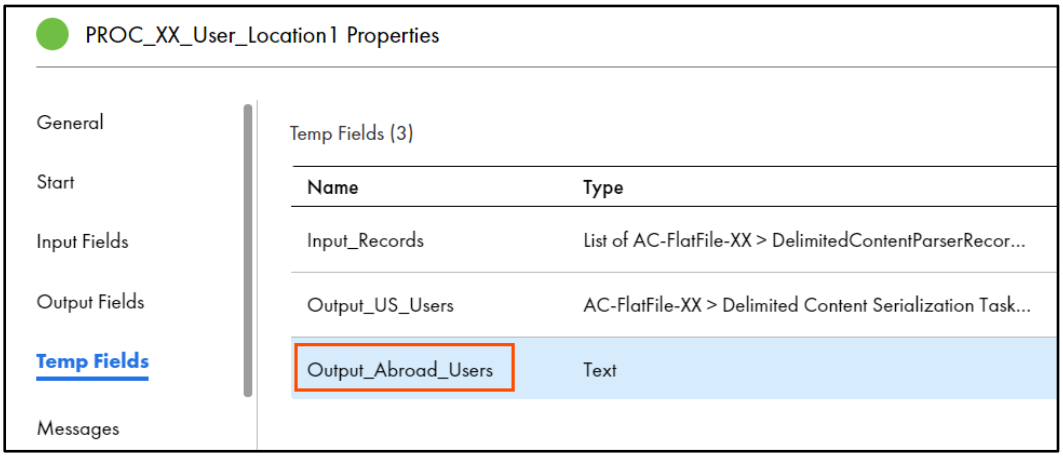


22. From the Type drop-down, select More types, and from the Category drop down, select **Connection defined Types**.
23. Again, from the Connection drop down, select the application connection, **AC-FlatFile-XX**.
24. In the Type, check the **Delimited Content Serialization Task**. Click **OK**.

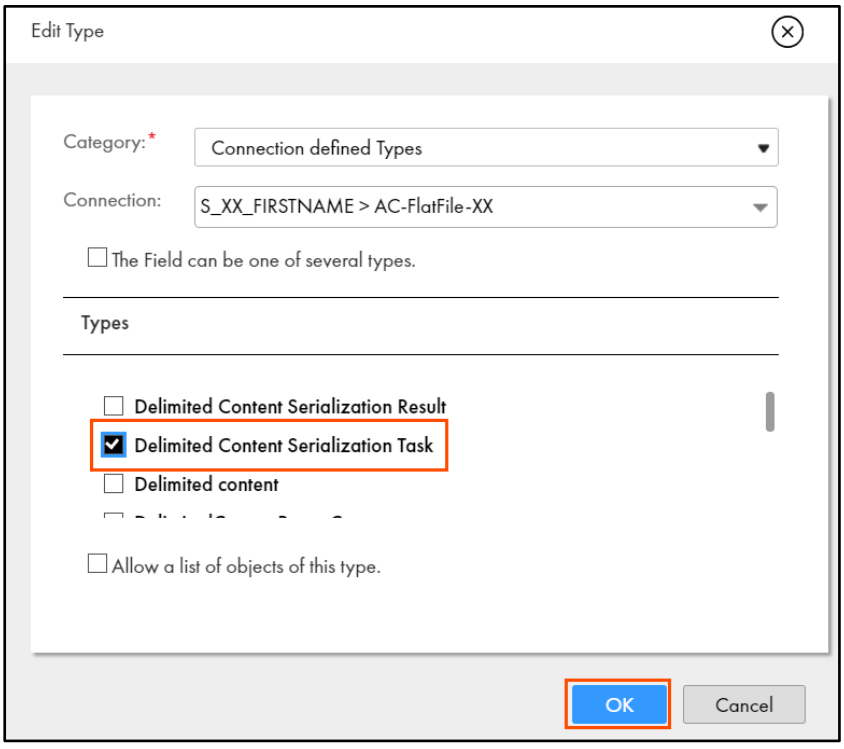


Note: Serialization is the term for generating output.

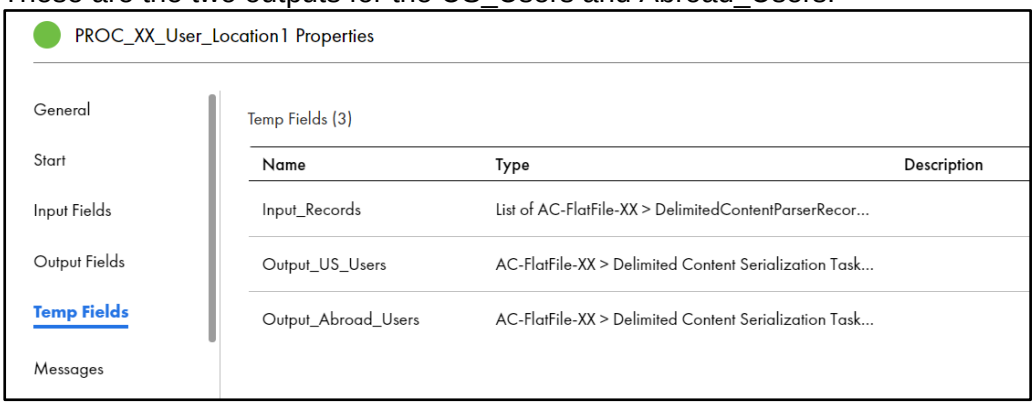
25. Add one more Temp field. Click and name it as **Output_Abroad_Users**.



26. In the In the Text drop-down, choose **More types**, then select **Connection defined Types** from the Category drop-down.
27. Next, select the application connection, AC-FlatFile-XX, from the Connection drop-down menu and select the option **Delimited Content Serialization Task**.
28. Click **OK**.

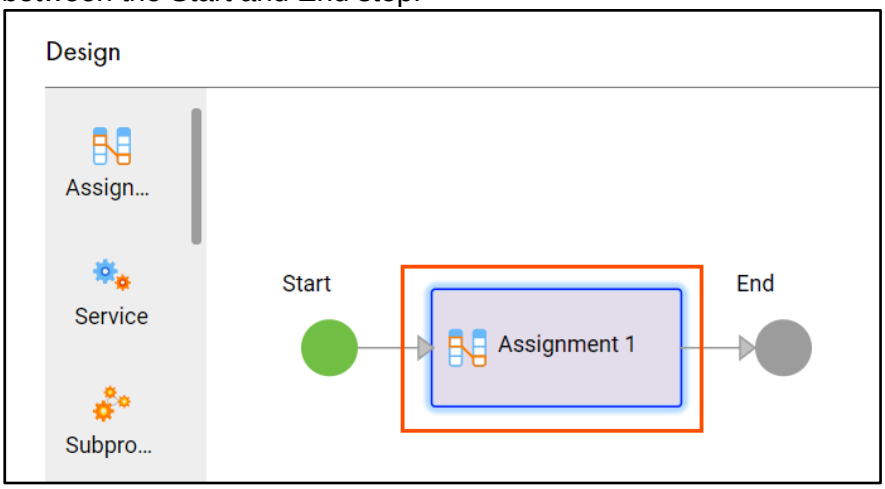


29. These are the two outputs for the US_Users and Abroad_Users.

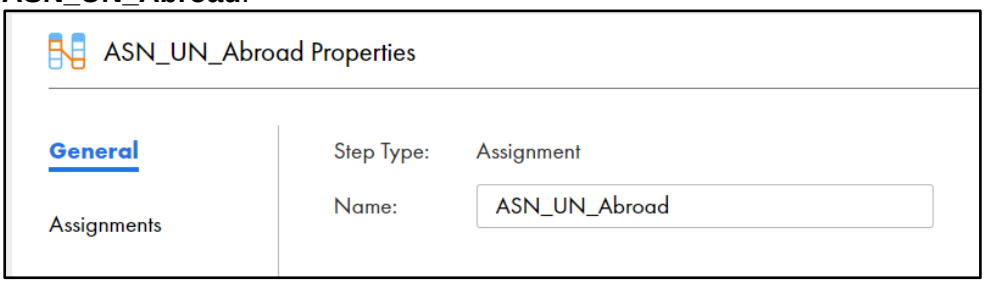


Add an Assignment Task

30. Now, from the Design panel, drag and drop an **Assignment task** on the canvas, between the Start and End step.

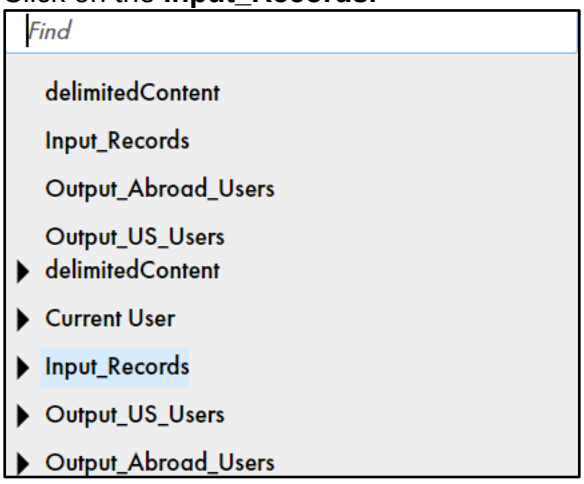


31. From the General properties pane of the Assignment Task, enter the Name as **ASN_UN_Abroad**.

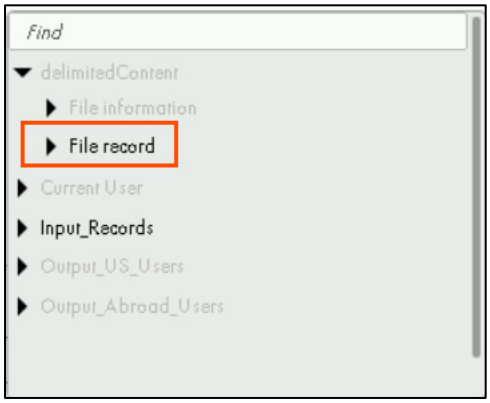


32. Go to the Assignments tab, from the properties pane. Click .

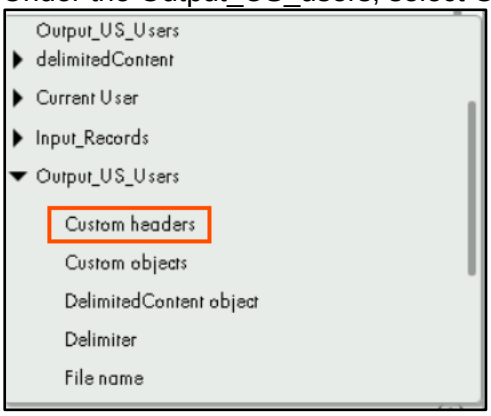
33. Click on the **Input_Records.**



34. In the From field, from the **Click to Select Field**, expand **delimitedContent** and select **File record**.



35. To add another field, click .
36. Under the Output_US_users, select **Custom headers**.



37. From the Assigned Using field drop down, select **Formula.**

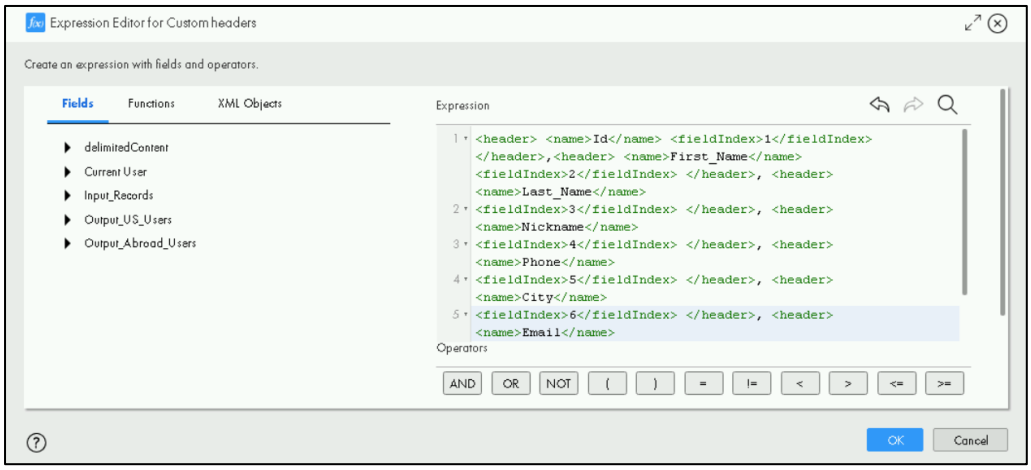
38. Click the to open the formula editor and enter the following formula:

    <header> <name>Id</name> <fieldIndex>1</fieldIndex> </header>,<header> <name>First_Name</name><fieldIndex>2</fieldIndex> </header>, <header> <name>Last_Name</name>
    <fieldIndex>3</fieldIndex> </header>, <header> <name>Nickname</name>
    <fieldIndex>4</fieldIndex> </header>, <header><name>Phone</name>
    <fieldIndex>5</fieldIndex> </header>, <header> <name>City</name>
    <fieldIndex>6</fieldIndex> </header>, <header> <name>Email</name>
    <fieldIndex>7</fieldIndex> </header>, <header> <name>Password</name>
    <fieldIndex>8</fieldIndex> </header>

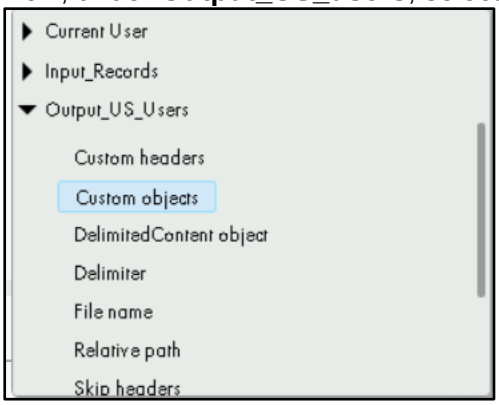
    OR,

    From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **18_LabGuide_CAI_Create a Process with File Listener Using Flat File Connection_7-8**. Copy the command mentioned under **Step A** and paste it into the body field.

39. Click **OK**.



40. To add another field, Click .
41. Now, under **Output_US_users**, select **Custom objects**.



42. In the Assigned Using field, select **Formula.**

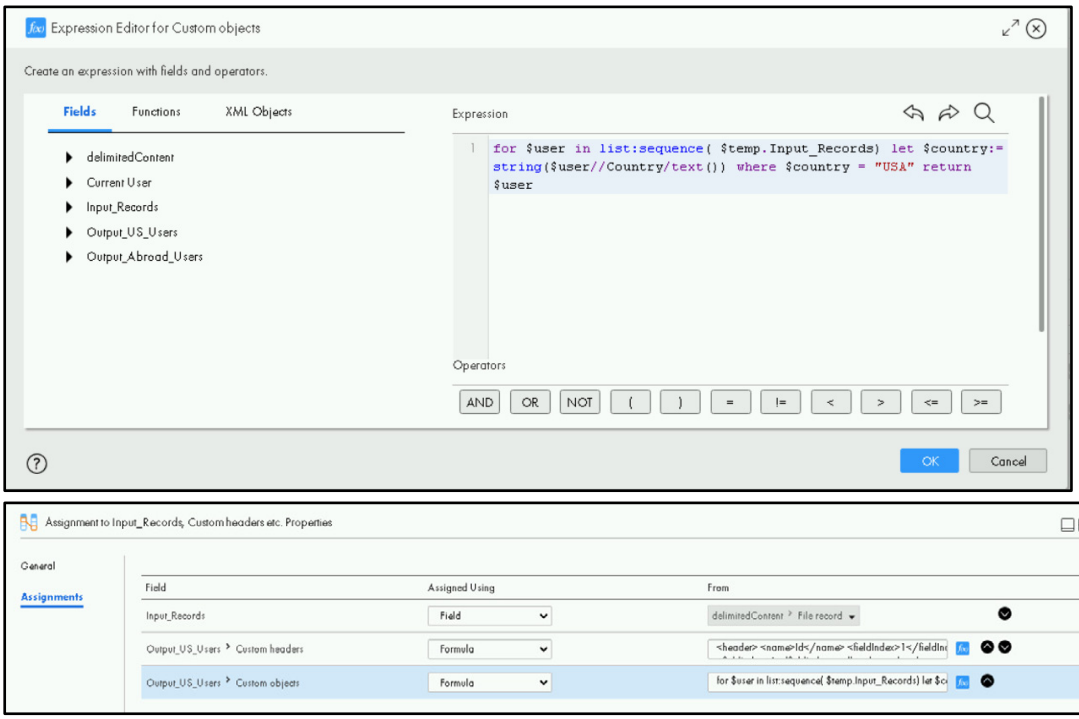
43. Click the to open the formula editor and enter the following formula:

    for $user in list:sequence($temp.Input_Records) let $country:= string($user//Country/text()) where ($country = "USA") return $user

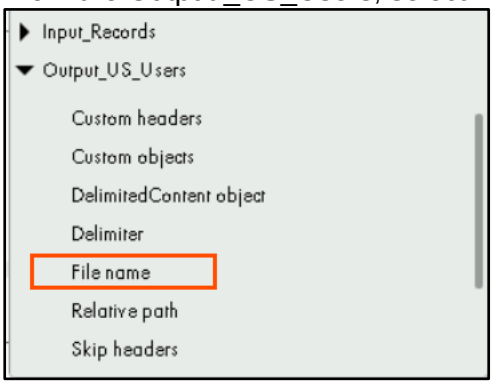
    OR,

    From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **18_LabGuide_CAI_Create a Process with File Listener Using Flat File Connection_7-8**. Copy the command mentioned under **Step B** and paste it into the body field.

44. Click **OK**.

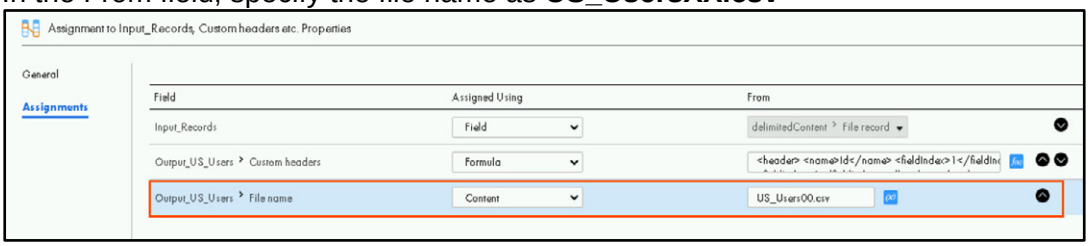


45. To add another assignment, click .
46. From the Output _US_Users, select **File name**.



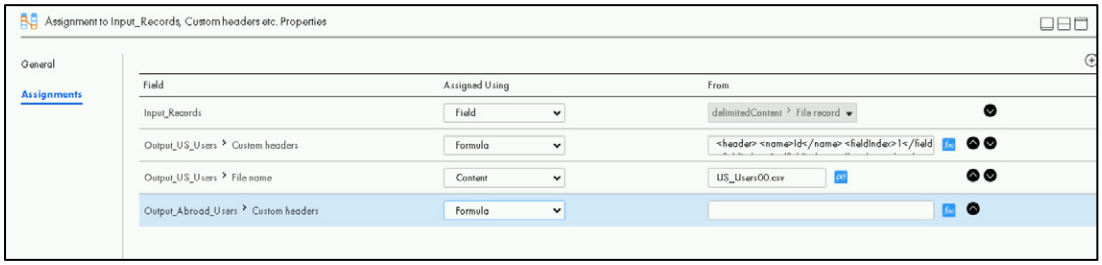
47. In the Assigned using field, retain the value as **Content**.

48. In the From field, specify the file name as **US_UsersXX.csv**



Note: Replace the XX, with the student's number.

49. Now, again to add the assignment, click .
50. Select the Output_Abroad_Users and click **Custom headers** under it.
51. In the Assigned Using field, select **Formula**.



52. In the Expression editor, enter the same XML code used before:

<header> <name>Id</name> <fieldIndex>1</fieldIndex> </header>,<header> <name>First_Name</name><fieldIndex>2</fieldIndex> </header>, <header> <name>Last_Name</name>

<fieldIndex>3</fieldIndex> </header>, <header> <name>Nickname</name>

<fieldIndex>4</fieldIndex> </header>, <header><name>Phone</name>

<fieldIndex>5</fieldIndex> </header>, <header> <name>City</name>

<fieldIndex>6</fieldIndex> </header>, <header> <name>Email</name>

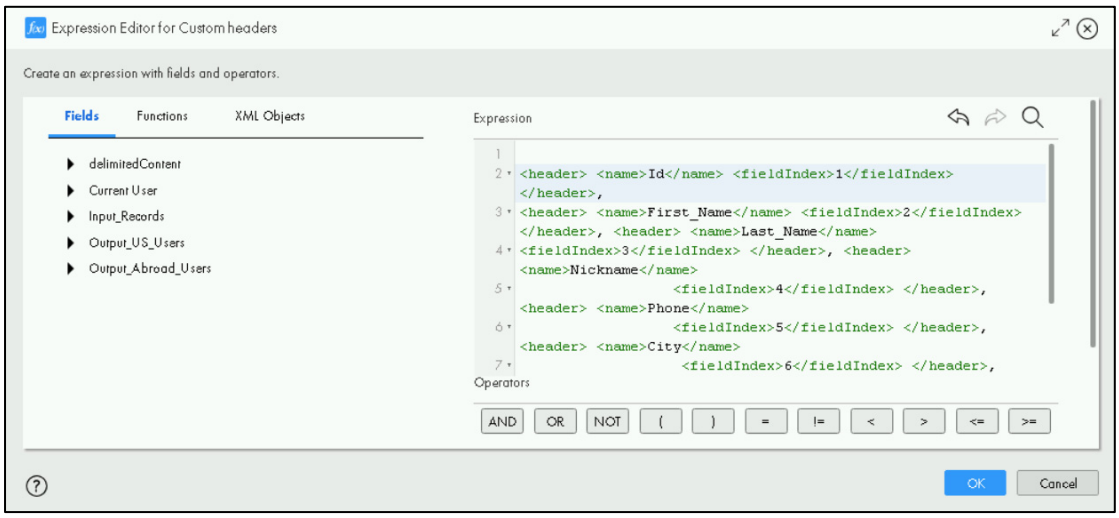
<fieldIndex>7</fieldIndex> </header>, <header> <name>Password</name>

<fieldIndex>8</fieldIndex> </header>

OR,

From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **18_LabGuide_CAI_Create a Process with File Listener Using Flat File Connection_7-8**. Copy the command mentioned under **Step C** and paste it into the body field.

53. Click **OK**.

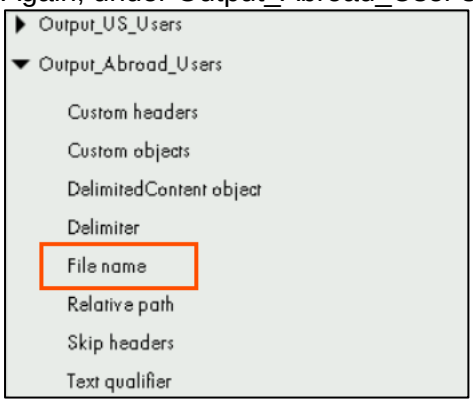


54. Add another field - **Output_Abroad_User** as **Custom Objects**.
55. In the Assigned Using field, select **Formula**.
56. In the Expression editor, enter the code as shown below:

    for $user in list:sequence($temp.Input_Records) let $country:= string($user//Country/text()) where not ($country = "USA") return $user

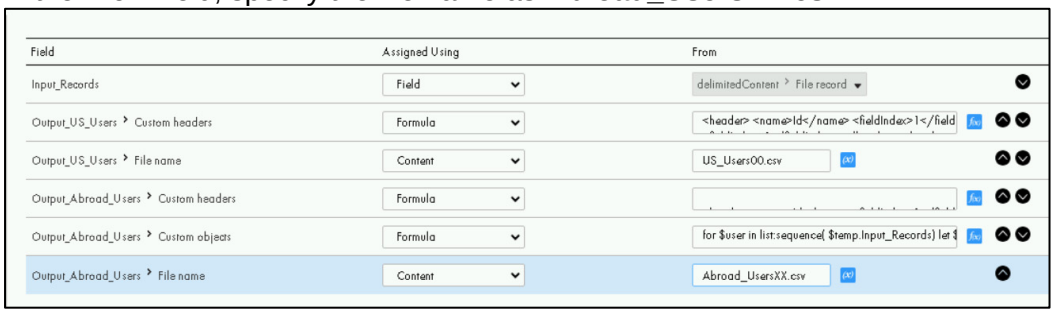
    OR,

    From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **18_LabGuide_CAI_Create a Process with File Listener Using Flat File Connection_7-8**. Copy the command mentioned under **Step D** and paste it into the body field.
57. Click **OK**.
58. Again, under Output_Abroad_User select **File name**.



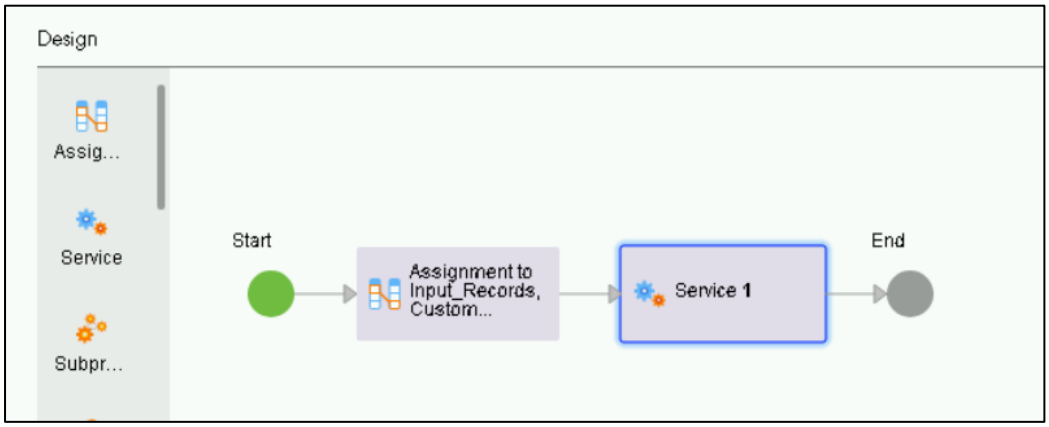
59. Retain the Assigned Field as **Content**.

60. In the From field, specify the file name as **Abroad_UsersXX.csv**.



Note: Replace the XX with your student number.

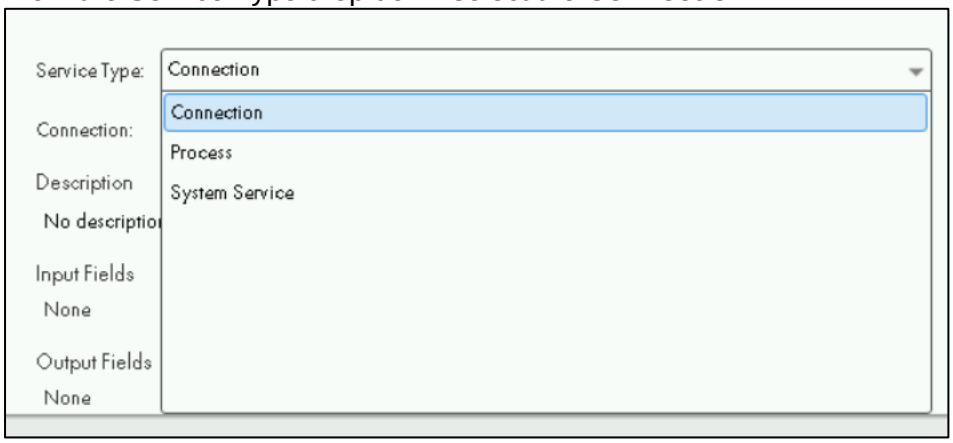
61. **Save** the file.
62. From the Design pane, drag and drop a **Service task** on the canvas between the **ASN_US_Abroad** and **End Task**.



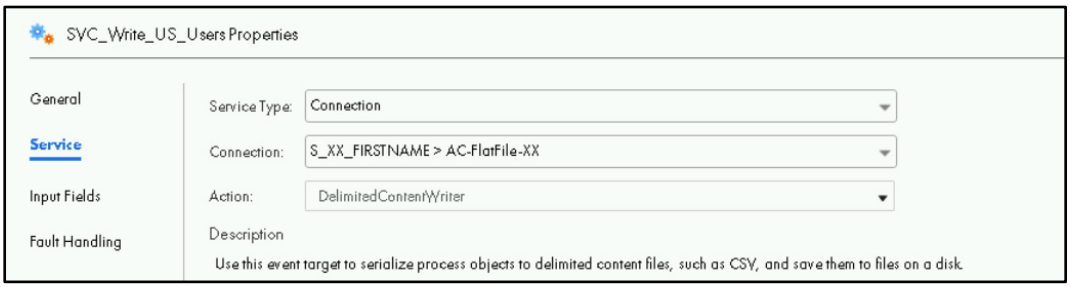
63. From the Properties pane, in the General section, enter the Name as **SVC_Write_US_Users**.

64. Select the **Service** tab from the properties pane.

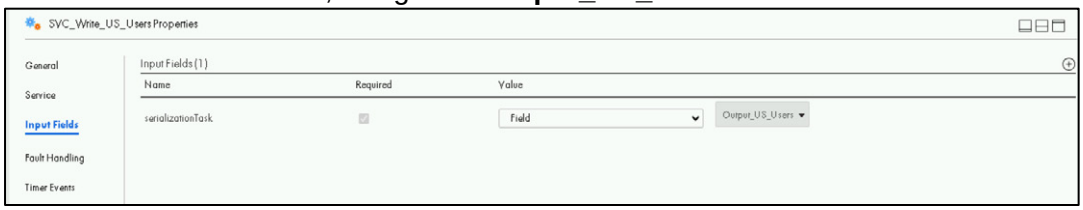
65. From the Service Type drop down select the **Connection.**



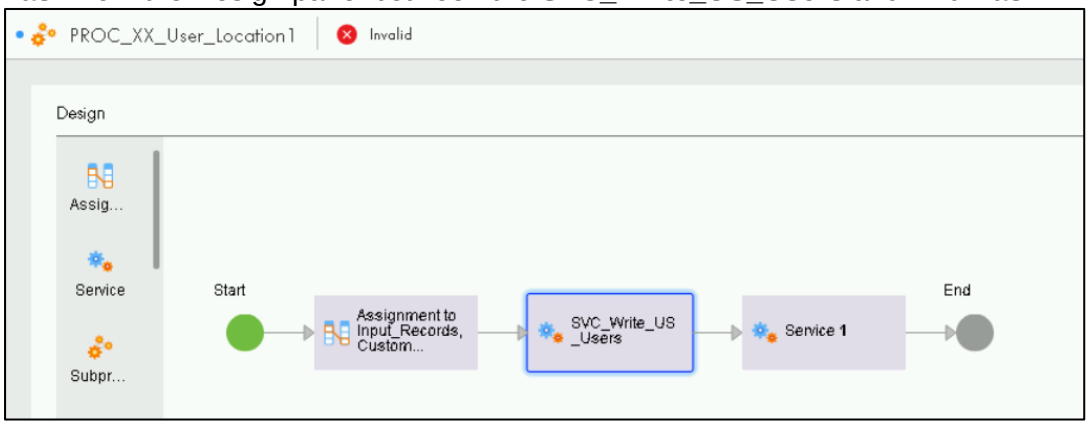
66. From the Connection drop down select, your flat file connection **AC-FlatFile-XX.**
67. Select the Action as **DelimitedContentWriter**.



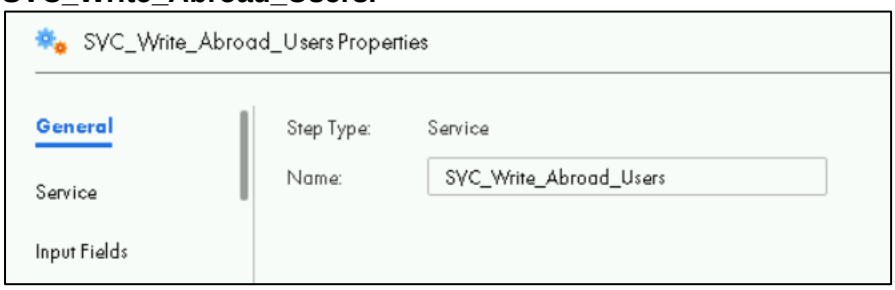
68. Go to the **Input Fields** from the properties pane.
69. For the Serialization task, assign the **Output_US_Users**.



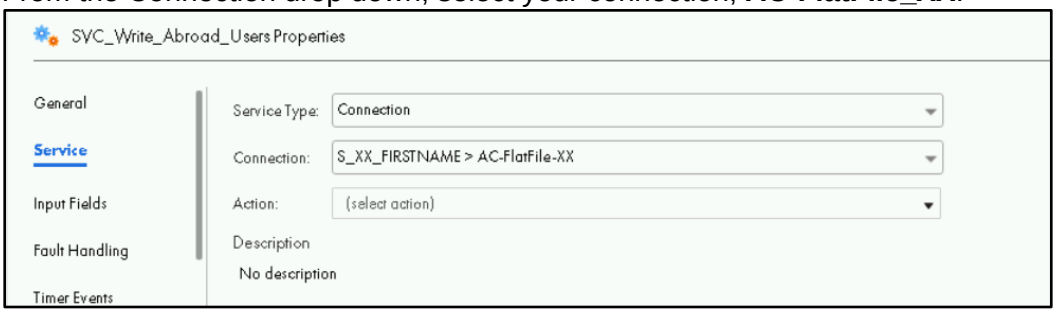
70. **Save** the Process.
71. To Write an Output file, you need to add another Service. Drag and drop the **Service Task** from the Design panel between the **SVC_Write_US_Users** and **End Task**.



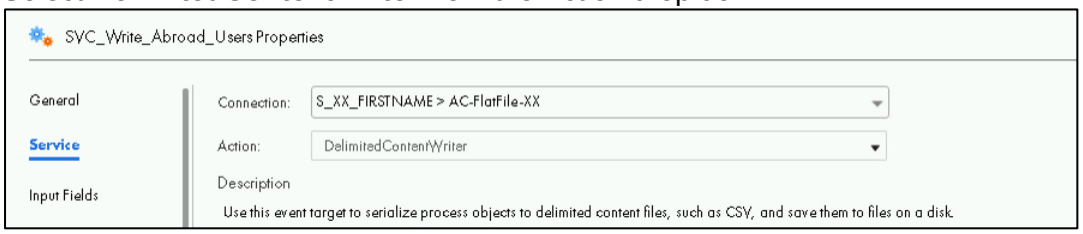
72. From the properties pane, in the General section, enter the name of the service as **SVC_Write_Abroad_Users.**



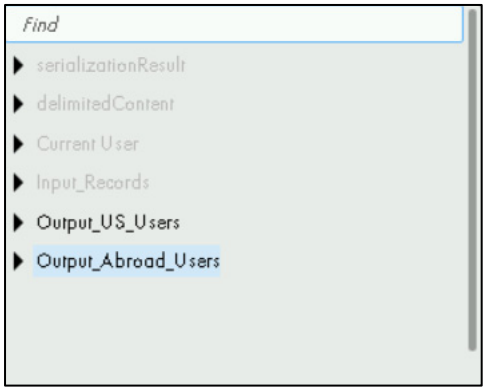
73. From the properties pane, select **Service**.
74. In the Service Type drop-down, select **Connection**.
75. From the Connection drop down, select your connection, **AC-FlatFile_XX**.



76. Select **DelimitedContentWriter** from the Action drop down.

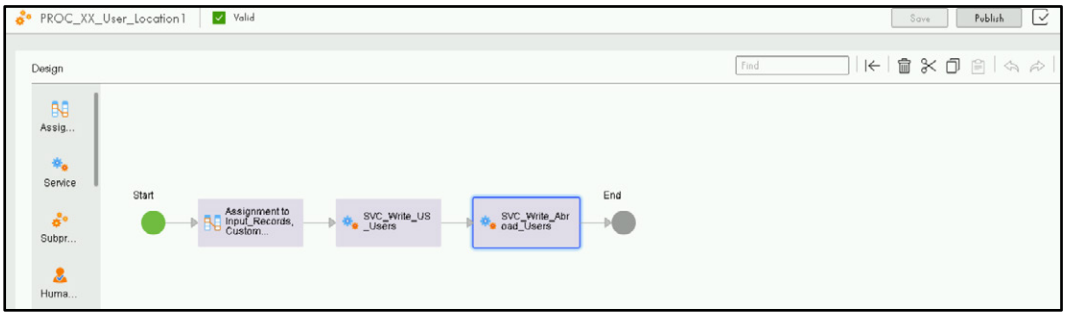


77. From the properties pane, select the **Input fields**.
78. Click on the Click to Select Field tab and select **Output_Abroad_Users.**



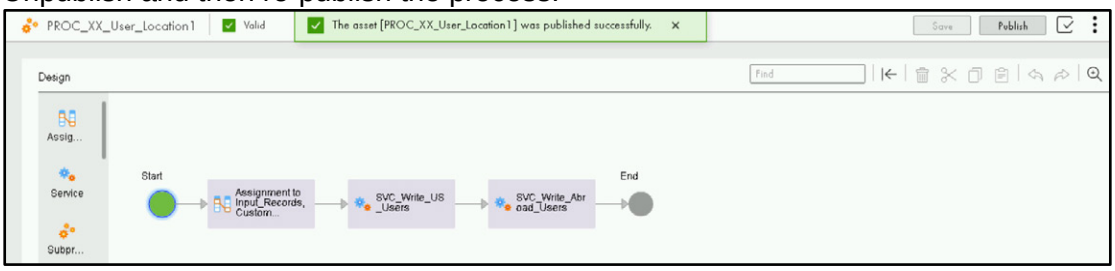
79. Save the Process and make sure it is **Valid**.

80. Click **Publish**.



81. Go to the Start task and select the Advanced tab from the properties pane.
82. From the Tracing level drop down field, select **Verbose**.

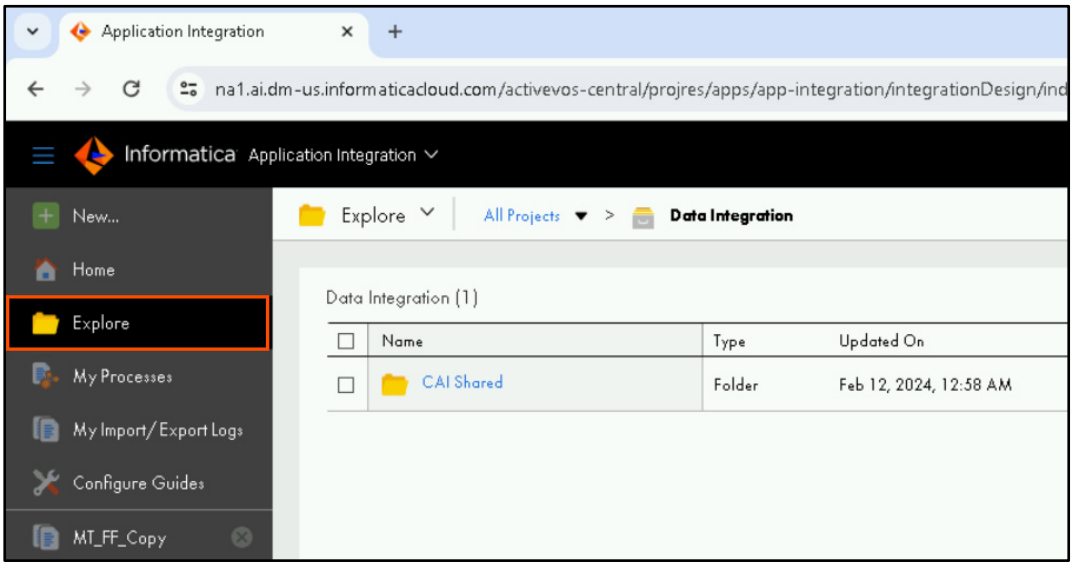
83. Save the Process.
84. Unpublish and then re-publish the process.



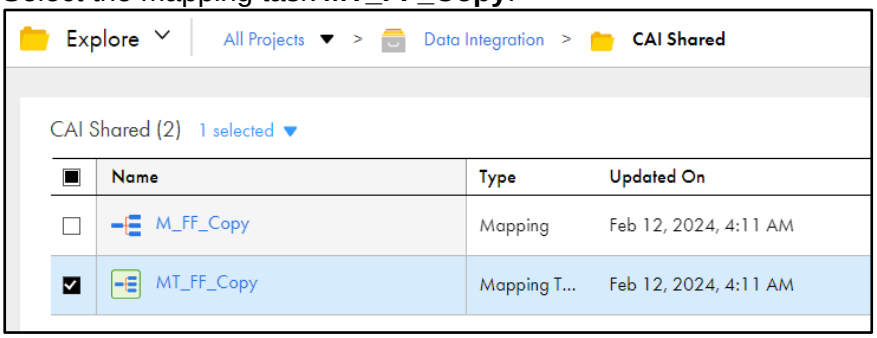
Note: The process is not a webservice. It is triggered by an Event, which is arrival of a file.

85. From the Navigation pane, click the **Explore** tab.

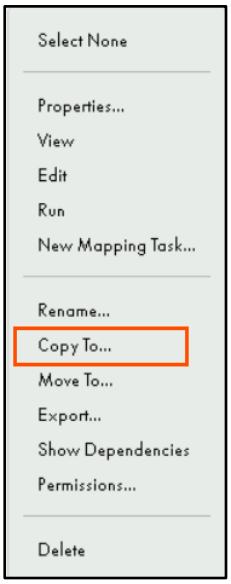
86. From the All Projects page, navigate to Data Integration, and go to the **CAI Shared** folder.



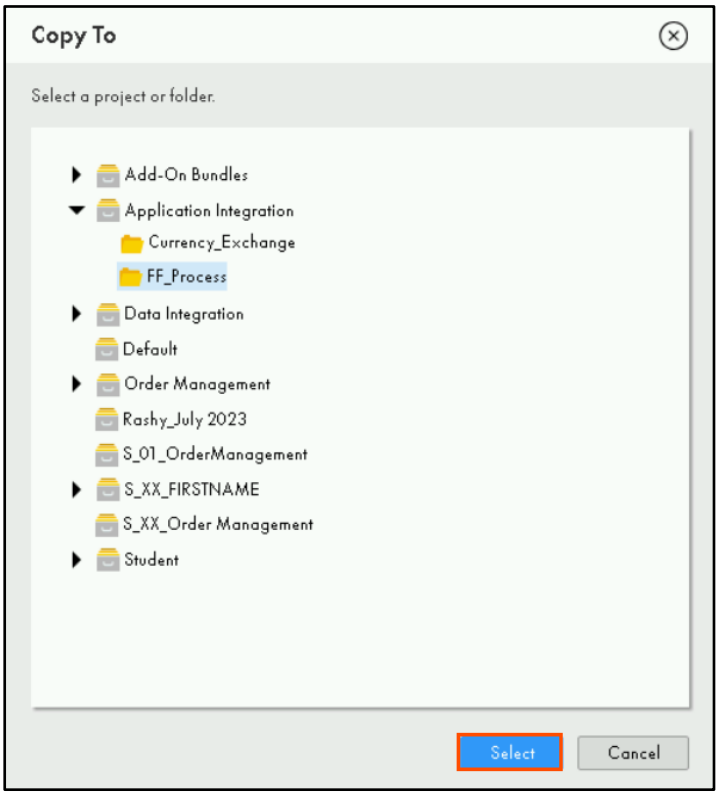
87. Select the mapping task **MT_FF_Copy**.



88. Right-click the asset and copy the mapping task MT_FF_copy into your working folder.

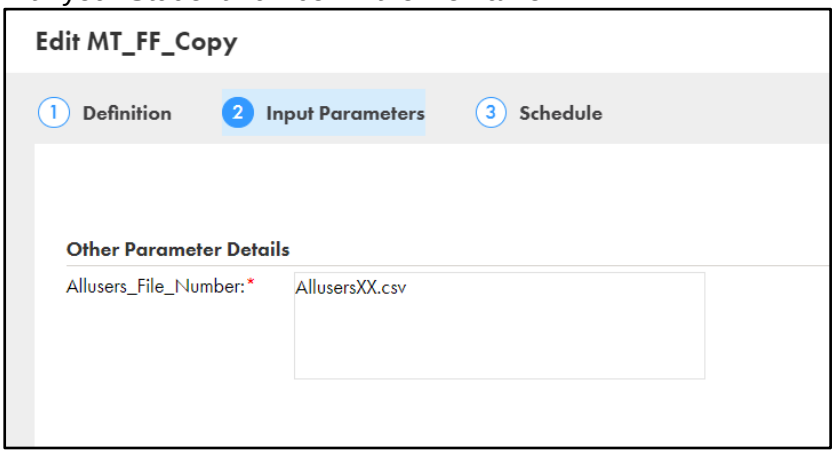


89. Click **Select**.

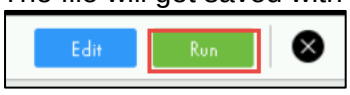


In your working folder, to edit the copied mapping task, click on the task name. This will open in the **Data Integration** service.

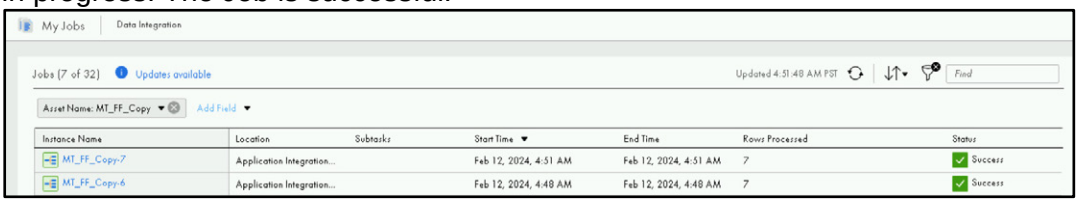
90. Click on **Edit** to edit the mapping task.
91. Navigate to the **Input Parameters** tab. In the Allusers_File_Number field, replace XX with your Student number in the file name.



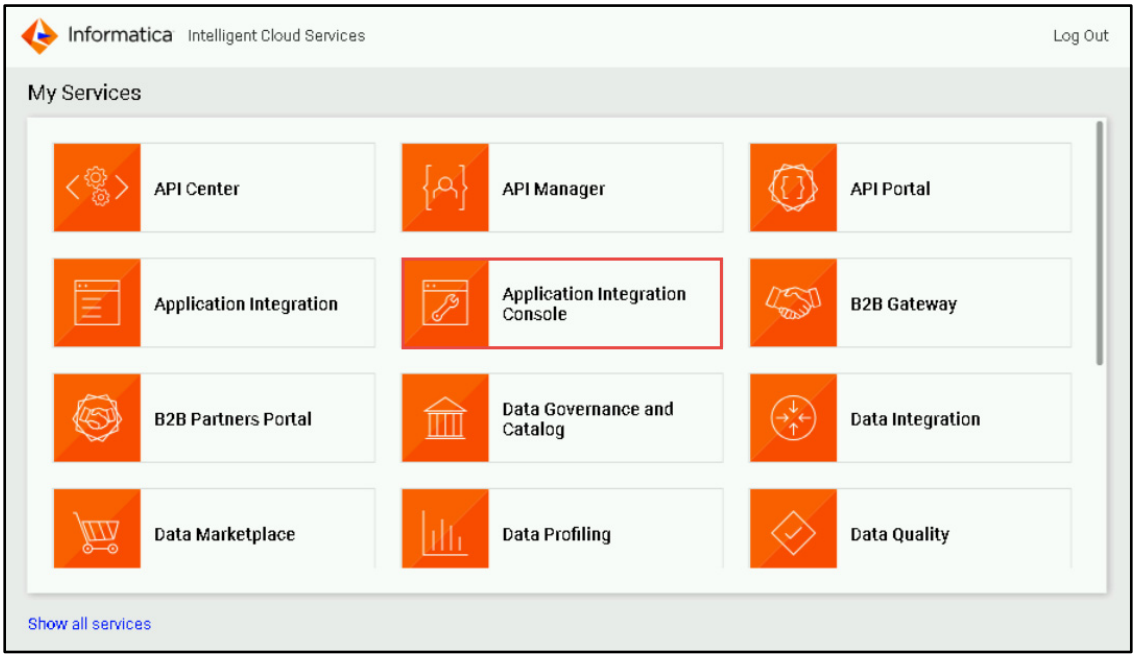
92. Click **Finish.**
93. The file will get saved with the name specified here. Click **Run.**



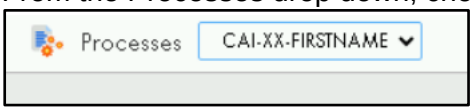
94. Navigate to the "My Jobs" page and select the option to refresh if the task is currently in progress. The Job is successful.



95. Click the arrow next to the Data Integration name and navigate to **Application Integration Console** service.

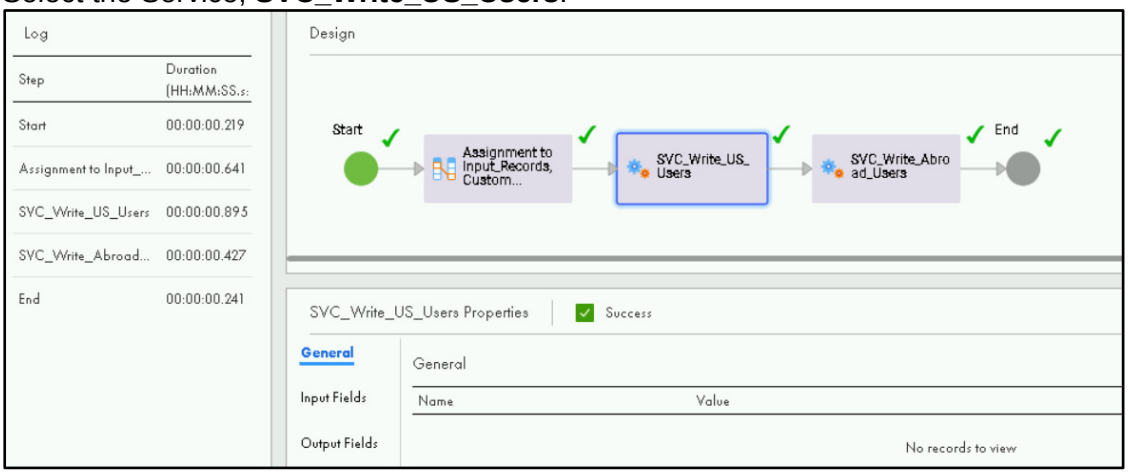


96. From the Processes drop down, choose your Secure Agent.

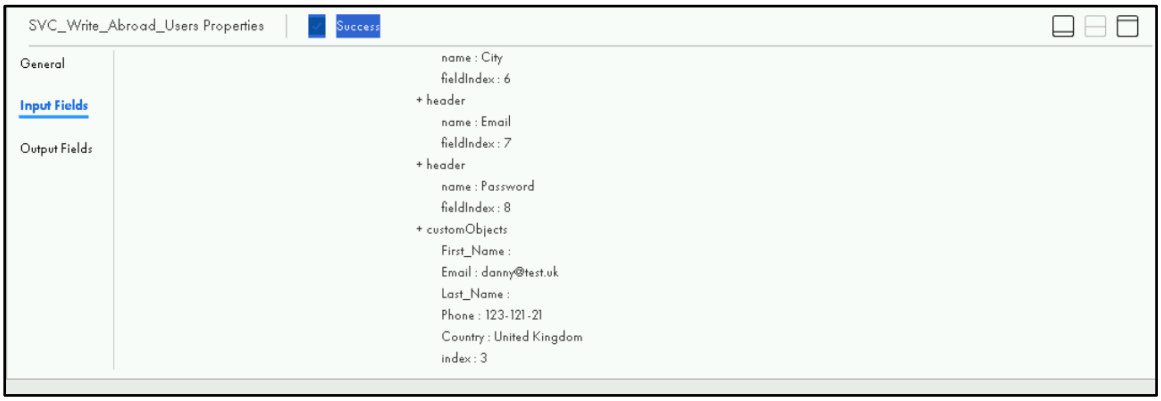


97. Click on the process ID for your process to open it.

98. Select the Service, **SVC_Write_US_Users**.



99. Click on the **Input Fields**, from the properties pane.
100. Scroll down and you will see the data.
101. Similarly, select the service, **SVC_Write_Abroad_users**.
102. Click on the Input Fields tab. Under the Custom objects parameter section, you can see the results.



This concludes the lab.

Module 8: Managing and Migrating Assets
Lab 8-1: Export an Asset

Overview:
When you export assets, Informatica Intelligent Cloud Services creates an export ZIP file that contains the assets that you selected for export.

You can select a single asset, multiple assets, or a project to export.

To include multiple assets, you can either select each asset within a folder or select a project or folder to export all of its assets. If you export a project, during import you can import the entire project or import only the assets that you select.

In this lab, you will export a single asset and view the log details.

Objectives:
- Export an asset
- View the log details

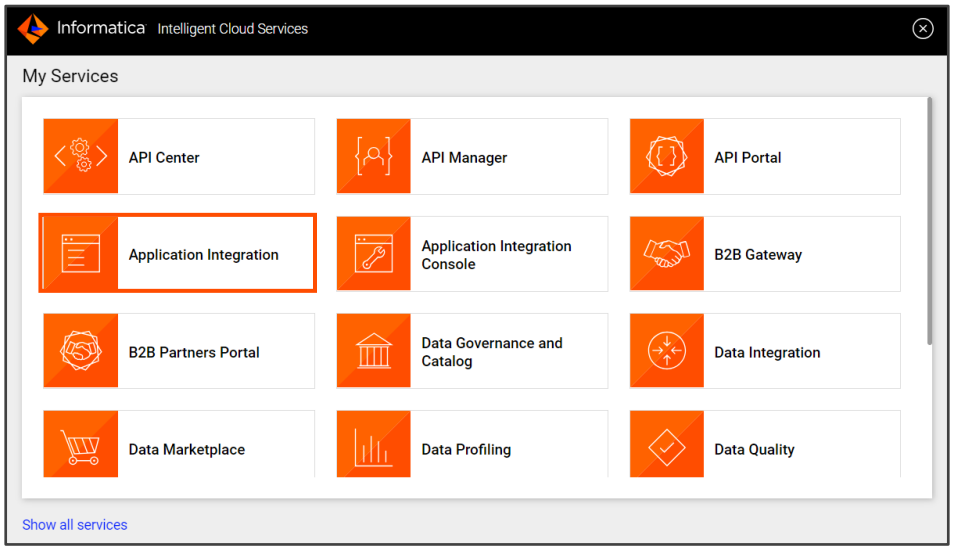
Duration:
10 Minutes

Tasks

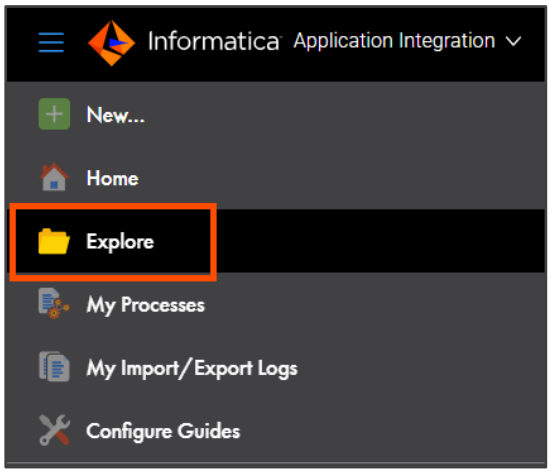
Note: The screen in your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

Export an Asset

1. Log into IICS and access the **Application Integration** service.



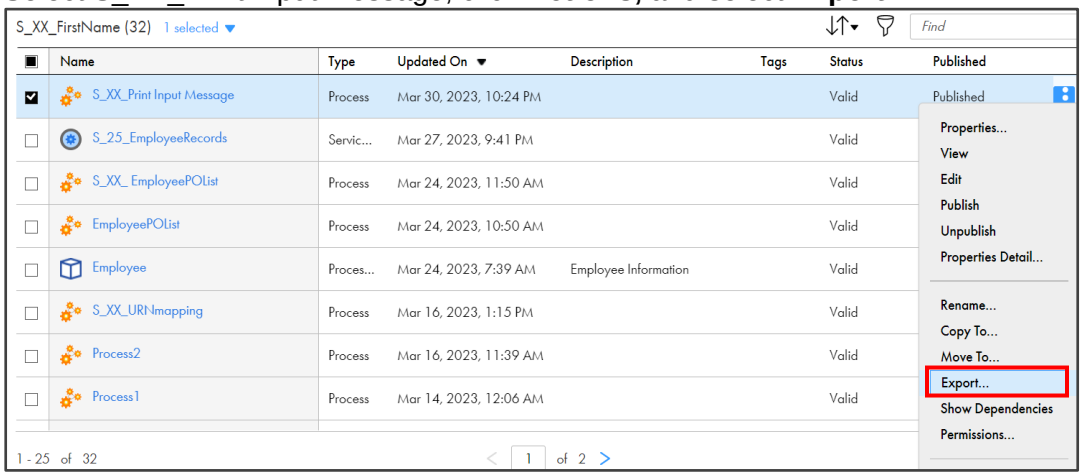
2. From the navigation pane, click **Explore**.



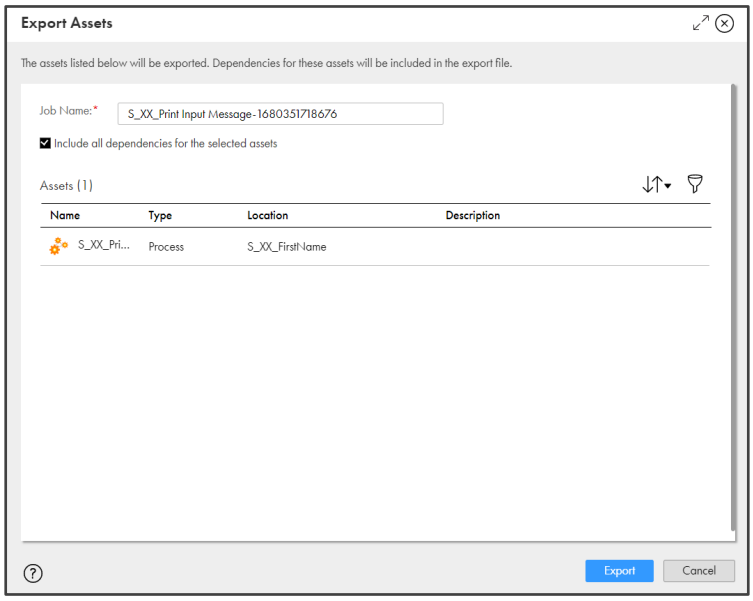
3. Open your project folder S_XX_FirstName.
4. Locate **S_XX_Print Input Message**.

   Note: In this lab, we are exporting S_XX_Print Input Message. You can select any other asset to export.

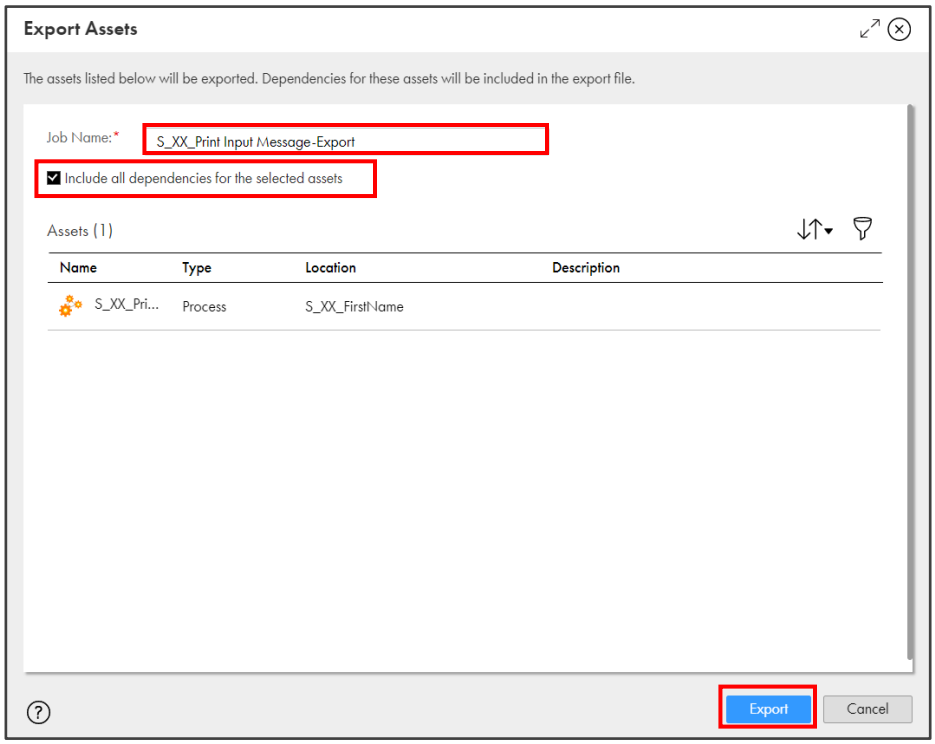
5. Select S_XX_Print Input Message, click **Actions,** and select **Export**.



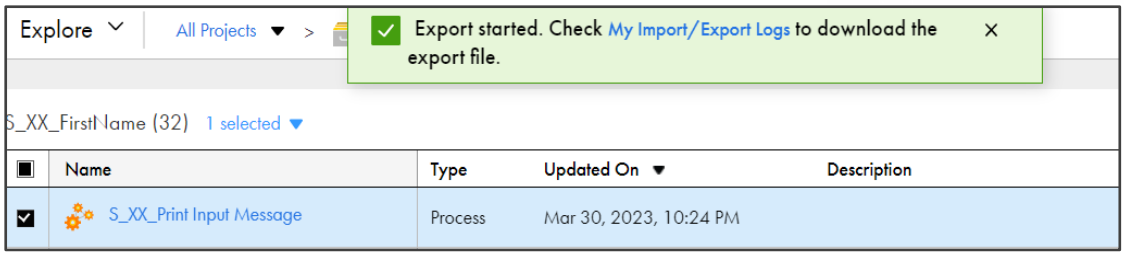
The **Export Assets** window appears.



6. Change Job Name to **S_XX_Print Input Message-Export**.
7. Select the **Include all dependencies for the select assets** checkbox.
8. Click **Export**.

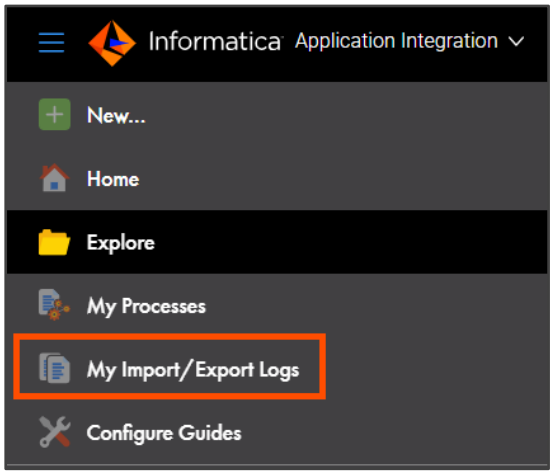


A message "**Export started. Check My Import/Export Logs to download the export file**" appears.

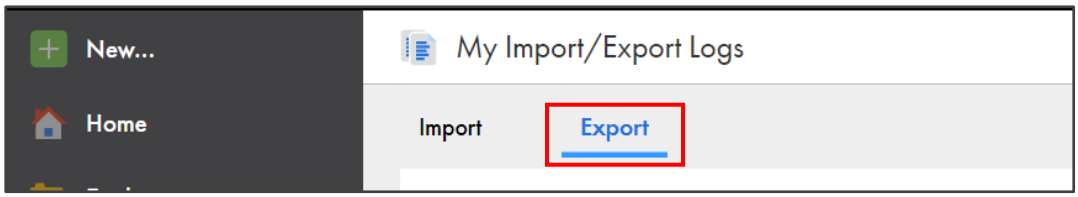


View My Import/Export Logs

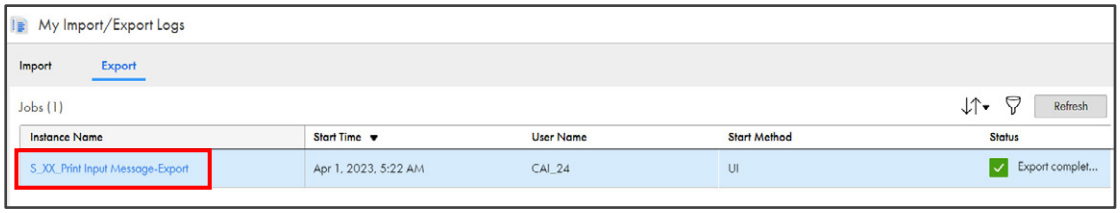
9. To see the progress of the job, from the navigation pane, select **My Import/Export Logs**.



10. From the menu at the top of the page, select **Export**.

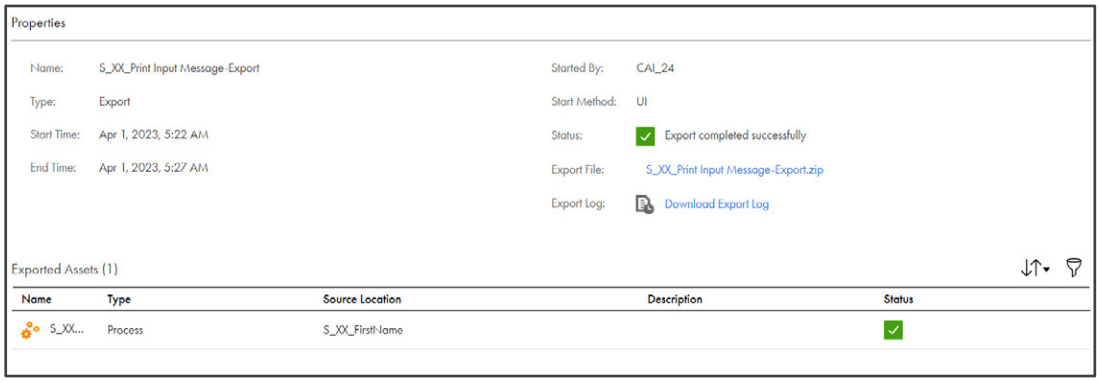


11. Click **S_XX_Print Input Message-Export** to open the log details page.

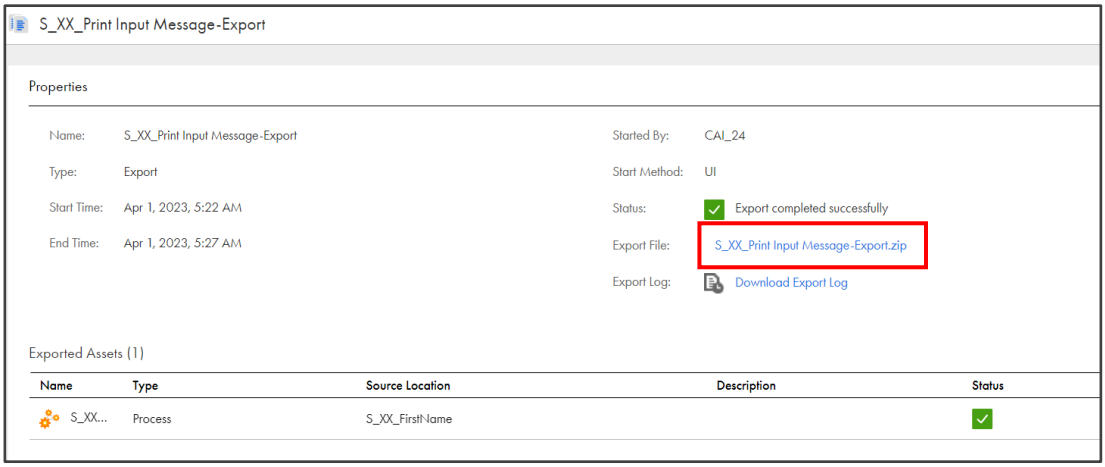


12. Observe the log details.

    Note: Check the Export File field. The file is exported as a ZIP file.



13. To download the export file, click **S_XX_Print Input Message-Export.zip**.



14. Close the asset tab.

This concludes the lab.

Module 9: Using XQuery in CAI
Lab 9-1: Create a Process Using Lists Function of XQuery

Overview:

XQuery is a functional language that is responsible for finding and extracting elements and attributes from XML documents. It can be used to extract information to use in a web service, generate summary reports, transform XML data to XHTML, and search web documents for relevant information.

Objectives:

- Create a Process Object – Employee
- Create a Process using Lists functions

Duration:

50 Minutes

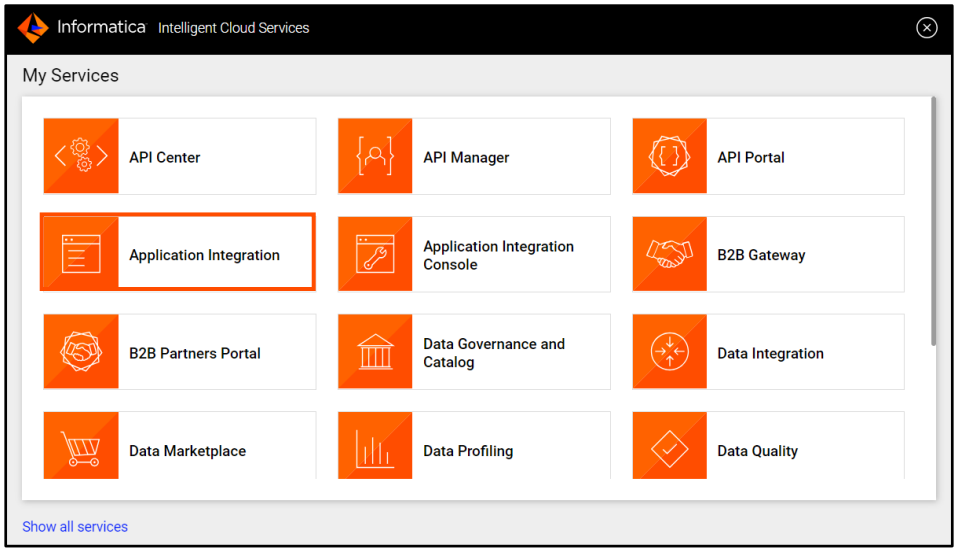
Tasks:

Note: The screen in your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

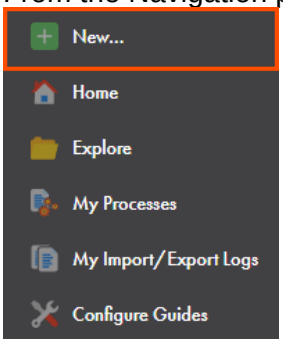
Create a Process Object – Employee

This process object includes the Employee ID, first name, and last name of the employee.

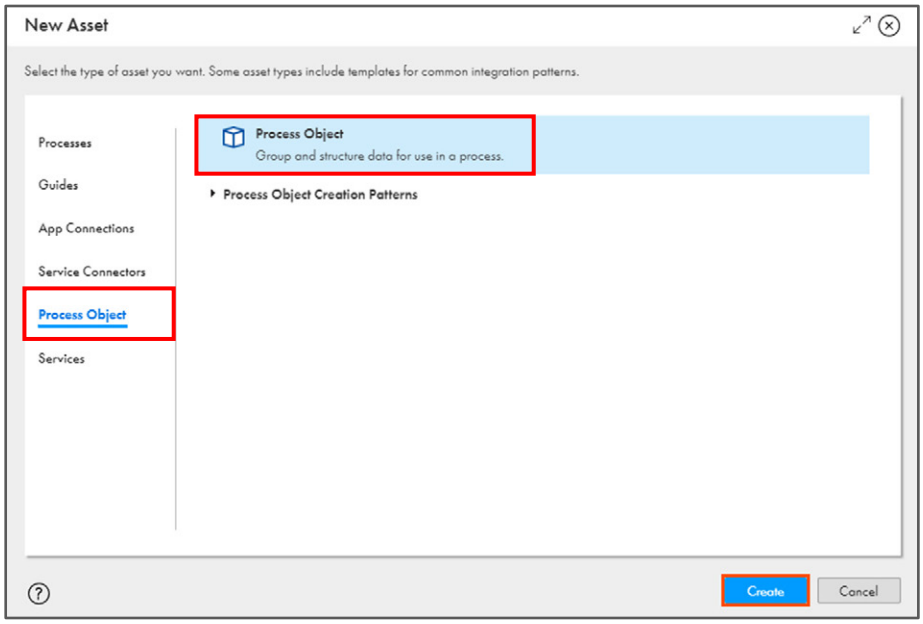
1. Log into IICS and access the Cloud **Application Integration** service.



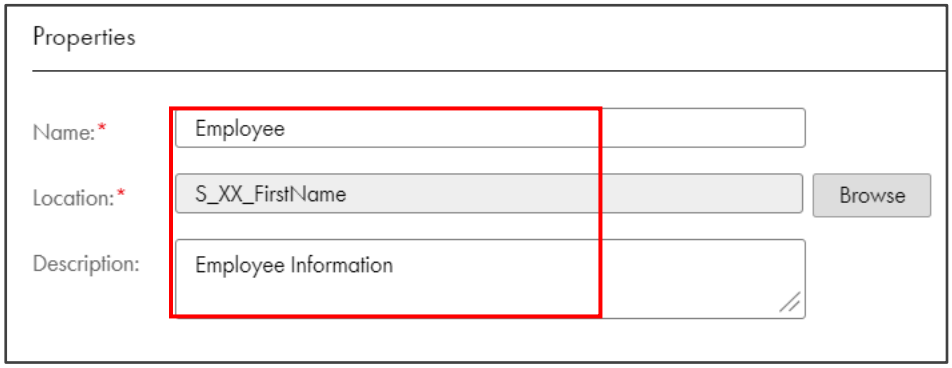
2. From the Navigation pane, click **New**.



3. In the New Asset window, from the Process Object tab, select **Process Object**, and click **Create**.



4. In the Process Object Properties page, in the Name field, enter **Employee**.
5. In the Location field, select your project folder **S_XX_FirstName**.
6. In the Description field, enter **Employee Information**.

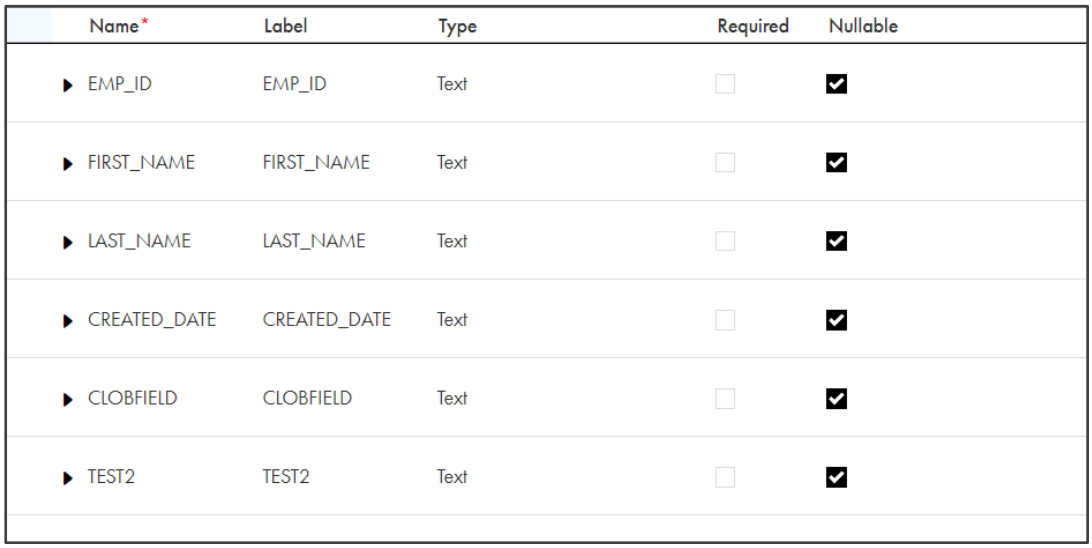


For the Employee Process Object, the required fields are – **EMP_ID**, **FIRST_NAME**, **LAST_NAME**, **CREATED_DATE**, **CLOBFIELD**, and **TEST2**.

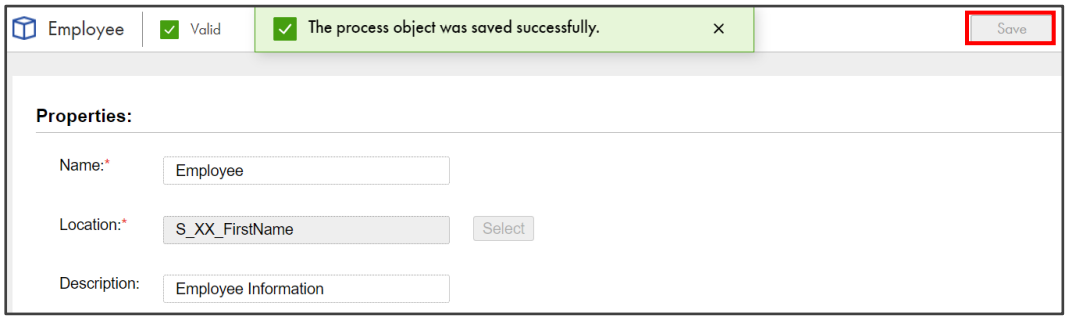
7. For the first field, enter Name and Label as **EMP_ID**.
8. From the Type drop-down, select **Text**.
9. Click **Add Field**.

10. Similarly, add the other fields as shown in the table below:

Name	Label	Type
FIRST_NAME	FIRST_NAME	Text
LAST_NAME	LAST_NAME	Text
CREATED_DATE	CREATED_DATE	Text
CLOBFIELD	CLOBFIELD	Text
TEST2	TEST2	Text



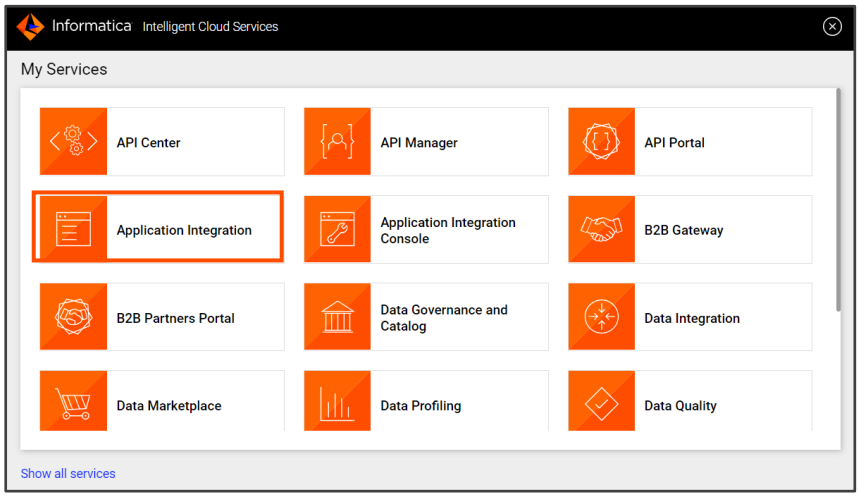
11. **Save** the Process Object.



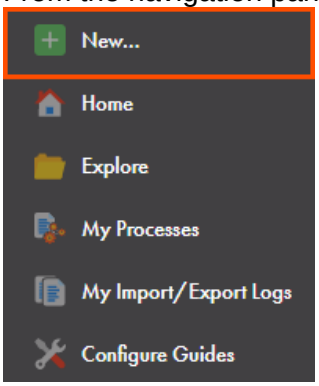
You have successfully designed a Process Object.

Create a Process S_XX_ EmployeePOList

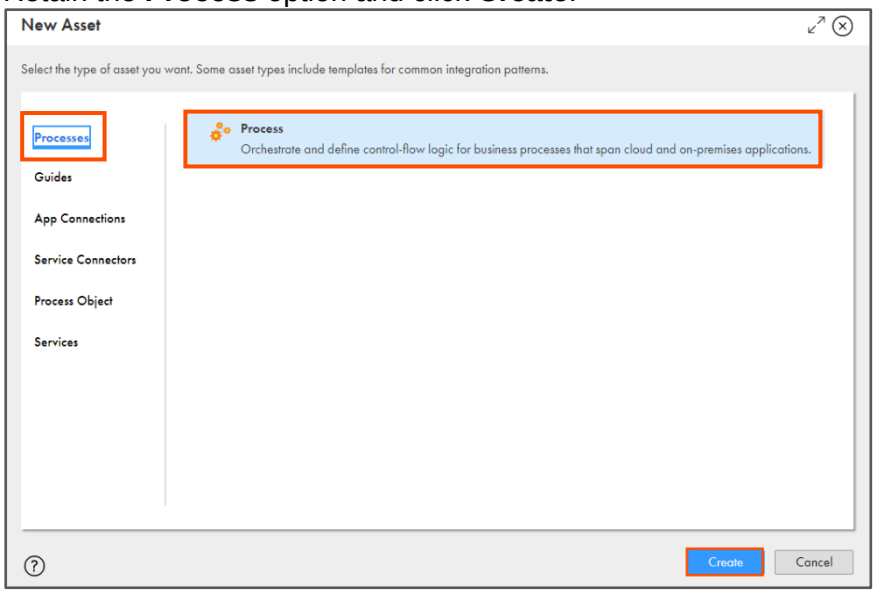
12. Navigate to **Application Integration**.



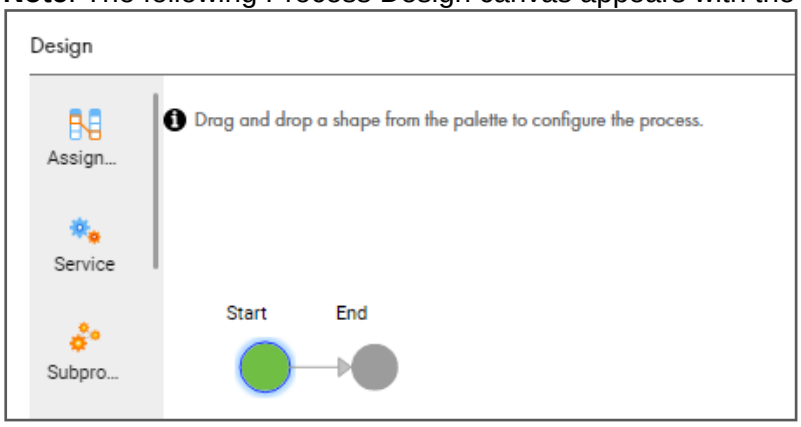
13. From the navigation pane, select **New**.



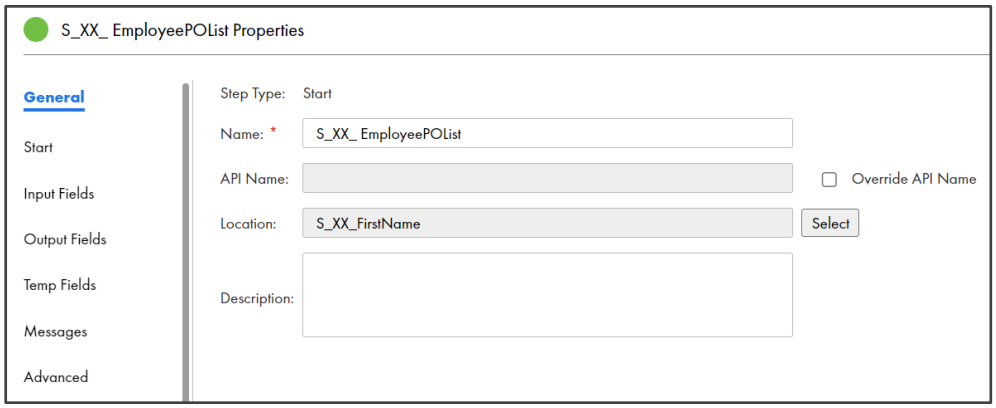
14. Retain the **Process** option and click **Create**.



Note: The following Process Design canvas appears with the Start and End steps.

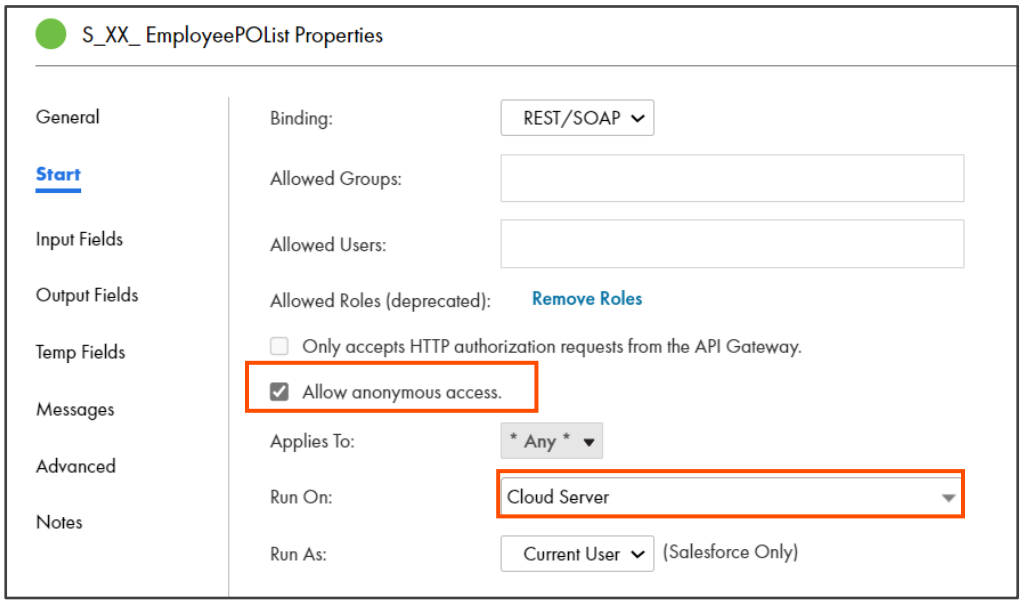


15. From the process canvas, select the **Start** step.
16. In the General properties tab, in the Name field, enter **S_XX_ EmployeePOList**.
17. Skip the API Name option.
18. In the Location field, select your project folder **S_XX_FirstName**.



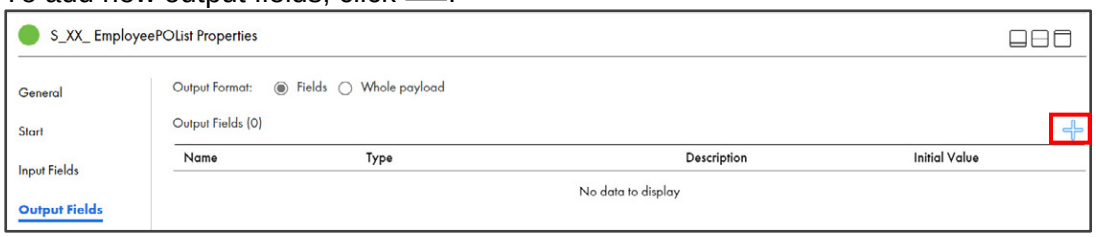
19. From the properties pane, select **Start**.
20. Select the **Allow anonymous access** option.

21. From the Run On drop-down, retain **Cloud Server**.



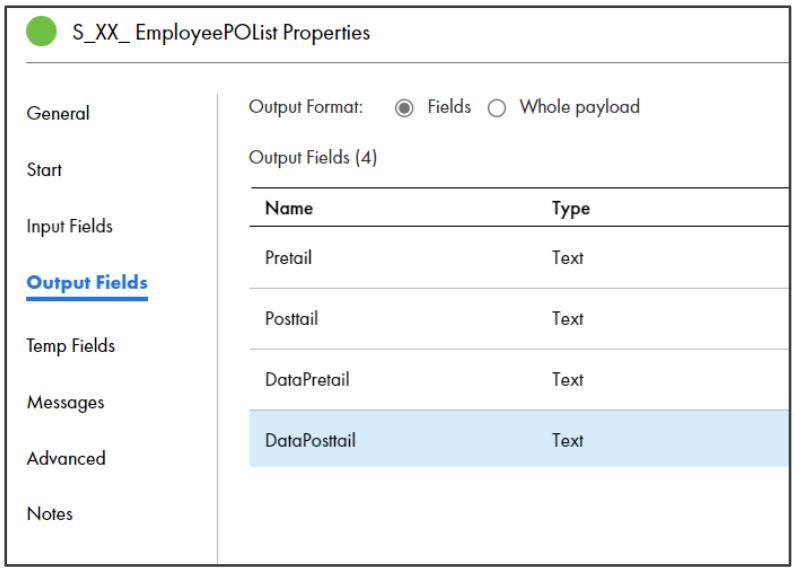
Next, you will create the output fields to display the results.

22. From the properties pane, select **Output Fields**.
23. To add new output fields, click .



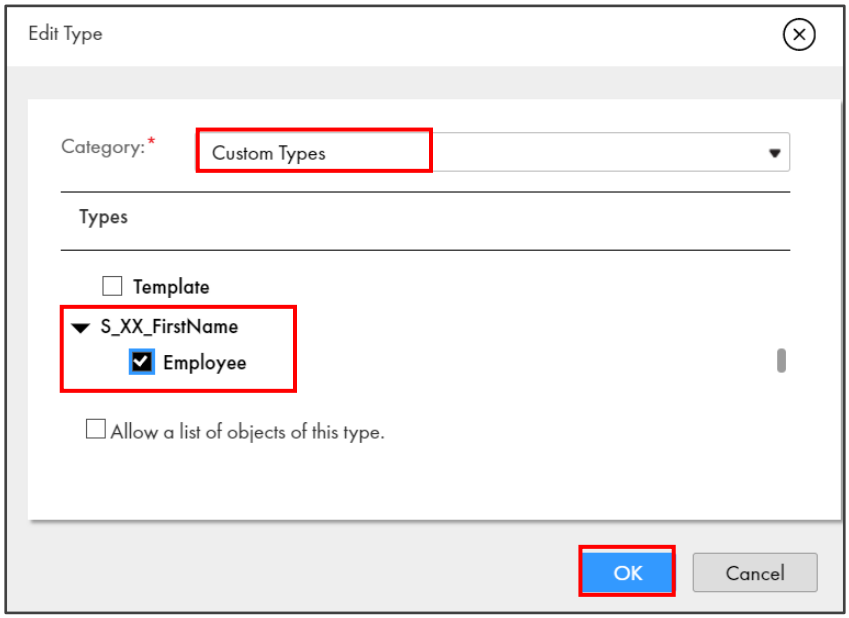
24. Create the output fields, as shown in the table below:

Name	Type
Pretail	Text
Posttail	Text
DataPretail	Text
DataPosttail	Text

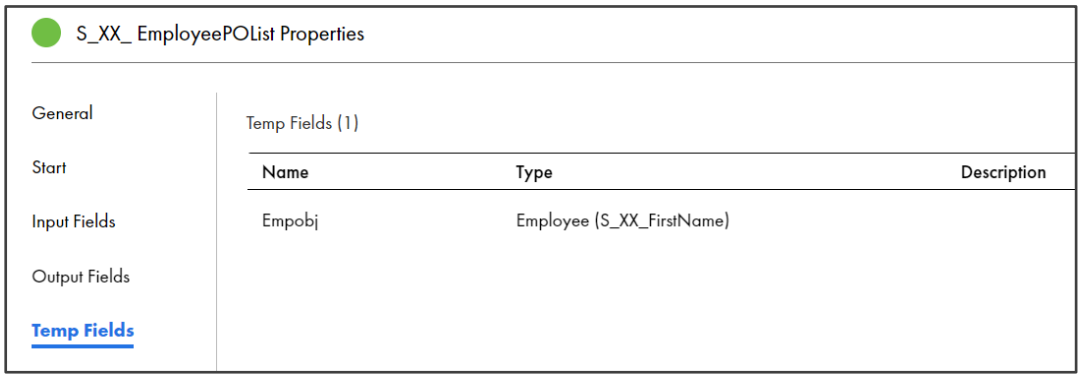


Next, you will create the Temp fields.

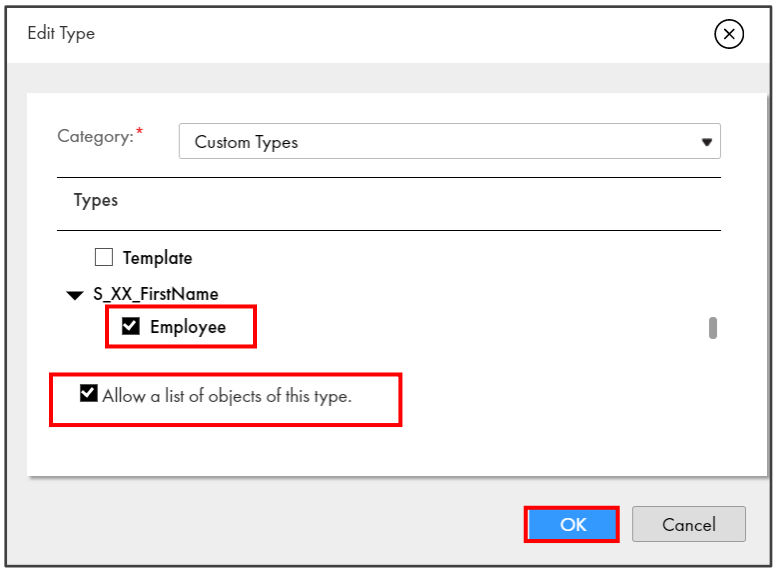
25. To add a new Temp field, click .
26. In the Name field, enter **Empobj**.
27. From the Type field drop-down, select **More Types**.
28. From the Category drop-down, select **Custom Types**.
29. Select your project folder.
30. Select **Employee**.
31. Click **OK**.



Note: Here, S_XX_FirstName is your project folder**.**

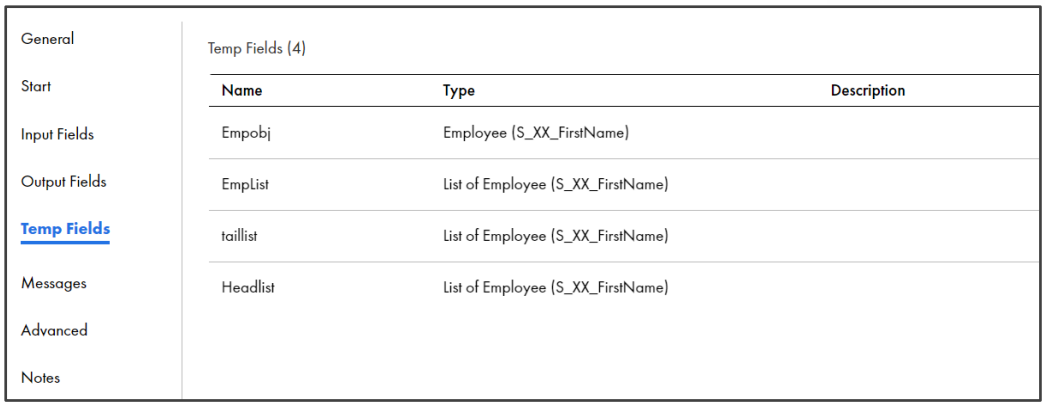


32. Create one more Temp field named as **EmpList**.
33. From the Type field drop-down, select **More Types**.
34. From the Category drop-down, select **Custom Types**.
35. Select your project folder.
36. Select **Employee**.
37. Select **Allow a list of objects of this type**.
38. Click **OK**.



39. Similarly, create three more Temp fields, as shown in the table below:

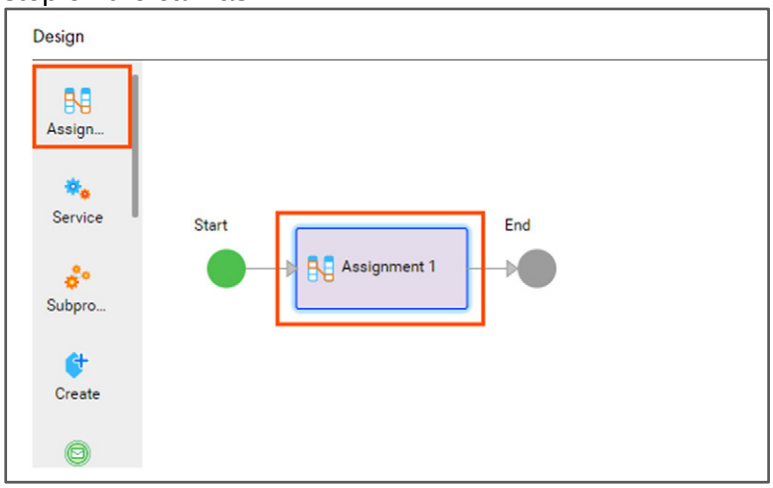
Name	Type
taillist	Employees (S_XX_FirstName)
Headlist	Employee (S_XX_FirstName)



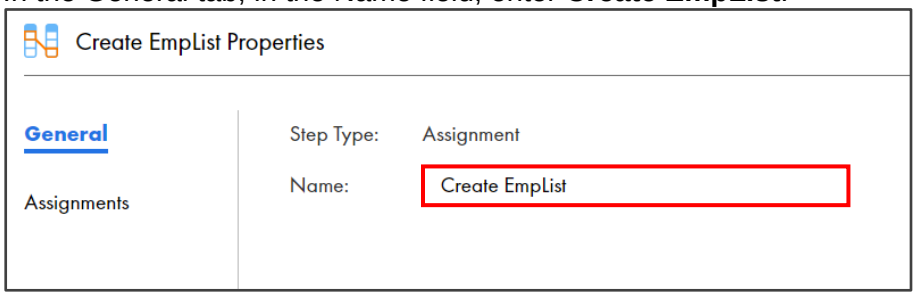
Add an Assignment step Create EmpList:

Now, you will add an Assignment step 'Create EmpList' to display the list of all elements.

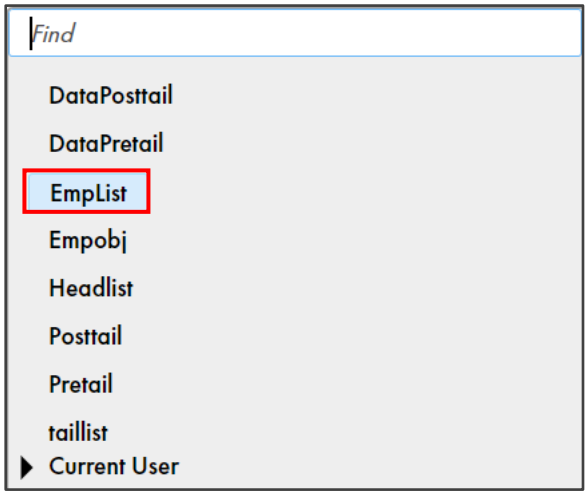
40. From the palette, drag and drop an **Assignment** step between the Start and the End step on the canvas.



41. Select the **Assignment** step.
42. In the General tab, in the Name field, enter **Create EmpList**.



43. In the Assignments tab, click the plus **(+)** icon, and select **EmpList**.



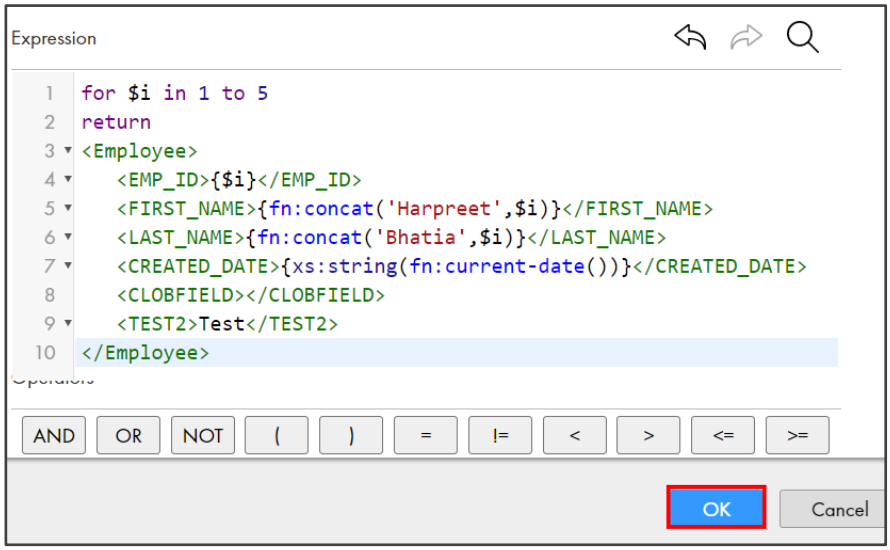
44. From the **Assigned Using** drop-down, select **Formula**.
45. To open the Formula Editor, click .
    In the Expression field, enter the following expression:

```
for $i in 1 to 5
return
<Employee>
  <EMP_ID>{$i}</EMP_ID>
  <FIRST_NAME>{fn:concat('Harpreet',$i)}</FIRST_NAME>
  <LAST_NAME>{fn:concat('Bhatia',$i)}</LAST_NAME>
  <CREATED_DATE>{xs:string(fn:current-date())}</CREATED_DATE>
  <CLOBFIELD></CLOBFIELD>
  <TEST2>Test</TEST2>
</Employee>
```

    OR

    From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **20_LabGuide_CAI_Create a Process Using Lists Function of XQuery_9-1**. Copy the command mentioned under **Step A** and paste it into the Expression field.

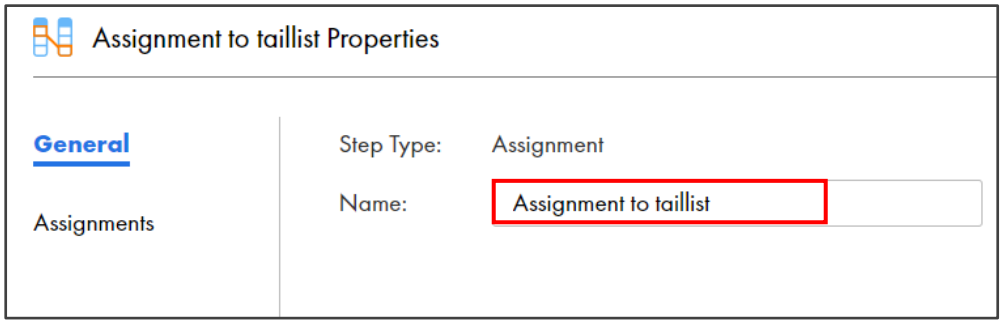
46. Click **OK**.



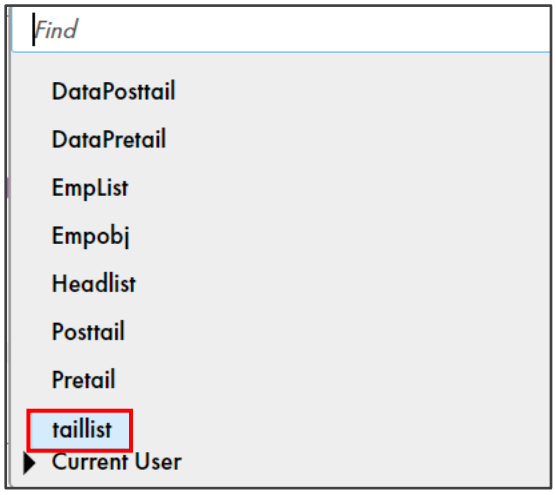
Add an Assignment step Assignment to taillist

Now, you will add an Assignment step to display the list of elements except the first one.

47. From the Design palette, drag and drop an **Assignment** step on the link between the Create EmpList and End step.
48. In the General section of Assignment properties, enter the Name as **Assignment to taillist**.



49. In the Assignments tab, click the plus **(+)** icon, and select **taillist**.



50. From the **Assigned Using** drop-down, select **Formula**.

51. To open the Formula Editor, click .

    In the Expression field, enter the following expression:

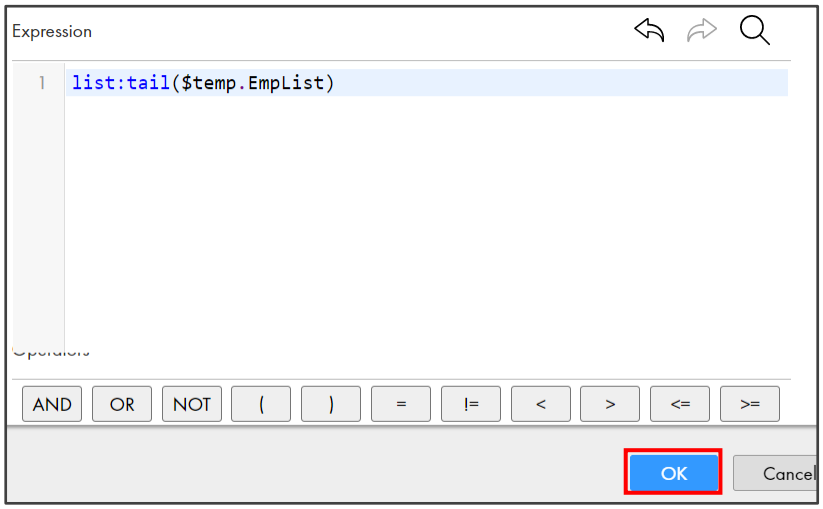
    list:tail($temp.EmpList)

    OR

    From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **20_LabGuide_CAI_Create a Process Using Lists Function of XQuery_9-1**. Copy the command mentioned under **Step B** and paste it into the Expression field.

    Note: The tail function returns an array containing all members except the first from a supplied array.

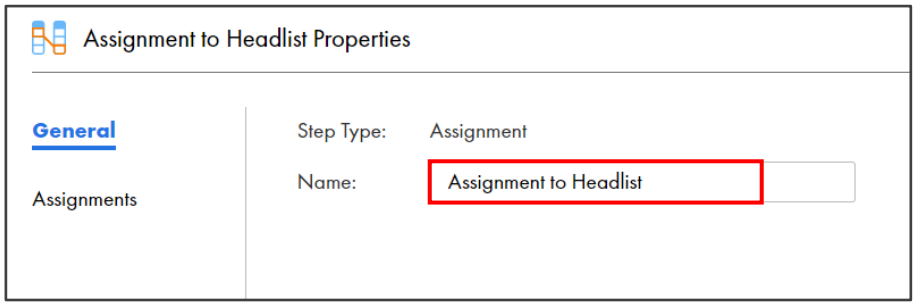
52. Click **OK**.



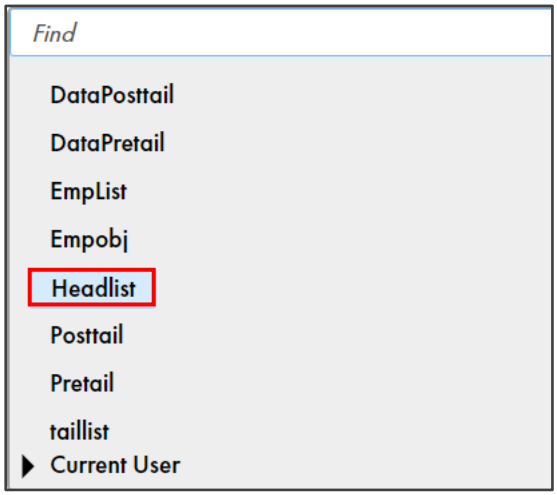
Add an Assignment step Assignment to Headlist

Now, you will add an Assignment step to display the first element of the list.

53. From the Design palette, drag and drop an **Assignment** step on the link between the Assignment to taillist and End step.
54. In the General section of Assignment properties, enter the Name as **Assignment to Headlist**.



55. In the Assignments tab, click the plus **(+)** icon, and select **Headlist**.



56. From the **Assigned Using** drop-down, select **Formula**.
57. To open the Formula Editor, click .

    In the Expression field, enter the following expression:

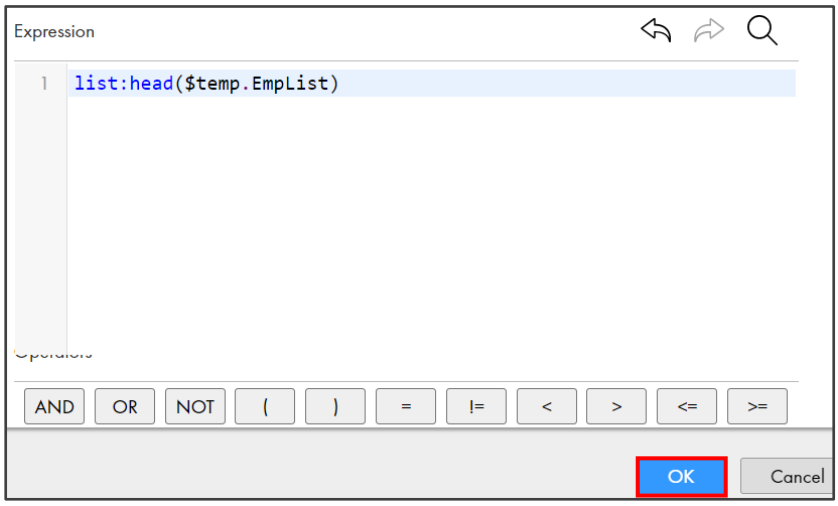
    list:head($temp.EmpList)

    OR

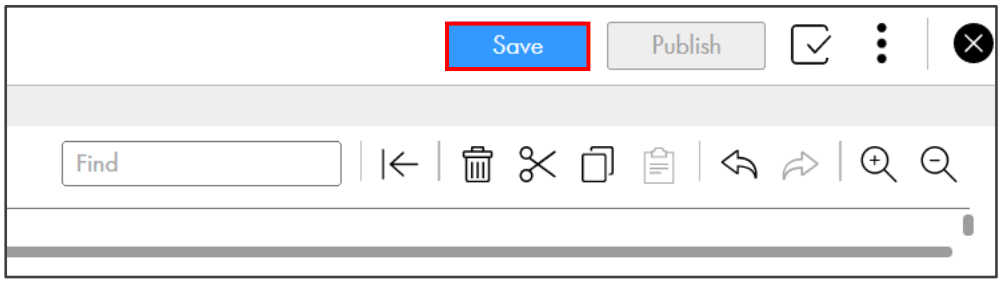
    From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **20_LabGuide_CAI_Create a Process Using Lists Function of XQuery_9-1**. Copy the command mentioned under **Step C** and paste it into the Expression field.

    Note: The head function returns the first element of an array.

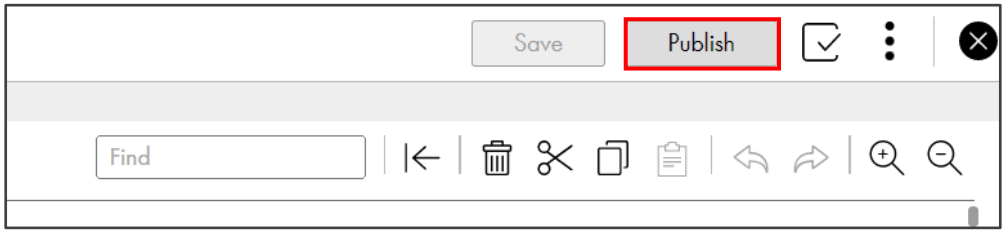
58. Click **OK**.

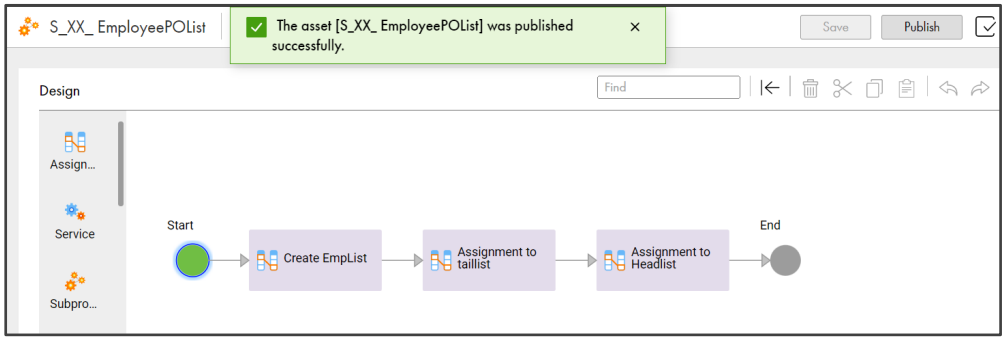


59. Click **Save**.



60. Click **Publish** to publish the process.

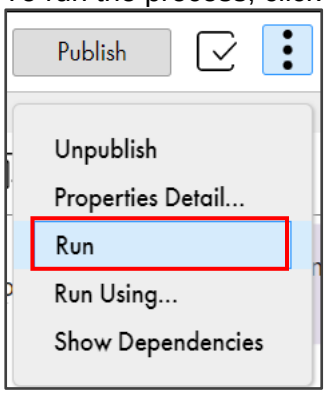




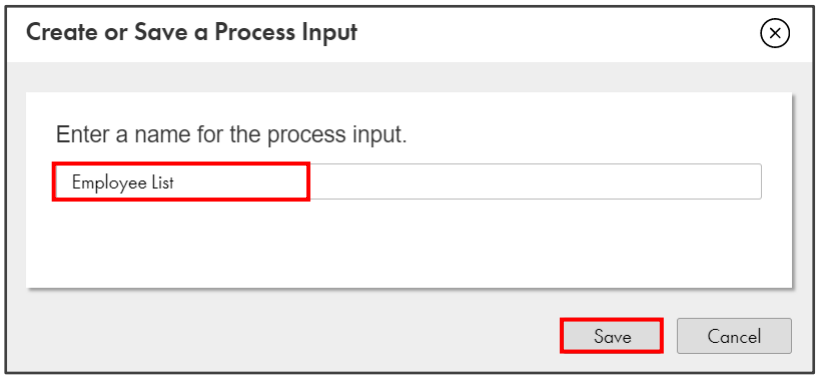
The process is published successfully.

Invoke the Process from the Interface

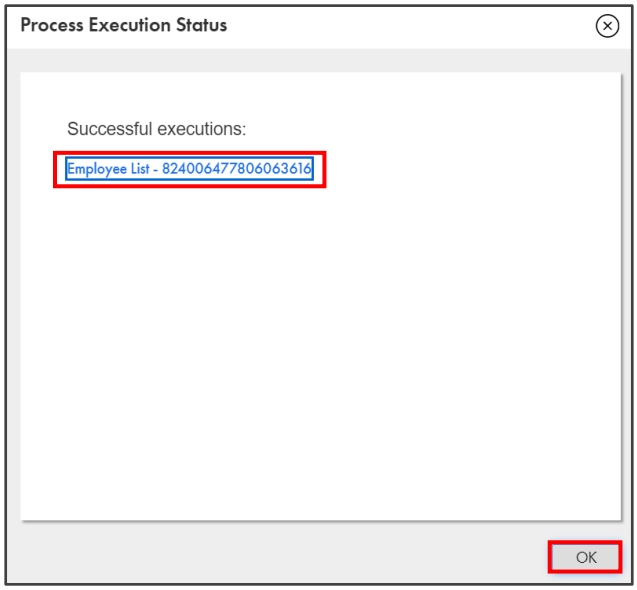
61. To run the process, click ⋮, and select **Run**.



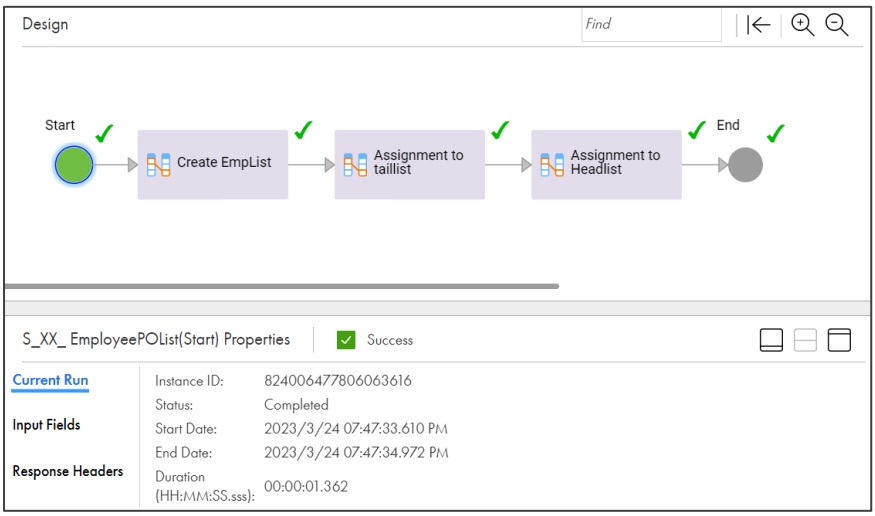
62. Click **Save As…** to Save the input.
63. Save the input as **Employee List**.



64. After the input is added and validated successfully, click **Run**.
65. Select the Process Execution link and click **OK**.

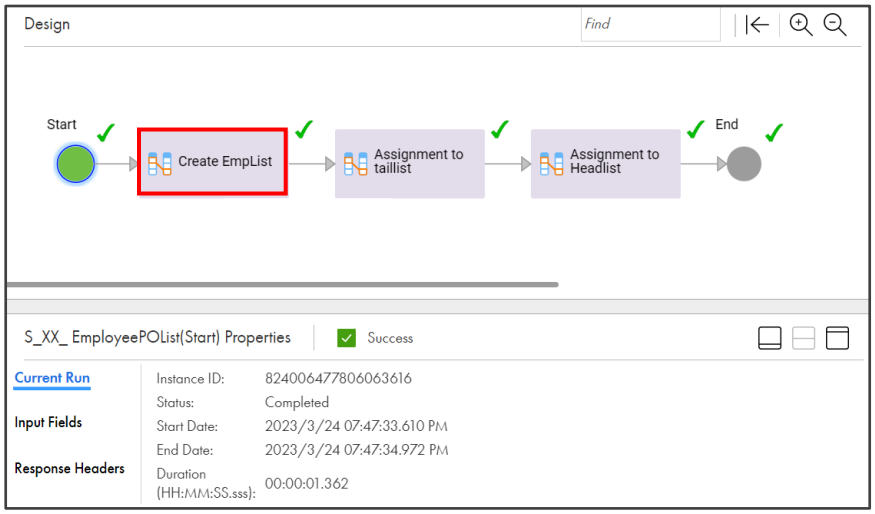


The process is executed successfully. The execution details are as shown below:

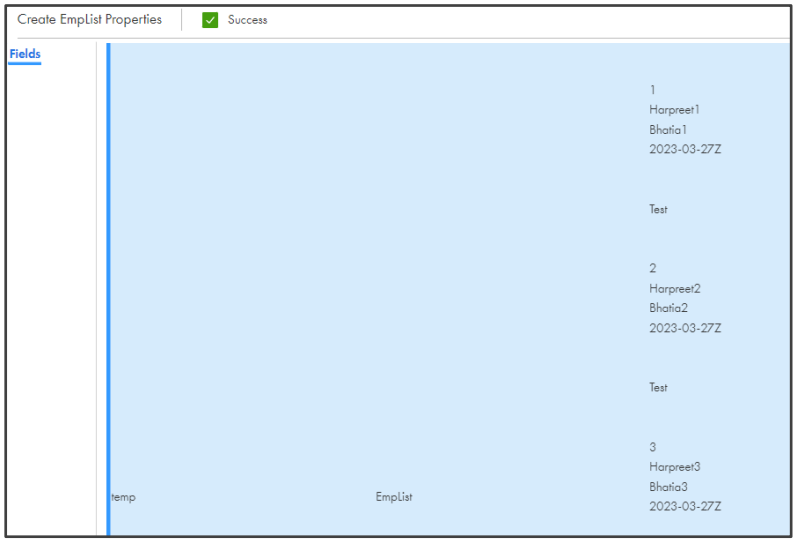


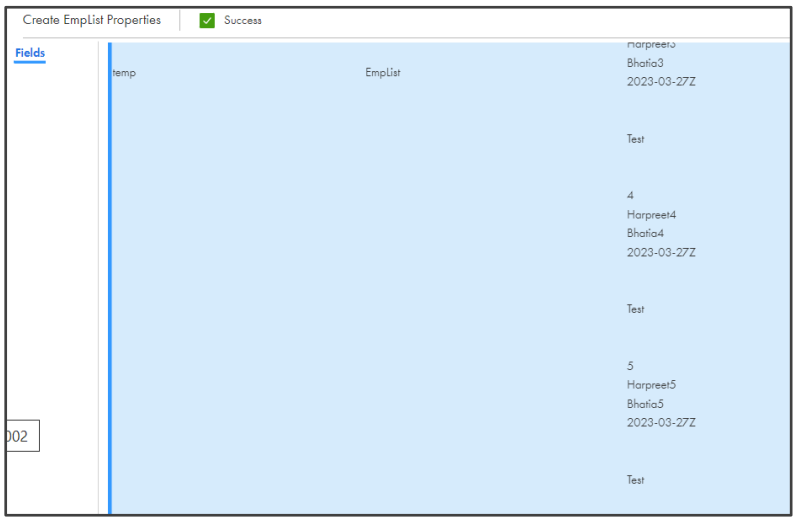
Note: If you are not able to see this screen, click **Refresh**.

66. From the Design selection, click **Create EmpList** to check the list of the employee.

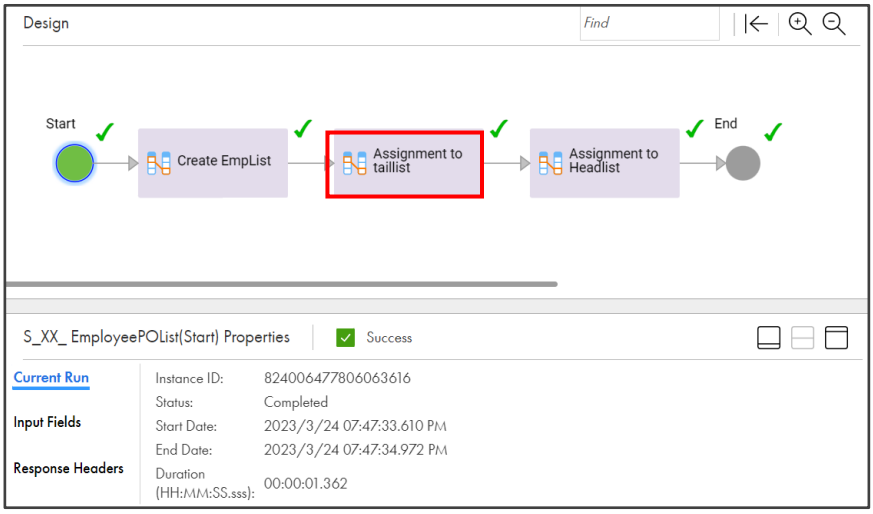


67. In the Create EmpList Properties section, observe the output. A list of five employees is created.



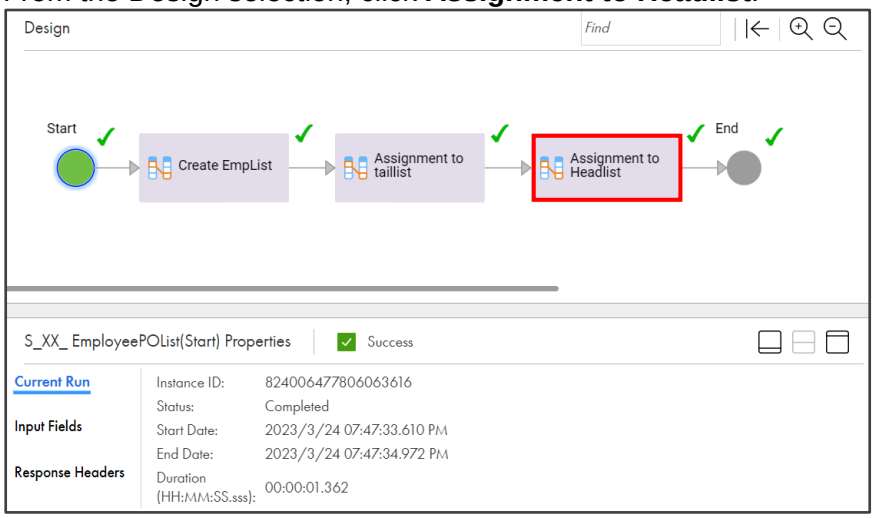


68. From the Design selection, click **Assignment to taillist**.



69. In the Create Assignment to taillist Properties section, observe the output. A list of all employees is displayed except the first one.

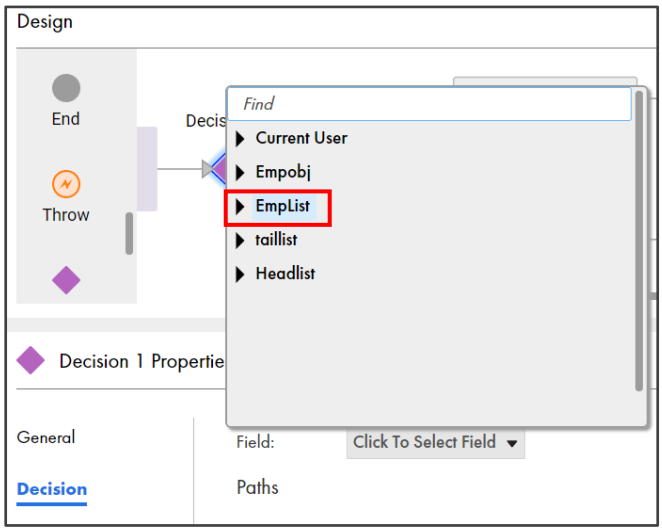
70. From the Design selection, click **Assignment to Headlist**.



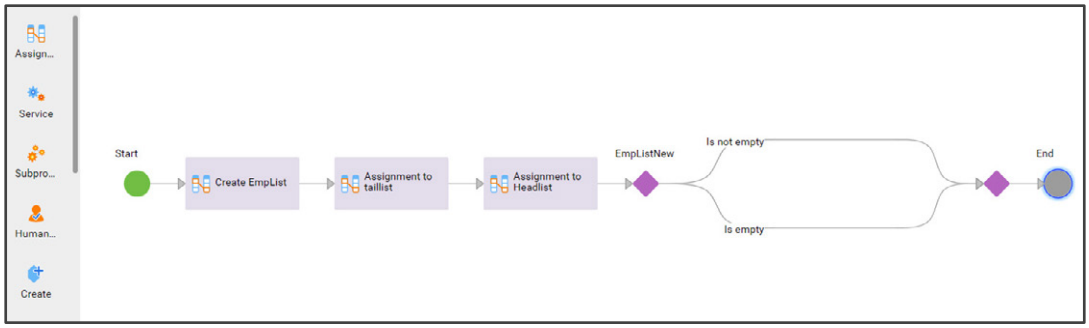
71. In the Create Assignment to Headlist Properties section, observe the output. Only the first employee is displayed.

Add a Decision step EmpListNew

72. From the Design palette, drag and drop a **Decision** step on the link between the Assignment to Headlist and End step.
73. To configure the step, from the process canvas, select the **Decision** step.
74. In the General section of Assignment properties, enter the Name as **EmpListNew**.
75. From the properties pane, select **Decision**.
76. From the Field drop-down, select **EmpList**.

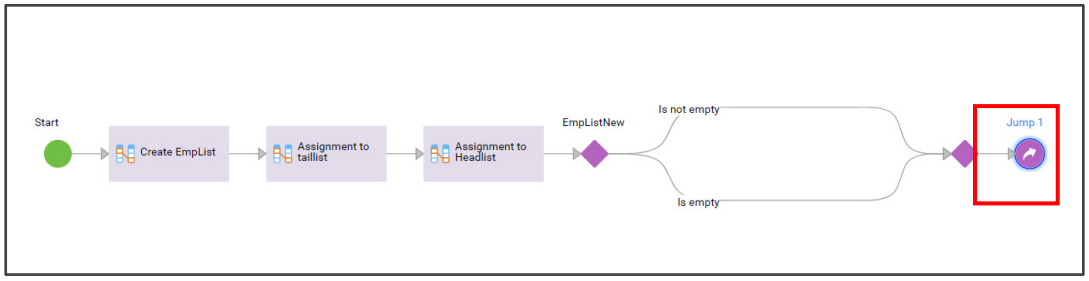


Your process should look like this:

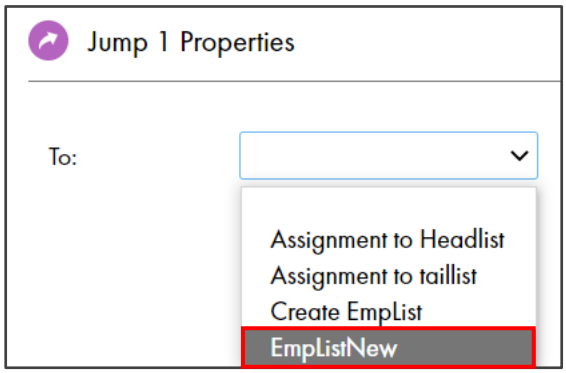


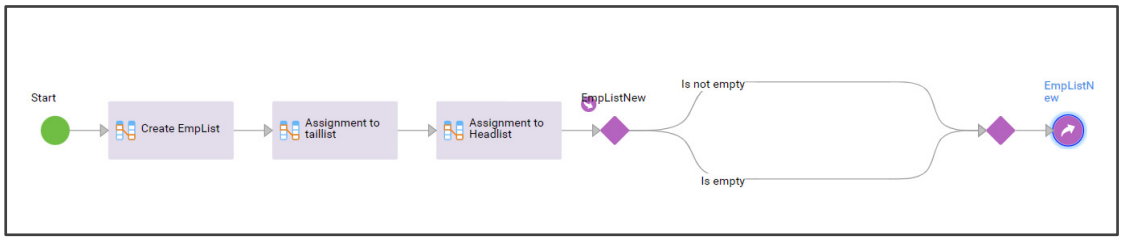
Next, you will add a Jump step.

77. From the Design palette, drag a **Jump** step and drop it at the **End** step.

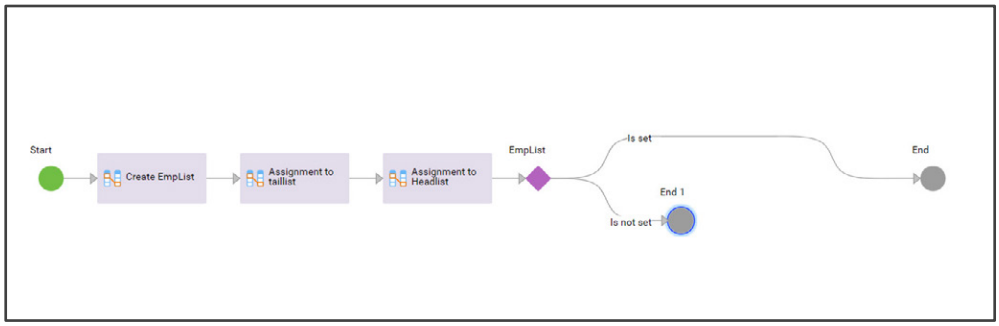


78. In the Jump properties, from the To drop-down, select **EmpListNew**.

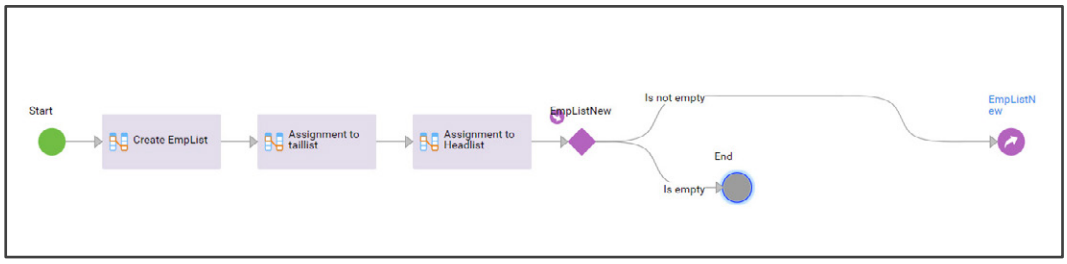




79. From the Design palette, drag and drop an **End** step to the **is empty** path of the Decision step.



80. In the General tab of the End properties, change the Name field to **End**.
    Your process should look like this:

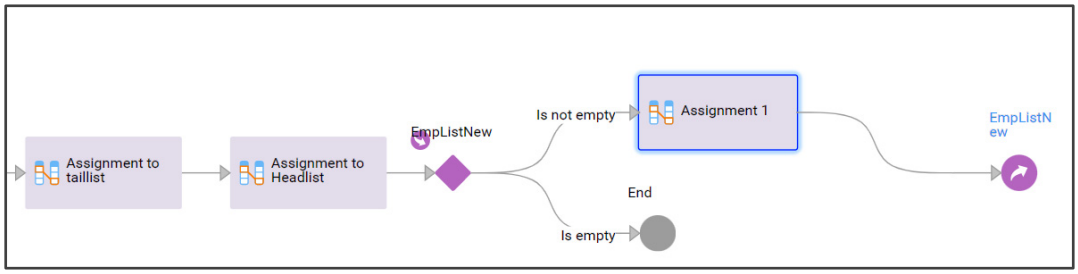


Next, you will add an Assignment step to display the last employee on the list.

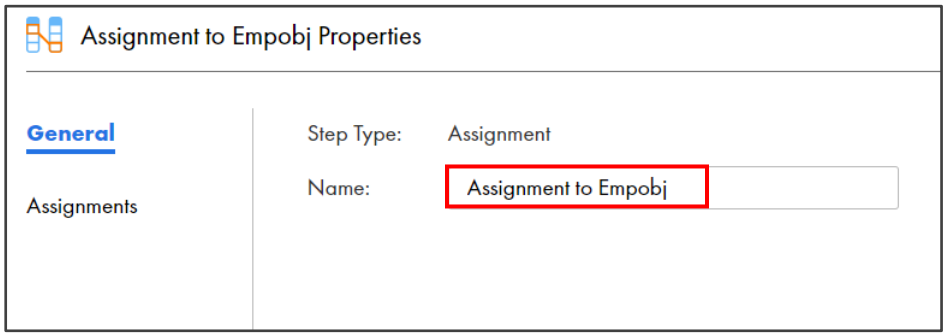
Add an Assignment step Assignment to Empobj

Now, you will display the last element of the list after the list is read.

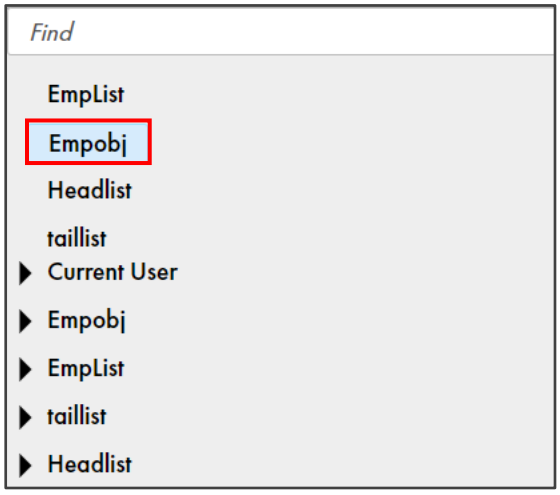
81. From the Design palette, drag and drop an **Assignment** step to the **is not empty** path of the Decision step.



82. In the General tab of the Assignment properties, enter the Name as **Assignment to Empobj**.



83. From the properties pane, select **Assignments**.
84. Click (**+**) icon.
85. From the list, select **Empobj**.



86. From the **Assigned Using** drop-down, select **Formula**.
87. To open the Formula Editor, click .

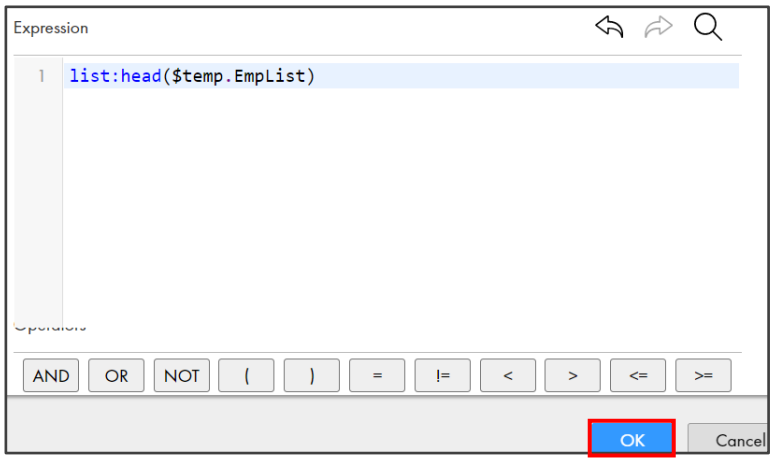
88. In the Expression field, enter the following expression:

    list:head($temp.EmpList)

    OR

    From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **20_LabGuide_CAI_Create a Process Using Lists Function of XQuery_9-1**. Copy the command mentioned under **Step D** and paste it into the Expression field.

89. Click **OK**.

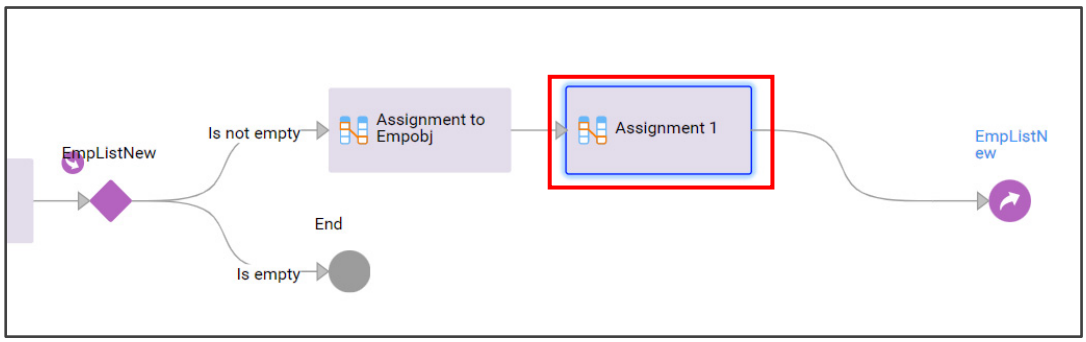


Note: After the list is read, the head function will return the last element of the list.

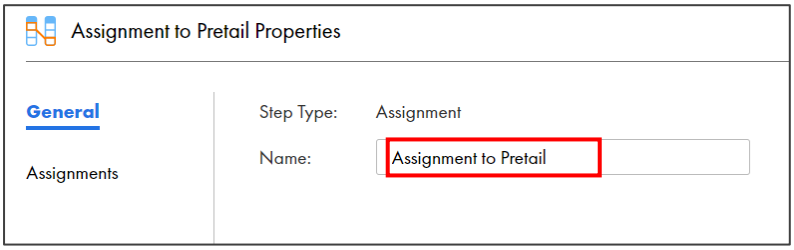
Add an Assignment step Assignment to Pretail

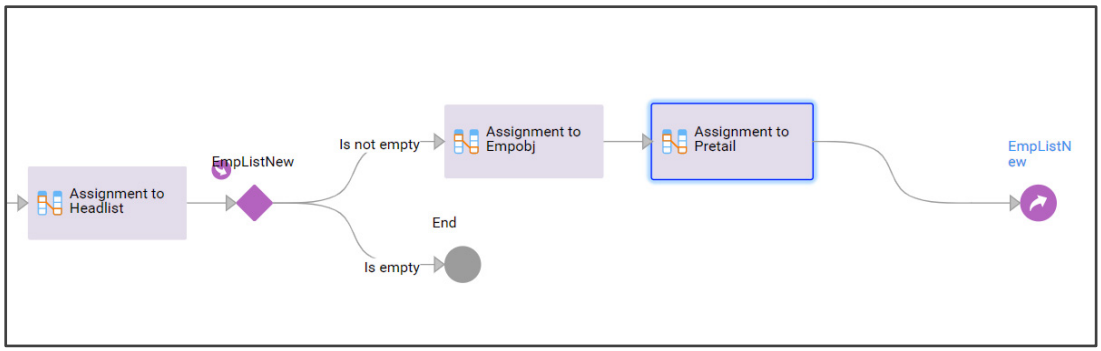
Now, you will add an Assignment step to count and return the total elements present in the list.

90. From the Design palette, drag and drop an **Assignment** step between Assignment to Empobj and Jump steps.

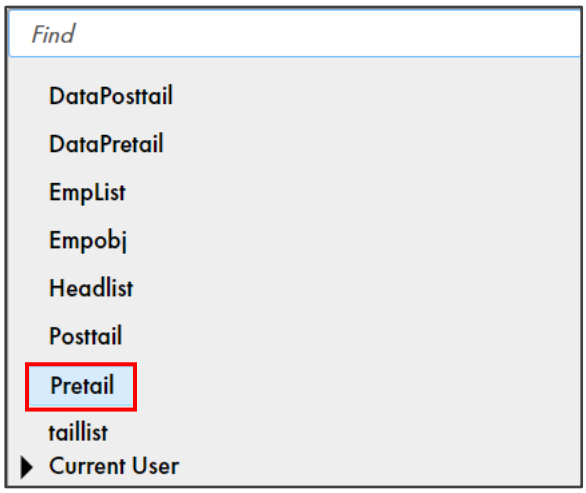


91. In the General tab of the Assignment properties, enter the Name as **Assignment to Pretail**.





92. From the properties pane, select **Assignments**.
93. Click (**+**) icon.
94. From the list, select **Pretail**.



95. From the **Assigned Using** drop-down, select **Formula**.
96. To open the Formula Editor, click .
97. In the Expression field, enter the following expression:

    if(empty($output.Pretail)) then
    **list:count($temp.EmpList)|| "
"**
    else
    **$output.Pretail || list:count($temp.EmpList) || "
"**

    OR

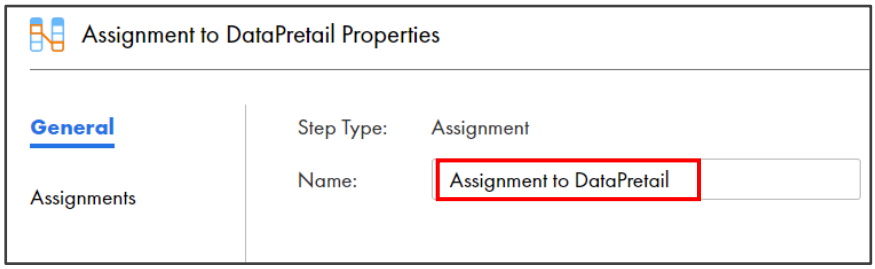
    From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **20_LabGuide_CAI_Create a Process Using Lists Function of XQuery_9-1**. Copy the command mentioned under **Step E** and paste it into the Expression field.

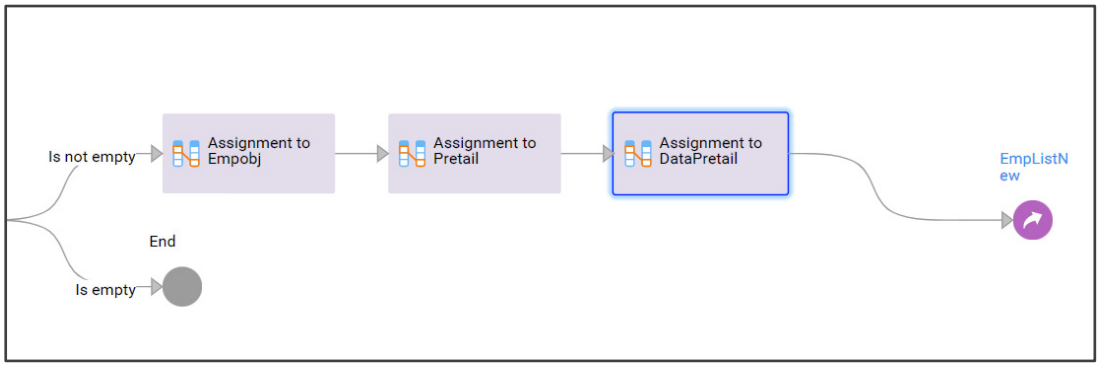
    Note: The count function counts and returns the total elements present in an array. Here, **&#10** is the carriage return which will insert a line break to the beginning of each element.
98. Click **OK**.

Add an Assignment step Assignment to DataPretail

Now, you will display all elements available in the list.

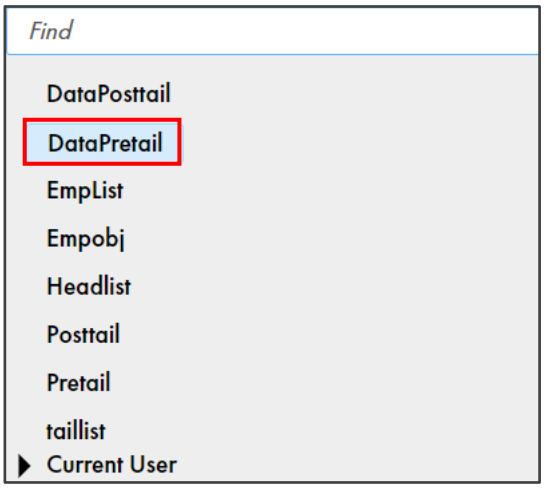
99. From the Design palette, drag and drop an **Assignment** step between the Assignment to Pretail and Jump steps.
100. In the General tab of the Assignment properties, enter the Name as **Assignment to DataPretail**.





101. From the properties pane, select **Assignments**.
102. Click (**+**) icon.

103. From the list, select **DataPretail**.



104. From the **Assigned Using** drop-down, select **Formula**.
105. To open the Formula Editor, click .
106. In the Expression field, enter the following expression:

if(empty($output.DataPretail)) then
**$temp.Empobj[1]/FIRST_NAME || "
"**
else
**$output.DataPretail || $temp.Empobj[1]/FIRST_NAME || "
"**

OR

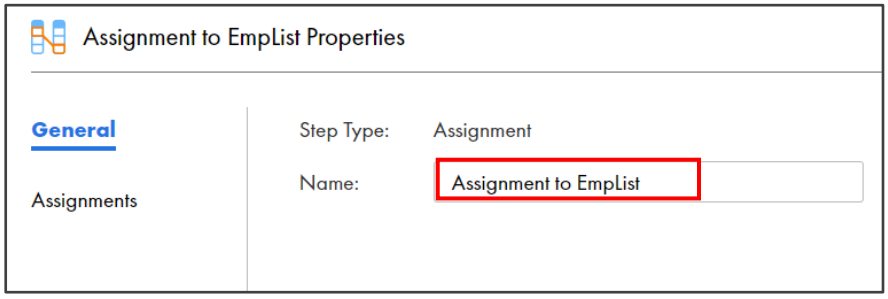
From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **20_LabGuide_CAI_Create a Process Using Lists Function of XQuery_9-1**. Copy the command mentioned under **Step F** and paste it in the Expression field.

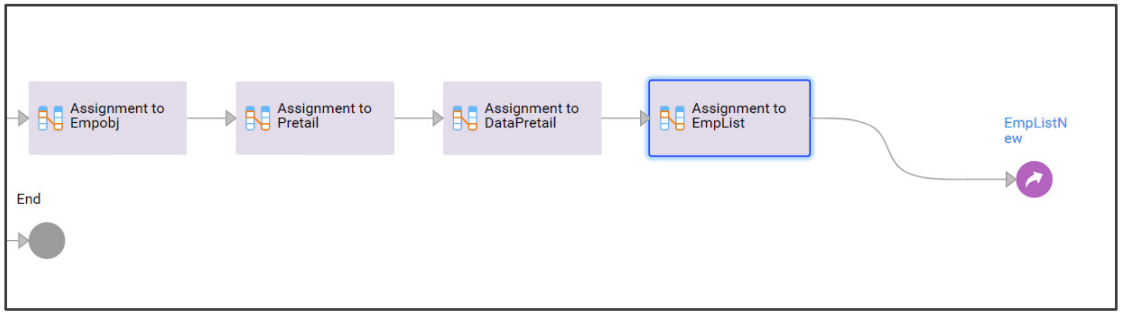
107. Click **OK**.

Add an Assignment step Assignment to EmpList

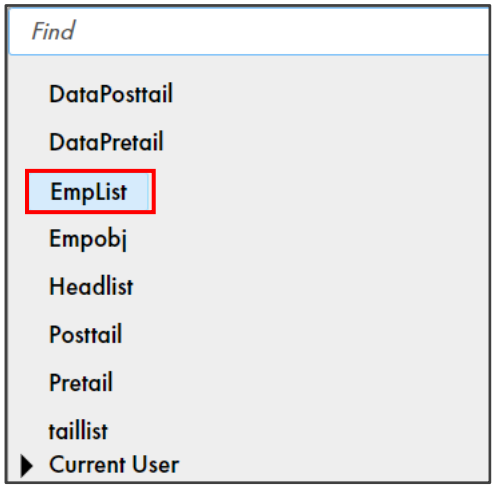
In this step, the entire list will be read, and the empty list will be displayed.

108. From the Design palette, drag and drop an **Assignment** step between Assignment to DataPretail and Jump steps.
109. In the General tab of the Assignment properties, enter the Name as **Assignment to EmpList**.





110. From the properties pane, select **Assignments**.
111. Click (**+**) icon.
112. From the list, select **EmpList**.



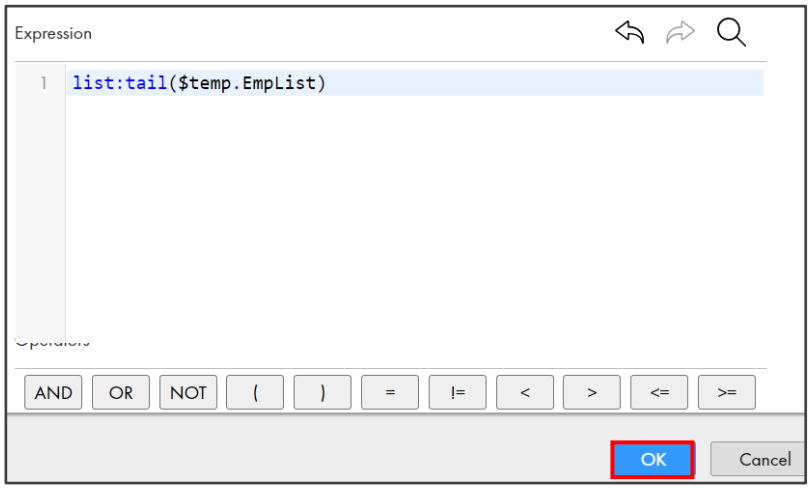
113. From the **Assigned Using** drop-down, select **Formula**.

114. To open the Formula Editor, click .
115. In the Expression field, enter the following expression:

     list:tail($temp.EmpList)

     OR

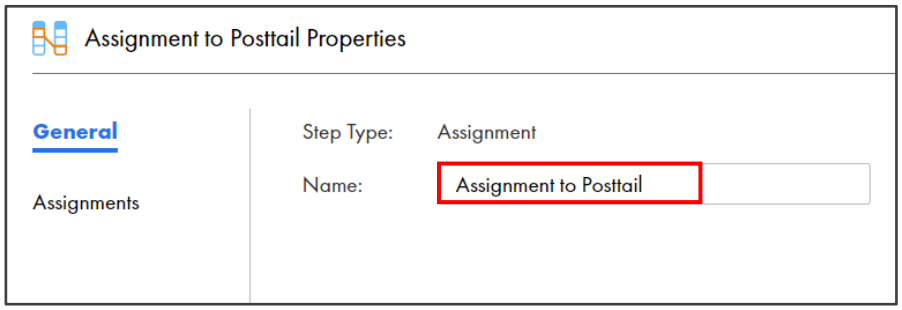
     From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **20_LabGuide_CAI_Create a Process Using Lists Function of XQuery_9-1**. Copy the command mentioned under **Step G** and paste it in the Expression field.
116. Click **OK**.

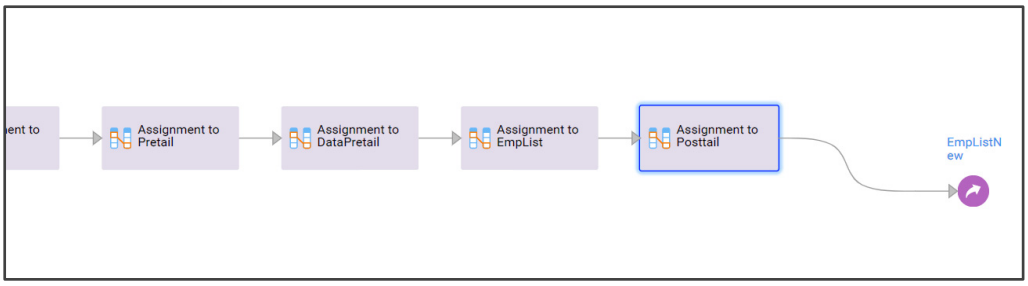


Add an Assignment step Assignment to Posttail

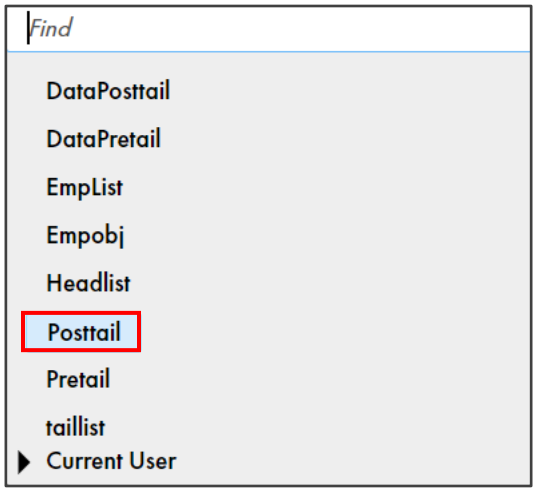
In this step, the order of elements in the tail will be read and displayed.

117. From the Design palette, drag and drop an **Assignment** step between Assignment to EmpList and Jump steps.
118. In the General tab of the Assignment properties, enter the Name as **Assignment to Posttail**.





119. From the properties pane, select **Assignments**.
120. Click (**+**) icon.
121. From the list, select **Posttail**.



122. From the **Assigned Using** drop-down, select **Formula**.
123. To open the Formula Editor, click .
124. In the Expression field, enter the following expression:

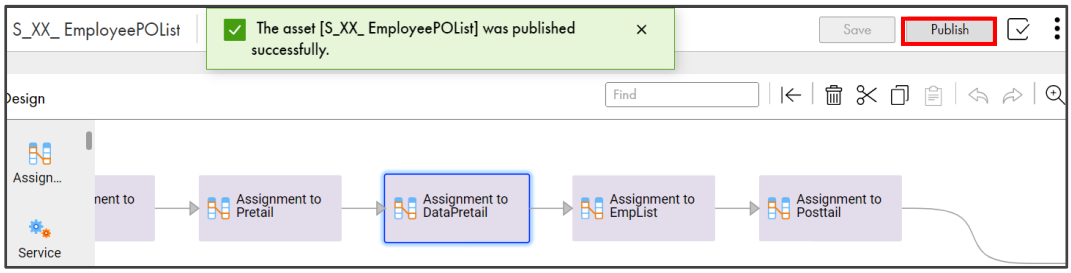
if(empty($output.Posttail)) then
**list:count($temp.EmpList)|| "
"**
else
**$output.Posttail || list:count($temp.EmpList) || "
"**

OR

From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **20_LabGuide_CAI_Create a Process Using Lists Function of XQuery_9-1**. Copy the command mentioned under **Step H** and paste it in the Expression field.

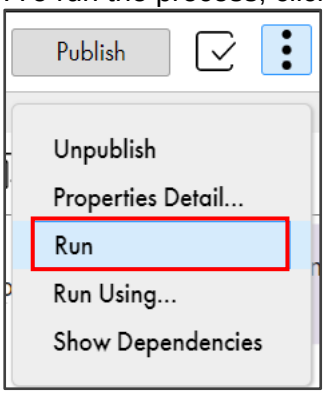
125. Click **OK**.

126. **Save** and **Publish** the process.

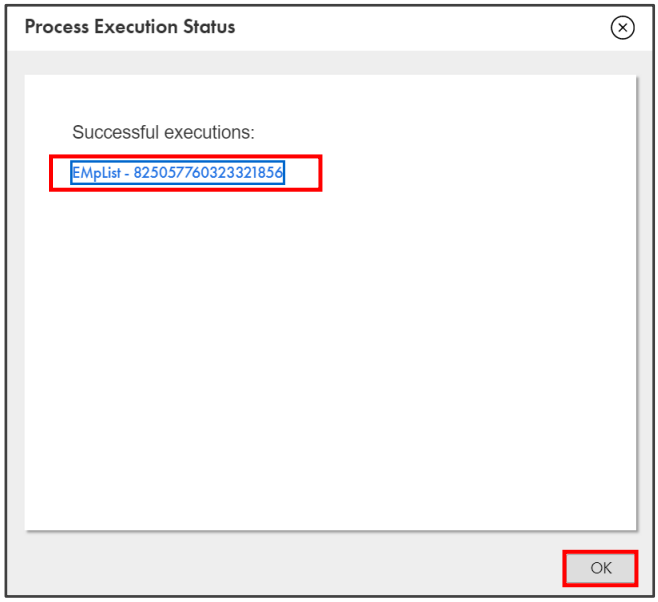


Invoke the Process from the Interface

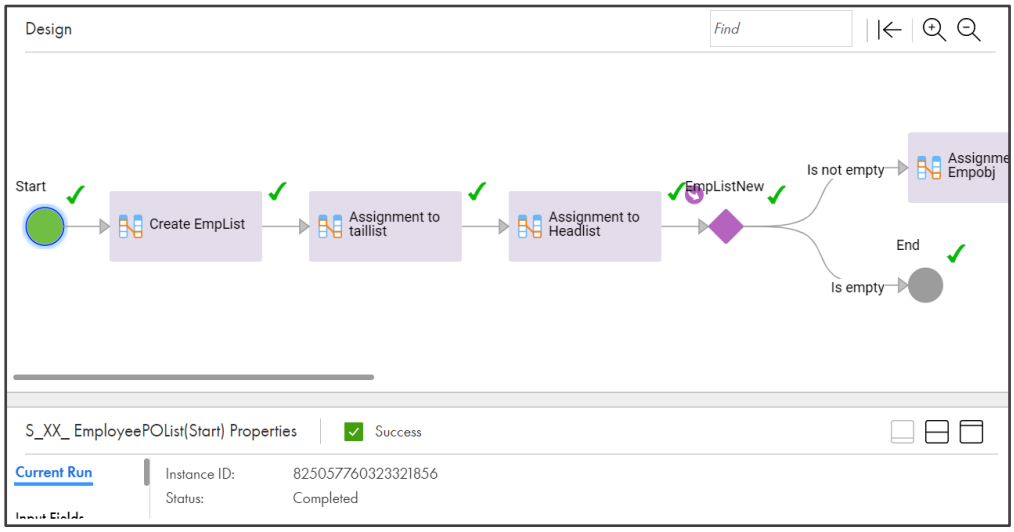
127. To run the process, click ⋮, and select **Run**.



128. Select the **Process Execution** link and click **OK**.

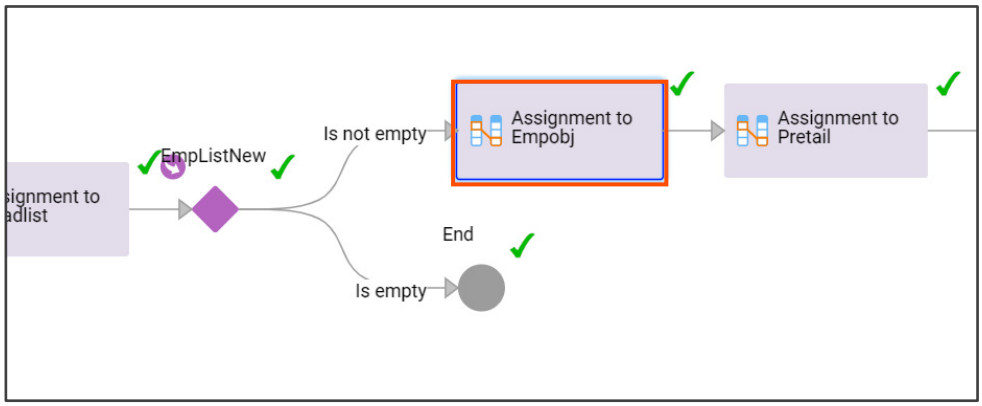


The process is executed successfully. The execution details are as shown below:

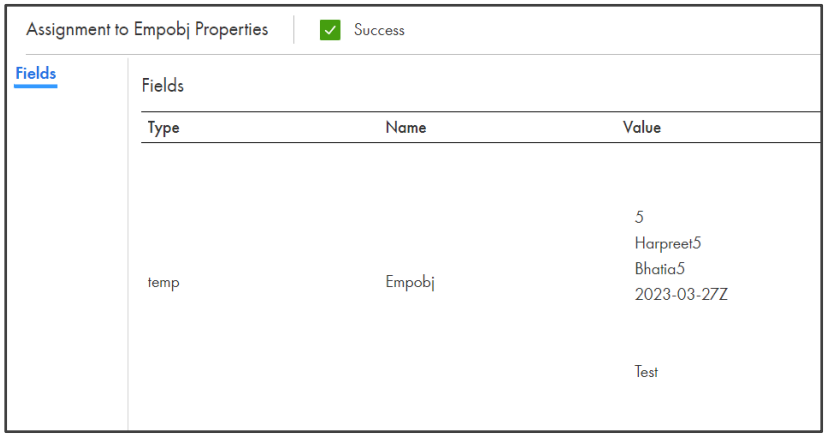


Note: If you are not able to see this screen, click **Refresh**.

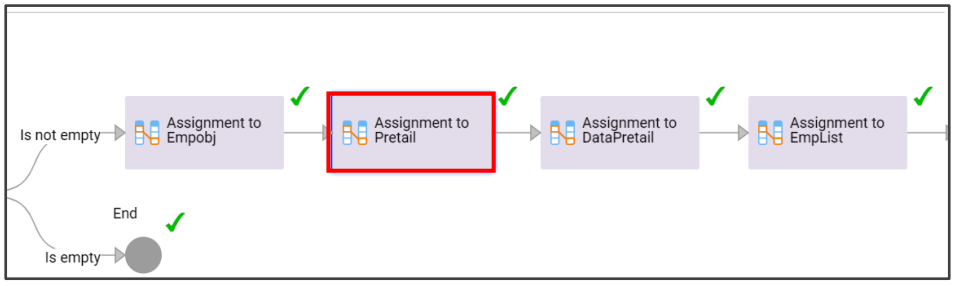
129. From the Design selection, click **Assignment to EmpObj**.



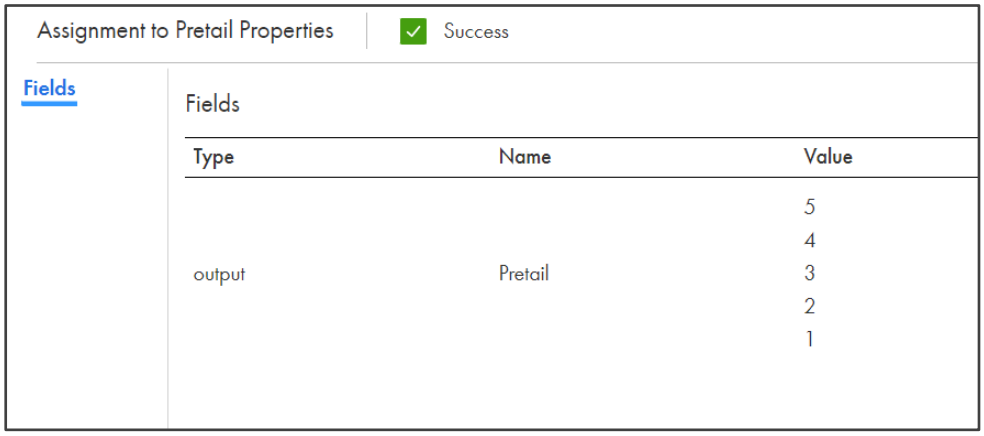
130. In the Assignment to EmpObj Properties section, observe the output. Only the last employee in the list is displayed.



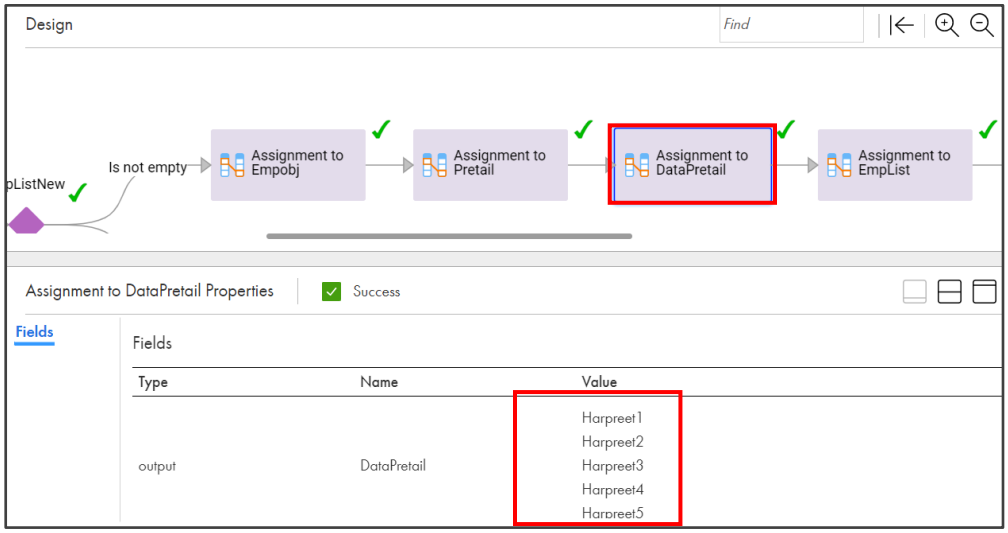
131. From the Design selection, click **Assignment to Pretail**.



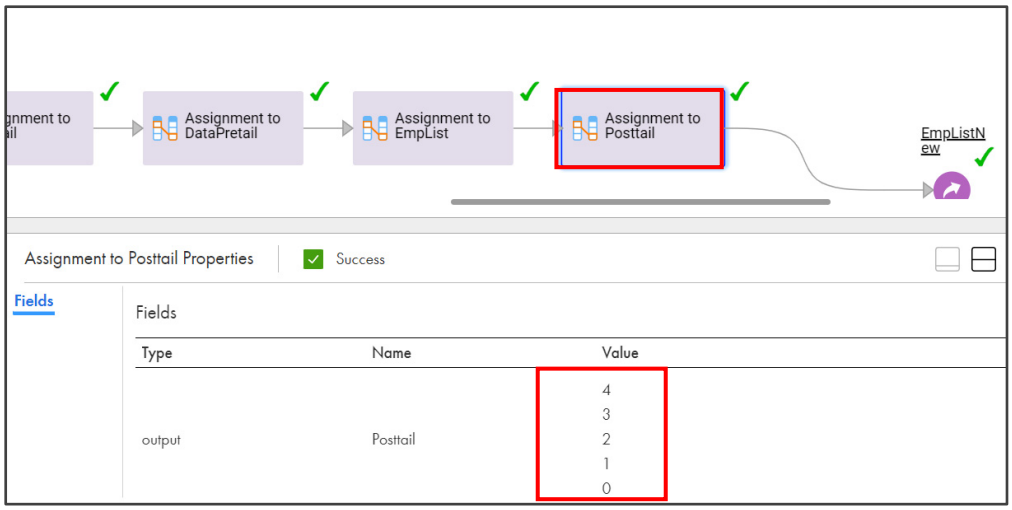
132. In the Assignment to Pretail Properties section, observe the output. You can see the total counts of the employees present in the array.



133. From the Design selection, click **Assignment to DataPretail.**
134. In the Assignment to DataPretail Properties section, observe the output. You can see the list of total employees present in the array.



135. From the Design selection, click **Assignment to Posttail.**
136. In the Assignment to Posttail Properties section, observe the output. The order of elements in the tail is displayed.



This concludes the lab.

Module 10: Adding Web Service to a Process
Lab 10-1: Invoke a Synchronous Web Service Call

Overview:
In this lab, you will create two processes. One process runs on the selected Secure Agent and the other process is executed on the cloud server. Using the Service URL generated by the process that executes on the cloud server, you will run the process using a synchronous web service call. The process objects and connections utilized in this process are already created in the previous modules.

Note: This lab is the last part of a set of three labs. Earlier, you created process objects (Lab 4-1: Create Process Objects) and then created a JDBC connection (Lab 7-2: Create a JDBC Connection). Using the assets created in the previous labs, you will create a process to obtain the exchange rates for the specified currencies.

Objectives:
- Create ExchangeRate Process
- Create ExchangeRateService Process
- Execute the ExchangeRateService process using a Rest Client

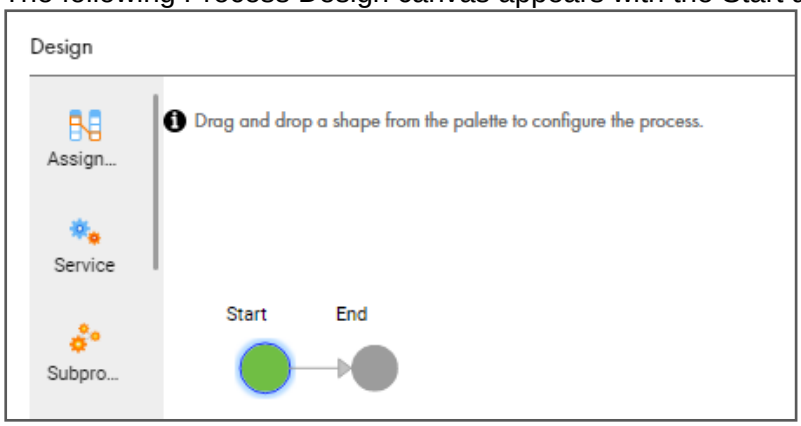
Duration:
30 Minutes

Tasks:

Note: The screen on your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

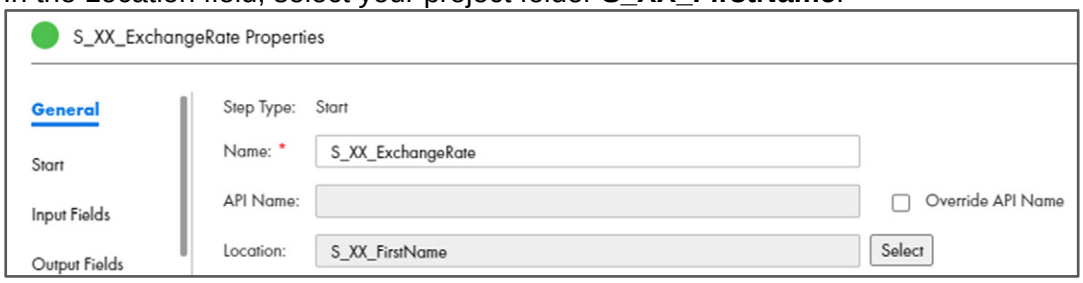
Create a new process

1. Create a new **Process**.
   The following Process Design canvas appears with the Start and End steps.

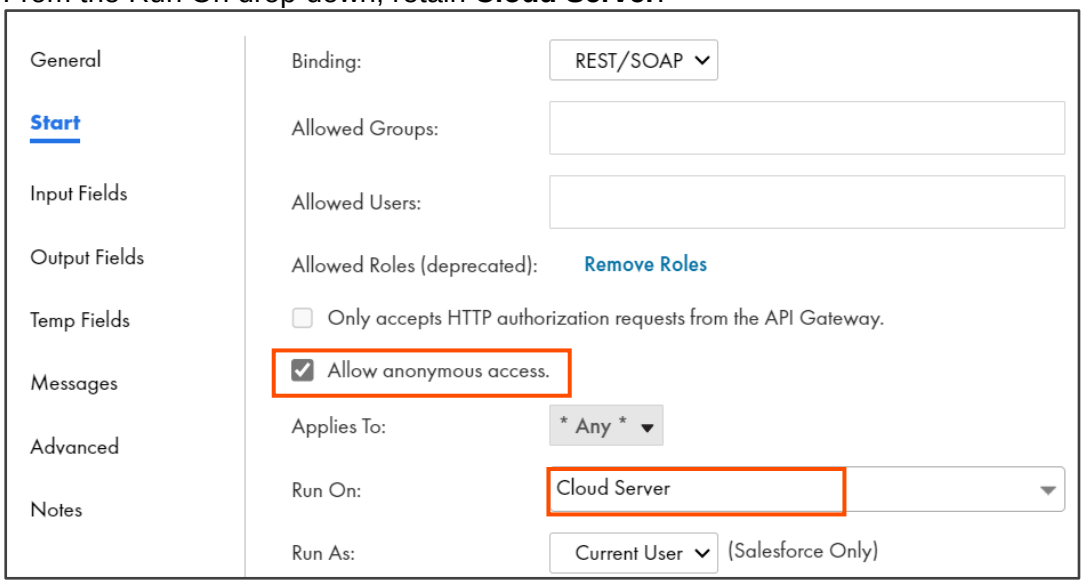


2. From the process canvas, select the **Start** step.
3. In the General tab, enter the name as **S_XX_ExchangeRate**.
4. Skip the API Name option.

5. In the Location field, select your project folder **S_XX_FirstName**.



6. From the properties pane, select **Start**.
7. Select the **Allow anonymous access** option.
8. From the Run On drop-down, retain **Cloud Server**.



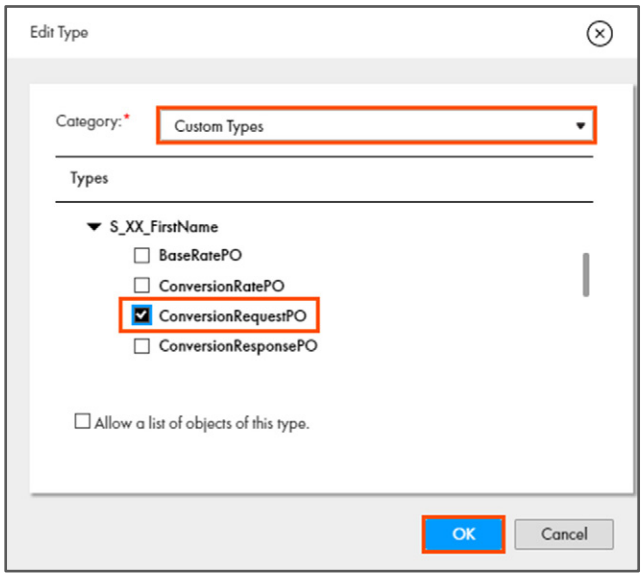
Note: Allowed Users and Allowed Groups textboxes remain blank as you have selected the Allow anonymous access checkbox.

9. From the properties pane, select **Input Fields**.
10. Add a new input field as shown in the table below:

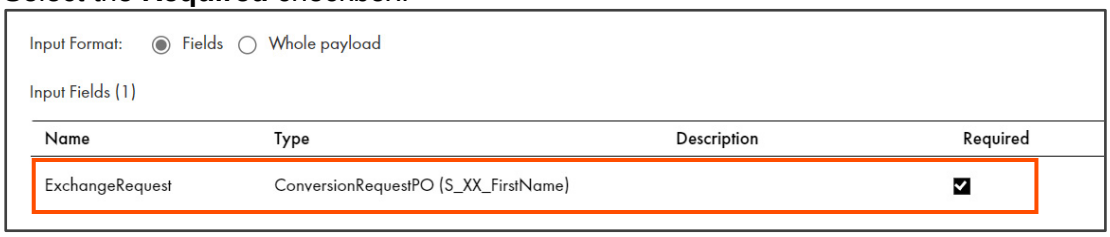
Name	Type
ExchangeRequest	More Types > Custom Types

11. Expand your project folder and select **ConversionRequestPO**.

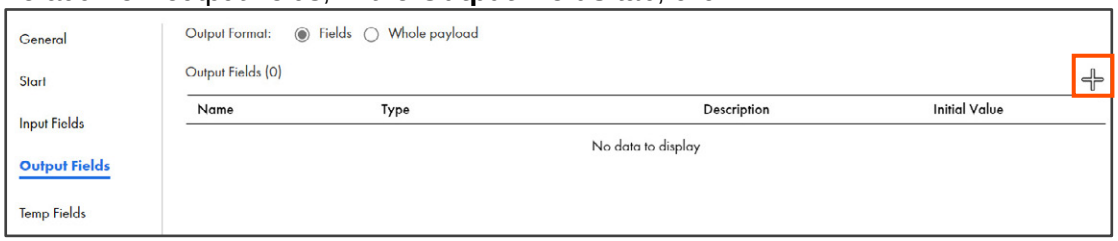
12. Click **OK**.



13. Select the **Required** checkbox.



14. To add new output fields, in the **Output Fields** tab, click .

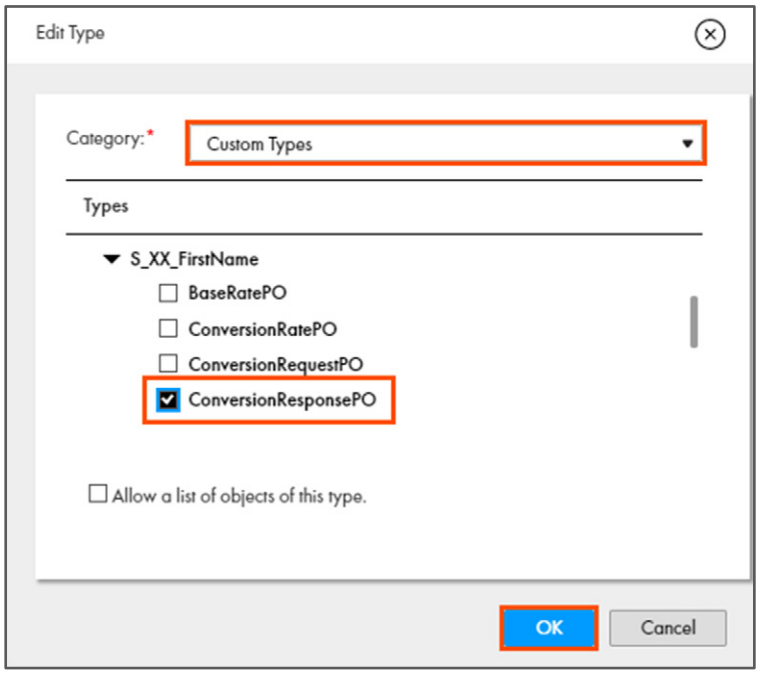


15. Create the output fields, as shown in the table below:

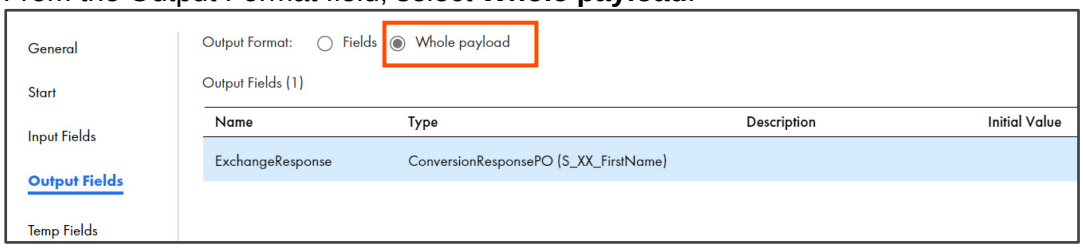
Name	Type
ExchangeResponse	More Types > Custom Types

16. Expand your project folder and select **ConversionResponsePO**.

17. Click **OK**.



18. From the Output Format field, select **Whole payload**.



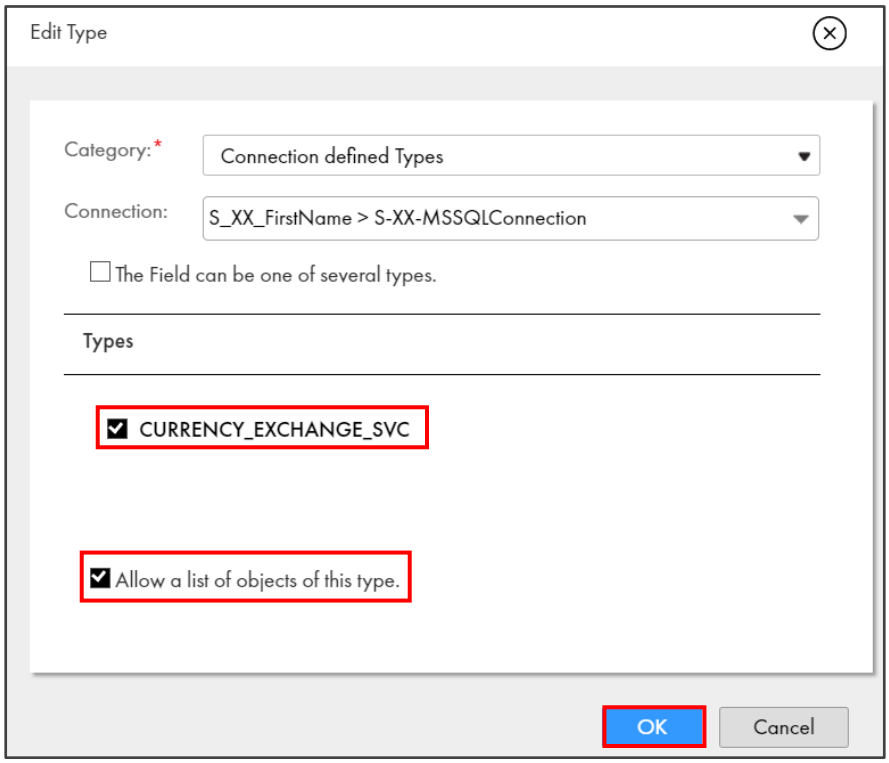
19. From the properties pane, select **Temp Fields**.
20. Add a new **Temp** field as shown in the table below:

Name	Type
ListofCurrencyExchangeRates	More Types > Connection defined Types

Note: This temp field is used to fetch the list of all currency exchange rates from the MS SQL table CURRENCY_EXCHANGE_SVC.

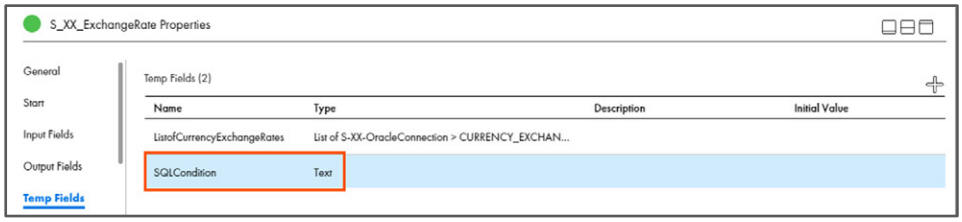
21. From the Connection drop-down, expand your project folder, and select **S-XX-MSSQLConnection**.
22. Select **CURRENCY_EXCHANGE_SVC** and **Allow a list of objects of this type** checkboxes.

23. Click **OK**.



24. Create another temp field, as shown in the table below:

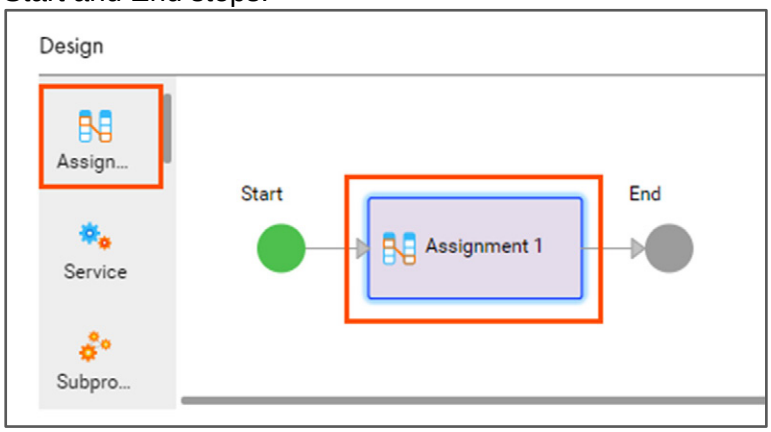
Name	Type
SQLCondition	Text



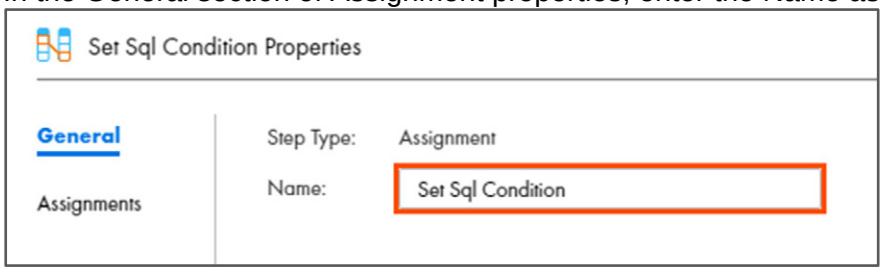
Note: This temp field is created to store the SQL condition that converts all the currency rates.

25. **Save** the process.

26. From the Design palette, drag and drop an **Assignment** step on the link between the Start and End steps.

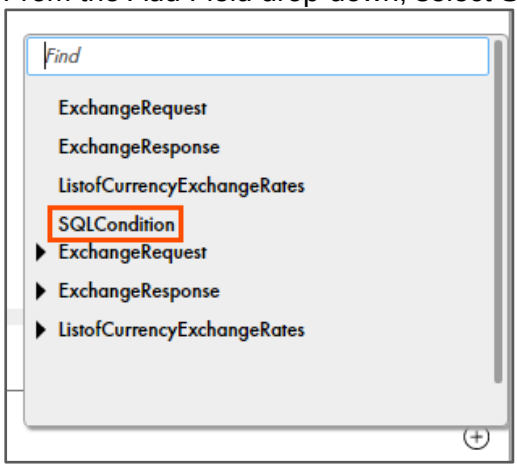


27. To configure the step, from the process canvas, select the **Assignment** step.
28. In the General section of Assignment properties, enter the Name as **Set Sql Condition**.



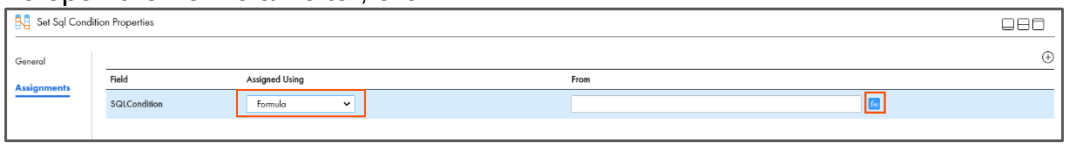
29. From the properties pane, select **Assignments**.
30. Click plus (**+**) icon.

31. From the Add Field drop-down, select **SQLCondition**.



32. From the Assigned Using drop-down, select **Formula**.

33. To open the Formula Editor, click .



Note: A new Expression Editor for the SQLCondition window appears.

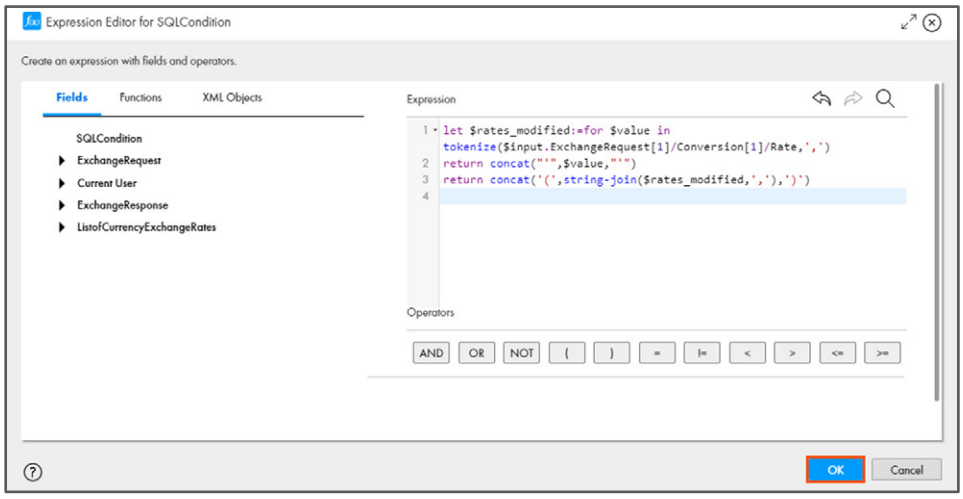
34. In the Expression field, enter the following expression:

    let $rates_modified:=for $value in
    tokenize($input.ExchangeRequest[1]/Conversion[1]/Rate,',')
    return concat("'",$value,"'")
    return concat('(',string-join($rates_modified,','),')')

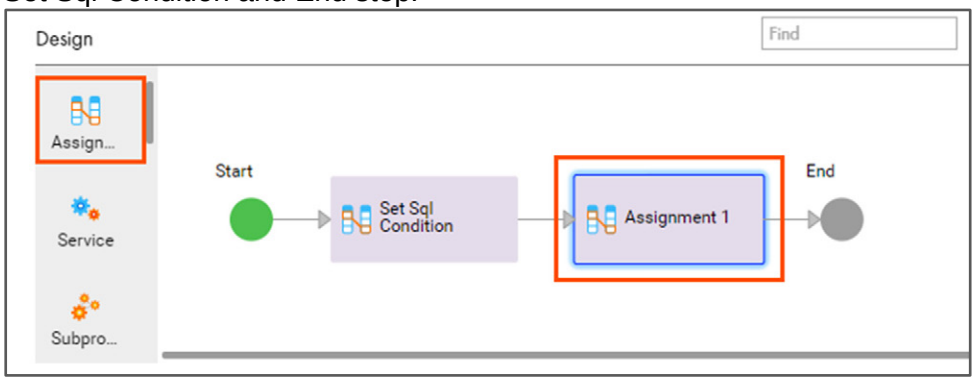
    OR

    From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **21_LabGuide_CAI_SynchronousWebServiceCall_10-1**. Copy the command mentioned under **Step A** and paste it into the Expression field.

35. Click **OK**.



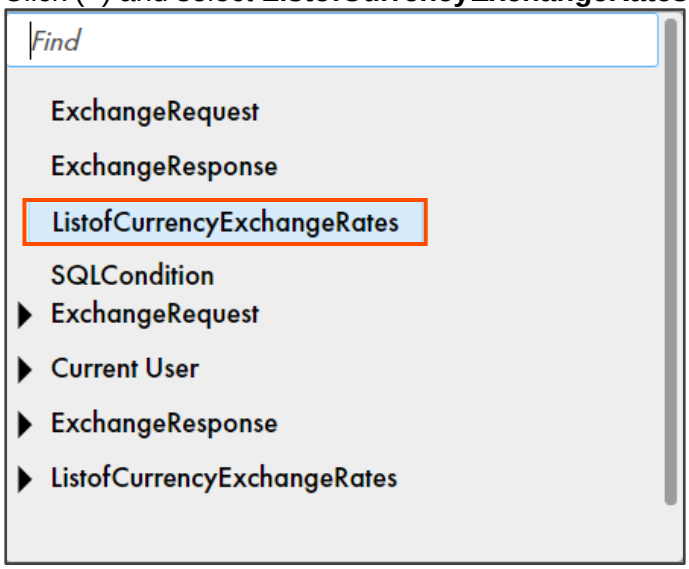
36. From the Design palette, drag and drop an **Assignment** step on the link between the Set Sql Condition and End step.



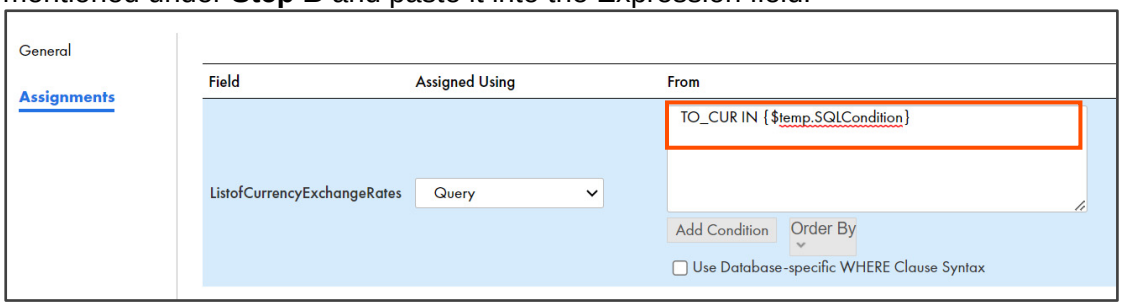
37. To configure the step, from the process canvas, select the **Assignment** step.

38. In the General section of Assignment properties, enter the Name as **Set Sql Condition_query**.

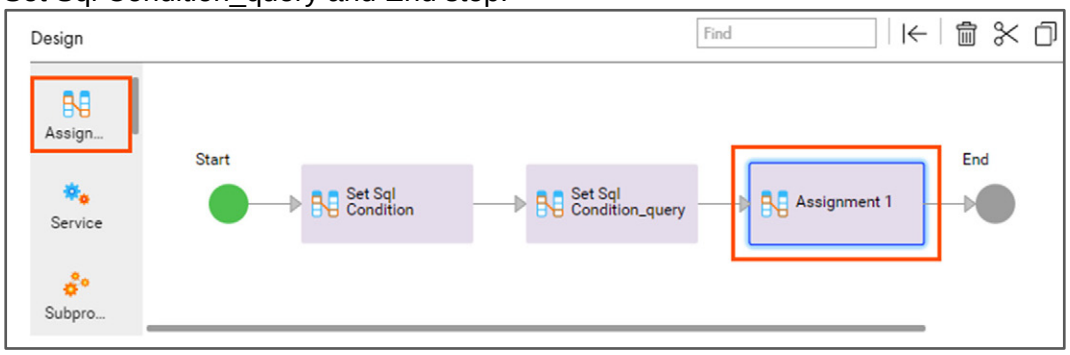
39. From the properties pane, select **Assignments**.
40. Click (**+**) and select **ListofCurrencyExchangeRates**.



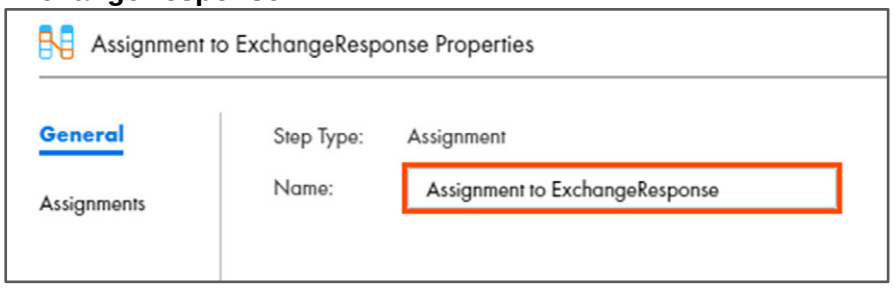
41. From the Assigned Using drop-down, select **Query**.
42. In the Query field, enter the following formula:
    TO_CUR IN {$temp.SQLCondition}
    OR
    From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **21_LabGuide_CAI_SynchronousWebServiceCall_10-1**. Copy the command mentioned under **Step B** and paste it into the Expression field.



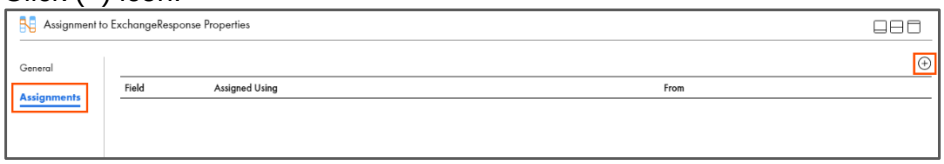
43. From the Design palette, drag and drop an **Assignment** step on the link between the Set Sql Condition_query and End step.



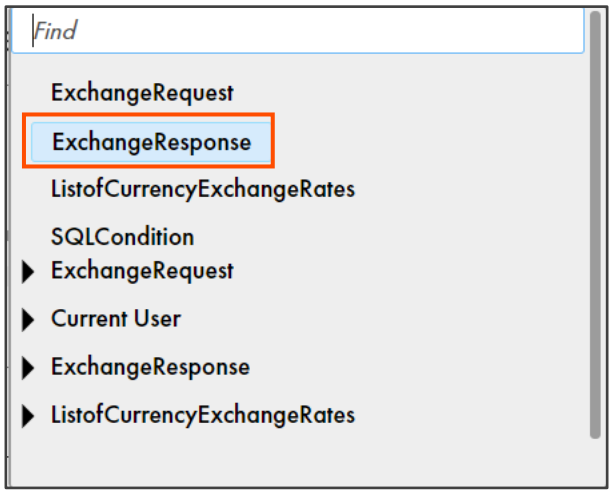
44. To configure the step, from the process canvas, select the **Assignment** step.
45. In the General section of Assignment properties, enter the Name as **Assignment to ExchangeResponse**.



46. From the properties pane, select **Assignments**.
47. Click (**+**) icon.

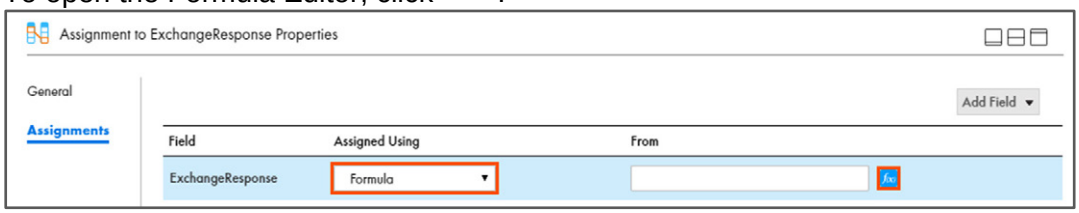


48. From the list, select **ExchangeResponse**.



49. From the Assigned Using drop-down, select **Formula**.

50. To open the Formula Editor, click .



Note: A new Expression Editor for the ExchangeResponse window appears.

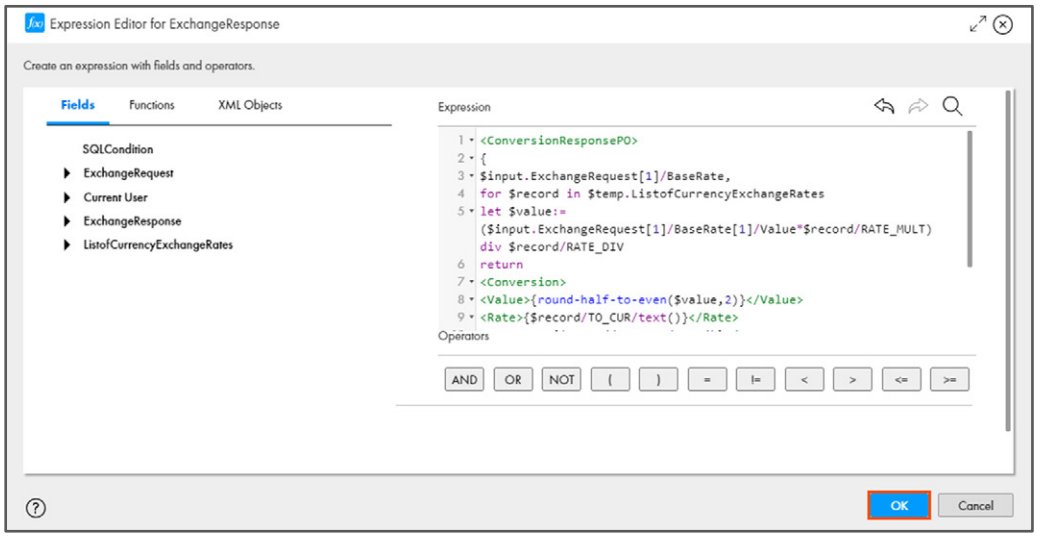
51. In the Expression field, enter the following expression:

```
<ConversionResponsePO>
{
$input.ExchangeRequest[1]/BaseRate,
for $record in $temp.ListofCurrencyExchangeRates
let $value:=($input.ExchangeRequest[1]/BaseRate[1]/Value*$record/RATE_MULT)
    div $record/RATE_DIV
return
<Conversion>
<Value>{round-half-to-even($value,2)}</Value>
<Rate>{$record/TO_CUR/text()}</Rate>
<RateType>{$record/RT_TYPE/text()}</RateType>
</Conversion>
}
</ConversionResponsePO>
```

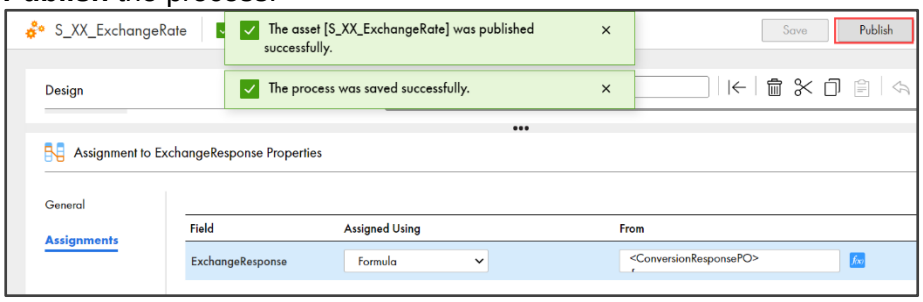
OR

From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **21_LabGuide_CAI_SynchronousWebServiceCall_10-1**. Copy the command mentioned under **Step C** and paste it into the Expression field.

52. Click **OK**.



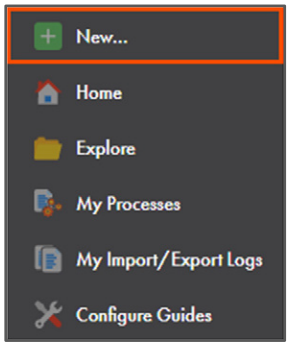
53. **Save** the process.
54. **Publish** the process.

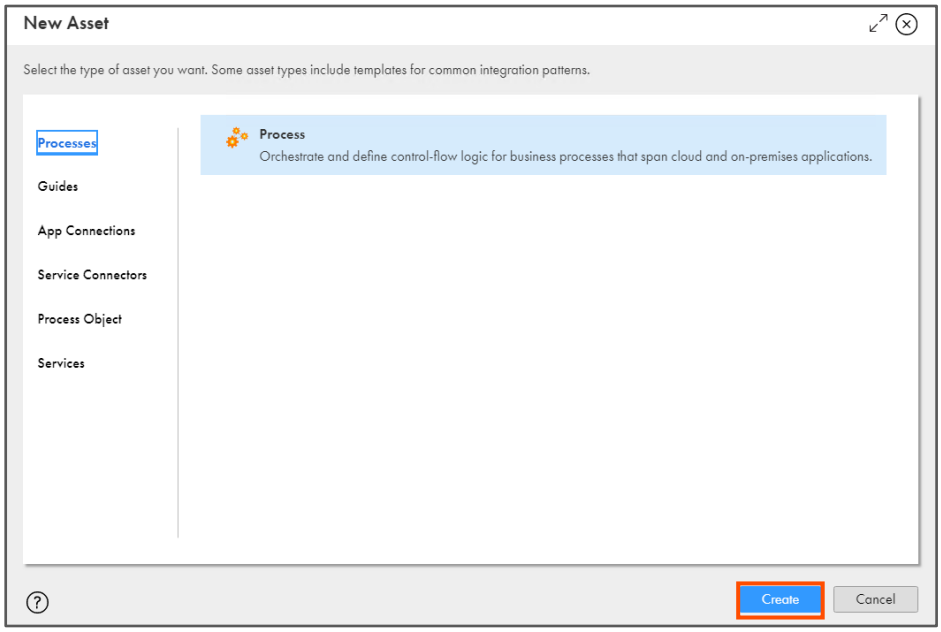


You will get a notification that '**The asset [S-XX-ExchangeRate] was published successfully.**'.

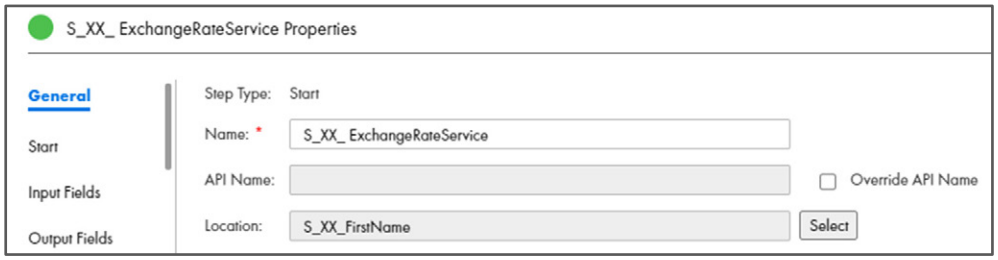
Configure ExchangeRateService Process Properties:
This is the parent process that runs on the Cloud Server and has the ExchangeRate process embedded in it. This process will be exposed as both REST and SOAP webservices for third-party access.

55. Create a new **Process**.

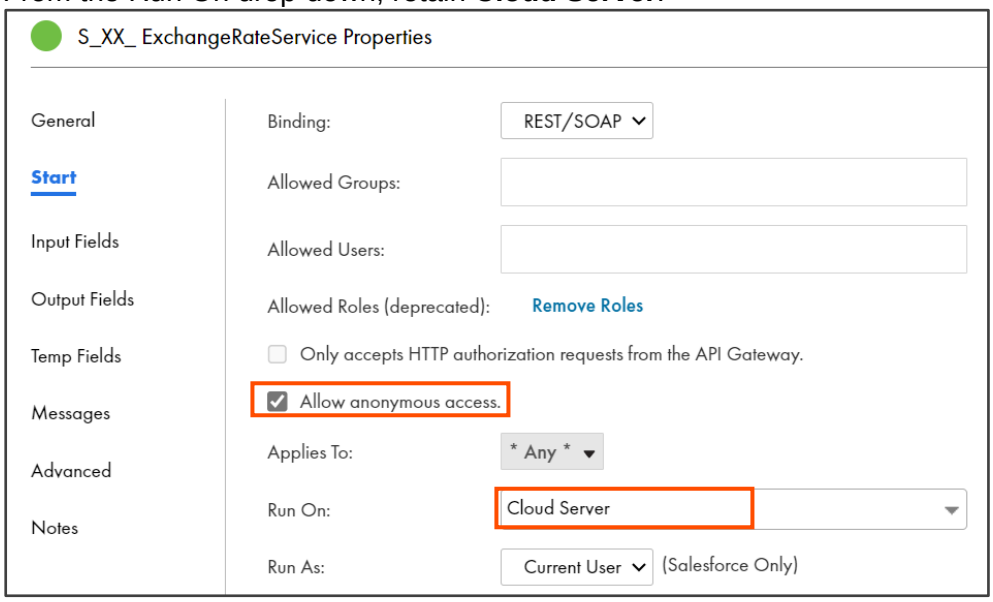




56. From the process canvas, select the **Start** step.
57. In the General tab, enter the name as **S_XX_ ExchangeRateService**.
58. Skip the API Name option.
59. In the Location field, select your project folder **S_XX_FirstName**.



60. From the properties pane, select **Start**.
61. Select the **Allow anonymous access** option.
62. From the Run On drop-down, retain **Cloud Server**.



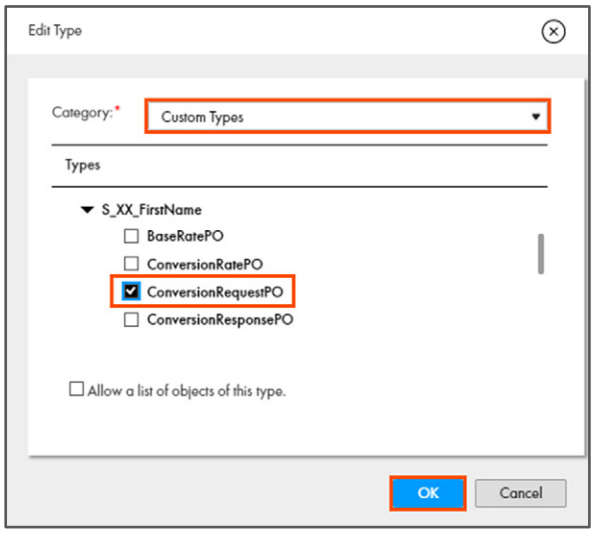
Note: Allowed Users and Allowed Groups textboxes remain blank as you have selected the Allow anonymous access checkbox.

63. From the properties pane, select **Input Fields**.
64. To add a new input field, click .
65. Create an input field, as shown in the table below:

Name	Type
ExchangeRequest	More Types > Custom Types

66. Expand your project folder and select **ConversionRequestPO**.

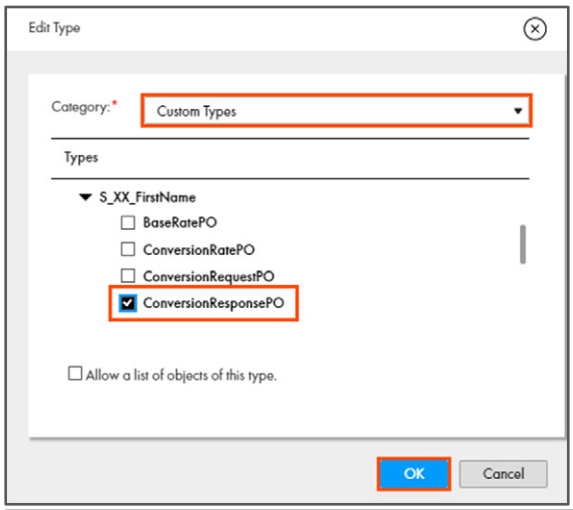
67. Click **OK**.

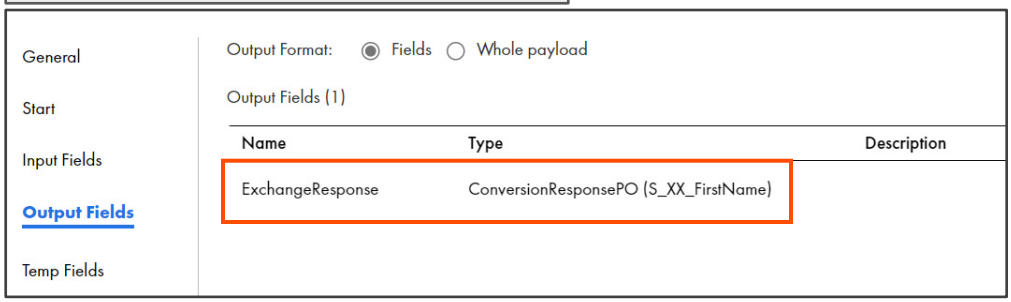


68. From the properties pane, select **Output Fields**.
69. To add a new output field, click .
70. Create an output field, as shown in the table below:

Name	Type
ExchangeResponse	More Types > Custom Types

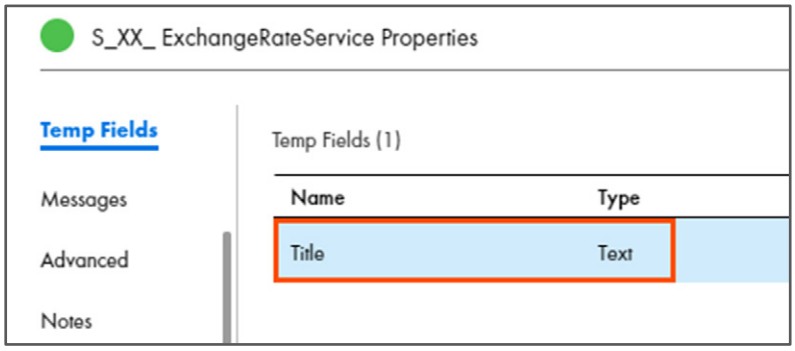
71. Expand your project folder and select **ConversionResponsePO**.
72. Click **OK**.



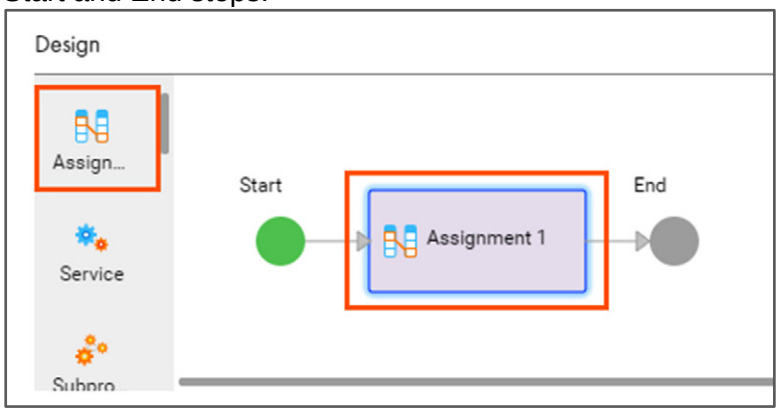


73. From the properties pane, select **Temp Fields**.
74. To add a new temporary field, click .
75. Create a temp field, as shown in the table below:

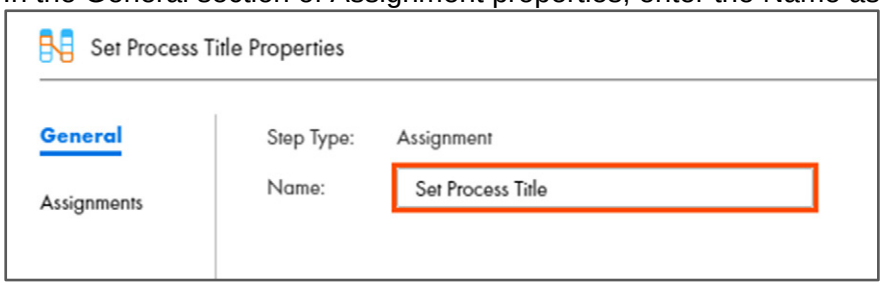
Name	Type
Title	Text



76. From the Design palette, drag and drop an **Assignment** step on the link between the Start and End steps.

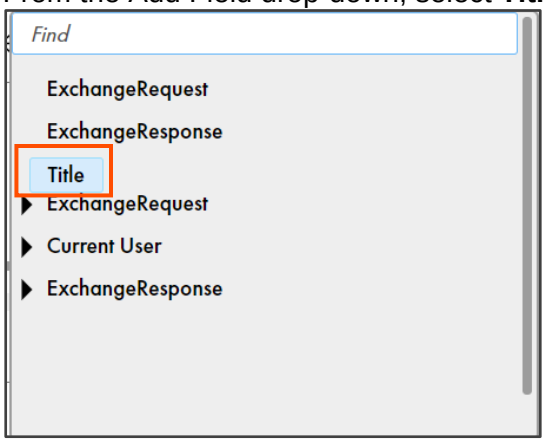


77. To configure the step, from the process canvas, select the **Assignment** step.
78. In the General section of Assignment properties, enter the Name as **Set Process Title**.

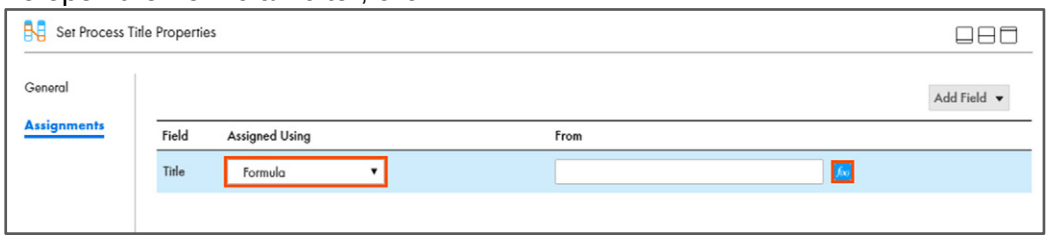


79. From the properties pane, select **Assignments**.

80. From the Add Field drop-down, select **Title**.



81. From the Assigned Using drop-down, select **Formula**.
82. To open the Formula Editor, click .

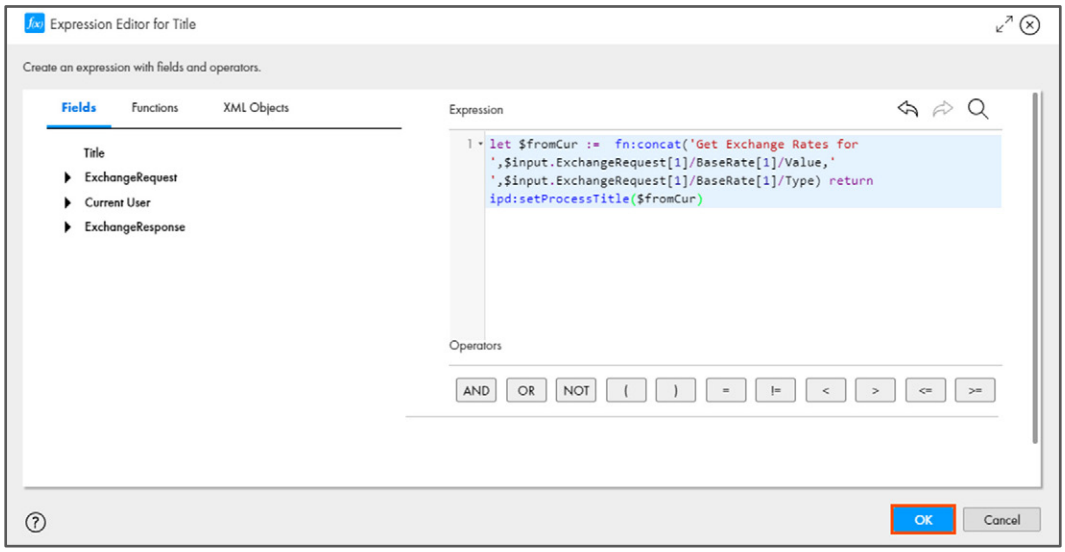


83. In the Expression field, enter the following formula:
    let $fromCur := fn:concat('Get Exchange Rates for ',$input.ExchangeRequest[1]/BaseRate[1]/Value,' ',$input.ExchangeRequest[1]/BaseRate[1]/Type) return ipd:setProcessTitle($fromCur)

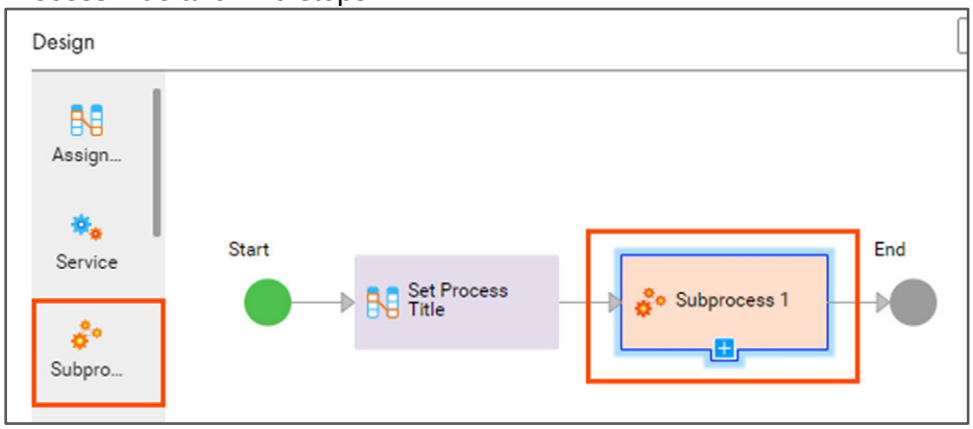
    OR

    From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **21_LabGuide_CAI_SynchronousWebServiceCall_10-1**. Copy the command mentioned under **Step D** and paste it into the Expression field.

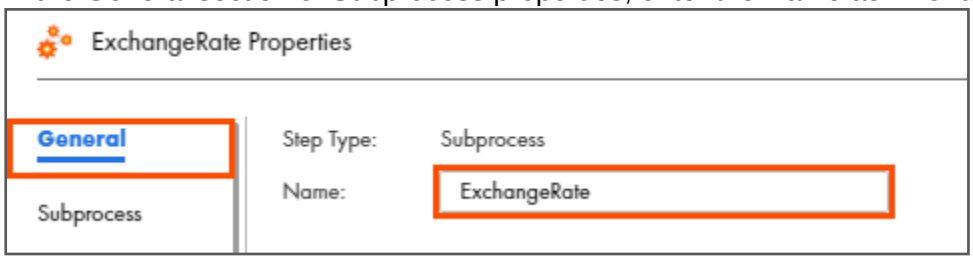
84. Click **OK**.



85. From the Design palette, drag and drop a **Subprocess** step on the link between the Set Process Title and End steps.

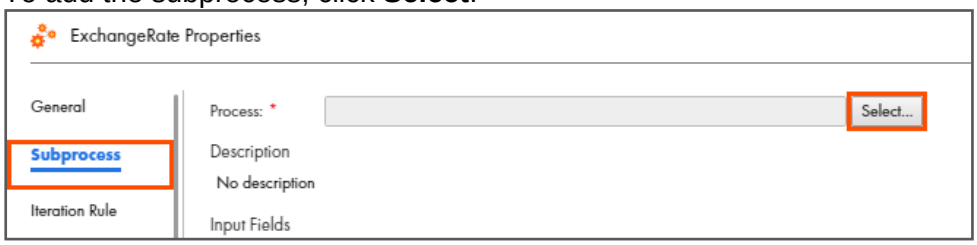


86. To configure the step, from the process canvas, select the **Subprocess** step.
87. In the General section of Subprocess properties, enter the Name as **ExchangeRate**.



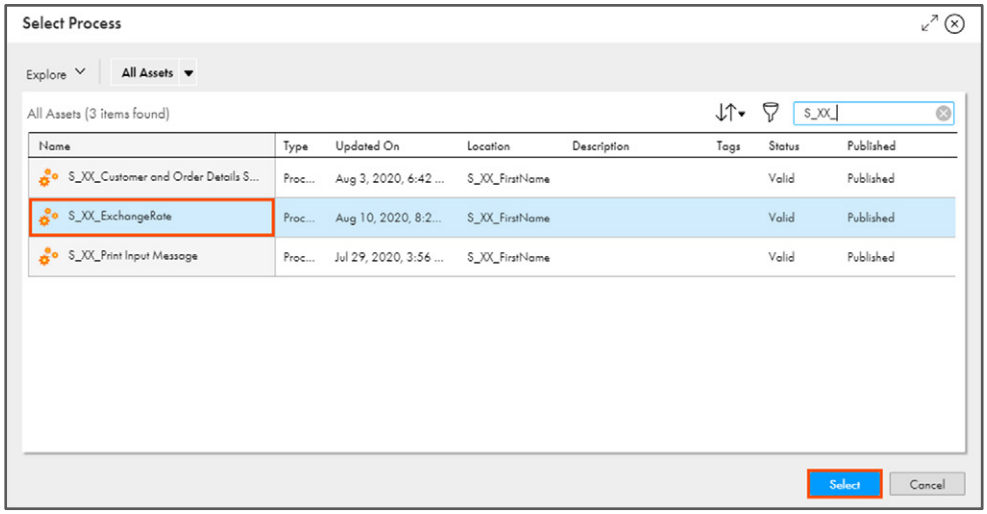
88. From the properties pane, select **Subprocess**.

89. To add the subprocess, click **Select**.

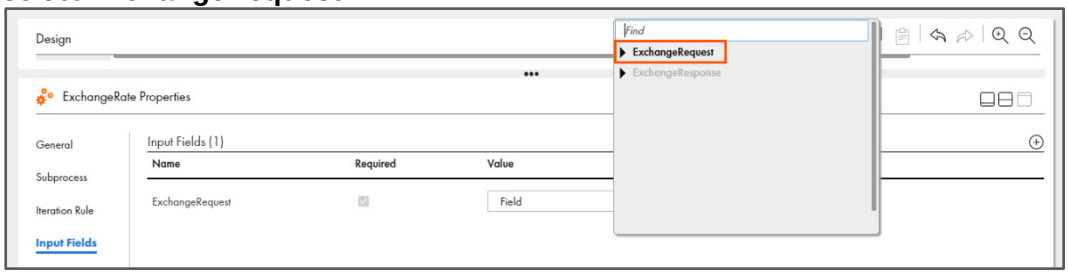


Note: A new Select Process window appears.

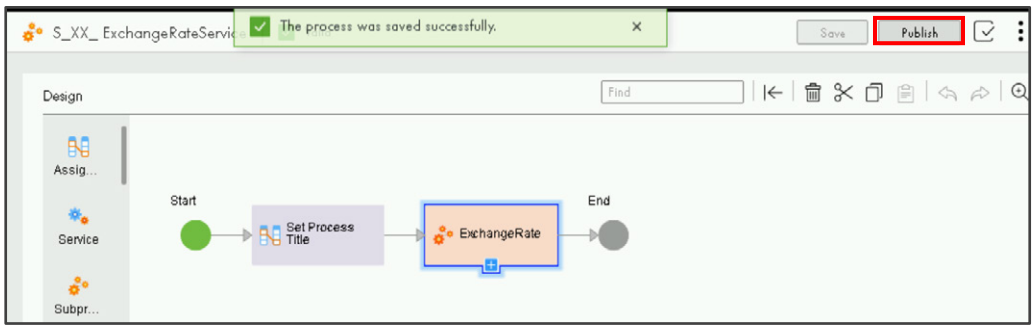
90. From the list, select the **S_XX_ExchangeRate** process.
Recommended: To quickly locate the process, use the **Search** field.
91. Click **Select**.



92. From the properties pane, select **Input Fields**.
93. From the ExchangeRequest field, select the **Click To Select Field** drop-down, and select **ExchangeRequest**.



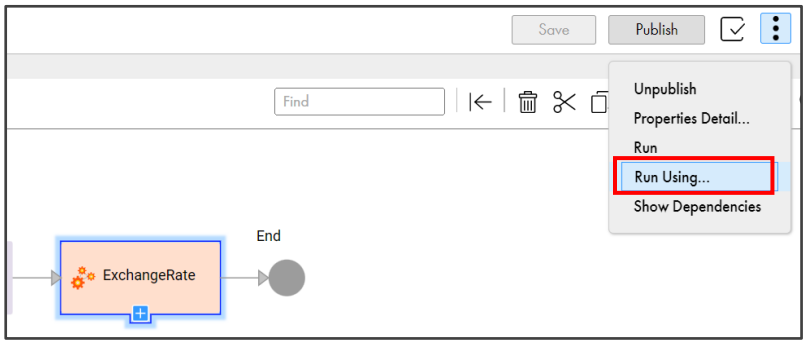
94. **Save** and **Publish** the process.



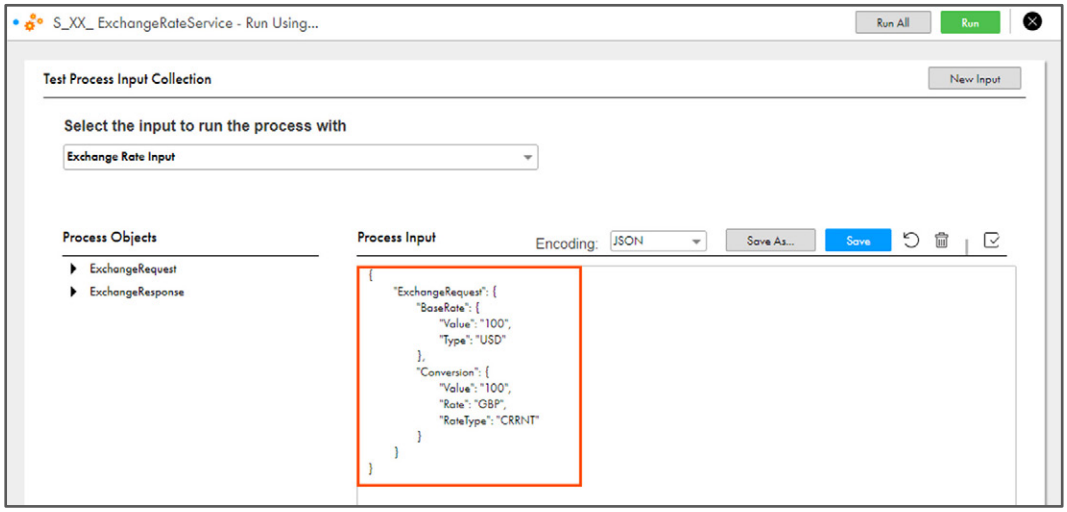
Execute the Process

You will now see how to run the process in two ways. Firstly, you will run it from the interface.

95. From the Actions menu, select **Run Using...**.

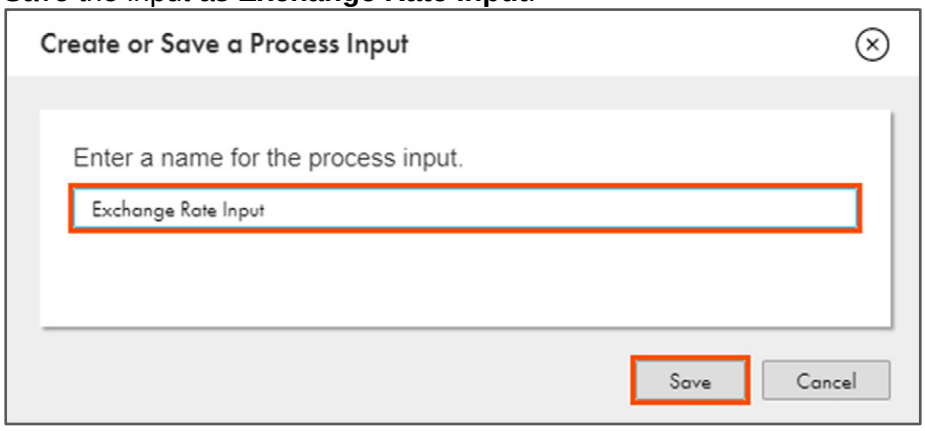


96. In the Process Input section, for the BaseRate object, add the Value as **100** and Type as **USD**.
97. For the Conversion object, add the Value as **100**, the Rate as **GBP**, and the RateType as **CRRNT**.

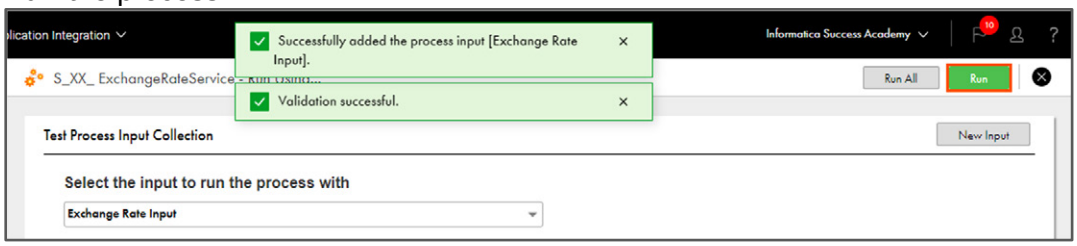


Note: This enables you to find the conversion rate of the Great Britain Pound against the US Dollar. You can replace GBP with AUD, CAD, EUR, and so on to obtain conversion rates for other currencies.

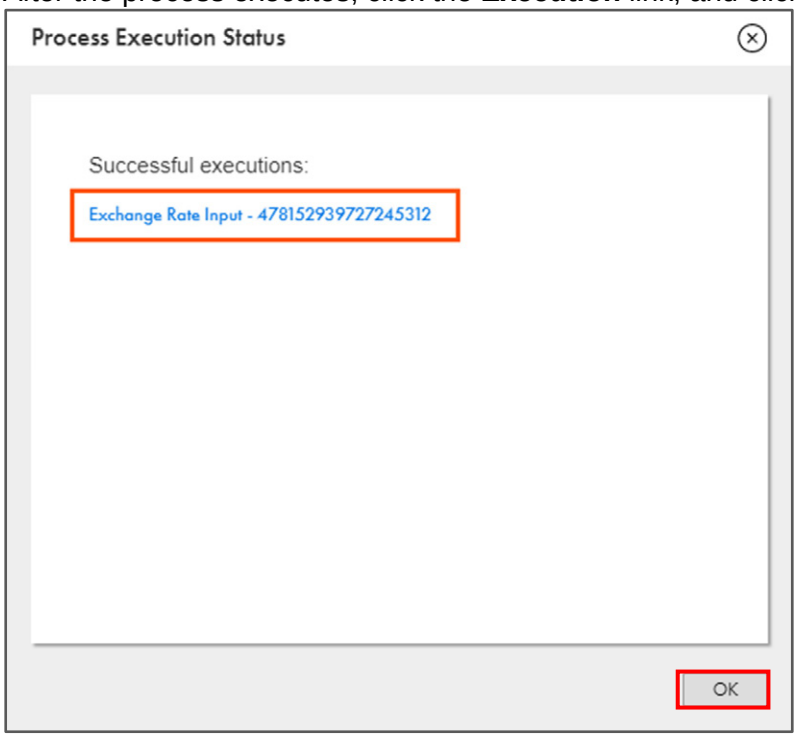
98. Save the input as **Exchange Rate Input**.



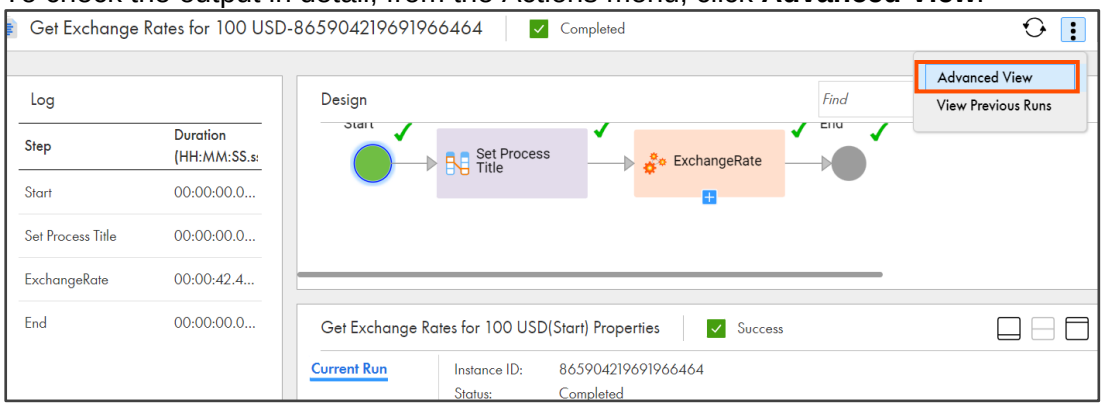
99. **Run** the process.



100. After the process executes, click the **Execution** link, and click **OK**.



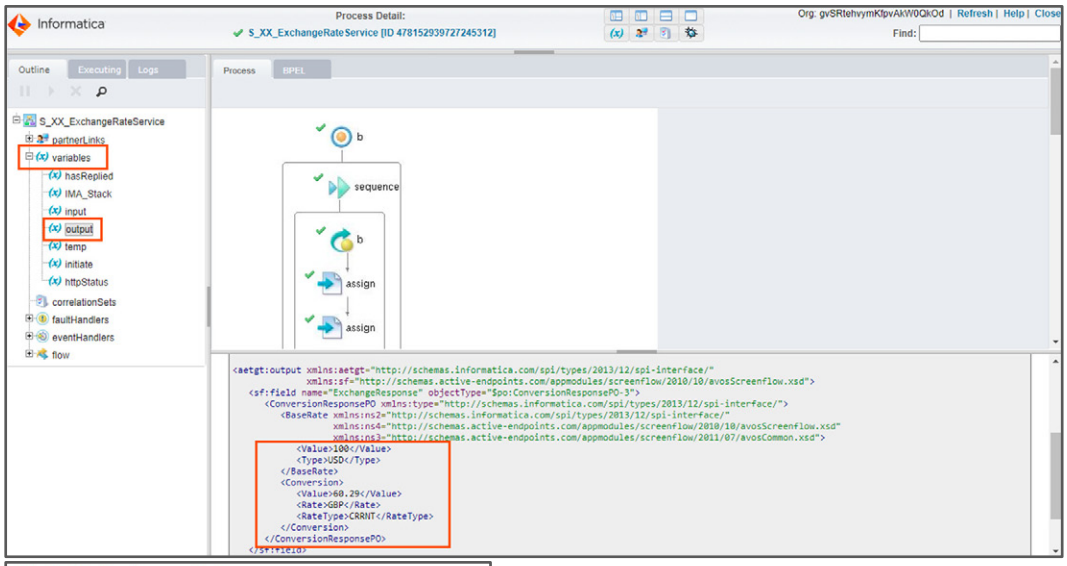
101. To check the output in detail, from the Actions menu, click **Advanced View**.

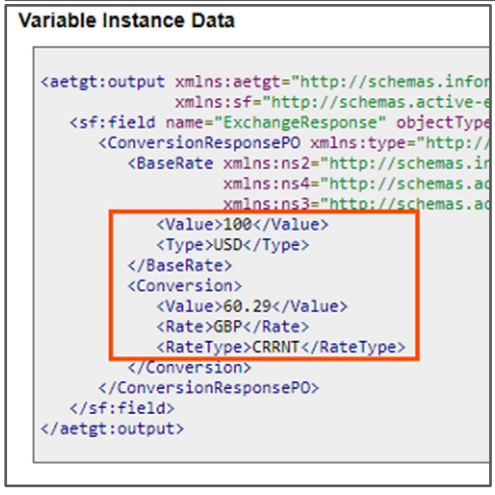


This directs you to Application Integration Console.

102. From the variables section, select **output**.

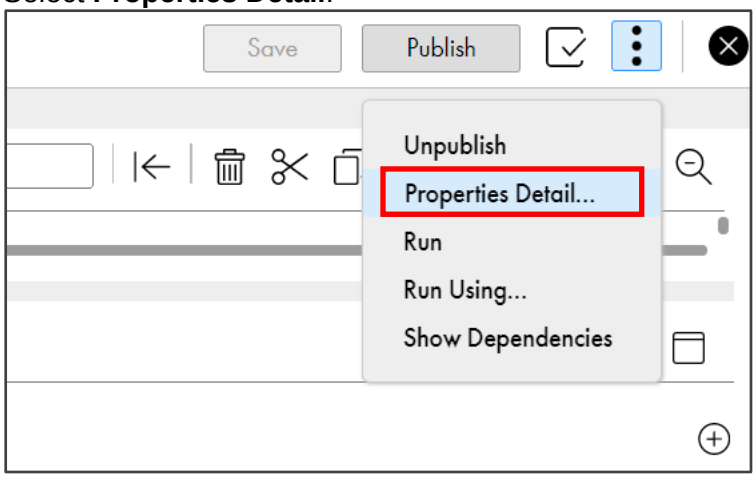
103. In the Variable Instance Data section, observe that you get the conversion value for **GBP**.



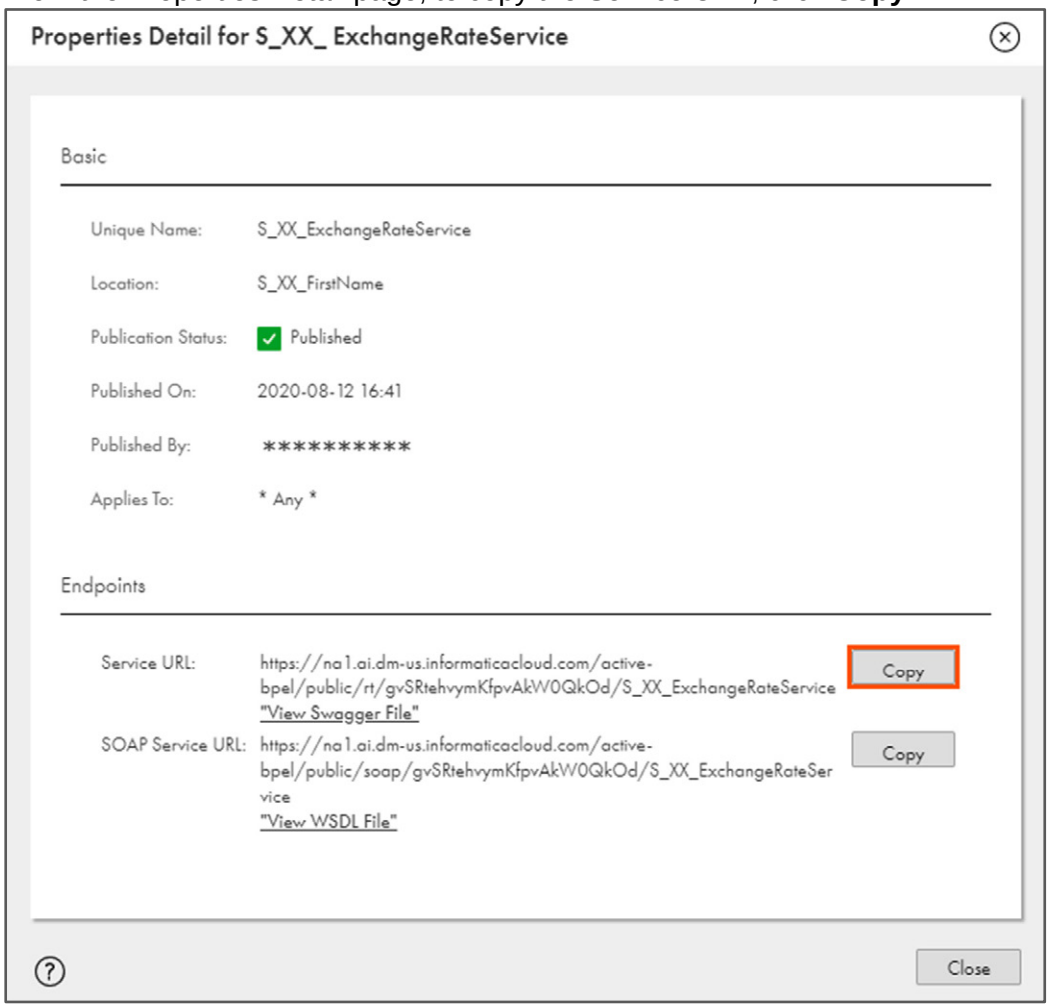


You can also execute the process from a third-party application, such as Postman. You will use the Service URL to execute it.

104. In CAI, navigate back to the S_XX_ExchangeRateService process, and click .
105. Select **Properties Detail**.

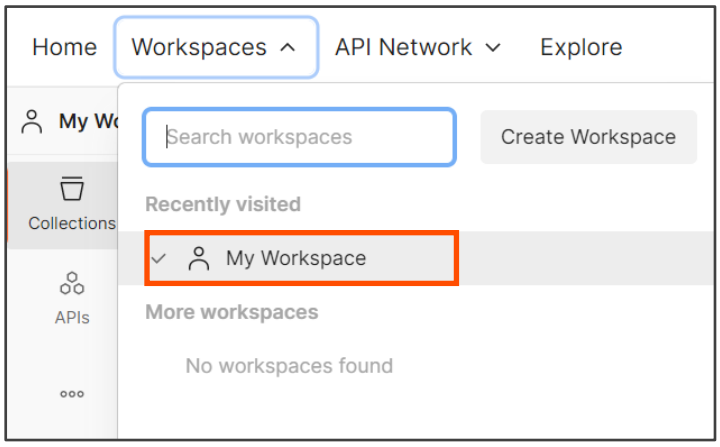


106. From the Properties Detail page, to copy the Service URL, click **Copy**.

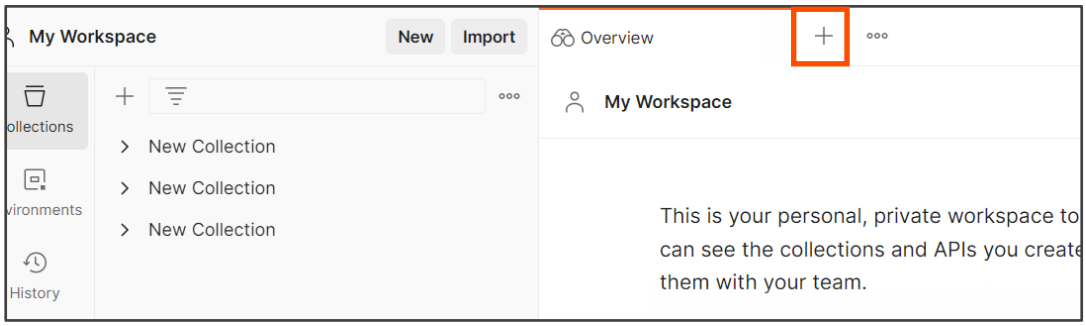


107. Copy the Service URL on a notepad file.

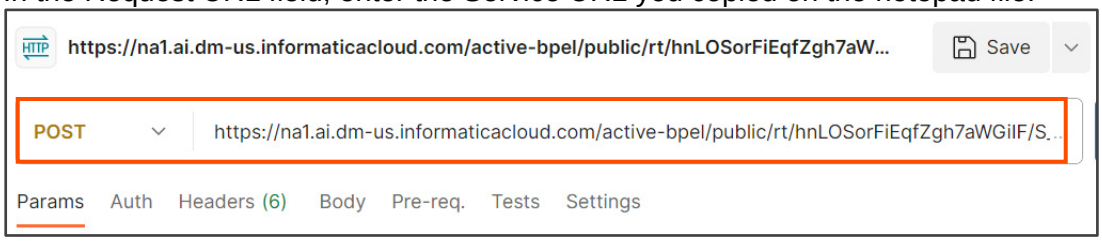
108. Open the web browser and enter the following link to open Postman: https://identity.getpostman.com/
109. Login to Postman using your credentials.
110. From the Workplace drop-down, select **My Workspace**.



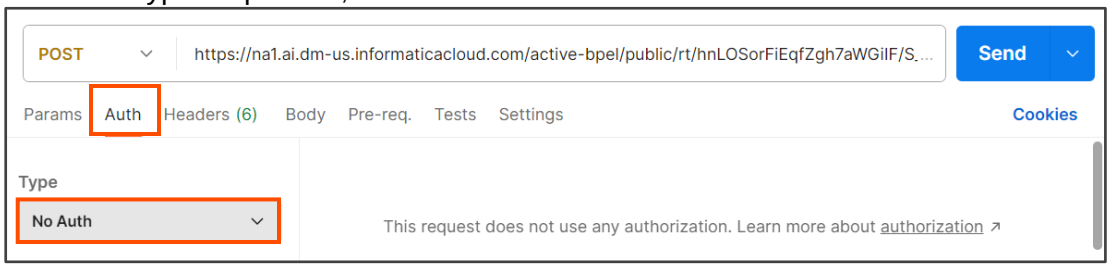
111. Click the plus (**+**) icon.



112. In the Workspace section, from the drop-down, select **POST**.
113. In the Request URL field, enter the Service URL you copied on the notepad file.

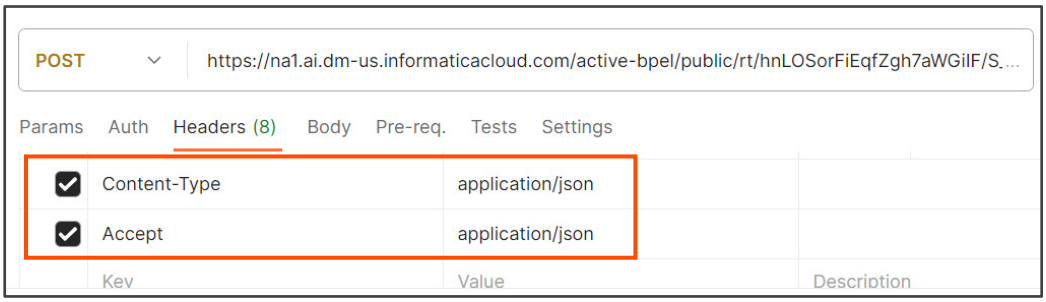


114. Select the **Auth** tab.
115. From the Type drop-down, select **No Auth**.

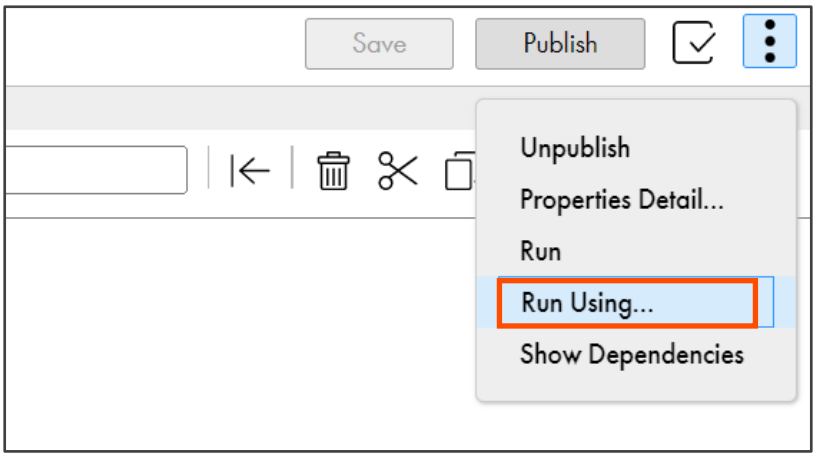


116. Select the **Headers** tab.
117. Enter Key and Value as shown in the table below:

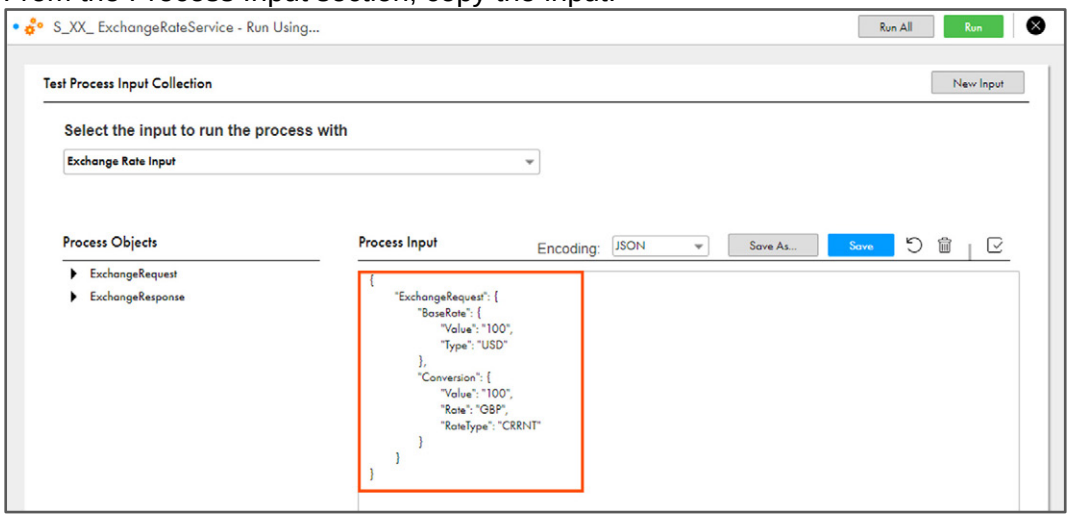
Key	Value
Content-Type	application/json
Accept	application/json



118. Select the **Body** tab and select **raw**.
119. In Cloud Application Integration, in S_XX_ ExchangeRateService process, from the **Actions** Menu, select **Run Using…**.



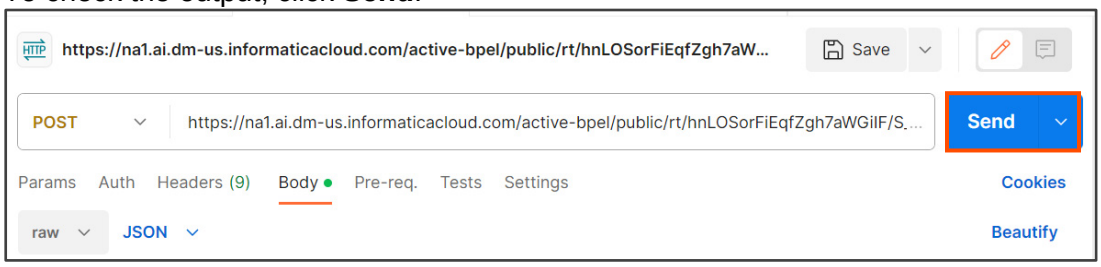
120. From the Process Input section, copy the input.



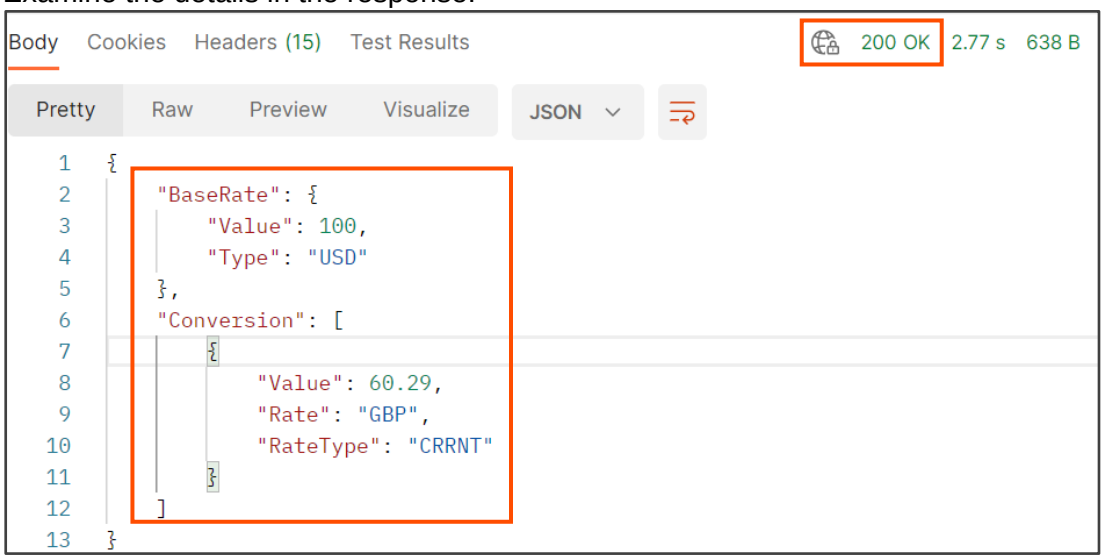
Note: While copying, make sure that input values are available.

121. Navigate back to Postman.
122. In the Body tab, select **raw** and paste the input.

123. To check the output, click **Send**.



124. Scroll down and verify that the status of the response is **200 OK**.
125. Examine the details in the response.



126. **Close** the Postman application.

In this way, you can add a web service to a process and execute it from the interface or a third-party application.

This concludes the lab.

Module 10: Adding Web Service to a Process
Lab 10-2: Invoke an Asynchronous Web Service Call

Overview:
In this lab, you will create a process that incorporates the asynchronous web service call and invoke the process.

Objectives:
- Install and setup a JMS Provider
- Configure a JMS Service in CAI
- Create a process with an asynchronous web service call
- Run the process from a third-party application
- View the contents of the queue after the message has been written

Duration:
40 Minutes

Tasks:

IMPORTANT: In this lab, the instructor needs to Install and set up a JMS provider on the Server Agent (Please check **Steps 1 to 9**). Once the JMS provider is installed and set up, learners can configure the JMS service in CAI using their IICS credentials provided by the instructor.

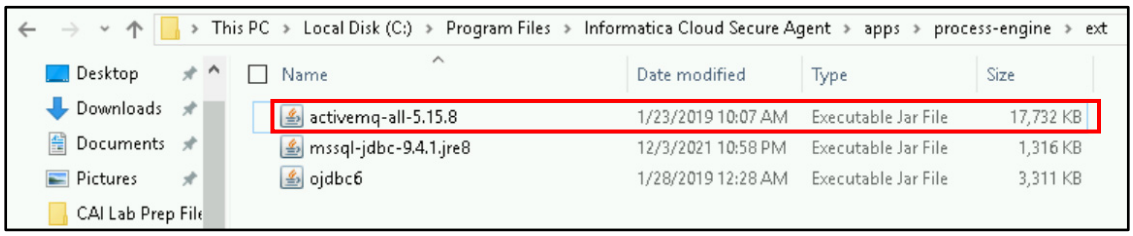
By the end of this lab (Steps 117 to 119), there is a Demo that needs to be performed by the instructor.

Note: The screen on your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

STEPS FOR INSTRUCTOR

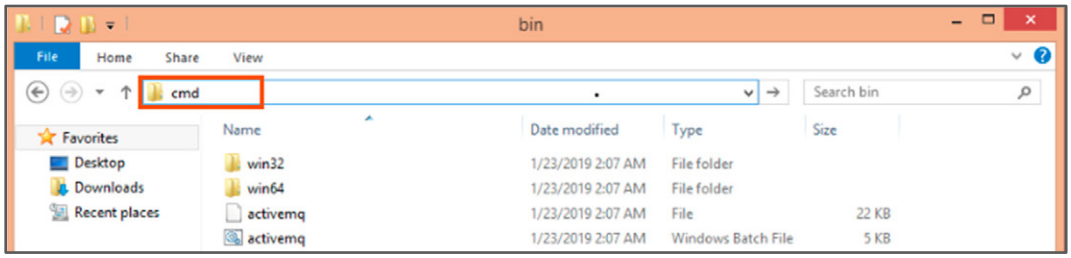
Install and set up a JMS provider:

1. On the desktop, from the **CAI Lab Prep Files** folder, open the **apache-activemq-5.15.8** folder.
2. Copy the jar file **activemq-all-5.15.8** and paste it in the **C:\Program Files\Informatica Cloud Secure Agent\apps\process-engine\ext** folder of the Secure Agent installation directory.



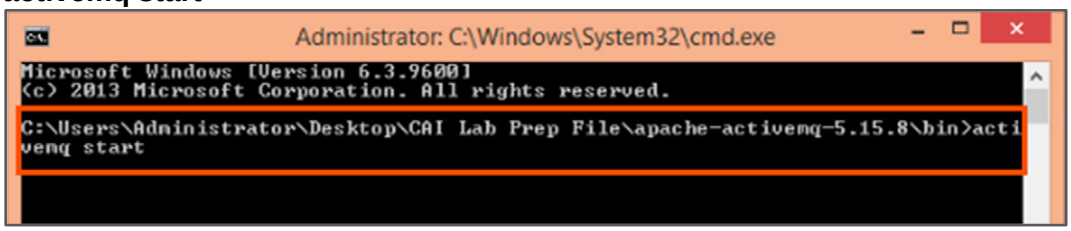
3. After you paste the jar file in the Secure Agent Installation directory, you must **Restart** the Secure Agent and wait for it to come up and running.

4. In the CAI Lab Prep Files folder, navigate to the **apache-activemq-5.15.8\bin** folder, and from the folder path bar, open the command prompt as shown in the screenshot below:

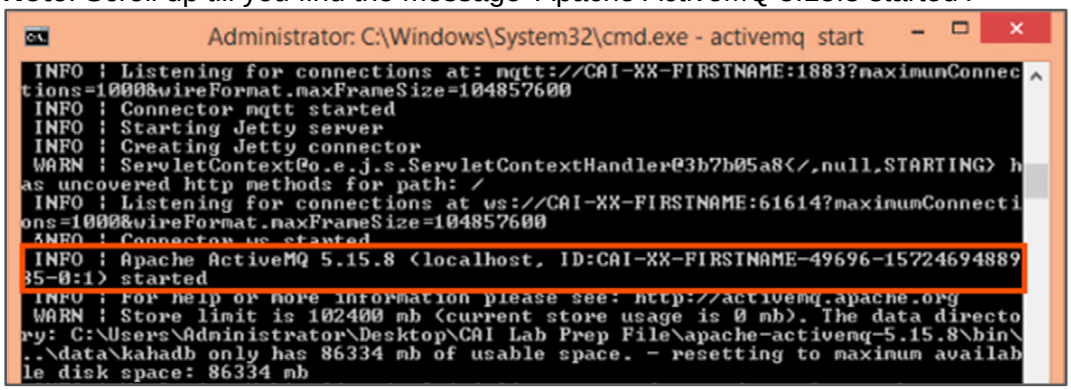


5. To start the activemq service, enter the following command:

   activemq start



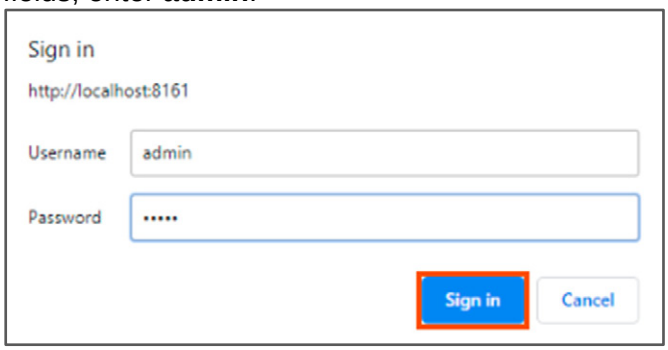
   Note: Scroll up till you find the message 'Apache ActiveMQ 5.15.8 started'.



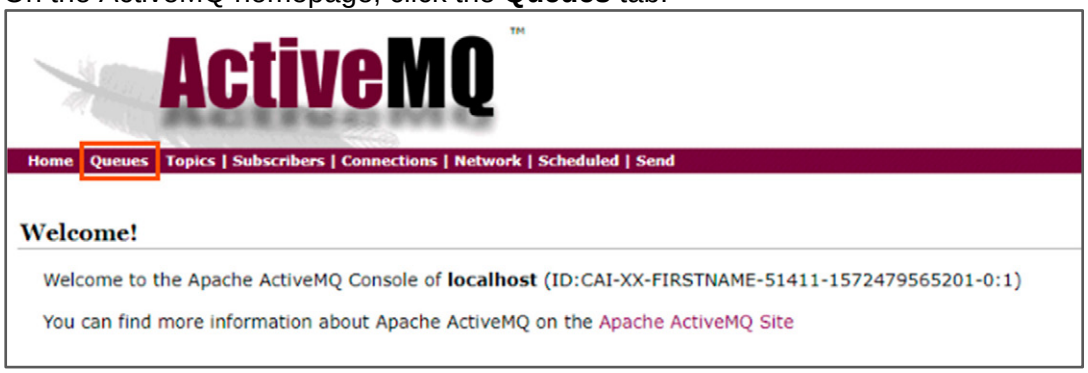
6. Open a web browser and enter the following URL:

   http://localhost:8161/admin

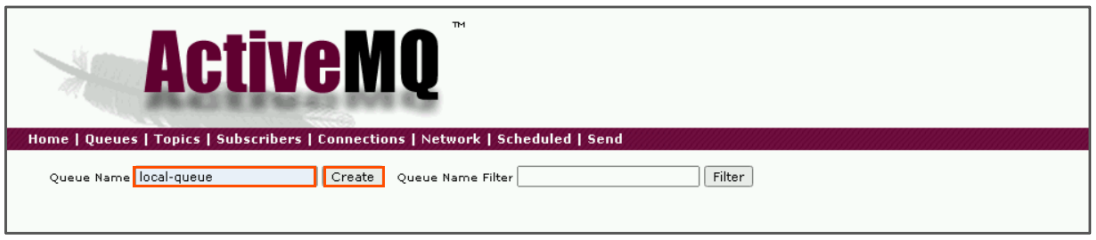
   Note: If the localhost prompts for login credentials, in the Username and Password fields, enter **admin**.



7. On the ActiveMQ homepage, click the **Queues** tab.



8. In the Queue Name field, enter **local-queue**.
9. Click **Create**.



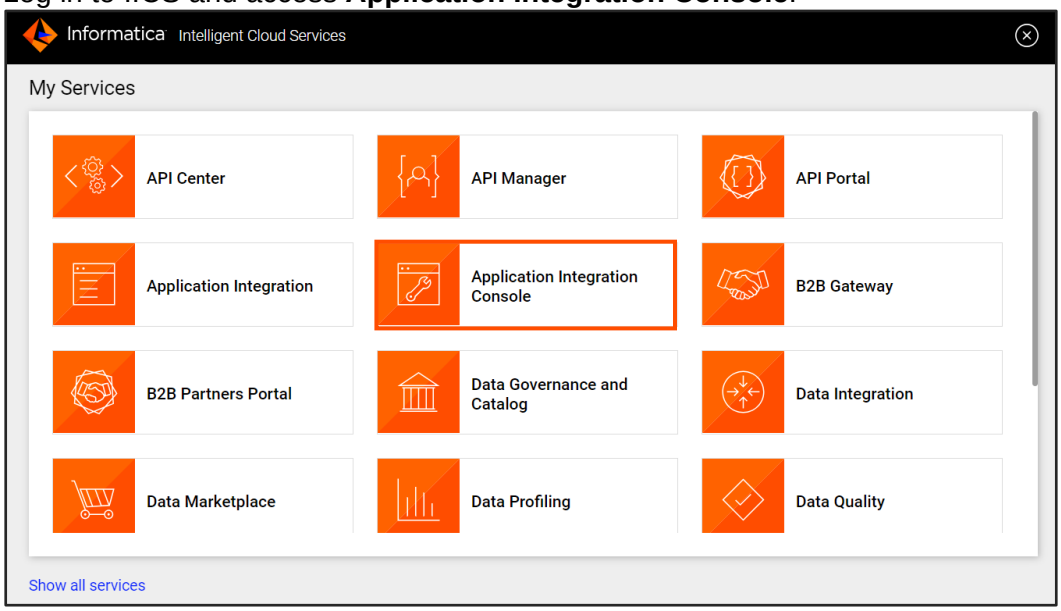
Note: You can see 'local-queue' in the list of available queues.

Note: You have successfully created the JMS queue to read messages from or write messages to it.

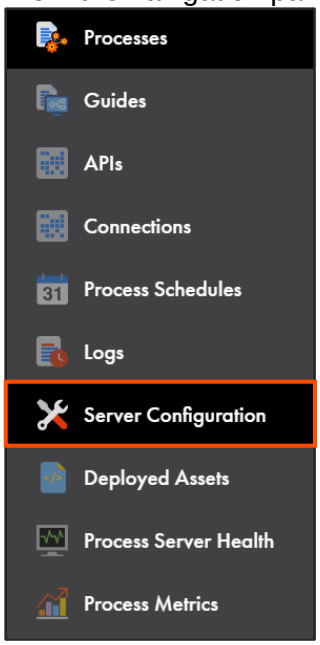
STEPS FOR LEARNERS

Configure JMS service:

10. Log in to IICS and access **Application Integration Console**.

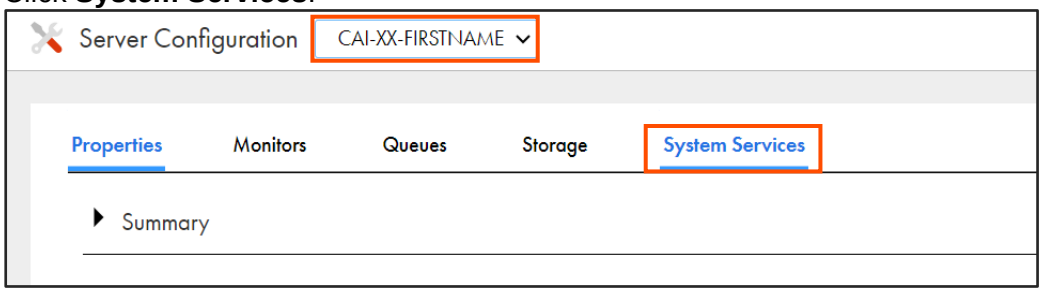


11. From the navigation pane, select **Server Configuration**.

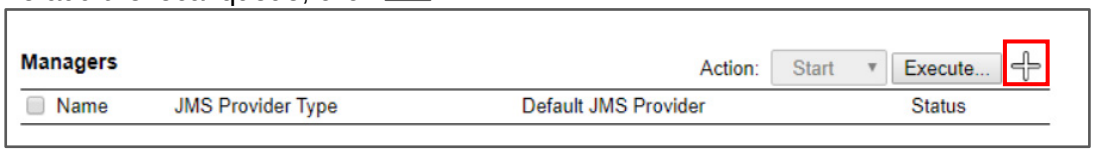


12. To configure the JMS service, from the Server Configuration drop-down, select your secure agent group.

13. Click **System Services**.



14. From the System Services menu, expand **Messaging Service**.
15. To add the local queue, click .

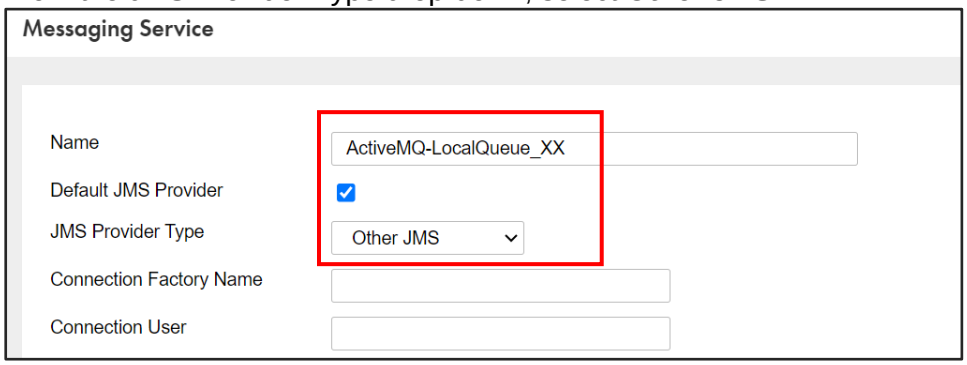


A new Messaging Service window appears.

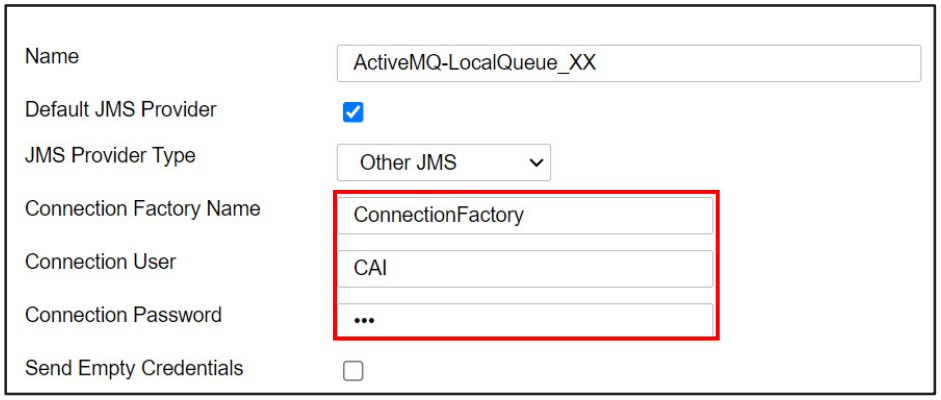
16. In the Name field, enter **ActiveMQ-LocalQueue_XX**.

Note: Here, XX is the unique student number.

17. Select the **Default JMS Provider** option.
18. From the JMS Provider Type drop-down, select **Other JMS**.



19. In the Connection Factory Name field, enter **ConnectionFactory**.
20. In the Connection User and Connection Password fields, enter **CAI**.



Configure ExchangeRateService Process Properties:

To enable the secure agent to receive JMS messages, you must set the **Initial Context Properties**.

In the Initial Context Properties, the following four properties are prepopulated.

java.naming.provider.url: Specifies the URL that the application client code uses to look up objects on the server.

java.naming.factory.initial: Specifies which initial context factory is used to create a new initial context object.

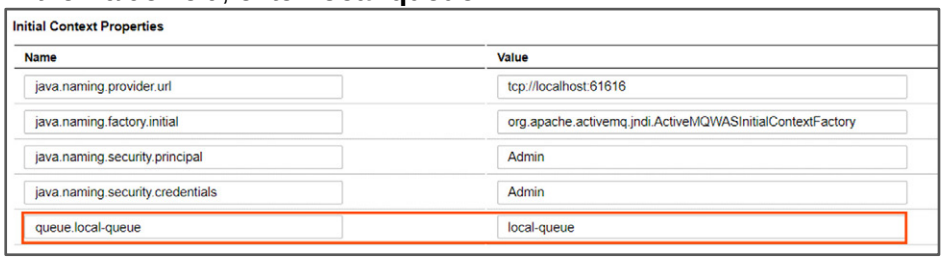
java.naming.security.principal: Specifies the user name that is required to authenticate the user that connects to the JMS provider.

java.naming.security.credentials: Specifies the password that is required to authenticate the user that connects to the JMS provider.

21. Enter the values for the fields as shown in the table below:

Name	Value
java.naming.provider.url	tcp://localhost:61616
java.naming.factory.initial	org.apache.activemq.jndi.ActiveMQWASInitialContextFactory
java.naming.security.principal	Admin
java.naming.security.credentials	Admin

22. To add another field, in Initial Context Properties, click .
23. In the Name field, enter **queue.local-queue**.
24. In the Value field, enter **local-queue**.

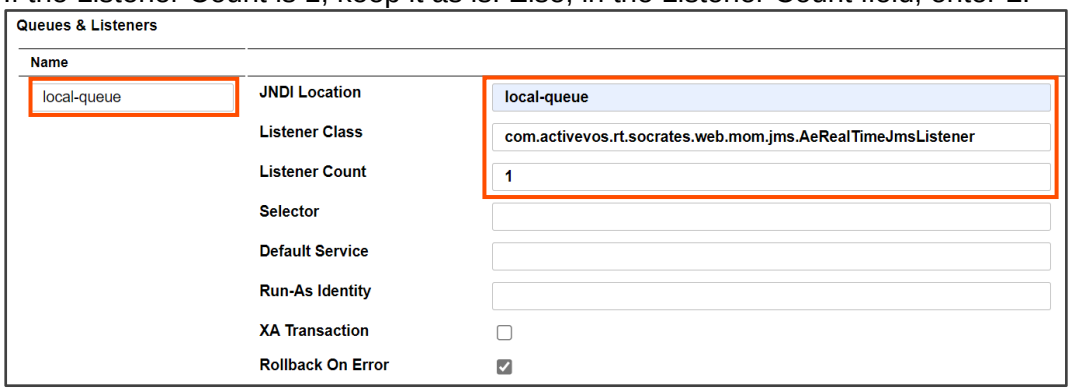


Note: The property name must be in the format: **queue.<name of queue>**, where the name of the queue is the queue name that you create in the ActiveMQ client, which is local-queue for this lab.

Configure the Queues & Listeners of the JMS service:

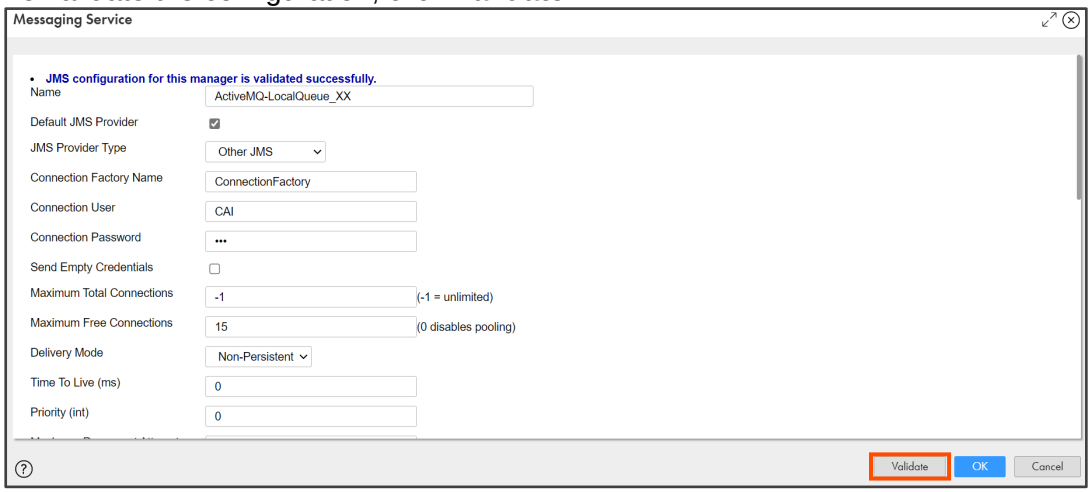
25. To add a queue in the **Queues & Listeners** section, click .
26. In the Name and JNDI Location fields, enter **local-queue**.

27. If the Listener Count is 1, keep it as is. Else, in the Listener Count field, enter **1**.



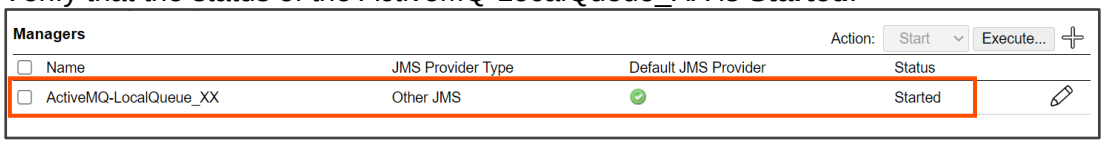
Note: Notice that here the **listener count** is **1**. This is the number of connections to keep open. When the server starts, the JMS manager creates instances of this class that serve as asynchronous consumers on the destination. Each asynchronous consumer has its own connection to the JMS server. The number of consumers and connections created is controlled by the listener count.

28. Retain default values in the other fields.
29. To validate the configuration, click **Validate**.



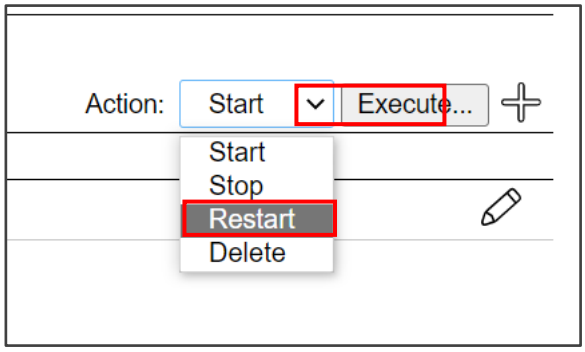
Note: A message, '**JMS Configuration for this manager is validated successfully.**' appears on top of the page.

30. Click **OK**.
31. Verify that the status of the ActiveMQ-LocalQueue_XX is **Started**.



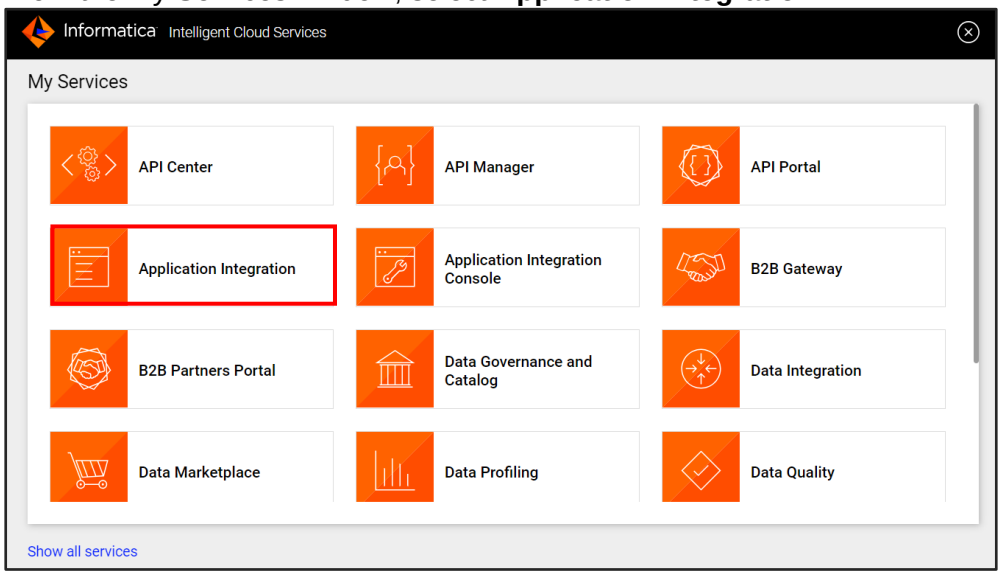
Note: If the status of the ActiveMQ-LocalQueue_XX is **Stopped**, check if the Apache ActiveMQ 5.15.8 is started. If not, then navigate to the **apache-activemq-5.15.8\bin** folder. From the folder path bar, open the command prompt and re-execute the **activemq start** command.

Then, from the Action drop-down, select **Restart** and click **Execute**.

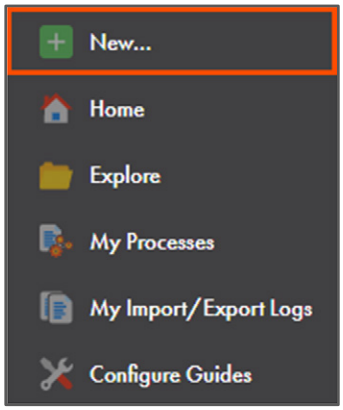


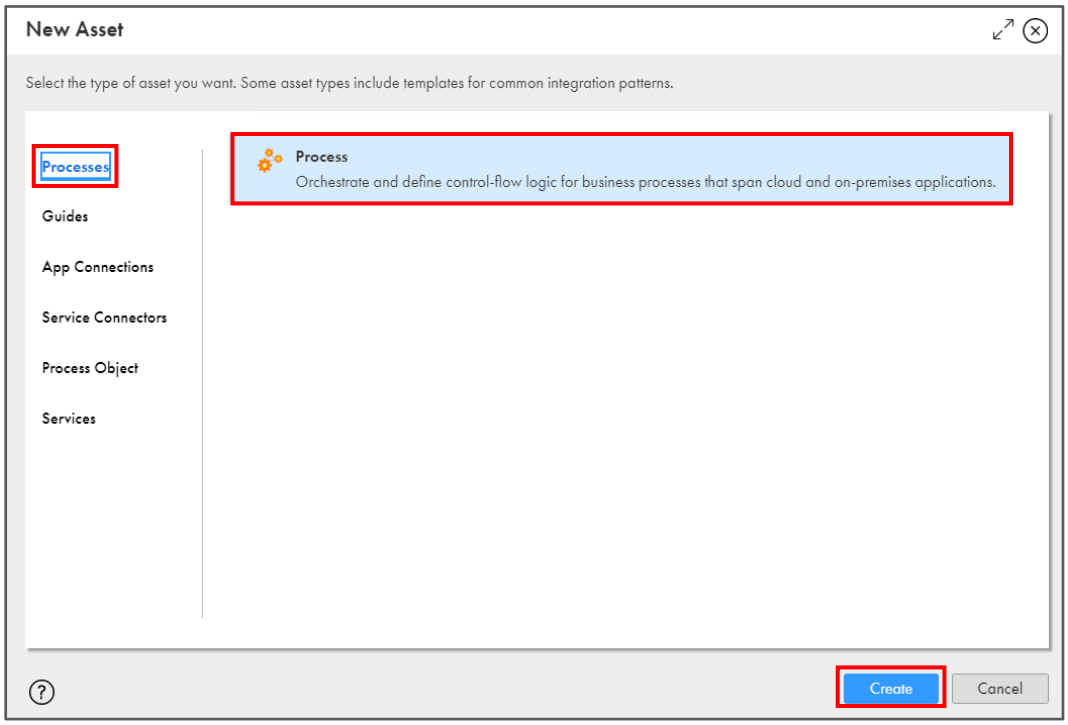
Create a process to publish a message to the queue:

32. To switch between the available services, from the toolbar, select the drop-down next to Application Integration Console.
33. From the My Services window, select **Application Integration**.



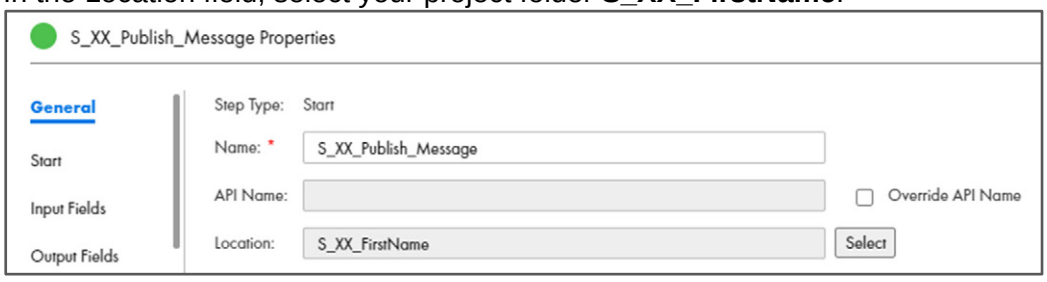
34. Create a new process.



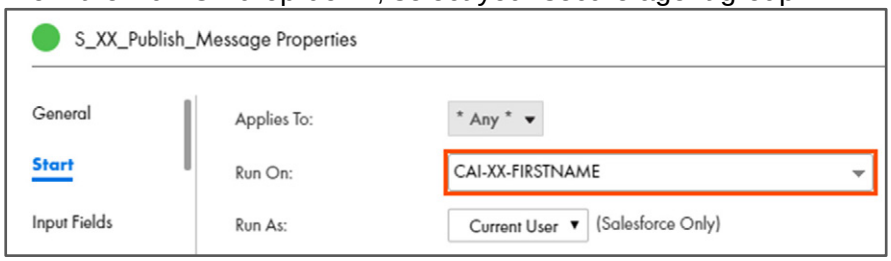


A new Process window appears.

35. From the process canvas, select the **Start** step.
36. In the General tab, enter the name as **S_XX_Publish_Message**.
37. Skip the API Name option.
38. In the Location field, select your project folder **S_XX_FirstName**.



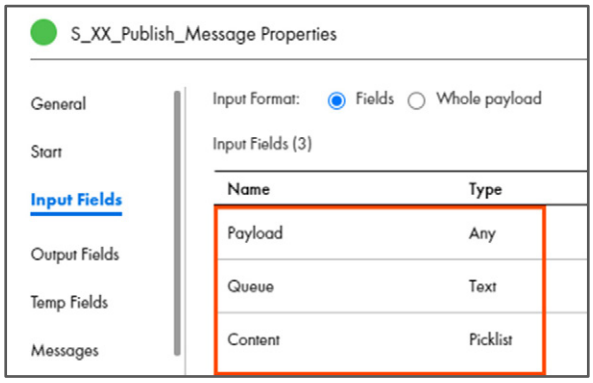
39. From the properties pane, select **Start**.
40. From the Applies To drop-down, retain **Any**.
41. From the Run On drop-down, select your secure agent group.



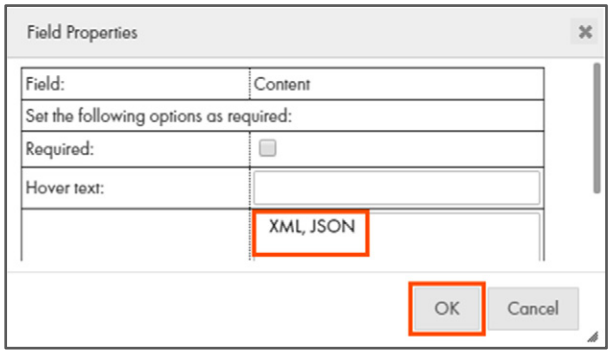
42. From the properties pane, select **Input Fields**.
43. From the Input Format field, retain **Fields**.

44. To add the Input Fields, click .
45. Add the Input Fields listed in the following table:

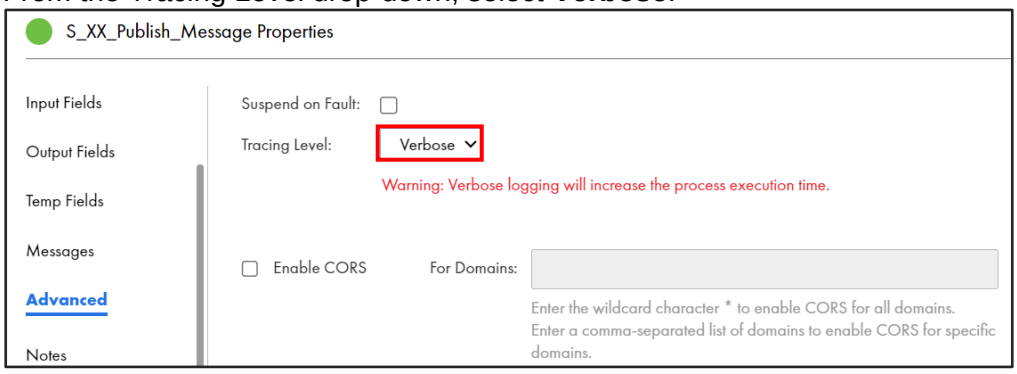
Name	Type
Payload	More Types > Custom Types > Any
Queue	Text
Content	More Types > Simple Types > Picklist



46. To edit the **Content** field, click .
    Note: A new Field Properties window appears.
47. In the Comma Separated List field, enter **XML, JSON**.
48. Click **OK**.



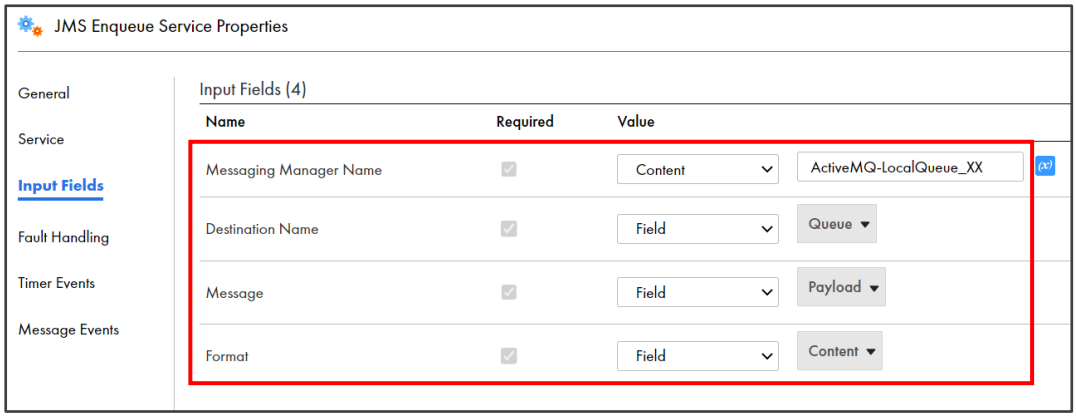
49. From the properties pane, select **Advanced**.
50. From the Tracing Level drop-down, select **Verbose**.



51. From the Design palette, drag and drop a **Service** step on the link between Start and End steps.
52. Retain the name in the General tab and from the properties pane, select **Service**.
53. From the Service Type drop-down, select **System Service**.
54. From the Action drop-down, select **JMS Enqueue Service**.

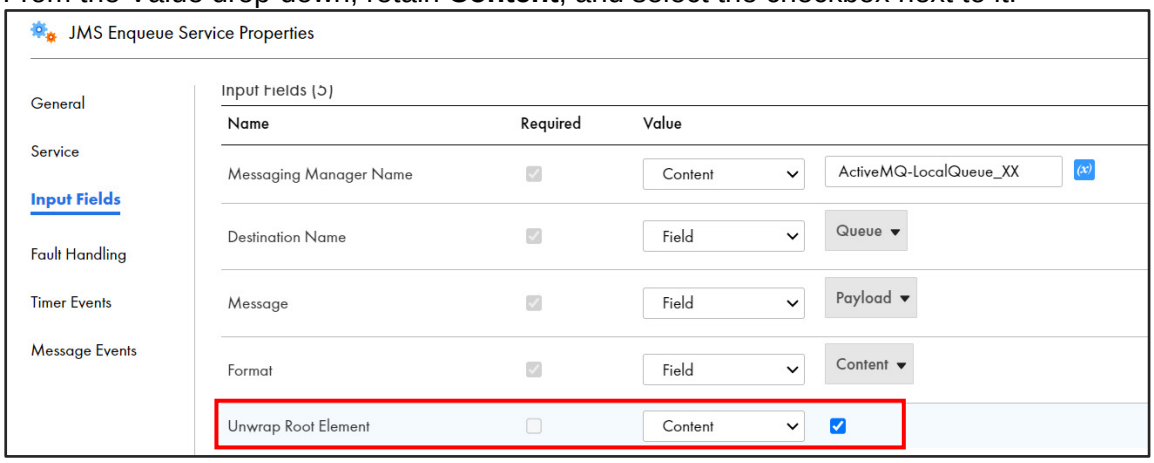
55. From the properties pane, select **Input Fields**.
56. Configure the input fields as shown in the table below:

Name	Value	
Messaging Manager Name	Content	ActiveMQ-LocalQueue_XX
Destination Name	Field	Queue
Message	Field	Payload
Format	Field	Content



57. To add another input field, click plus **(+)** icon.
58. From the Add Field drop-down, select **Unwrap Root Element**.

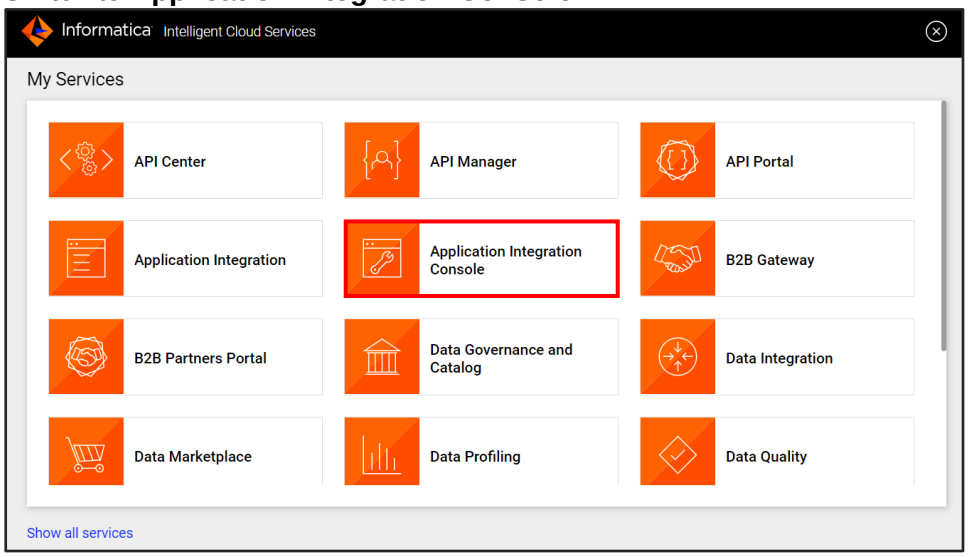
59. From the Value drop-down, retain **Content**, and select the checkbox next to it.



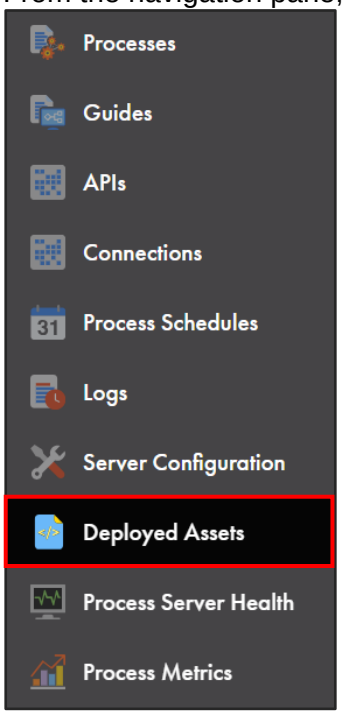
60. **Save** and **Publish** the process.

Create a process to publish a message to the queue:

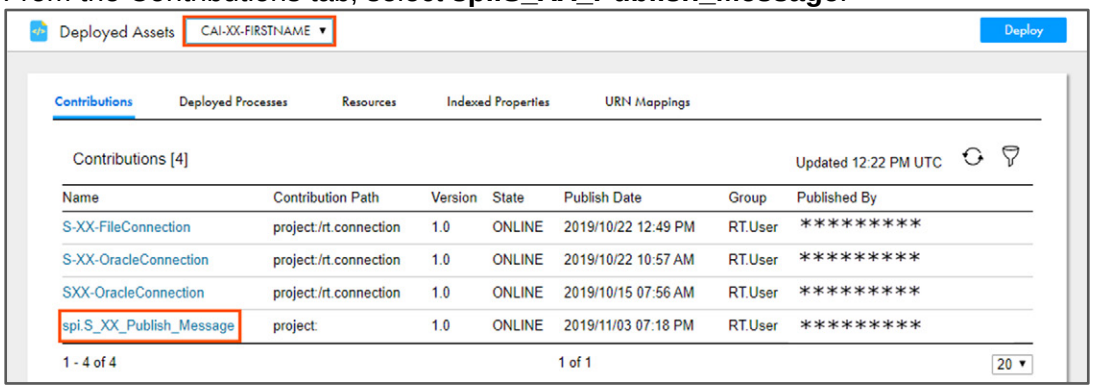
61. Switch to **Application Integration Console**.



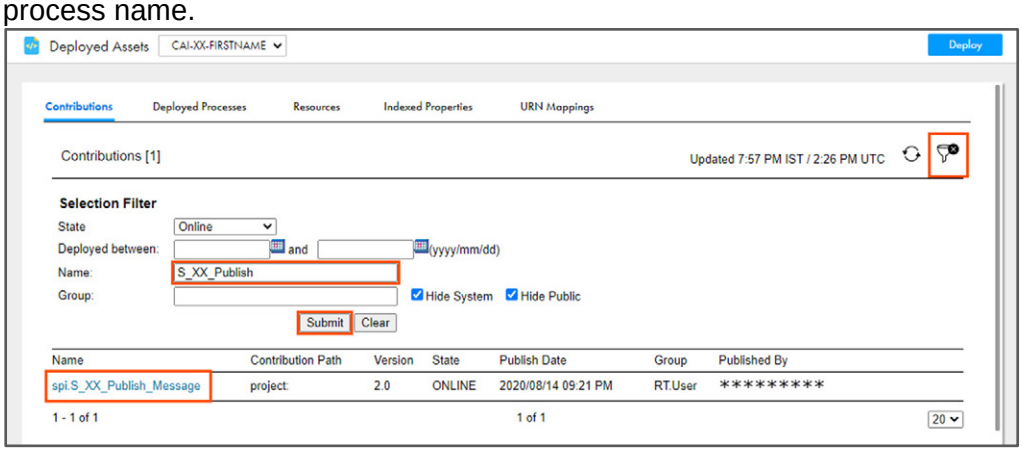
62. From the navigation pane, select **Deployed Assets**.



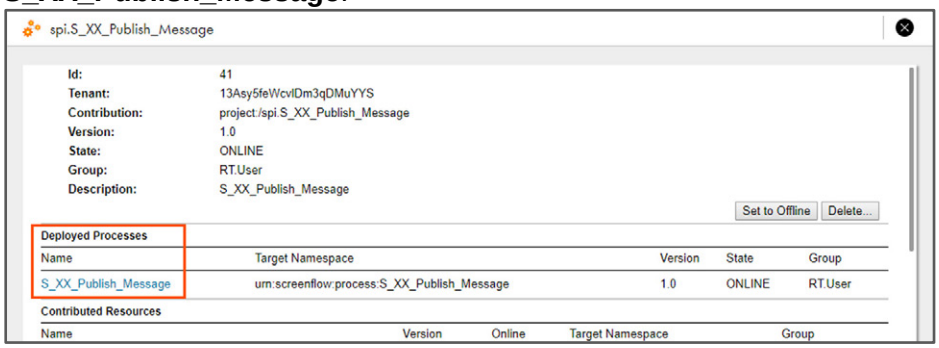
63. From the Deployed Assets drop-down, select your Secure Agent Group.
64. From the Contributions tab, select **spi.S_XX_Publish_Message**.



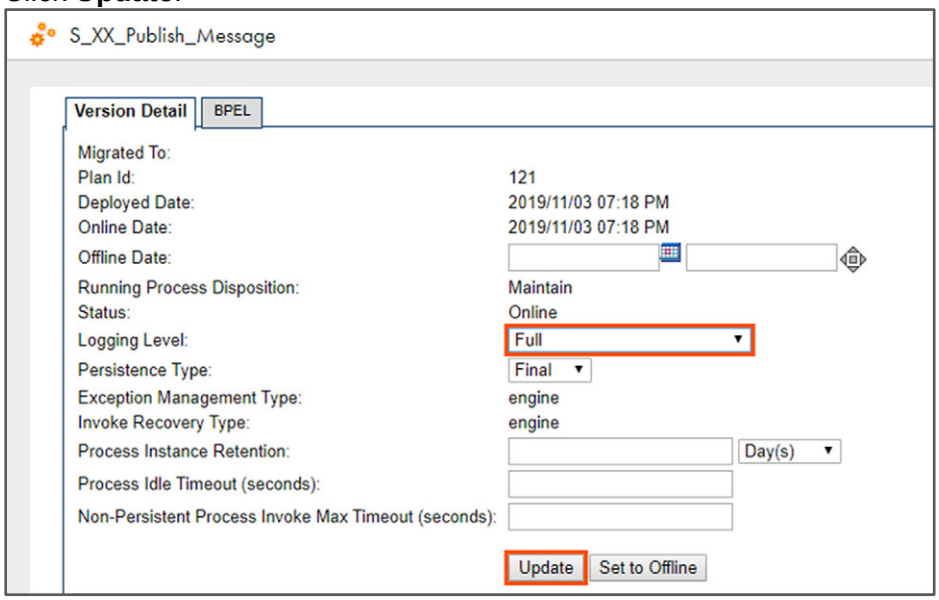
Recommended: In case of multiple assets, you can filter the search based on the process name.



65. On the spi.Publish_Message page, from the Deployed Processes section, select **S_XX_Publish_Message**.



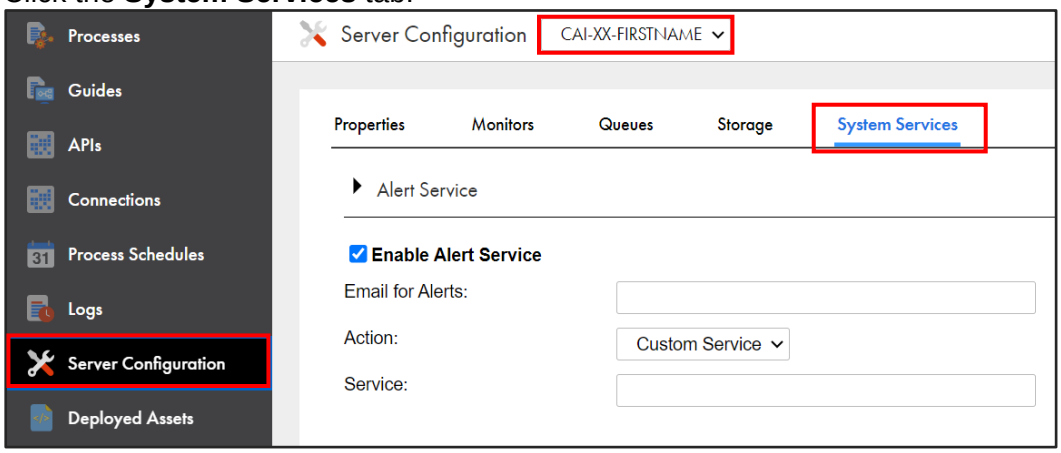
66. From the Logging Level drop-down, select **Full**.
67. Click **Update**.



Set the Listener Count for the messaging service to 0:

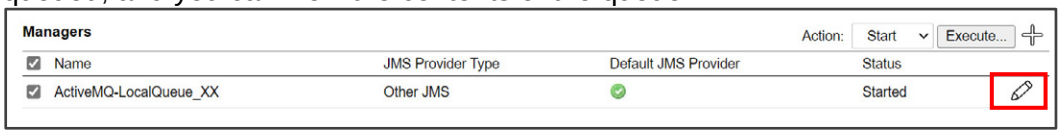
68. From the navigation pane, select **Server Configuration**.
69. From the Server Configuration drop-down, ensure that your Secure Agent Group is selected.

70. Click the **System Services** tab.



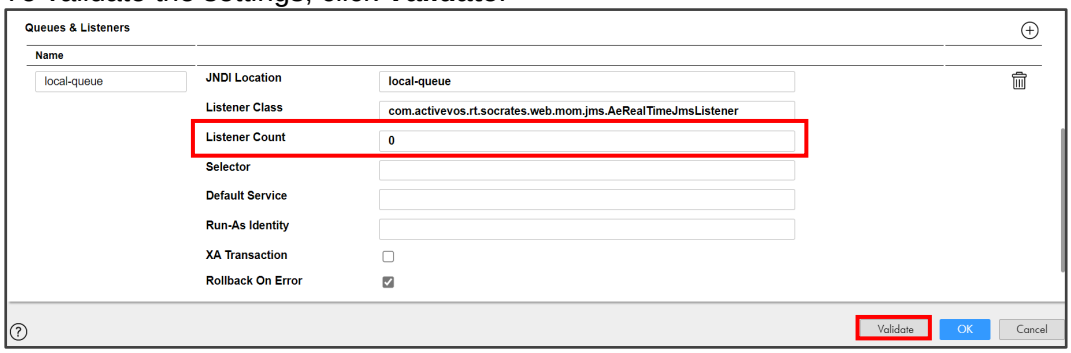
71. Expand **Messaging Service** and under Managers, select **ActiveMQ-LocalQueue_XX**.

72. Click to edit the number of consumers so that the message is not automatically de-queued, and you can view the contents of the queue.



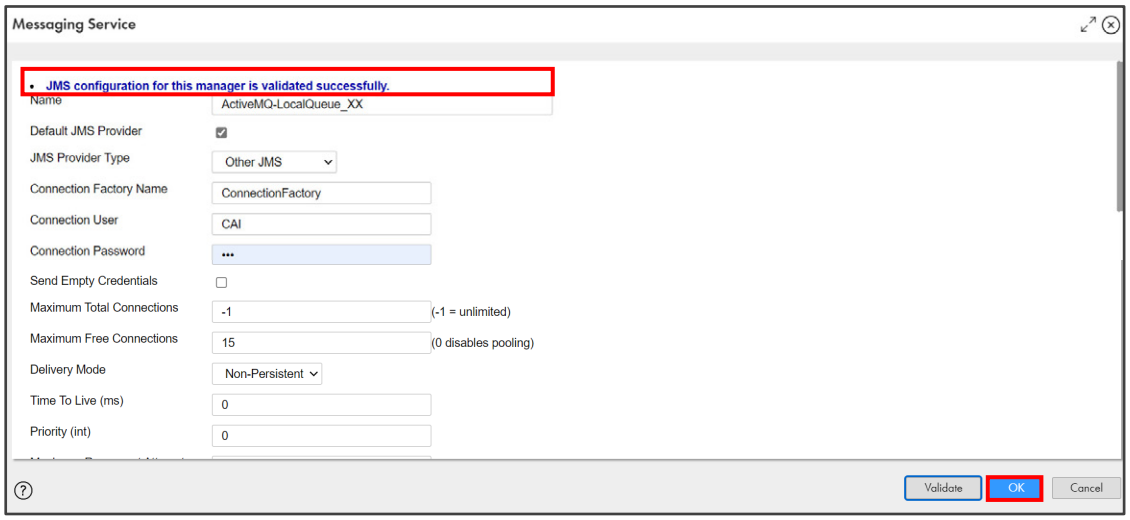
73. Under the **Queues & Listeners** section, set the Listener Count to **0**.

74. To validate the settings, click **Validate**.

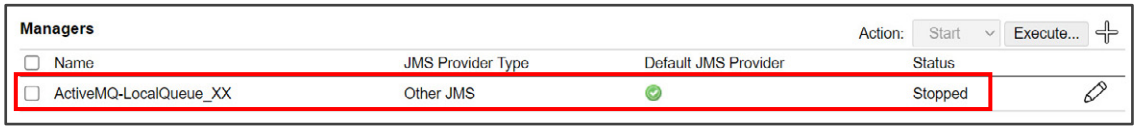


Note: A message '**JMS configuration for this manager is validated successfully**' appears at the top of the Messaging Service page.

75. Click **OK**.

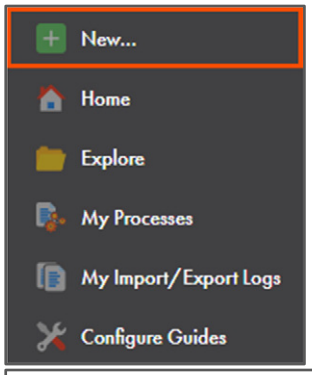


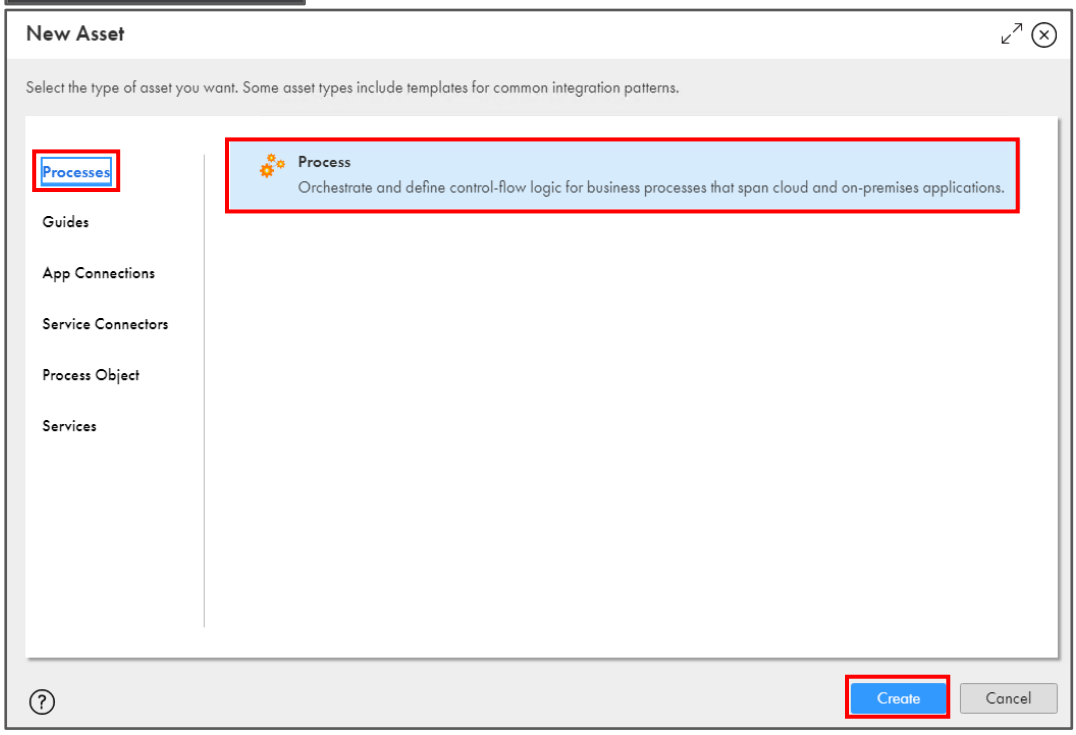
Note: You can notice that the ActiveMQ-LocalQueue_XX status is Stopped.



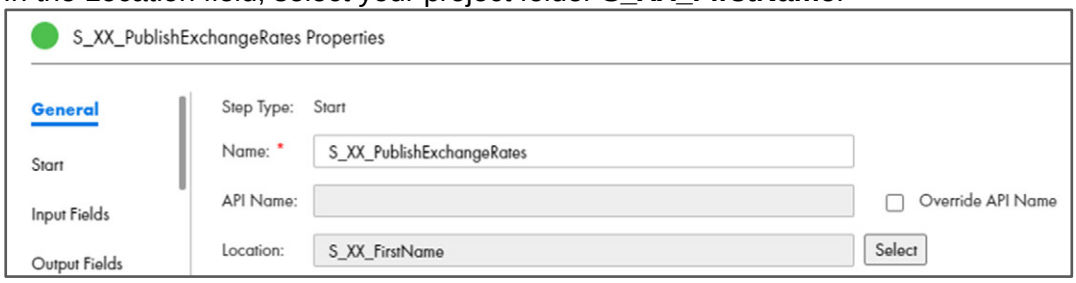
Create a Parent Process to Publish the Exchange Rate to a Queue:

76. Navigate to the **Application Integration** service.
77. Create a new **Process**.



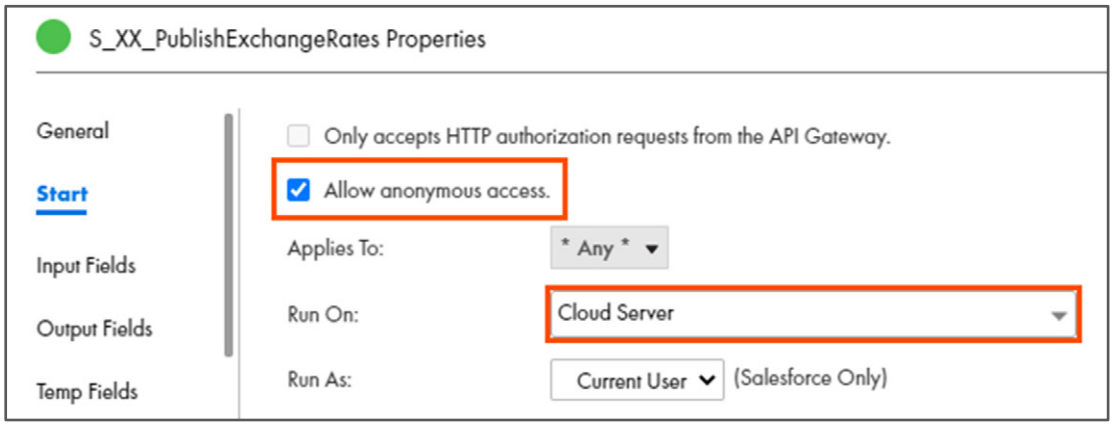


78. From the process canvas, select the **Start** step.
79. In the General tab, enter the name as **S_XX_PublishExchangeRates**.
80. Skip the API Name option.
81. In the Location field, select your project folder **S_XX_FirstName**.



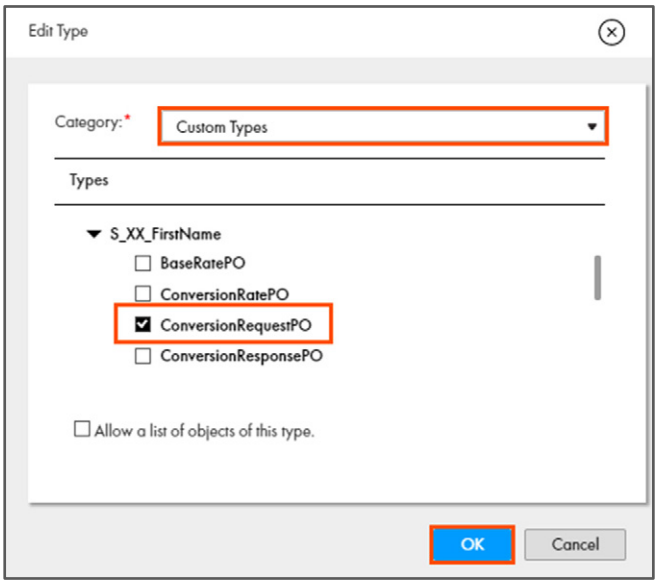
82. From the properties pane, select **Start**.

83. Select the **Allow anonymous access** checkbox.
84. Retain the Run On field as **Cloud Server**.



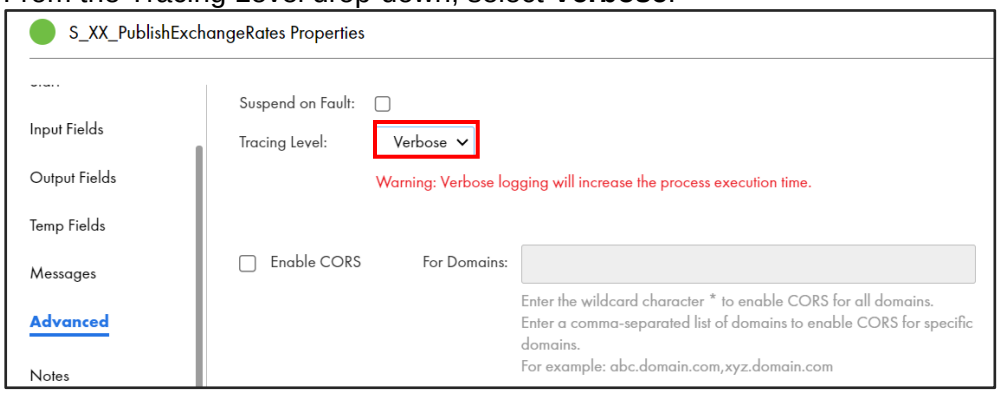
85. From the properties pane, select **Input Fields**.
86. To add input fields, click .
87. Create the input field, as shown in the table below:

Name	Type
ExchangeRequest	More types > Custom Types > S_XX_FirstName > ConversionRequestPO

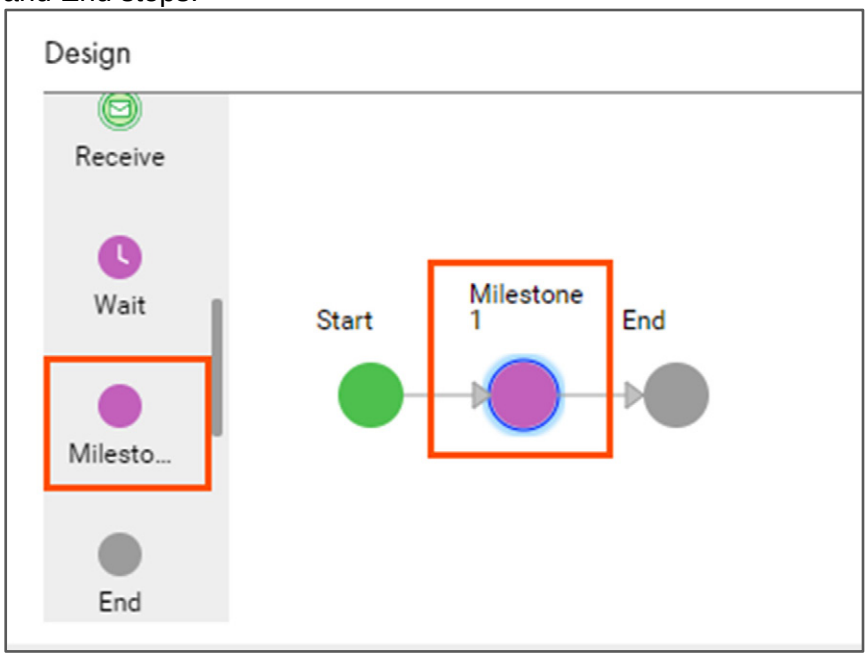


88. From the properties pane, select **Advanced**.

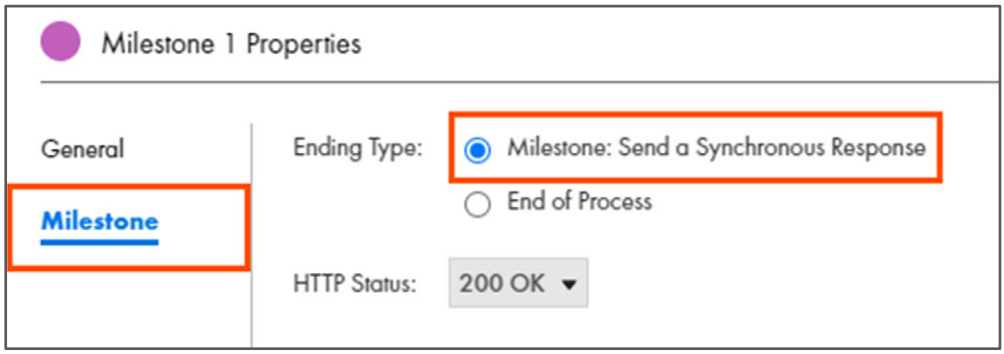
89. From the Tracing Level drop-down, select **Verbose**.



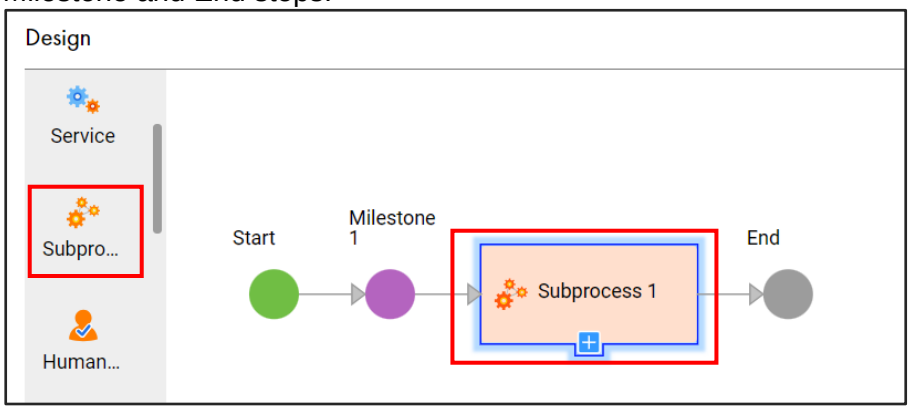
90. From the Design palette, drag and drop a **Milestone** step on the link between the Start and End steps.



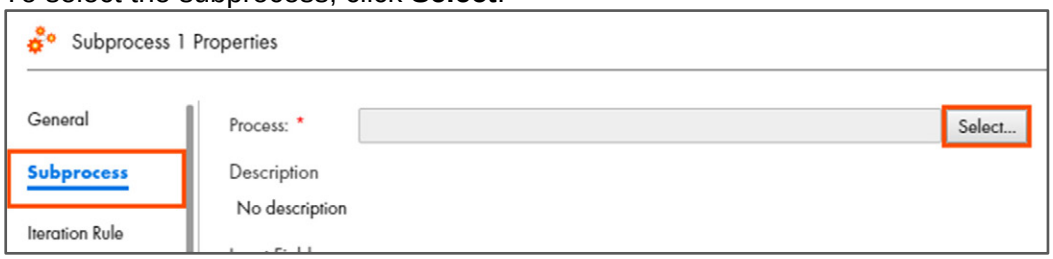
91. From the properties pane, select **Milestone**.
92. Verify that the Ending Type is set as **Milestone: Send a Synchronous Response** and HTTP Status is **200 OK**.



93. From the Design palette, drag and drop a **Subprocess** step on the link between the Milestone and End steps.

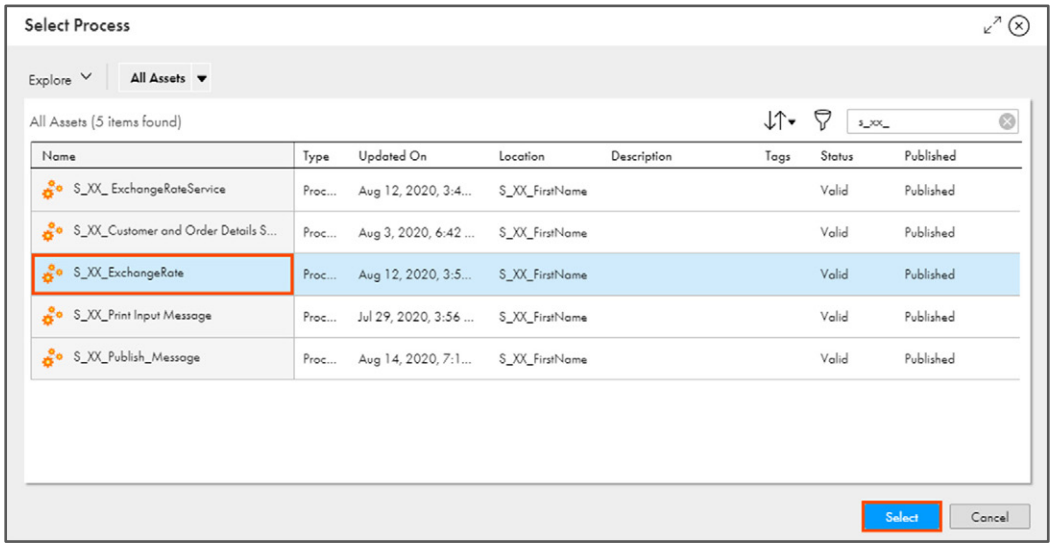


94. From the properties pane, select **Subprocess**.
95. To select the subprocess, click **Select**.



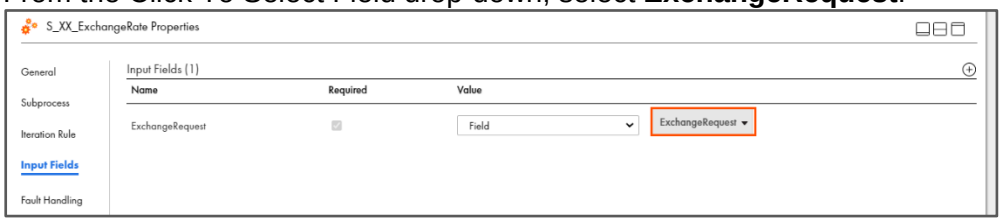
Note: A Select Process window appears.

96. From the list, select the **S_XX_ExchangeRate** process created the previous lab.
97. Click **Select**.

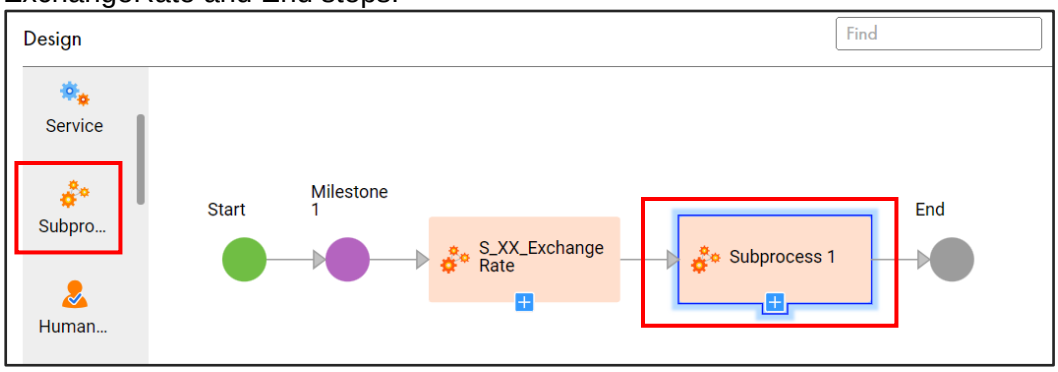


98. From the properties pane, select **Input Fields**.
99. To add the input field, click plus **(+)** icon and select **ExchangeRequest**.

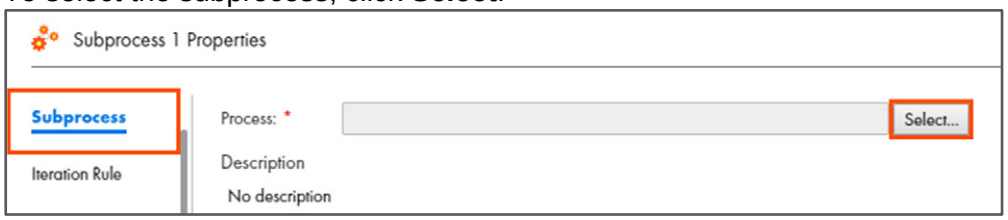
100. From the Click To Select Field drop-down, select **ExchangeRequest**.



101. From the Design palette, drag and drop a **Subprocess** step on the link between the ExchangeRate and End steps.

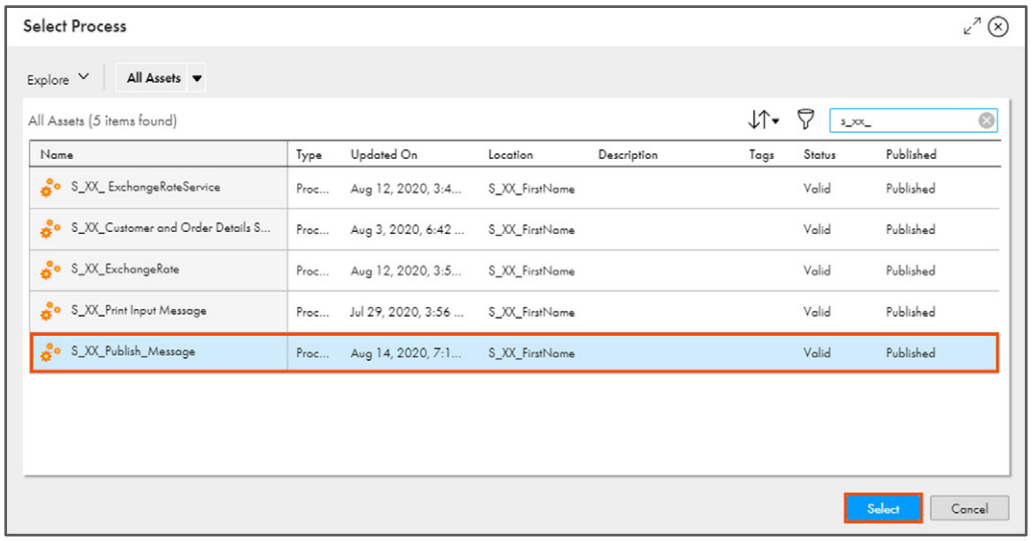


102. From the properties pane, select **Subprocess**.
103. To select the subprocess, click **Select**.



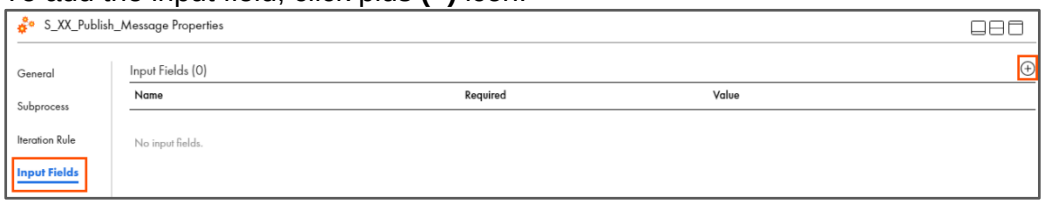
Note: A Select Process window appears.

104. From the list, select **S_XX_Publish_Message** process.
105. Click **Select**.



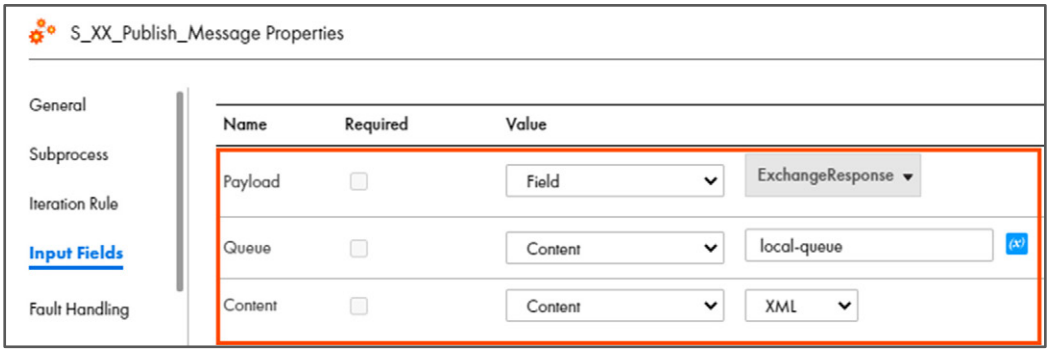
Note: This step adds the process that will publish the response to the JMS queue.

106. From the properties pane, select **Input Fields**.
107. To add the input field, click plus **(+)** icon.

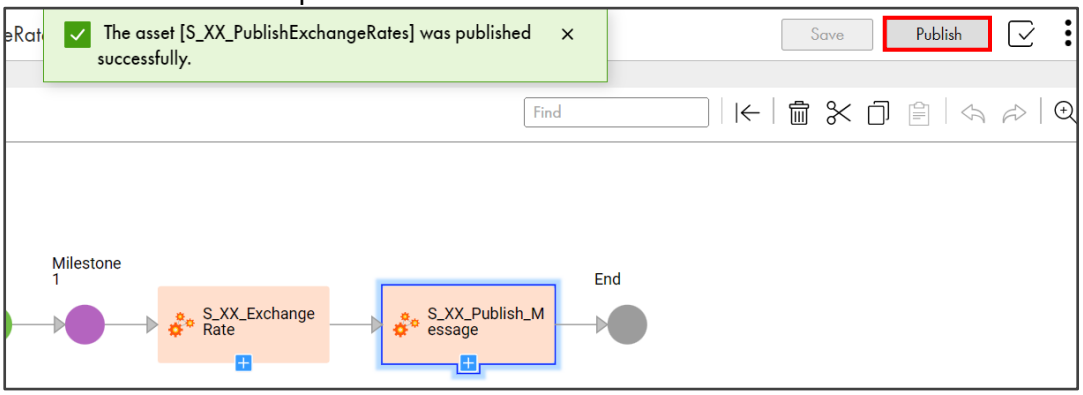


108. Add the fields as shown in the table below:

Name	Value	Type
Payload	Field	ExchangeResponse
Queue	Content	local-queue
Content	Content	XML

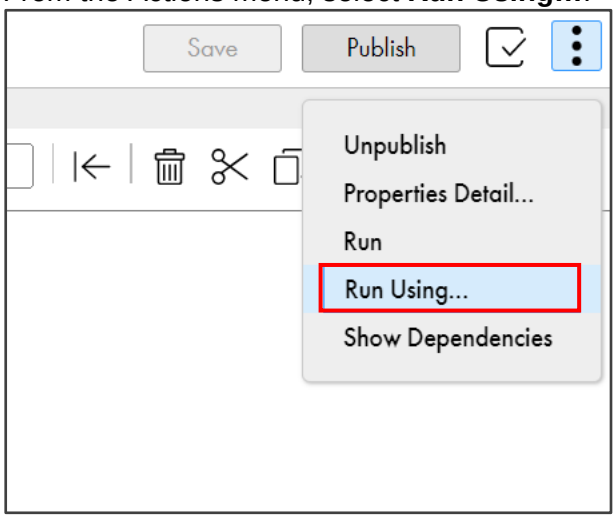


109. **Save** and **Publish** the process.

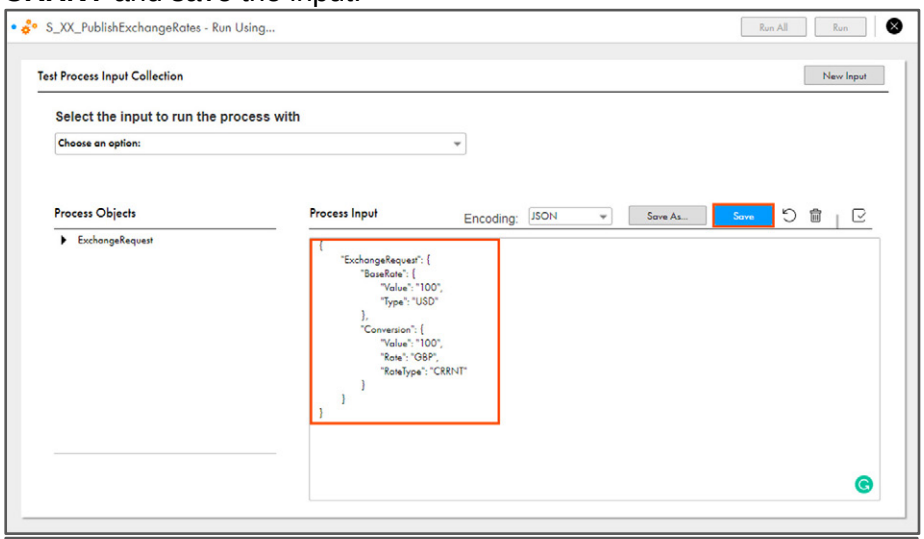


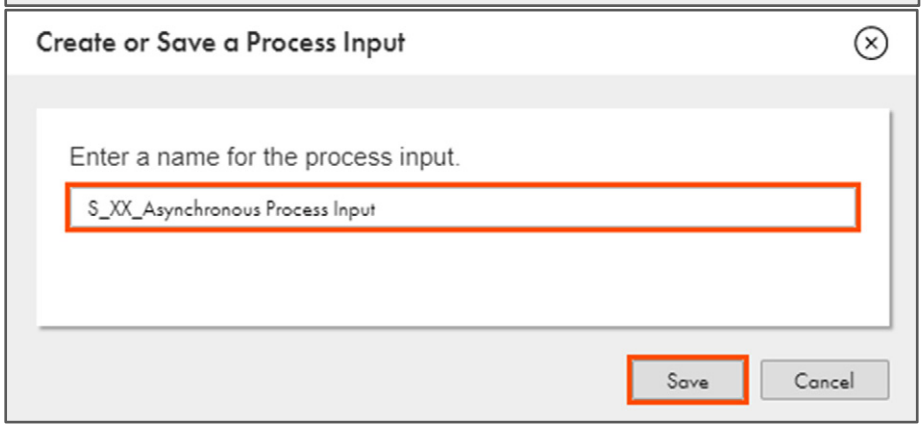
Execute the process:

110. From the Actions menu, select **Run Using…**.

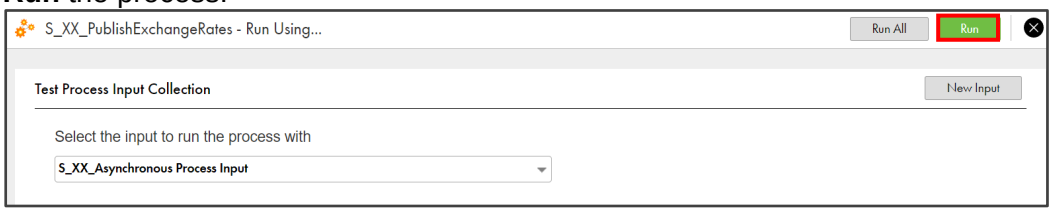


111. In the Process Input section, for the BaseRate object, add the Value as **100** and Type as **USD**.
112. For the Conversion object, add the Value as **100**, Rate as **GBP**, and the RateType as **CRRNT** and save the input.

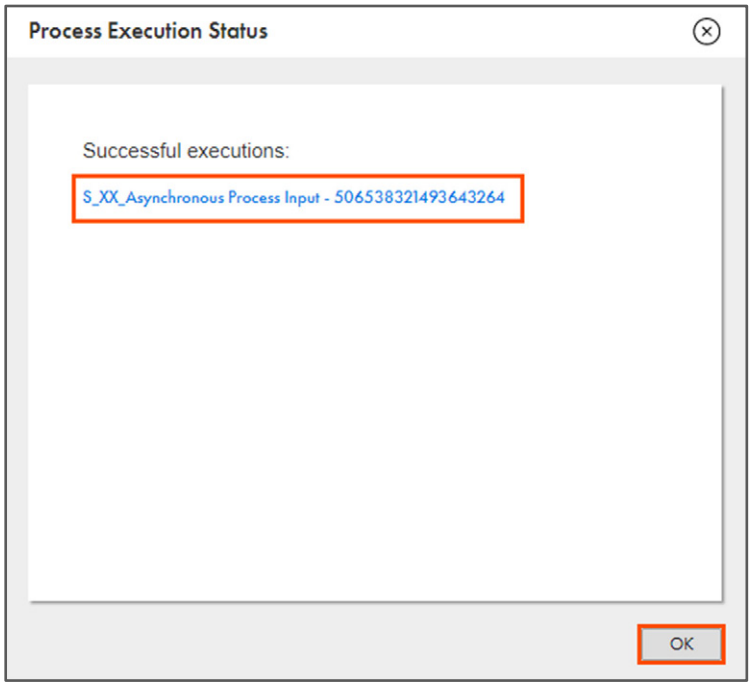




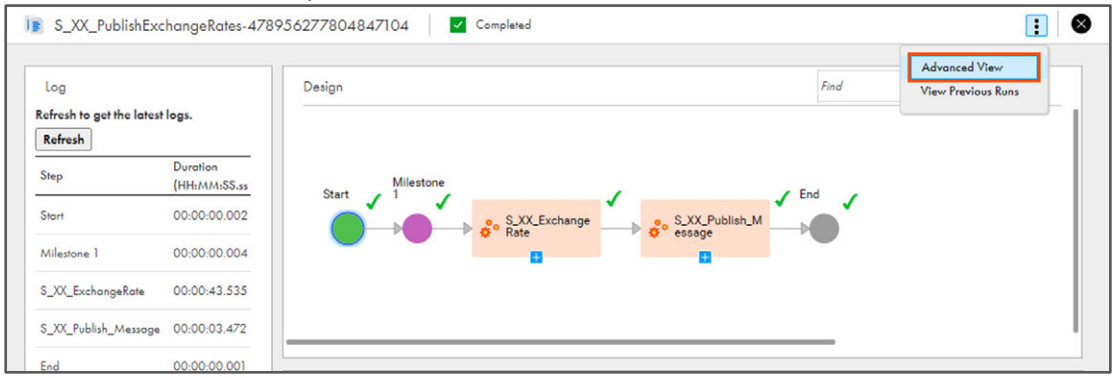
113. **Run** the process.



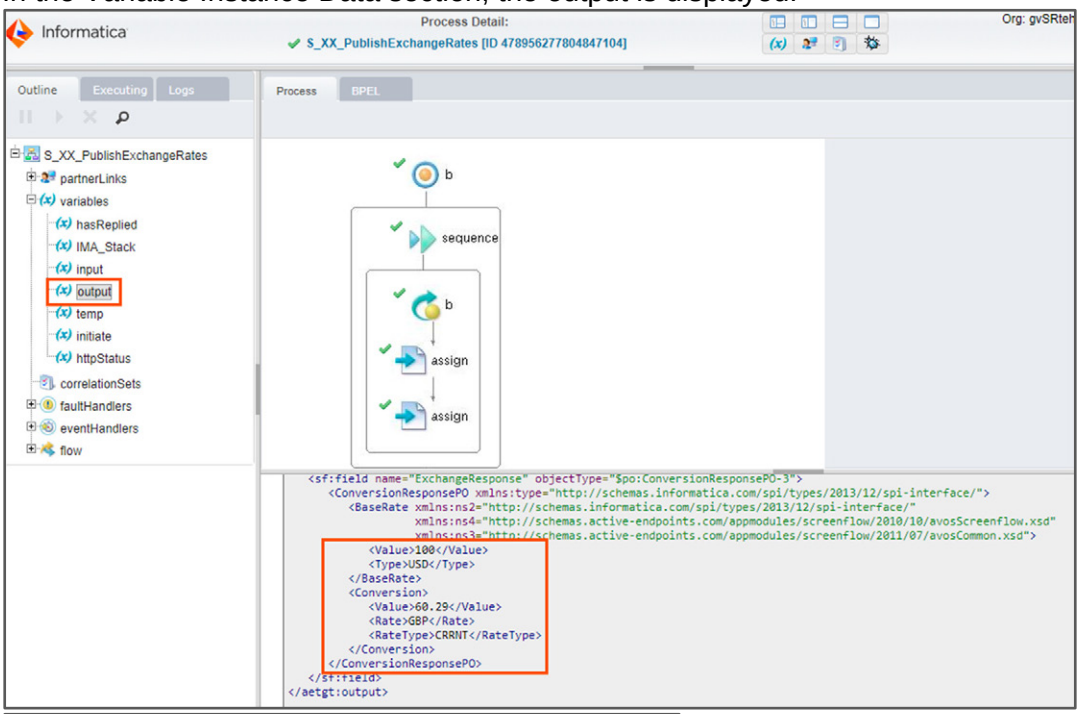
114. When the process executes successfully, to check the output, click the execution link, and click **OK**.



115. From the Actions menu, click **Advanced View**.



116. In the Advanced View, expand the **variables** section, and click **output**.
117. In the Variable Instance Data section, the output is displayed.

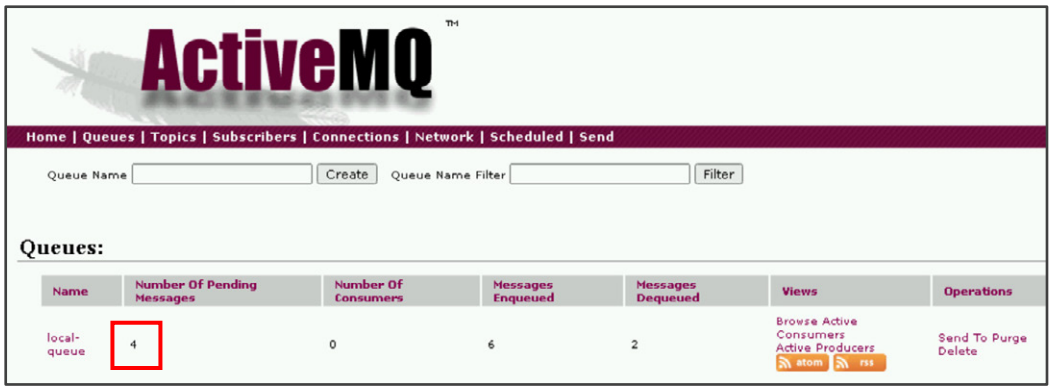


In this way, you can invoke a process using an asynchronous web service call and view the response stored in the JMS queue.

Demo

118. Go back to the ActiveMQ page, and click the **Queues** tab.

     Note: Refresh the ActiveMQ page if the Queues tab is already open.
119. On the ActiveMQ homepage, click the **Queues** tab.
120. Observer the **Number of Pending Messages** column.



Note: The value of the Number of Pending Messages will change each time when the S_XX_PublishExchangeRates process will run in the local-queue.

This concludes the lab.

Module 11: Fault Handling
Lab 11-1: Handle Sub-process Failures

Overview:
In this lab, you will configure processes that handle process execution faults. You will then run the process and view failed processes and error messages. For the process, you will customize the error messages so that business users receive simple messages with failure reasons, instead of long technical messages that are difficult to decode.

Objectives:
- Configure error handling for sub-process failure
- Run the process
- Check failed process instance

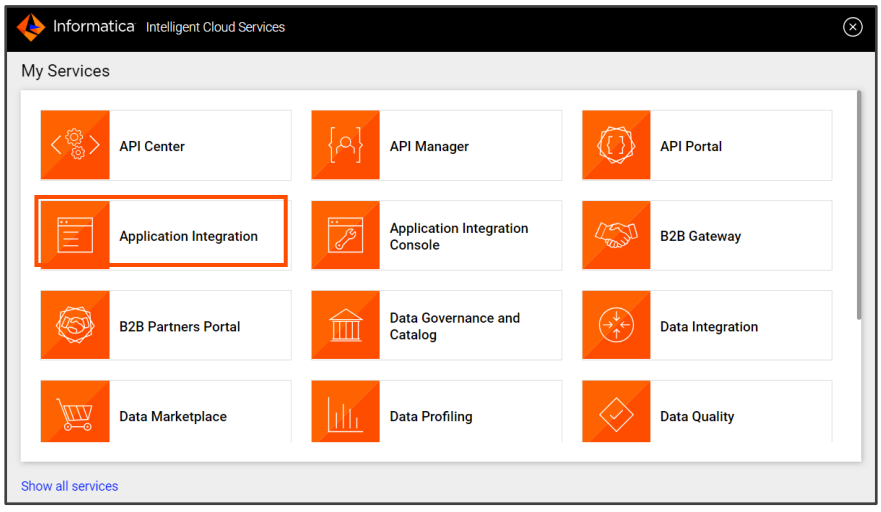
Duration:
15 Minutes

Tasks:

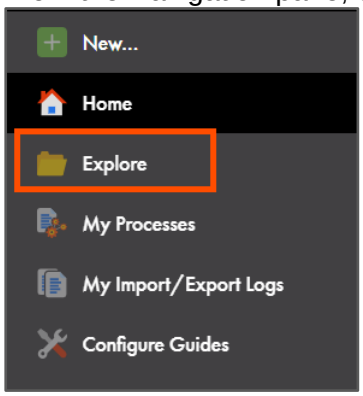
Note: The screen on your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

Configure error handling for sub-process failure

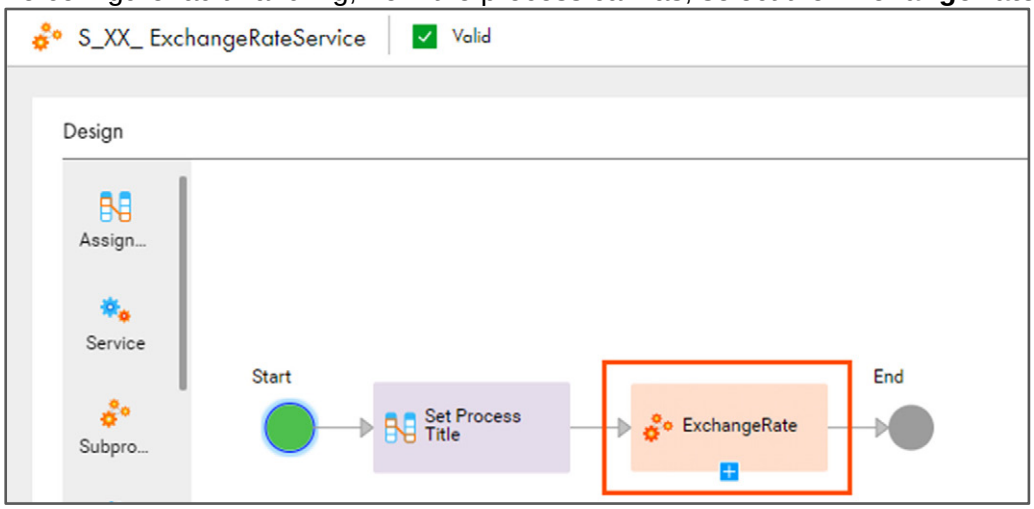
1. Log in to IICS and access **Application Integration**.



2. From the Navigation pane, click **Explore**.

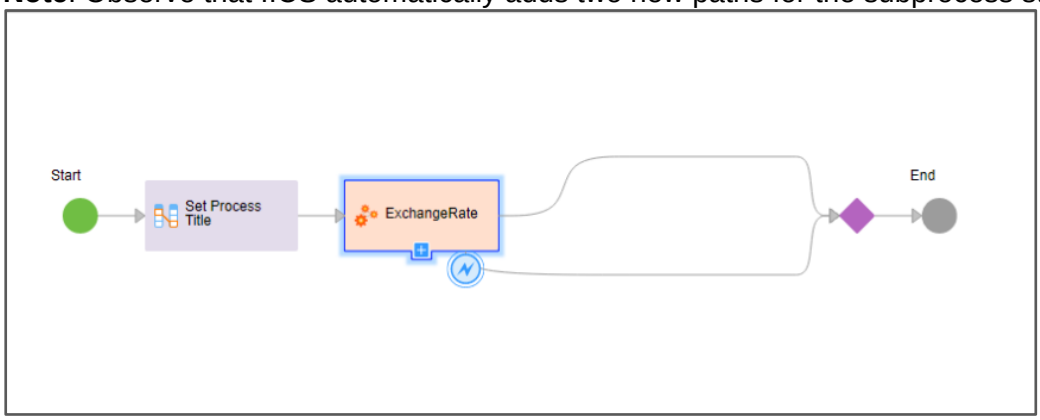


3. Navigate to your project folder and open the **S_XX_ExchangeRateService** process.
4. To configure fault handling, from the process canvas, select the **ExchangeRate** step.

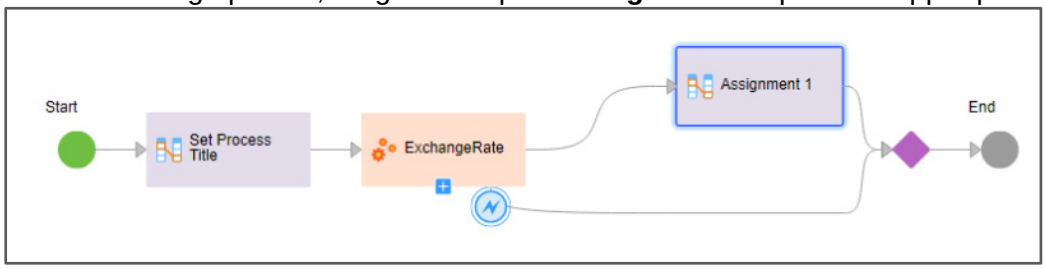


5. From the properties pane, select **Fault Handling**.
6. Select the **Catch faults** option.

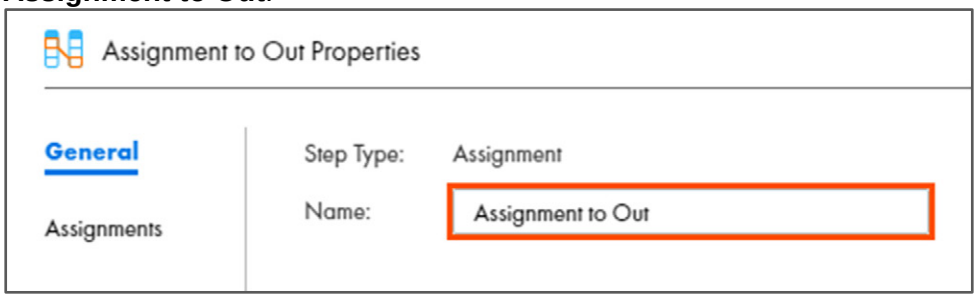
Note: Observe that IICS automatically adds two new paths for the subprocess step.



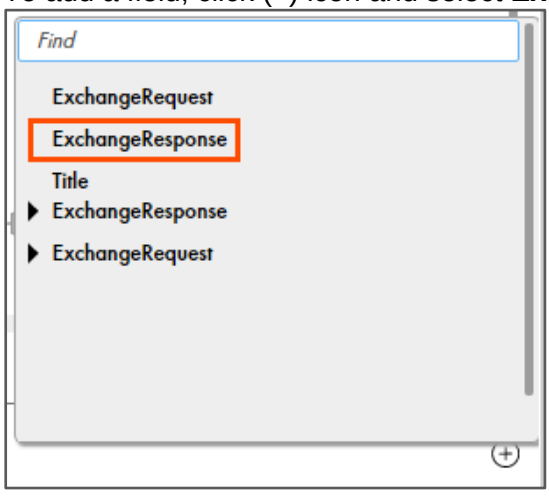
7. From the Design palette, drag and drop an **Assignment** step on the upper path.



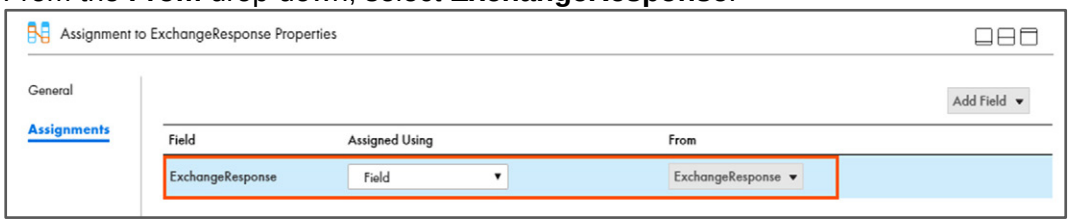
8. From the Assignment properties pane, in the General tab, enter the Name as **Assignment to Out**.



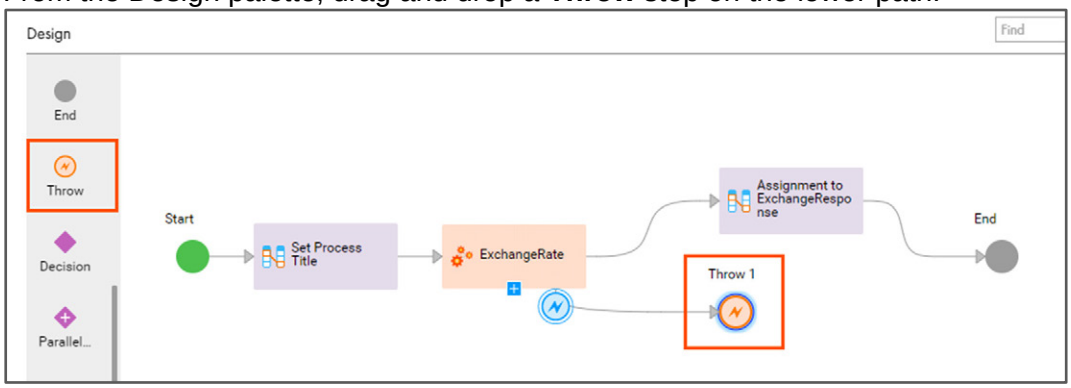
9. From the properties pane, select **Assignments**.
10. To add a field, click (**+**) icon and select **ExchangeResponse**.



11. From the Assigned Using drop-down, retain **Field**.
12. From the **From** drop-down, select **ExchangeResponse**.



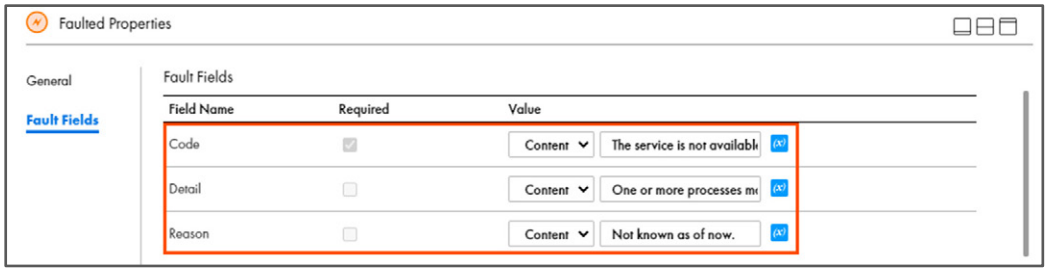
13. From the Design palette, drag and drop a **Throw** step on the lower path.



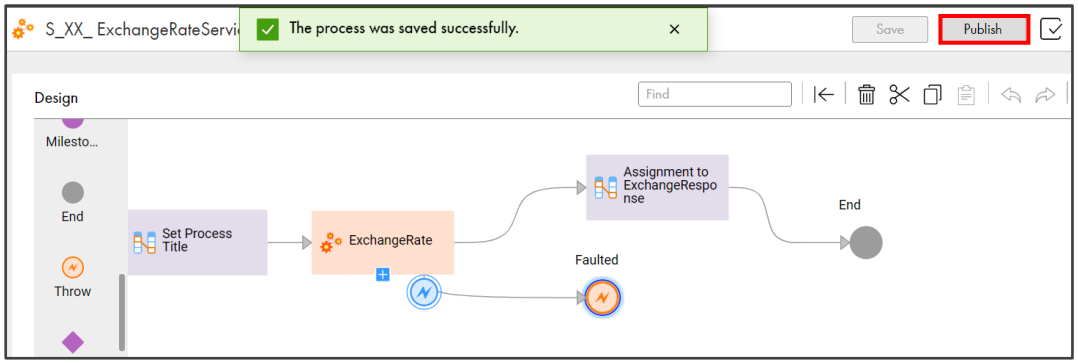
14. From the Throw properties pane, in the General tab, enter the Name as **Faulted**.

15. From the properties pane, select **Fault Fields**.
16. For the Fault Fields, configure the values as shown in the table below:

Field Name	Value	Type
Code	Content	The service is not available. Please try after some time.
Detail	Content	One or more processes may not be executing, hence the fault.
Reason	Content	Not known as of now.



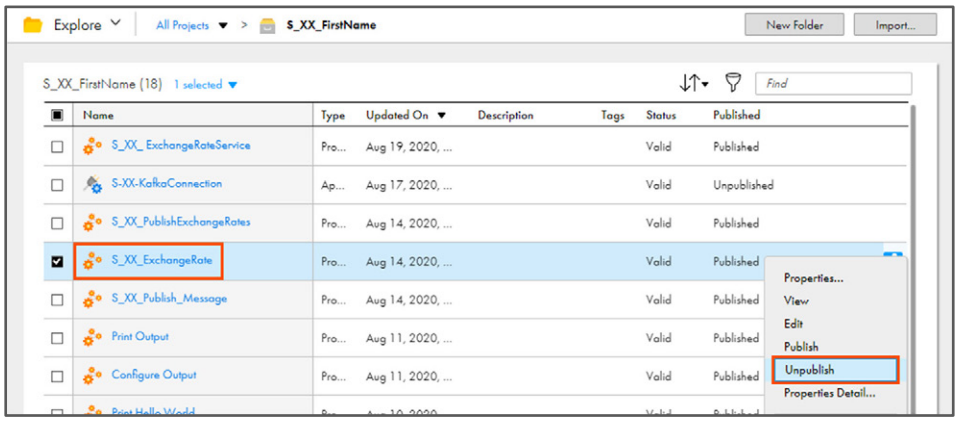
17. **Save** and **Publish** the process.



Note: If the Publish option is disabled, unpublish the process, save it, and then re-publish the process.

Now, to understand fault handling, you will introduce an error to the process.

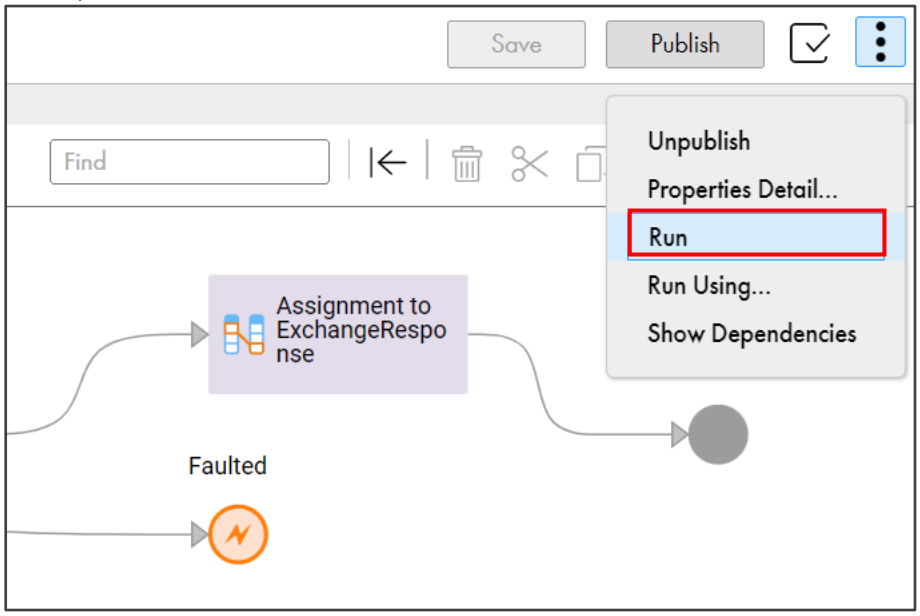
18. Navigate to your project folder and locate the **S_XX_ExchangeRate** process.
19. To unpublish the process, click ⋮ and select **Unpublish**.



Note: Unpublishing the ExchangeRate process introduces an error when you run the ExchangeRateService Process. You can use this error to understand the fault handling techniques used in CAI.

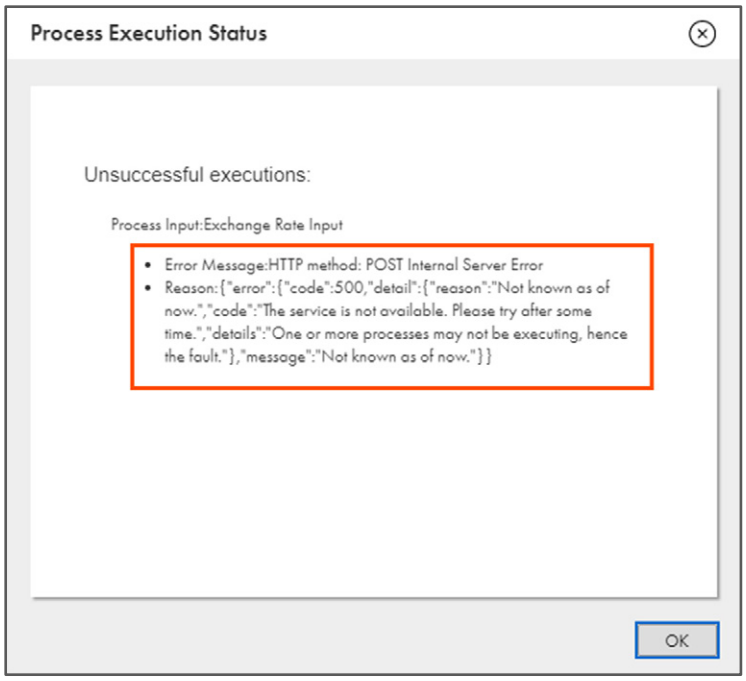
Execute the Process

20. Navigate back to the **S_XX_ExchangeRateService** process and from the **Actions** menu, click **Run**.



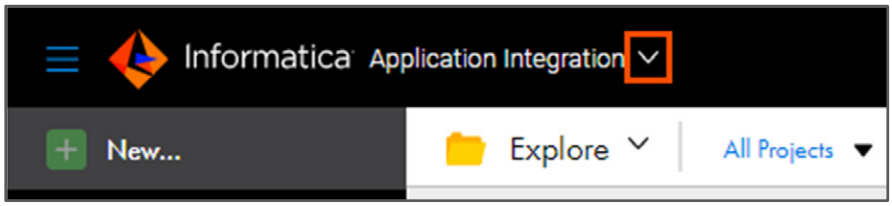
Note: The Run option executes the process based on the previous process input provided to it.

Once you run the process, you may or may not get the **Process Execution Status** window. If you get this window, click **OK**. Otherwise, continue with the next step to observe the fault.

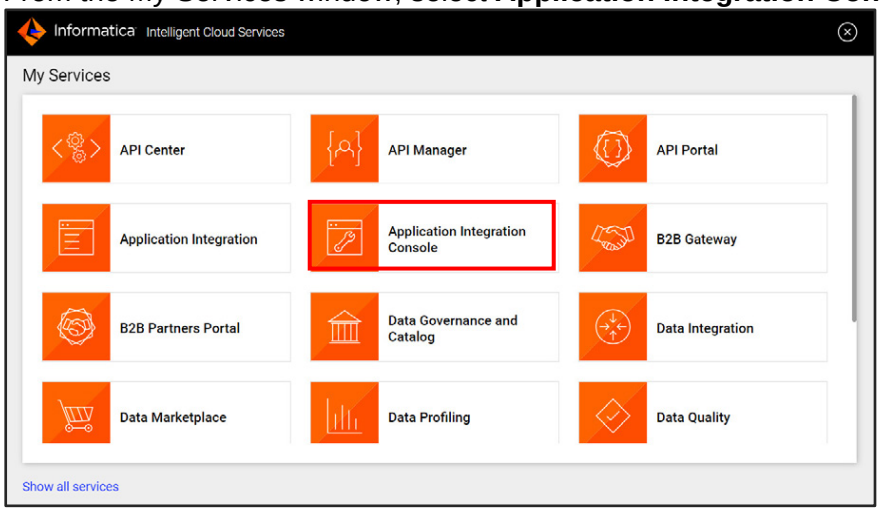


Observe the Fault Process

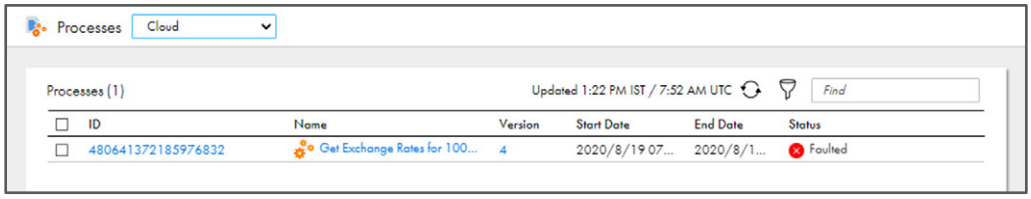
21. To switch between the available services, from the toolbar, select the drop-down next to **Application Integration**.



22. From the My Services window, select **Application Integration Console**.



23. In the Processes page, refresh the page periodically using the **Refresh** icon and observe the status of the service as **Faulted**.



It may take 2 to 3 minutes for the process to finish running.

This concludes the lab.

Module 11: Fault Handling
Lab 11-2: Handle Credential Failures

Overview:
In this lab, you will introduce an error in the Salesforce connection by changing your account password in Salesforce. You will then create a process that uses the Salesforce connection and using the Catch Faults feature, you will customize the error message in the process.

Objectives:
- Change the Salesforce password to introduce an error
- Create a process
- Enable Fault Handling to simplify the error message

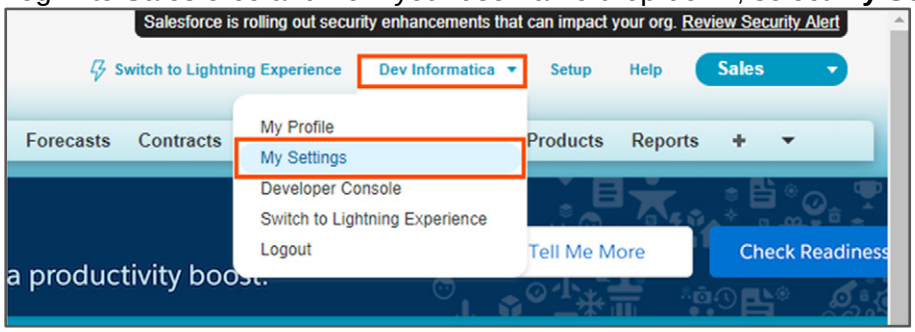
Duration:
15 Minutes

Tasks

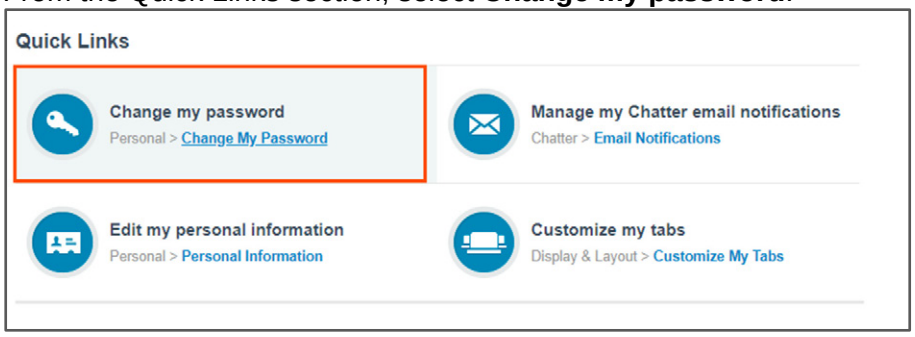
Note: The screen on your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

Change Salesforce Password to Introduce Error

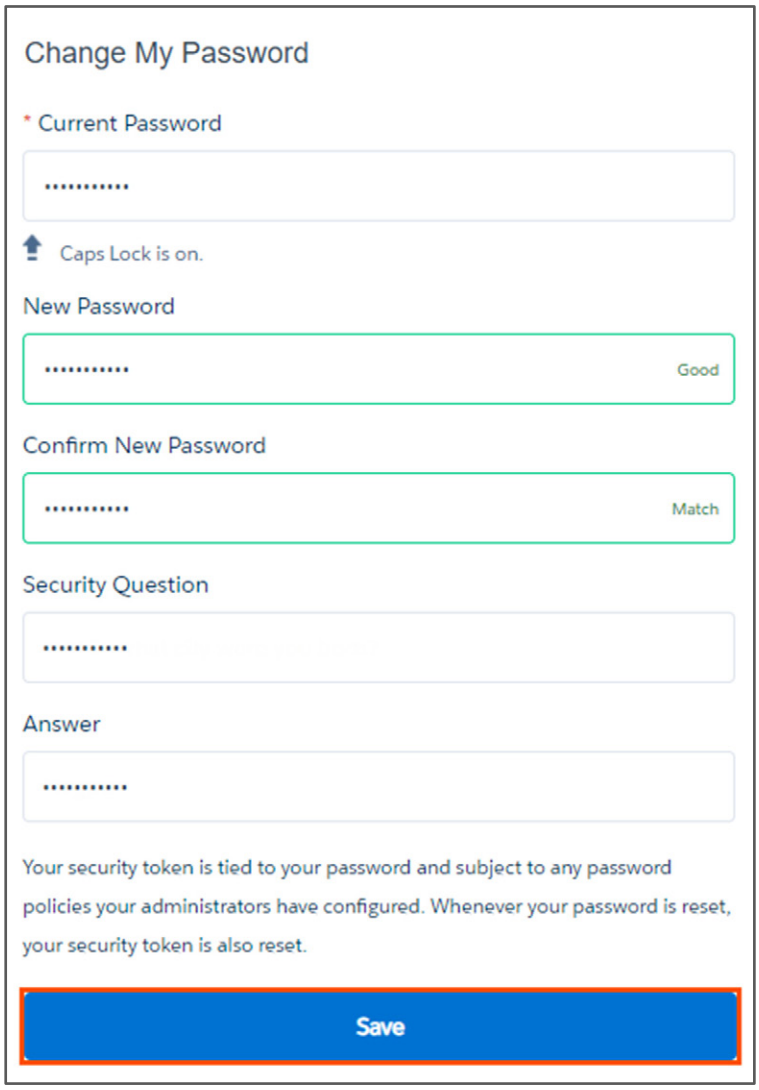
1. In a web browser, open Salesforce using the following URL: https://login.salesforce.com/
2. Log in to Salesforce and from your username drop-down, select **My Settings**.



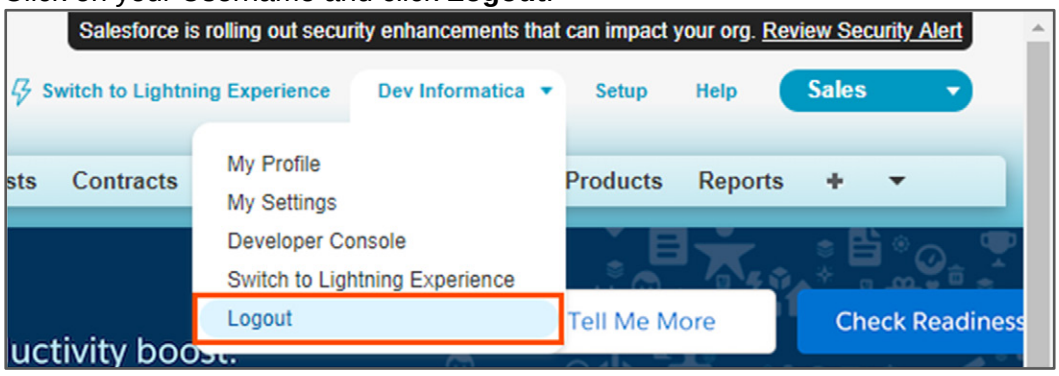
3. From the Quick Links section, select **Change my password**.



4. Enter the details and click **Save**.

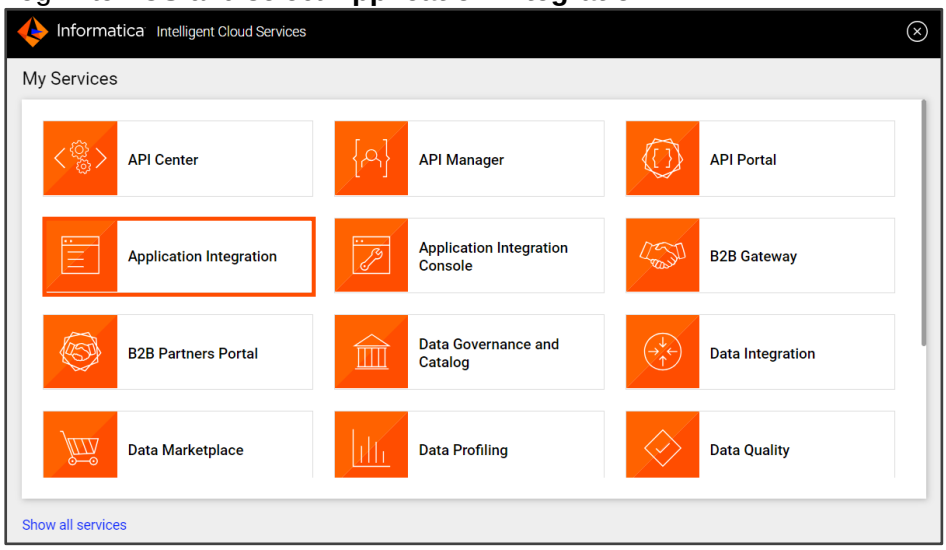


5. Click on your Username and click **Logout**.

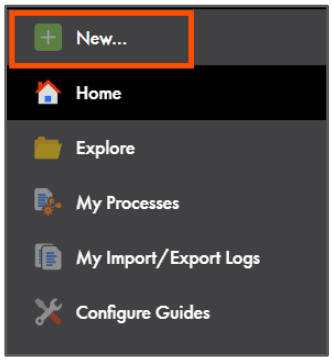


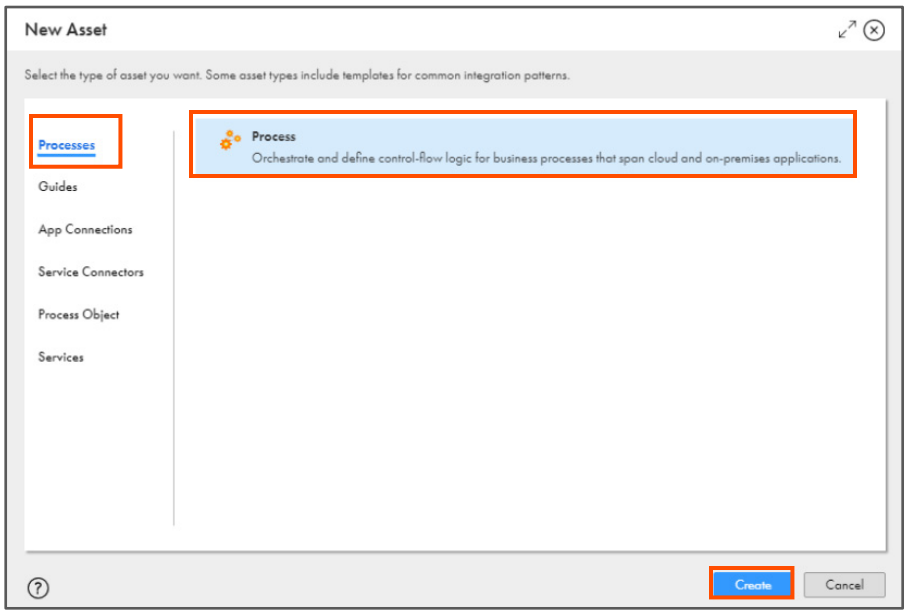
Create a process

6. Log in to IICS and select **Application Integration**.



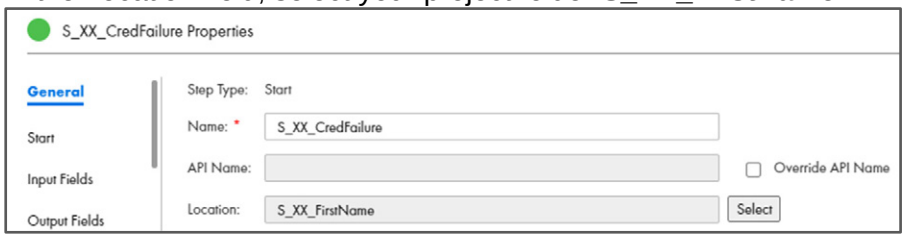
7. Create a new **Process**.



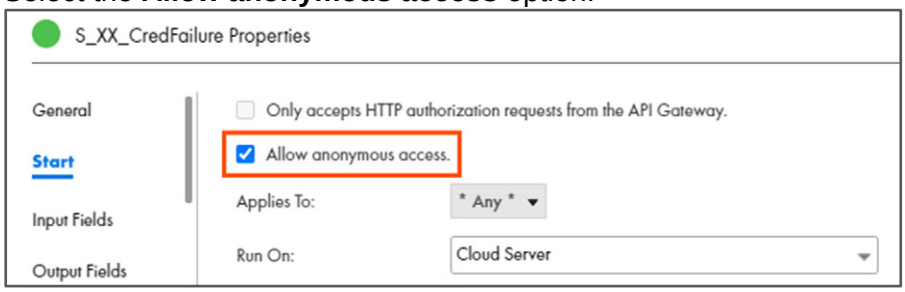


A new Process window appears.

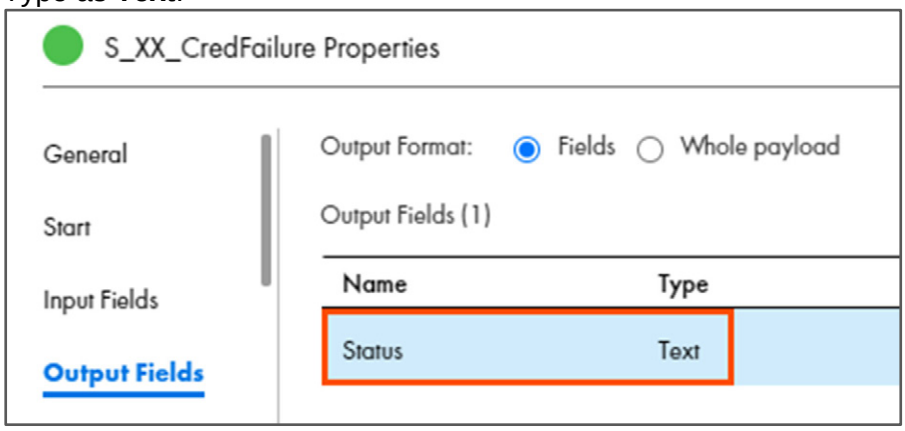
8. From the Process Canvas, select the **Start** step.
9. In the General tab, enter the name as **S_XX_CredFailure**.
10. Skip the API Name option.
11. In the Location field, select your project folder **S_XX_FirstName**.



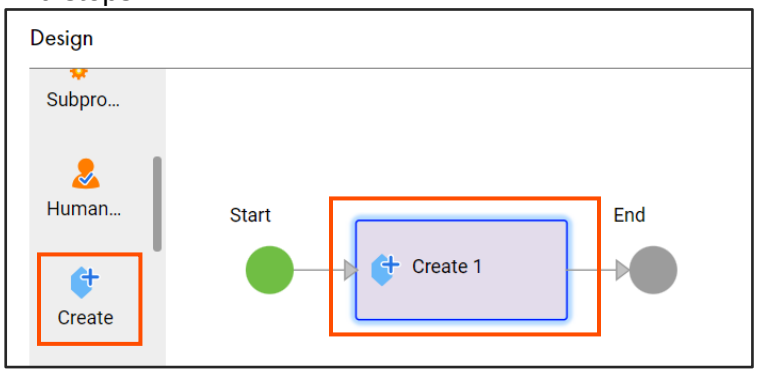
12. From the Start properties pane, select **Start**.
13. Select the **Allow anonymous access** option.



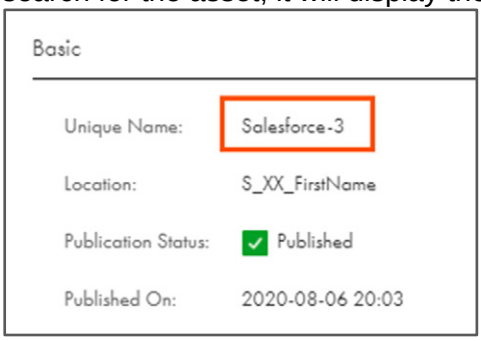
14. In the Output Fields tab, click , and in the Name field, enter Name as **Status** and Type as **Text**.



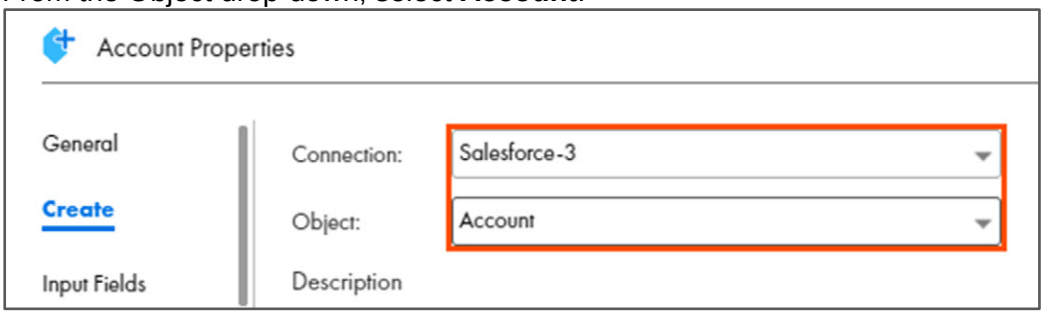
15. From the Design palette, drag and drop a **Create** step on the link between the Start and End steps.



16. In the Create tab, from the Connection drop-down, select **Salesforce**.
    Important: If you see multiple assets with the same name, check the Unique Name of your Salesforce connection. Navigate to **Explore** > **your_project_folder (for example, S_XX_FirstName)** > **Salesforce** connection. Click the Ellipsis icon and from the Properties Detail page of the connection, check the unique name as shown in the screenshot. The unique name and the asset name can be different. So, when you search for the asset, it will display the unique name.

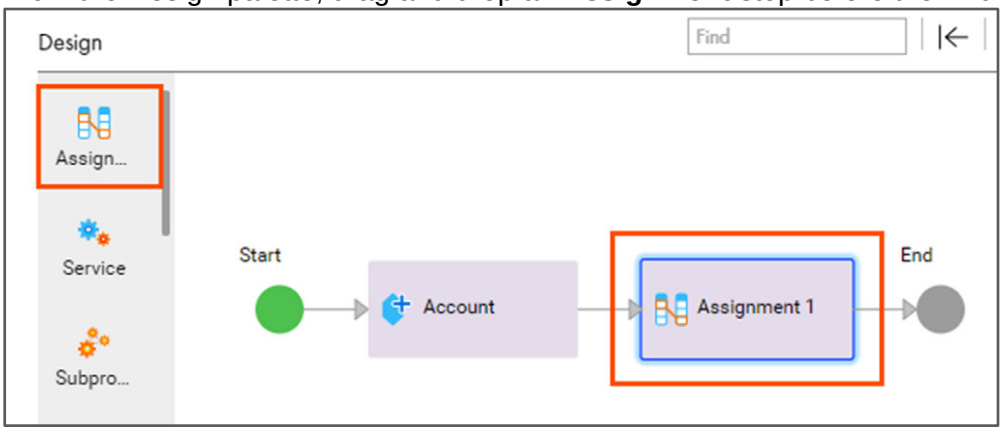


17. From the Object drop-down, select **Account**.

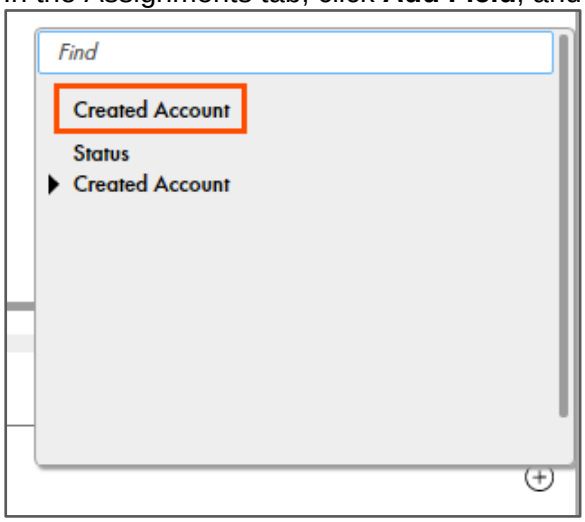


18. In the Input Fields tab, for the Account Name field, enter **S-XX-Test**.

19. From the Design palette, drag and drop an **Assignment** step before the End step.

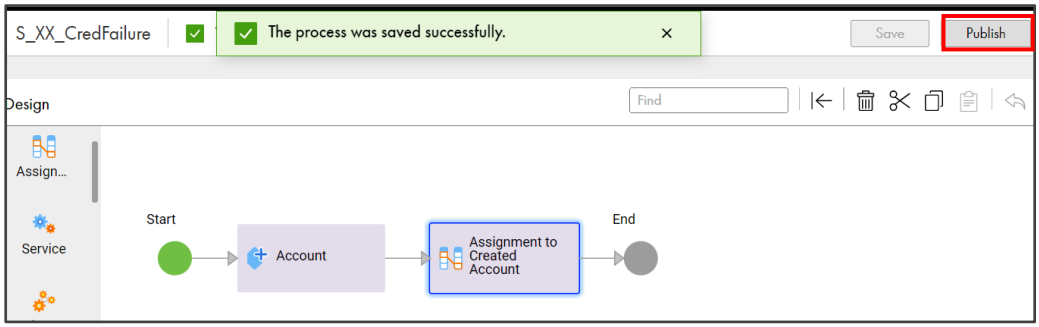


20. In the Assignments tab, click **Add Field**, and select **Created Account**.

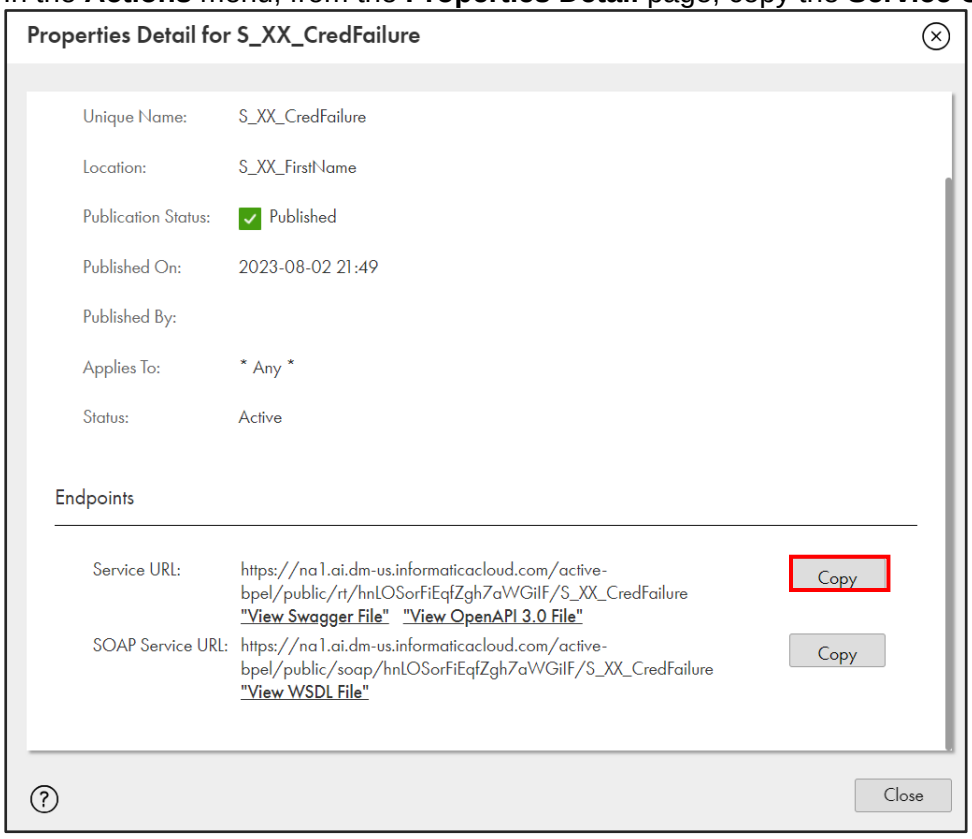


21. From the Assigned Using drop-down, select **Field**, and from the **From** drop-down, select **Created Account** > **Account ID**.

22. **Save** and **Publish** the process.

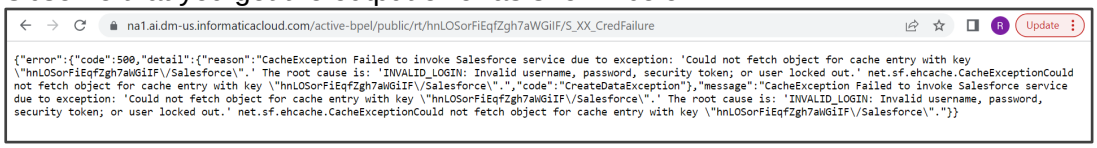


23. In the **Actions** menu, from the **Properties Detail** page, copy the **Service URL**.



24. In a web browser, enter the copied Service URL.

25. Observe that you get the output error as shown below:

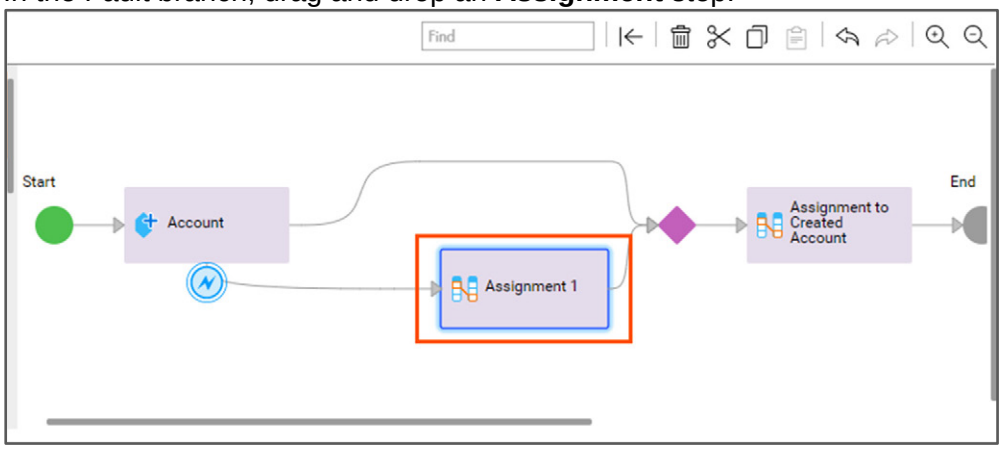


Note: To catch the faults, you must enable Fault Handling.

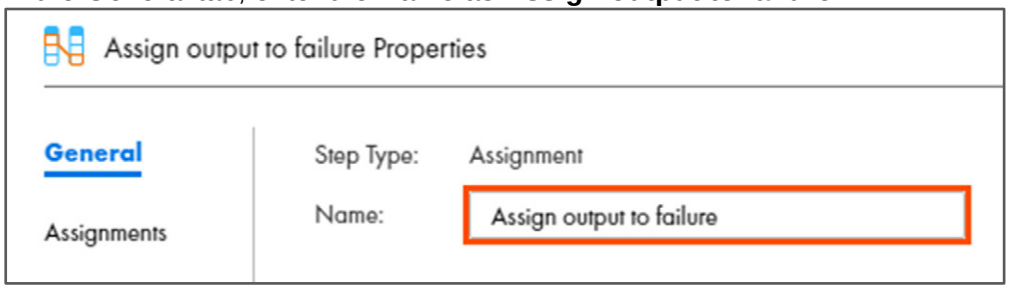
26. Navigate back to the CAI console and close the Properties Detail window.
27. In the **S_XX_CredFailures** process, select the **Create** (Account) step.

28. From the Properties Pane, select **Fault Handling** and select the **Catch faults** checkbox.

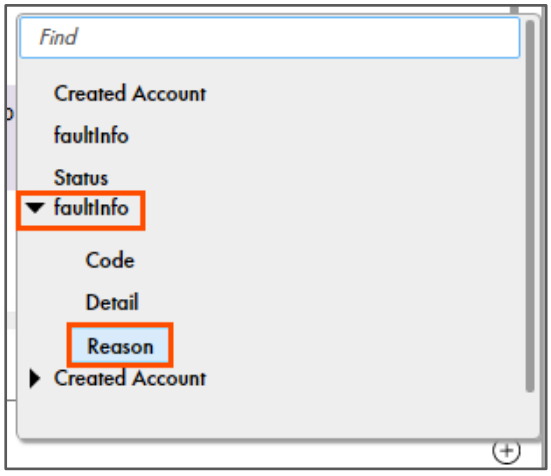
29. In the Fault branch, drag and drop an **Assignment** step.



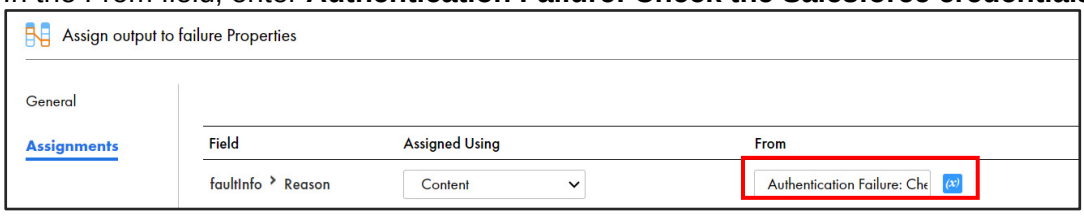
30. In the General tab, enter the Name as **Assign output to failure**.



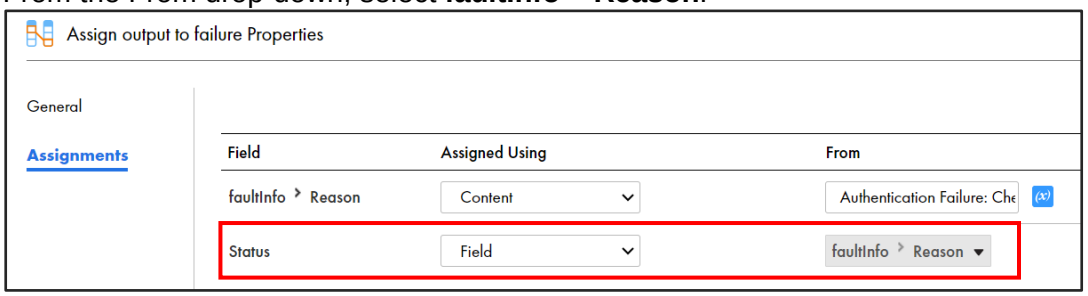
31. In the Assignments tab, click plus (+) icon, and from the **faultInfo** drop-down, select **Reason**.



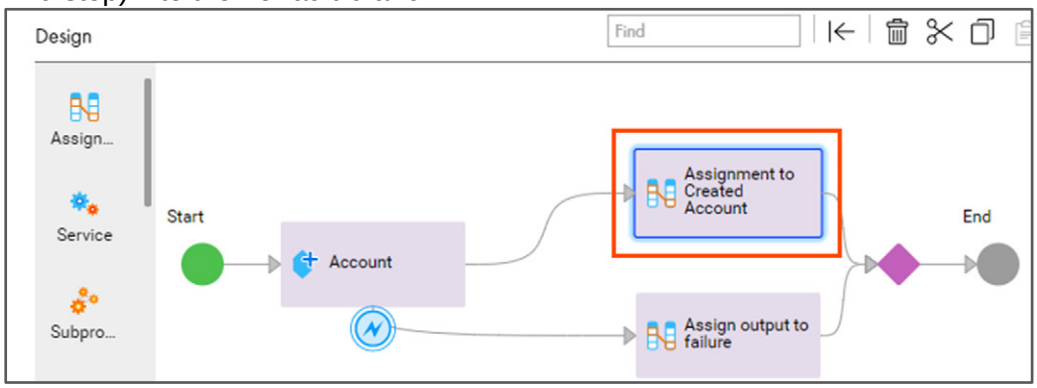
32. In the From field, enter **Authentication Failure: Check the Salesforce credentials**.



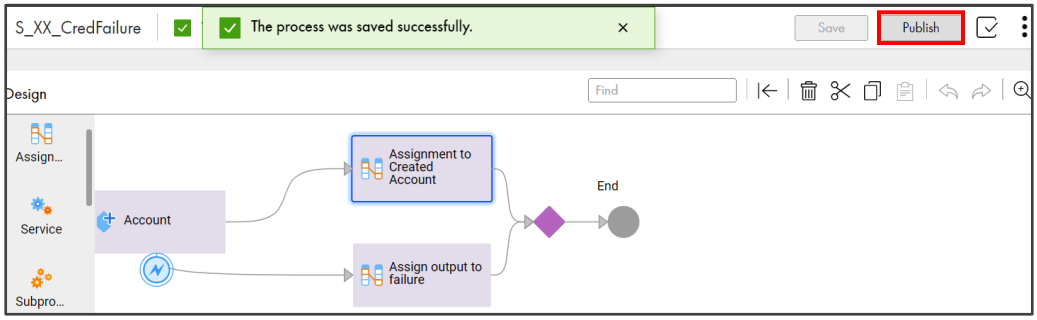
33. To add the Status field, click **Add Field**, and select **Status**.
34. From the Assigned Using drop-down, select **Field**
35. From the From drop-down, select **faultInfo** > **Reason**.



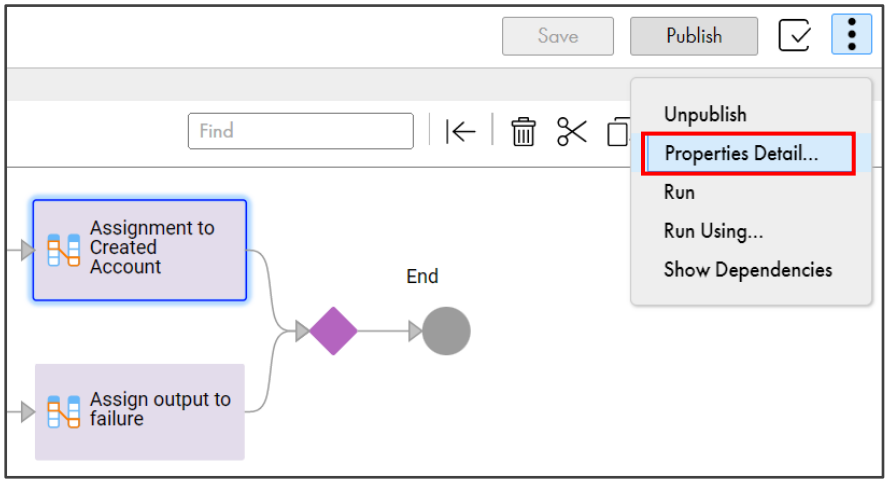
36. Drag and drop the **Assignment to Created Account** step (that is present before the End step) into the no-fault branch.



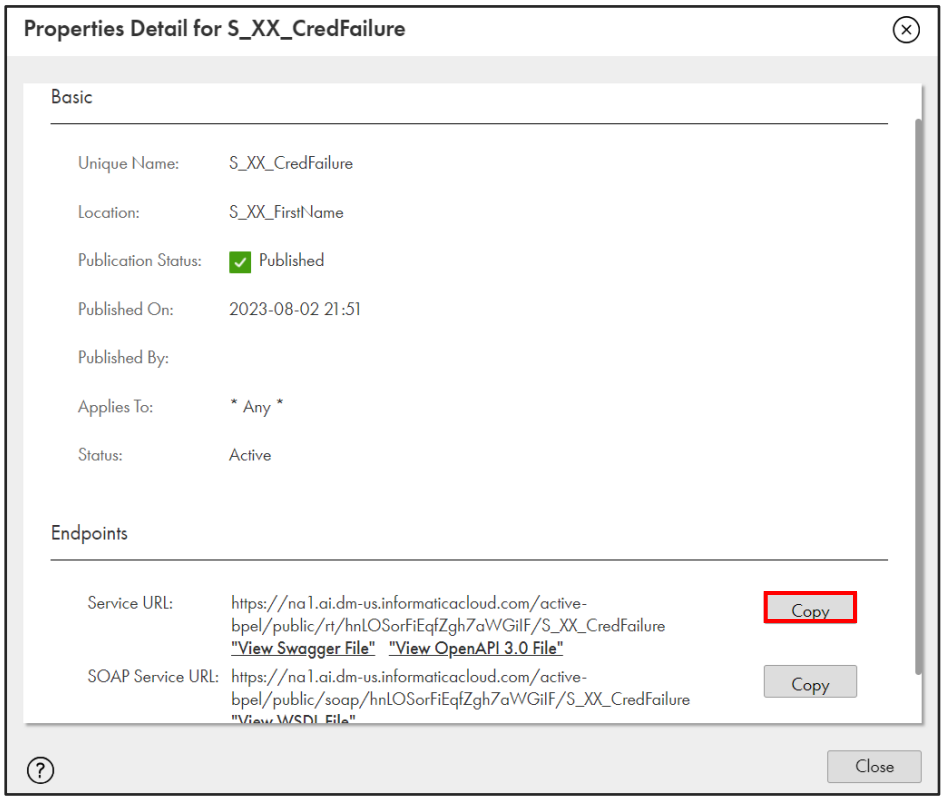
37. **Save** and **Publish** the process.



38. From the Action menu, select **Properties Detail…**.

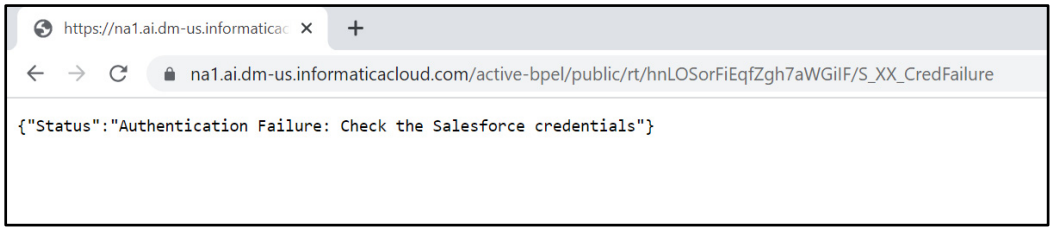


39. From the **Properties Detail** page, copy **Service URL**.



40. In a web browser, enter the copied Service URL.

You get the customized error message as shown below:



41. **Close** the browser tab.

This concludes the lab.

Module 12: Introduction to Guides Designer
Lab 12-1: Create a Simple Guide

Overview:
A Guide is a set of screens that helps a user to implement business logic. It prompts the users to enter, review, or confirm data. Guides run within mobile apps or on traditional platforms such as a PC or a Mac.

In this lab, you will create a simple calculator guide and run it through the interface. You will specify the operations to be performed on the specified inputs and obtain an output based on the selected operation.

Objectives:

- Create a Guide
- Run the Guide through the Interface

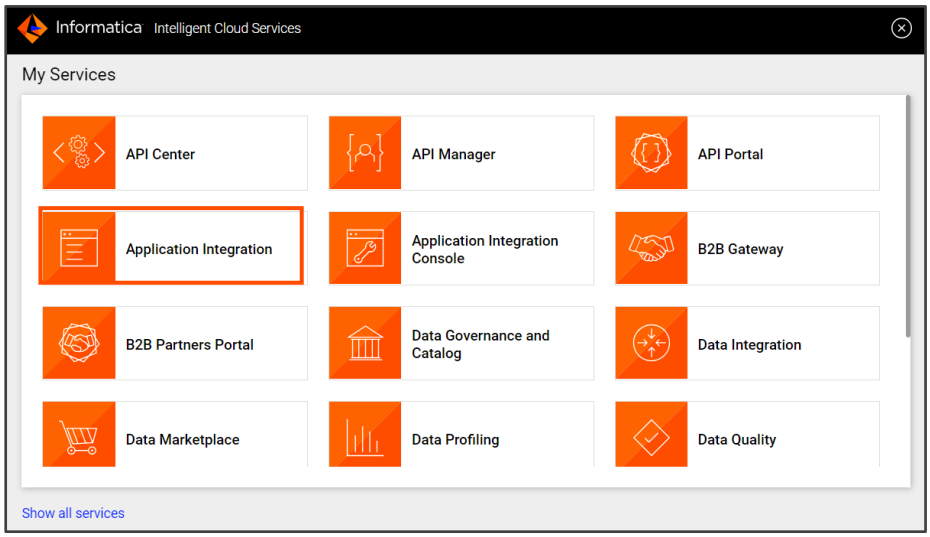
Duration:
20 Minutes

Tasks

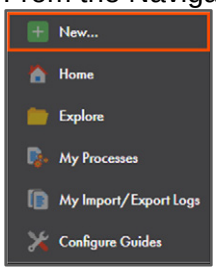
Note: The screen on your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

Create a New Guide

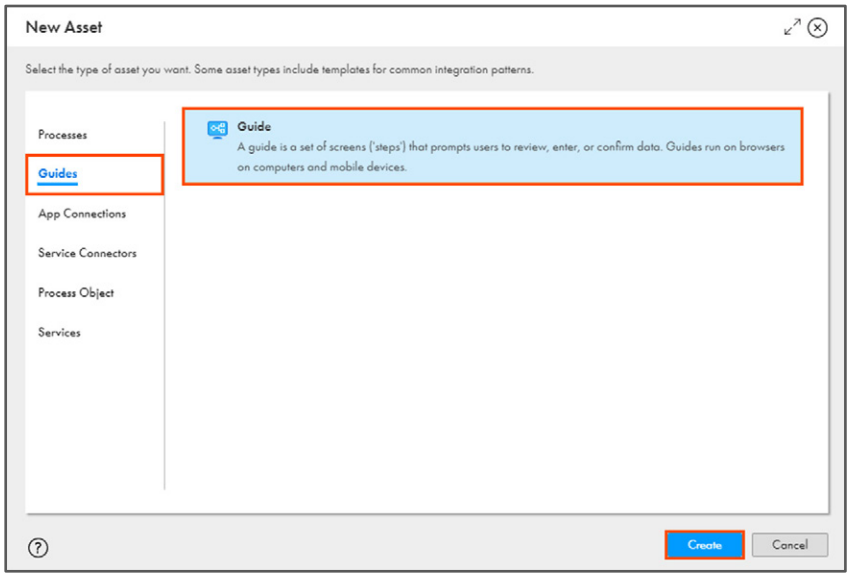
1. Log in to IICS and access the **Application Integration** service.



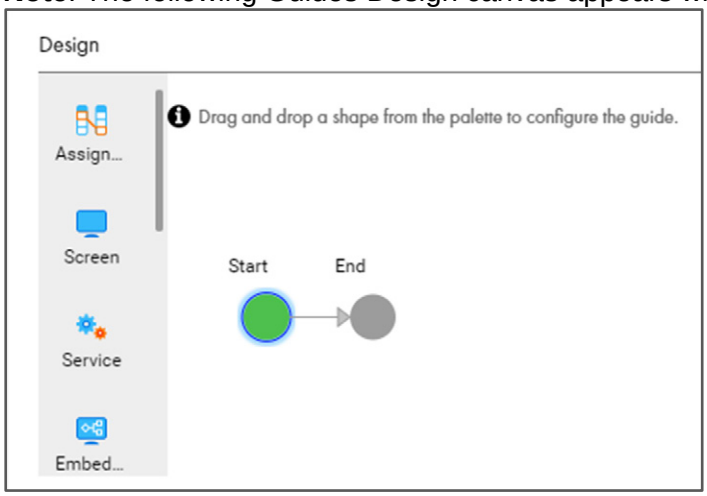
2. From the Navigation pane, click **New**.



3. In the New Asset window, from the **Guides** tab, retain the **Guide** option, and click **Create**.

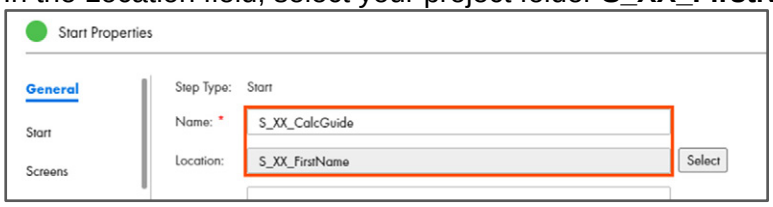


Note: The following Guides Design canvas appears with the Start and End steps.

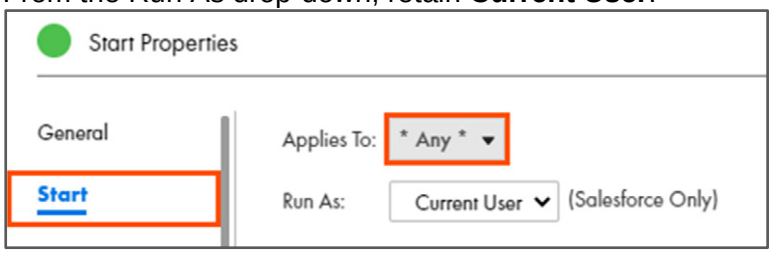


4. From the canvas, select the **Start** step.
5. In the General tab, enter the name as **S_XX_CalcGuide**.

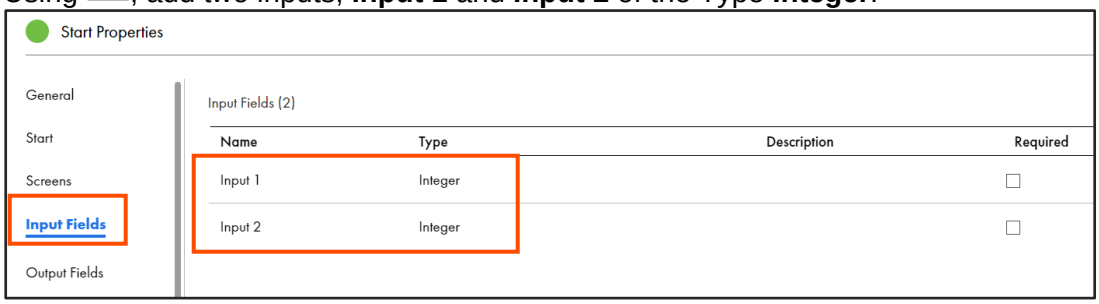
6. In the Location field, select your project folder **S_XX_FirstName**.



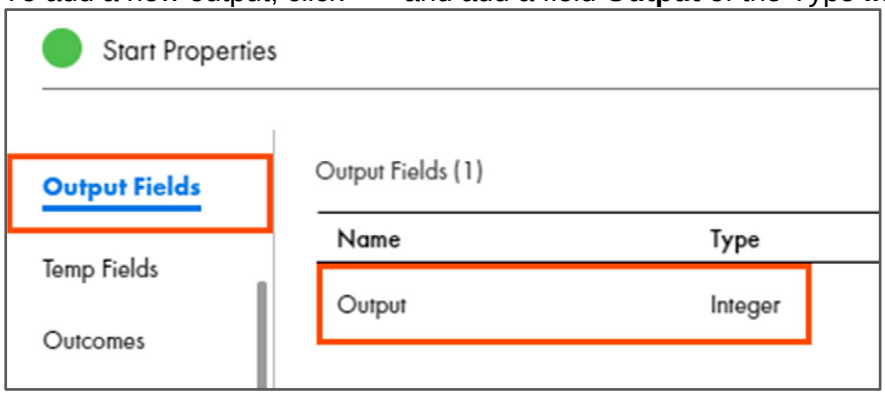
7. From the Start properties pane, select **Start**.
8. From the Applies To drop-down, select **Any**.
9. From the Run As drop-down, retain **Current User**.



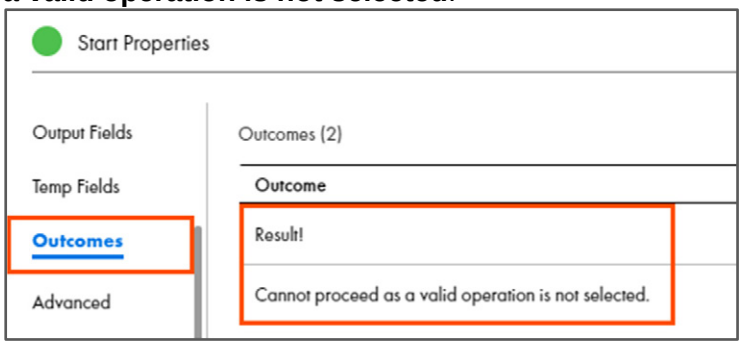
10. To configure the input fields, select the **Input Fields** tab.
11. Using , add two inputs, **Input 1** and **Input 2** of the Type **Integer**.



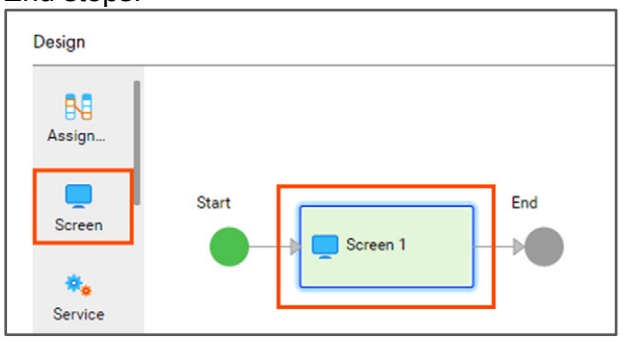
12. From the properties pane, select **Output Fields**.
13. To add a new output, click and add a field **Output** of the Type **Integer**.



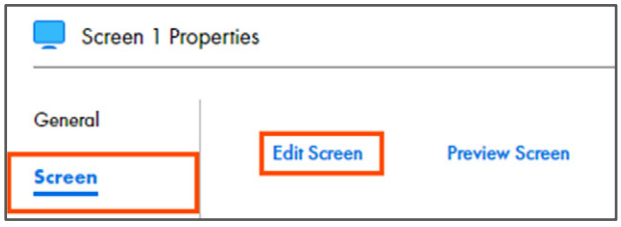
14. In the Outcomes tab, click and add two outcomes: **Result!** and **Cannot proceed as a valid operation is not selected**.



15. From the Design palette, drag and drop a **Screen** step on the link between Start and End steps.

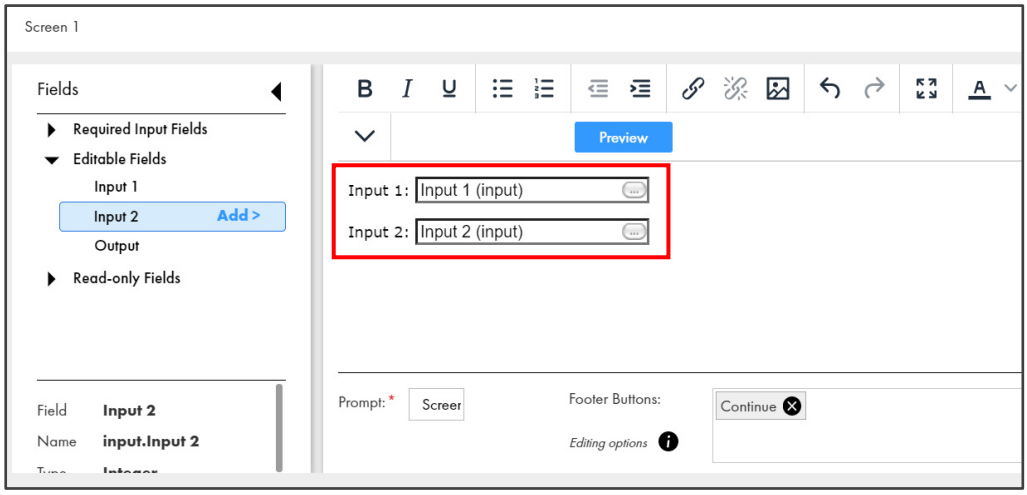


16. To configure the step, select the **Screen** step.
17. From the properties pane, select **Screen**.
18. Select **Edit Screen**.



19. In the screen editor, from the Fields section, expand the **Editable Fields** option.
20. Select **Input 1** and click **Add**.

21. Similarly, add **Input 2** and configure the screen as shown in the screenshot:

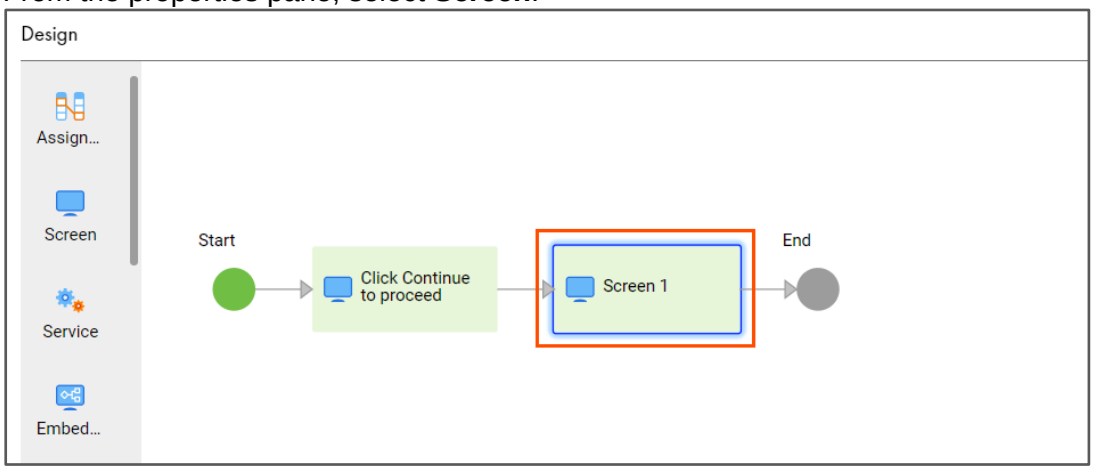


Note: To bring Input 2 to the next line, place the cursor after the Input 1 (input) box and press **Enter**. Add a **:** between the field name and the field. The screen editor looks as shown in the screenshot above.

22. In the Prompt field, enter **Click Continue to proceed.**
23. In the Footer Buttons field, retain the **Continue** button and click **OK**.

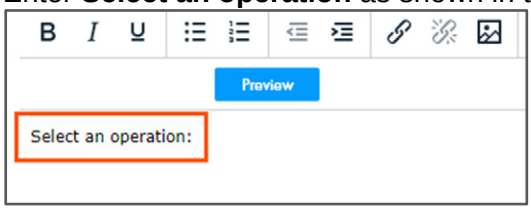
Note: If you notice multiple Continue buttons, remove them by clicking the ⊗ icon.

24. Add another Screen step.
25. From the properties pane, select **Screen**.

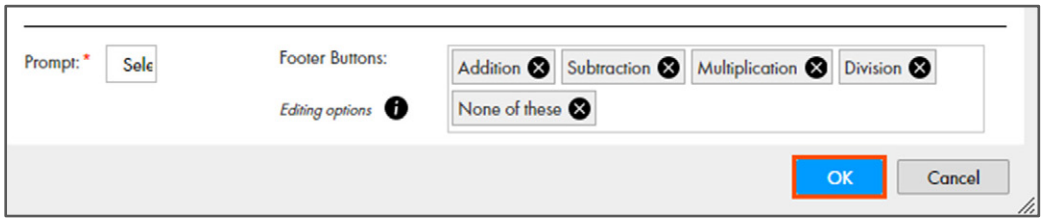


26. Select **Edit Screen**.

27. Enter **Select an operation** as shown in the screenshot:

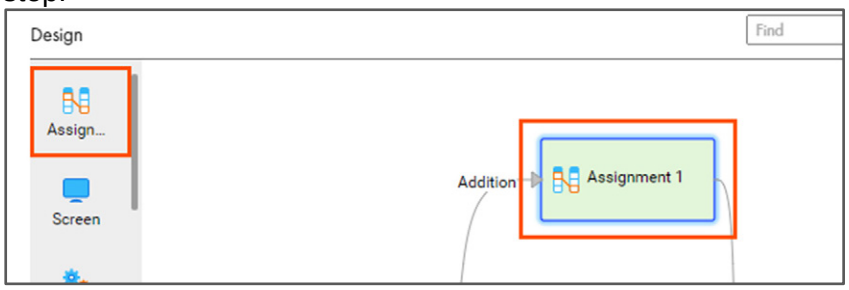


28. In the Prompt field, enter **Select any one of the following**:
29. In the Footer Buttons field, remove the **Continue** button.
30. To add a button, in the Footer Buttons field, enter **Addition**, and press **enter**.
31. Similarly, add **Subtraction**, **Multiplication**, **Division**, and **None of these** buttons and click **OK**.

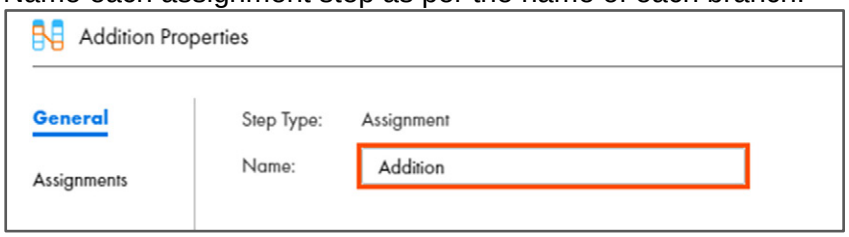


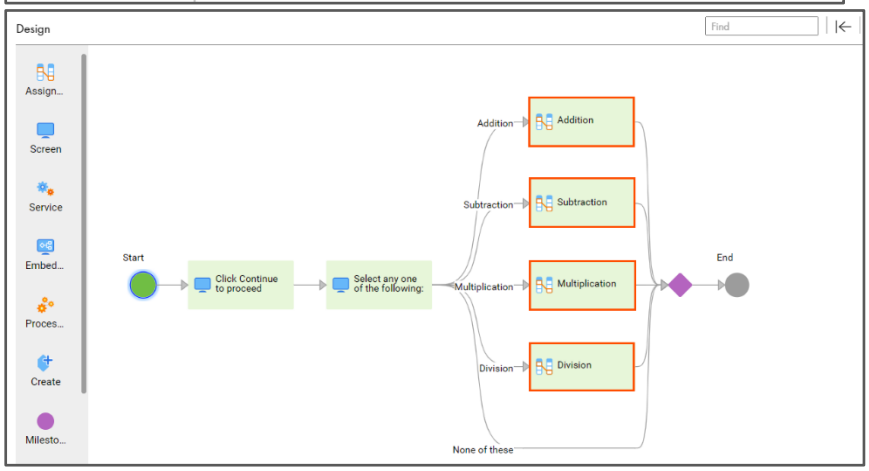
Notes: If the Continue button still appears, go back to the Edit Screen step, remove the Continue button, and click **OK**. Also, observe that the Guide creates a branch for each button.

32. For each branch, except the None of these branches, drag and drop an **Assignment** step.

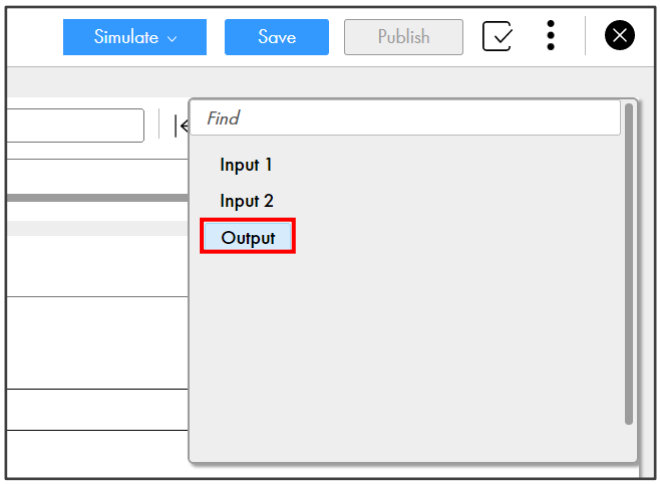


33. Name each assignment step as per the name of each branch.

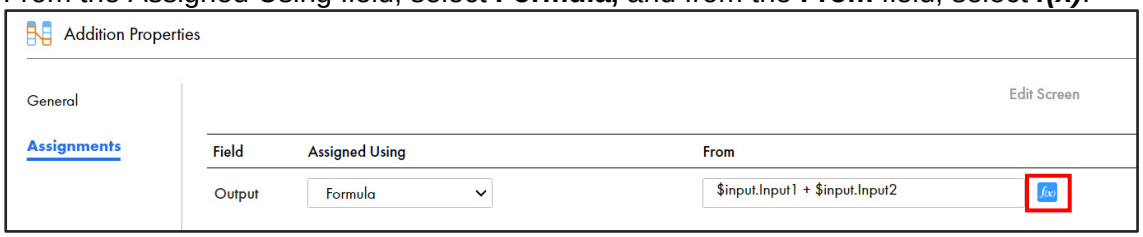




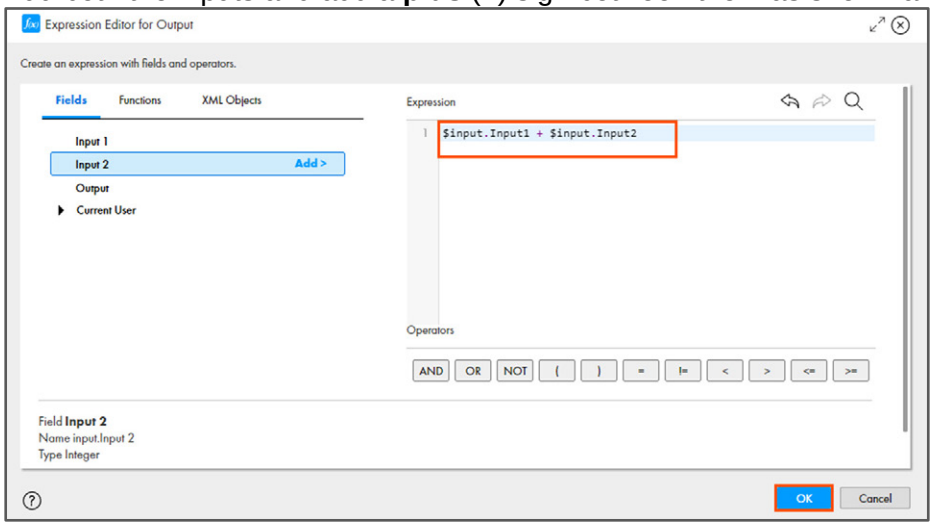
34. Select the **Addition** assignment step.
35. In the Assignments tab, click the plus **'+'** icon and select **Output**.



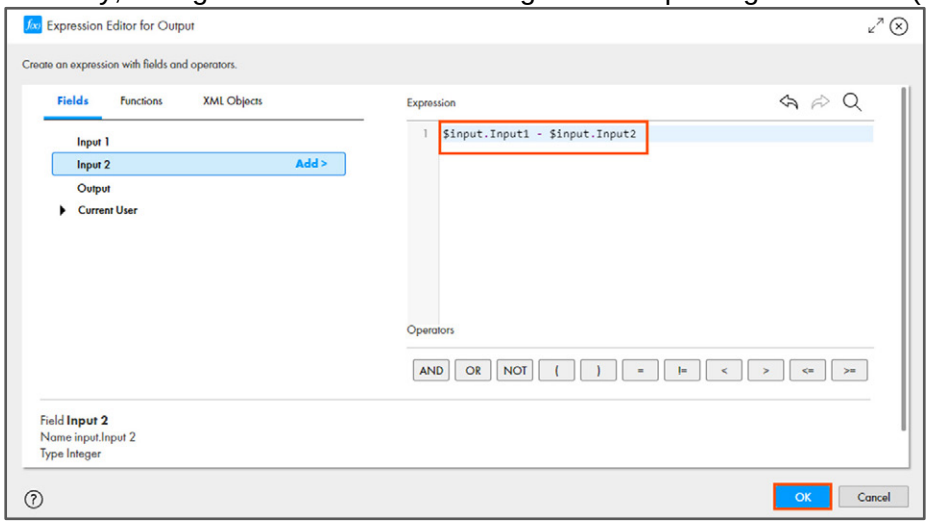
36. From the Assigned Using field, select **Formula,** and from the **From** field, select ***f(x)***.



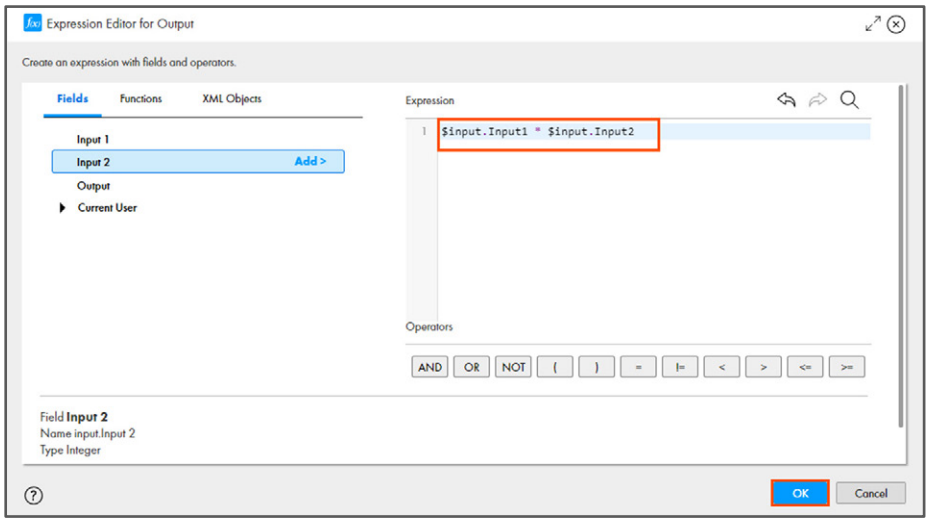
37. Add both the inputs and add a **plus** (+) sign between them as shown and click **OK**.



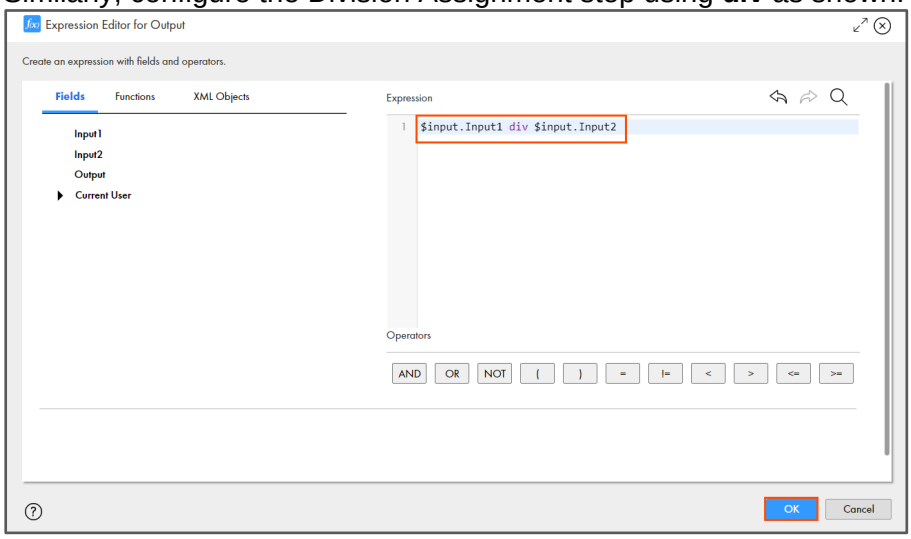
38. Similarly, configure the Subtraction Assignment step using the **minus** (-) sign as shown:



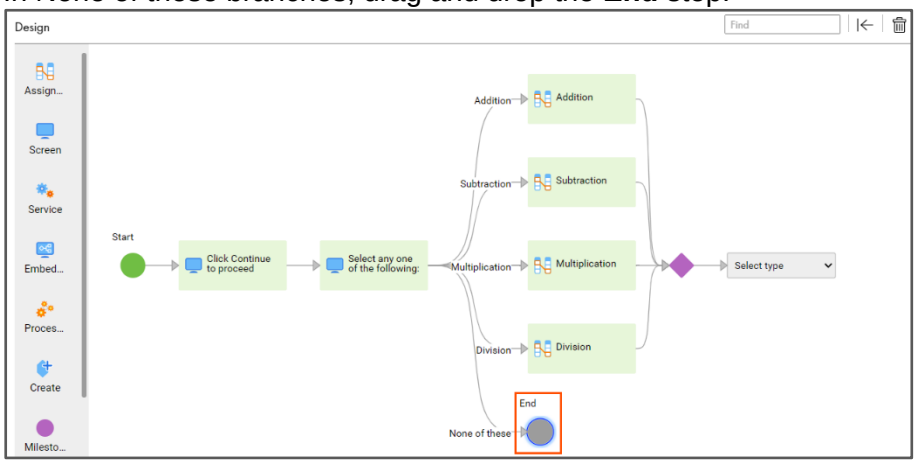
39. Similarly, configure the Multiplication Assignment step using the **asterisk** (*) sign as shown:



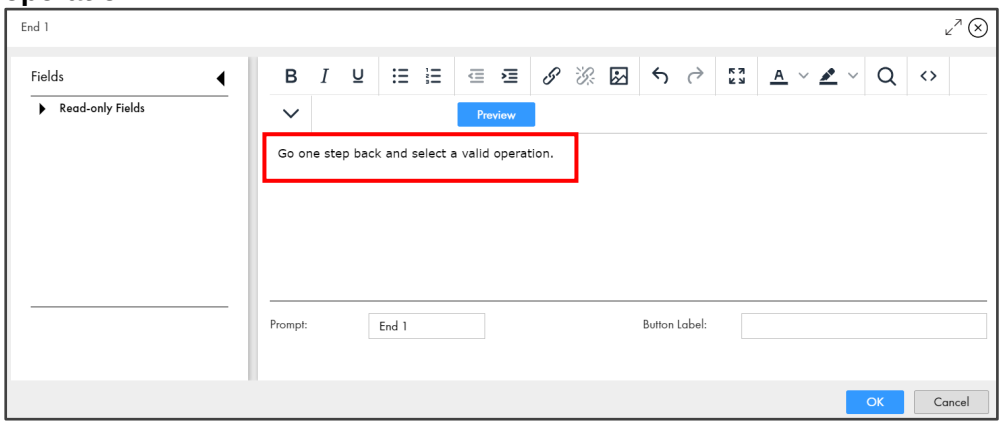
40. Similarly, configure the Division Assignment step using **div** as shown:



41. In None of these branches, drag and drop the **End** step.

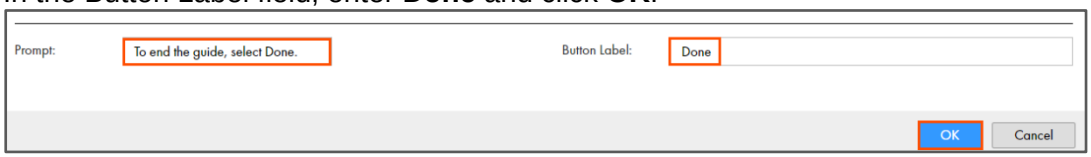


42. In the End tab, from the Outcome drop-down, select **Cannot proceed as a valid operation is not selected**.
43. Select **Edit Screen** and configure the screen as **Go one step back and select a valid operation**.

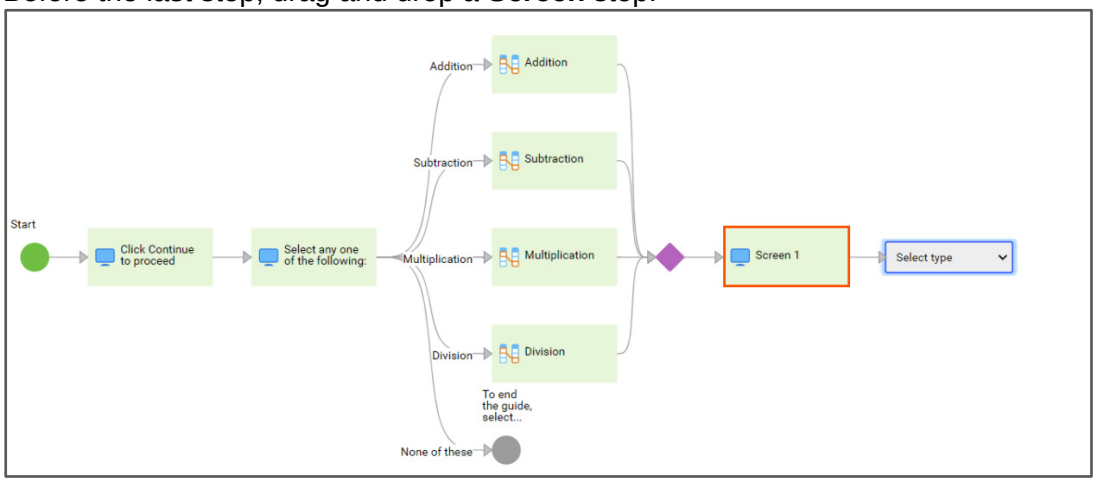


44. In the Prompt field, enter **To end the guide, select Done**.

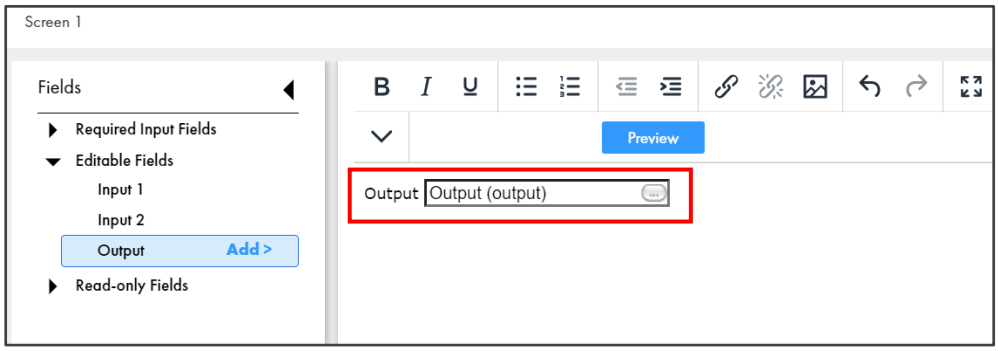
45. In the Button Label field, enter **Done** and click **OK**.



46. Before the last step, drag and drop a **Screen** step.

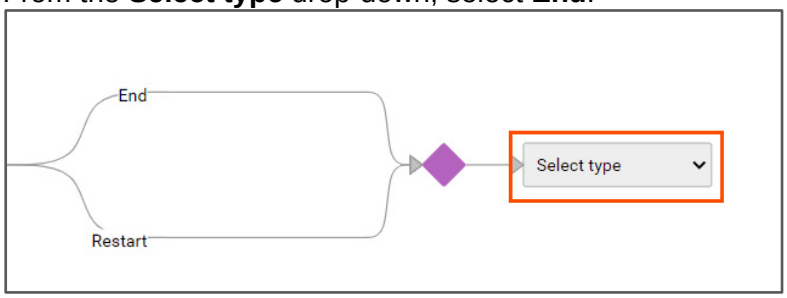


47. In the Screen tab, select **Edit Screen**.
48. From the Editable fields drop-down, select **Output**, click **Add**, and configure the screen as shown:

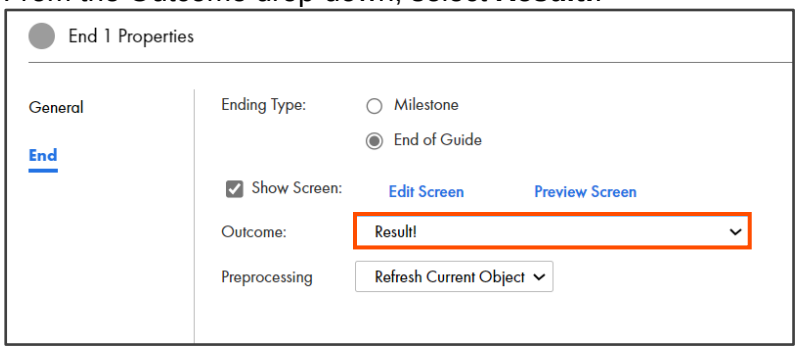


49. In the Prompt field, enter **Select an option to proceed**.
50. In the Footer Buttons field, remove **Continue** and add two buttons, **End** and **Restart** and click **OK**.

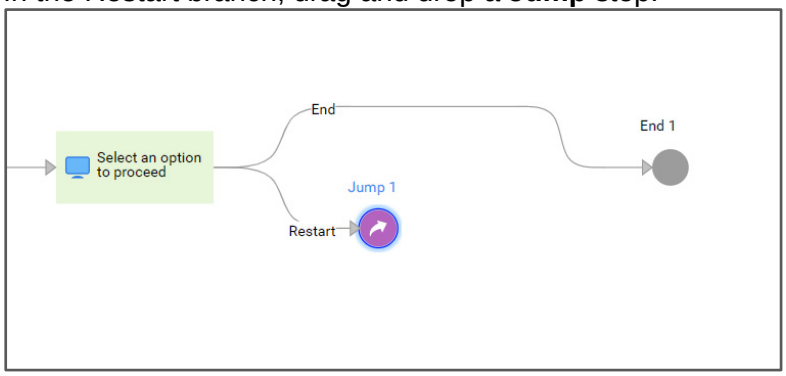
51. From the **Select type** drop-down, select **End**.



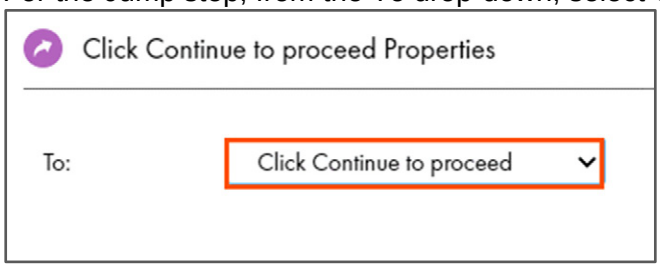
52. For the End branch, select the **End** tab.
53. From the Outcome drop-down, select **Result!**.



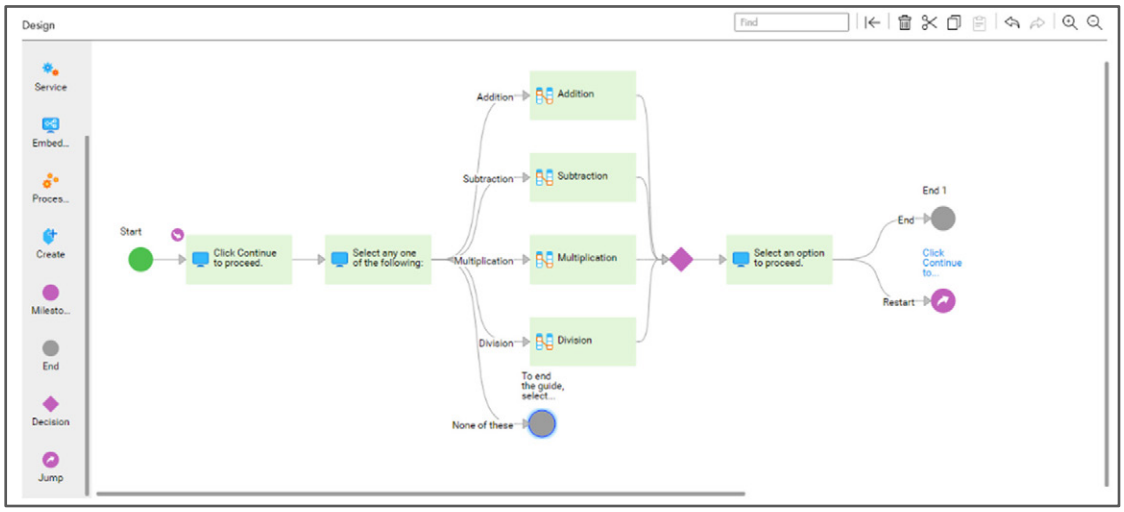
54. In the Restart branch, drag and drop a **Jump** step.



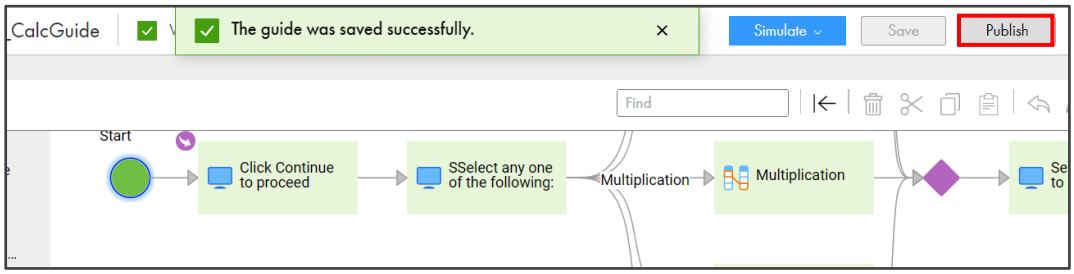
55. For the Jump step, from the To drop-down, select **Click Continue to proceed**.



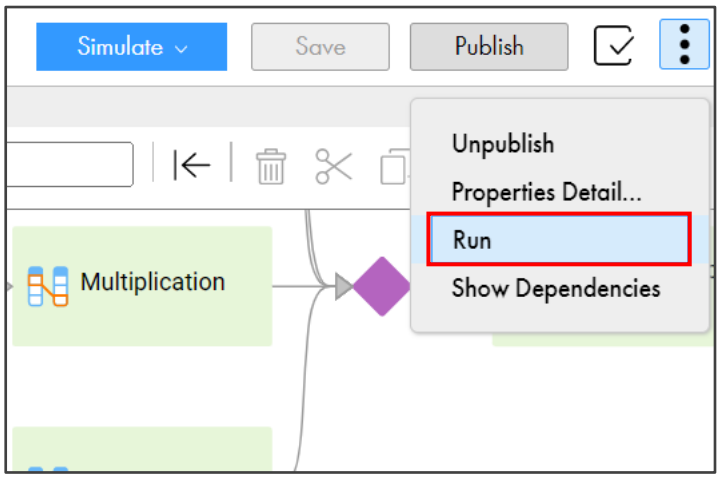
56. The Guide should look as shown in the screenshot:



57. **Save** and **Publish** the Guide.

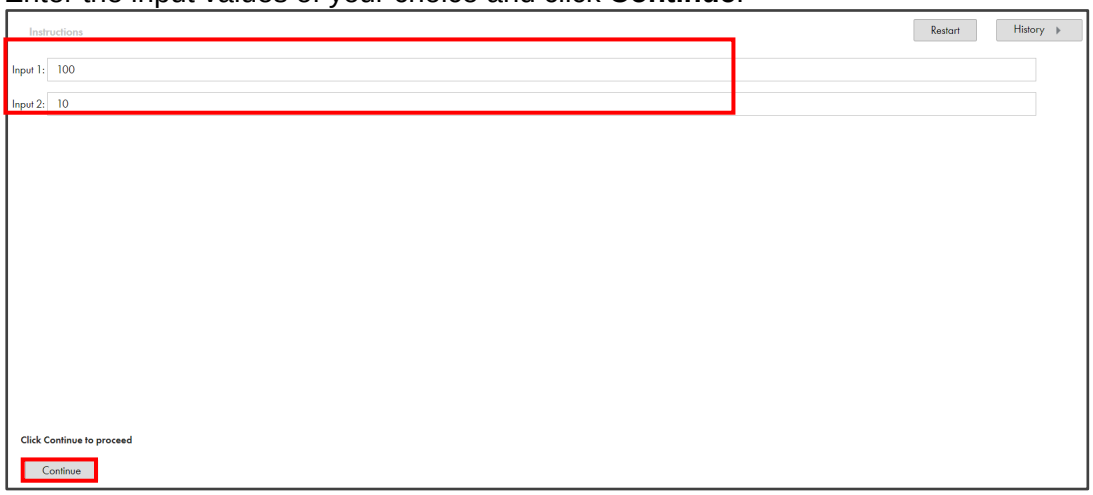


58. To execute the Guide, from the Actions menu, click **Run**.

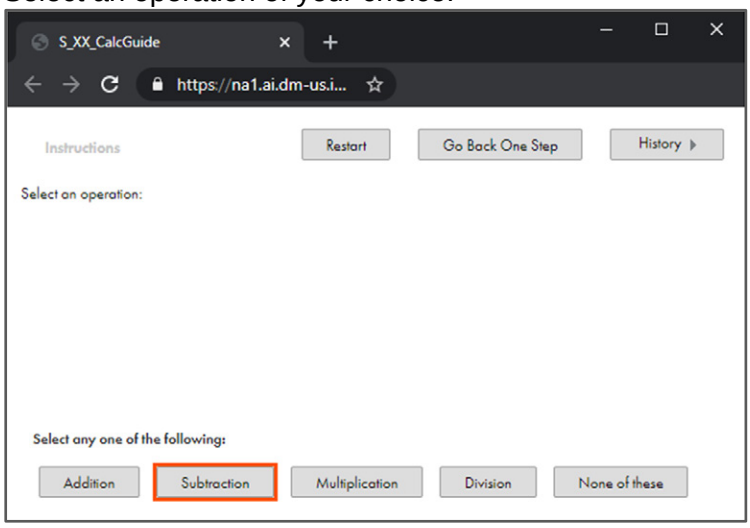


This opens the Guide in a new browser tab.

59. Enter the input values of your choice and click **Continue**.

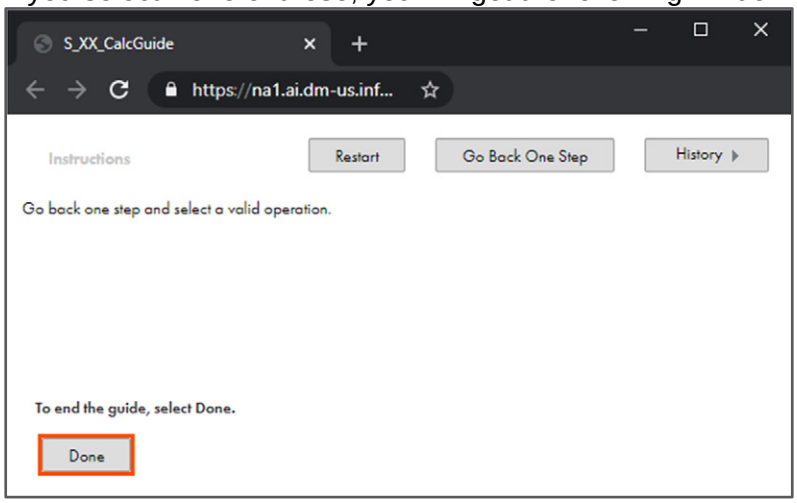


60. Select an operation of your choice.



61. In the next screen, you will get the output.

62. If you select None of these, you will get the following window:



This concludes the lab.

Module 12: Introduction to Guides Designer
Lab 12-2: Create a Guide to Invoke the Exchange Rate Process

Overview:

In this lab, you will create a Guide for the currency conversion using the exchange rate process that you have created in the **Lab 6-3: Using Currency Exchange Data Access Service Connector in a Process**. To do this, first you will create a Process Object. You will use this Process Object to create a Guide.

Objectives:

- Create a Process Object
- Create a Guide using the exchange rate process

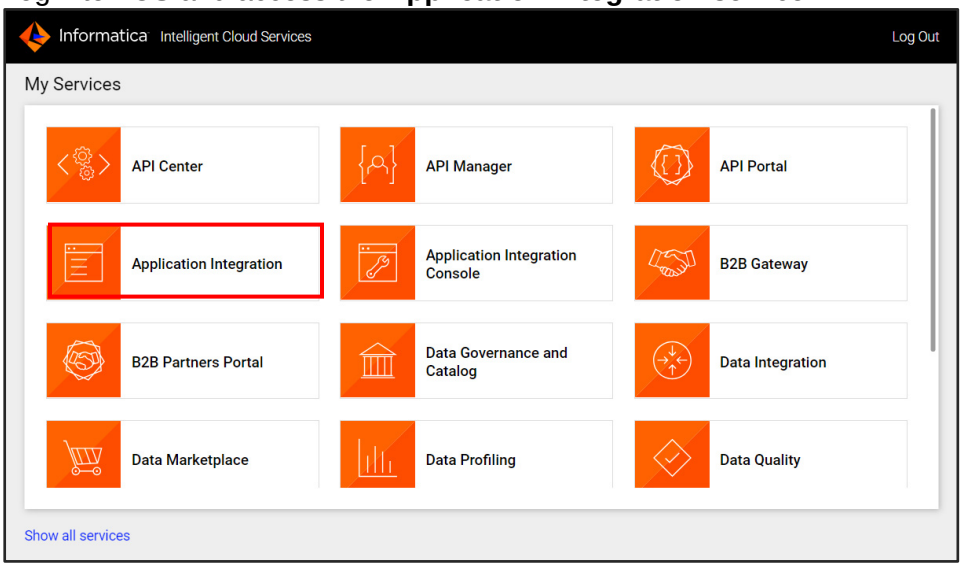
Duration:

45 Minutes

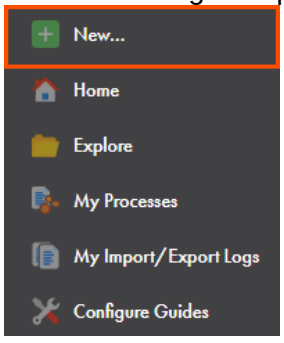
Tasks

Create a Process Object:

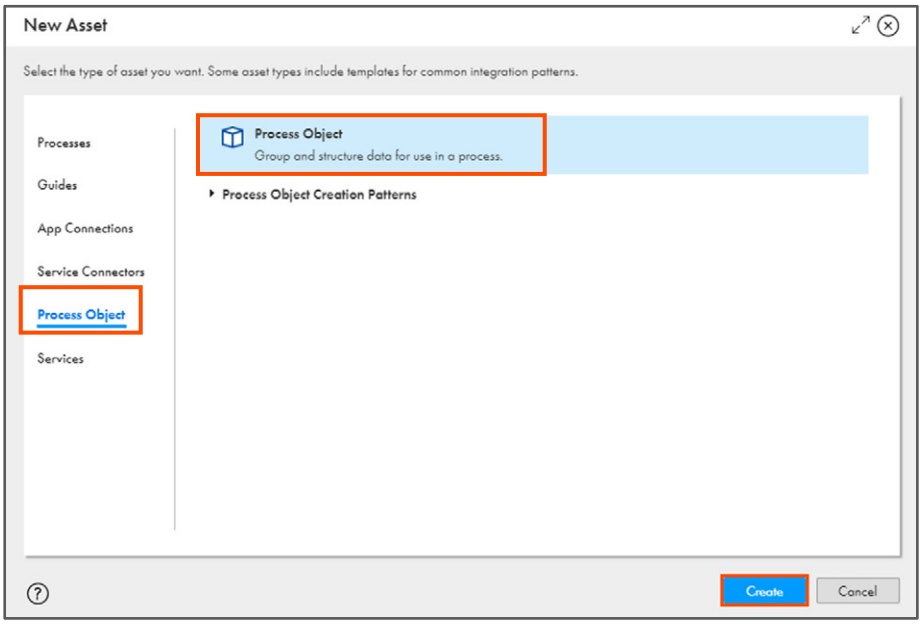
1. Log into IICS and access the **Application Integration** service.



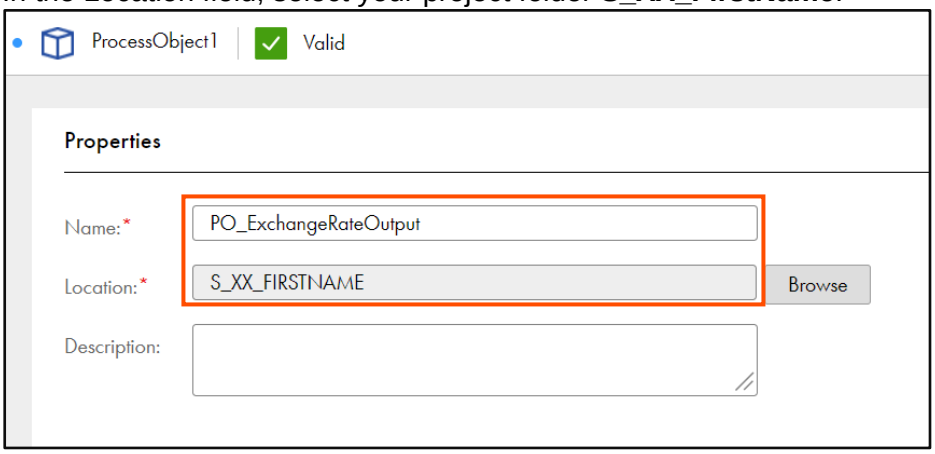
2. From the Navigation pane, click **New**.



3. In the New Asset window, from the Process Object tab, select **Process Object**, and click **Create**.

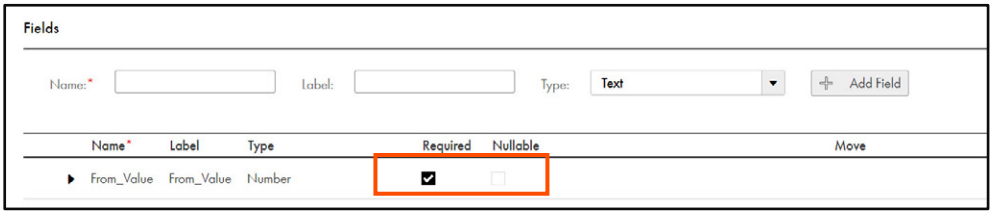


4. In the Process Object Properties page, in the Name field, enter **PO_ExchangeRateOutput_XX**. Replace XX with your student number.
5. In the Location field, select your project folder **S_XX_FirstName**.



6. For the first field, enter Name and Label as **From_Value**.
7. From the Type drop-down, select **Number**.
8. Click **Add Field**.

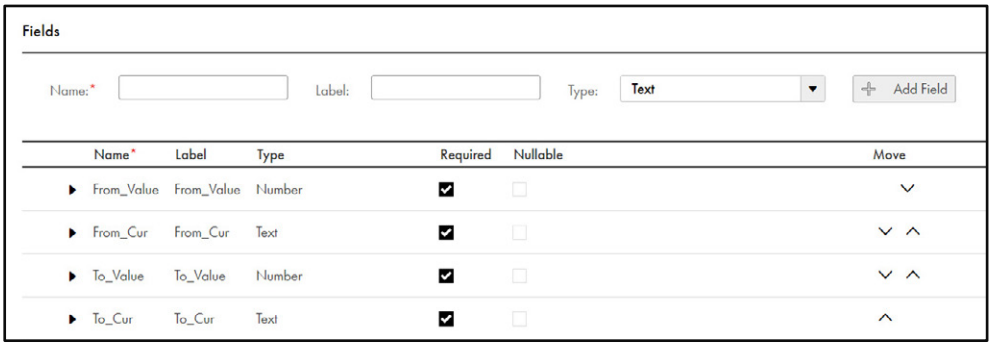
9. Check the **Required** checkbox.
10. Uncheck the **Nullable** checkbox.



11. Similarly, add three more fields.

Name	Label	Type	Required	Nullable
From_Cur	From_Cur	Text	Check	Uncheck
To_Value	To_Value	Number	Check	Uncheck
To_Cur	To_Cur	Text	Check	Uncheck

Your fields should look like this:



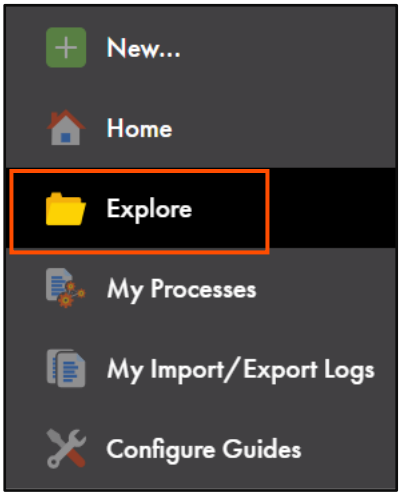
12. **Save** the Process Object.

You have successfully designed a Process Object.

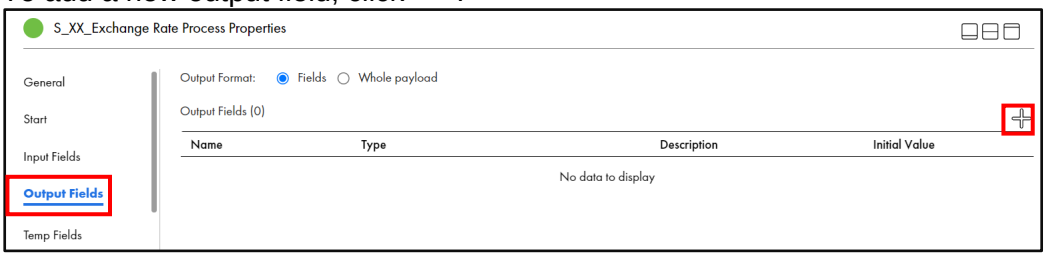
13. Close the Process Object.

Edit S_XX_Exchange Rate Process

14. From the navigation pane, click **Explore**.

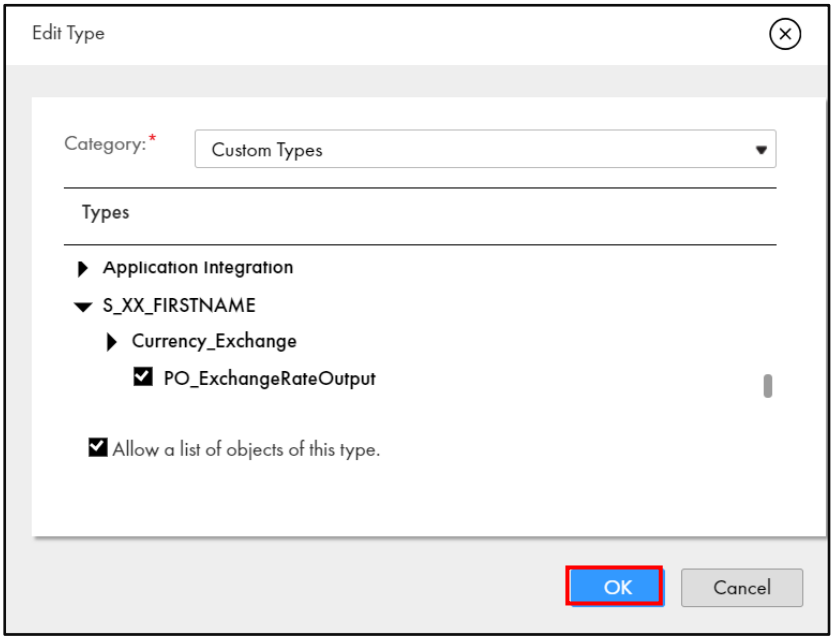


15. From your working folder, open the process **S_XX_Exchange Rate Process**.
16. From the process canvas, select the **Start** step.
17. From the properties pane, select **Output Fields**.
18. To add a new output field, click .



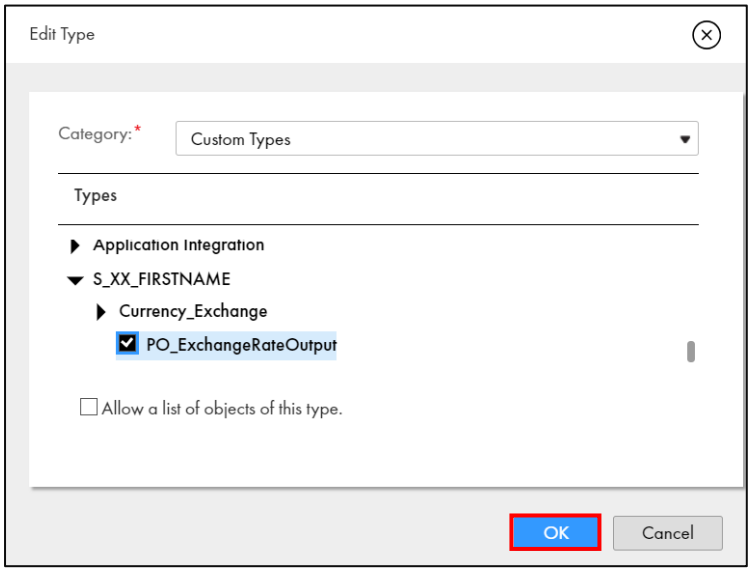
19. In the Name field, enter **Exchange_Response_List**.
20. From the Type drop-down, select **More types**.
21. From the category drop-down, select **Custom Types**.
22. Expand your project folder S_XX_FIRSTNAME and select the **PO_ExchangeRateOutput** Process Object.
23. Check the **Allow a list of objects of this type** checkbox.

24. Click **OK**.



Now, you will add a temp field.

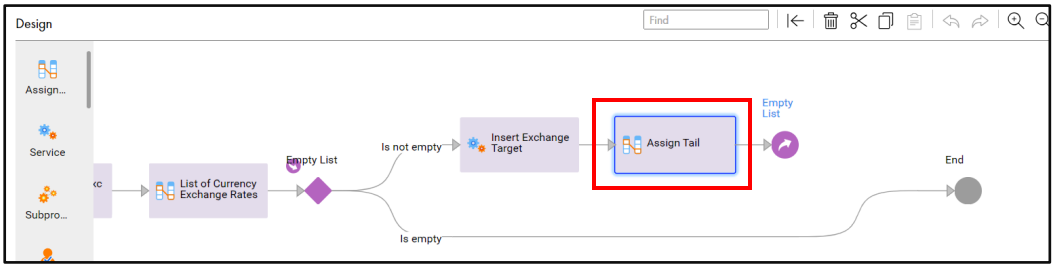
25. From the navigation pane, click **Temp Fields**.
26. To add a new temp field, click .
27. In the Name field, enter **Exchange_Response_Object**.
28. From the Type drop-down, select **More types**.
29. From the category drop-down, select **Custom Types**.
30. Expand your project folder S_XX_FIRSTNAME and select the **PO_ExchangeRateOutput** Process Object.
31. Click **OK**.



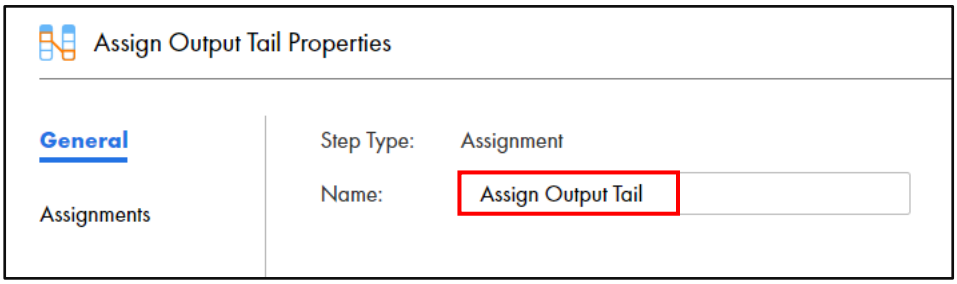
32. Click **Save** to save the process.

Now, we want to display each object of Exchange Response List in the guide. To do this, you need to modify the **Assign Tail** Assignment step.

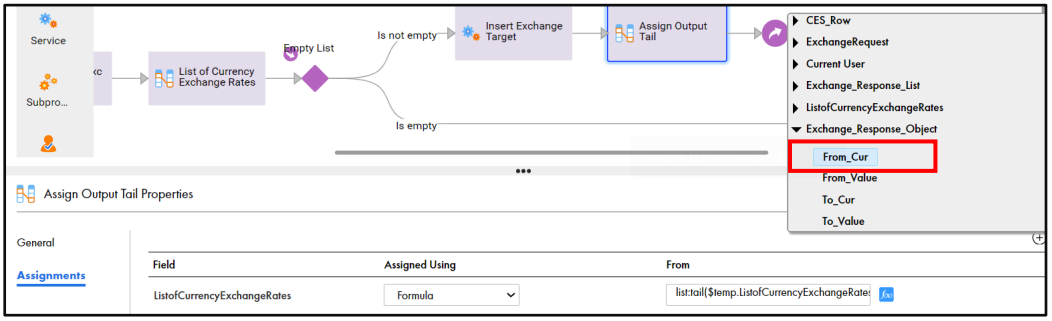
33. From the process canvas, select the **Assign Tail** step.



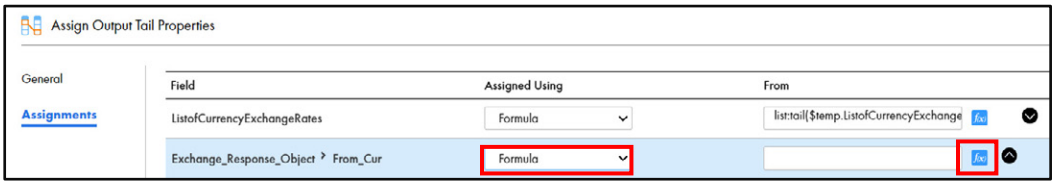
34. In the General tab, replace the Name with **Assign Output Tail**.



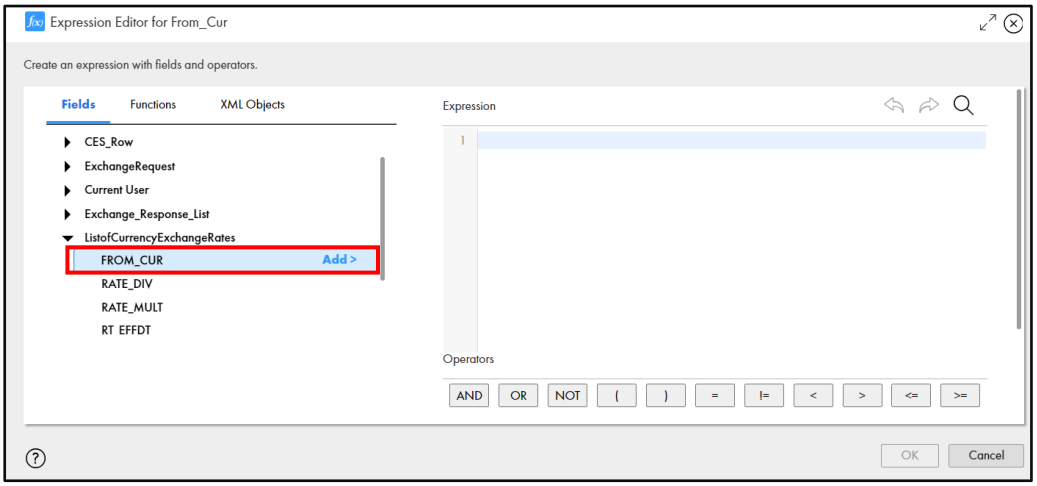
35. From the properties pane, select **Assignments**.
36. To add new field, click .
37. From the drop-down, select **Exchange_Response_Object** > **From_Cur**.



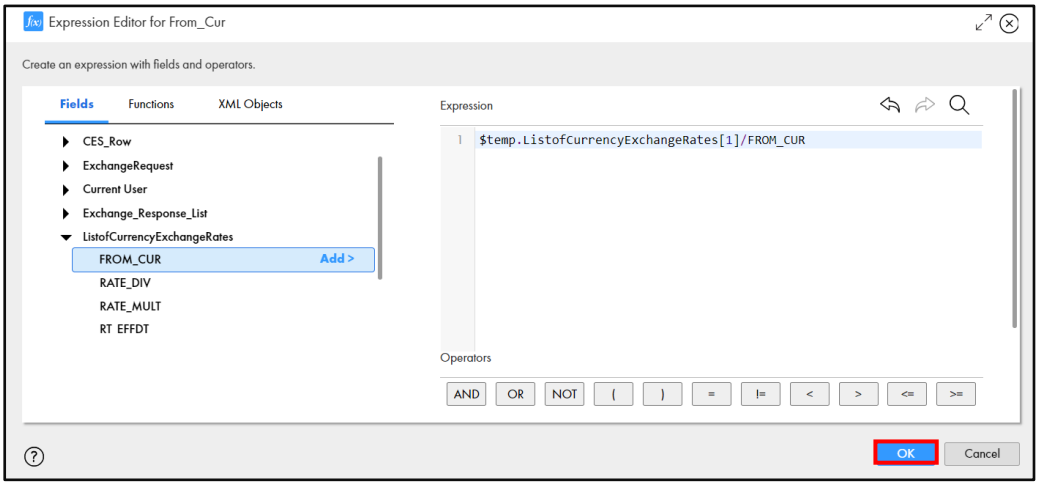
38. From the Assigned Using drop-down, select **Formula**.
39. To open the Formula Editor, click .



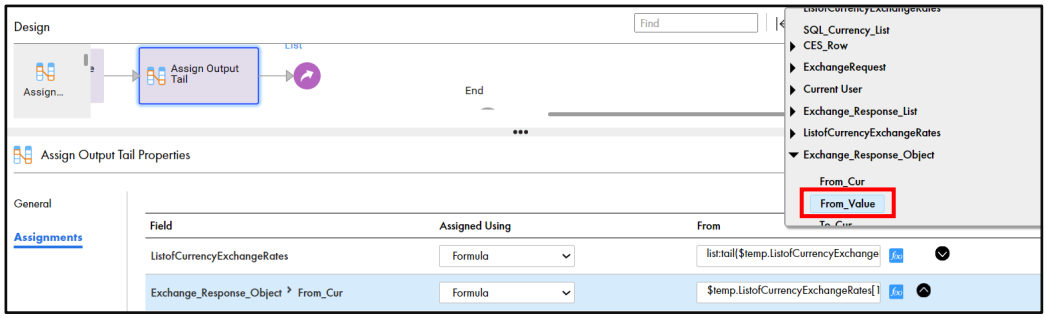
40. In the Expression Editor window, expand **ListofCurrencyExchangeRates** and select **FROM_CUR**.
41. Click **Add>**.



42. Click **OK**.

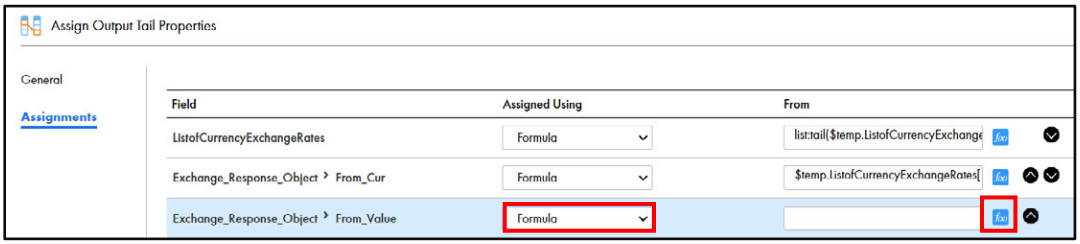


43. To add the second field, click .
44. From the drop-down, select **Exchange_Response_Object** > **From_Value**.

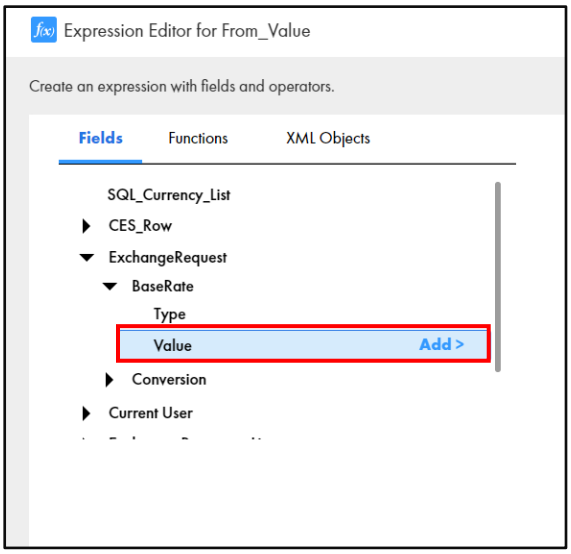


45. From the Assigned Using drop-down, select **Formula**.

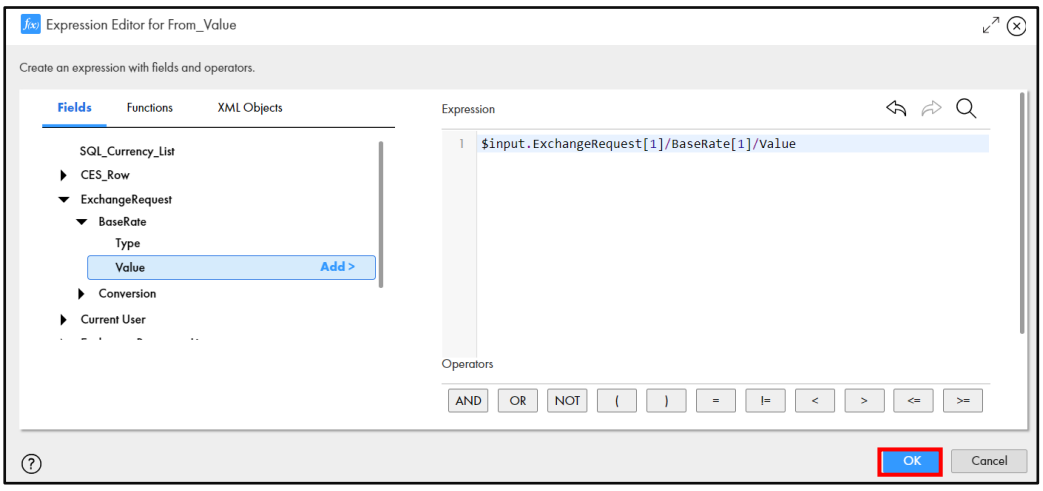
46. To open the Formula Editor, click .



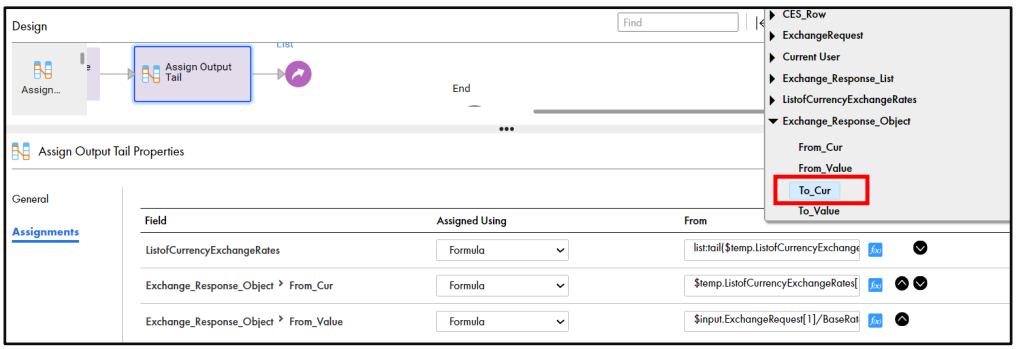
47. From the Fields tab, select **ExchangeRequest** > **BaseRate** > **Value**.
48. Click **Add>**.



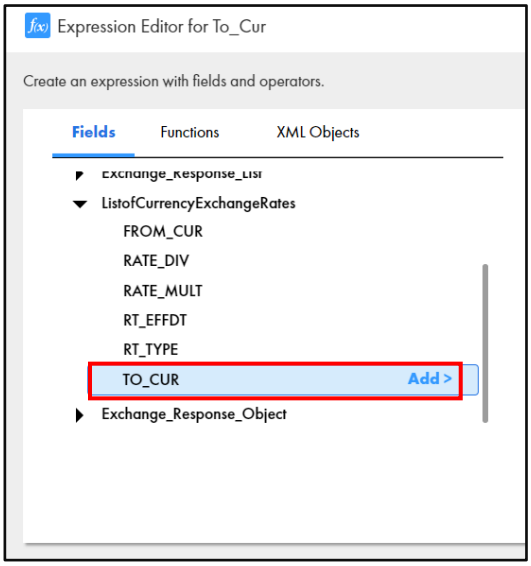
49. Click **OK**.



50. Click to add the third field.
51. From the drop-down, select **Exchange_Response_Object** > **To_Cur**.

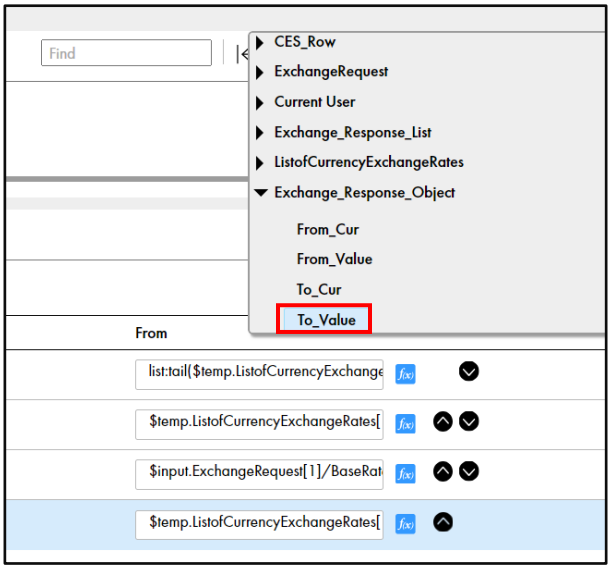


52. From the Assigned Using drop-down, select **Formula** and click .
53. From the Fields tab, select **ListofCurrencyExchangeRates > To_Cur**.
54. Click **Add>**.

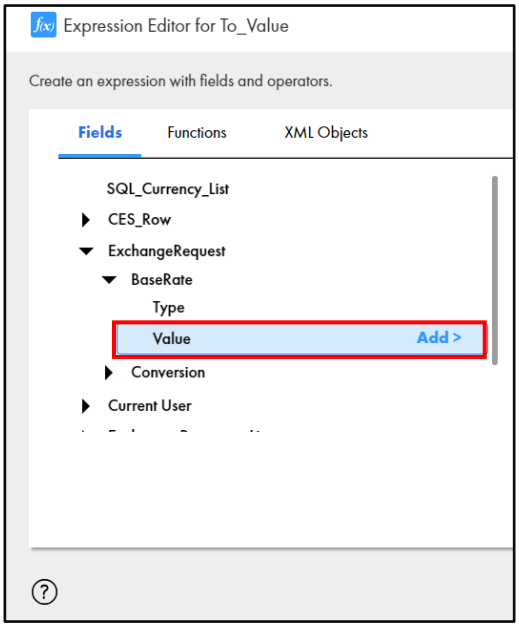


55. Click **OK**.

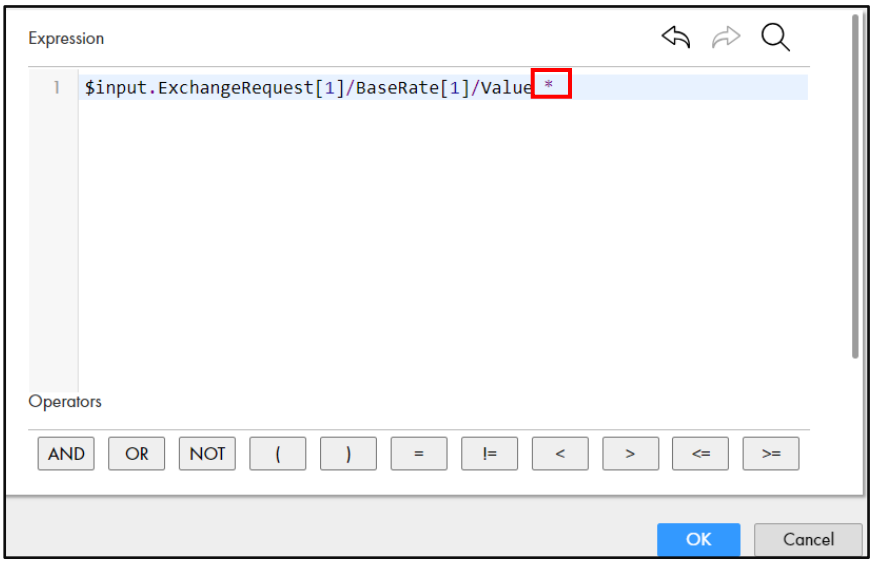
56. Click to add the fourth field.
57. From the drop-down, select **Exchange_Response_Object** > **To_Value**.



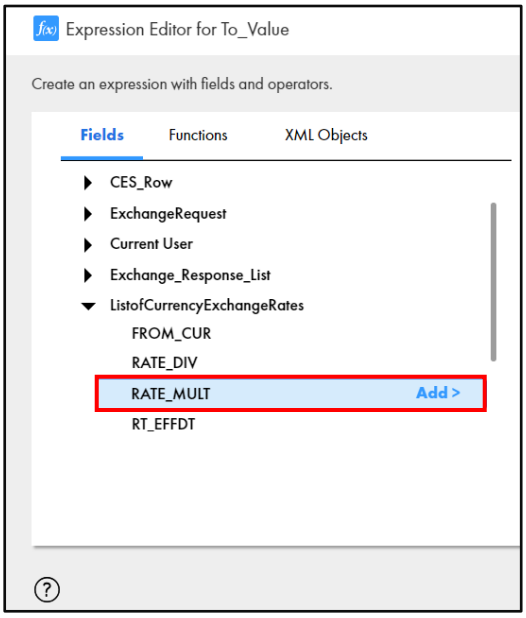
58. From the Assigned Using drop-down, select **Formula** and click .
59. From the Fields tab, select **ExchangeRequest** > **BaseRate** > **Value**.
60. Click **Add>**.



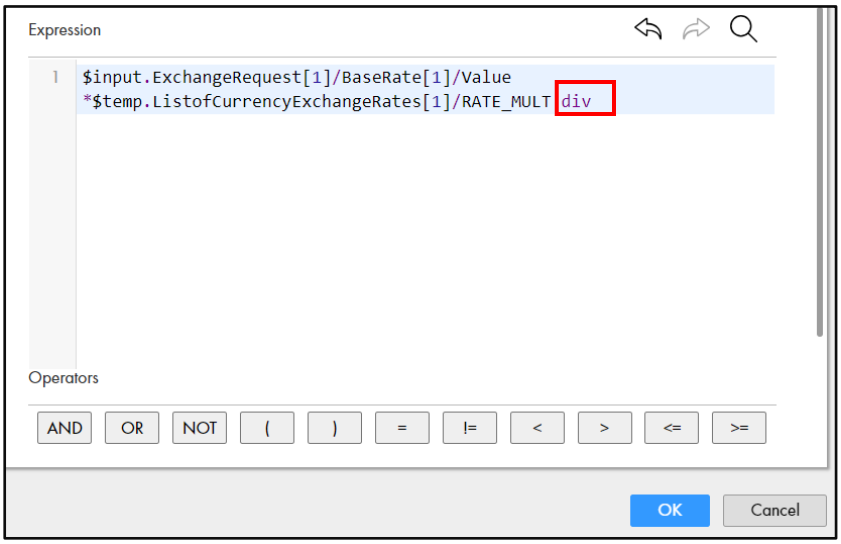
61. In the Expression window, enter ***** in the end of the expression.



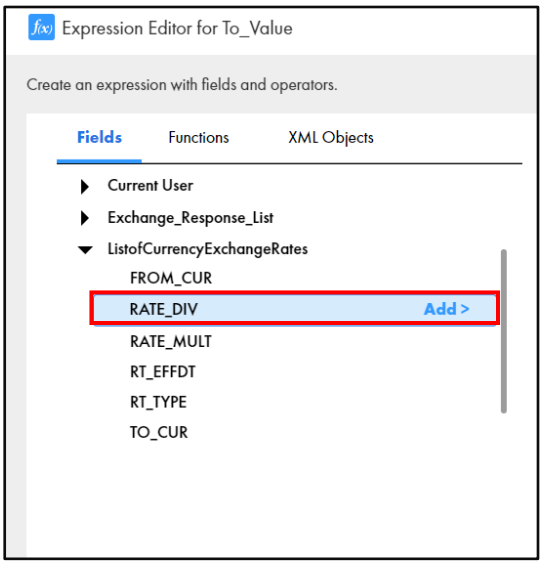
62. From the Fields tab, select **ListofCurrencyExchangeRates** > **RATE_MULT** and click **Add>**.



63. In the Expression window, enter **div** in the end of the expression.



64. From the Fields tab, select **ListofCurrencyExchangeRates** > **RATE_DIV** and click **Add>**.

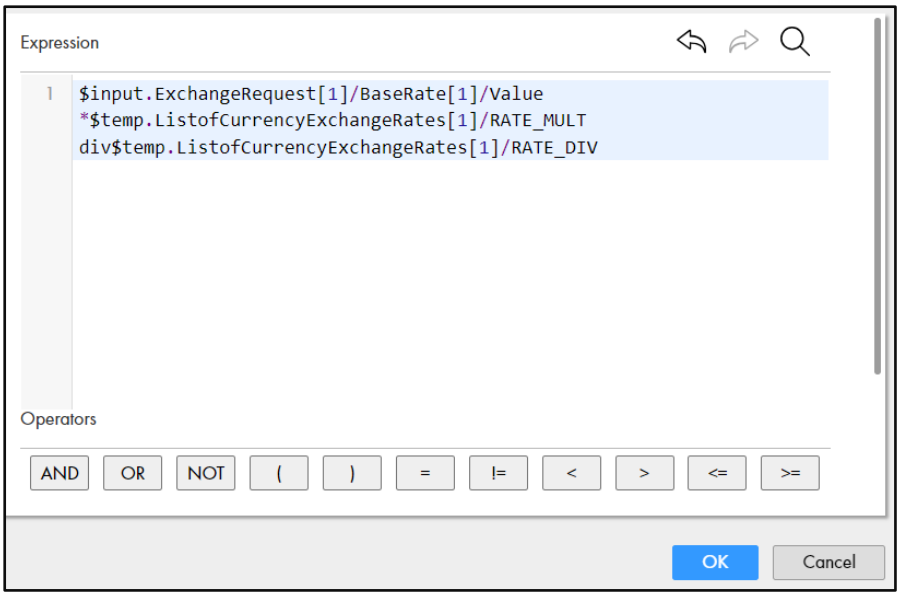


Note: in the end, your expression should look like this:
$input.ExchangeRequest[1]/BaseRate[1]/Value
***$temp.ListofCurrencyExchangeRates[1]/RATE_MULT**
div$temp.ListofCurrencyExchangeRates[1]/RATE_DIV

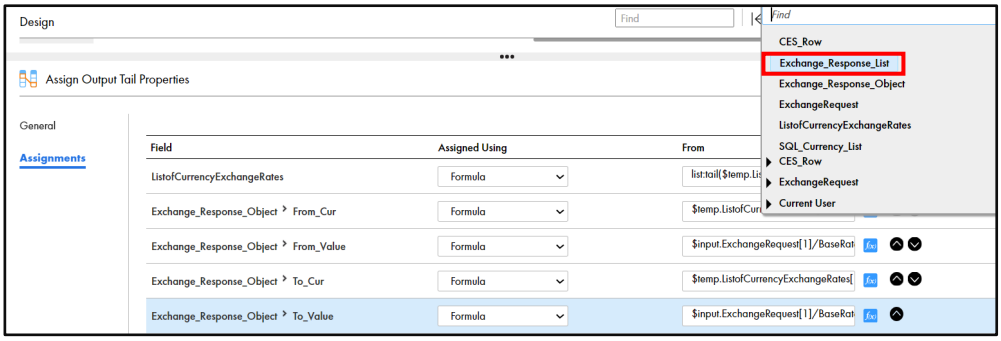
OR

From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **Using Currency Exchange Data Access Service Connector in a Process_12-2**. Copy the command mentioned under **Step A** and paste it into the Expression field.

65. Click **OK**.



66. Click to add the fifth field.
67. From the drop-down, select **Exchange_Response_List**.

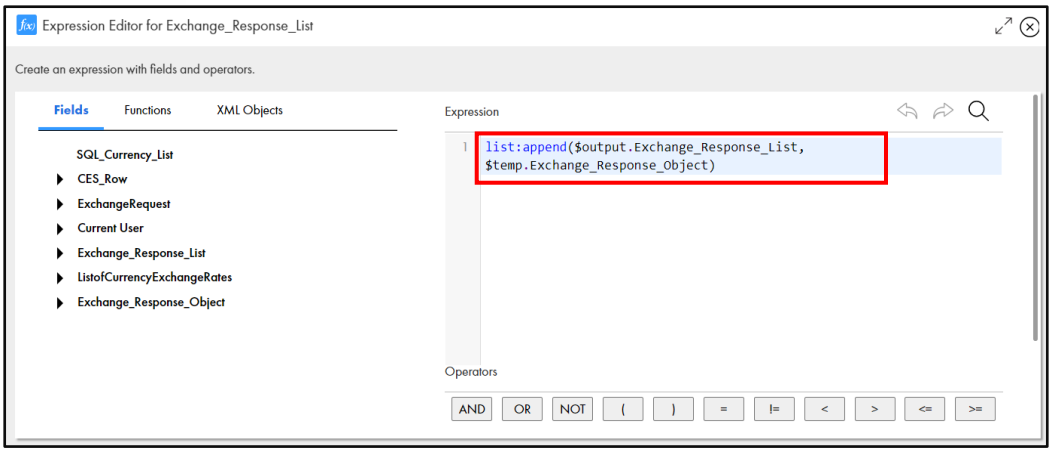


68. From the Assigned Using drop-down, select **Formula** and click .
69. In the Expression window, enter the following:

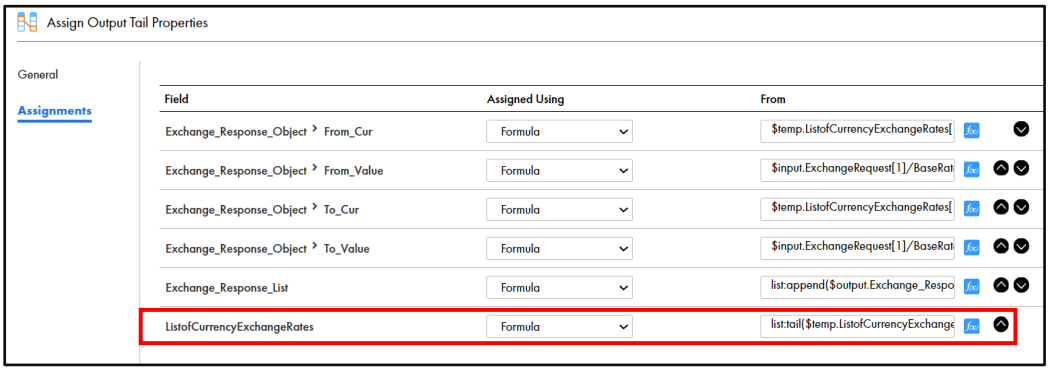
    list:append($output.Exchange_Response_List, $temp.Exchange_Response_Object)

    OR

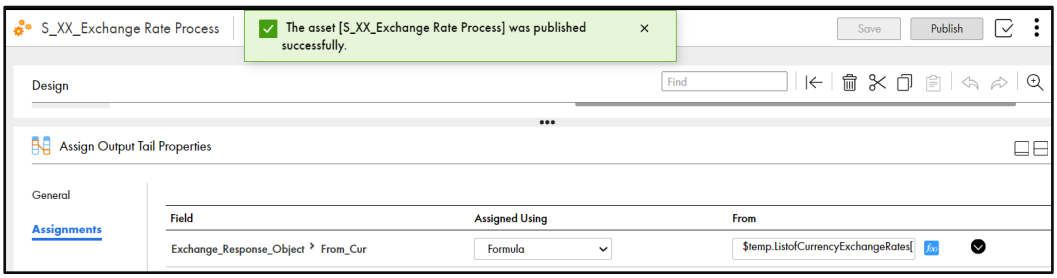
    From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **Using Currency Exchange Data Access Service Connector in a Process_12-2**. Copy the command mentioned under **Step B** and paste it into the Expression field.



70. Click **OK**.
71. Click the Move down arrow to move **ListofCurrencyExchangeRates** at the bottom.



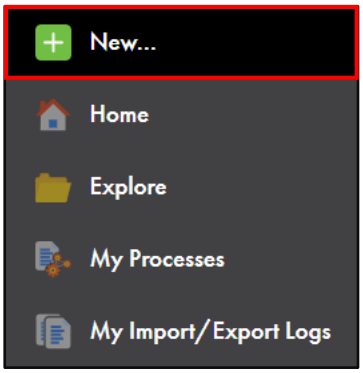
72. Save and publish the process.



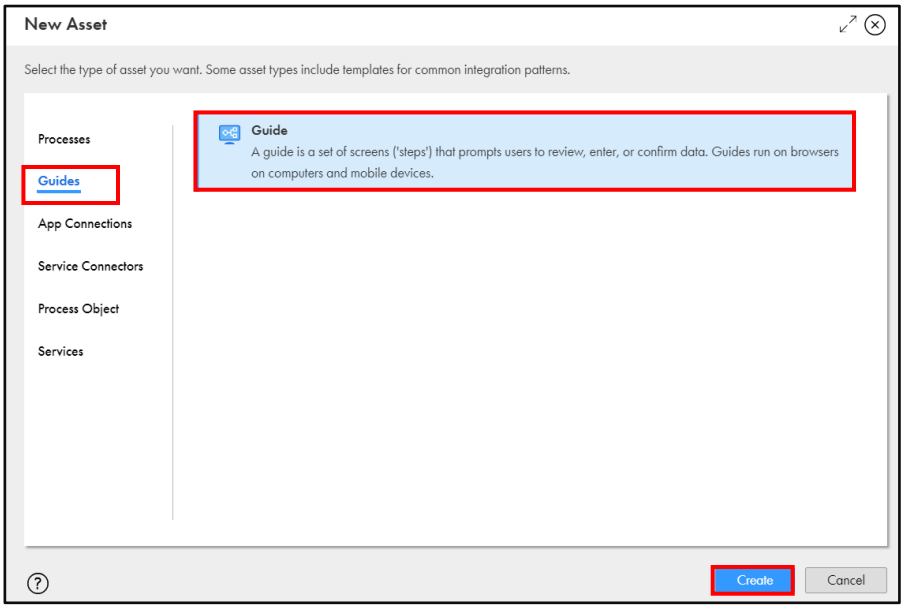
73. Close the process.

Create a Guide

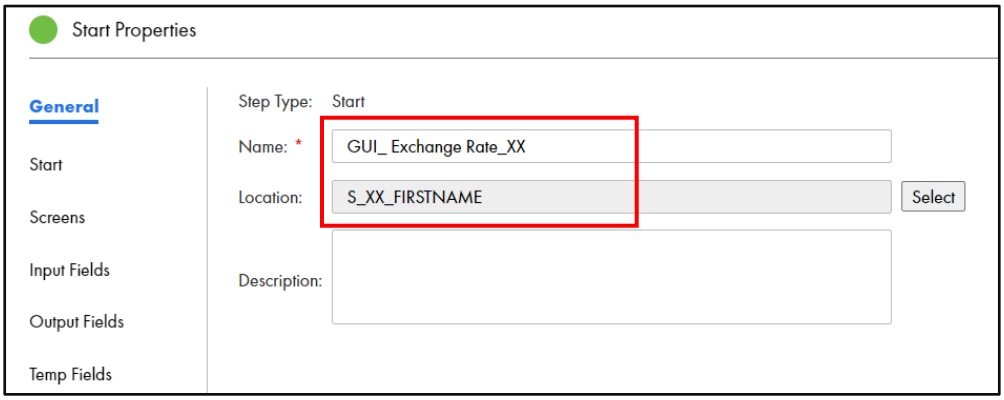
74. From the navigation pane, select New to create a Guide.



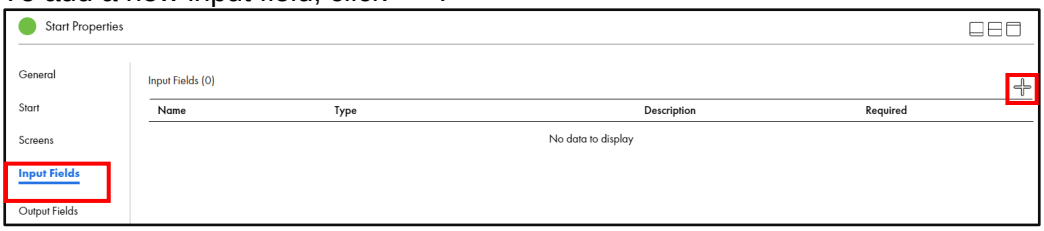
75. Select **Guides** > **Guide** and click **Create**.



76. From the process canvas, select the **Start** step.
77. In the General tab, enter the name as **GUI_ Exchange Rate_XX**.
    Note: Replace XX with your student number.
78. In the Location field, select your project folder **S_XX_FirstName**.

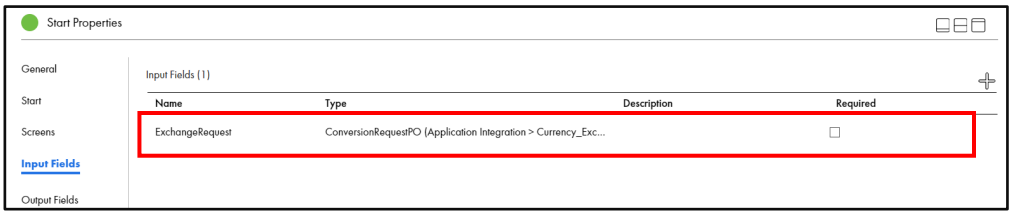


79. From the properties pane, select **Input Fields**.
80. To add a new input field, click .

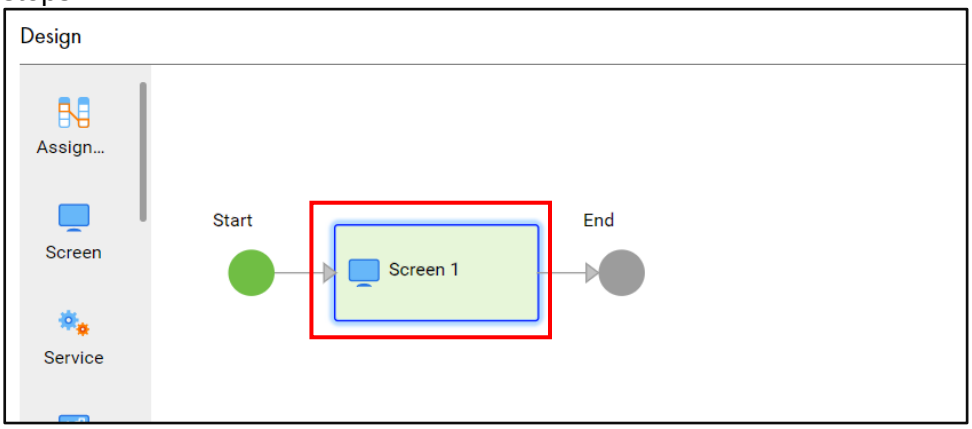


81. Create the input field, as shown in the table below:

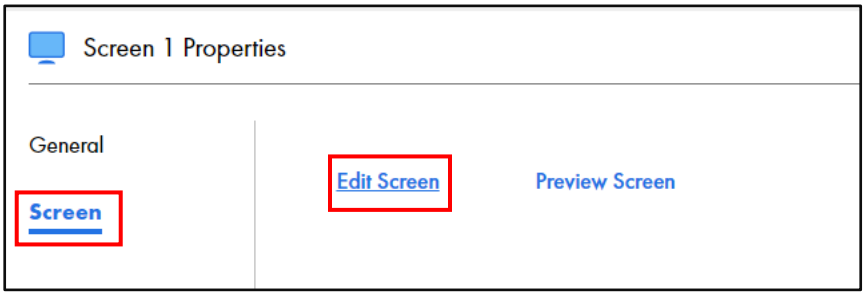
Name	Type
ExchangeRequest	More types>Custom Types> S_XX_FIRSTNAME> ConversionRequestPO



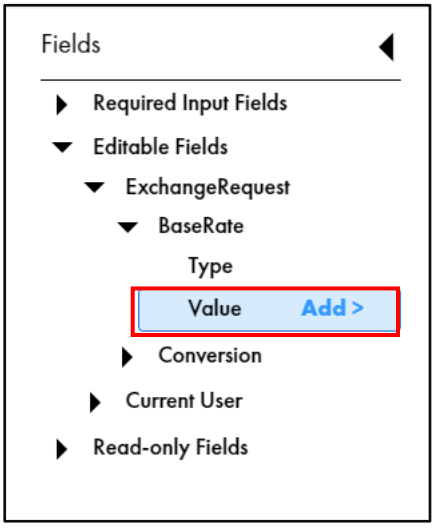
82. From the Design palette, drag and drop a **Screen** step between the Start and End steps.



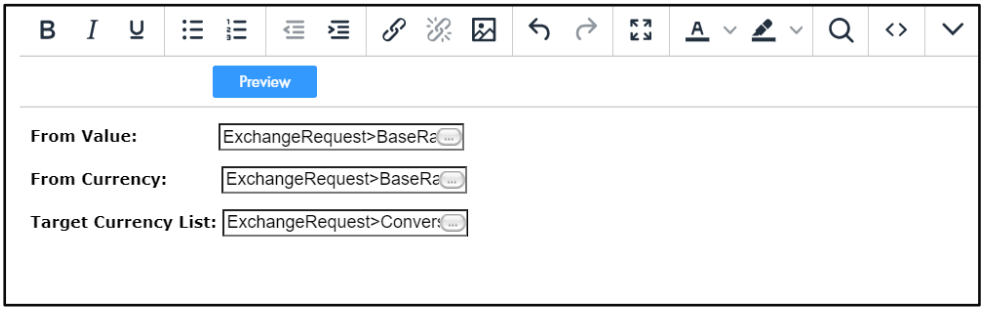
83. From the Screen properties pane, select **Screen** and click **Edit Screen**.



84. Expand **Editable Fields** and select **ExchangeRequest** > **BaseRate** > **Value**.
85. Click **Add>**.



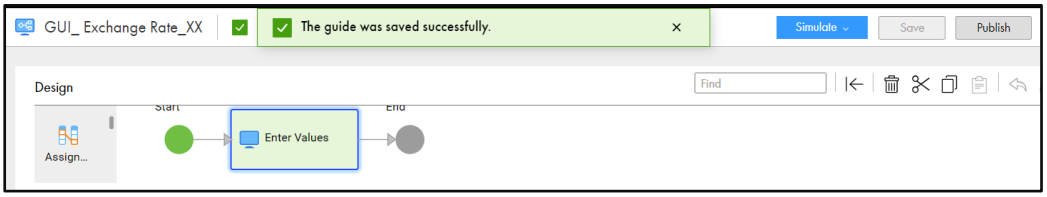
86. In the Expression section, replace ExchangeRequest>BaseRate>Value with **From Value:** .
87. Hit Enter to go to the next line.
88. Similarly, add one more field. Expand **Editable Fields** and select **ExchangeRequest** > **BaseRate** > **Type**.
89. Click **Add>**.
90. In the Expression section, replace ExchangeRequest>BaseRate>Type with **From Currency:** .
91. Hit Enter to go to the next line.
92. Finally, add one more field. Expand **Editable Fields** and select **ExchangeRequest** > **Conversion** > **Rate**.
93. Click **Add>**.
94. In the Expression section, replace ExchangeRequest>Conversion>Rate with **Target Currency List:** .
95. Align all the text boxes. Your Expression section should look like this:



96. In the Prompt field, enter **Enter Values**.

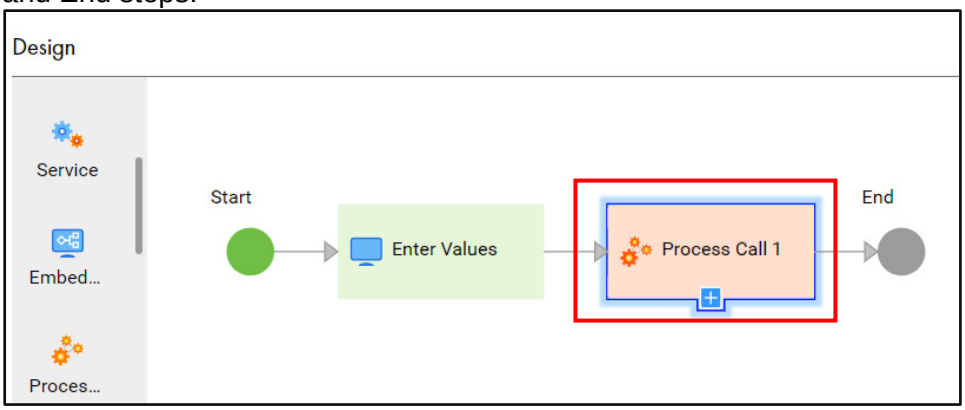
Note: If you notice more than one **Continue** Button in the Footer Button section, then click cross to delete them.

97. Click **OK**.
98. Click **Save** to save the Guide.

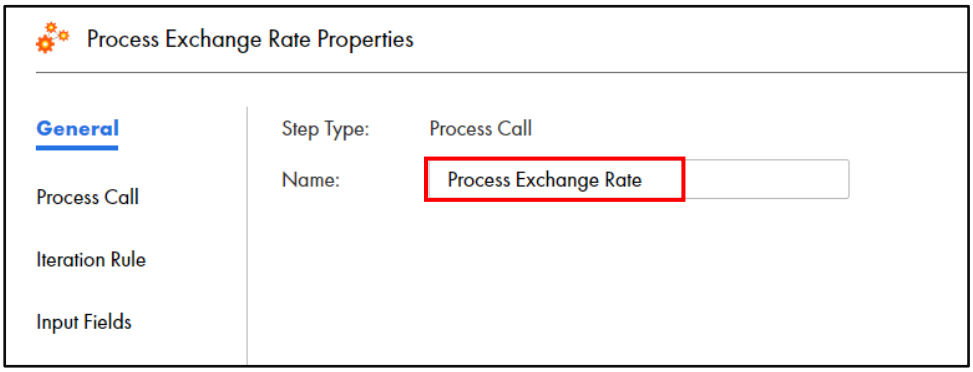


Invoke the Process in the Guide

99. From the Design palette, drag and drop a **Process** step between the Enter Values and End steps.

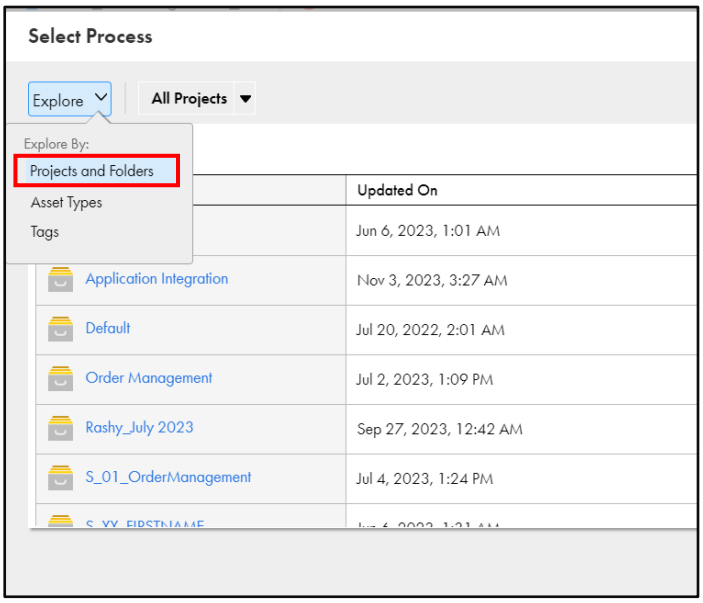


100. In the General tab, enter the name as **Process Exchange Rate**.

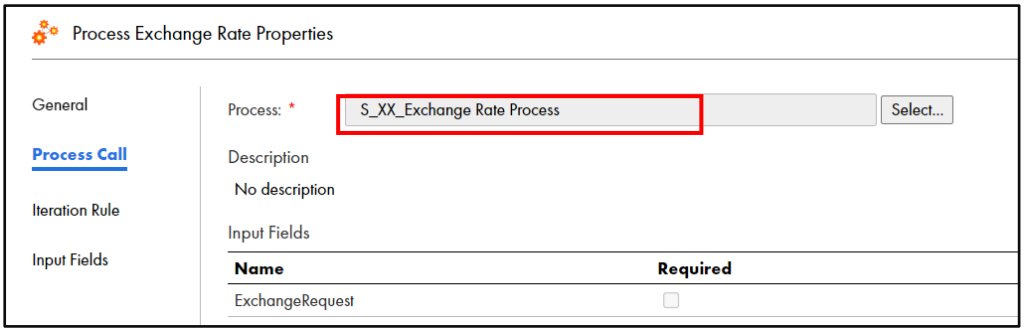


101. From the properties pane, select **Process Call** and click **Select**.

102. Click **Explore** and select **Project and Folder**.

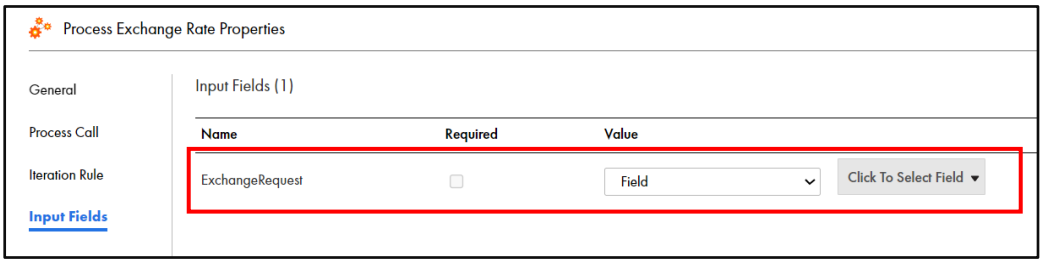


103. Locate your project folder and select **S_XX_Exchange Rate Process**.

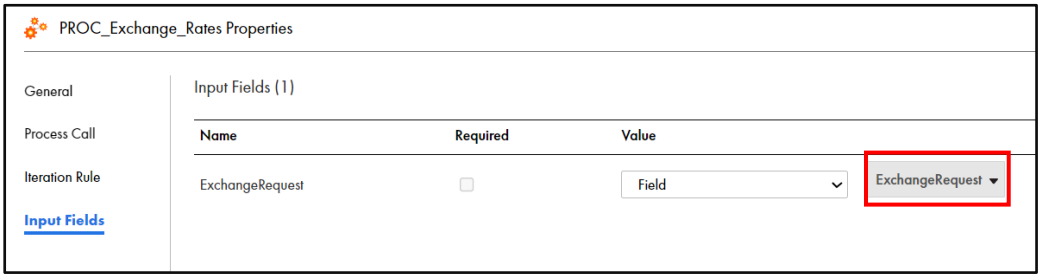


104. From the Properties pane, select **Input Fields**.

105. Click the '**+**' icon to add a new input and select **ExchangeRequest**.

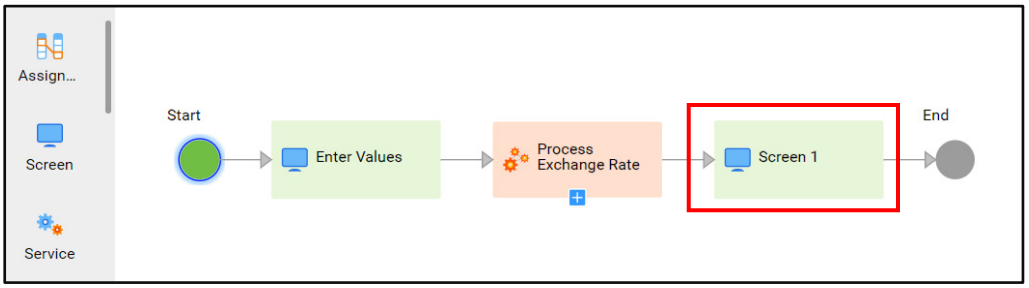


106. From the Click To Select Field drop-down, select **ExchangeRequest**.

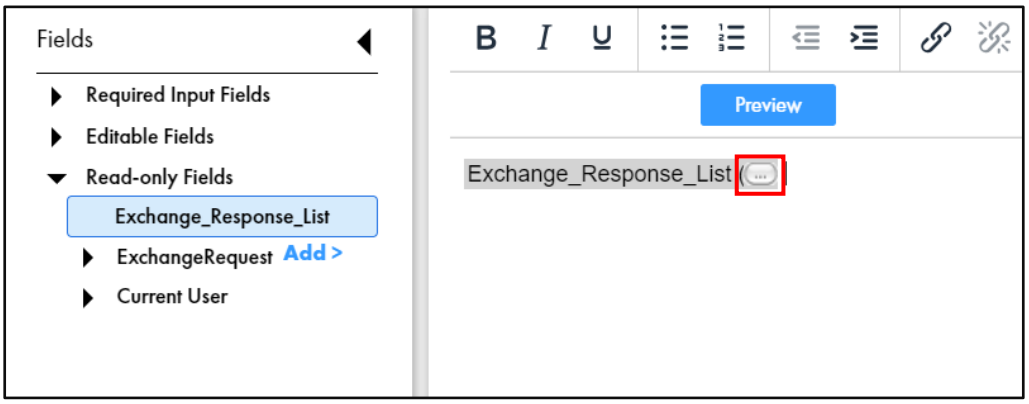


Add the Screen Step to Display the Results

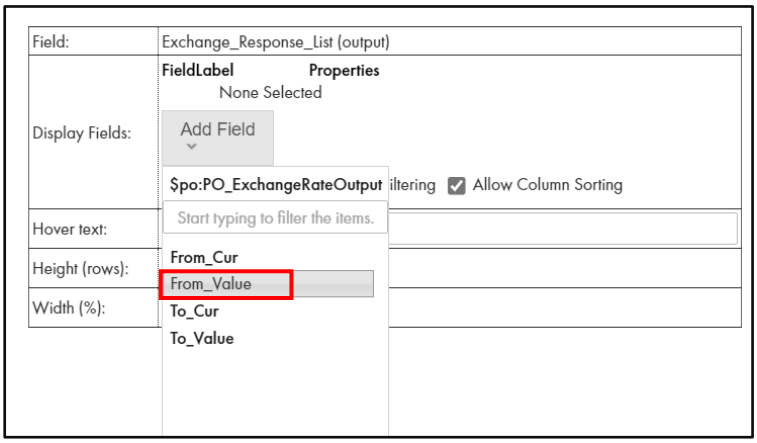
107. From the Design palette, drag and drop a **Screen** step between the Process Exchange Rate and End steps.



108. From the Screen properties pane, select **Screen** and click **Edit Screen**.
109. Select **Read-only Fields** > **Exchange_Response_List** and click **Add>**.
110. Click ellipse .

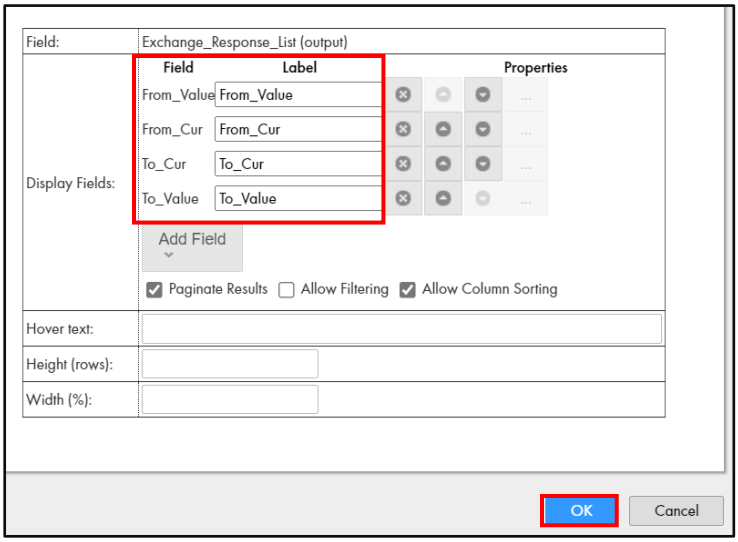


111. From the Add Field drop-down, select **From_Value**.



112. Similarly, add **From_Cur**, **To_Cur**, and **To_Value** fields.

113. Click **OK**.

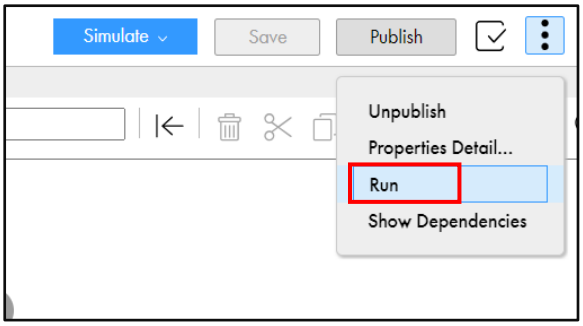


114. In the Prompt field, enter **View Results** and click **OK**.

115. Save and publish the Guide.

Execute the Guide

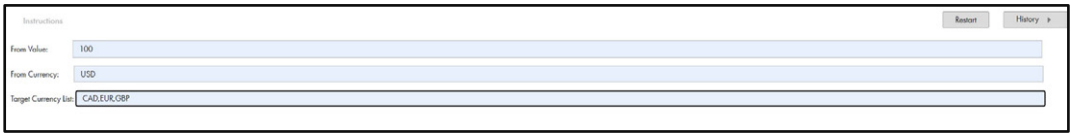
116. click ⋮ and select **Run**.



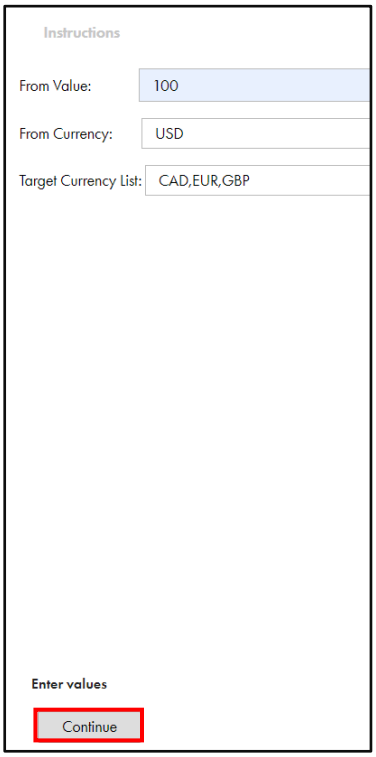
Note: If the **Run** button is not enabled, close and re-open the asset.

117. The Guide will be opened in a new browser tab. Enter the following values in the input fields:

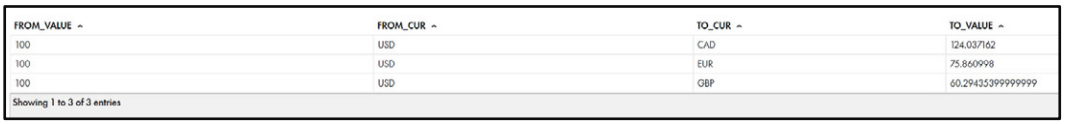
From Value	100
From Currency	USD
Target Currency List	CAD,EUR,GBP



118. Click the **Continue** button.



The result is displayed as shown below:



FROM_VALUE	FROM_CUR	TO_CUR	TO_VALUE
100	USD	CAD	124.037162
100	USD	EUR	75.860998
100	USD	GBP	60.29435399999999

Showing 1 to 3 of 3 entries

This concludes the lab.

Module 13: Monitoring
Lab 13-1: Create and Use URN Mapping

Overview:
You can assign a physical address to a universal resource name (URN). The URN is a logical address of a partner link, specified in a deployment resource. URN mappings provide a flexible and dynamic way to define target endpoint references.

By mapping a URN to a URL, you do not have to rely on invoking a statically defined endpoint address. URN mappings give you flexibility, for example, to deploy the same BPR files for testing and production environments. Also, if you specify a URL, you can replace the URL by mapping it to a different URL.

In this lab, you will create a URN to a URL mapping and then use this mapping in a process.

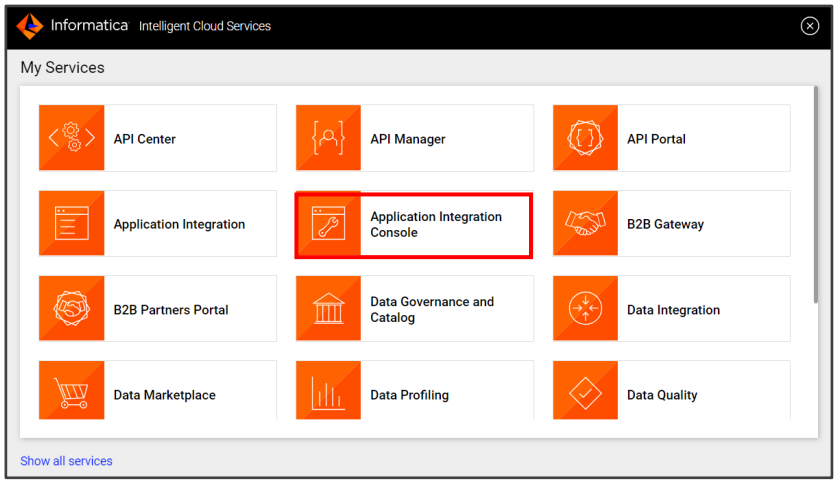
Objectives:
- Create a URN to URL Mapping
- Use the URN Mapping in a process

Duration:
15 Minutes

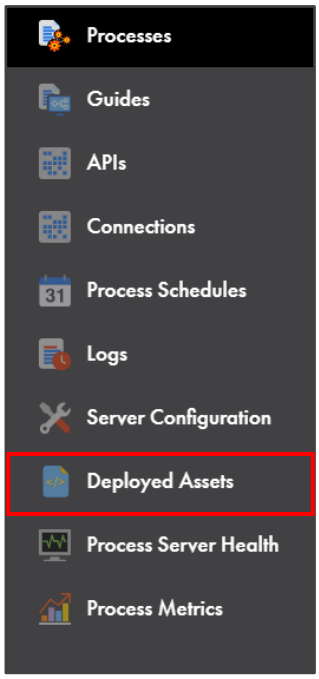
Tasks

Create a URN Mapping

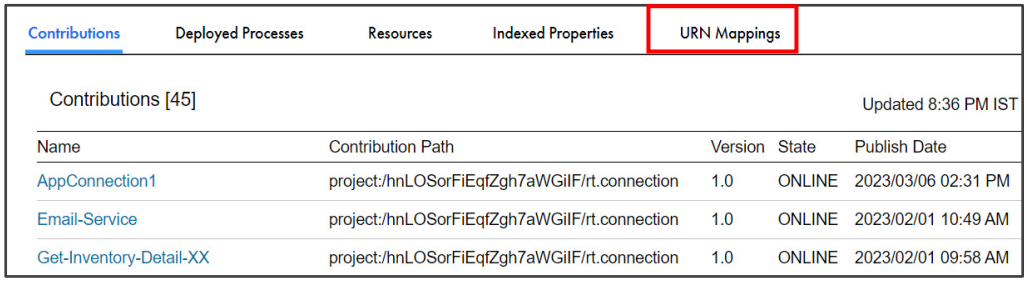
1. Log into IICS and access the **Application Integration Console** service.



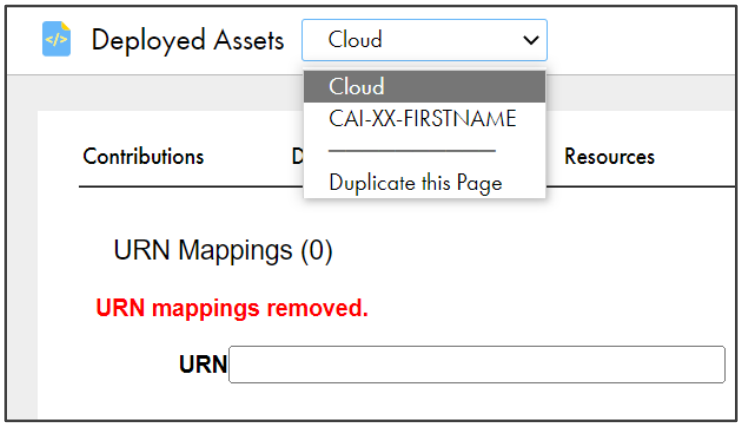
2. From the navigation pane, click **Deployed Assets**.



3. Click **URN Mappings**.

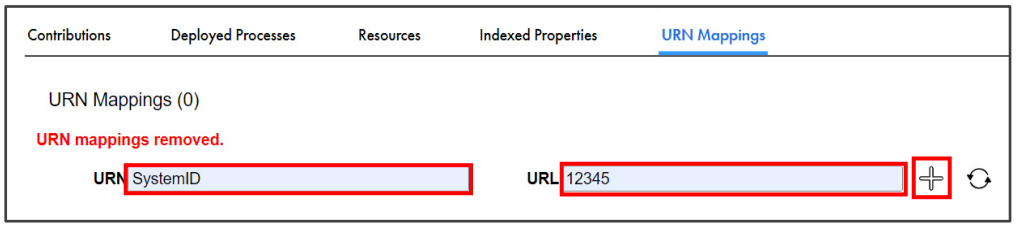


4. From the Deployed Assets drop-down, select **Cloud**.

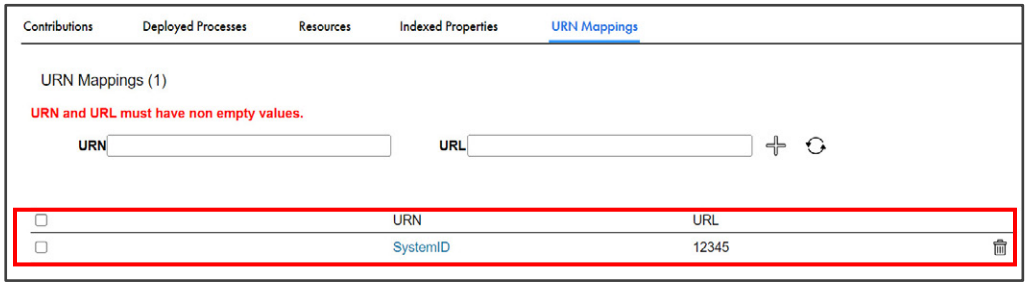


5. In the URN field, enter **SystemID**.
6. In the URL field, enter **12345**.

7. Click the '**+**' icon.

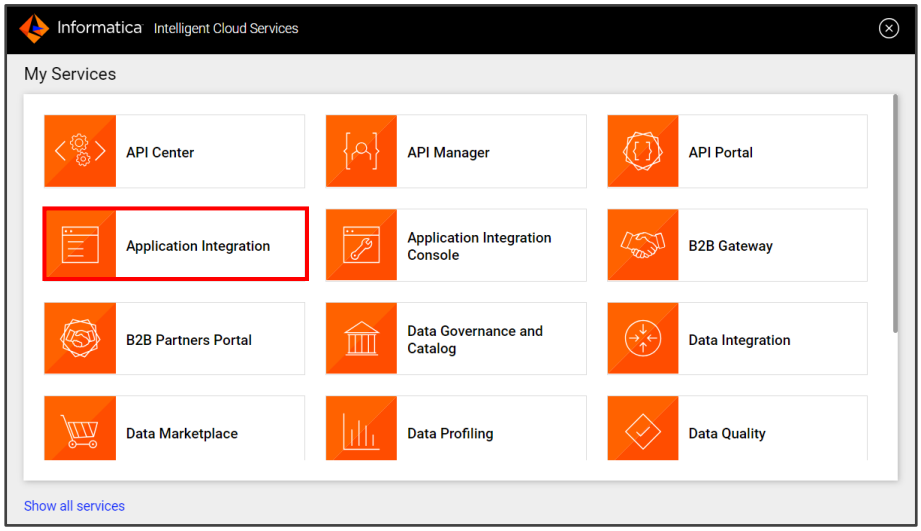


URN mapping is created.

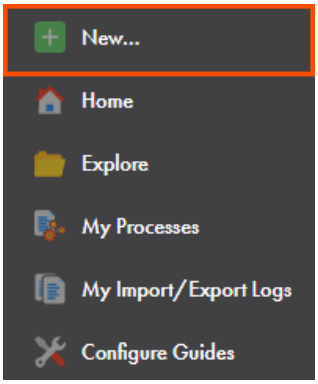


Create a Process

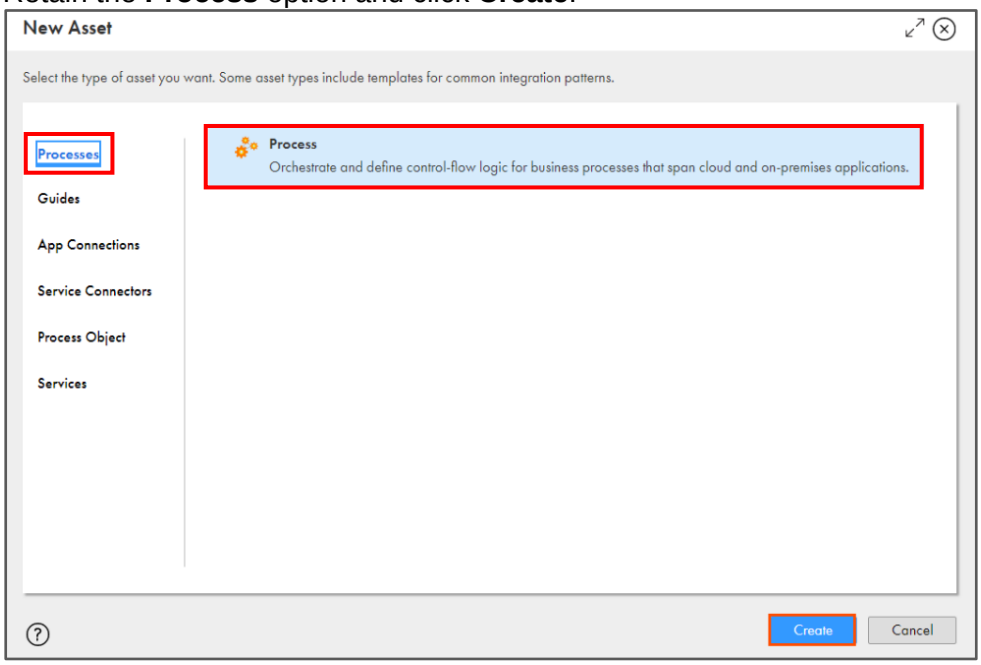
8. Navigate to the **Application Integration** service.



9. From the navigation pane, select **New**.



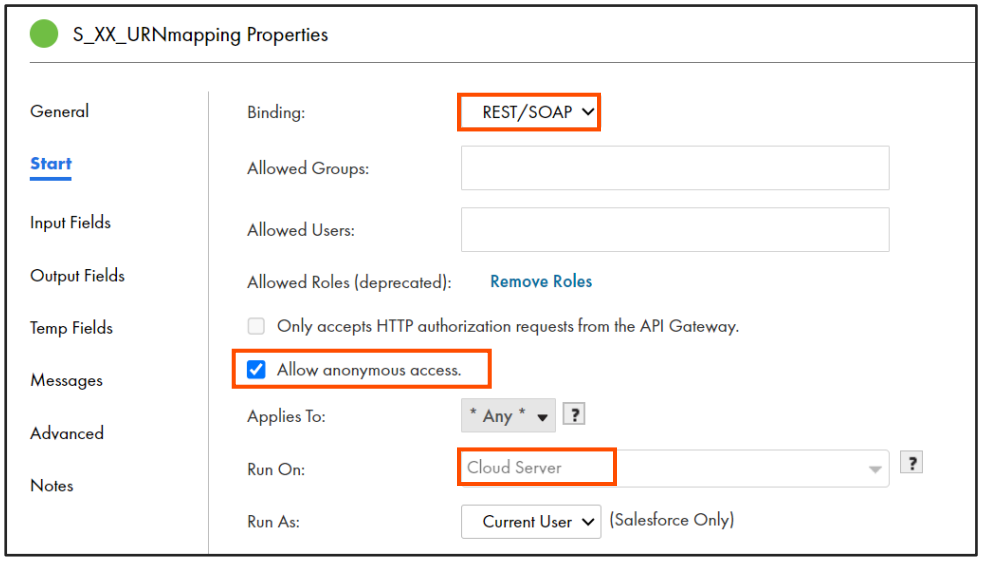
10. Retain the **Process** option and click **Create**.



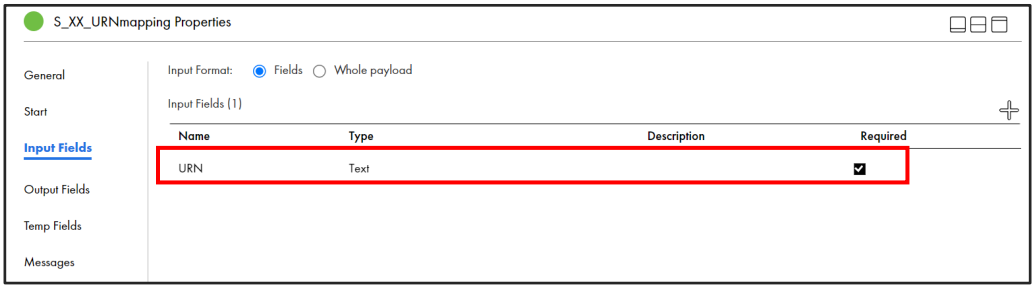
11. In the General properties tab, in the Name field, enter **S_XX_URNmapping**.
12. In the Location field, select your project folder **S_XX_FirstName**.

13. In the Start tab, retain the Binding option as **REST/SOAP**.
14. Select the **Allow anonymous access** option.

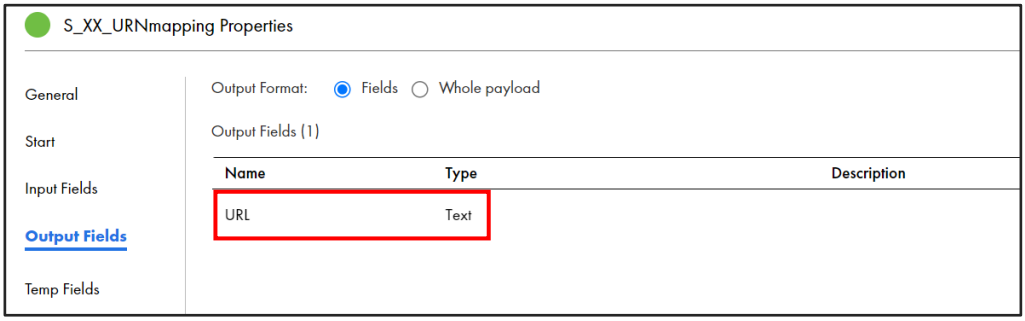
15. From the Run On drop-down, retain **Cloud Server**.



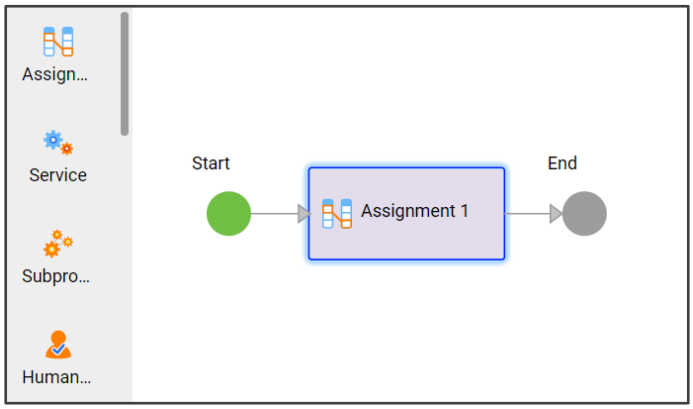
16. In the Input Fields tab, to add a new field, click .
17. In the Name field, enter **URN** and retain the Type as **Text**.
18. Select the **Required** checkbox.



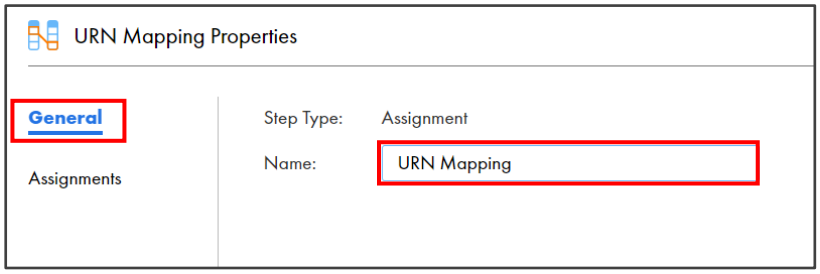
19. In the Output Fields tab, to add a new output field, click .
20. In the Name field, enter **URL** and retain the Type as **Text**.



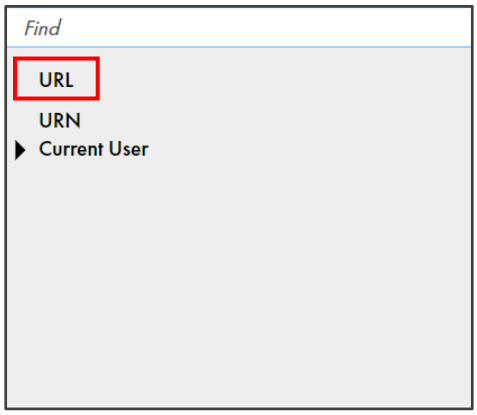
21. From the palette, drag and drop an **Assignment** step between the Start and the End step on the canvas.



22. Select the **Assignment** step.
23. In the General tab, in the Name field, enter **URN Mapping**.

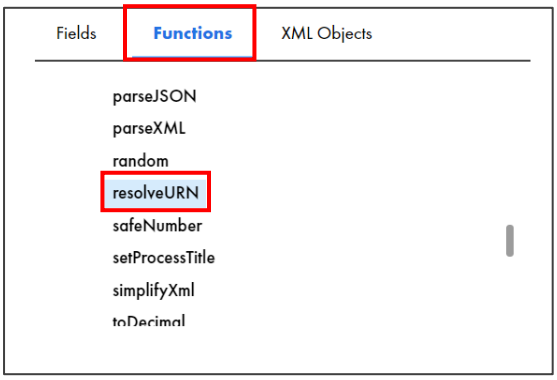


24. In the Assignments tab, click the plus **(+)** icon, and select **URL**.

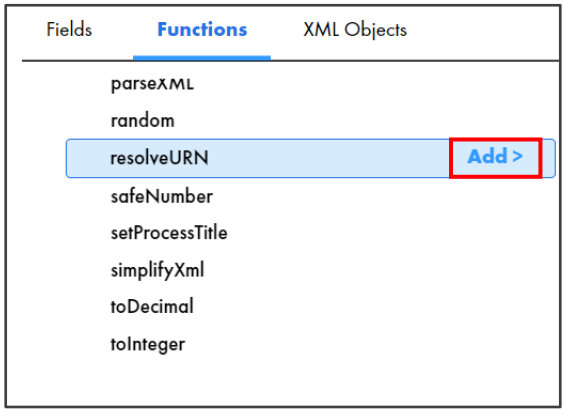


25. From the **Assigned Using** drop-down, select **Formula**.
26. Click to open the Expression Editor.

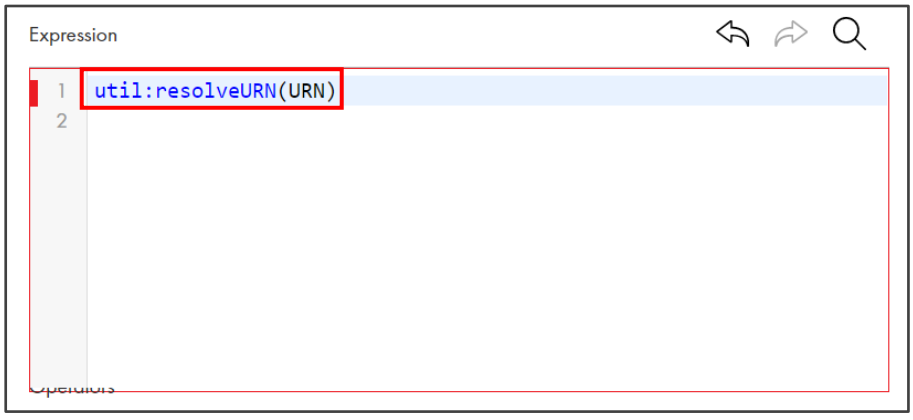
27. Click **Functions** and select **Miscellaneous** > **resolveURN**.



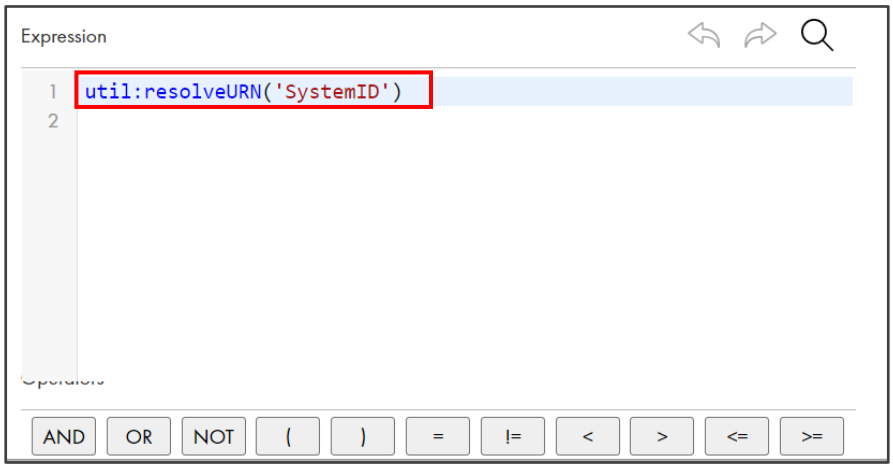
28. Click **Add >**.



29. In the Expression section, observe the **util:resolveURN(urn)** function.

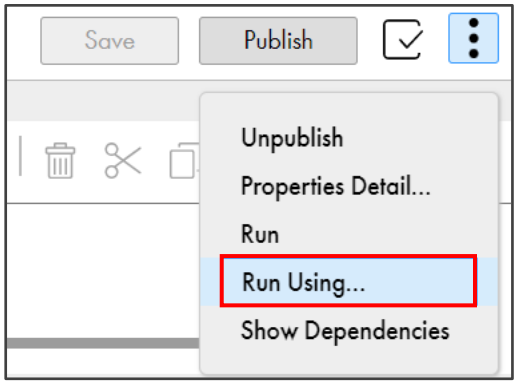


30. Replace URN with **'SystemID'** as displayed here:

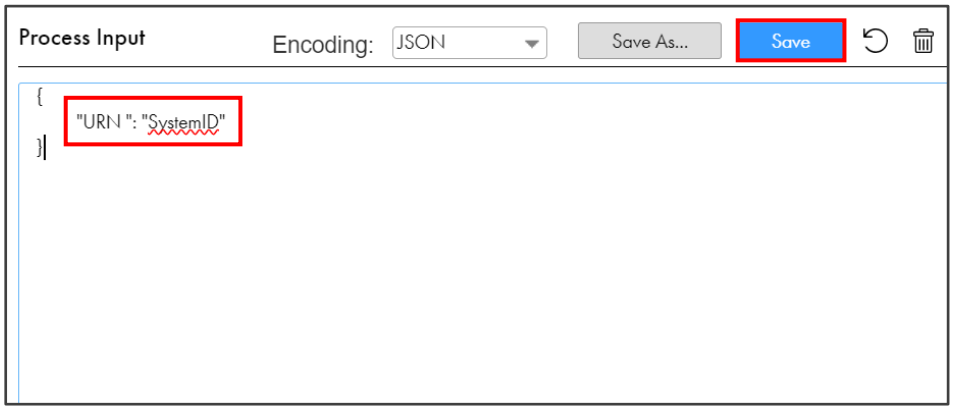


Note: SystemID should be in parentheses.

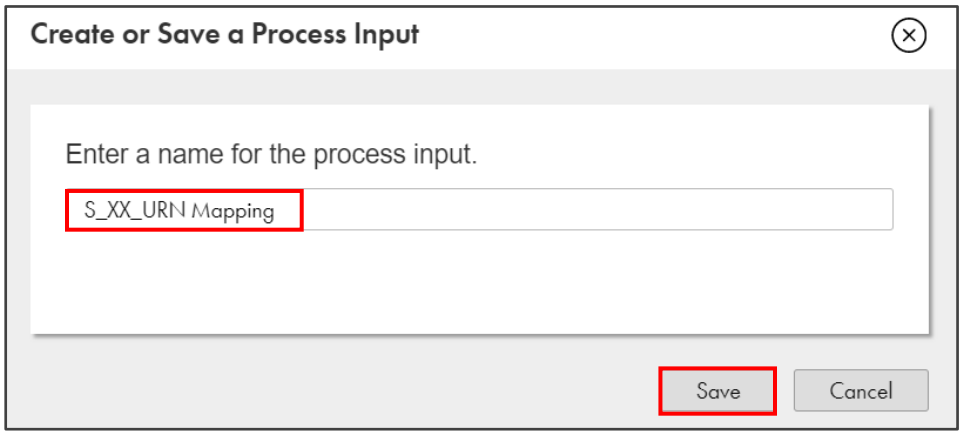
31. Click **OK**.
32. **Save** and **Publish** the process.
33. From the Action menu, select **Run Using…**.



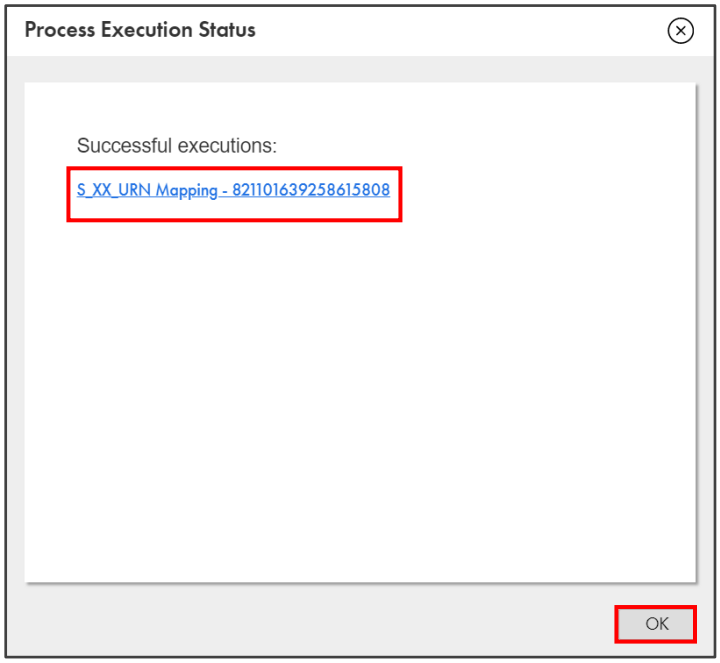
34. In the Process Input section, enter the input as **SystemID**, and click **Save**.



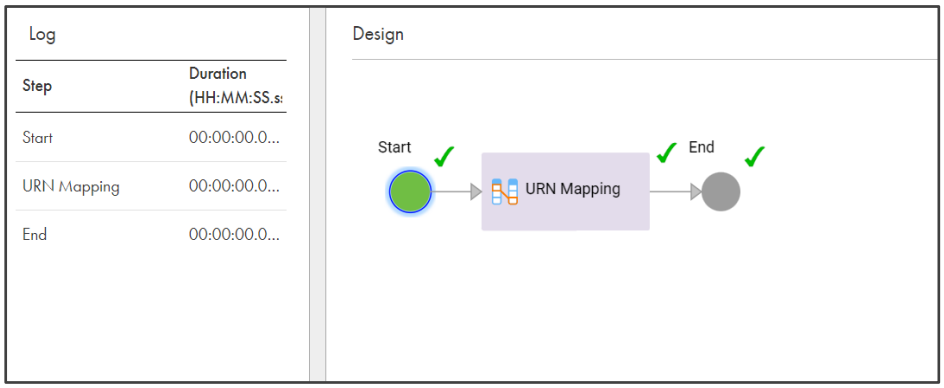
35. To save the input, enter the input name as **S_XX_URN Mapping** and **Save** it.



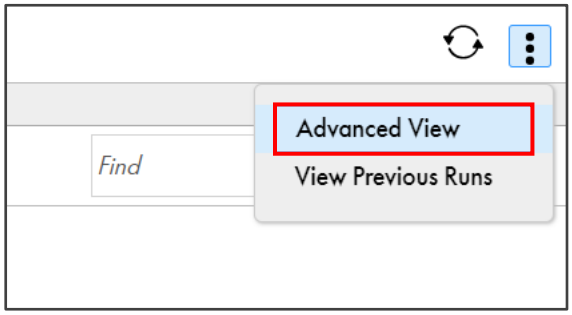
36. After the input is added and validated successfully, click **Run**.
37. To check the Process Execution Status, click the **Process Execution link**, and then click **OK**.



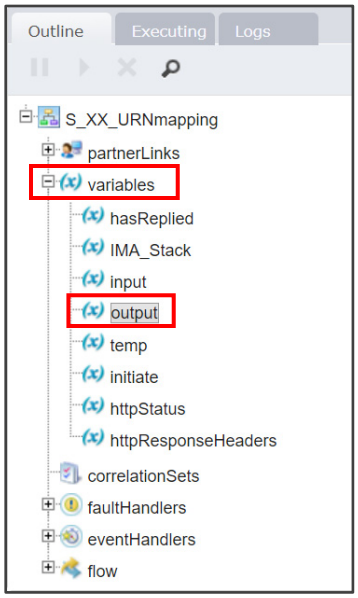
The execution details are displayed as shown below:



38. To check the output, from the **Actions** menu, select **Advanced View**.



39. Expand the **variables** section and select **output**.



40. In the Variable Instance Data section, observe the value of URL.

Note: Notice the value of URL. It is the same value that you have given while creating the URN mapping.

This concludes the lab.

Module 13: Monitoring
Lab 13-2: Create a New Process Schedule

Overview:

A Process Schedule can be created to perform routine maintenance or another recurring task. If desired, you can deploy the process and then set a schedule to run it, such as weekly or on a specific day of the month. You can set a fine-grained schedule (intraday, seconds) if desired.

In this lab, you will create a Process Schedule for an existing process.

Objectives:

- Create a Process Schedule for an existing process

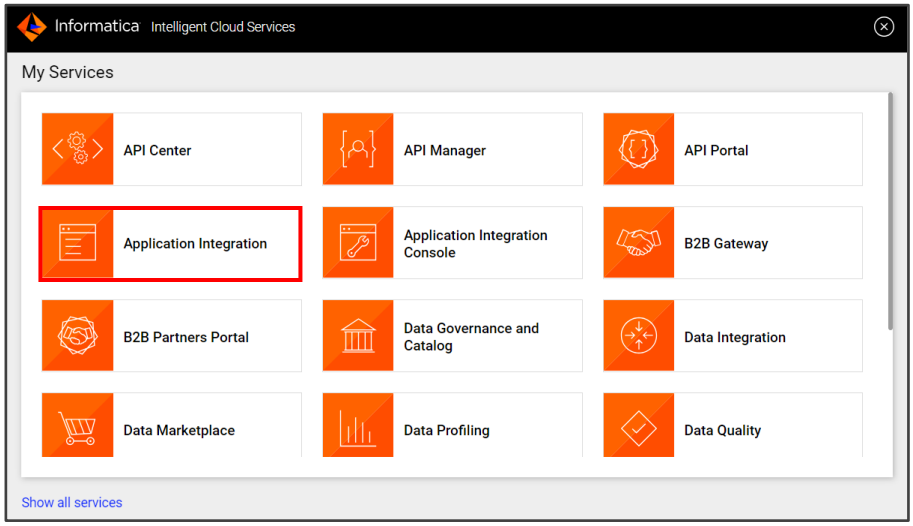
Duration:

15 Minutes

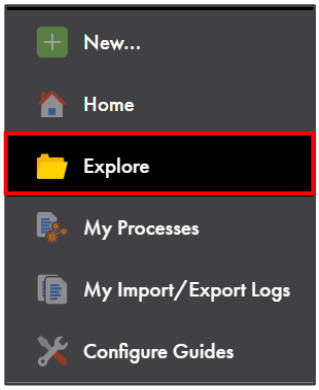
Tasks

Copy the Process Input

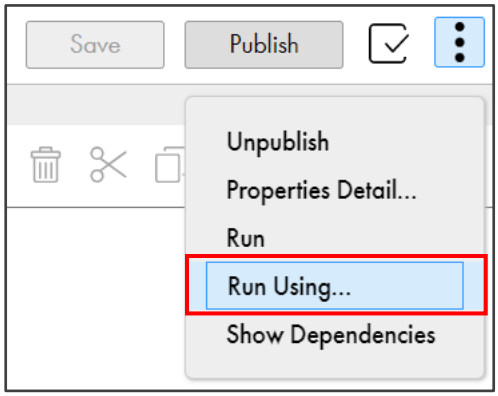
1. Navigate to the **Application Integration** service.



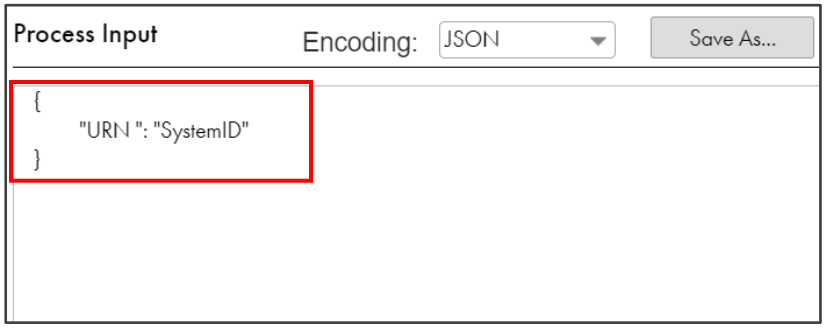
2. From the navigation pane, select **Explore**.



3. Navigate to your folder **S_XX_Firstname**.
4. Locate and open the **S_XX_URNmapping** process.
   Note: Use the Search box to find the process.
5. From the Actions menu, select **Run Using…**.



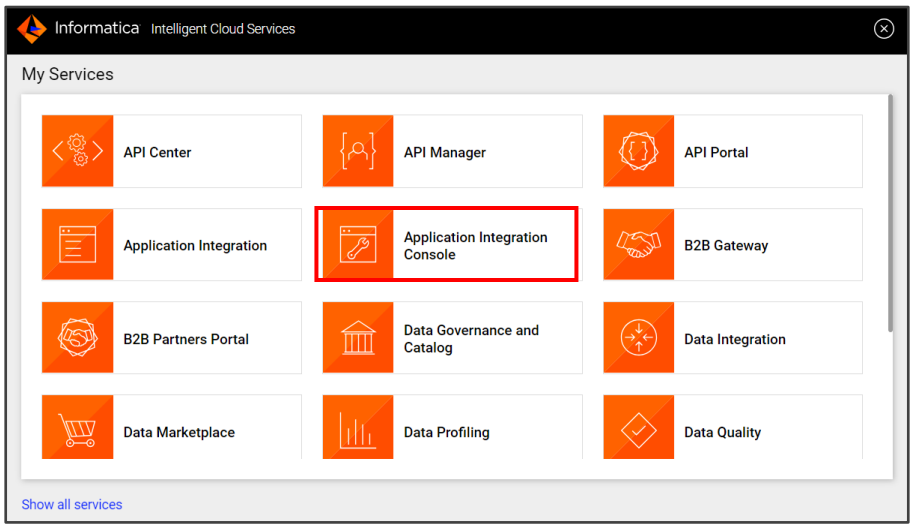
6. From the Process Input field, copy the code and paste it into a notepad file.



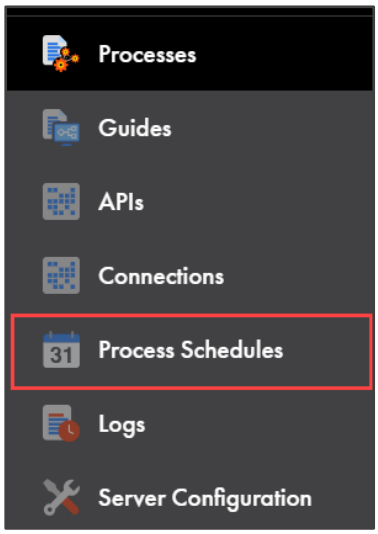
Create a Process Schedule

In this step, we will schedule an execution for the S_XX_URNmapping process.

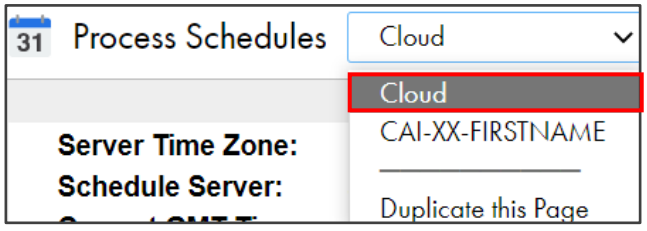
7. Access the **Application Integration Console** service.



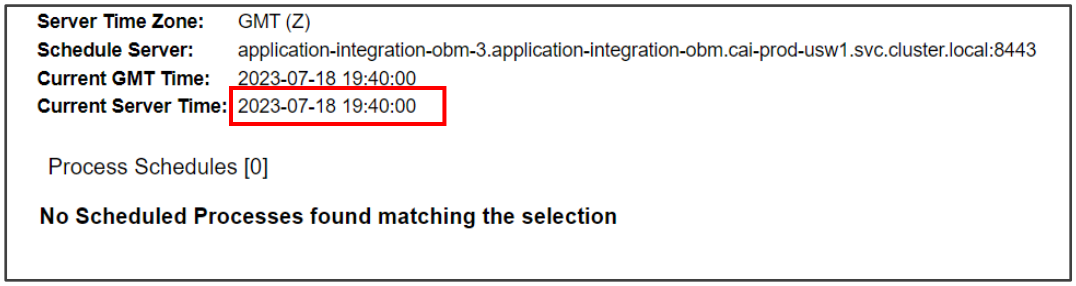
8. From the navigation pane, click **Process Schedules**.



9. From the Process Schedules drop-down, select **Cloud.**

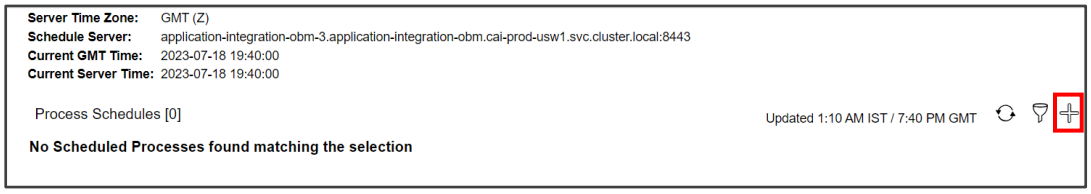


10. Check the **Current Server Time** and make a note of it.

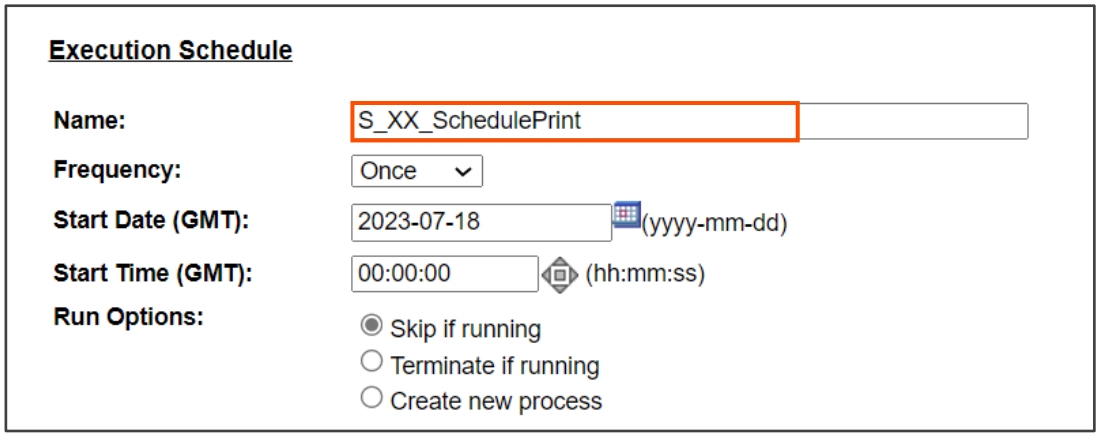


Note: You will use this time while creating the Process Schedule.

11. Click the '**+**' icon to add a new Process Schedule.



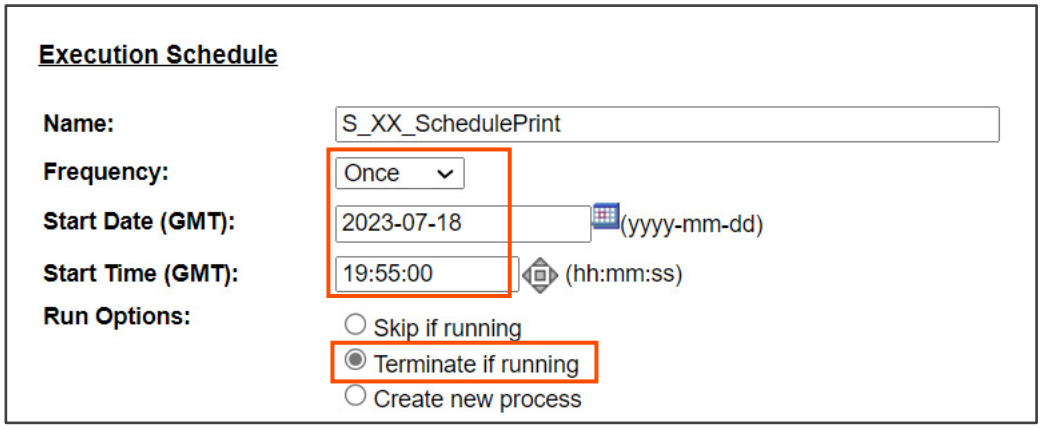
12. In the Execution Schedule window, in the Name field, enter **S_XX_SchedulePrint**.



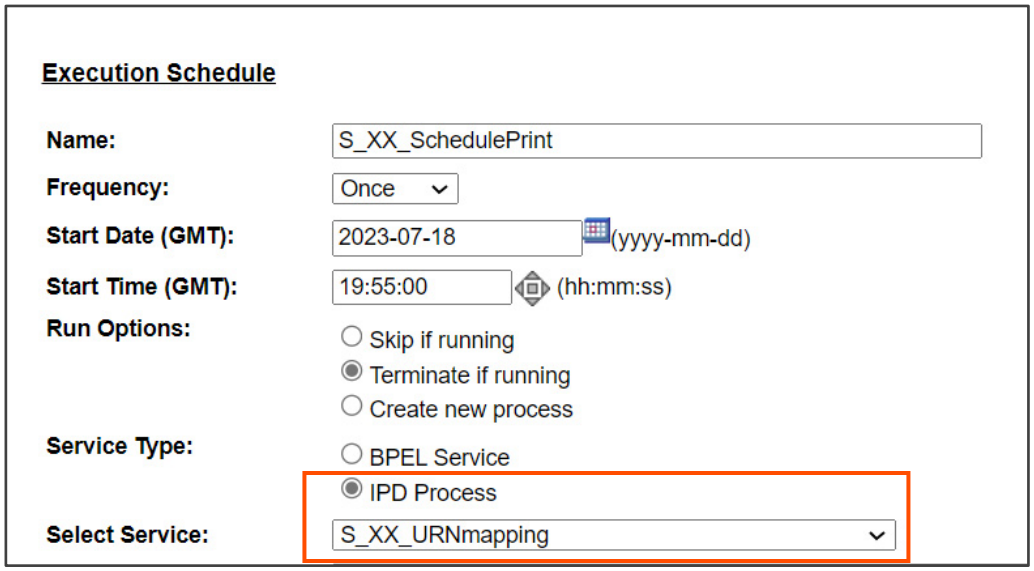
13. From the Frequency drop-down, select **Once**.
14. In the Start Date field, select today's date.
15. In the Start Time field, enter 15 minutes delayed time from the Current Server Time. **For example**, if the Current Server Time is 19:40:00, then Start Time should be 19:55:00.

    Note: This is the time when you want to schedule the process execution.

16. From Run Options, select **Terminate if running**.

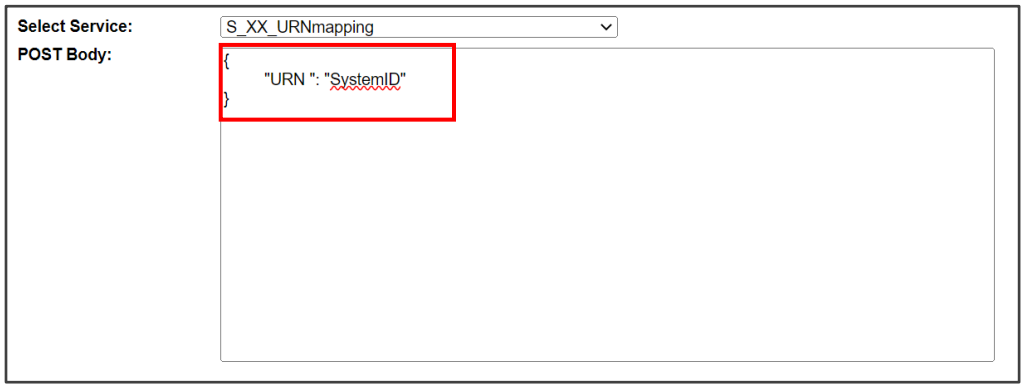


17. From Service Type, select **IPD Process**.
18. From the Select Service drop-down, select **S_XX_URNmapping**.



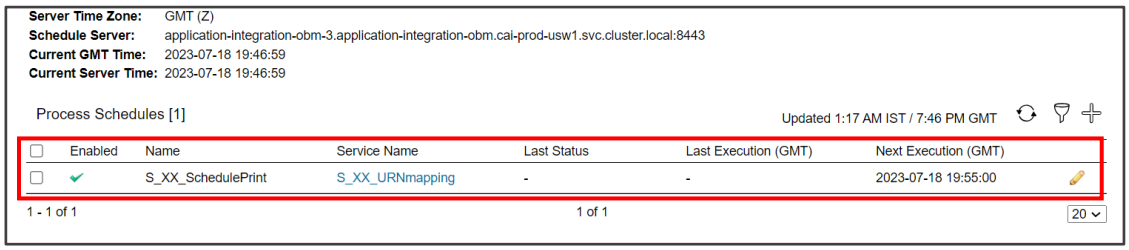
19. In the Post Body field, enter the following code:

```
{
 "URN ": "SystemID"
}
```



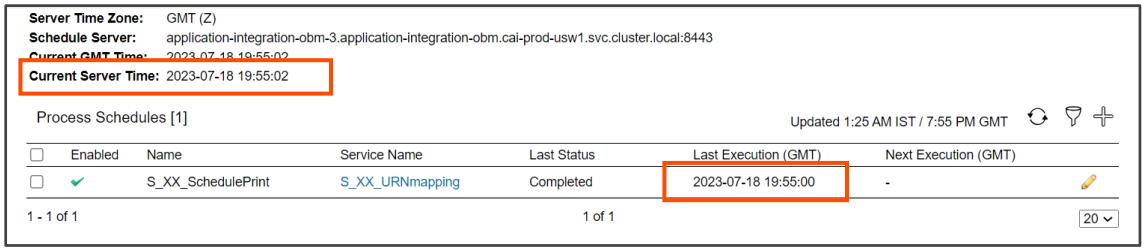
20. Click **OK**.

    The Process Schedule is created.



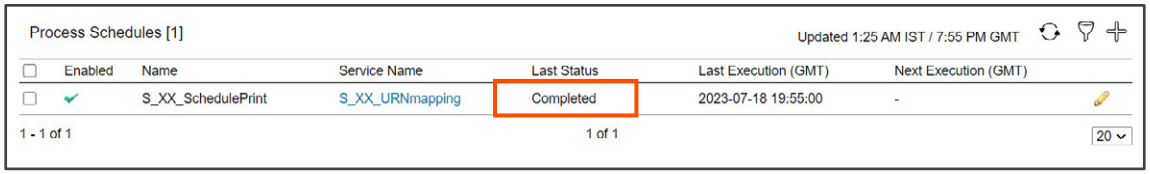
    Note: Observe the **Last Status** field. It is blank.

21. Wait until the **Next Execution** time matches the **Current Server Time**.



    Note: Keep clicking the refresh icon to check the Current Server Time.

22. Observe the Last Status field. It should be **Completed**.



    The process is executed successfully.

This concludes the lab.

Module 13: Monitoring
Lab 13-3: Start and Stop an Event Source in a Listener-based Connection

Overview:
You can start and stop event sources in listener-based connections that run on a Secure Agent or a Secure Agent group. With this option, you do not have to unpublish and republish the connection to stop and start the event-based processes.

In Lab 7-5: Create a Kafka Connection, you have created two processes: **S-XX-KafkaWriteDemo** and **S-XX-KafkaReadDemo**. The S-XX-KafkaWriteDemo process writes a message to the topic. Whereas the S-XX-KafkaReadDemo is an Even-driven process that reads the data from the topic.

In this lab, you will stop and start the Event Source in the Kafka connection and observe that S-XX-KafkaReadDemo is executing without republishing the process.

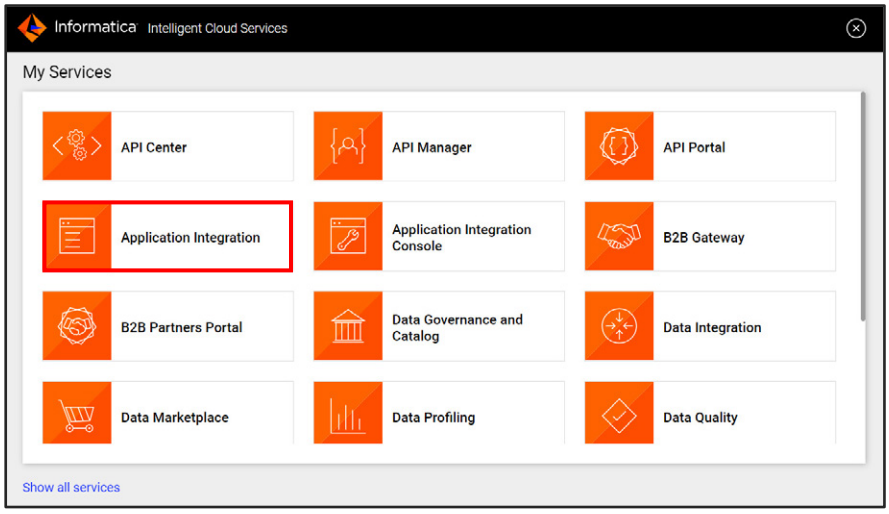
Objectives:
- Stop the Event Source in the Kafka connection
- Start the Event Source in the Kafka connection

Duration:
20 Minutes

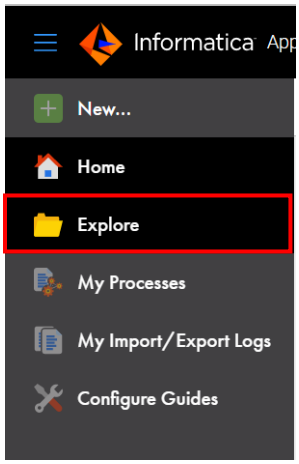
Tasks

Execute an Event-driven Process

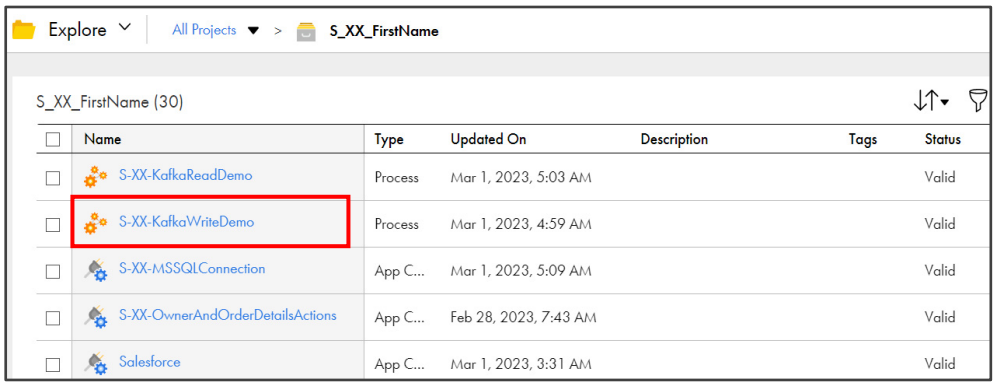
1. Log into IICS and access the **Application Integration** service.



2. From the navigation pane, click **Explore**.

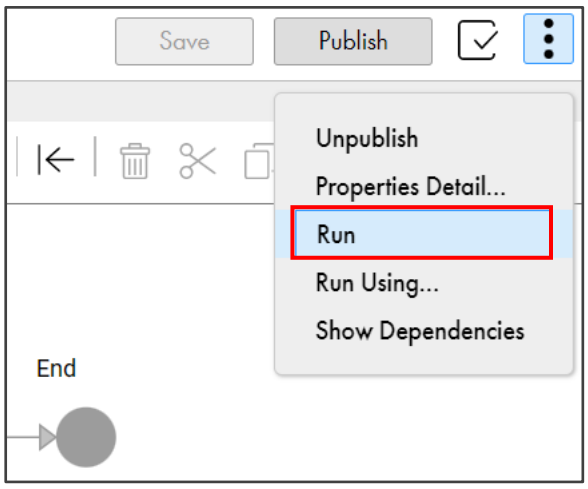


3. Open your project folder S_XX_FirstName.
4. Search and open **S-XX-KafkaWriteDemo**.

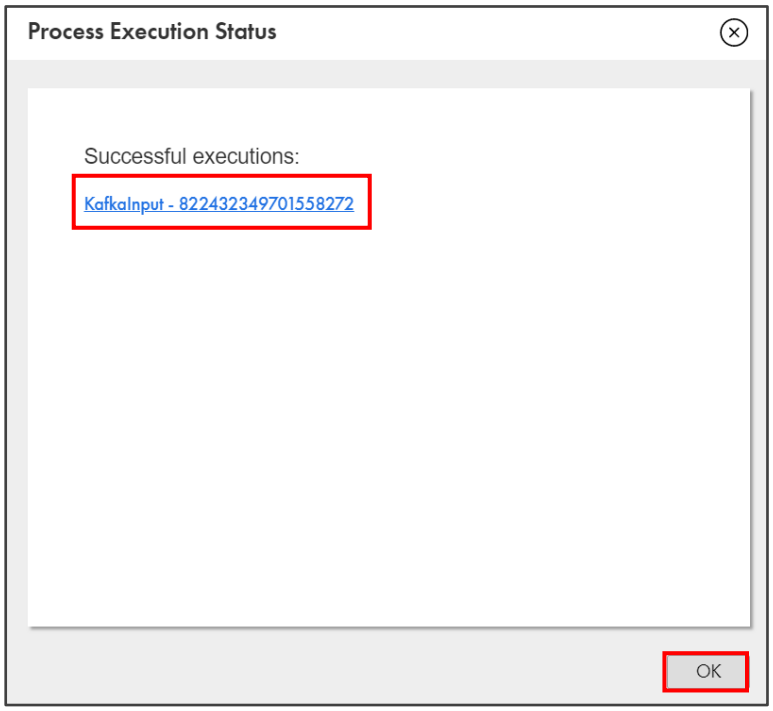


Note: you can use the Search box to find S-XX-KafkaWriteDemo.

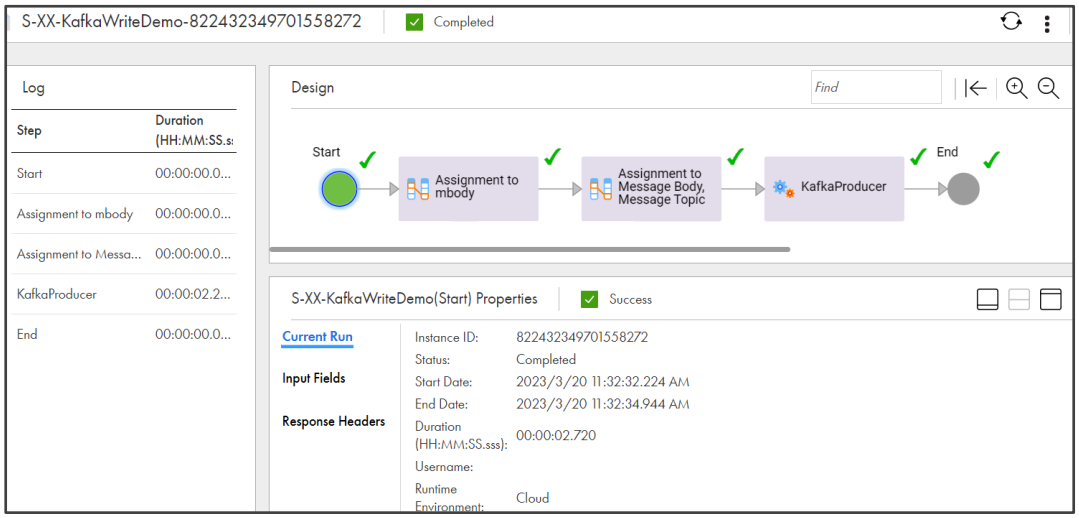
5. Click the Actions menu and select **Run**.



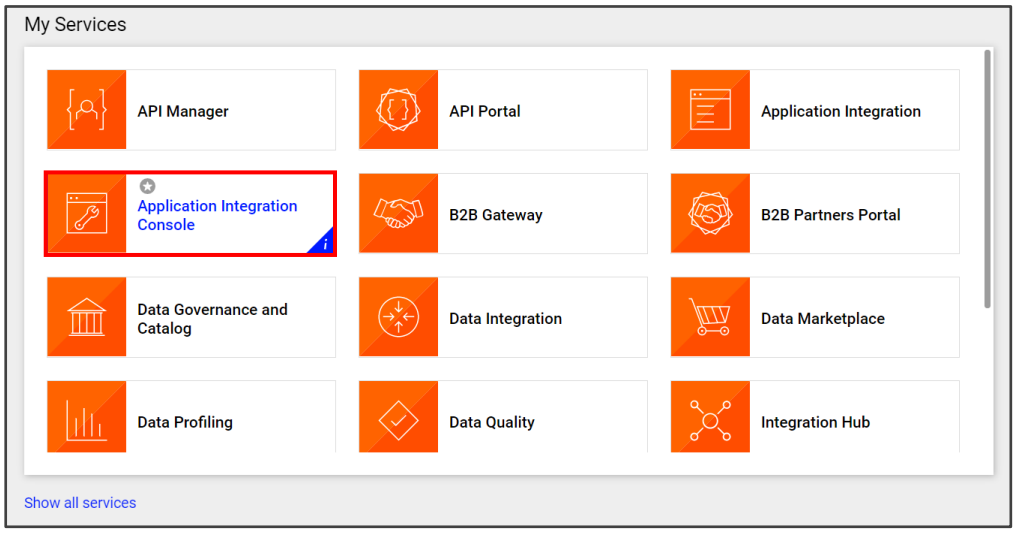
6. Select the execution link and click **OK**.



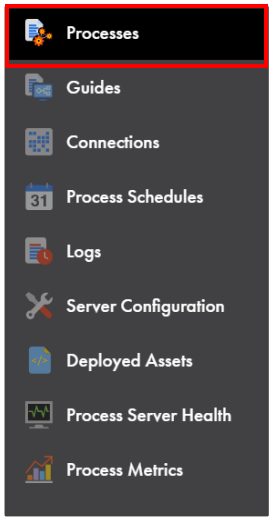
The S-XX-KafkaWriteDemo process was executed successfully.



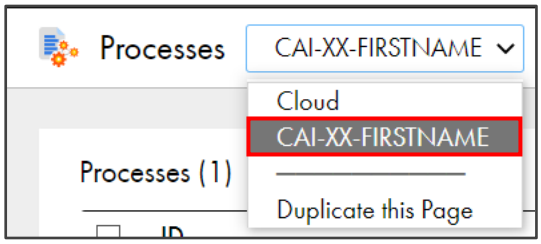
7. Access the **Application Integration Console** service.



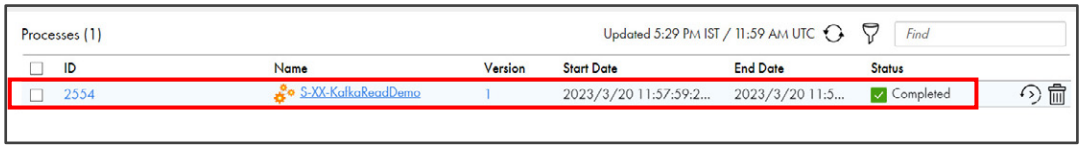
8. From the navigation pane, click **Processes** if it is not selected.



9. From the Process drop-down on the top, select your Secure Agent.

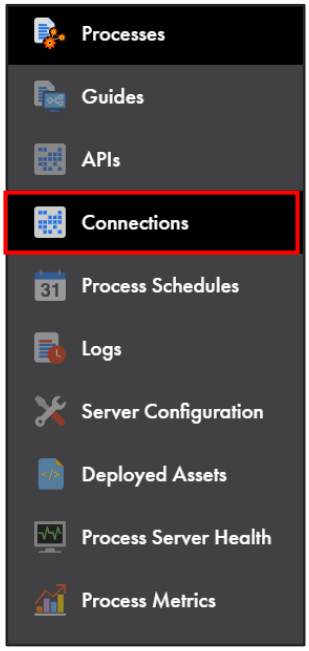


10. Observe that S-XX-KafkaReadDemo is executed.

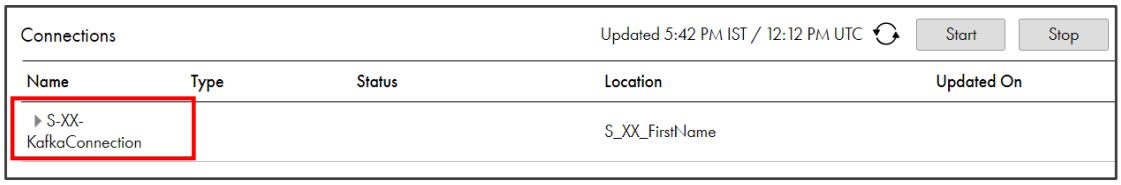


Stop the Event Source in the Kafka Connection

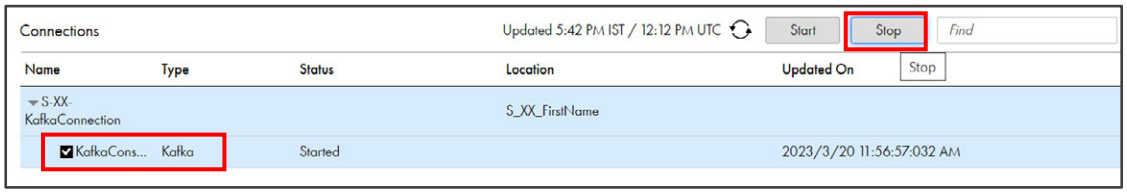
11. From the navigation pane, click **Connections**.



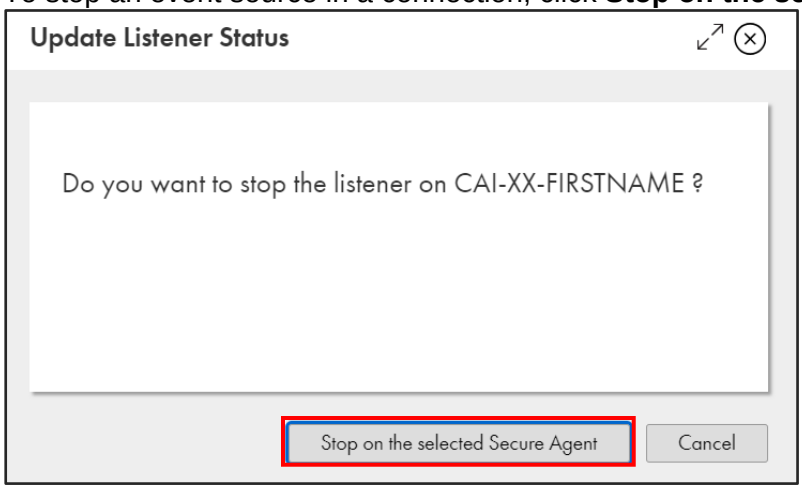
12. From the Connections drop-down on the top, select your Secure Agent to see the connections.
13. Expand S-XX-KafkaConnection to view the Event Source.



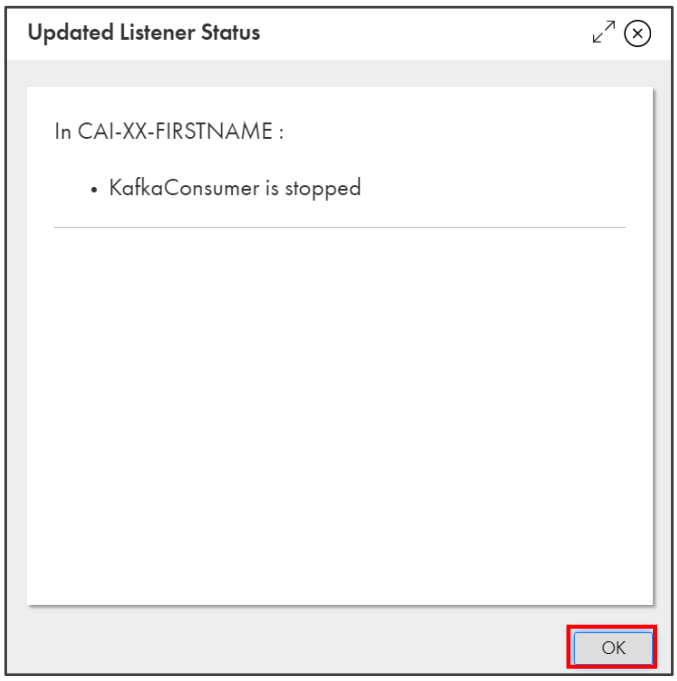
14. Select the **KafkaConsumer** event source and click **Stop**.



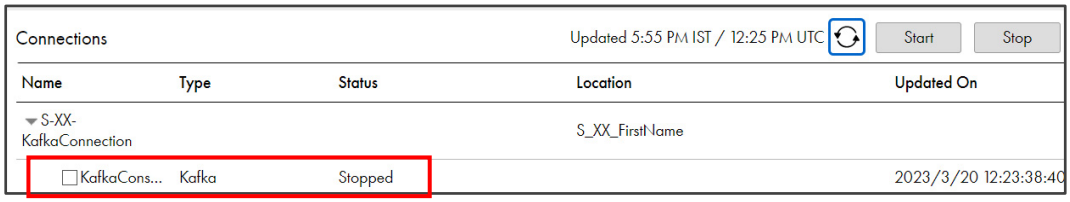
15. To stop an event source in a connection, click **Stop on the selected Secure Agent**.



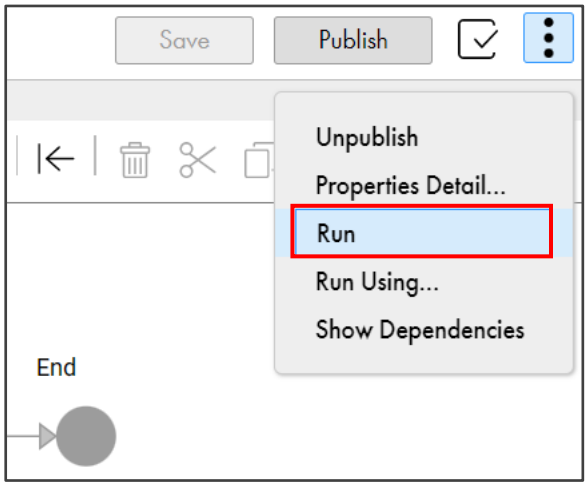
16. Click **OK**.



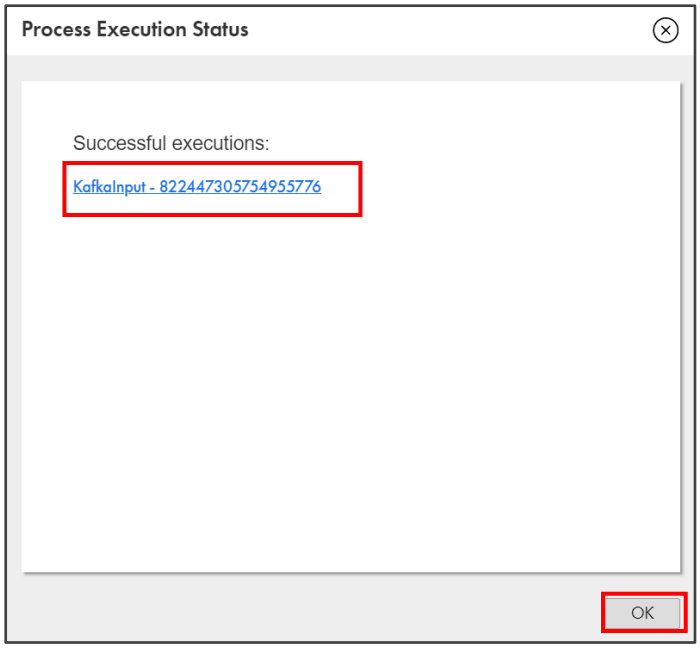
The status of the KafkaConsumer event source is updated to Stopped on the Connections page.



17. Go back to the **Application Integration** service.
18. Open the **S-XX-KafkaWriteDemo** process if it is closed.
19. Click the Actions menu and select **Run**.

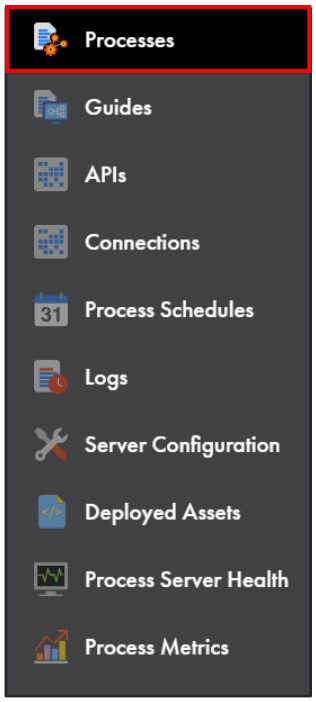


20. Select the execution link and click **OK**.

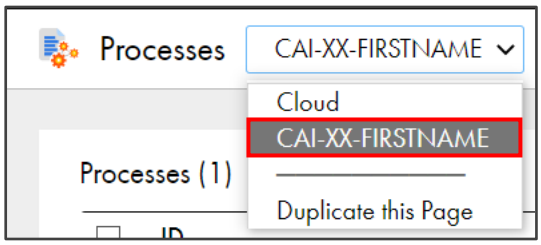


The S-XX-KafkaWriteDemo process was executed successfully.

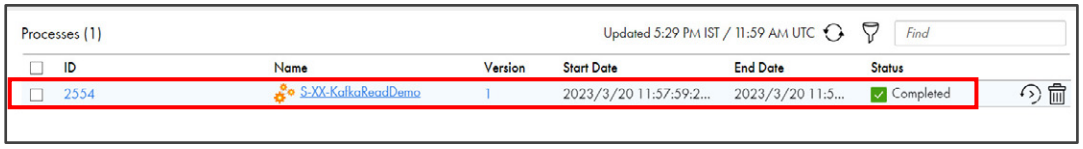
21. Access the **Application Integration Console** service and from the navigation pane, click **Processes**.



22. From the Process drop-down on the top, select your Secure Agent.

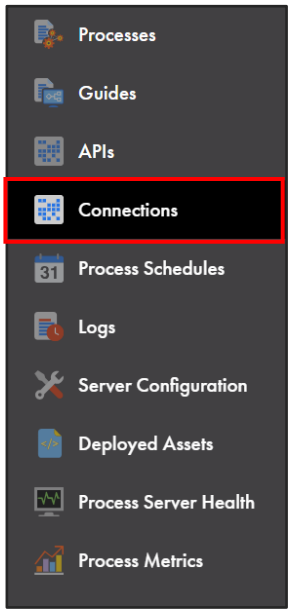


23. Observe that S-XX-KafkaReadDemo is not executed this time. This is because you have stopped the KafkaConsumer event source.

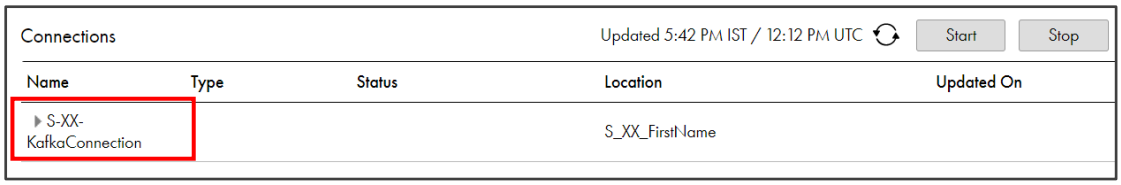


Start the Event Source in the Kafka Connection

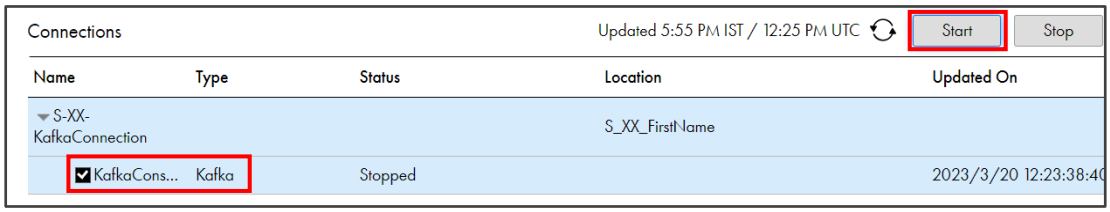
24. From the navigation pane, click **Connections**.



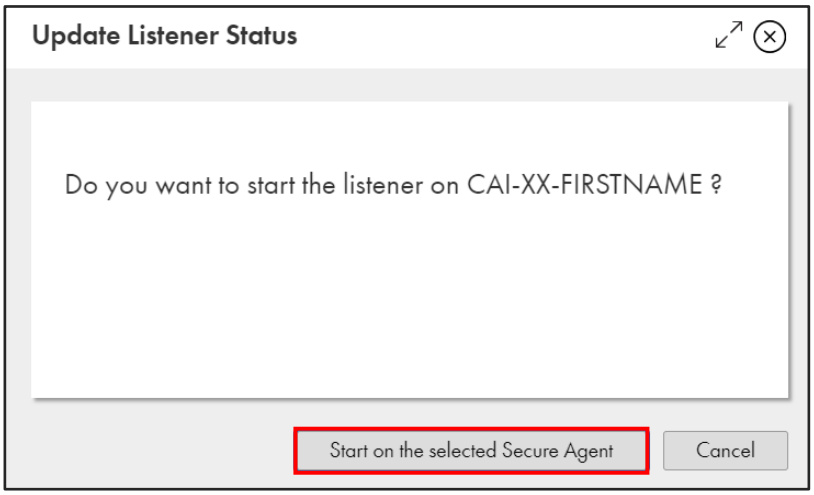
25. Make sure that Secure Agent is selected in the Connections drop-down on the top.
26. Expand S-XX-KafkaConnection to view the Event Source.



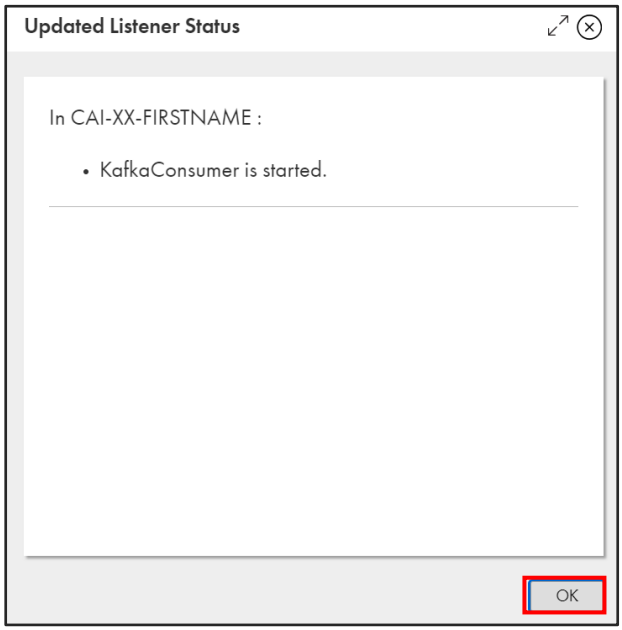
27. Select the **KafkaConsumer** event source and click **Start**.



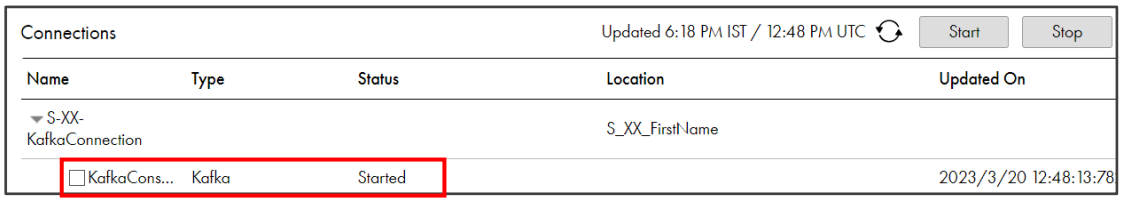
28. To start the Event Source, click **Start on the selected Secure Agent**.



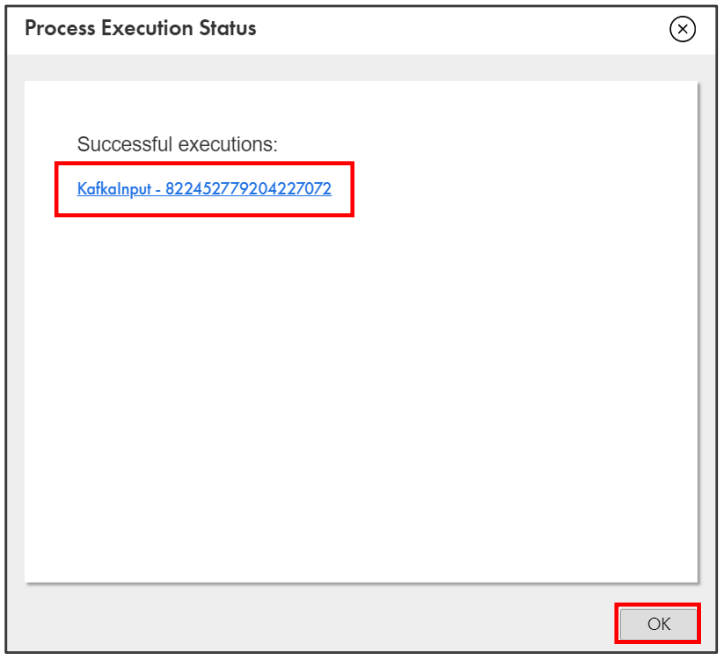
29. Click **OK**.



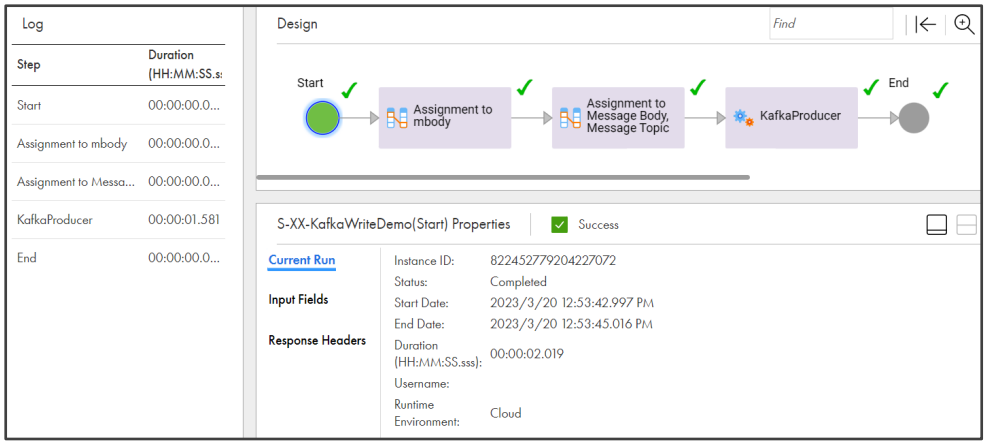
The status of the KafkaConsumer event source is updated to Started on the Connections page.



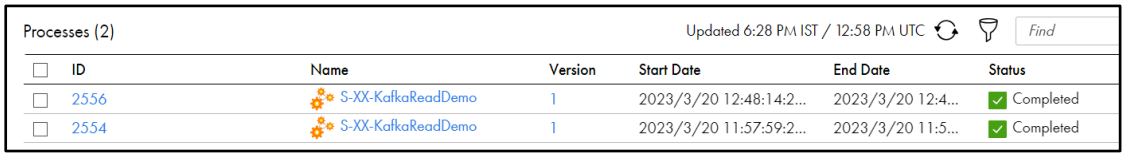
30. Go back to the **Application Integration** service and run the **S-XX-KafkaWriteDemo** process once again.
31. Select the execution link and click **OK**.



The S-XX-KafkaWriteDemo process was executed successfully.



32. Switch back to the **Application Integration Console** service and from the navigation pane, click **Processes**.
33. Click the icon to refresh the Processes page.
34. Observe that S-XX-KafkaReadDemo is executed this time. This is because you have started the KafkaConsumer event source.



This concludes the lab.

Module 14: API Management
Lab 14-1: API Manager

Overview:
You can create a managed API for any Informatica Cloud Application Integration service. By default, the managed API is active and the API Portal view for the API is enabled. The name that you assign to the API is part of the API URL.
In this lab, you will create a managed API for the ExchangeRate process.

Objectives:

- Create a Managed API for the ExchangeRate process

Duration:
15 Minutes

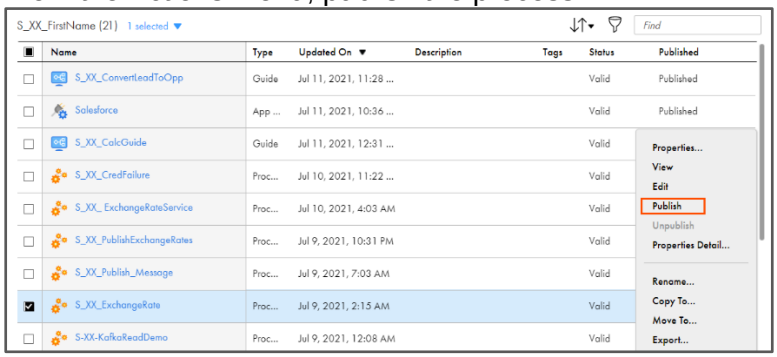
Tasks

Note: The screen on your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

Publish the ExchangeRateService Process

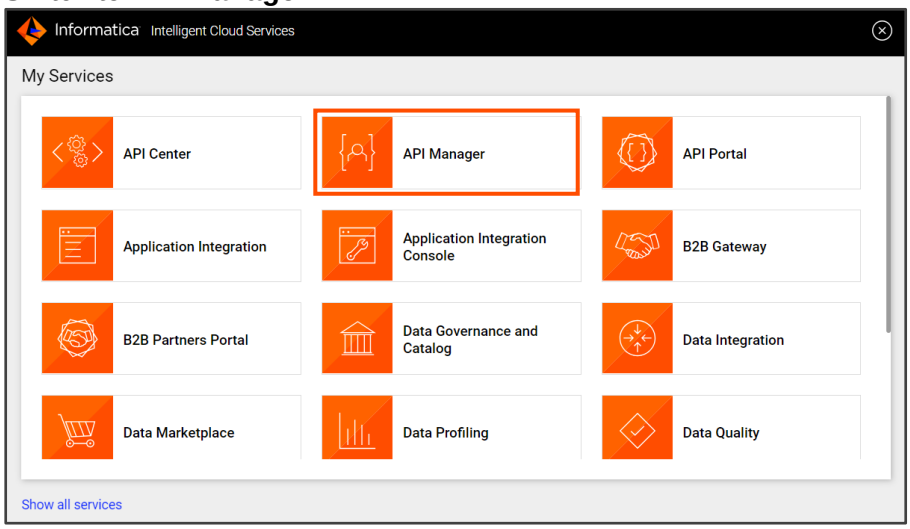
While executing the sub-process failure lab (11-1), you unpublished the ExchangeRate process. For this lab, you must publish the process.

1. Navigate to the **Application Integration** service.
2. Open your project folder and locate the **S_XX_ExchangeRate** process.
3. From the Actions menu, publish the process.



Create a Managed API for the ExchangeRate process

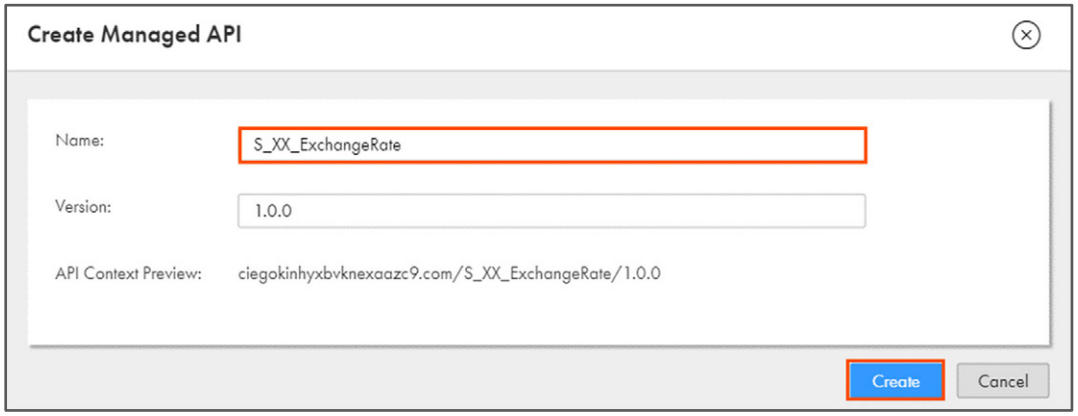
4. Switch to **API Manager**.



Note: If you get the API Domain Name window, click **Save**.

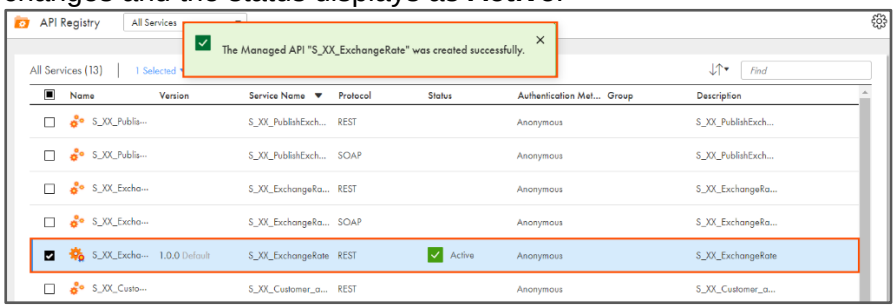
5. In the **API Registry** tab, for the ExchangeRate (REST) process, from the **Actions** menu, select **Create Managed API**.

6. Click **Create**.

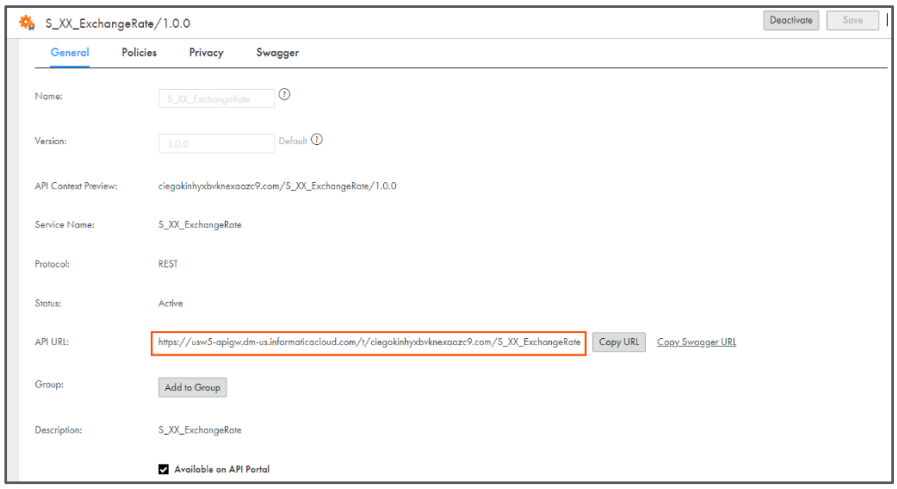


A message '**Managed API is created successfully**' appears.

7. Locate the **S_XX_ExchangeRate** REST service and observe that the icon of this service changes and the status displays as **Active**.



8. Open the service and check the **API URL** field. This is a proxy URL provided for the service.



This concludes the lab.

Module 15: CAI and CDI Integration

Lab 15-1: Create a Flat File Connection

Overview:

Flat file connections store the information to create, access, and store flat files.

In this lab, you will set up a connection to access flat files on your local machine.

Objective:

- Create a flat file connection

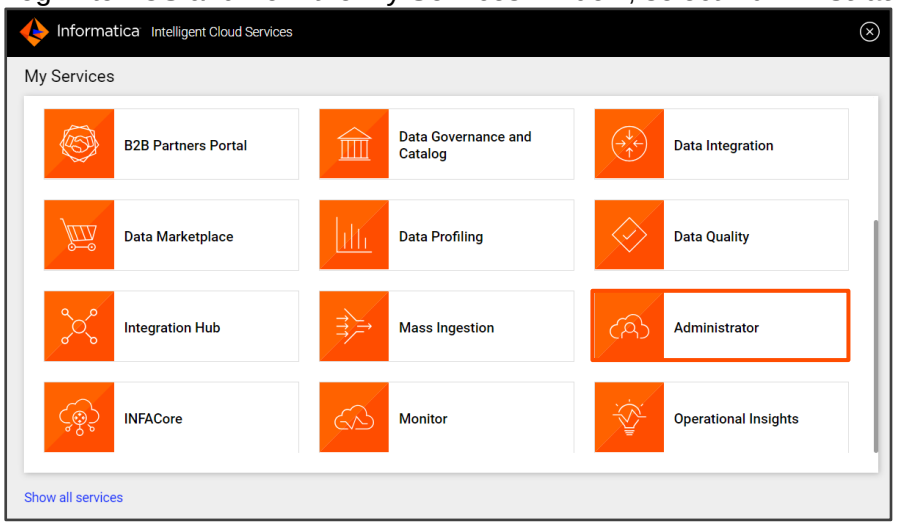
Duration:
10 minutes

Tasks

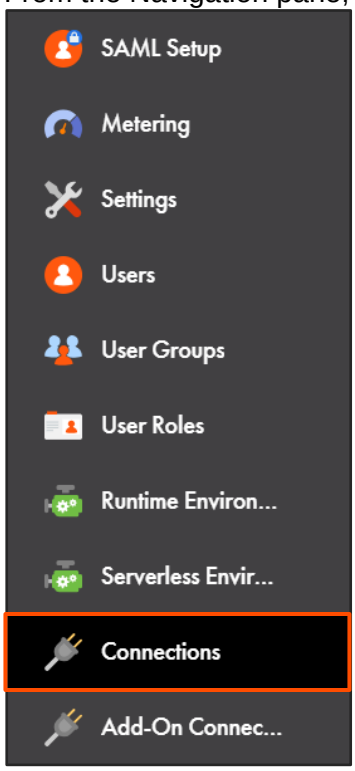
Note: The screen in your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

Create a Flat File Connection in IICS

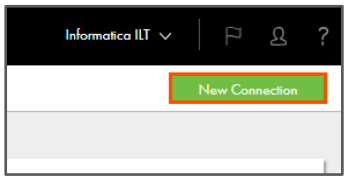
1. Log in to IICS and from the My Services window, select **Administrator**.



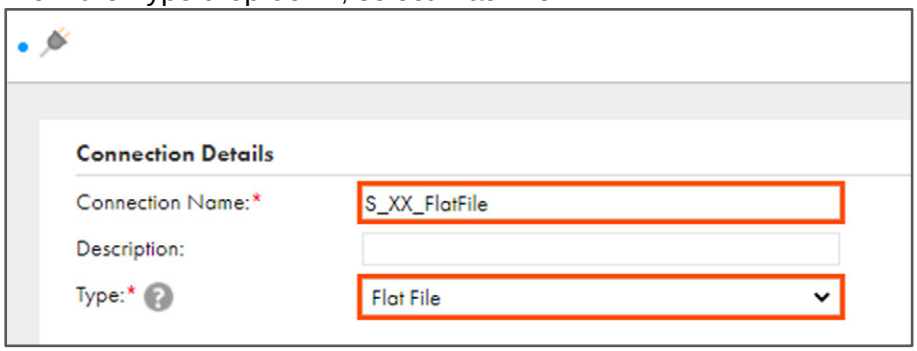
2. From the Navigation pane, select **Connections**.



3. To create a new connection, select **New Connection**.

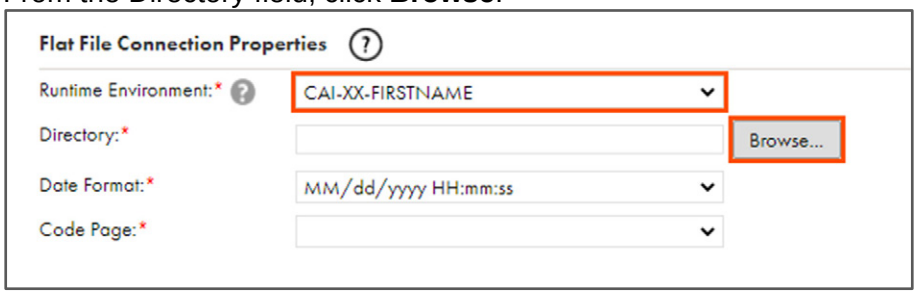


4. Enter the Connection Name as **S_XX_FlatFile**.
   Note: Replace XX with your student number.
5. From the Type drop-down, select **Flat File**.

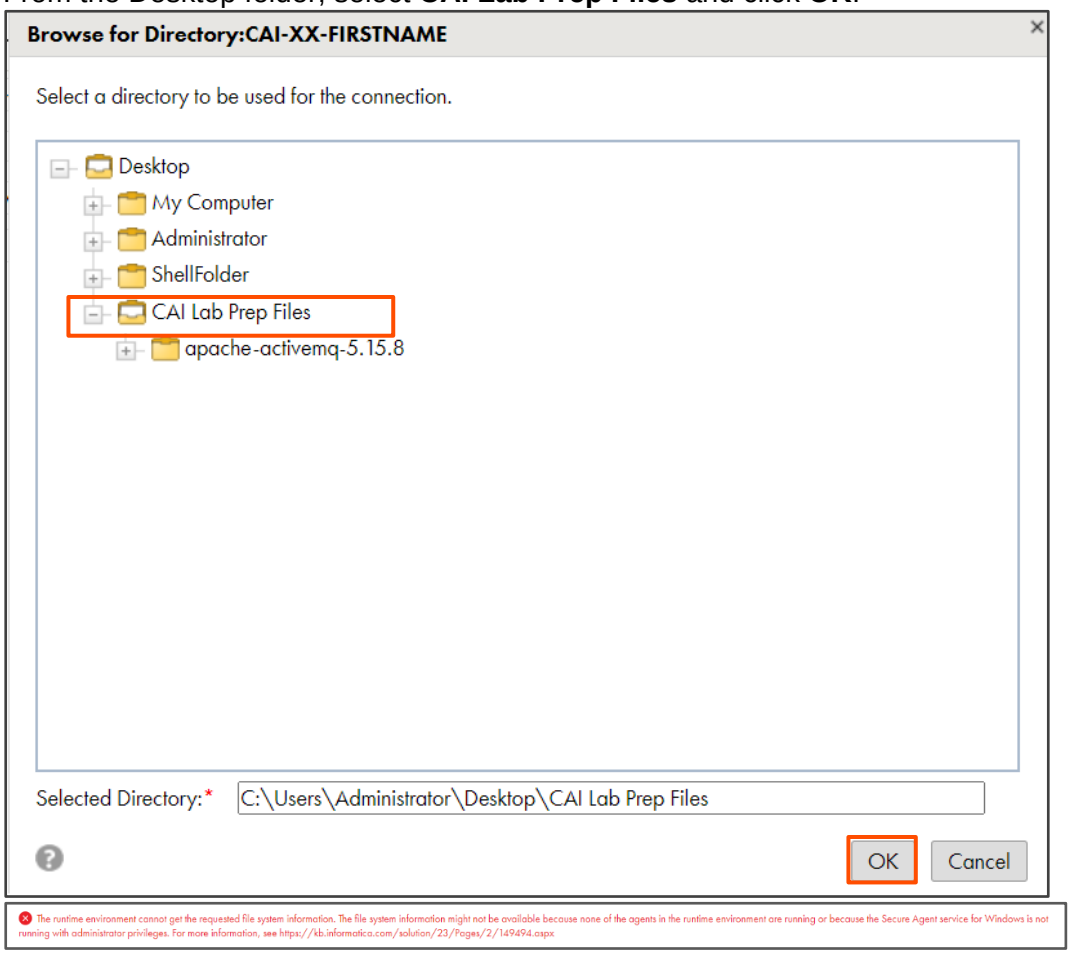


6. From the **Runtime Environment** drop-down, select your secure agent group.

7. From the Directory field, click **Browse**.



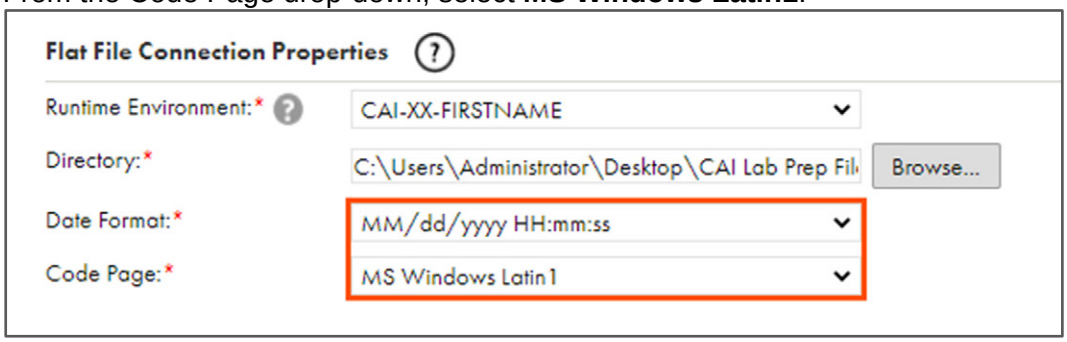
8. From the Desktop folder, select **CAI Lab Prep Files** and click **OK**.



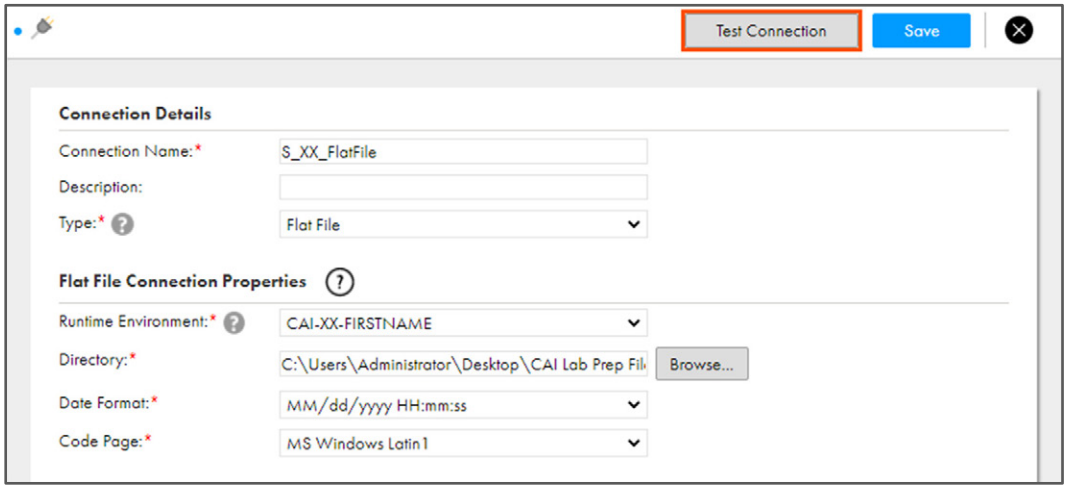
Note: If you get the above error, copy the **CAI Lab Prep Files** location, and paste it into the Selected Directory field.

9. Retain the value in the **Date Format** field.

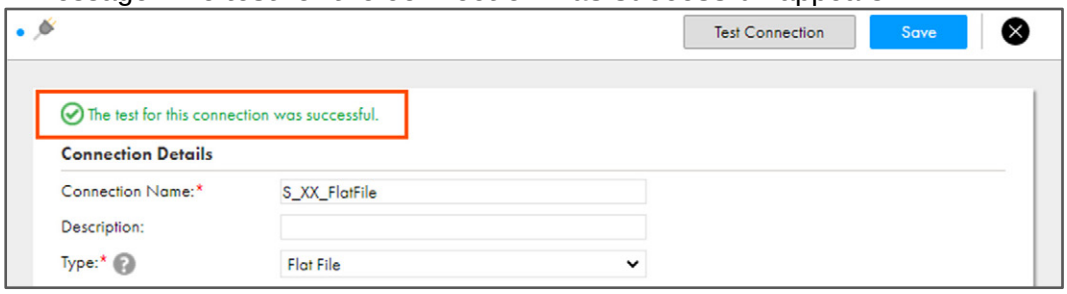
10. From the Code Page drop-down, select **MS Windows Latin1**.



11. To test the connection, click **Test Connection**.

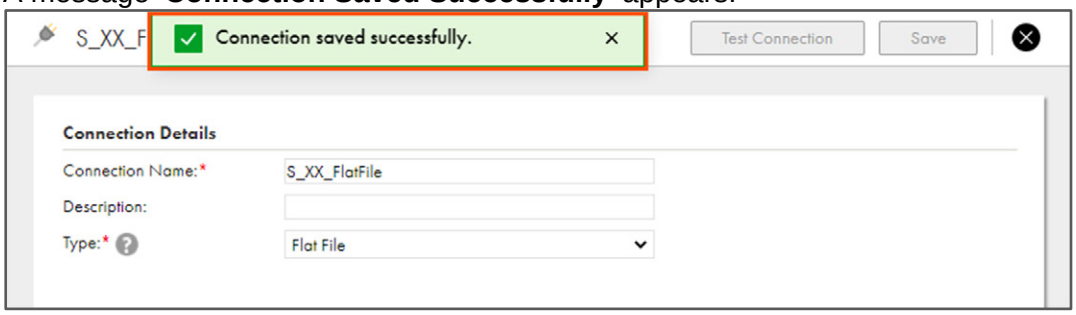


A message '**The test for the connection was successful**' appears.



12. **Save** the connection.

13. A message '**Connection Saved Successfully**' appears.



This concludes the lab.

Module 15: CAI and CDI Integration

Lab 15-2: Create a Basic Mapping and a Mapping Task

Overview:
In IICS, the Mapping designer creates a mapping that you can use in several mapping tasks. In this lab, you will create a mapping to filter the Order data and create a mapping task.

Objective:

- Use the Mapping Designer to create a mapping
- Create a Mapping task

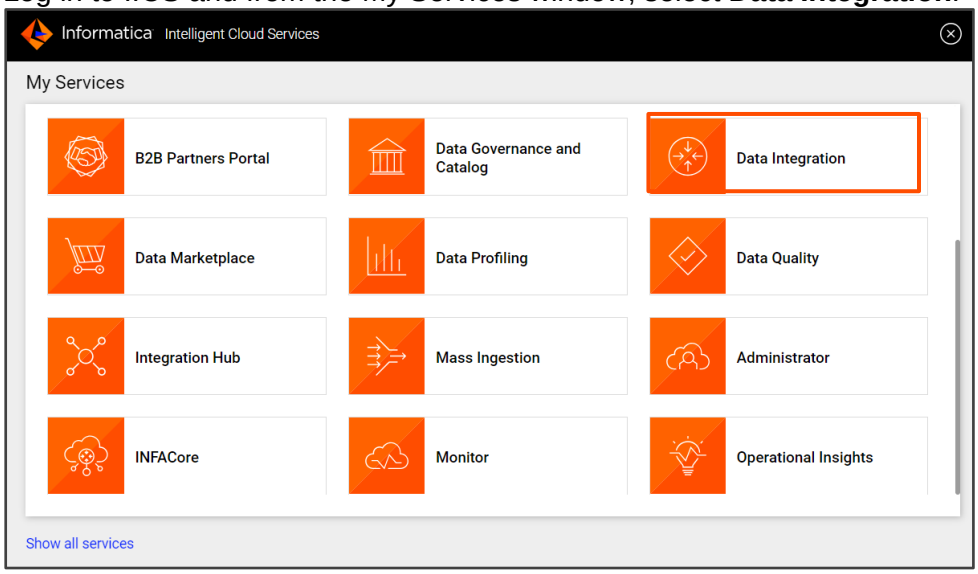
Duration:
20 minutes

Tasks

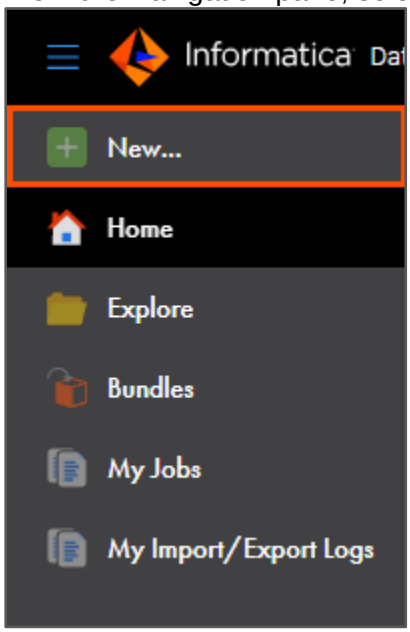
Note: The screen in your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

Create a Mapping

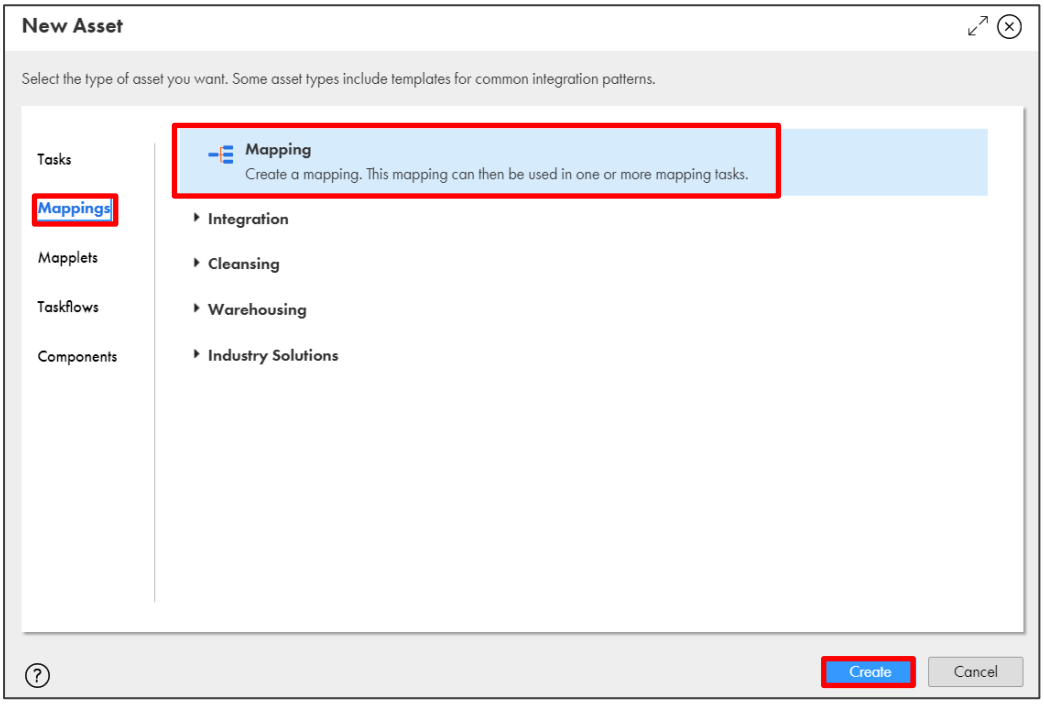
1. Log in to IICS and from the My Services window, select **Data Integration**.



2. From the Navigation pane, select **New**.

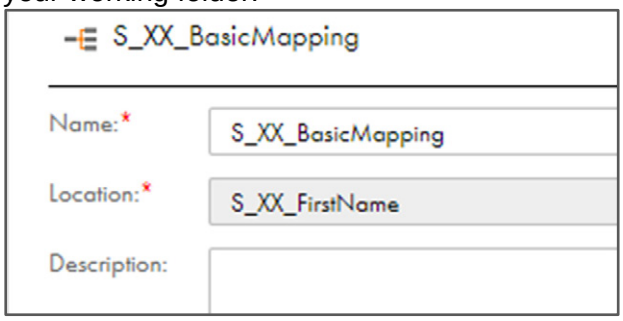


3. From the New Asset window, select the **Mappings** tab.
4. Select **Mapping** and click **Create**.

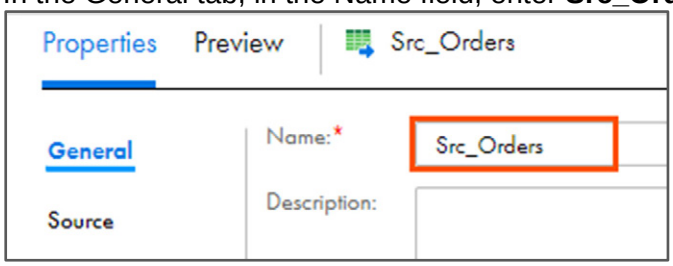


The Mapping Designer opens.

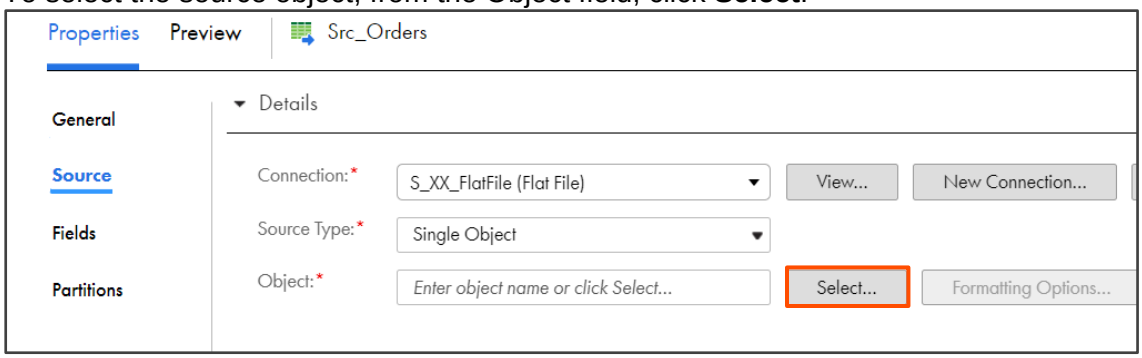
5. In the Name field, enter **S_XX_BasicMapping** and verify that the asset location is set to your working folder.



6. To configure the source, from the mapping canvas, click the **Source** transformation.
7. In the General tab, in the Name field, enter **Src_Orders**.

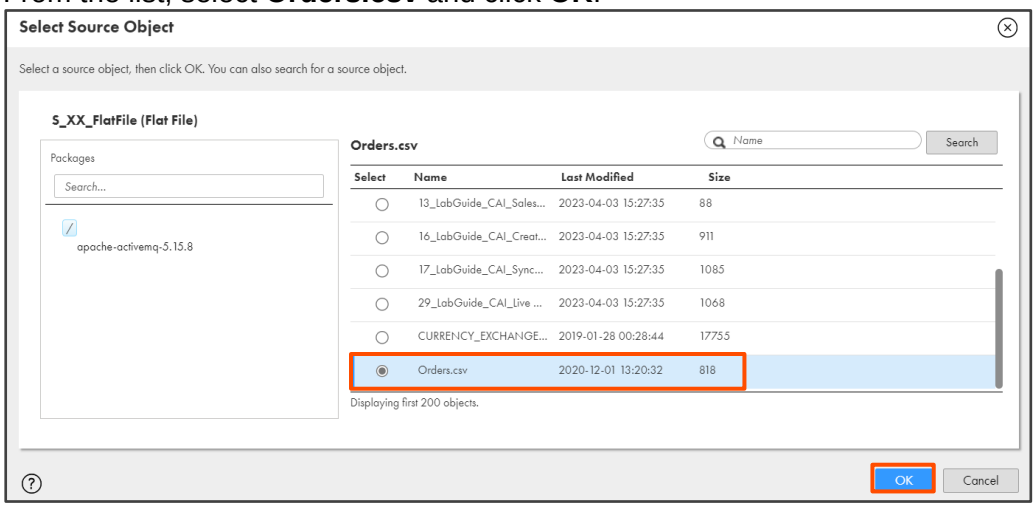


8. From the properties pane, click **Source**.
9. From the Connection drop-down, select **S_XX_FlatFile** and retain Source Type as **Single Object**.
10. To select the source object, from the Object field, click **Select**.



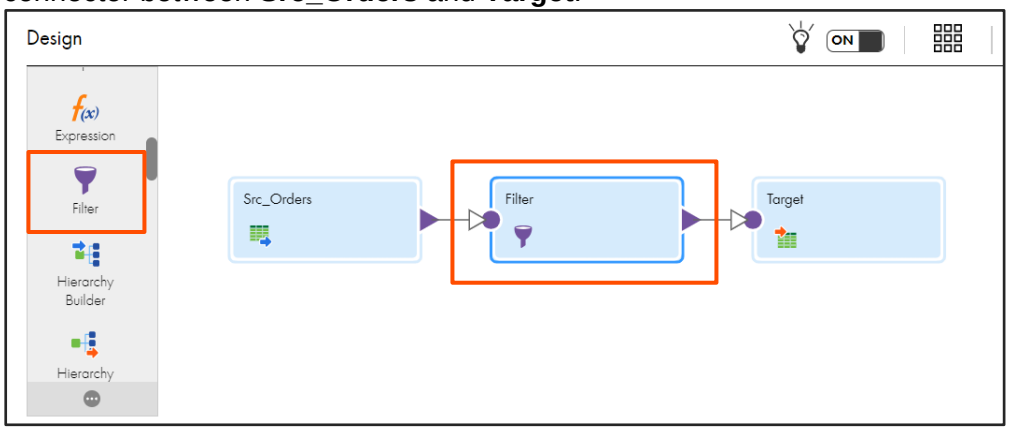
The Select Source Object window appears.

11. From the list, select **Orders.csv** and click **OK**.

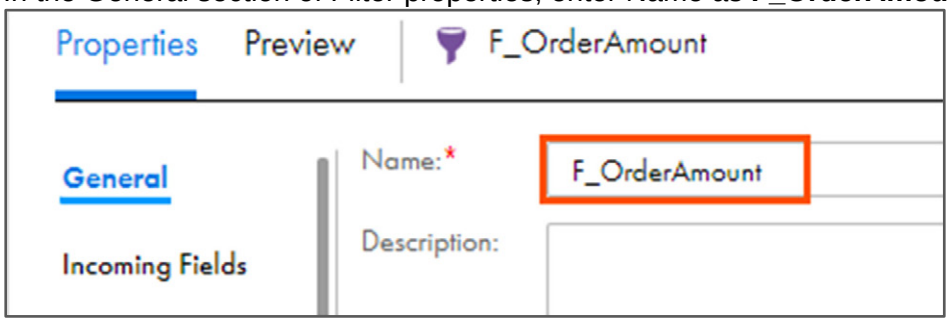


Add Filter Transformation

12. From the list of available transformations, drag and drop **Filter** transformation onto the connector between **Src_Orders** and **Target**.

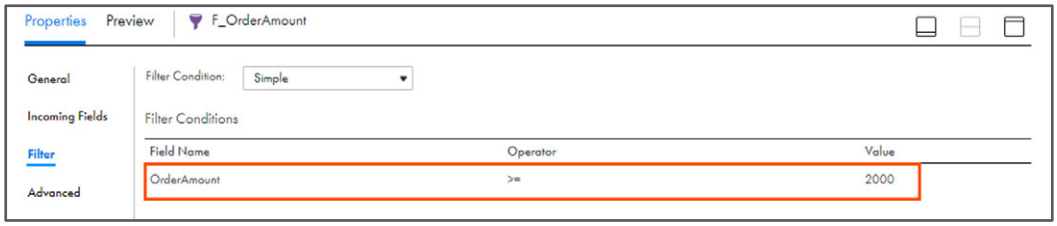


13. In the General section of Filter properties, enter Name as **F_OrderAmount**.

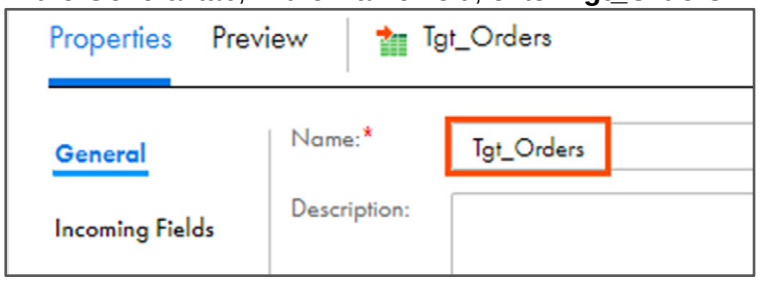


14. From the properties pane, click **Filter**.
15. Retain the Filter Condition as **Simple** and to add a new filter condition, click ⊕.
16. Enter the filter condition as shown in the table below:

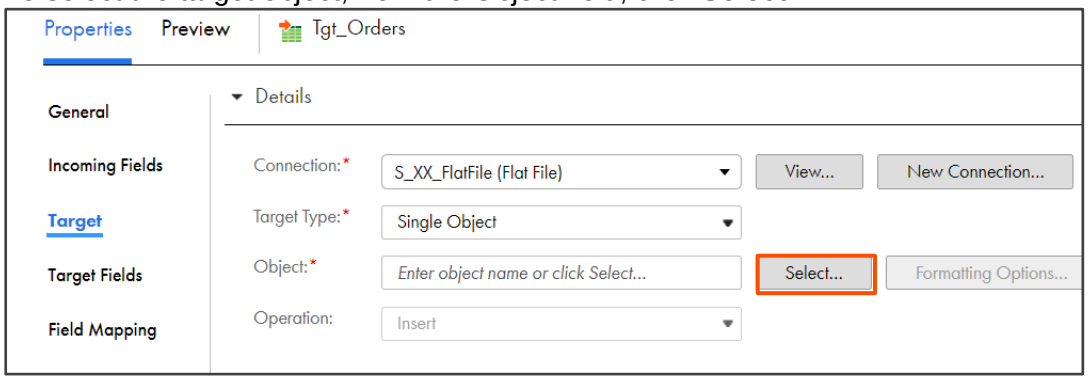
Field Name	Operator	Value
OrderAmount	>= (Greater than or equals)	2000



17. To configure the target, from the mapping canvas, click the **target** transformation.
18. In the General tab, in the Name field, enter **Tgt_Orders**.



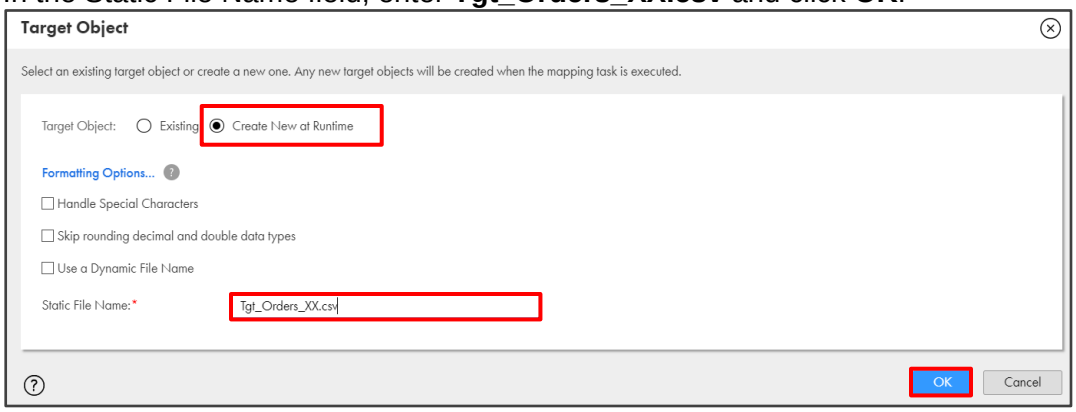
19. From the properties pane, click **Target**.
20. From the Connection drop-down, select **S_XX_FlatFile** and retain the Target Type as **Single Object**.
21. To select the target object, from the Object field, click **Select**.



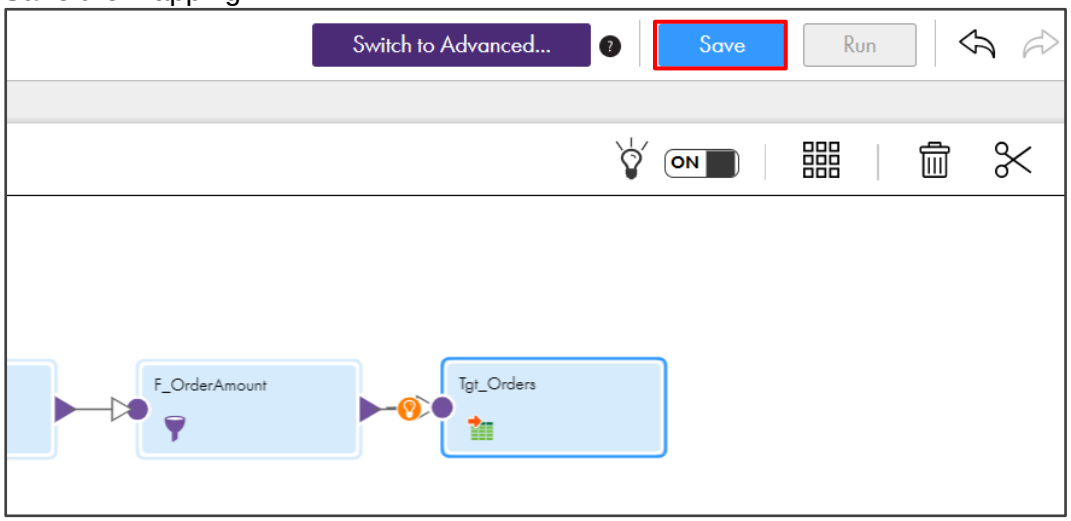
The Target Object window appears.

22. To create the target at runtime, select **Create New at Runtime**.

23. In the Static File Name field, enter **Tgt_Orders_XX.csv** and click **OK**.



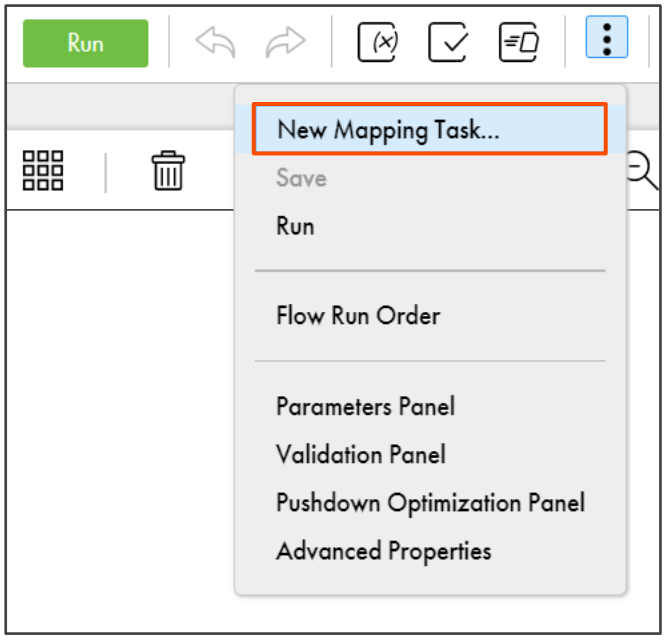
24. **Save** the mapping.



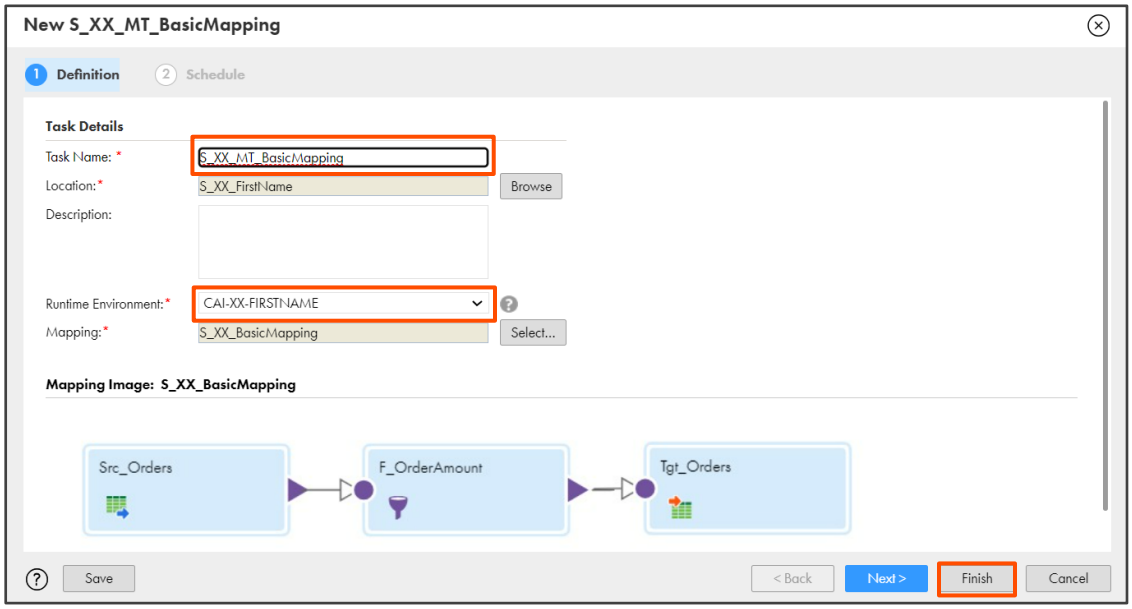
Note: After you save the mapping, the mapping status (**Valid or Invalid**) is automatically displayed next to the mapping name. To check for the validity of a mapping while working with the mapping, click .

You can create a mapping in advanced mode when you want to process multilevel hierarchical data, embedded code snippets, and workloads at any scale. The Mapping Designer updates the mapping canvas to include the transformations and functions that enable advanced functionality. Mappings in advanced mode can run on advanced clusters in addition to the Secure Agent.

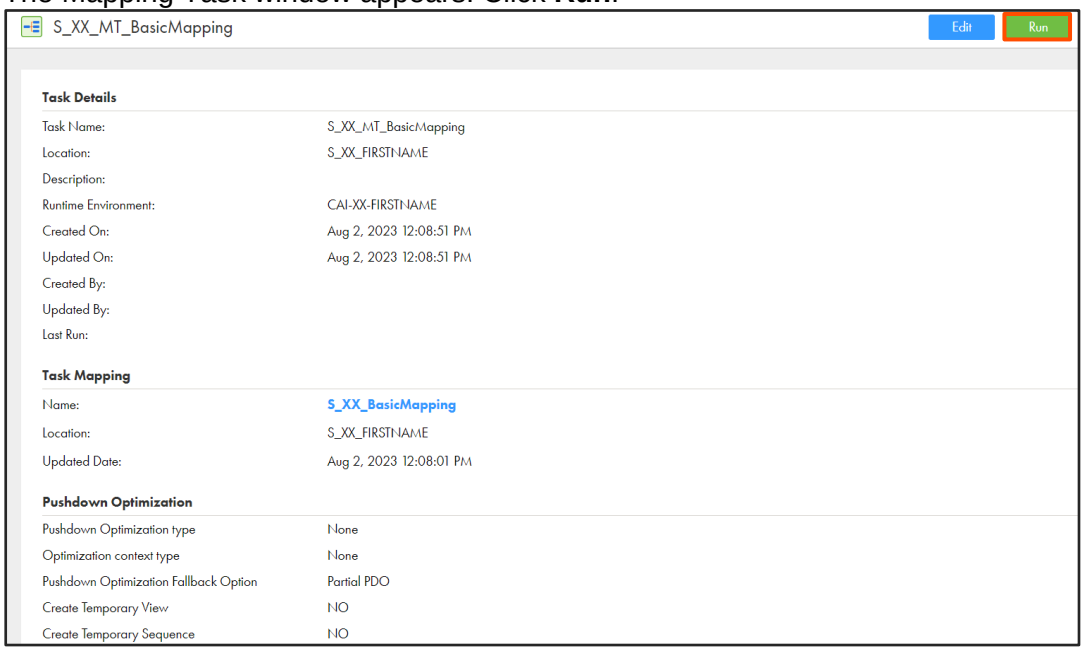
25. From the Actions menu, select **New Mapping Task**.



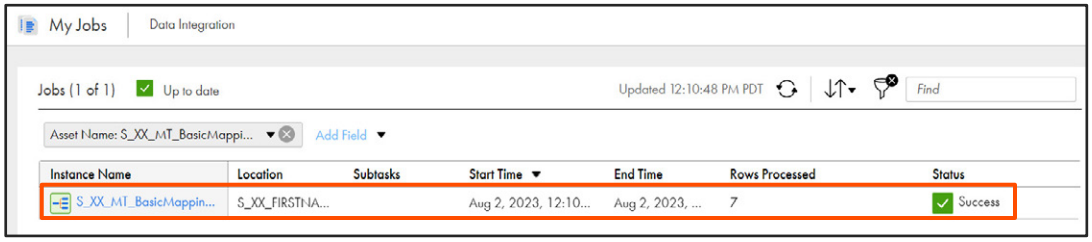
26. In the Task Name field, enter **S_XX_MT_BasicMapping**.
27. From the Runtime Environment drop-down, select your secure agent group and click **Finish**.

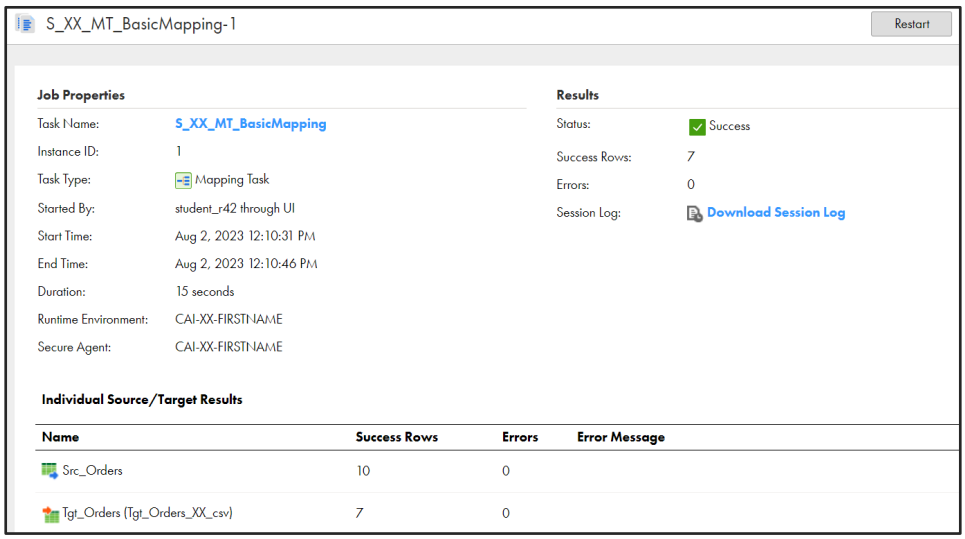


28. The Mapping Task window appears. Click **Run**.



29. Navigate to **My Jobs** and click the task to check the results.





This concludes the lab.

Module 15: CAI and CDI Integration
Lab 15-3: Invoke a Mapping in CAI

Overview:
In this lab, you will create a process to invoke a mapping in CAI. In the previous labs, you created a flat file connection, a basic mapping, and a mapping task. You will now utilize this mapping task in a Service step to invoke the mapping.

Objectives:
- Create a Process
- Use a Service step and add a Cloud task
- Execute the Process

Duration:
20 Minutes

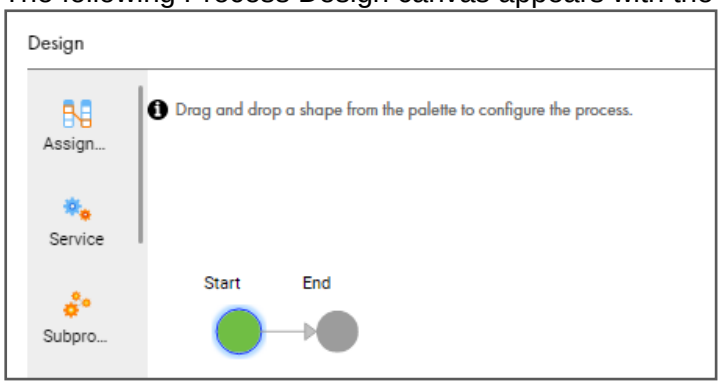
Tasks:

IMPORTANT: At the end of this lab, there is a demo that should be performed by the instructor.

Note: The screen on your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

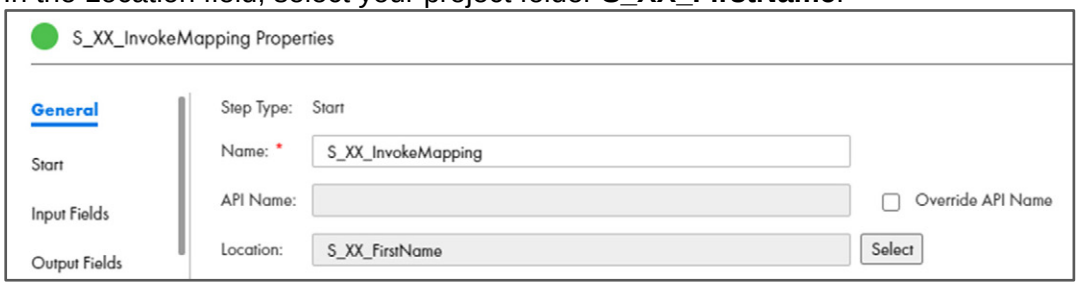
Create a new process:

1. Switch to the **Application Integration** service and create a new **Process**.
   The following Process Design canvas appears with the Start and End steps.

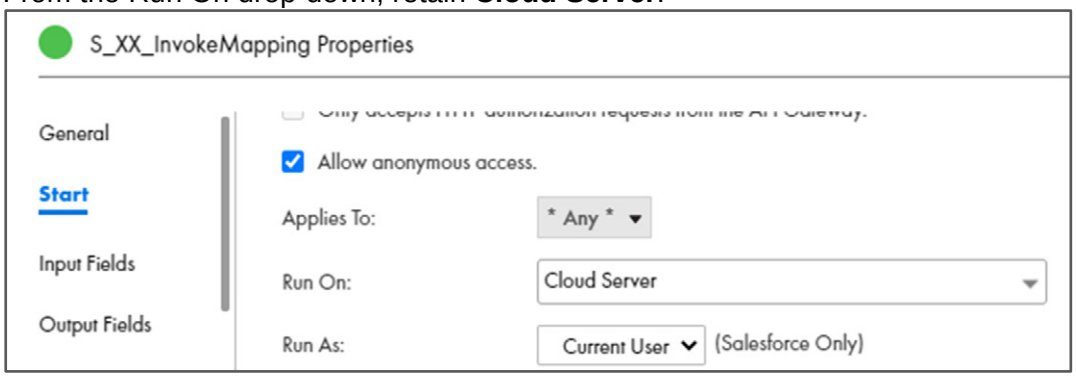


2. From the process canvas, select the **Start** step.
3. In the General tab, enter the name as **S_XX_InvokeMapping**.
4. Skip the API Name option.

5. In the Location field, select your project folder **S_XX_FirstName**.

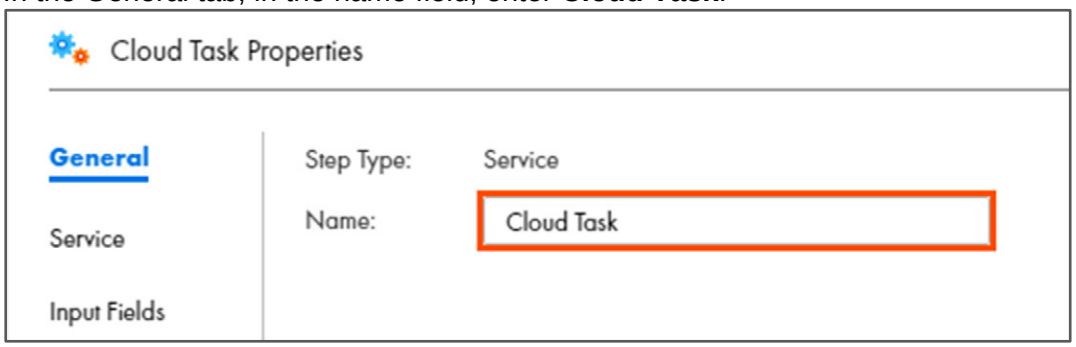


6. From the properties pane, select **Start**.
7. Select the **Allow anonymous access** option.
8. From the Run On drop-down, retain **Cloud Server**.

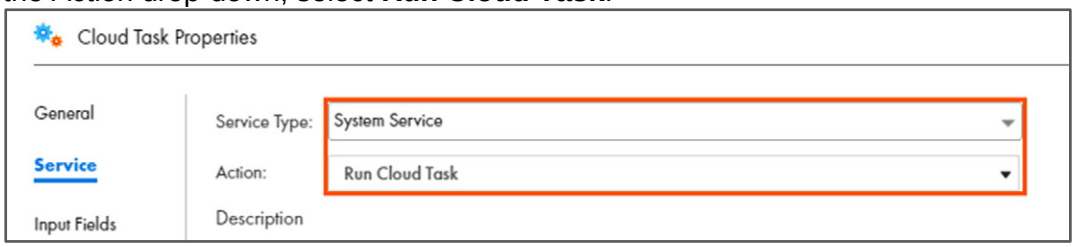


Note: Allowed Users and Allowed Groups textboxes remain blank as you have selected the Allow anonymous access checkbox.

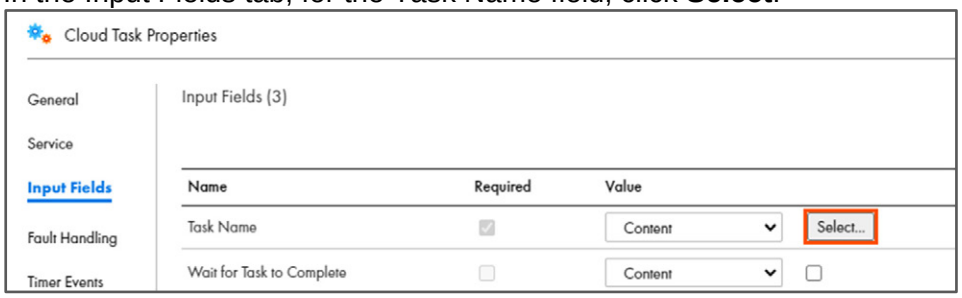
9. From the process canvas, drag and drop the **Service** step between the Start and the End step.
10. In the General tab, in the name field, enter **Cloud Task**.



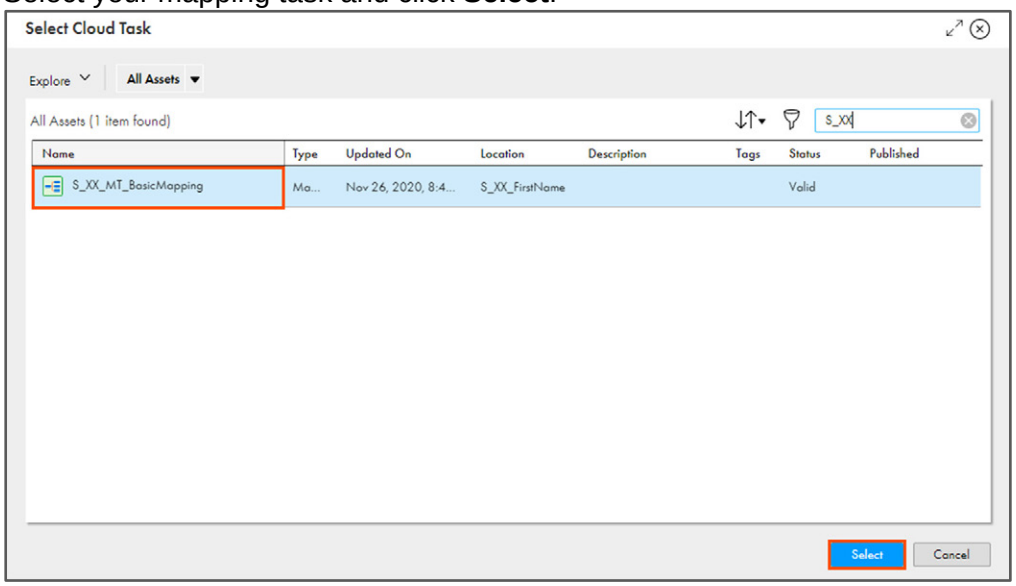
11. In the Service tab, from the Service Type drop-down, select **System Service** and from the Action drop-down, select **Run Cloud Task**.



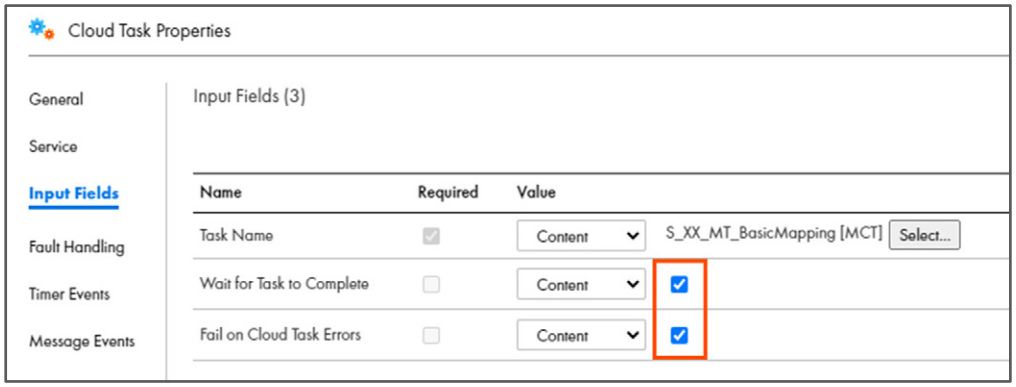
12. In the Input Fields tab, for the Task Name field, click **Select**.



13. Select your mapping task and click **Select**.

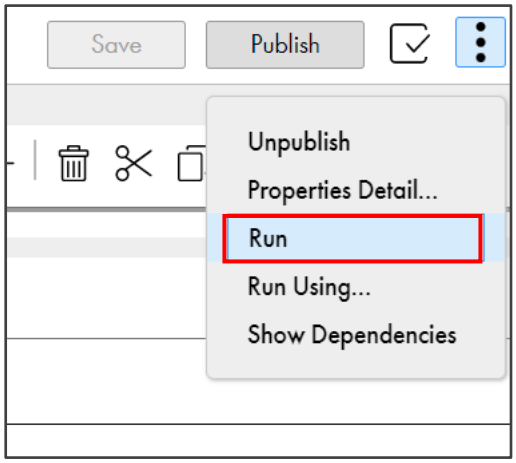


14. For the **Wait for Task to Complete** and **Fail on Cloud Task Errors** fields, select the **Content** checkboxes.

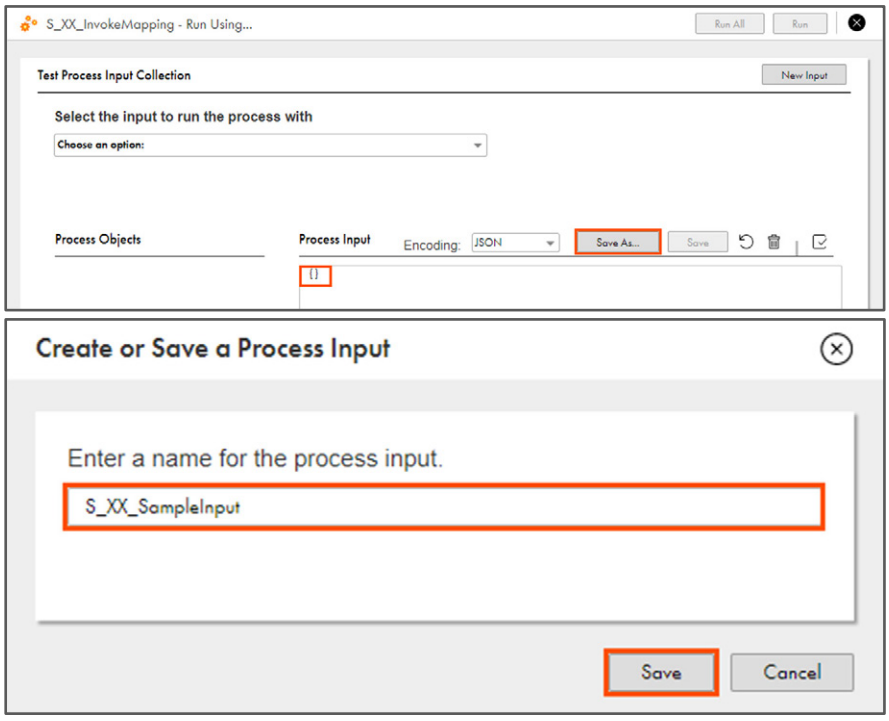


15. **Save** and **Publish** the process.

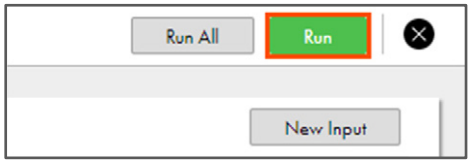
16. To execute the process, click **Run**.



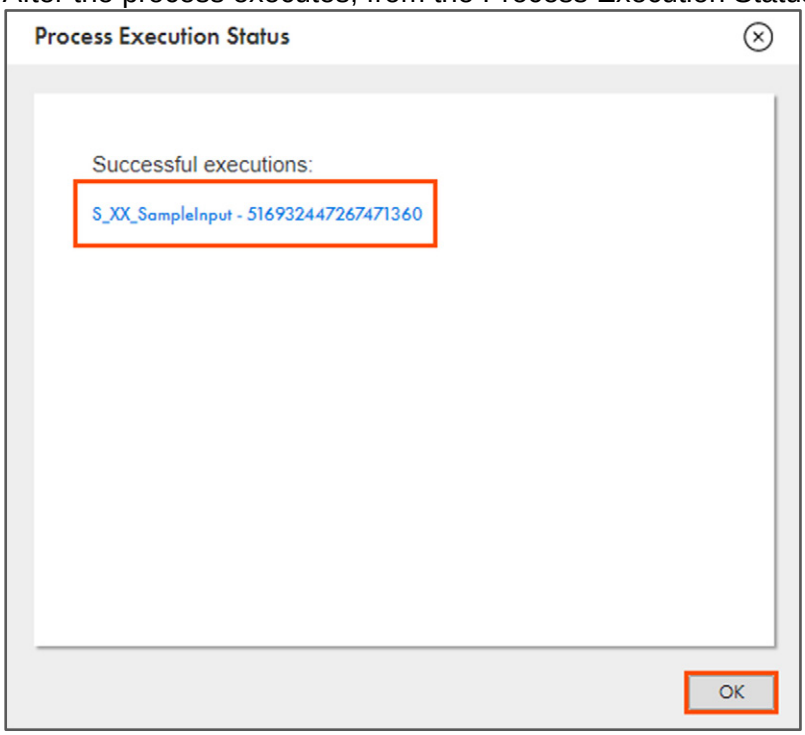
17. To save the sample input, click **Save As**.



18. Run the process.



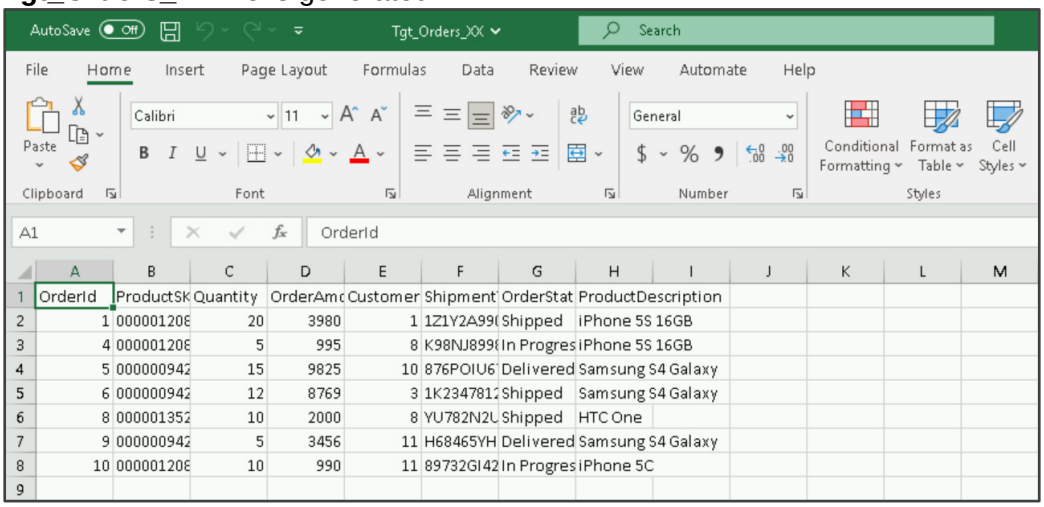
19. After the process executes, from the Process Execution Status window, click **OK**.



Demo

This demo should be presented by the instructor.

- On your system's desktop, in the CAI Lab Prep Files folder, observe that the **Tgt_Orders_XX** file is generated.



	A	B	C	D	E	F	G	H	I	J	K	L	M
1	OrderId	ProductSK	Quantity	OrderAmo	Customer	Shipment	OrderStat	ProductDescription					
2	1	00000120	20	3980	1	1Z1Y2A99	Shipped	iPhone 5S 16GB					
3	4	00000120	5	995	8	K98NJ899	In Progres	iPhone 5S 16GB					
4	5	00000094	15	9825	10	876POIU6	Delivered	Samsung S4 Galaxy					
5	6	00000094	12	8769	3	1K234781	Shipped	Samsung S4 Galaxy					
6	8	00000135	10	2000	8	YU782N2U	Shipped	HTC One					
7	9	00000094	5	3456	11	H68465YH	Delivered	Samsung S4 Galaxy					
8	10	00000120	10	990	11	89732GI42	In Progres	iPhone 5C					
9													

This concludes the lab.

Module 17: CAI Live Project Walk-Through
Lab 17-1: CAI Live Project- Order Management

Overview:

In this lab, you will use the IICS Application Integration service to create a process that fetches inventory details and enables you to obtain margin and commission details. You will also create a managed API using the API Manager Service.

Objectives:

- Create and invoke the Order Management Process
- Connect to a Margin Service
- Modify the Process to add the Margin Details Service
- Create a Service Connector to the Email Service
- Create an Email Connection
- Create a process that uses the email connection to send emails
- Invoke the Order Management Process
- Enable Fault Handling
- Create a Managed API

Duration:
90 Minutes

Tasks

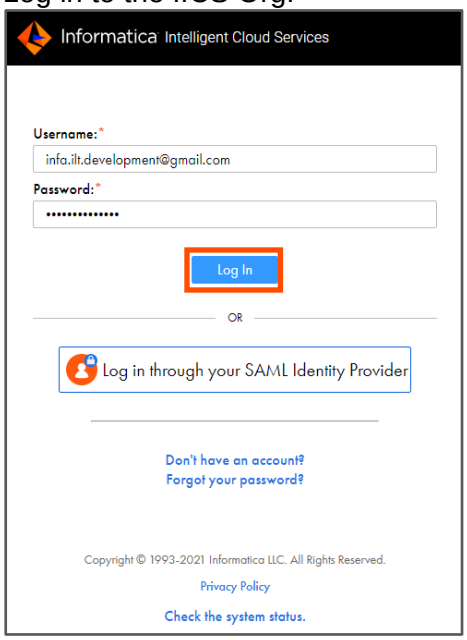
Note: The screen on your lab environment and the images in the lab guide may differ in folder names, asset names, and asset numbers. However, the steps remain the same.

Import a Service Connector and Connection

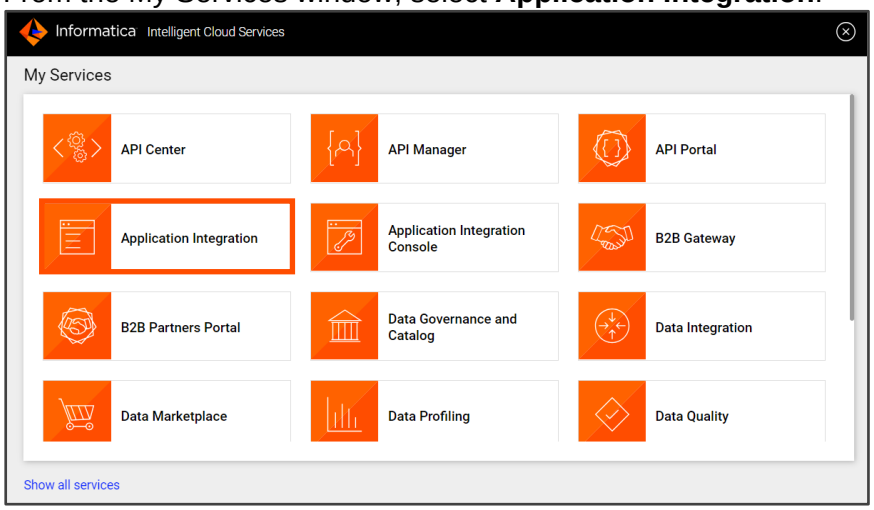
1. In a web browser, enter the following URL:
   https://network.informatica.com/docs/DOC-17733
2. Download the **Get Inventory Detail Connector.zip** file and save it on your desktop.

   Note: The Get Inventory Detail Connector.zip file is also available in the **CAI Lab Prep Files_Learners** folder.

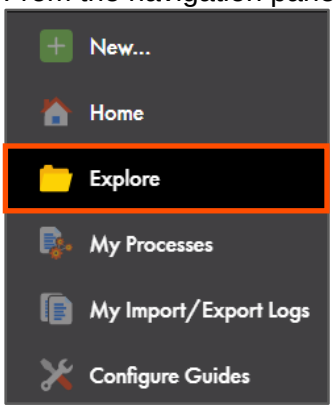
3. Log in to the IICS Org.



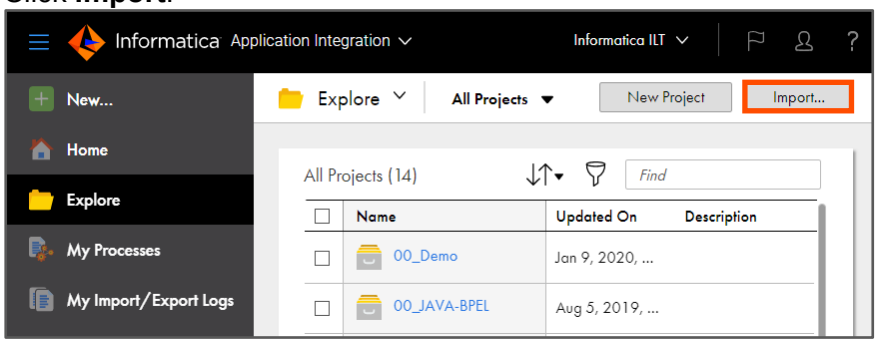
4. From the My Services window, select **Application Integration**.



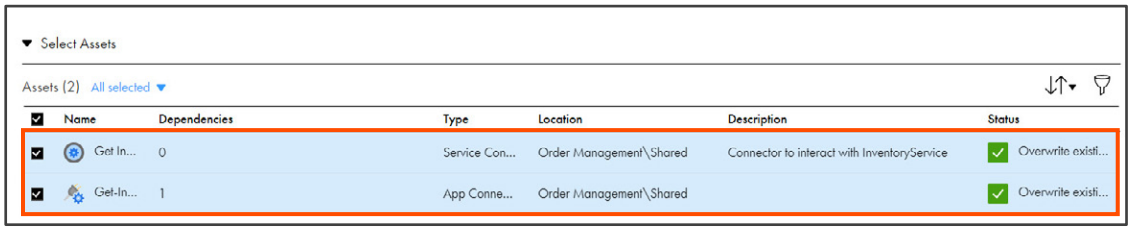
5. From the navigation pane, select **Explore**.



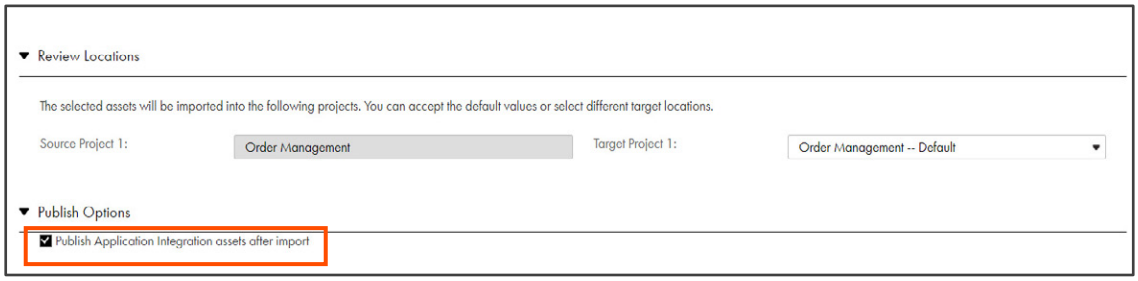
6. Click **Import**.



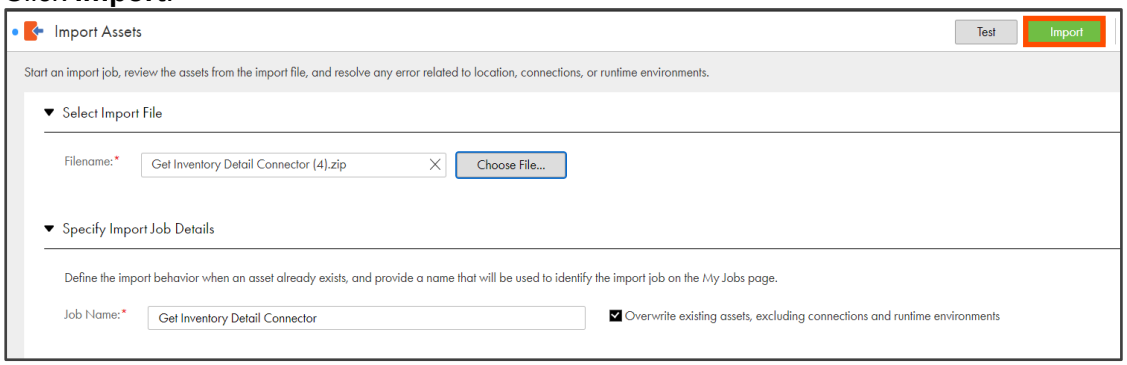
7. Click **Choose File**.
8. From your desktop, select **Get Inventory Detail Connector.zip** and click **Open**.
   The following assets are available for import:
   - Get Inventory Detail (a service connector)
   - Get-Inventory-Detail (a connection)



9. Scroll down and select **Publish Application Integration assets after import**.

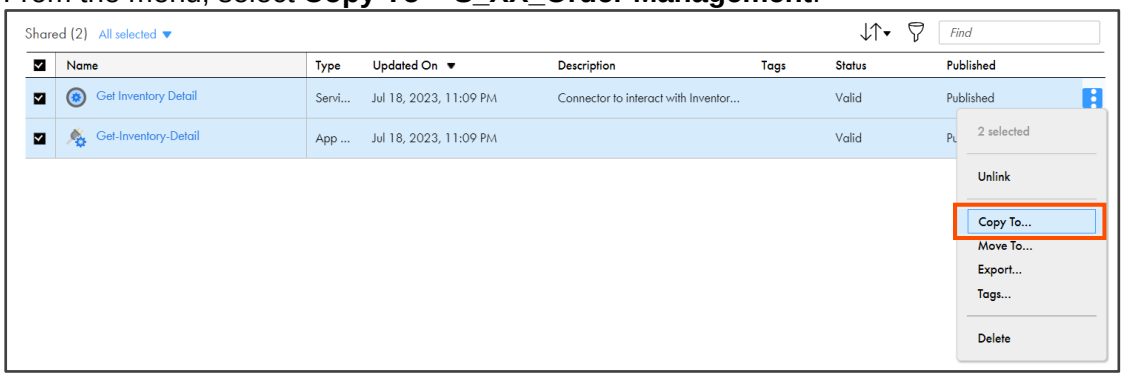


10. Click **Import**.

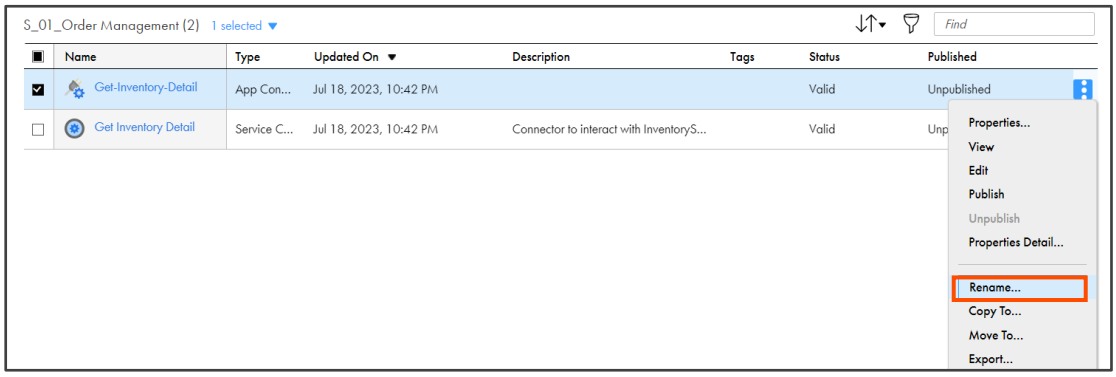


11. In the Explore tab, create Project **S_XX_Order Management**.
12. Go to the folder **Order Management** > **Shared** and select **Get Inventory Detail** and **Get-Inventory-Detail**.

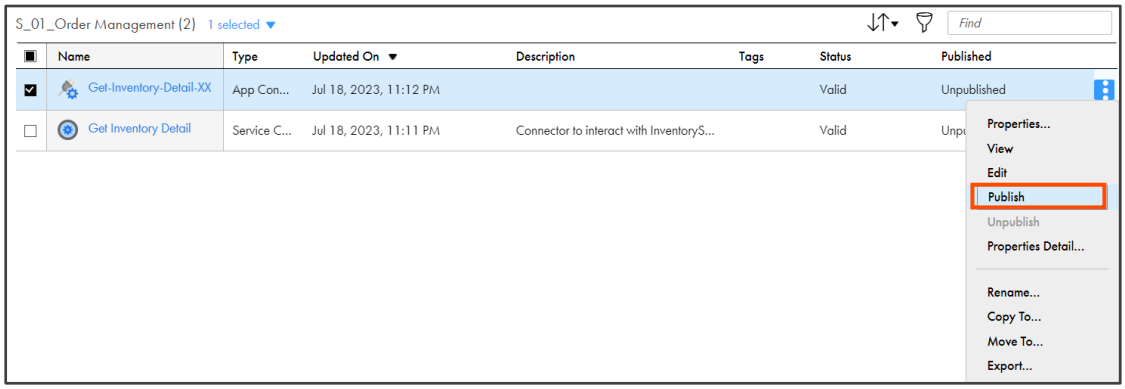
13. From the menu, select **Copy To** > **S_XX_Order Management**.



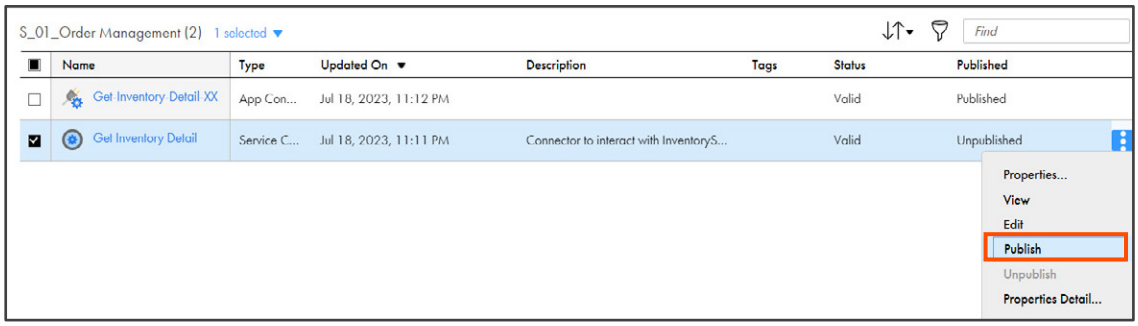
14. Rename the **Get-Inventory-Detail** connection to **Get-Inventory-Detail-XX**.



15. Select **Get-Inventory-Detail-XX**, and from the Actions menu, select **Publish**.



16. Select the **Get Inventory Detail** service connector, and from the Actions menu, select **Publish**.



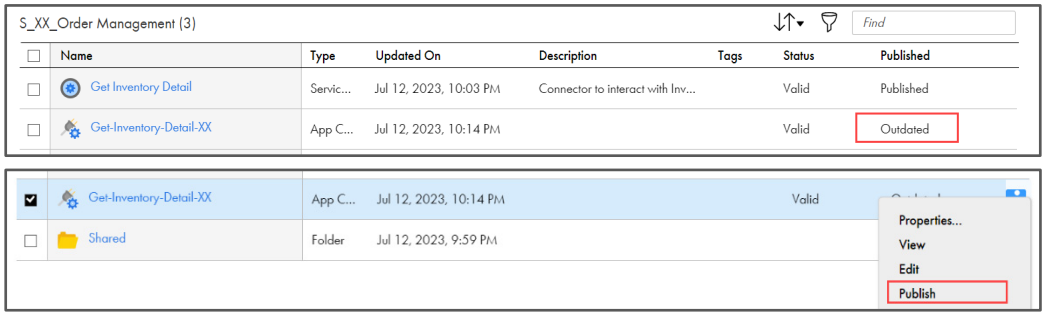
A message '**The asset [Get Inventory Detail] was published successfully.**' appears.

17. Open **Get-Inventory-Detail-XX** connection. Verify the connection details and **Save** the connection if it is not saved already.
18. **Publish** the **Get-Inventory-Detail-XX** again.

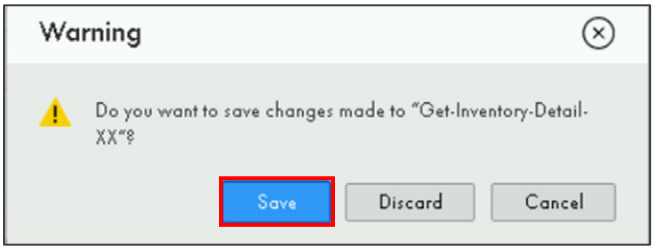
Note: Get Inventory Detail must be populated in the Type field.

19. Close the assets from the navigation pane.

Note: If the status of **Get-Inventory-Detail-XX** is outdated, click ellipses, and select **Publish**.

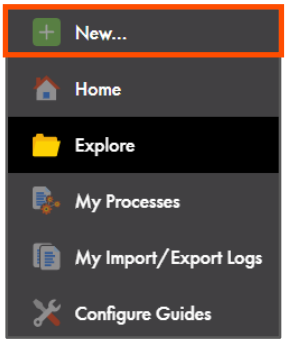


If the Warning window pops up, click **Save**.

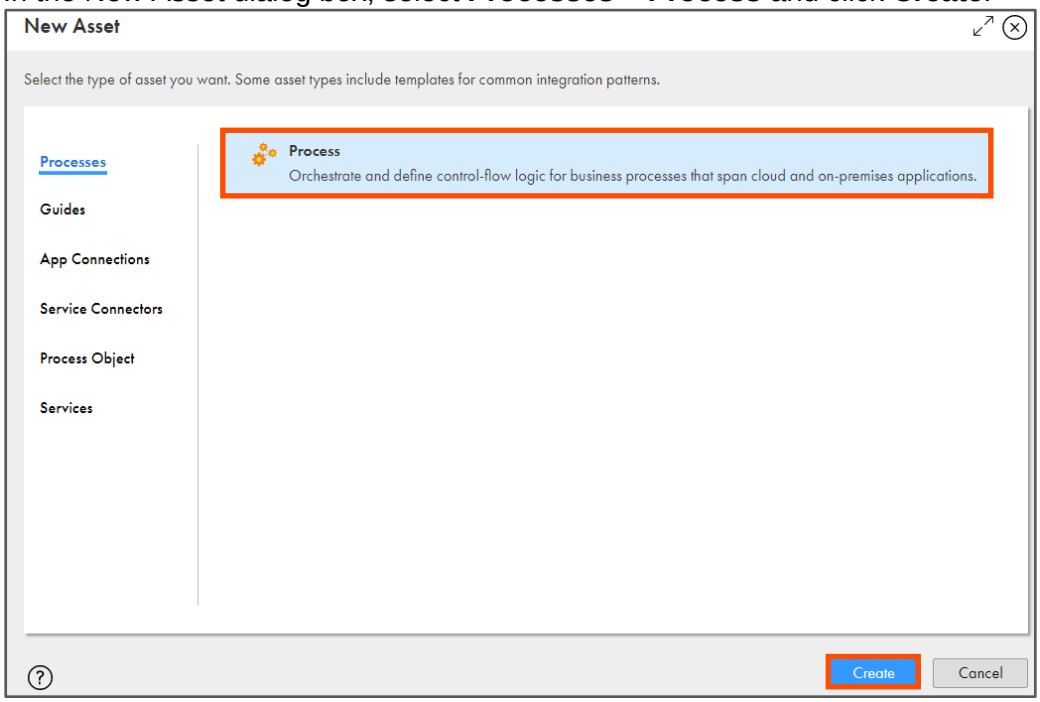


Create a New Process

20. From the Navigation pane, click **New**.

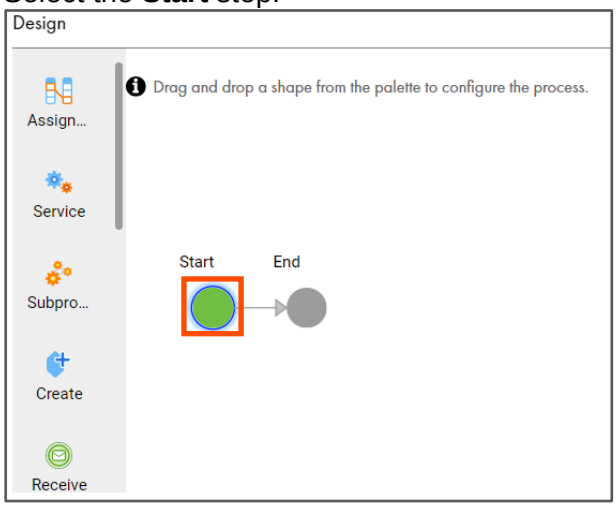


21. In the New Asset dialog box, select **Processes** > **Process** and click **Create**.

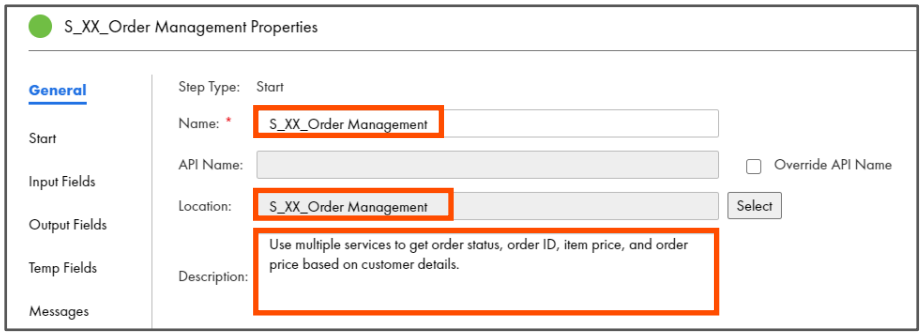


Process Designer opens with a process template.

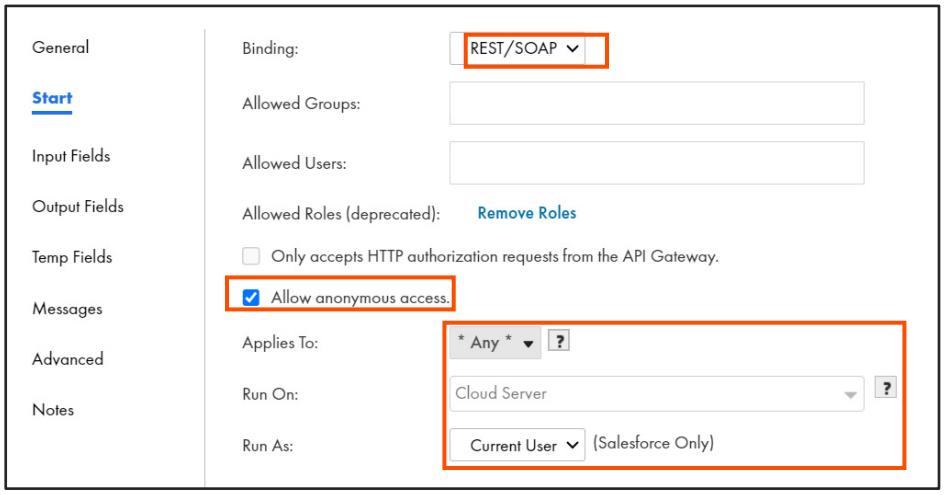
22. Select the **Start** step.



23. In the General tab, in the Name field, enter **S_XX_Order Management**.
24. In the Location field, select your project folder **S_XX_Order Management**.
25. In the Description field, enter **Use multiple services to get order status**, **order ID**, **item price**, **and order price based on customer details.**

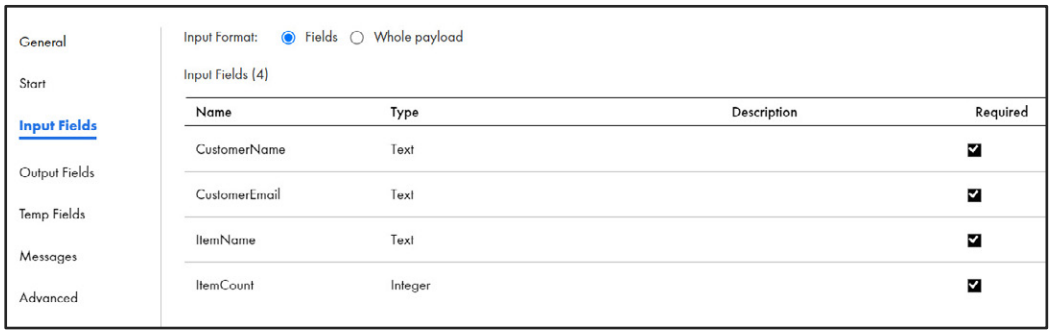


26. In the Start tab, for the Binding field, select **REST/SOAP**.
27. For Allowed Roles, select the **Allow anonymous access** option.
28. Keep the default selection for **Applies To**, **Run On**, and **Run As**.



29. In the Input Fields tab, to add a new field, click .
30. Add the following fields:

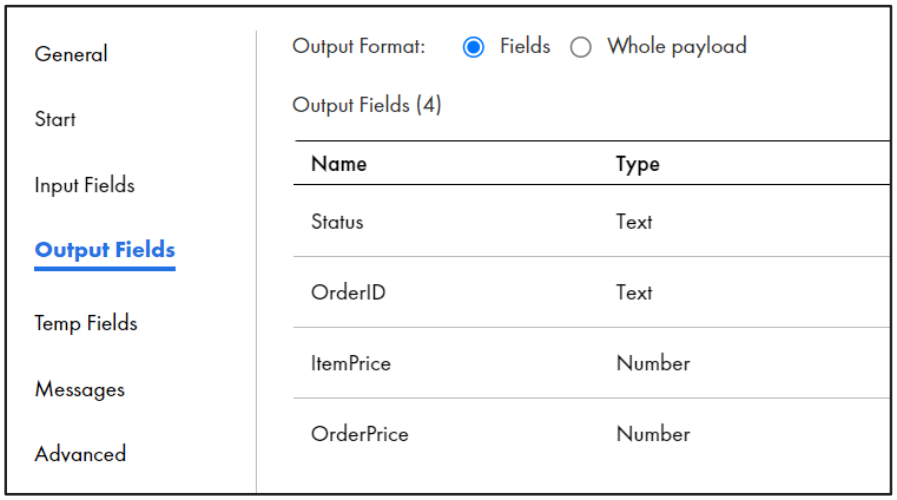
Name	Type	Required
CustomerName	Text	Yes
CustomerEmail	Text	Yes
ItemName	Text	Yes
ItemCount	Integer	Yes



Note: You enter values for the above Input Fields at runtime when you invoke the process.

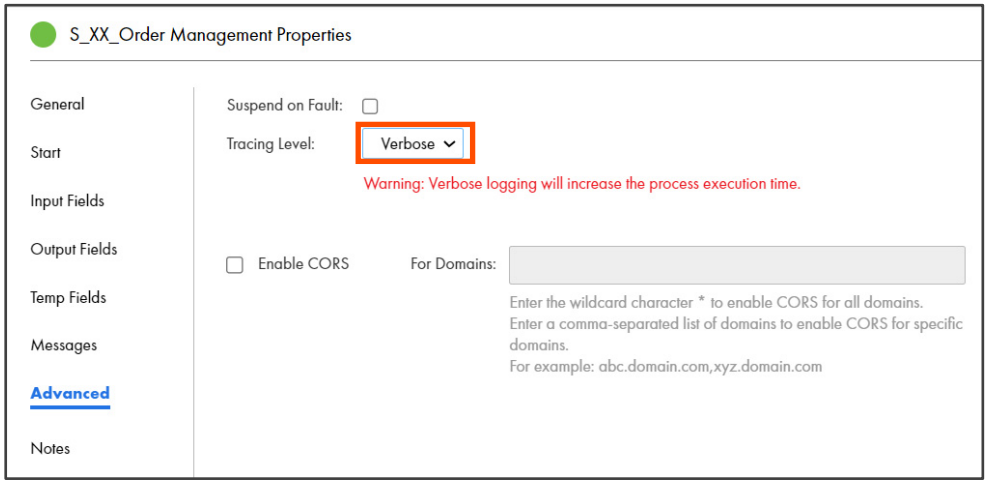
31. In the Output Fields tab, to add a new field, click .
32. Add the following fields:

Name	Type
Status	Text
OrderID	Text
ItemPrice	More types>Number
OrderPrice	Number



Note: This is the output you see after you invoke the process.

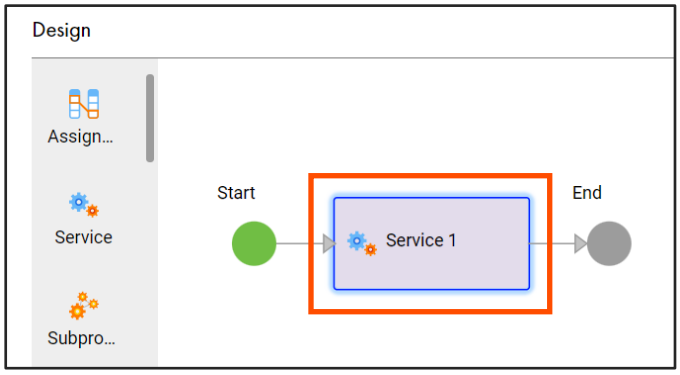
33. In the Advanced tab, from the **Tracing Level** drop-down, select **Verbose**.



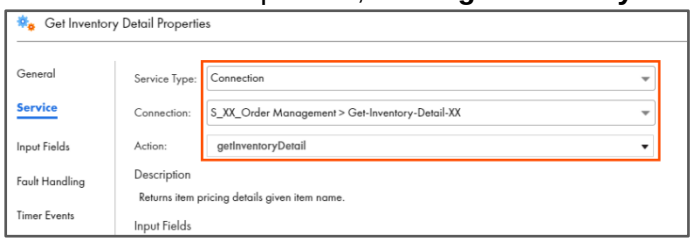
Note: you will notice a warning Verbose logging will increase the process execution time. Ignore this warning. When the Tracing Level is Verbose, the Application Integration Console logs all the steps. This is useful if you need to debug the process.

Get Inventory Details

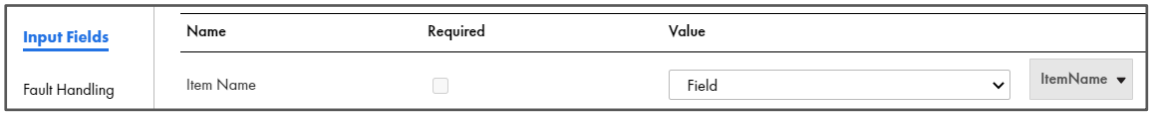
34. Between the Start and End steps, drag and drop a **Service** step.



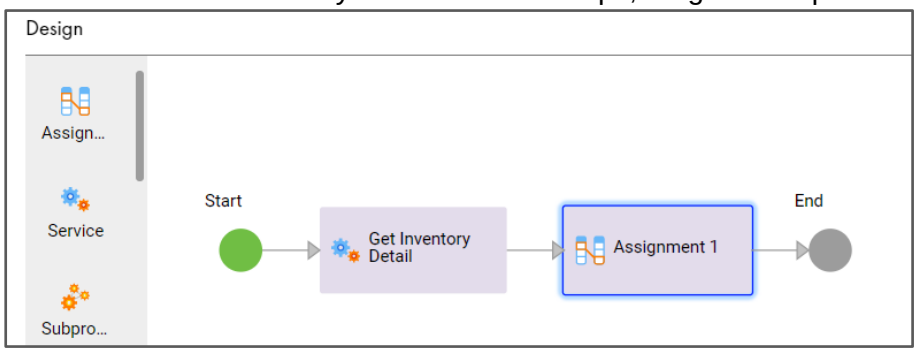
35. Select the **Service** step.
36. In the General tab, in the Name field, enter **Get Inventory Detail**.
37. In the Service tab, from the Service Type drop-down, select **Connection**.
38. From the Connection drop-down, select **S_XX_Order Management** > **Get-Inventory-Detail-XX**.
39. From the Action drop-down, select **getInventoryDetail**.



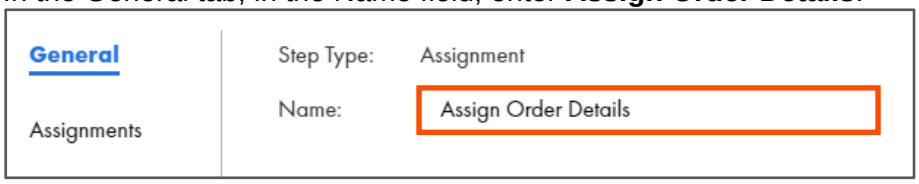
40. In the Input Fields tab, click the plus (**+**) icon. Select **Item Name**.
41. From the Value drop-down, select **Field** > **ItemName**.



42. Between the Get Inventory Detail and End steps, drag and drop an **Assignment** step.



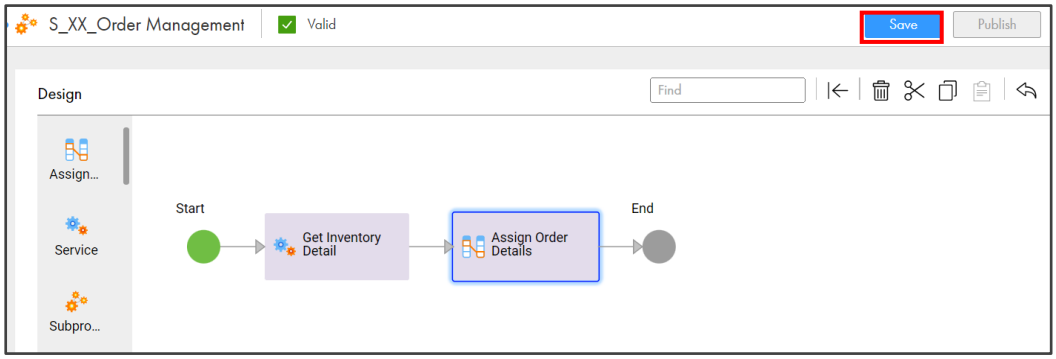
43. Select the **Assignment** step.
44. In the General tab, in the Name field, enter **Assign Order Details**.



45. In the **Assignments** tab, assign values to the following output fields:

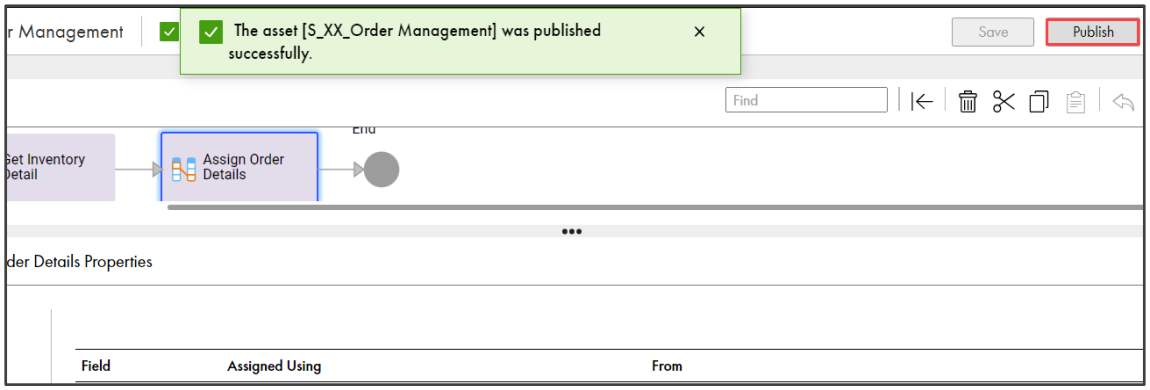
Target	Value Type	Value
Status	Content	Your order is accepted.
OrderID	Formula	infa:getProcessId()
ItemPrice	Field	item>sellprice
OrderPrice	Formula	$output.ItemPrice*$input.ItemCount

46. Click **Save**.

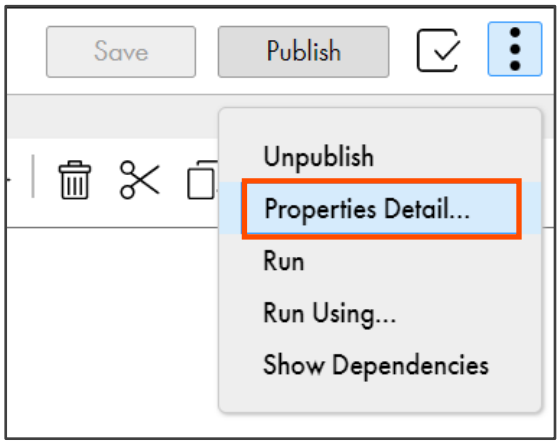


A message '**The process was saved successfully**' appears.

47. Click **Publish**.

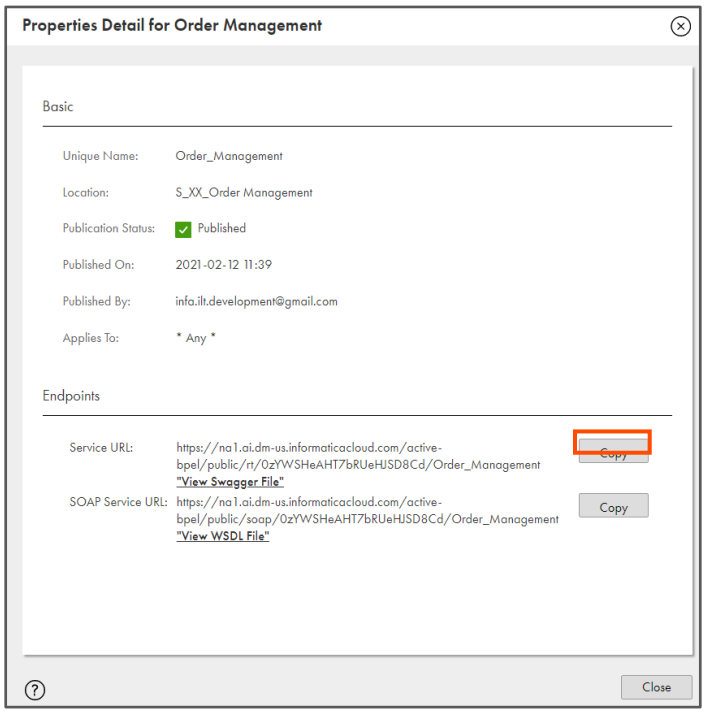


48. Click **Actions** > **Properties Detail**.



The Properties Detail for Order Management window appears.

49. Copy the Service URL.



50. Open a notepad file, paste the Service URL, and add the following input parameters to the end of the service URL:

    <Service URL>?CustomerName=TestConsumer&CustomerEmail=testconsumer@mailinator.com&ItemName=item1&ItemCount=2

    OR

    From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **34_LabGuide_CAI_Live Project_17-1**. Copy the command mentioned under **Step A**.

    The following URL is a sample service URL with input parameters:

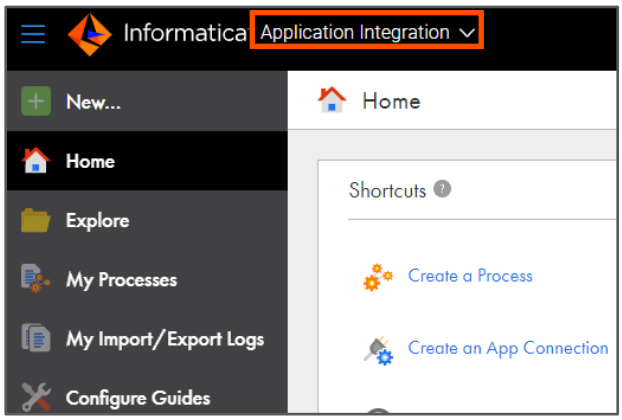
    https://na1.ai.dm-us.informaticacloud.com/active-bpel/public/rt/0zYWSHeAHT7bRUeHJSD8Cd/Order_Management?CustomerName=TestConsumer&CustomerEmail=testconsumer@mailinator.com&ItemName=item1&ItemCount=2

51. Open a browser and paste the service URL with input parameters.
52. Enter your Informatica Intelligent Cloud Services credentials, if prompted.

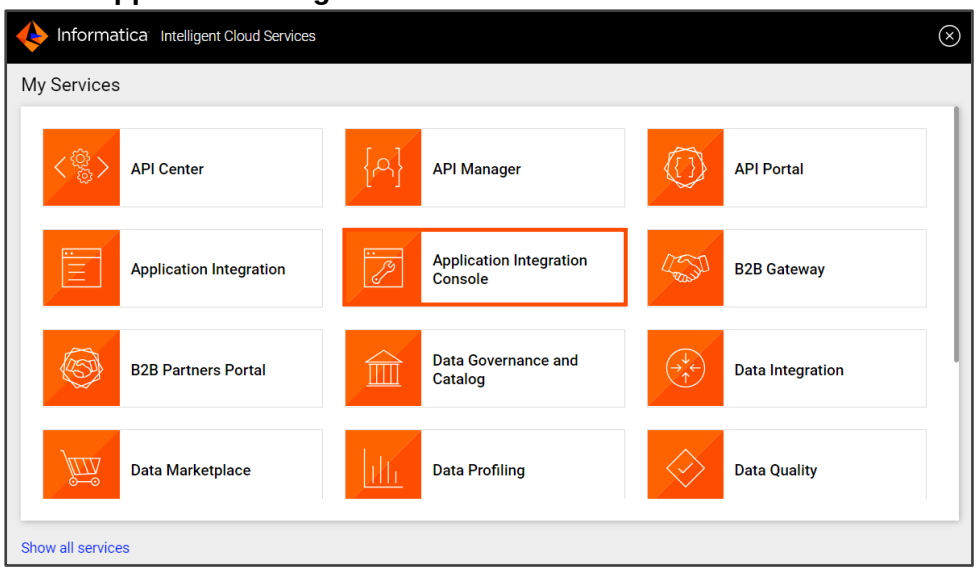
    You see the following output:

```
{"Status":"Your order is accepted","OrderID":"544763854068928512","ItemPrice":50.0,"OrderPrice":100.0}
```

53. Click the drop-down beside the Application Integration service.

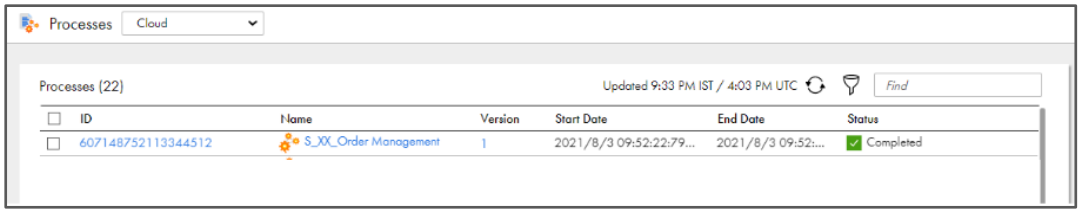


54. Select **Application Integration Console**.

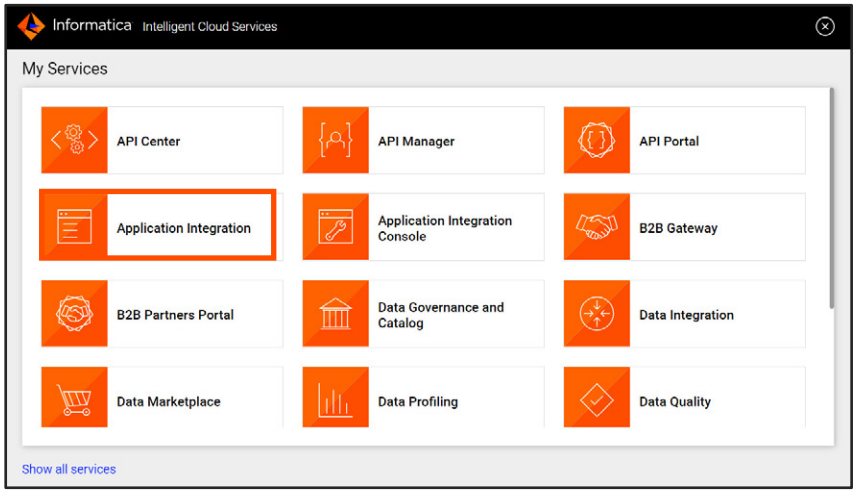


55. Click **Processes**.

56. You can see that the process status is Completed.

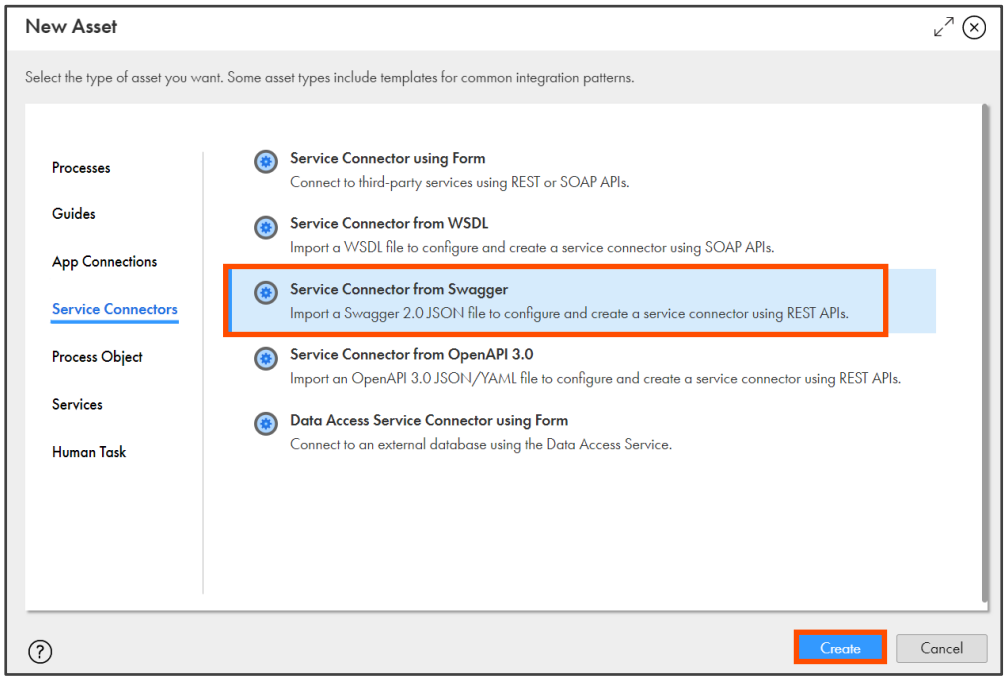


57. Navigate back to **Application Integration**.



Create a Service Connector to the Margin Service

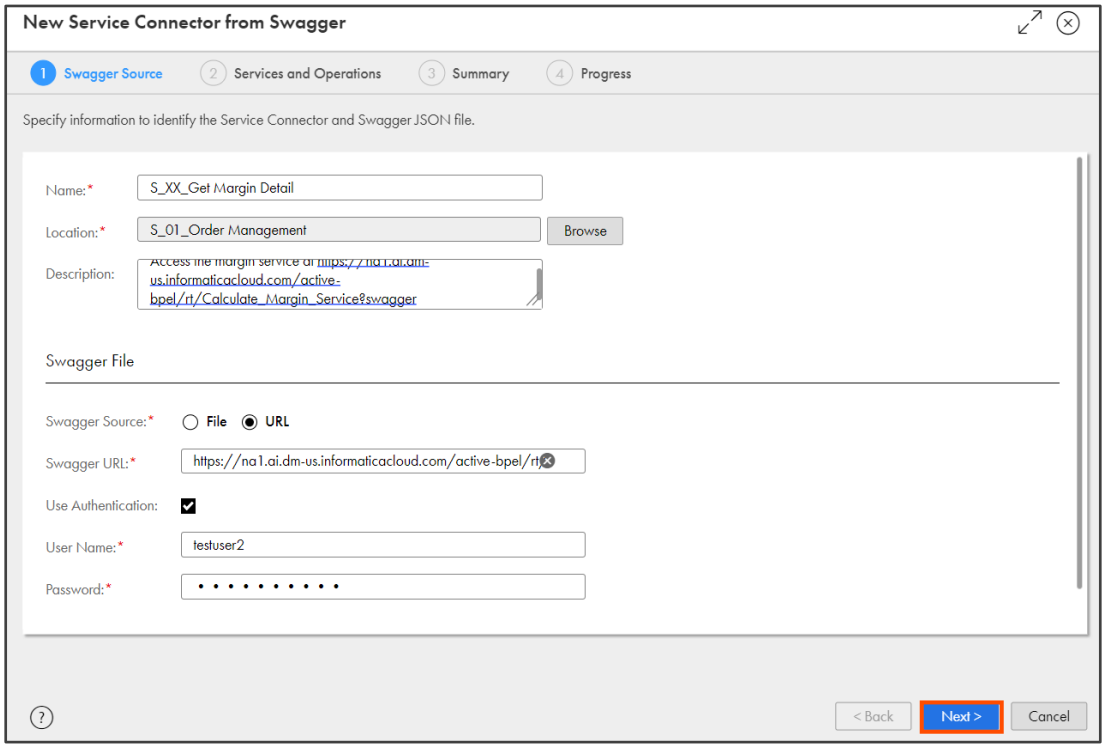
58. From the Navigation pane, click **New**.
59. In the New Asset dialog box, select **Service Connectors** > **Service Connector from Swagger**, and click **Create**.



60. In the New Connector from Swagger dialog box, enter the following properties in the Swagger Source tab:

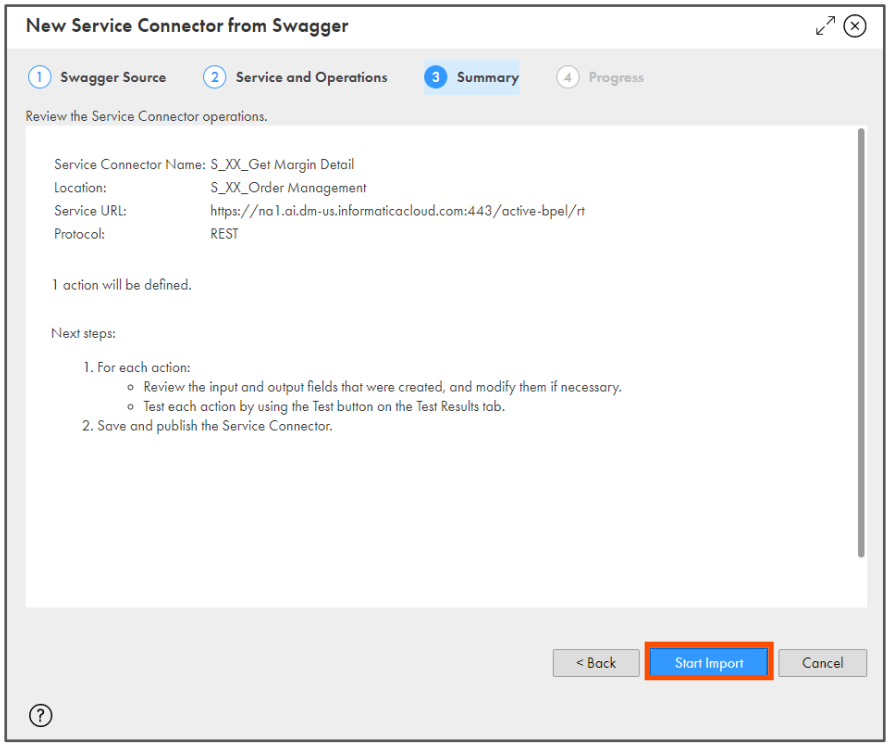
Property Name	Property Content
Name	S_XX_Get Margin Detail
Location	S_XX_Order Management
Description	Access the margin service at https://na1.ai.dm-us.informaticacloud.com/active-bpel/rt/Calculate_Margin_Service?swagger
Swagger Source	URL
Swagger URL	https://na1.ai.dm-us.informaticacloud.com/active-bpel/rt/Calculate_Margin_Service?swagger
User Name	testuser2
Password	password2#

61. Click **Next**.

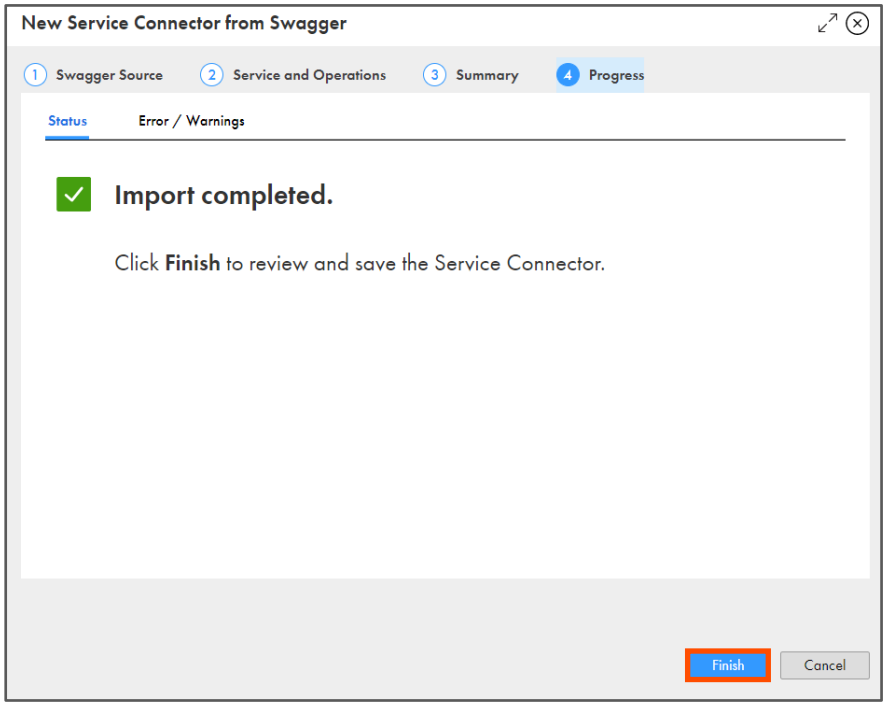


62. Retain the default selection for Service and Operations and click **Next**.

63. In the Summary tab, click **Start Import**.

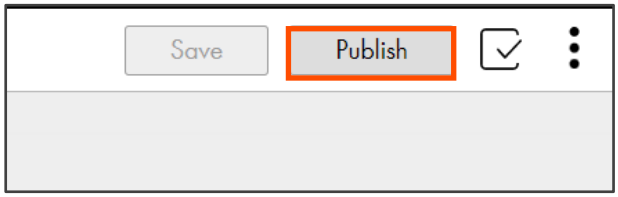


64. A message 'Import Completed.' appears, click **Finish**.



65. Click **Save.**

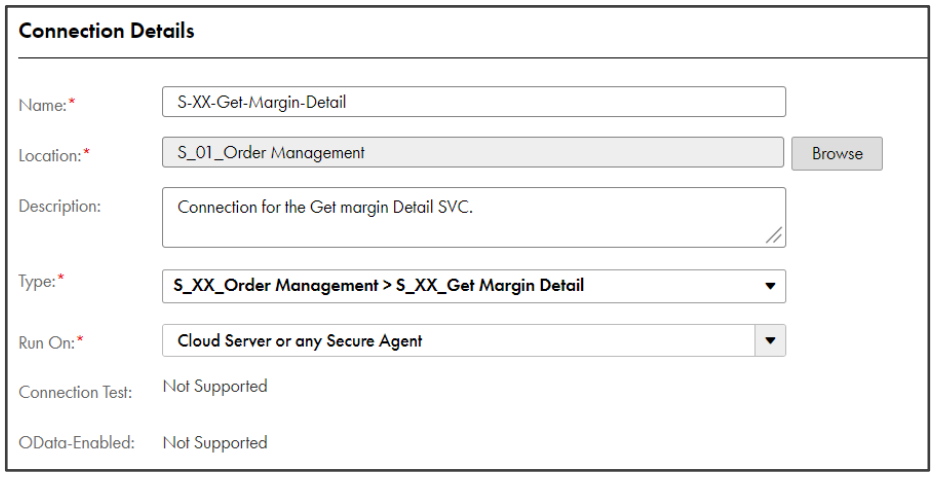
66. **Publish** the connector.



Create a Connection

67. From the Navigation Pane, click **New**.
68. In the New Asset dialog box, select **App Connections** > **App Connection**, and click **Create**.

Property Name	Property Content
Name	S-XX-Get-Margin-Detail
Location	S_XX_Order Management
Description	Connection for the Get margin Detail SVC.
Type	S_XX_Order Management>S_XX_Get Margin Detail
Run On	Cloud Server or any Secure Agent
Hostname	na1.ai.dm-us.informaticacloud.com:443
Username	testuser2
Password	password2#

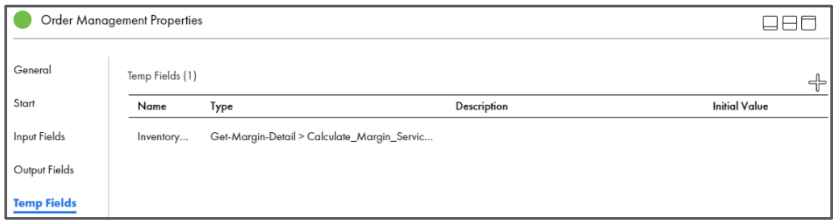


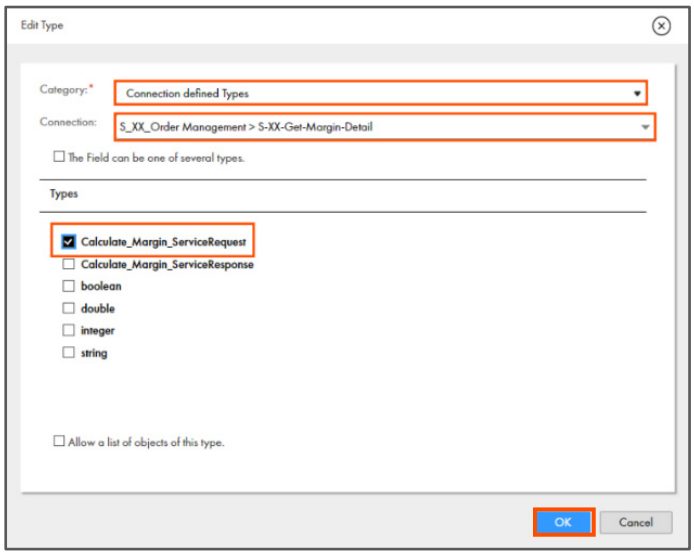
69. **Save** and **Publish** the asset.

    A message ‘The asset [Get-Inventory-Detail] was published successfully.’ appears.

Modify the Process to Add the Margin Details Service

70. In Application Integration, navigate to the **S_XX_Order Management** folder.
71. Reopen the S_XX_Order Management process and select the **Start** step.
72. In the Temp Fields tab, to add a new field, click .
73. In the field name, enter **InventoryDetails**.
74. From the Type drop-down, select **More Types**.
75. In the Edit Type dialog box, from the Category drop-down, select **Connection defined Types**.
76. From the Connection drop-down, select **S_XX_Order Management** > **S-XX-Get-Margin-Detail**.
77. Select the **Calculate_Margin_ServiceRequest** option and click **OK**.

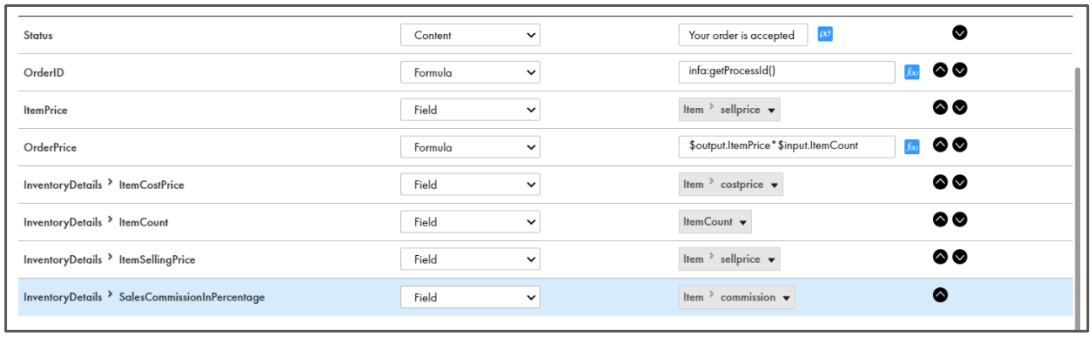




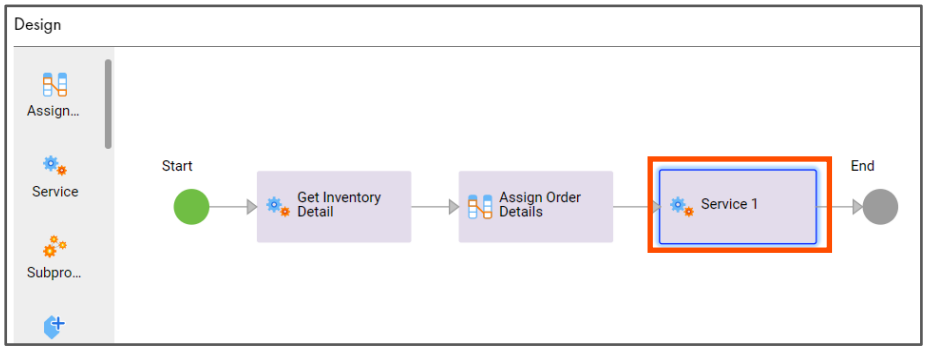
78. Click **Save**.

79. Select the **Assign Order Details** step and in the **Assignments** tab, add the following fields:

Target	Value Type	Value
InventoryDetails > ItemCostPrice	Field	Item>costprice
InventoryDetails > ItemCount	Field	ItemCount
InventoryDetails > ItemSellingPrice	Field	Item>sellprice
InventoryDetails > SalesCommissionInPercentage	Field	Item>commission



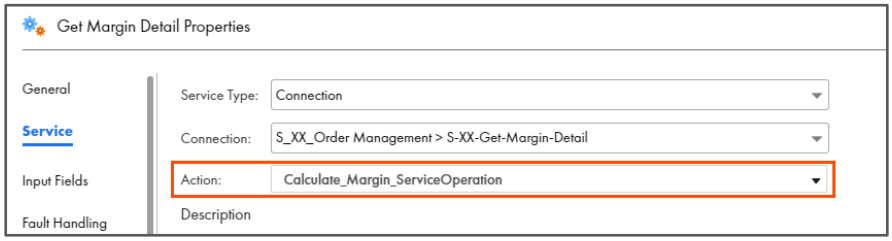
80. Click **Save**.
81. Between the Assign Order Details step and the End step, drag and drop a **Service** step.



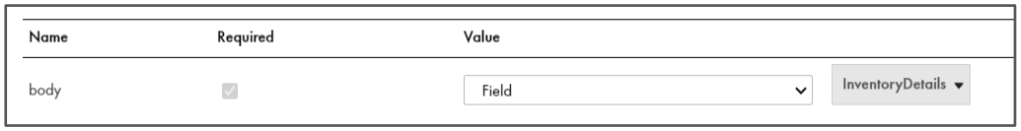
82. In the General tab, enter **Get Margin Detail**.

83. In the Service tab, from the Service Type drop-down, select **Connection**.
84. From the Connection drop-down, select **S_XX_Order Management > S-XX-Get-Margin-Detail**.

85. From the Action drop-down, select **Calculate_Margin_ServiceOperation**.



86. In the Input Fields tab, retain the default name, and assign the Value **Field** > **InventoryDetails**.

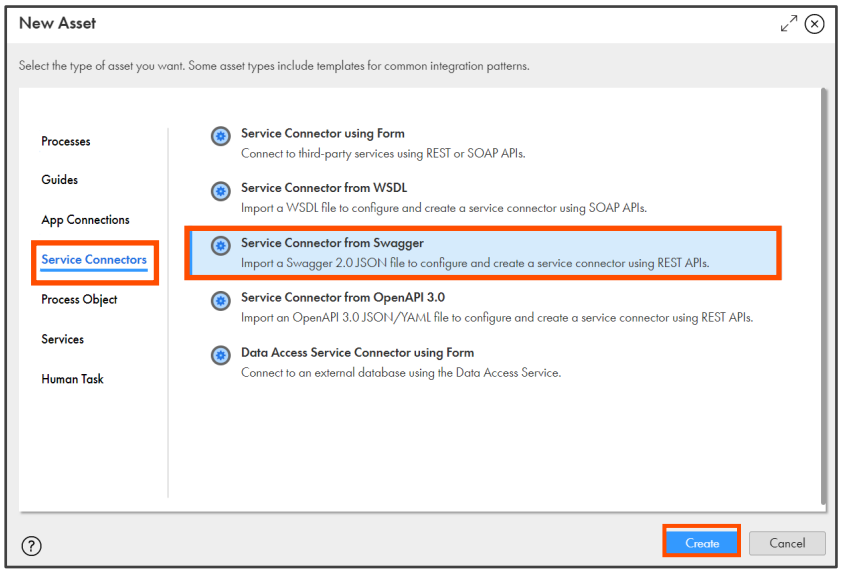


87. **Save** and **Publish** the process.

Now, you will create an email service connector and a connection. You later use these connections in the process to send emails to consumers and service providers.

Create a Service Connector to the Email Service

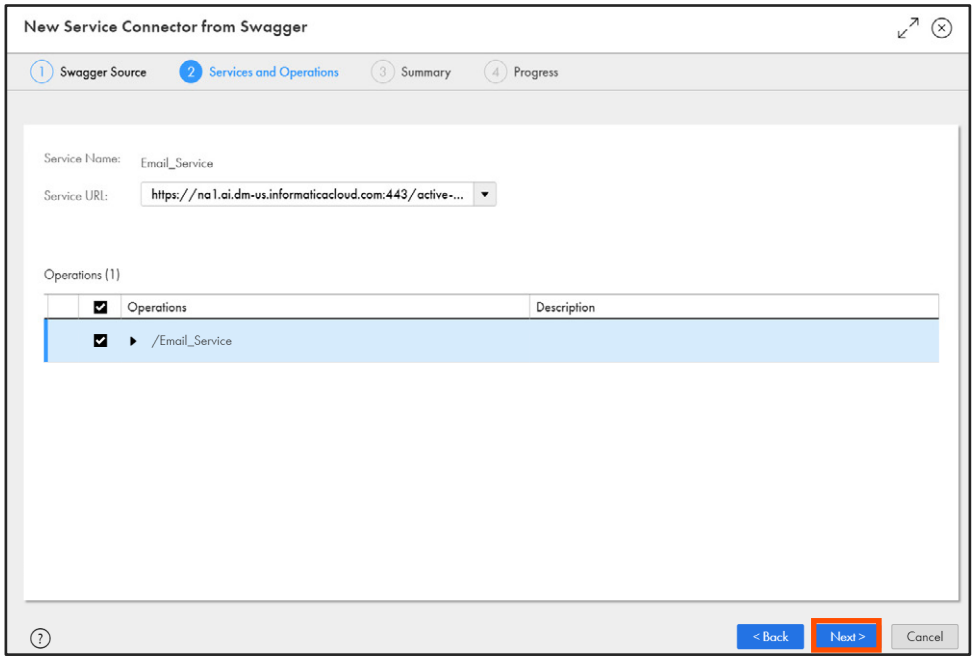
88. In Application Integration, click **New**.
89. In the New Asset dialog box, select **Service Connectors** > **Service Connector from Swagger** > **Create**.



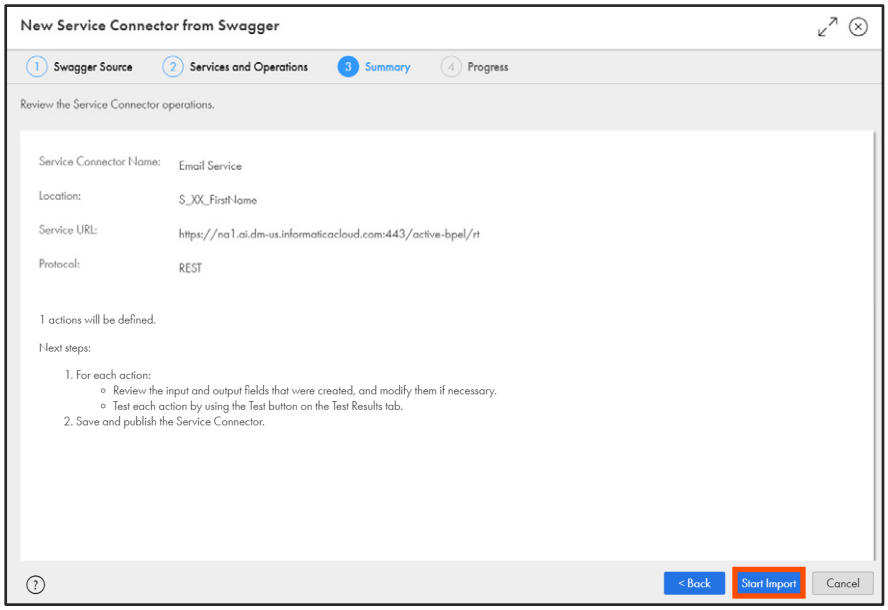
90. In the New Connector from Swagger dialog box, enter the following properties in the Swagger Source tab:

Property Name	Property Content
Name	Email Service
Location	S_XX_Order Management
Description	Sends emails using the https://na1.ai.dm-us.informaticacloud.com/active-bpel/rt/Email_Service service.
Swagger Source	URL
Swagger URL	https://na1.ai.dm-us.informaticacloud.com/active-bpel/rt/Email_Service?swagger
User Name	testuser2
Password	password2#

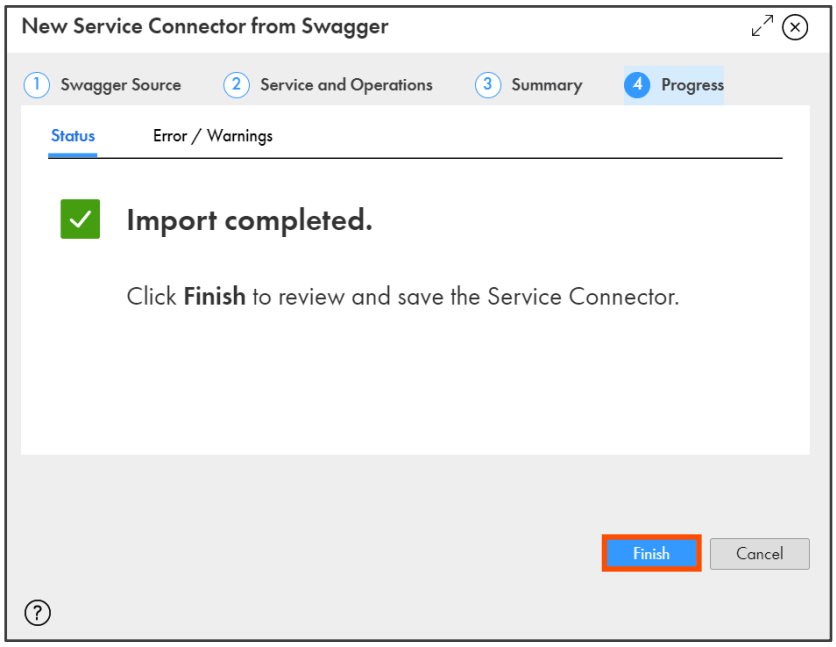
91. Click **Next**.
92. Review the **Service and Operations** tab and then click **Next**.



93. In the Summary tab, click **Start Import**.



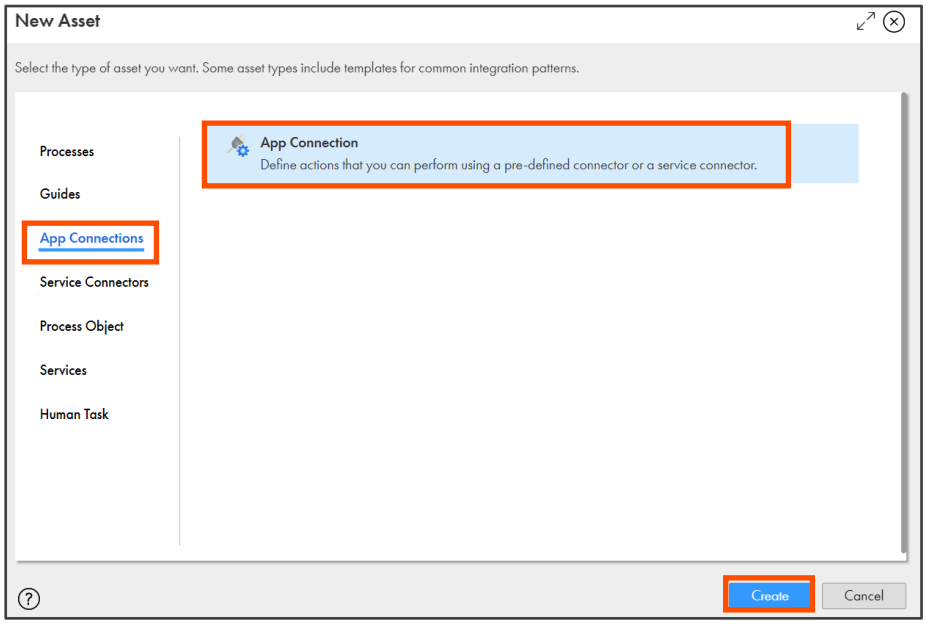
94. A message 'Import Completed.' appears, click **Finish**.



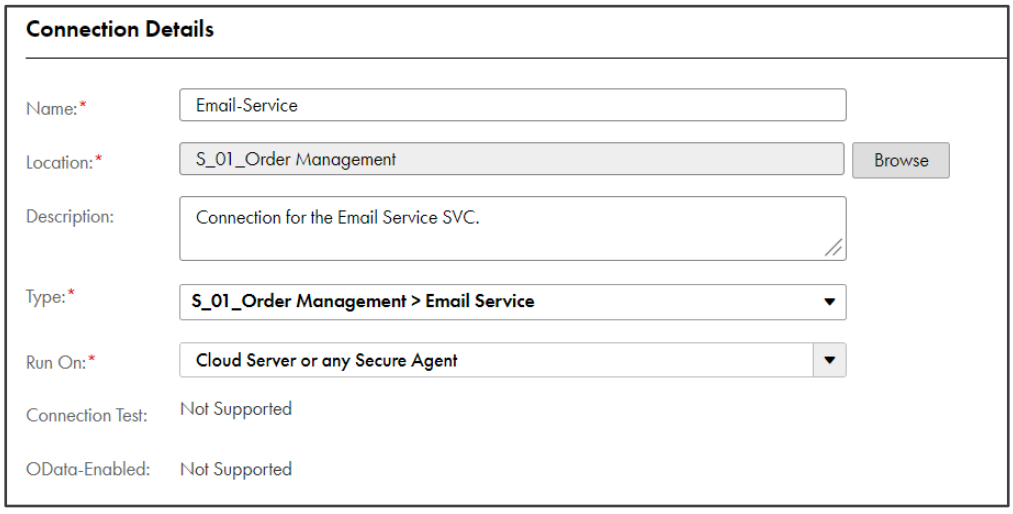
95. **Save** and **Publish** the process.

Create an Email Connection

96. In Application Integration, click **New**.
97. In the New Asset dialog box, select **App Connections** > **App Connection** > **Create**.



98. In the Name field, enter **Email-Service**.
99. In the Location field, select your project folder **S_XX_Order Management**.
100. In the Description field, enter **Connection for the Email Service SVC**.
101. In the Type field, enter **S_XX_Order Management** > **Email Service**.
102. From the Run On drop-down, select **Cloud Server or any Secure Agent**.



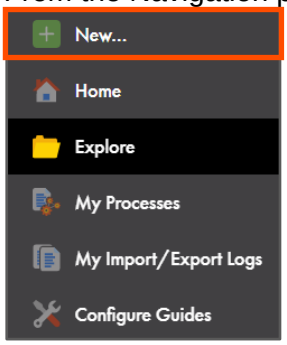
103. In the hostName field, enter **na1.ai.dm-us.informaticacloud.com:443**
104. In the username field, enter **testuser2**.
105. In the password field, enter **password2#**.

106. Click **Save** and then click **Publish**.

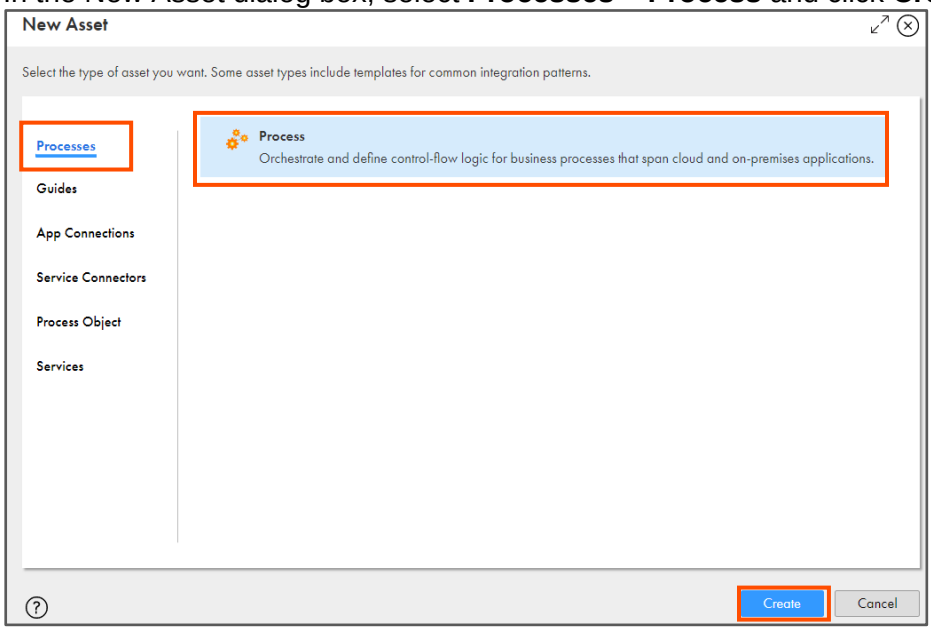
Create a Subprocess

Now, you will create a process that uses the email connection to send emails to consumers and vendors. When you add this process to the Order Management process, it becomes a subprocess.

107. From the Navigation pane, click **New**.

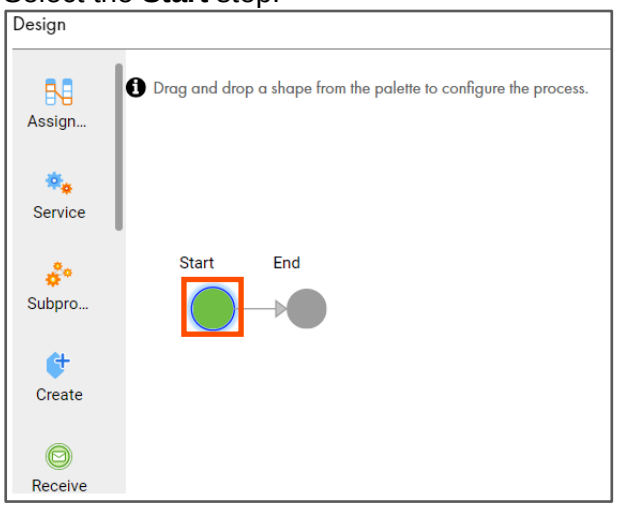


108. In the New Asset dialog box, select **Processes** > **Process** and click **Create**.

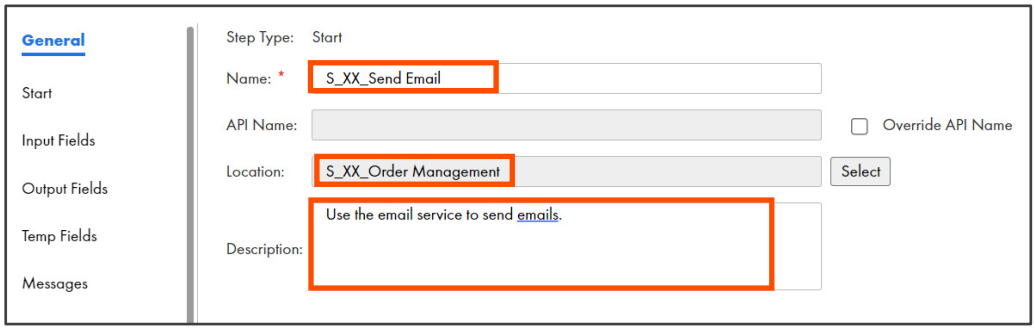


Note: Process Designer opens with a process template.

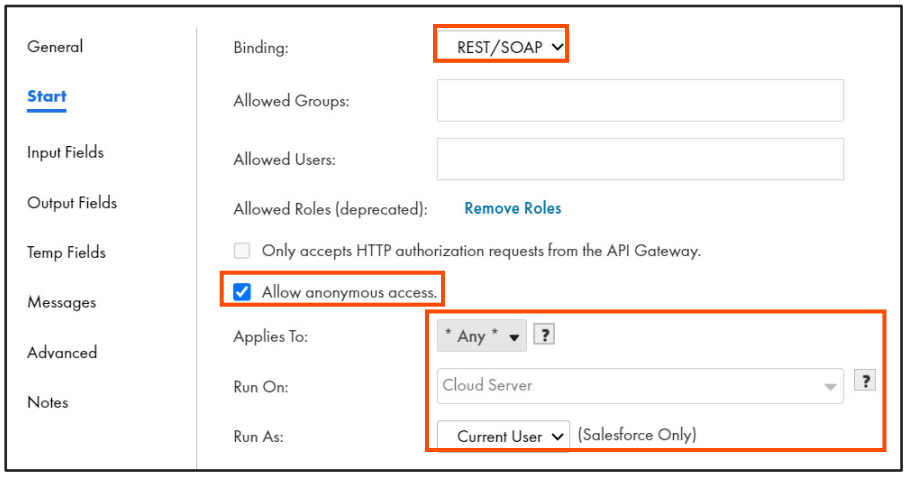
109. Select the **Start** step.



110. In the General tab, in the Name field, enter **S_XX_Send Email**.
111. In the Location field, select your project folder **S_XX_Order Management**.
112. In the Description field, enter **Use the email service to send emails**.

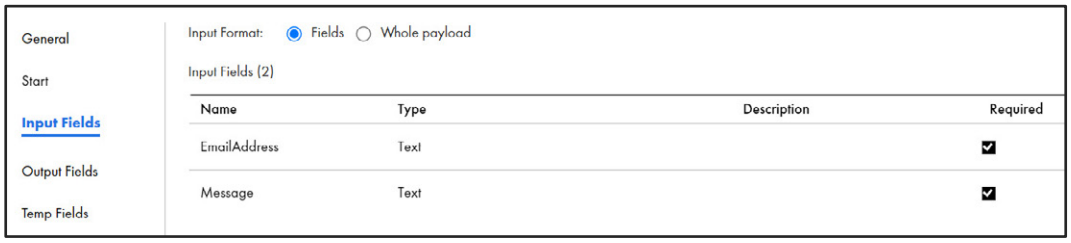


113. In the Start tab, for the Binding field, select **REST/SOAP**.
114. For Allowed Roles, select the **Allow anonymous access** option.
115. Keep the default selection for **Applies To**, **Run On**, and **Run As**.



116. In the Input Fields tab, to add a new field, click .
117. Add the following fields:

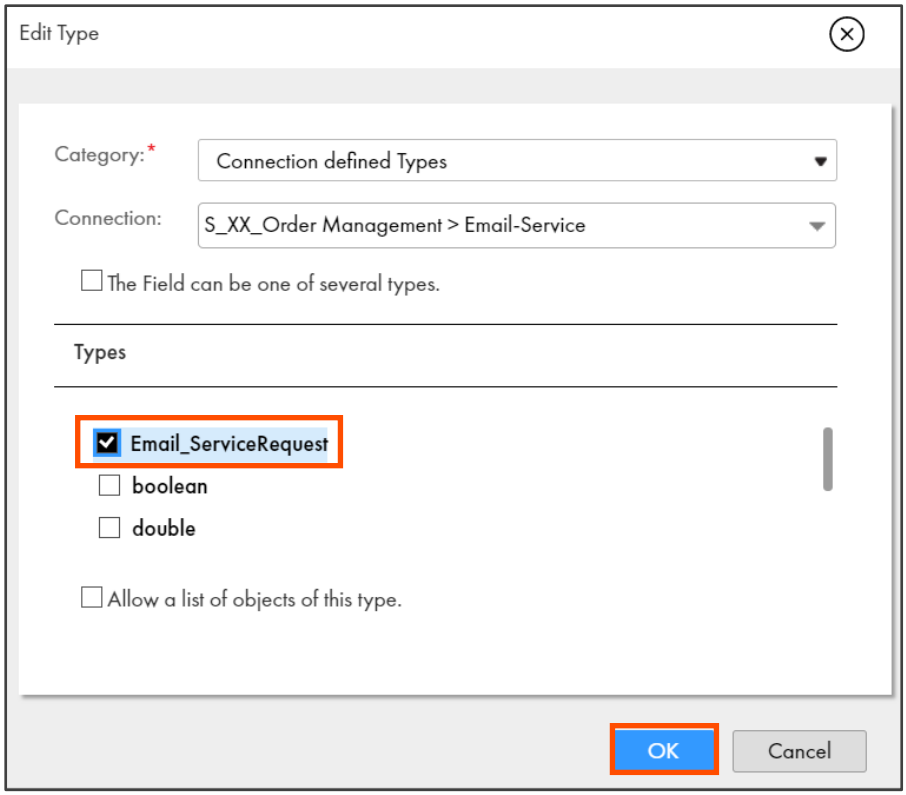
Name	Type	Required
EmailAddress	Text	Yes
Message	Text	Yes



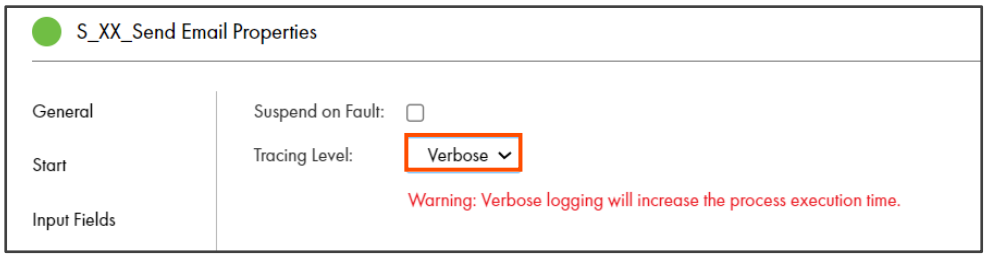
118. In the Temp Fields tab, to add a new field, click .

119. Add the following field:

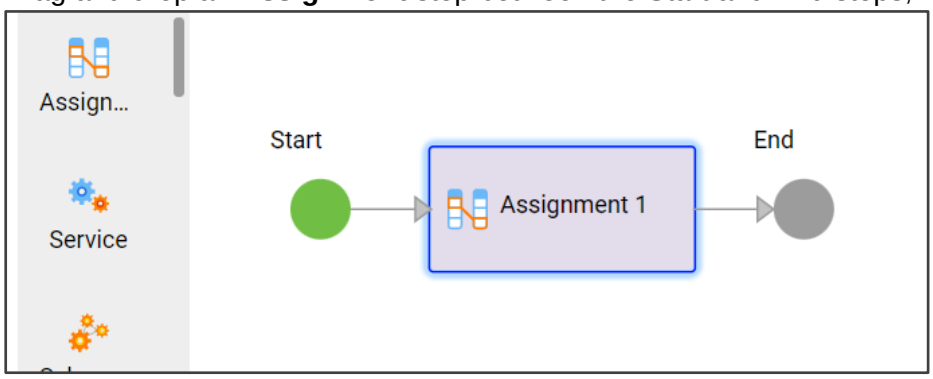
Name	Type
InventoryDetails	More Types > Connection Defined Types > S_XX_Order Management > Email-Service > Email_ServiceRequest



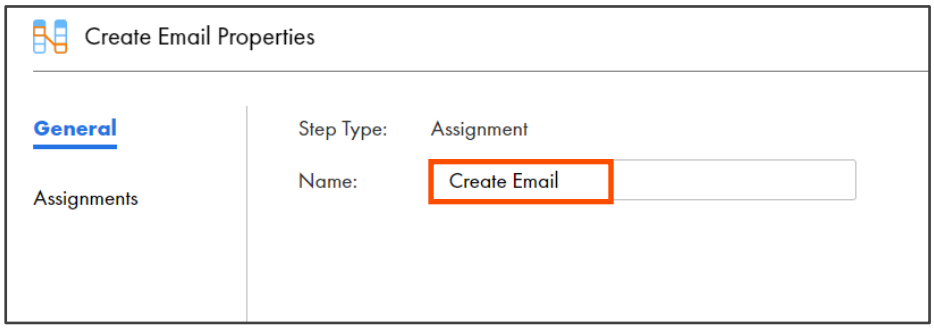
120. In the Advanced tab, from the **Tracing Level** drop-down, select **Verbose**.



121. Drag and drop an **Assignment** step between the Start and End steps,



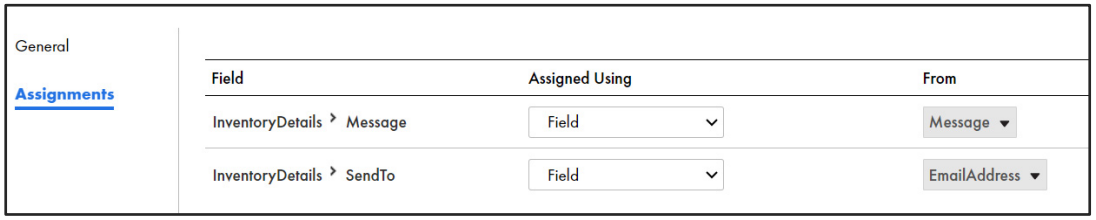
122. Select the General tab and enter **Create Email** in the Name field.



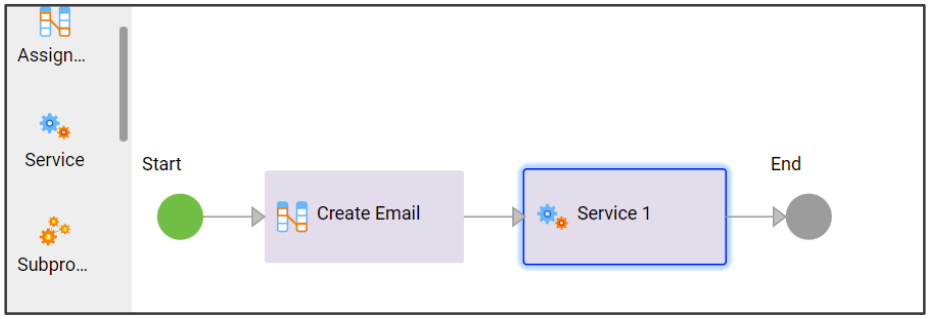
123. Select the **Assignment** step.

124. In the **Assignments** tab, assign values to the following output fields:

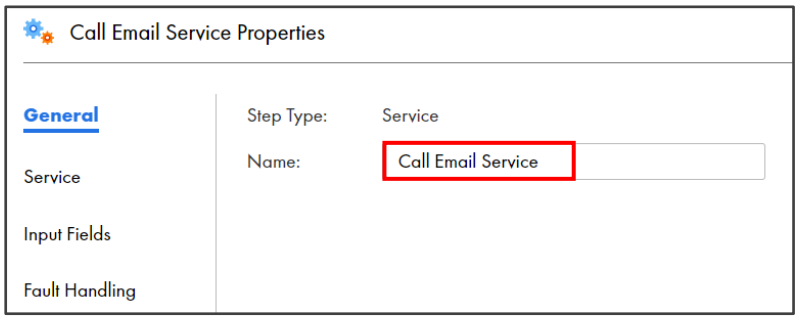
Field	Assigned Using	From
InventoryDetails > Message	Field	Message
InventoryDetails > SendTo	Field	EmailAddress



125. Drag and drop a service step onto the canvas in between the Create Email step and the End step.

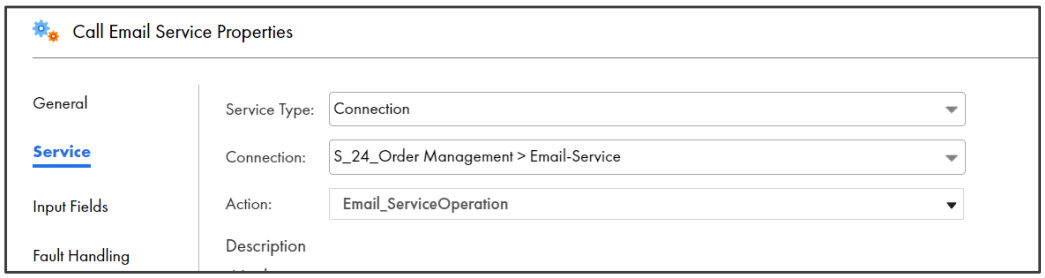


126. Select the General tab and enter **Call Email Service** in the Name field.

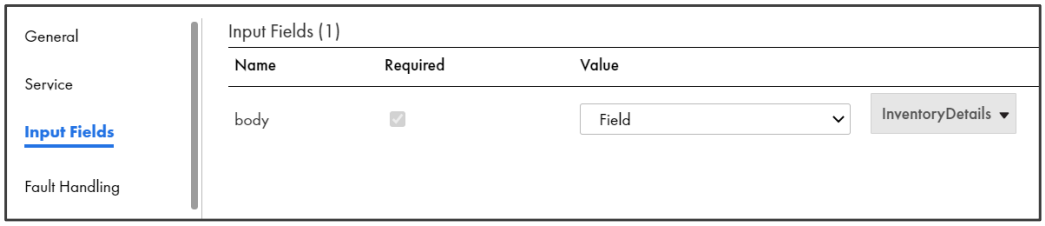


127. Select the **Service** tab.

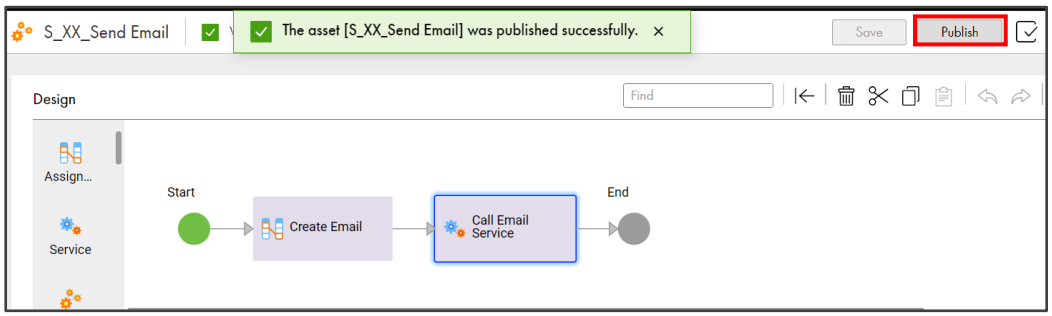
128. Set the Service Type to **Connection**.
129. Set the Connection to **S_XX_Order Management** > **Email-Service**.
130. Set the Action field to **Email_ServiceOperation**.



131. Select the **Input** tab and assign the value **Field** > **Inventory Details** to the body field.



132. Click **Save**.
133. Click **Publish**.



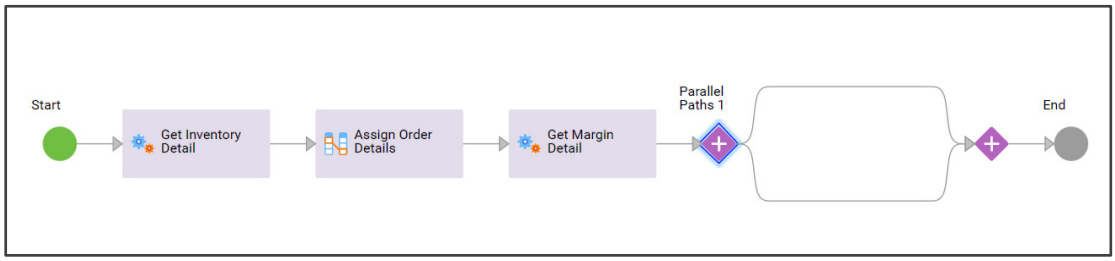
Add the Subprocess to the Main Process

Now, you will add the S_XX_Send Email process to the S_XX_Order Management process. You need to send emails to the consumer and the vendor. To do this, add a Parallel Paths step with two branches and add a service step on each branch. The first branch sends an email to the consumer and the second branch sends an email to the vendor.

134. In Application Integration, click **Explore**.
135. Open the **S_XX_Order Management** process.

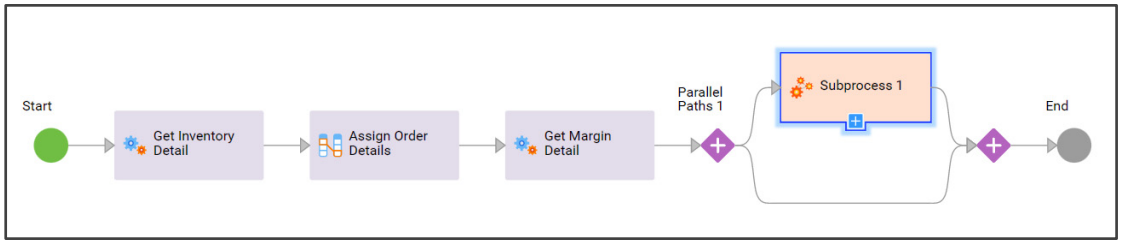
Note: You can use the Search box to find the S_XX_Order Management process.

136. Drag and drop a **Parallel** Path step onto the canvas in between the Get Margin Detail Service step and the End step.

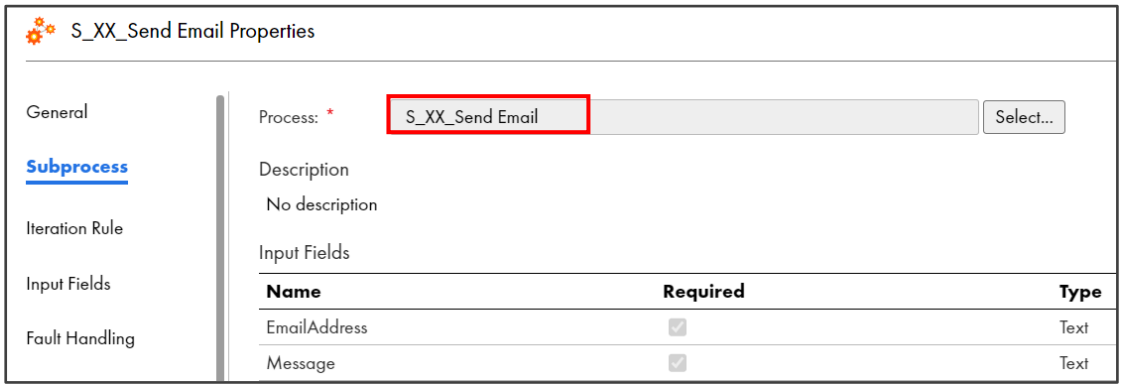


Next, you will add the S_XX_Send Email process to both branches. The first branch sends an email to the consumer. The second branch sends an email to the vendor.

137. Drag and drop a **Subprocess** step onto the first branch.

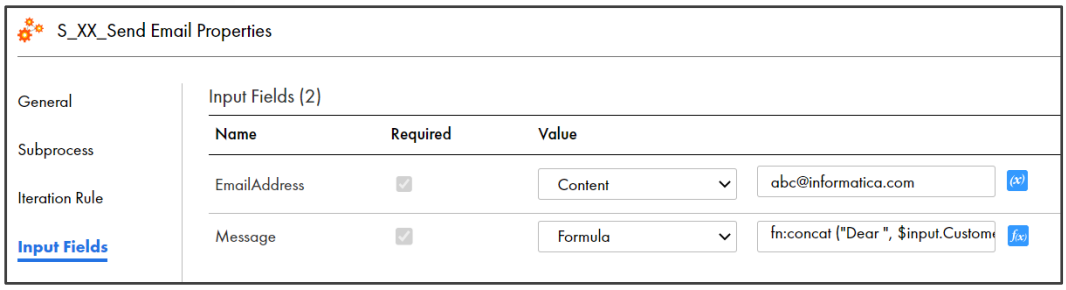


138. From the Subprocess tab, click **Select** and select the **S_XX_Send Email** process.



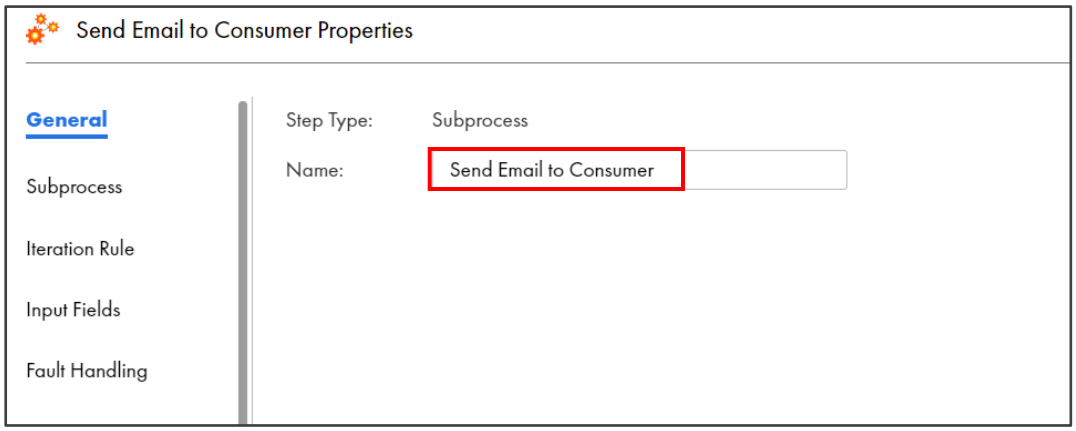
139. Select the **Input Fields** tab and add the following fields:

Name	Type	Value
EmailAddress	Content	Enter a valid email address.
Message	Formula	fn:concat ("Dear ", $input.CustomerName, ". Thanks for ordering.", ". We will deliver your order in 7 days.")

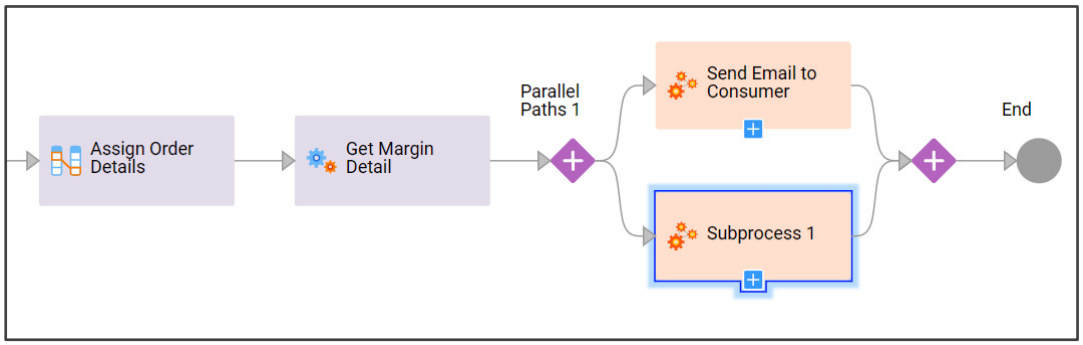


Note: To enter the value in the Expression field, from the **CAI Lab Prep Files_Learners** folder, open the notepad file named **34_LabGuide_CAI_Live Project_17-1**. Copy the command mentioned under **Step B** and paste it into the Expression field.

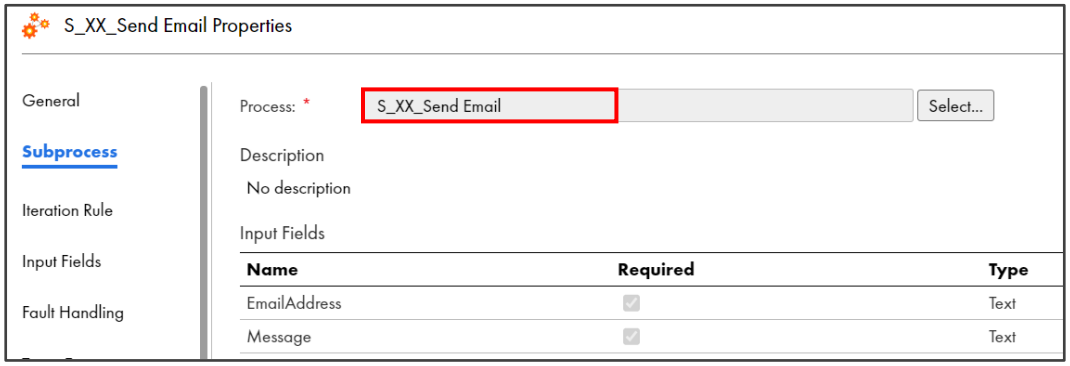
140. Select the **General** tab and in the Name field, enter **Send Email to Consumer**.



141. Drag and drop a **Subprocess** step onto the second branch.

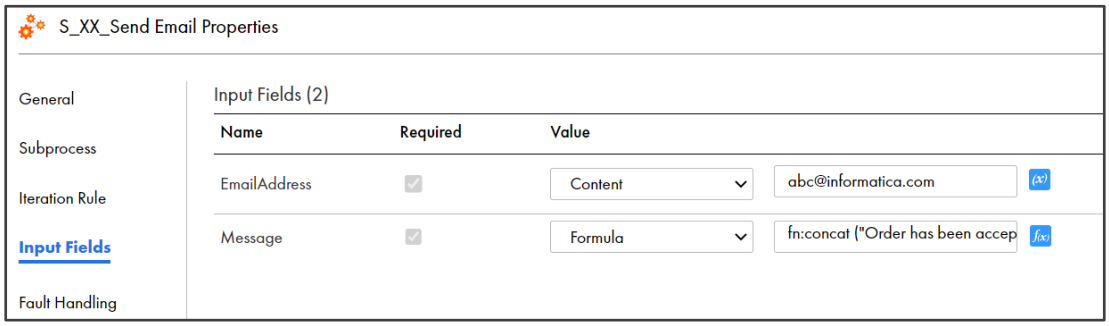


142. From the Subprocess tab, click **Select** and select the **S_XX_Send Email** process.

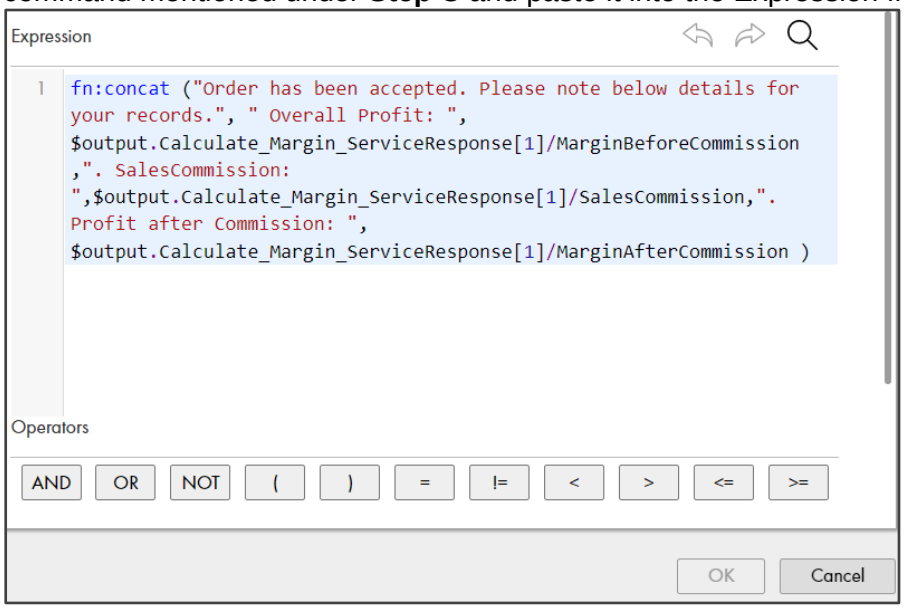


143. Select the Input Fields tab and add the following fields:

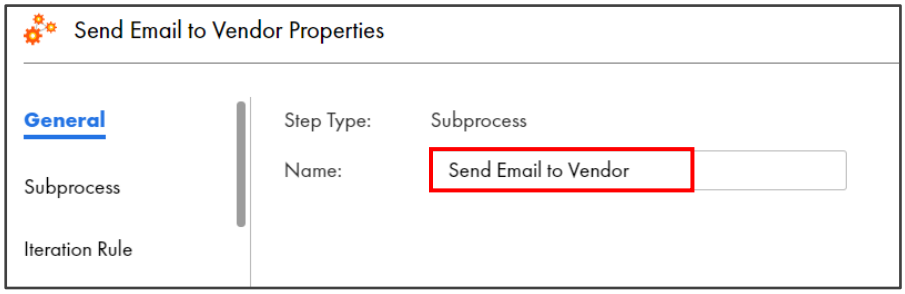
Name	Type	Value
EmailAddress	Content	Enter a valid email address.
Message	Formula	fn:concat ("Order has been accepted. Please note below details for your records.", " Overall Profit: ", $output.Calculate_Margin_ServiceResponse[1]/MarginBeforeCommission ,". SalesCommission: ",$output.Calculate_Margin_ServiceResponse[1]/SalesCommission,". Profit after Commission: ", $output.Calculate_Margin_ServiceResponse[1]/MarginAfterCommission)



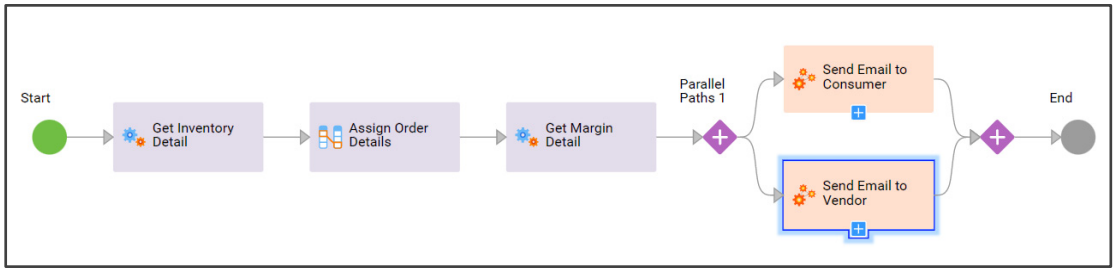
Note: To enter the value in the Expression field, from the **CAI Lab Prep Files_Learners** folder, open the notepad file named **34_LabGuide_CAI_Live Project_17-1**. Copy the command mentioned under **Step C** and paste it into the Expression field.



144. Select the **General** tab and in the Name field, enter **Send Email to Vendor**.



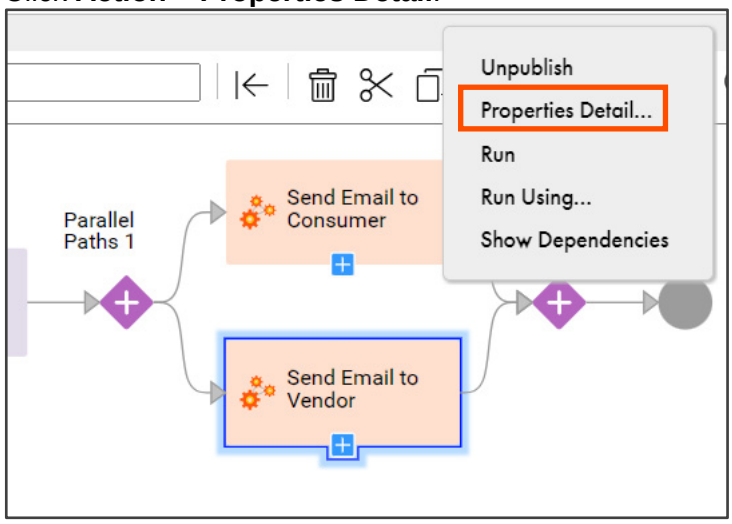
The following image shows the process canvas:



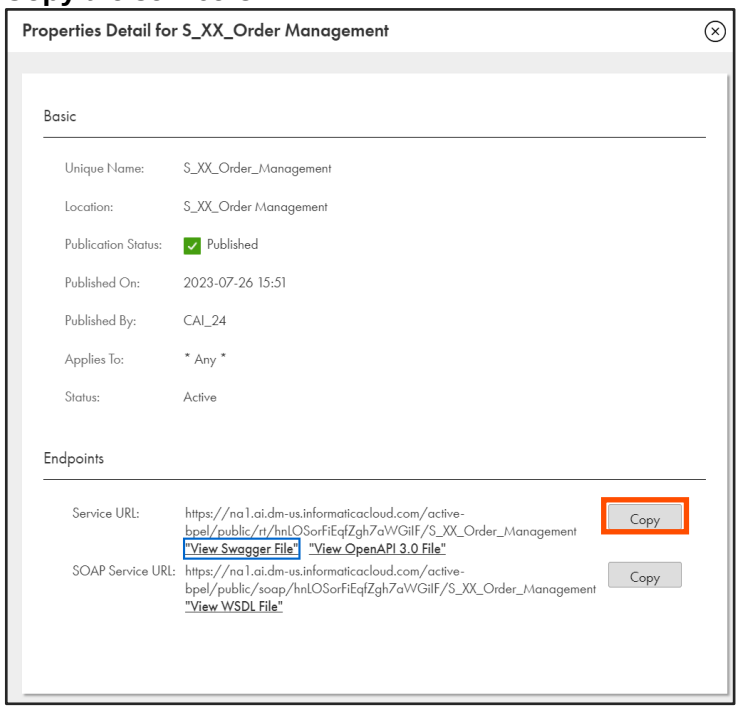
145. **Save** and **Publish** the process.

Invoke the Modified Order Management Process

146. Click **Action** > **Properties Detail**.



147. **Copy** the service URL.



148. Open a notepad file, paste the Service URL, and add the following input parameters to the end of the service URL:

<serviceURL>?CustomerName=TestConsumer&CustomerEmail=testconsumer@mailinator.com&ItemName=item1&ItemCount=2

OR

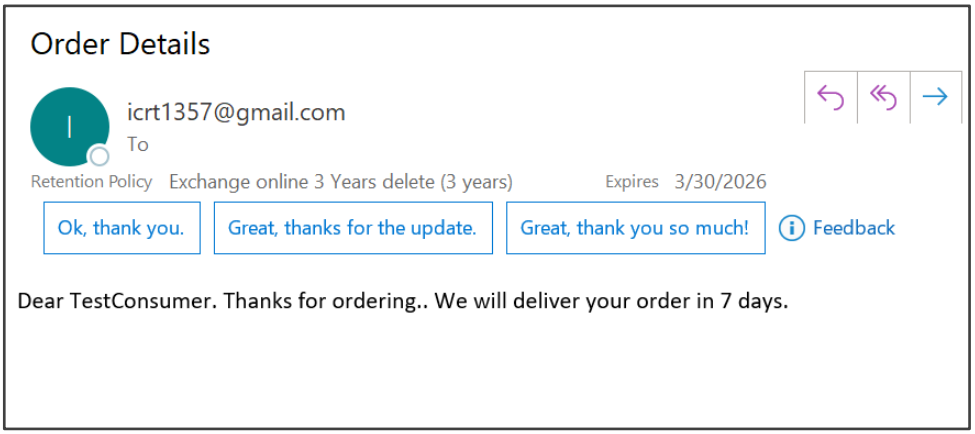
From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **34_LabGuide_CAI_Live Project_17-1**. Copy the command mentioned under **Step D**.

The following URL is a sample service URL with input parameters:

https://na1.ai.dm-us.informaticacloud.com/active-bpel/public/rt/hnLOSorFiEqfZgh7aWGiIF/S_XX_Order_Management-1?CustomerName=TestConsumer&CustomerEmail=testconsumer@mailinator.com&ItemName=item1&ItemCount=2

149. Open a browser and paste the service URL with input parameters.

Note: You see a successful response and an email to the given email address.



Invoke Process to Show a Fault

150. In a browser, paste the following parameterized service URL. Here, we have changed the item name to Newitem1, which is an invalid value.

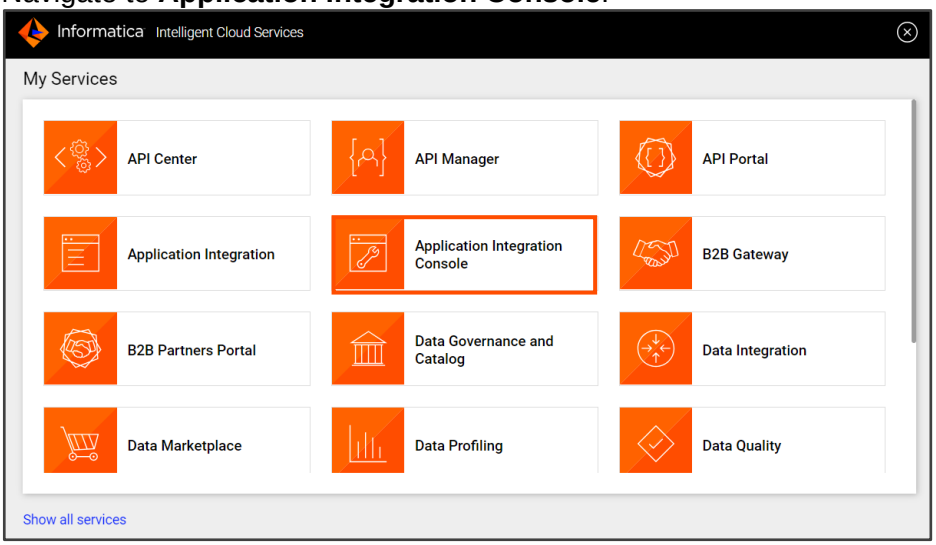
<serviceURL>?CustomerName=TestConsumer&CustomerEmail=testconsumer@mailinator.com&ItemName=Newitem1&ItemCount=2

OR

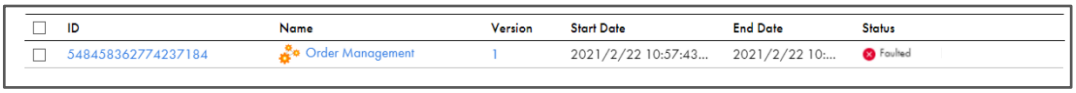
From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **34_LabGuide_CAI_Live Project_17-1**. Copy the command mentioned under **Step E** and paste it into the browser.

You see an HTTPS 400 error response.

151. Navigate to **Application Integration Console**.



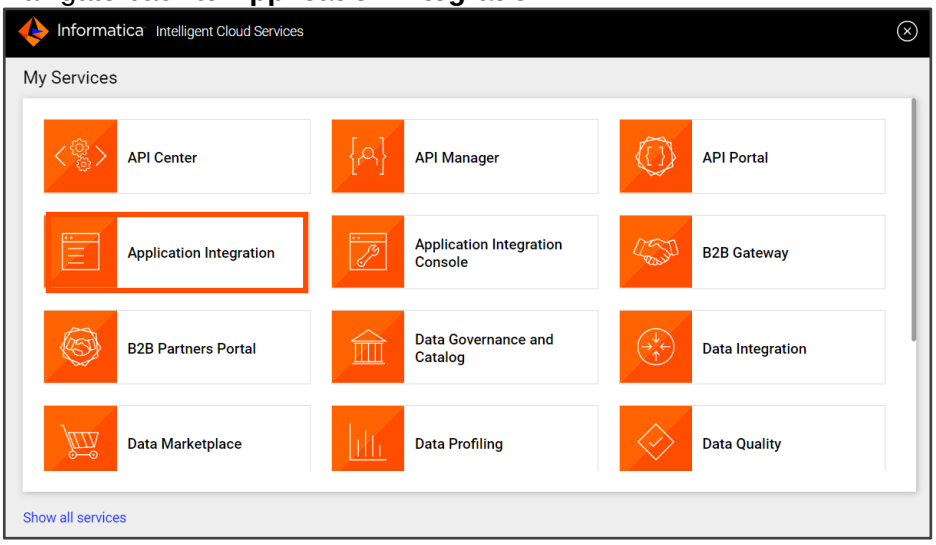
152. Click **Processes**.
153. You can see that the Process status is Faulted.



Note: The process has been faulted because "newitem1" does not exist.

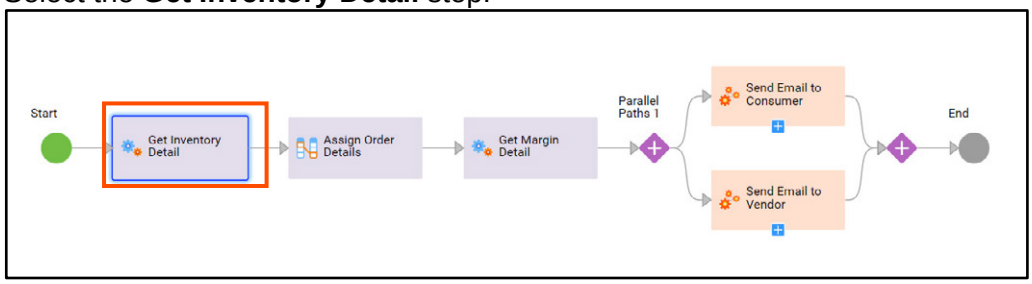
Enable Fault Handling

154. Navigate back to **Application Integration**.

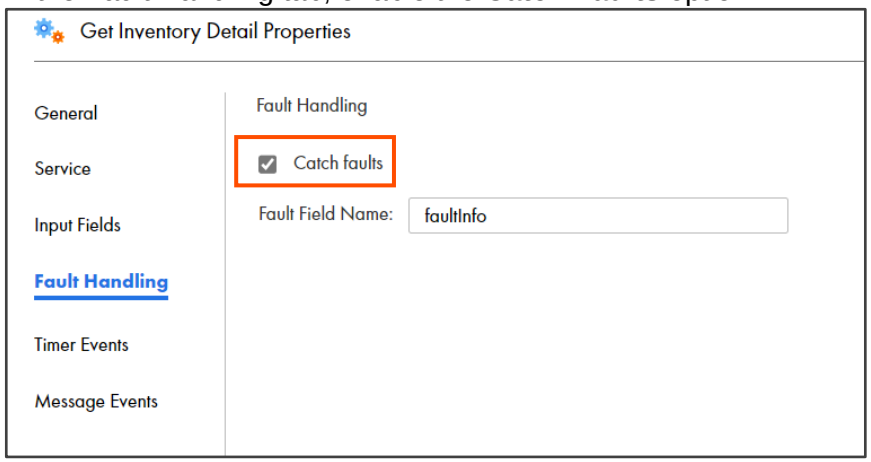


155. From the Explore window, select, and open the S_XX_Order Management process.

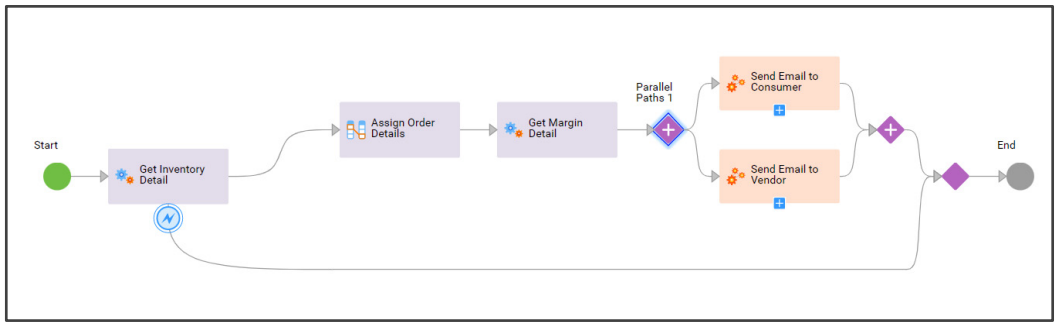
156. Select the **Get Inventory Detail** step.



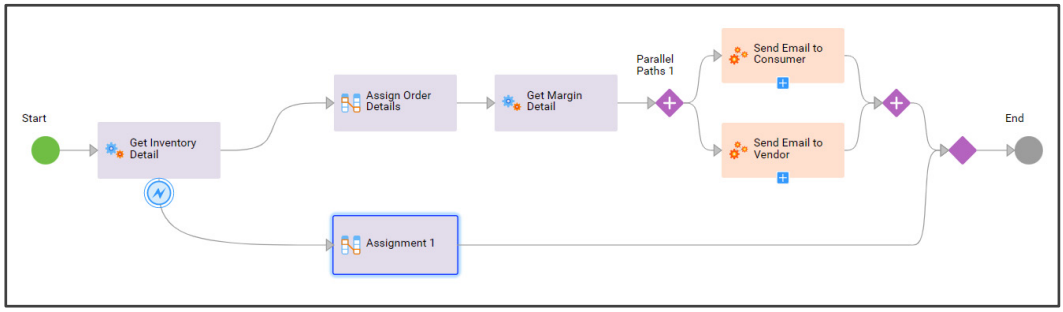
157. In the Fault Handling tab, enable the **Catch Faults** option.



158. Drag and drop the **Assign Order Details** step to the “no fault” branch.
159. Repeat the previous step for all steps except the End step. Do not change the order of the steps.

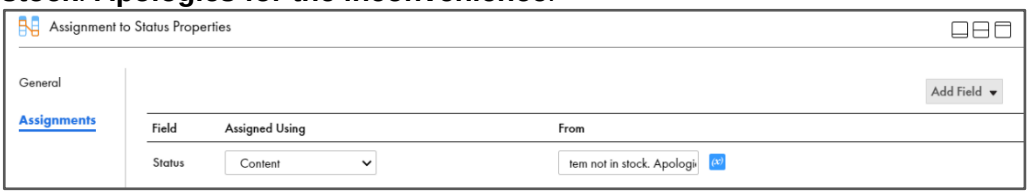


160. Drag and drop an Assignment step onto the Fault (bottom) branch.



161. In the General tab, in the Name field, enter **Assignment to Status**.

162. In the Assignments tab, add the Status field with the Value as **Content** > **Item not in stock**. **Apologies for the inconvenience**.



163. Save and publish the process.
164. Click **Actions** > **Properties Detail**.
165. **Copy** the Service URL.
166. Open a notepad file, paste the Service URL, and add the following parameters to the end of the service URL:

    <ServiceURL>?CustomerName=TestConsumer&CustomerEmail=testconsumer@mailinator.com&ItemName=newitem1&ItemCount=2

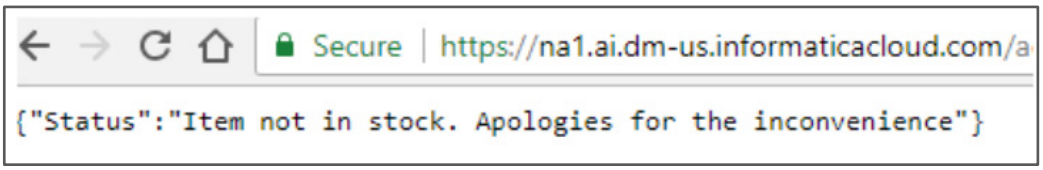
    OR

    From the **CAI Lab Prep Files_Learners** folder, open the notepad file named **34_LabGuide_CAI_Live Project_17-1**. Copy the command mentioned under **Step F**.

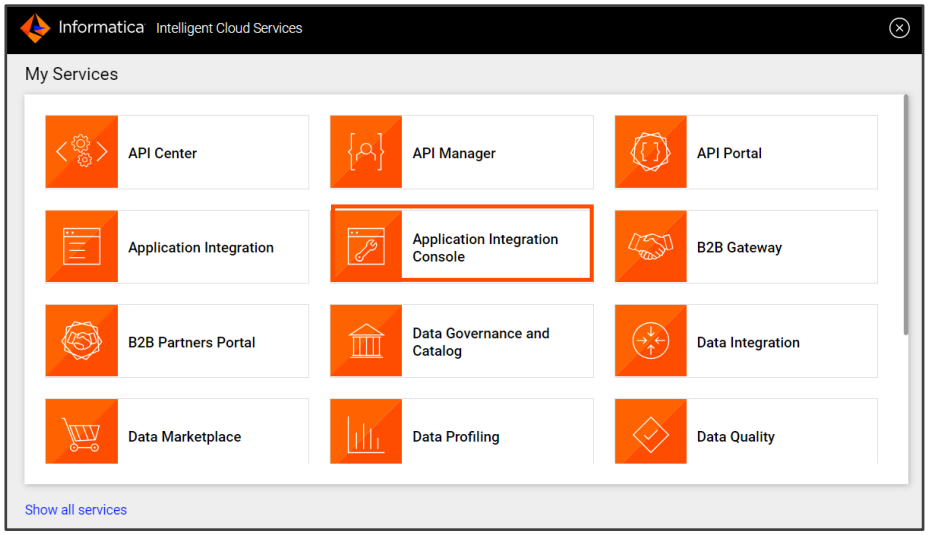
    The following URL is a sample service URL with input parameters:

    https://na1.ai.dm-us.informaticacloud.com/active-bpel/public/rt/hnLOSorFiEqfZgh7aWGiIF/S_XX_Order_Management-1?CustomerName=TestConsumer&CustomerEmail=testconsumer@mailinator.com&ItemName=newitem1&ItemCount=2

167. Open a browser and paste the service URL with input parameters.

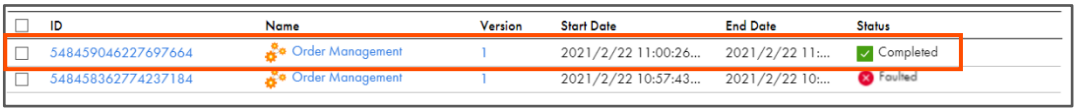


168. Navigate to **Application Integration Console**.



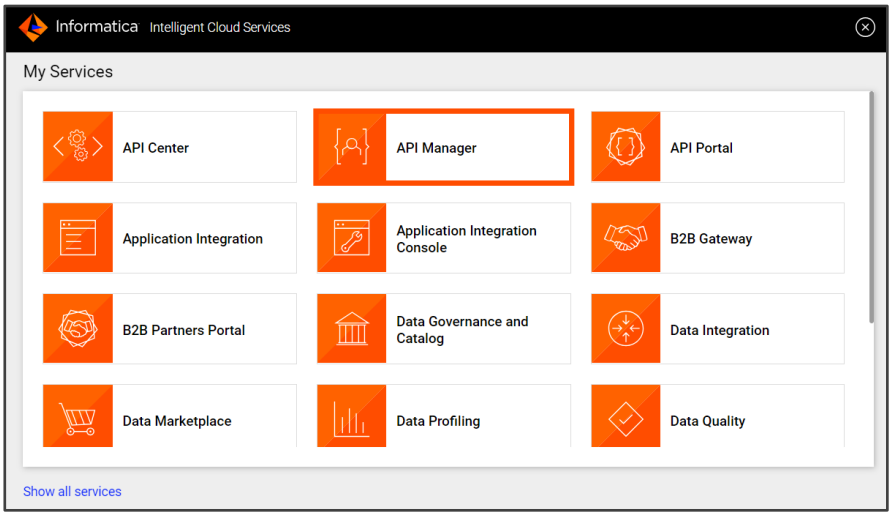
169. Click **Processes**.

170. You can see that the Process status is not Faulted**.**



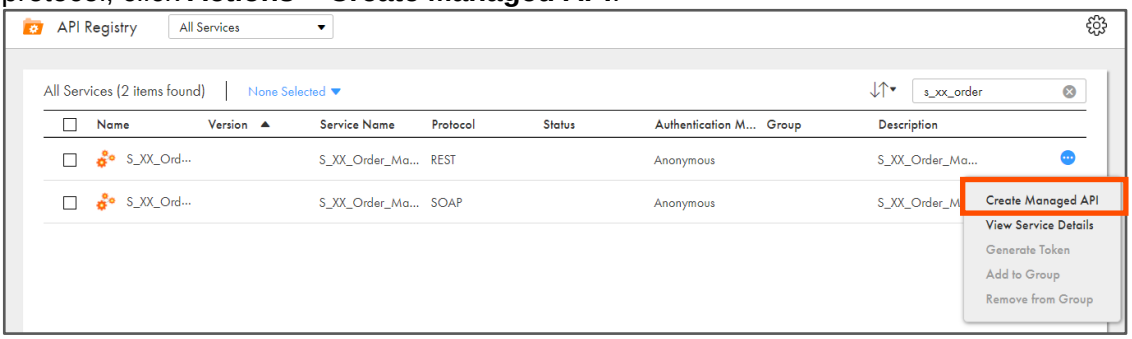
Create a Managed API

171. Navigate to **API Manager**.



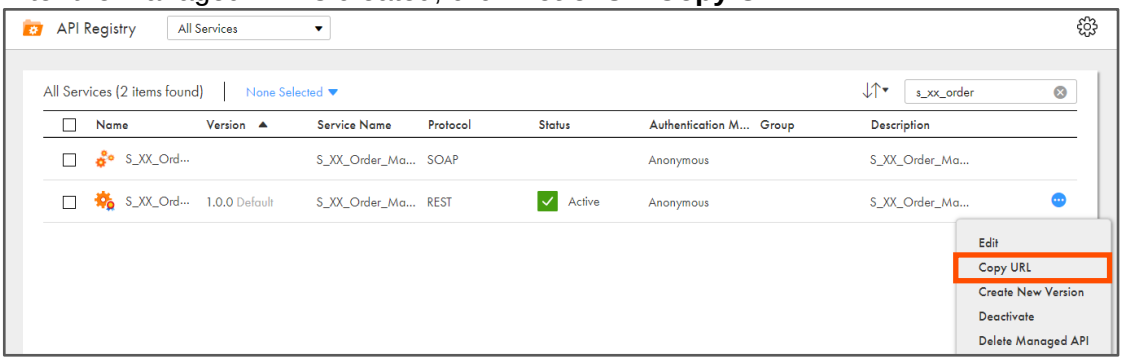
Note: If you are prompted to choose an API domain name. Retain the default name and click **Save**.

172. On the API Registry page, for the S_XX_Order_Management process with the REST protocol, click **Actions** > **Create Managed API**.



173. In the Create Managed API page, keep the default details and click **Create**. The managed API is successfully created.

174. After the Managed API is created, click **Actions** > **Copy URL**.



175. Open a notepad file, paste the URL, and add the following parameters to the end of the URL:

     <ServiceURL>?CustomerName=TestConsumer&CustomerEmail=testconsumer@mailinator.com&ItemName= item1&ItemCount=2

     OR

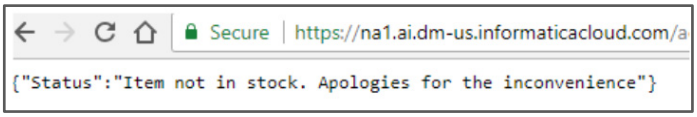
     From the **CAI Lab Prep Files_Learners**, open the notepad file named **34_LabGuide_CAI_Live Project_17-1**. Copy the command mentioned under **Step G**.

     The URL should look like the following URL:

     https://apigw-pod1.dm-us.informaticacloud.com/t/gvsrtehvymkfpvakw0qkod.com/Order_Management?CustomerName=TestConsumer&CustomerEmail=testconsumer@mailinator.com&ItemName=%20item1&ItemCount=2

176. Paste the URL in a browser.

     You see the same result as when you invoked the process with the Application Integration service URL.



     You have successfully created a process and invoked it.

This concludes the lab.